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(54) **BACTERIA ENGINEERED TO REDUCE
HYPERPHENYLALANINEMIA**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Genetically engineered bacteria, pharmaceutical composi-
tions thereof, and methods of modulating and treating dis-
eases associated with hyperphenylalaninemia are disclosed.

9 Claims, 90 Drawing Sheets

Specification includes a Sequence Listing.

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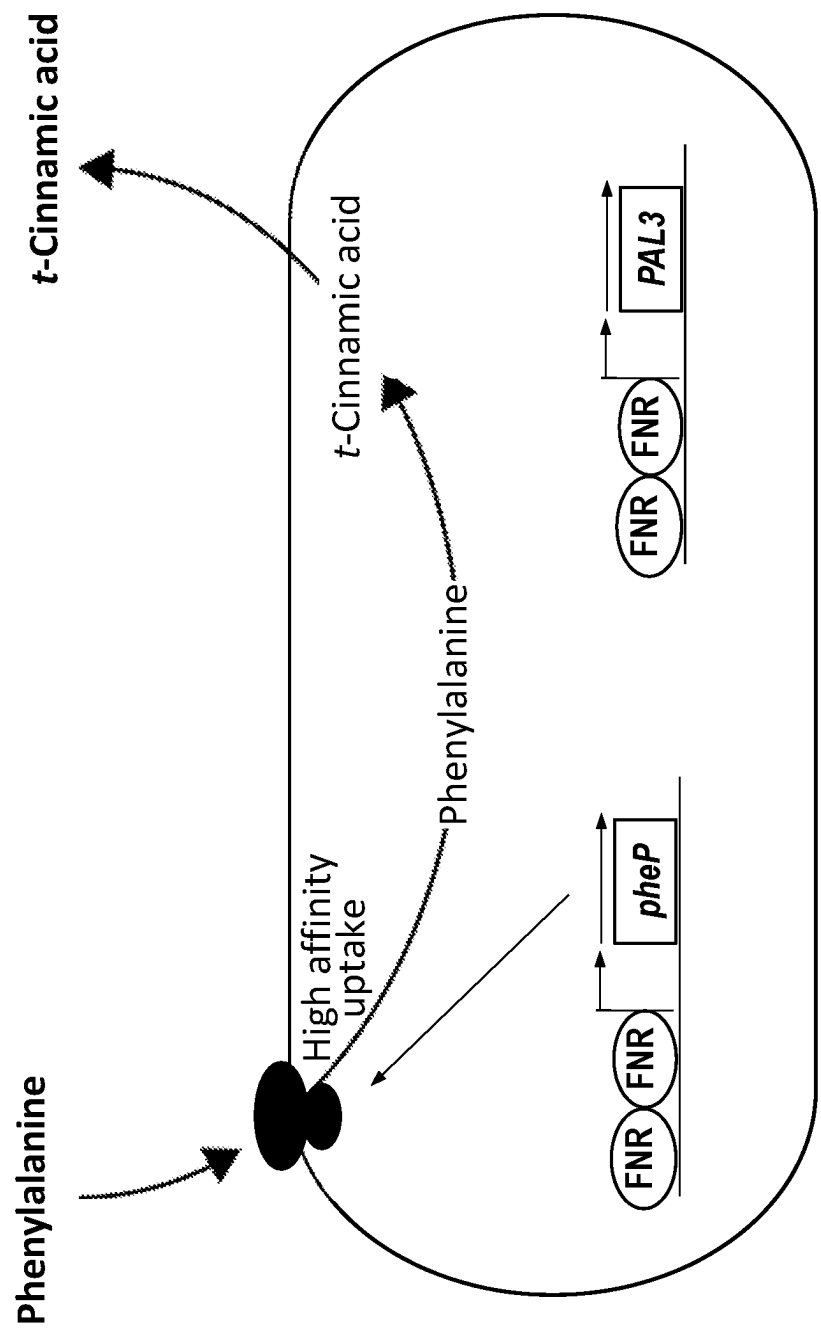
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PAL = Phenylalanine Ammonia Lyase
PheP = High Affinity Phenylalanine Transporter

Fig. 1

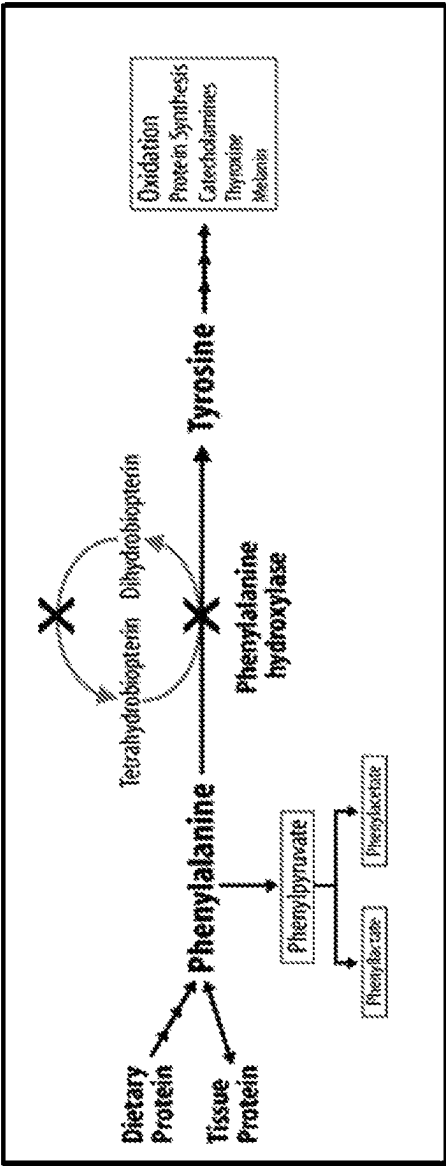


Fig. 2A

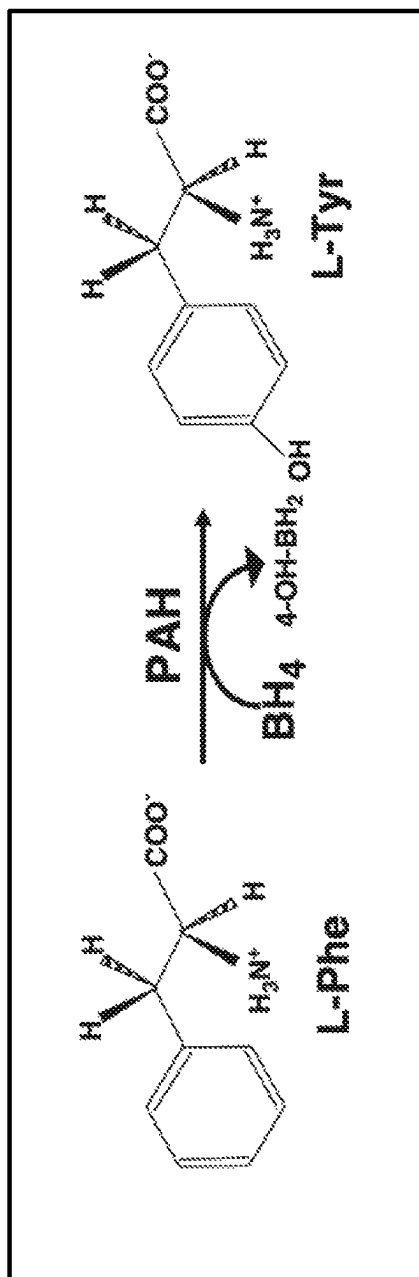


Fig. 2B

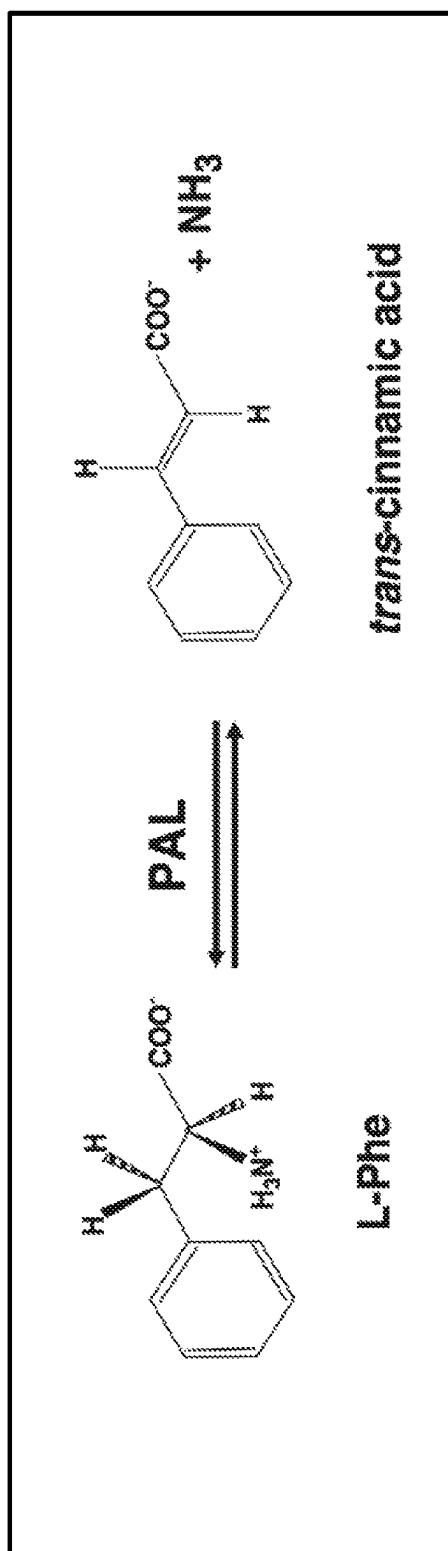


Fig. 2C

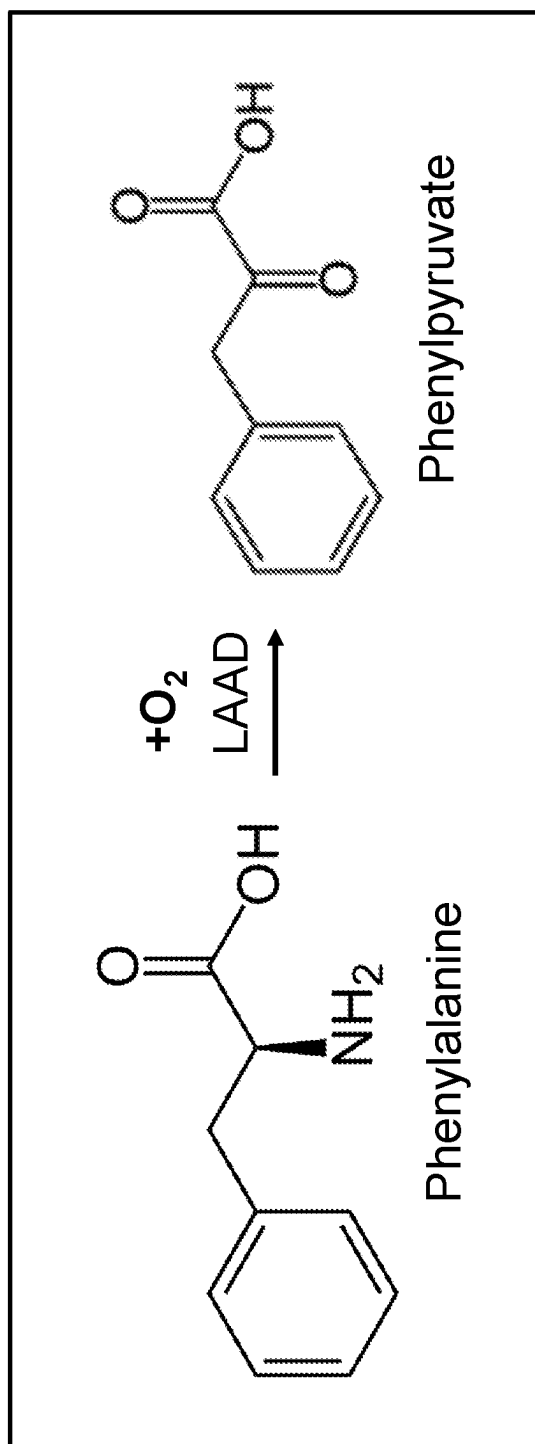
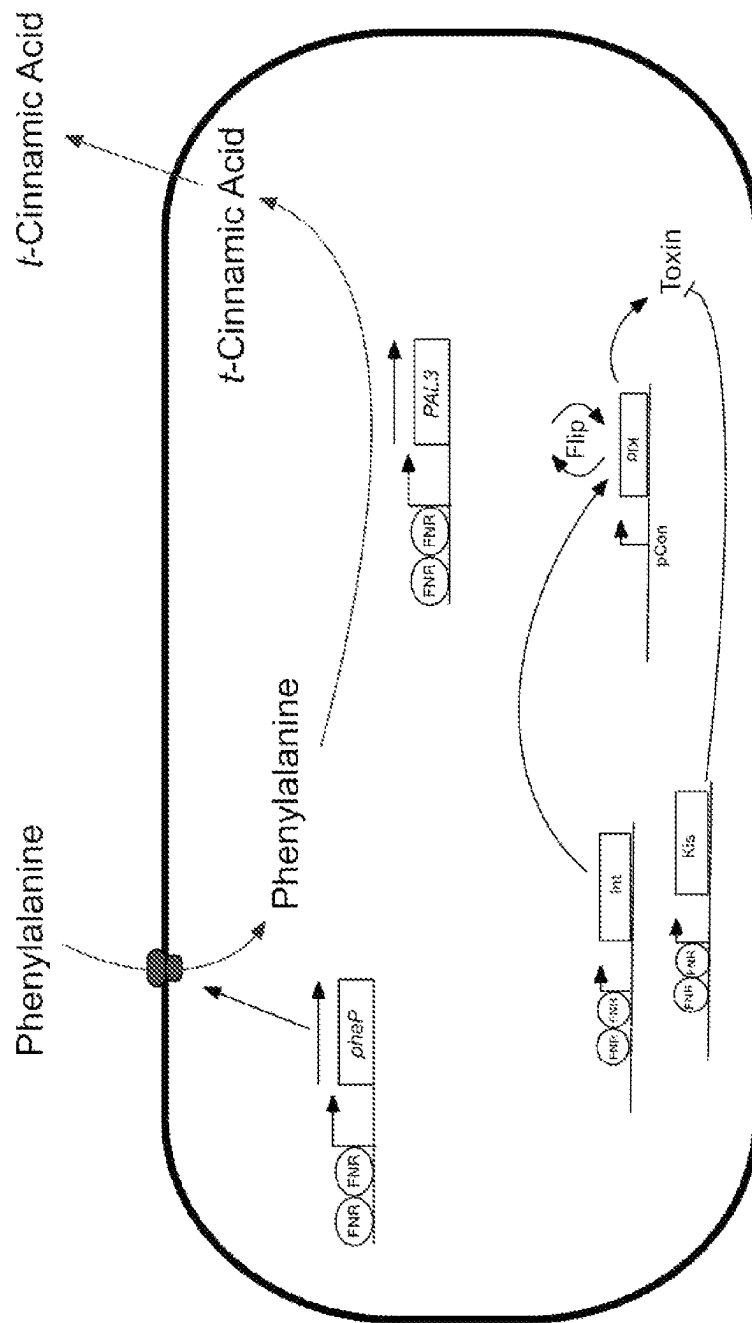


Fig. 2D



- PAL = phenylalanine ammonia lyase from an Enterobacteriaceae species
- PheP = high affinity phenylalanine transporter

Fi. 3

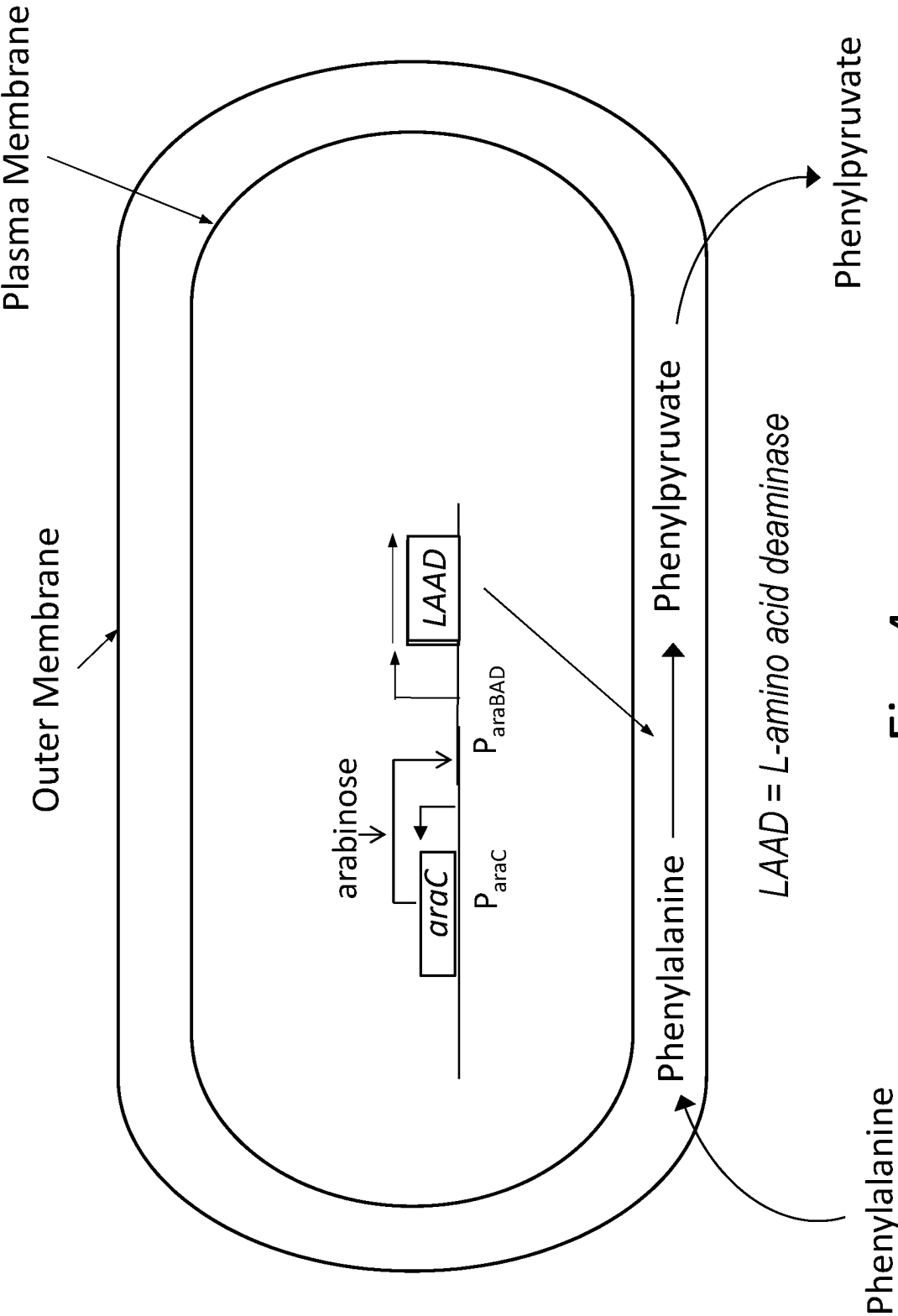
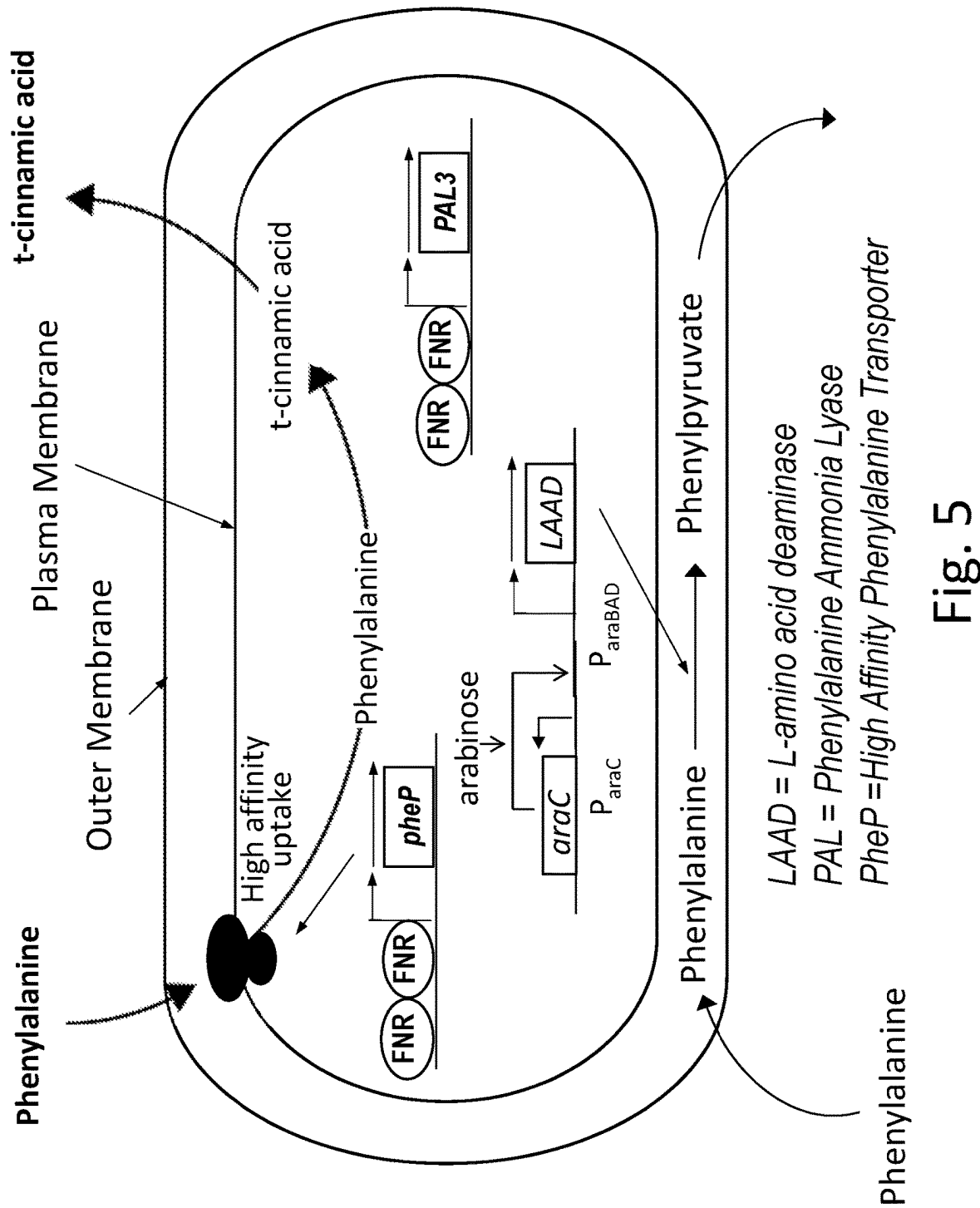


Fig. 4



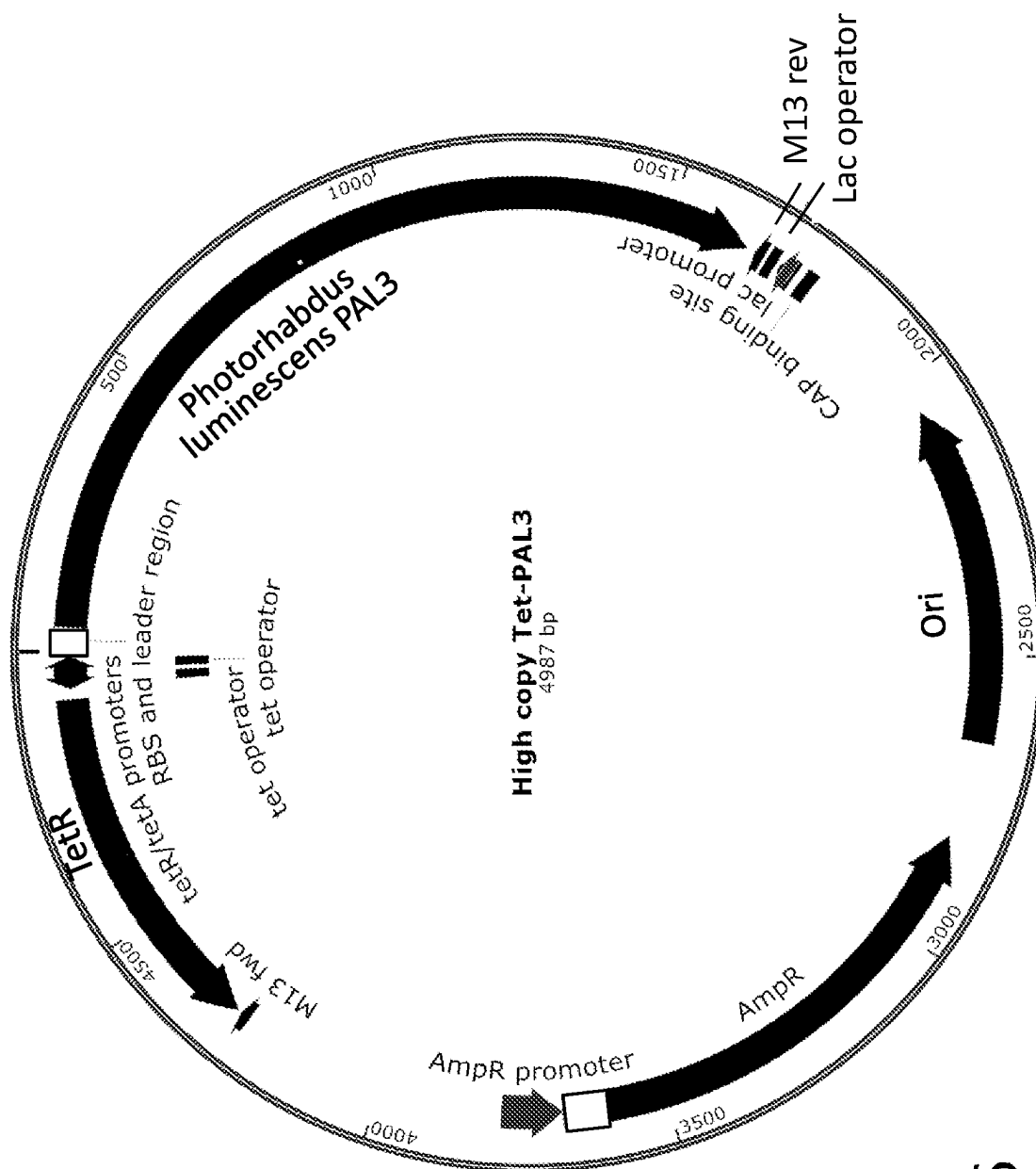


Fig. 6

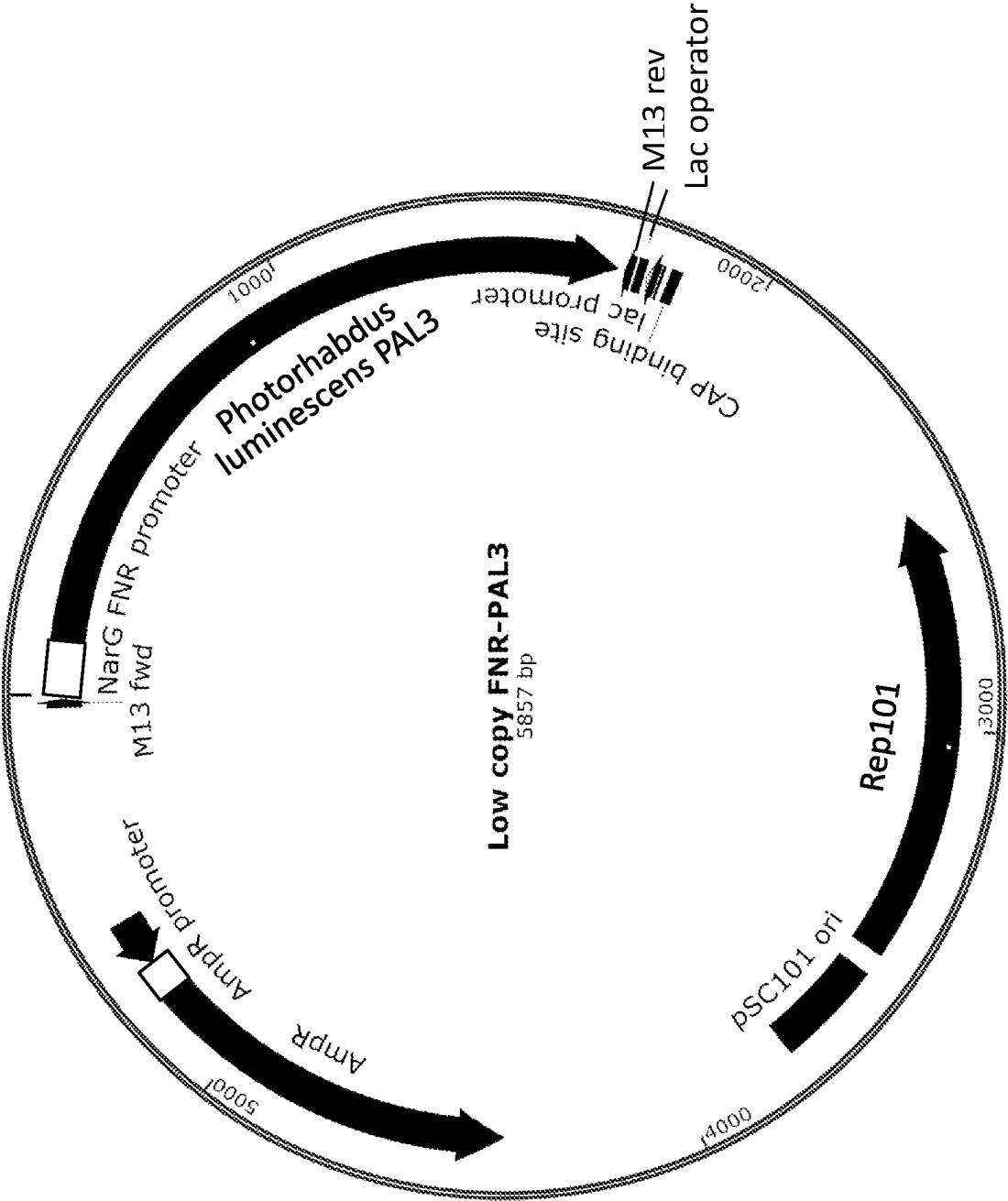


Fig. 7

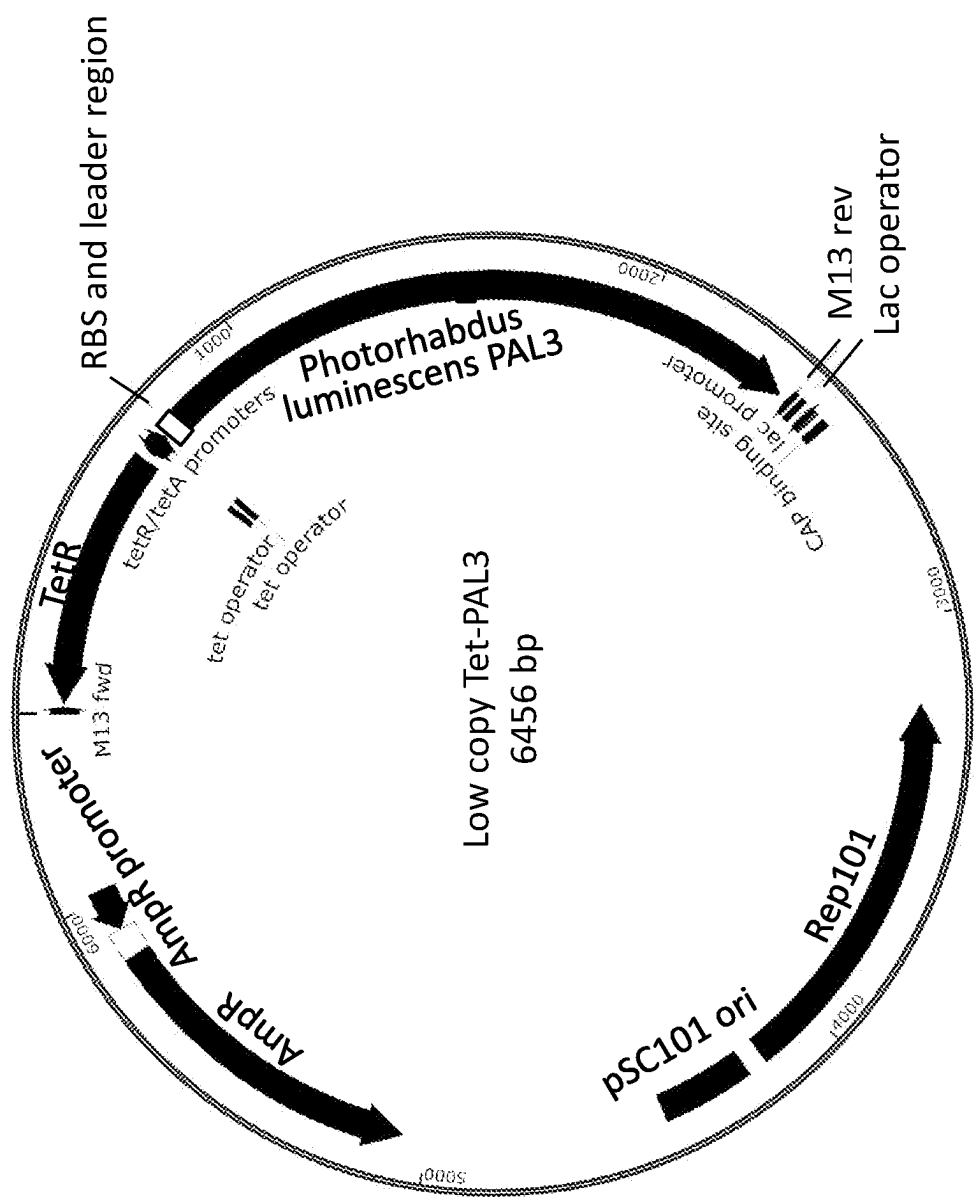


Fig. 8

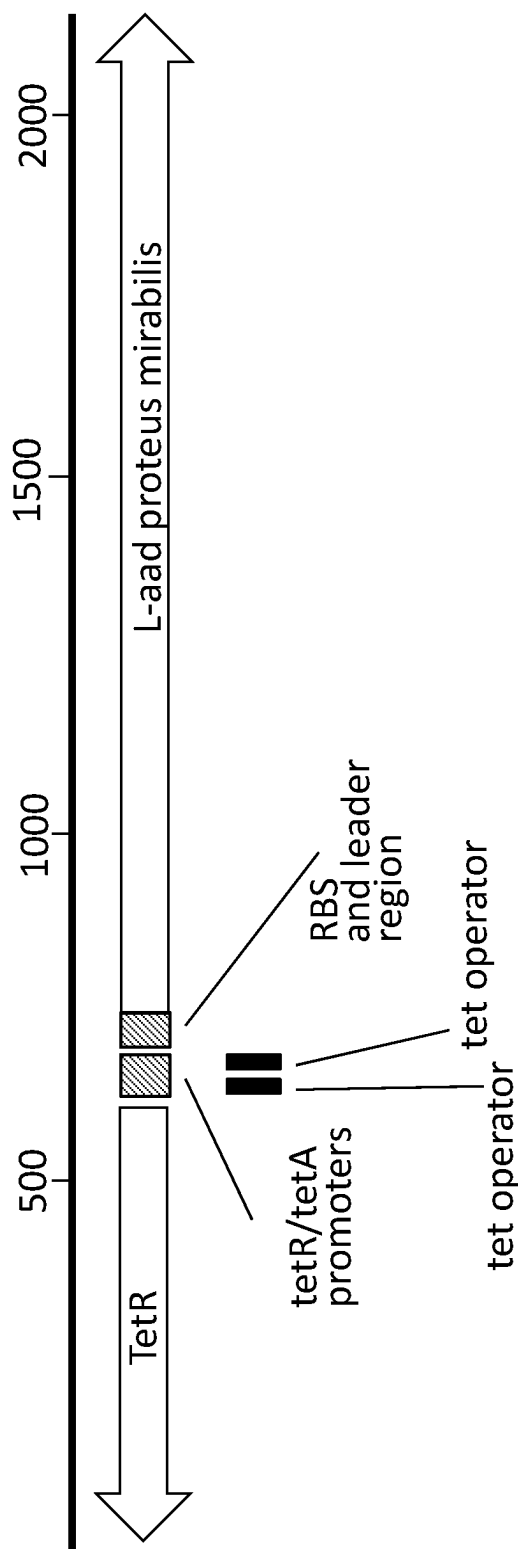


Fig. 9

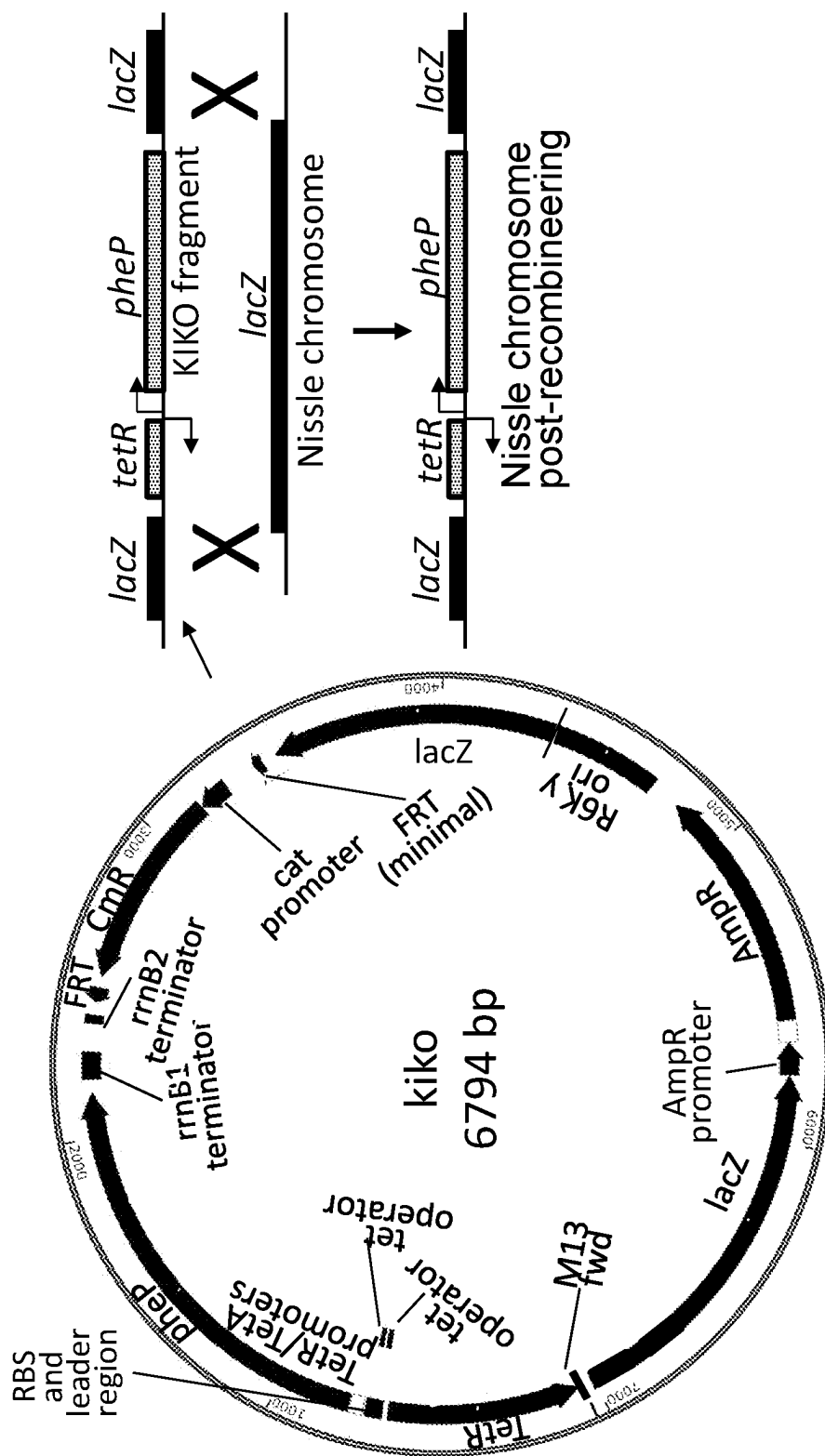


Fig. 10

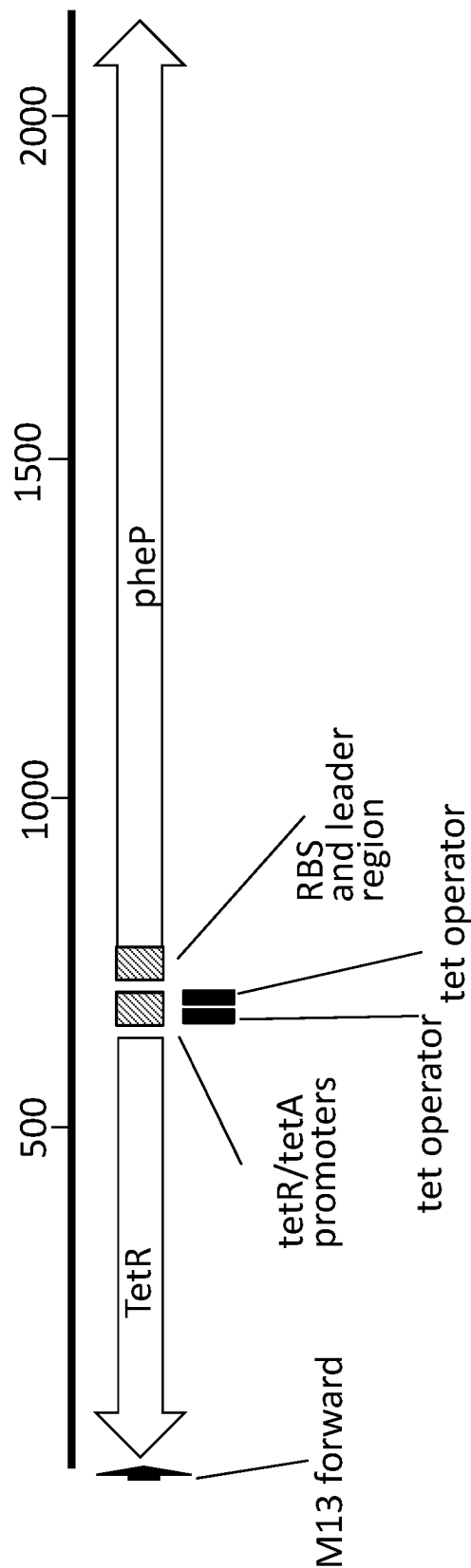


Fig. 11

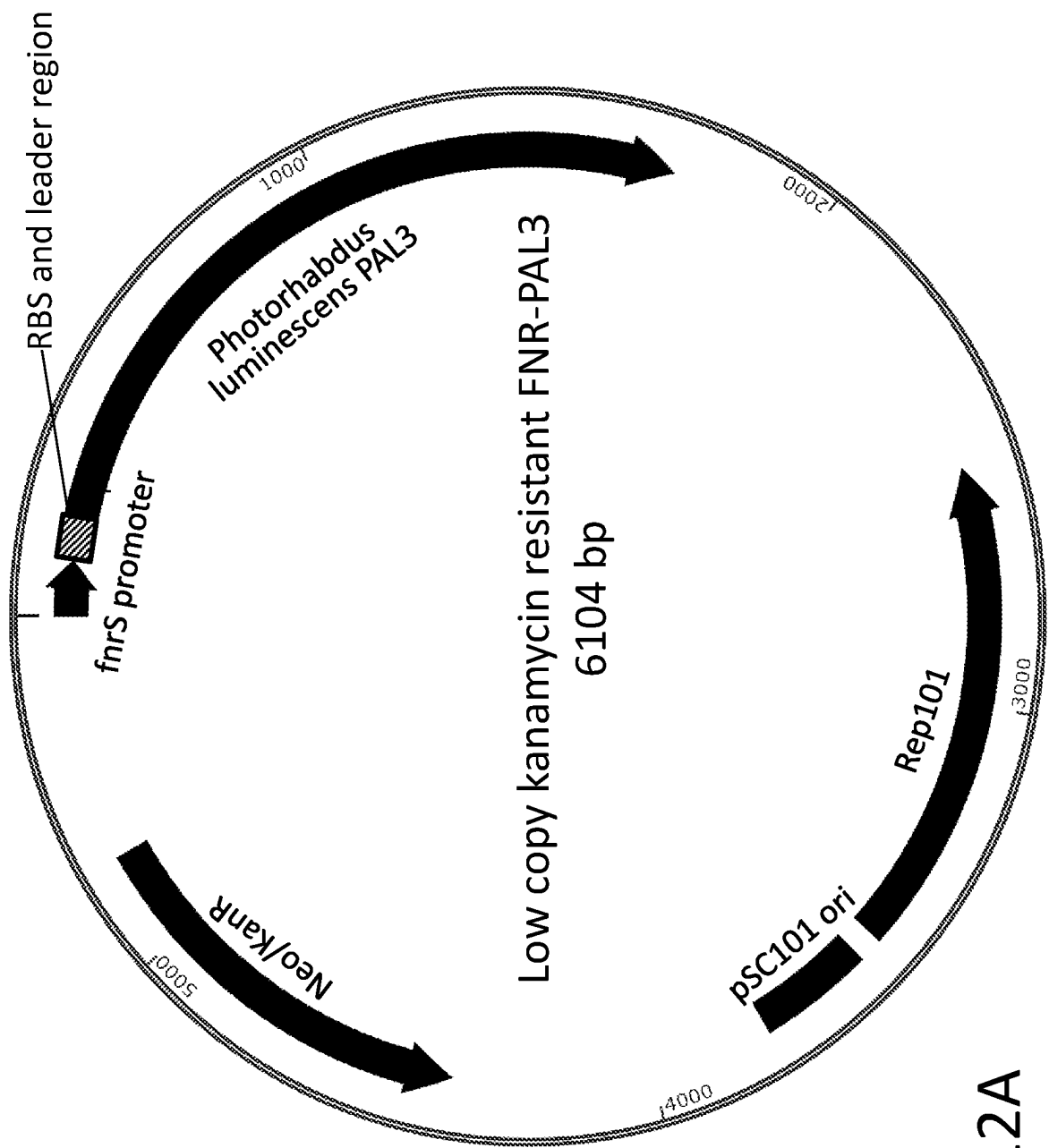


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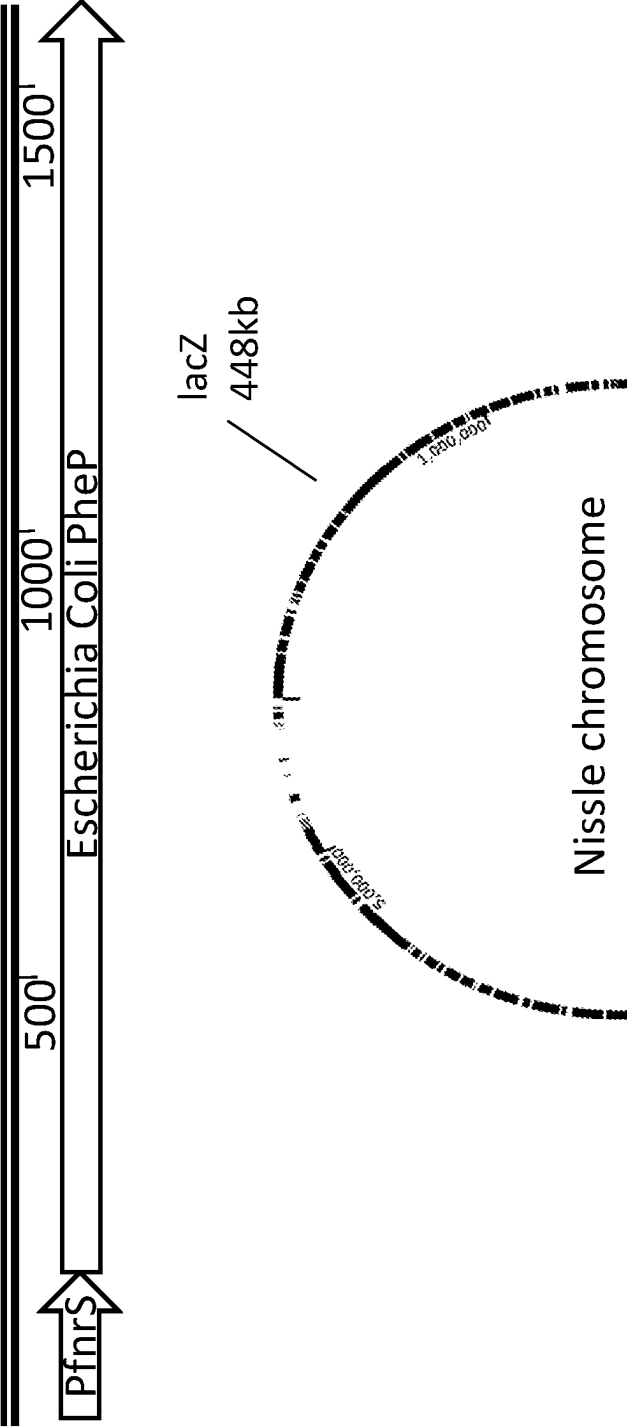


Fig. 12B

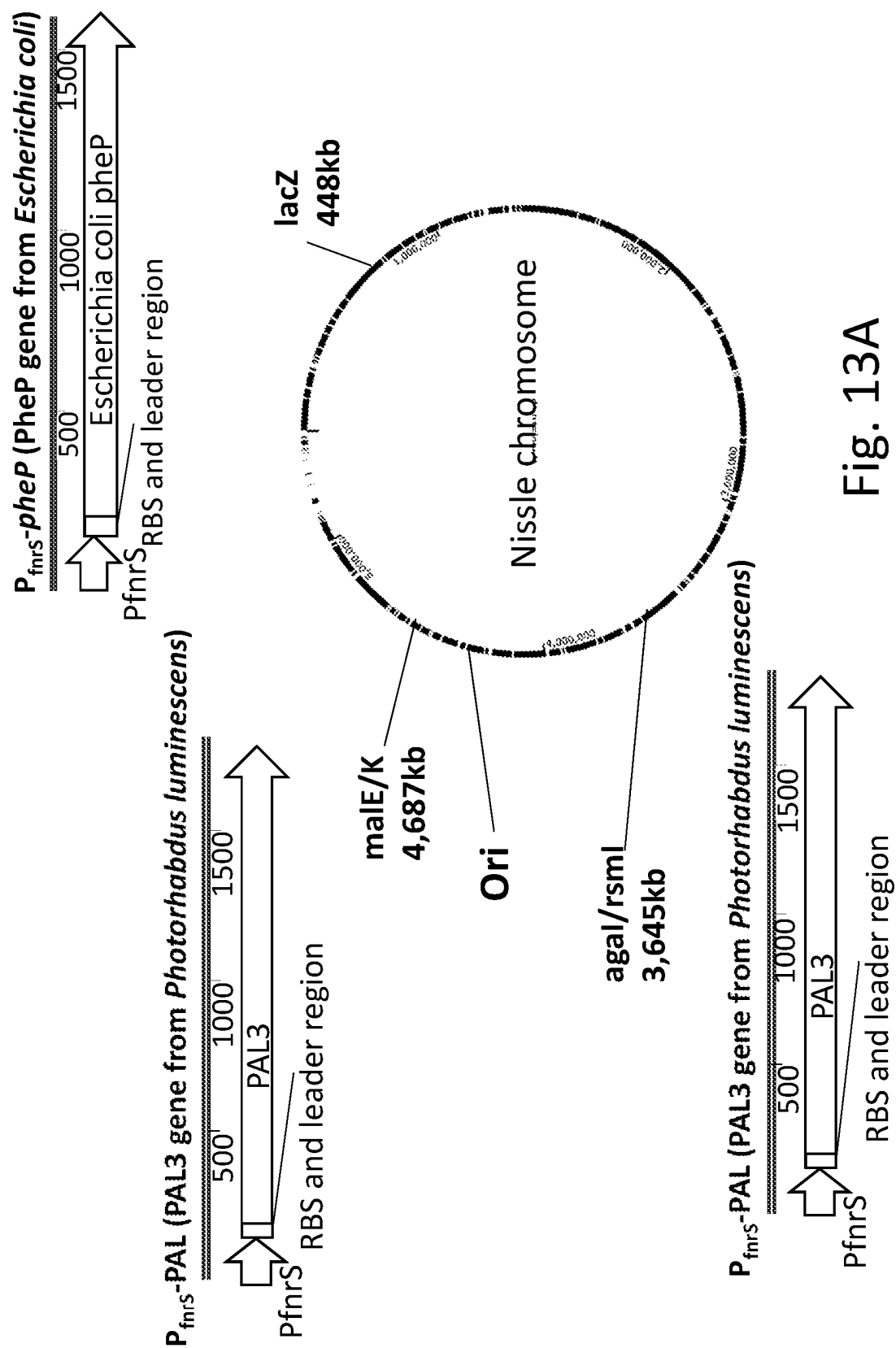
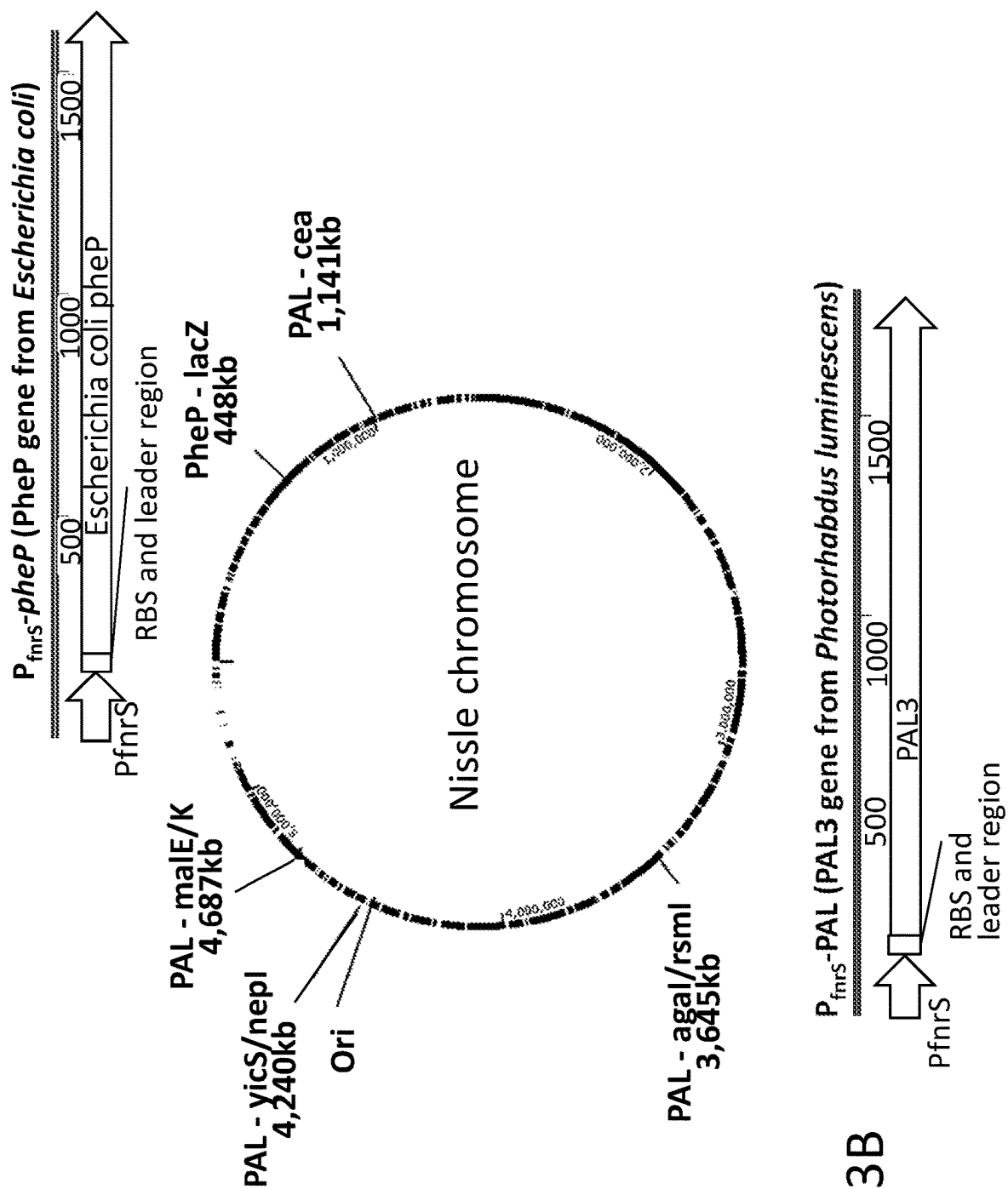


Fig. 13A



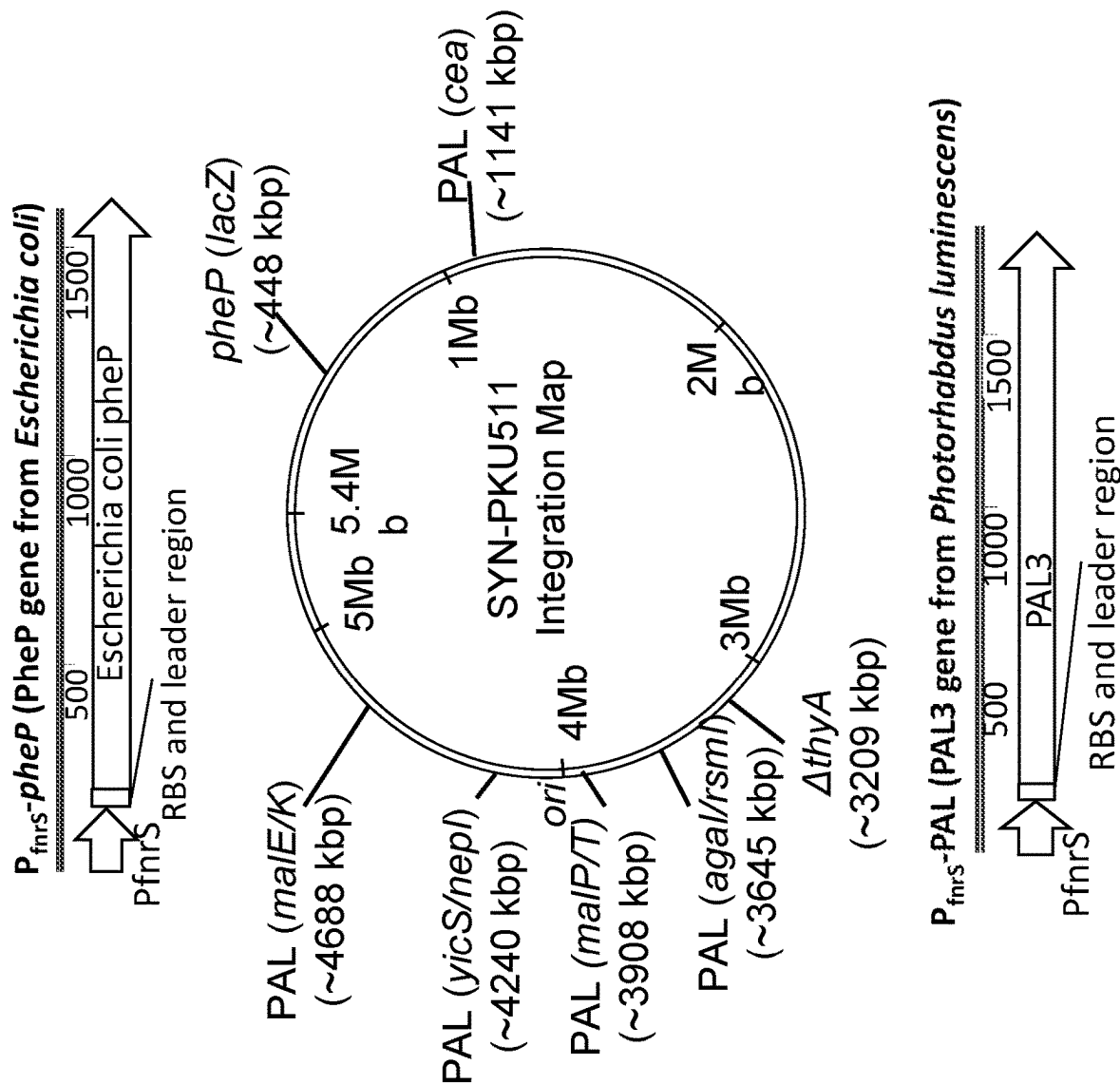


Fig. 13C

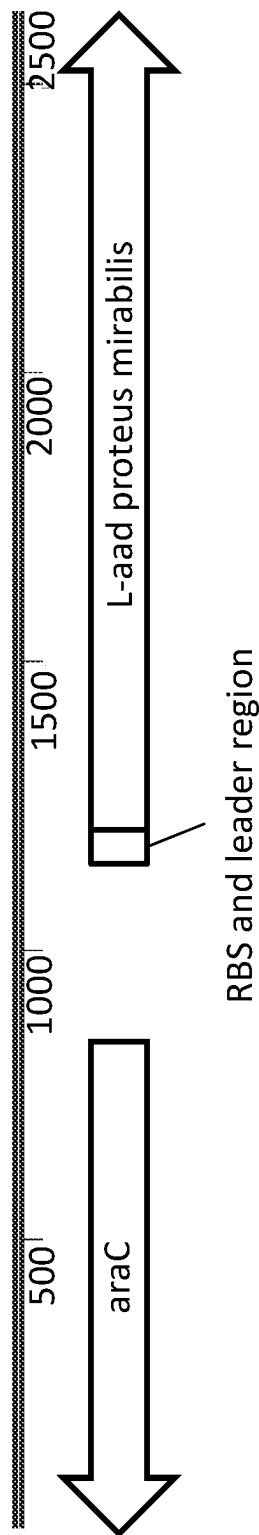


Fig. 14

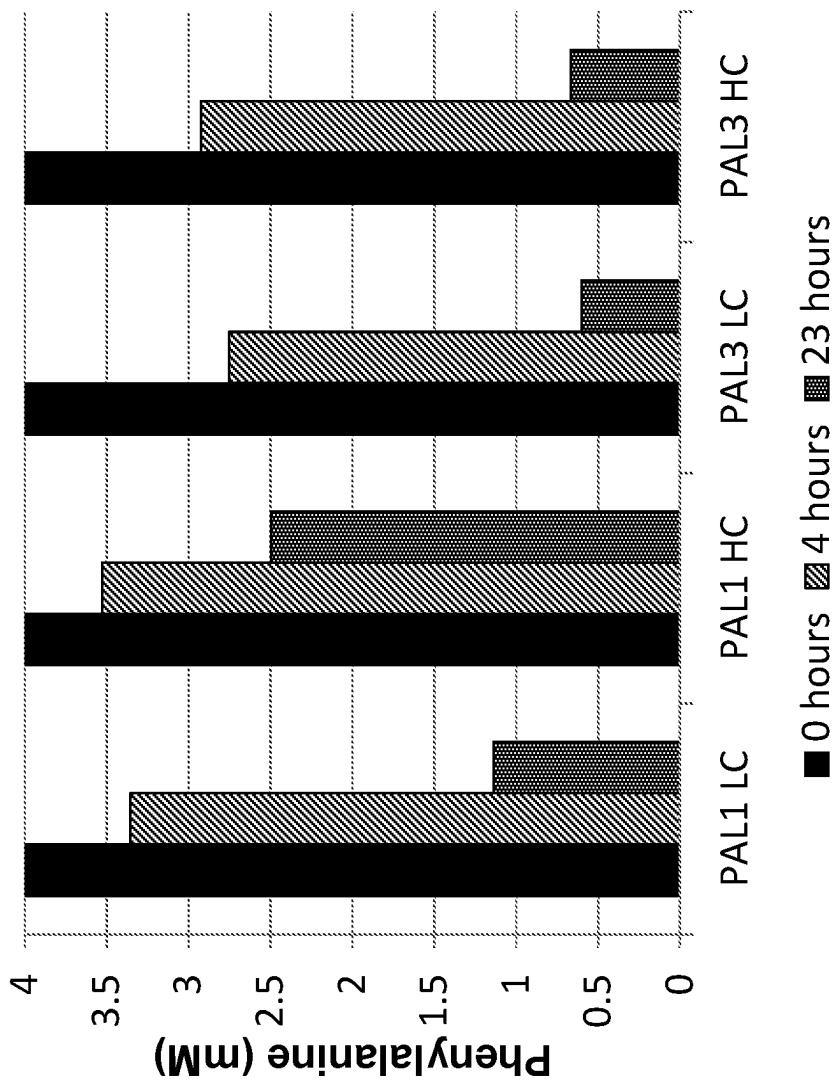


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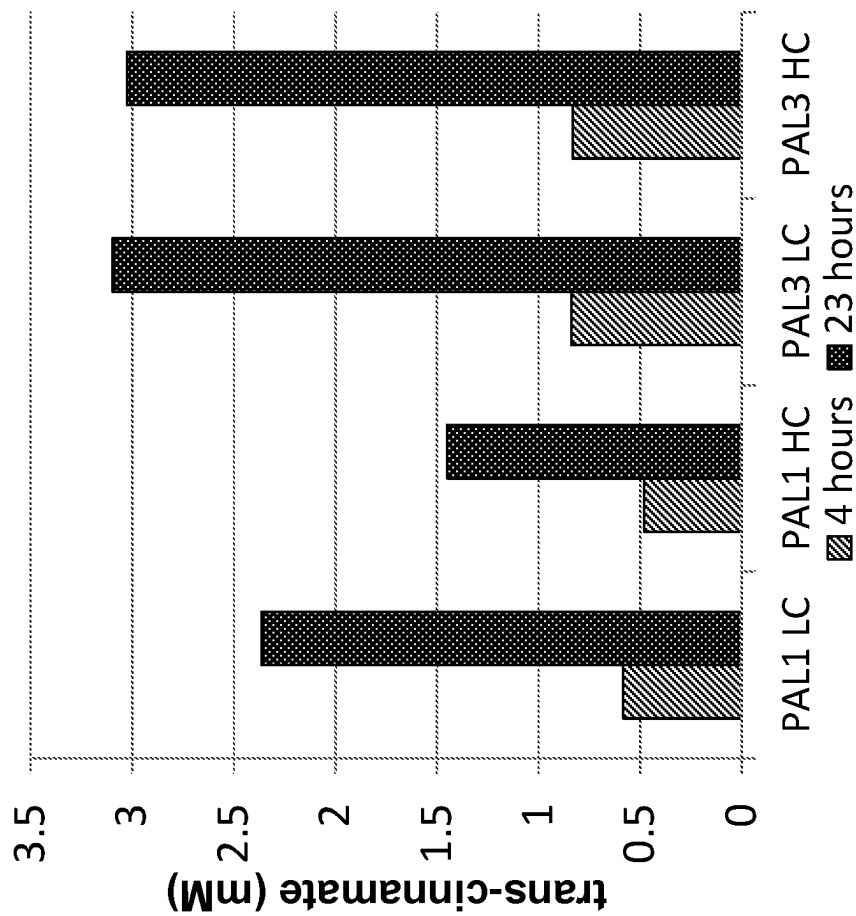


Fig. 15B

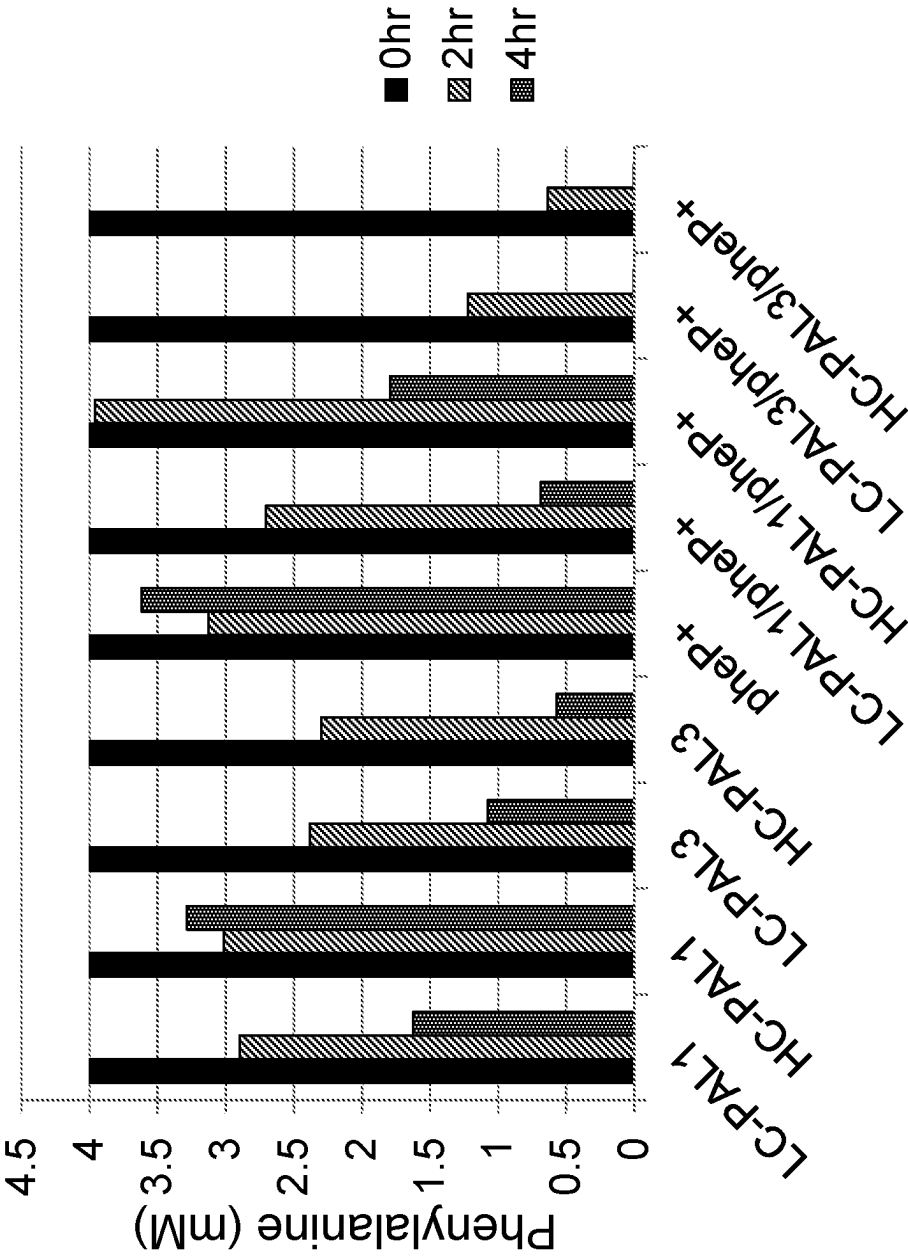


Fig. 16A

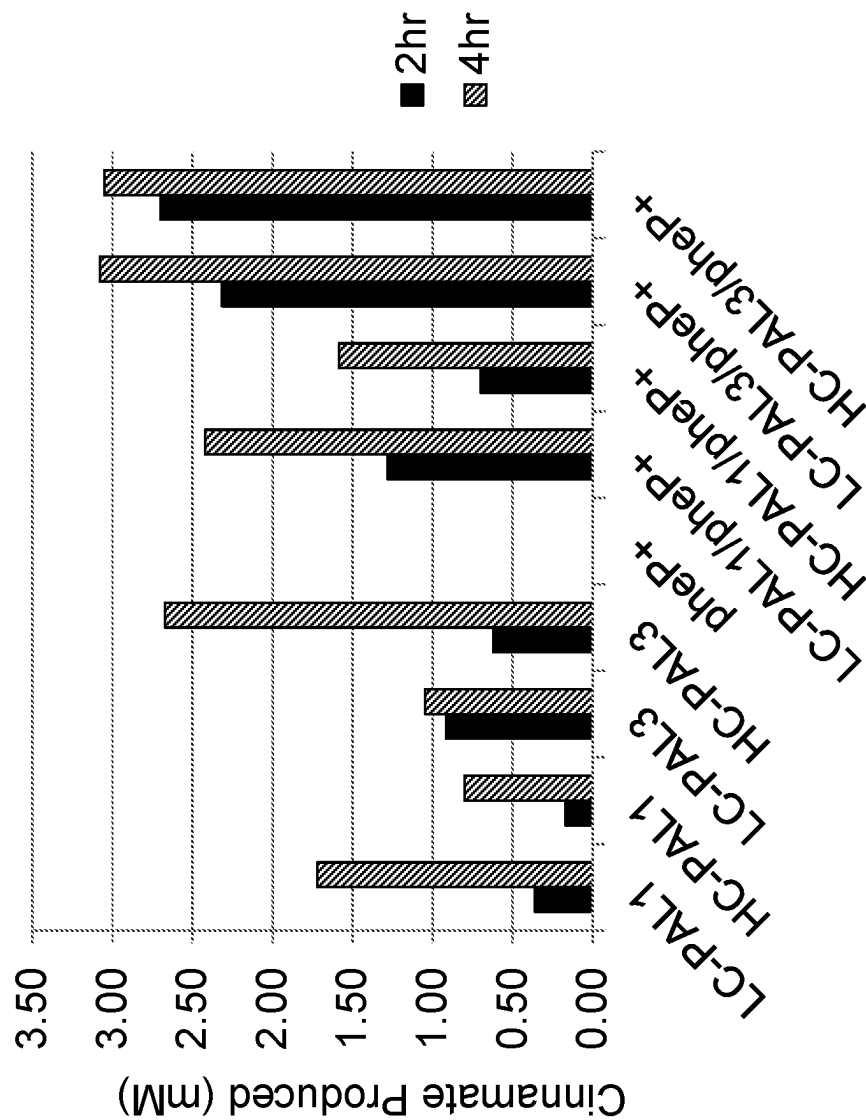


Fig. 16B

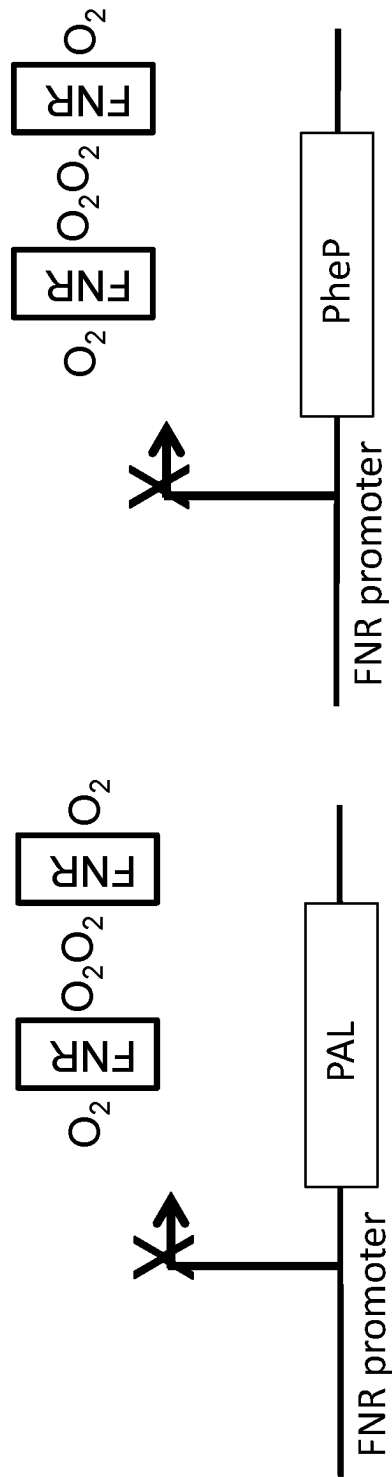


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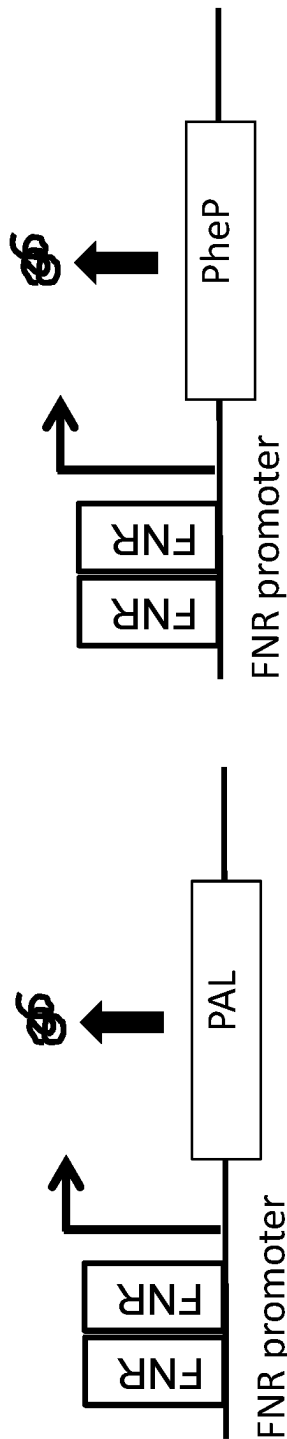


Fig. 17B

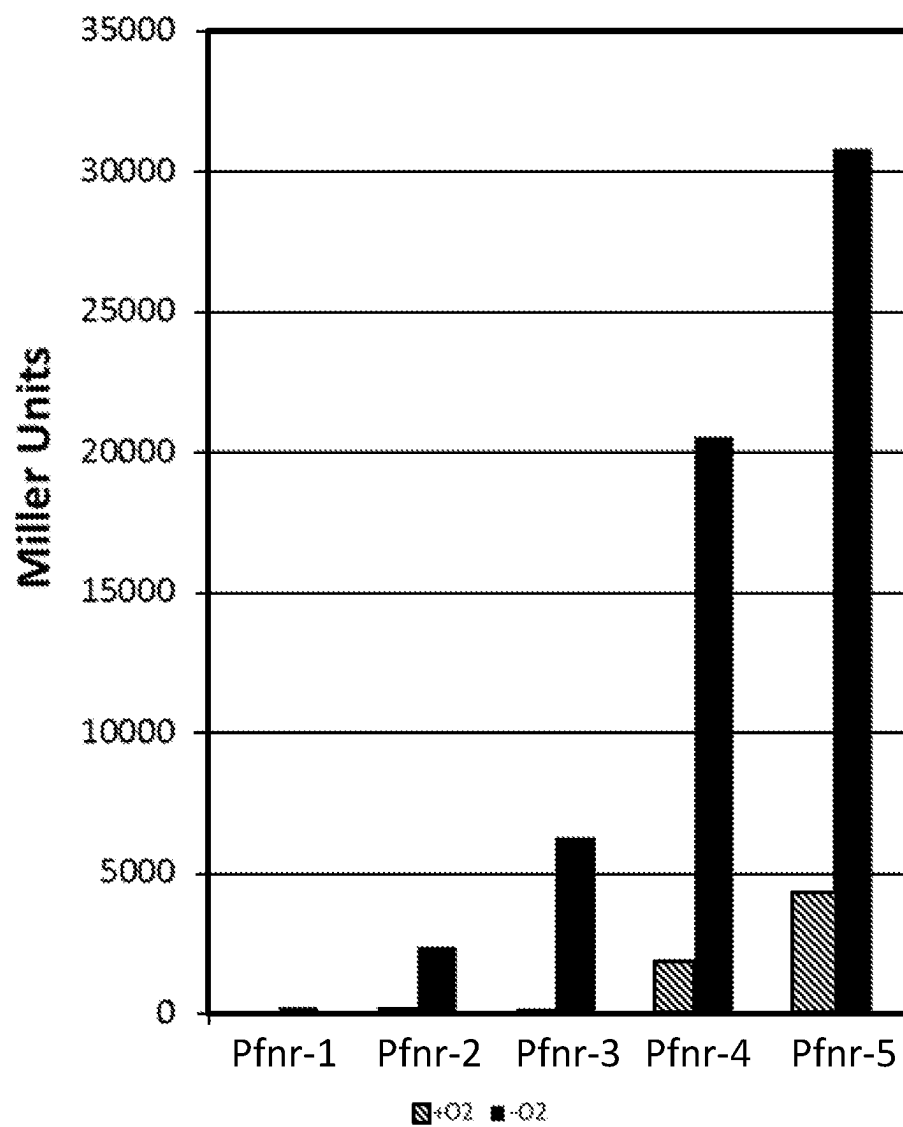


Fig. 18

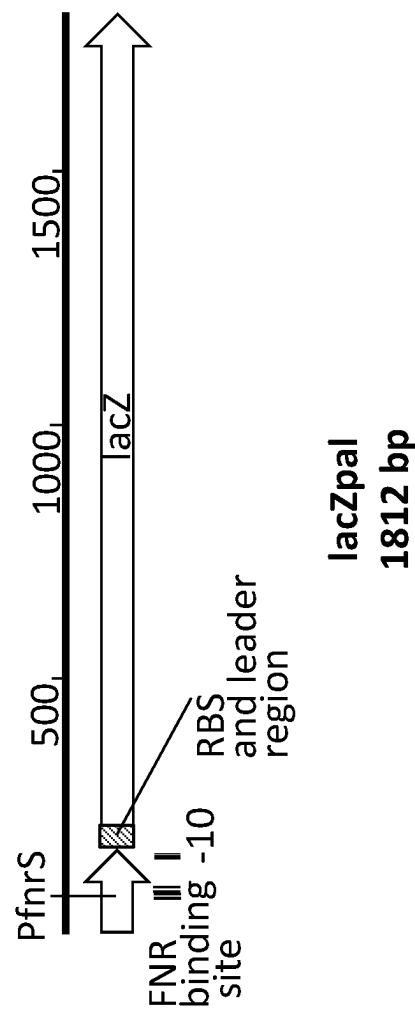


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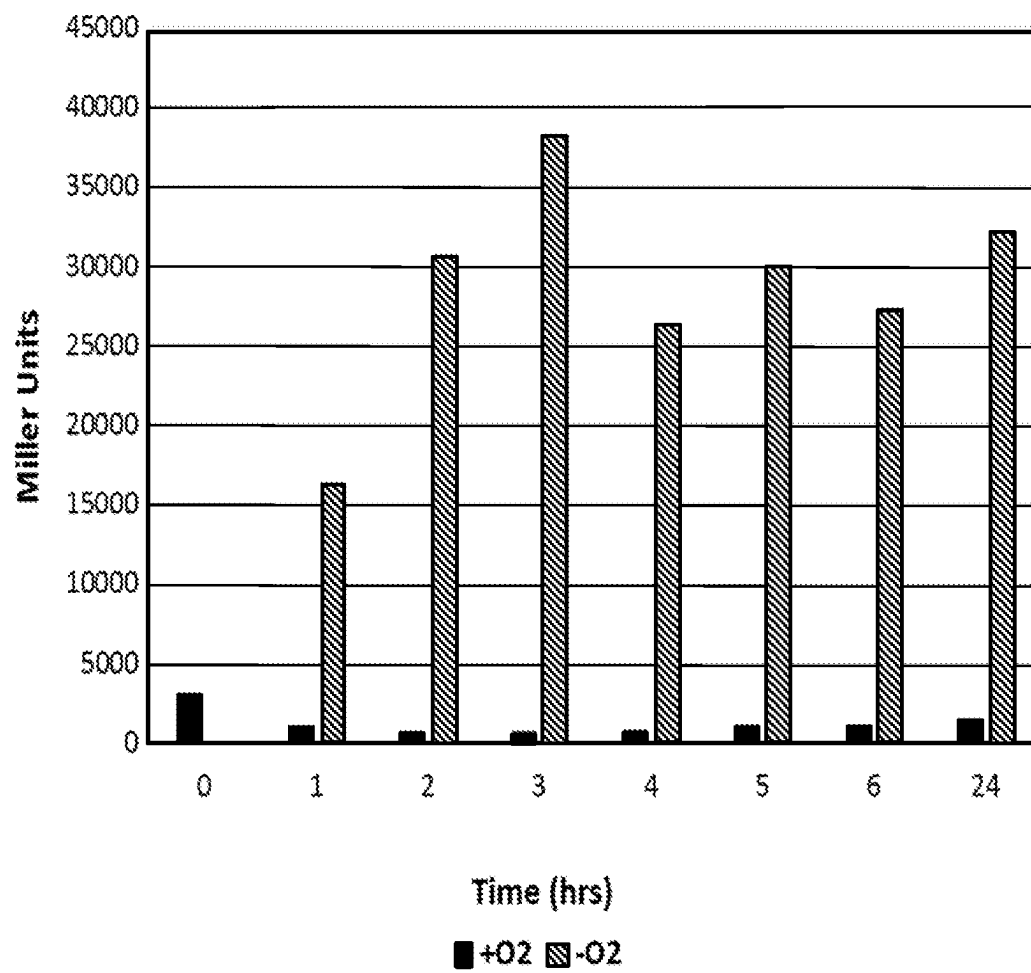


Fig. 19B

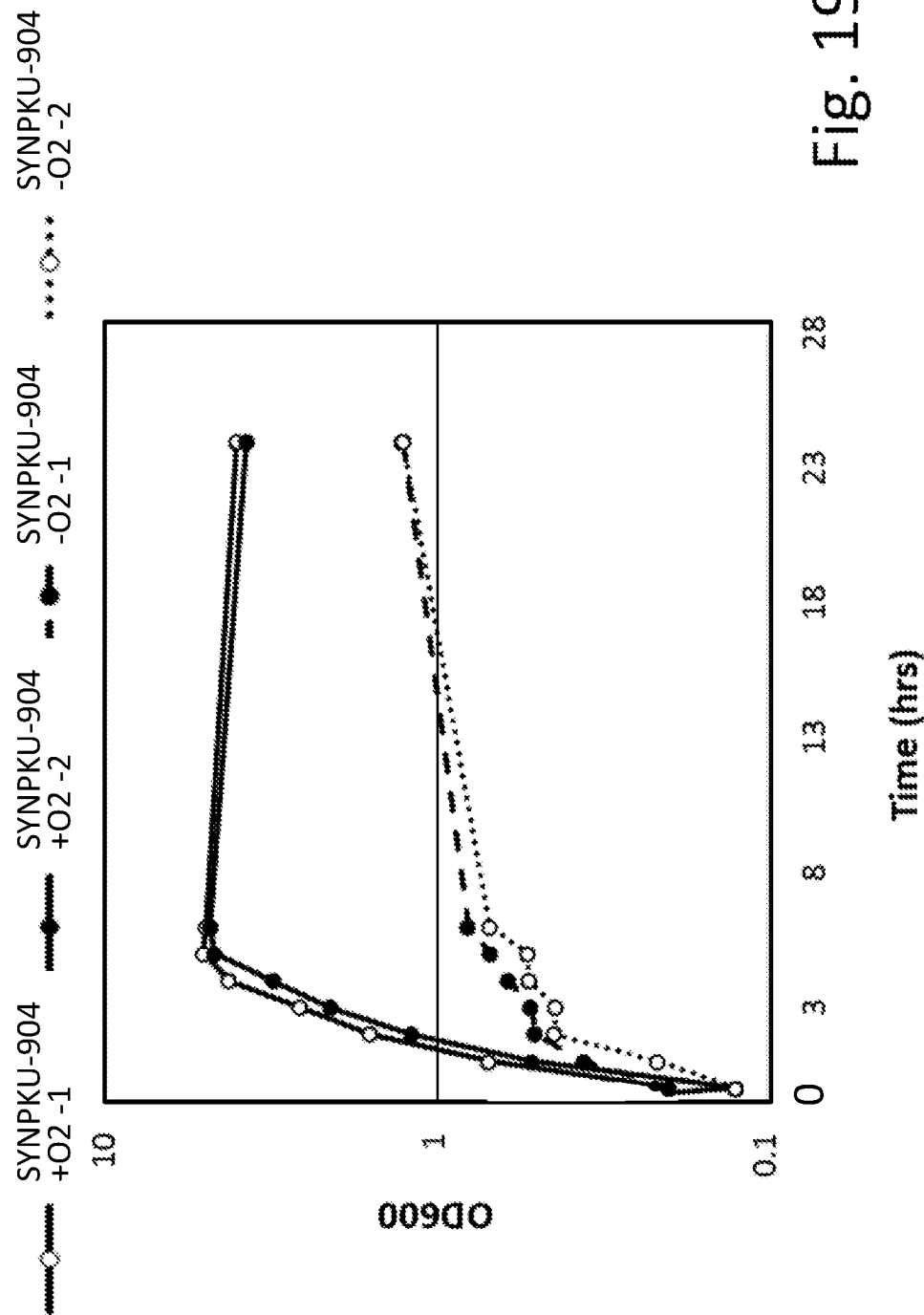


Fig. 19C

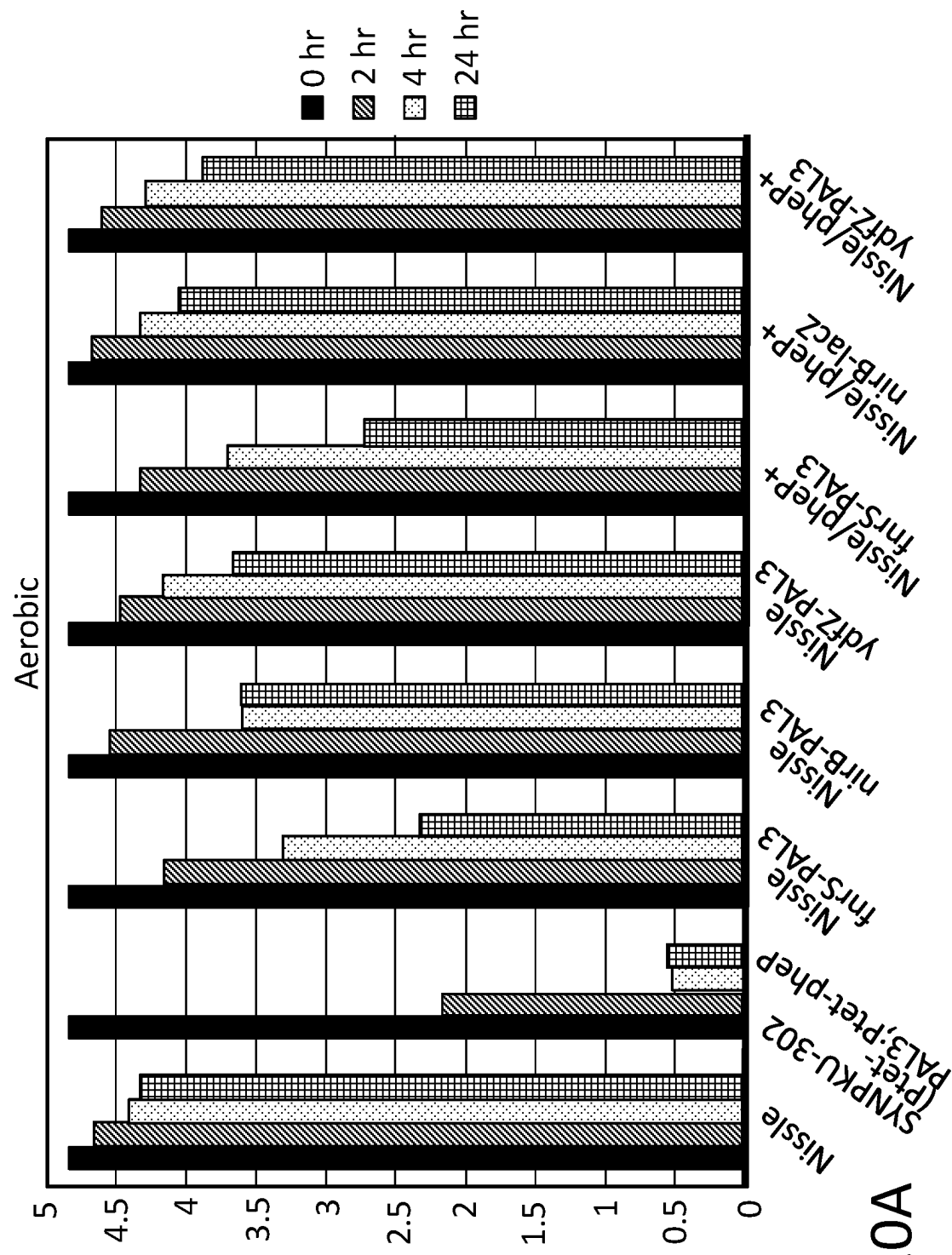


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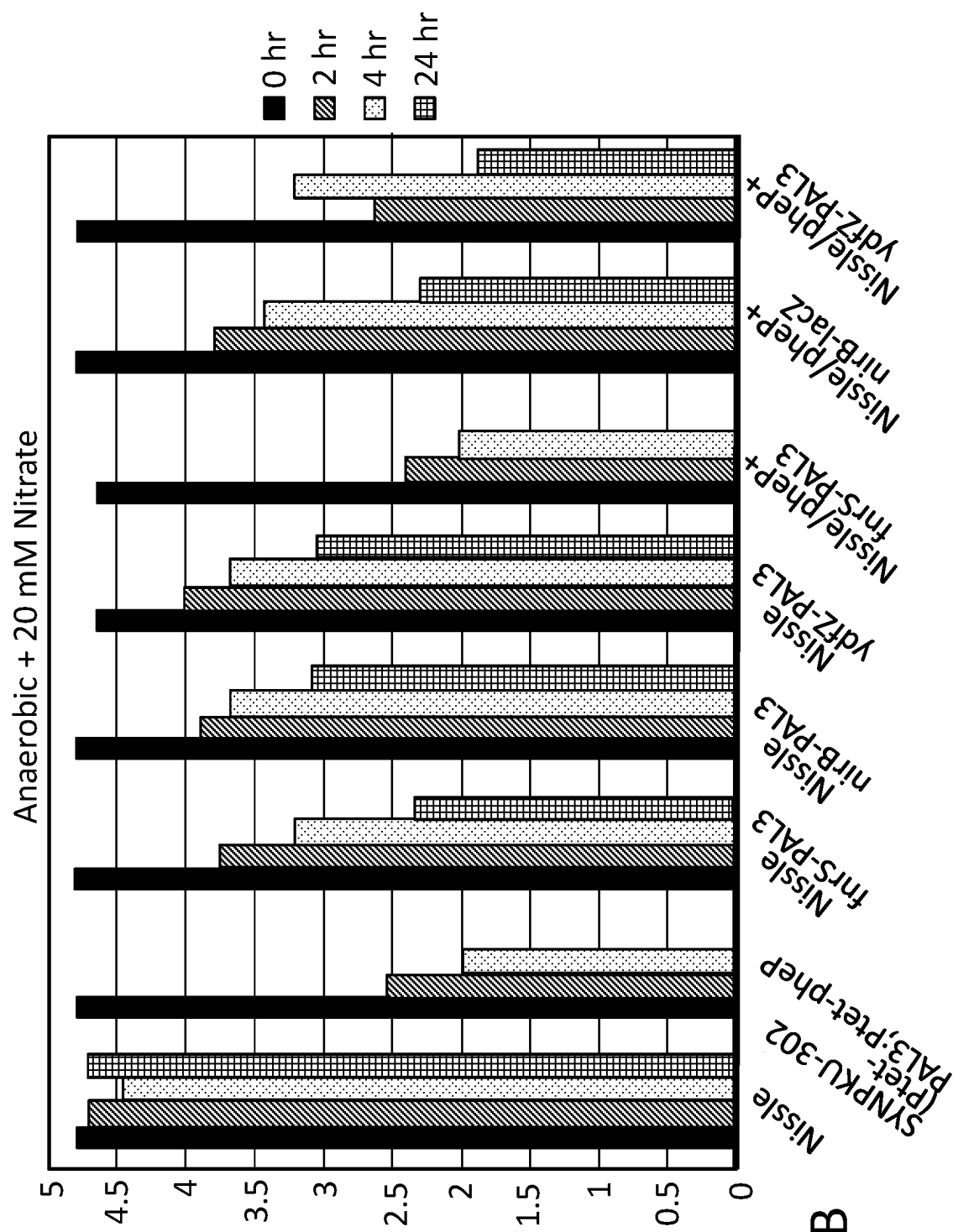


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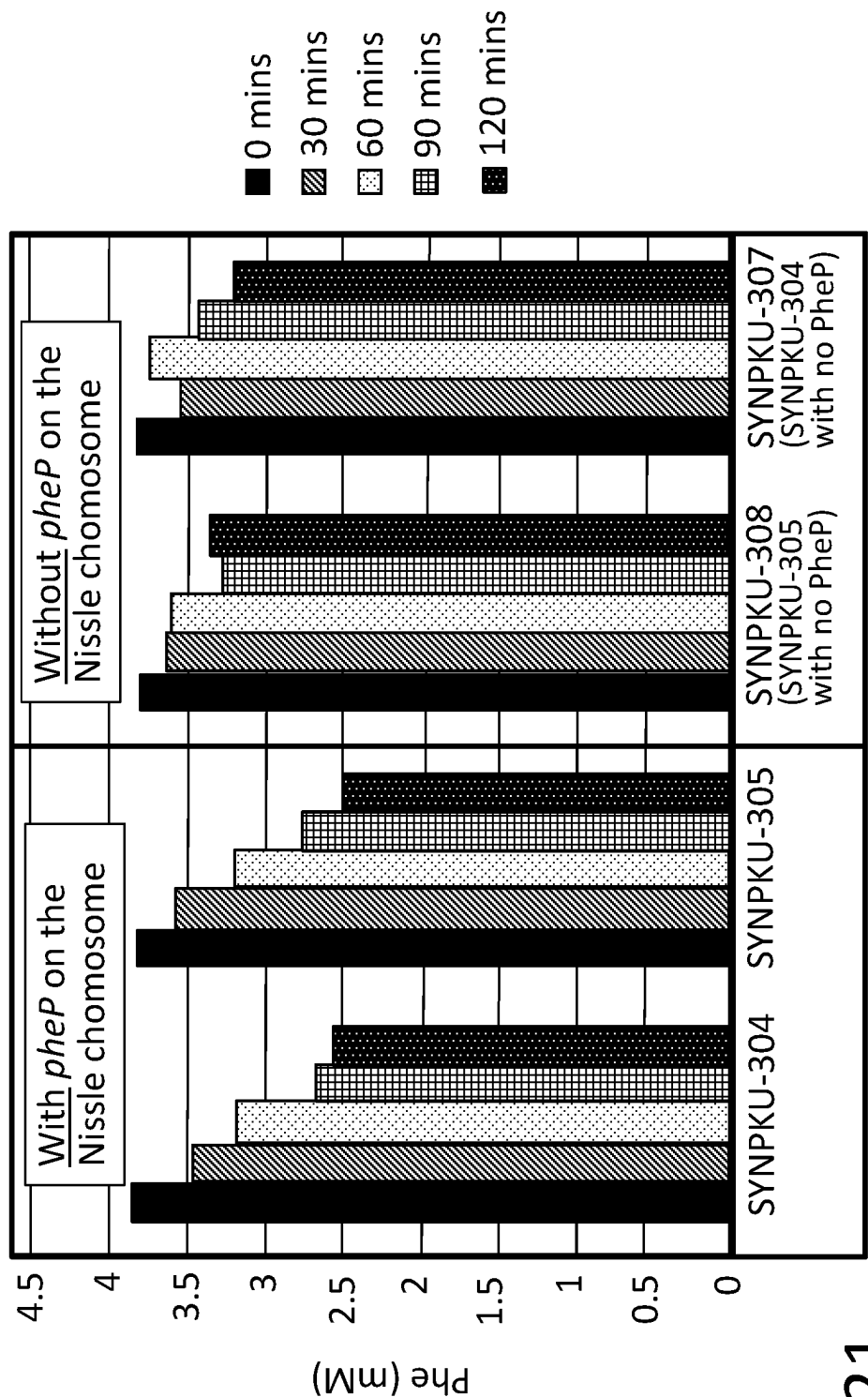


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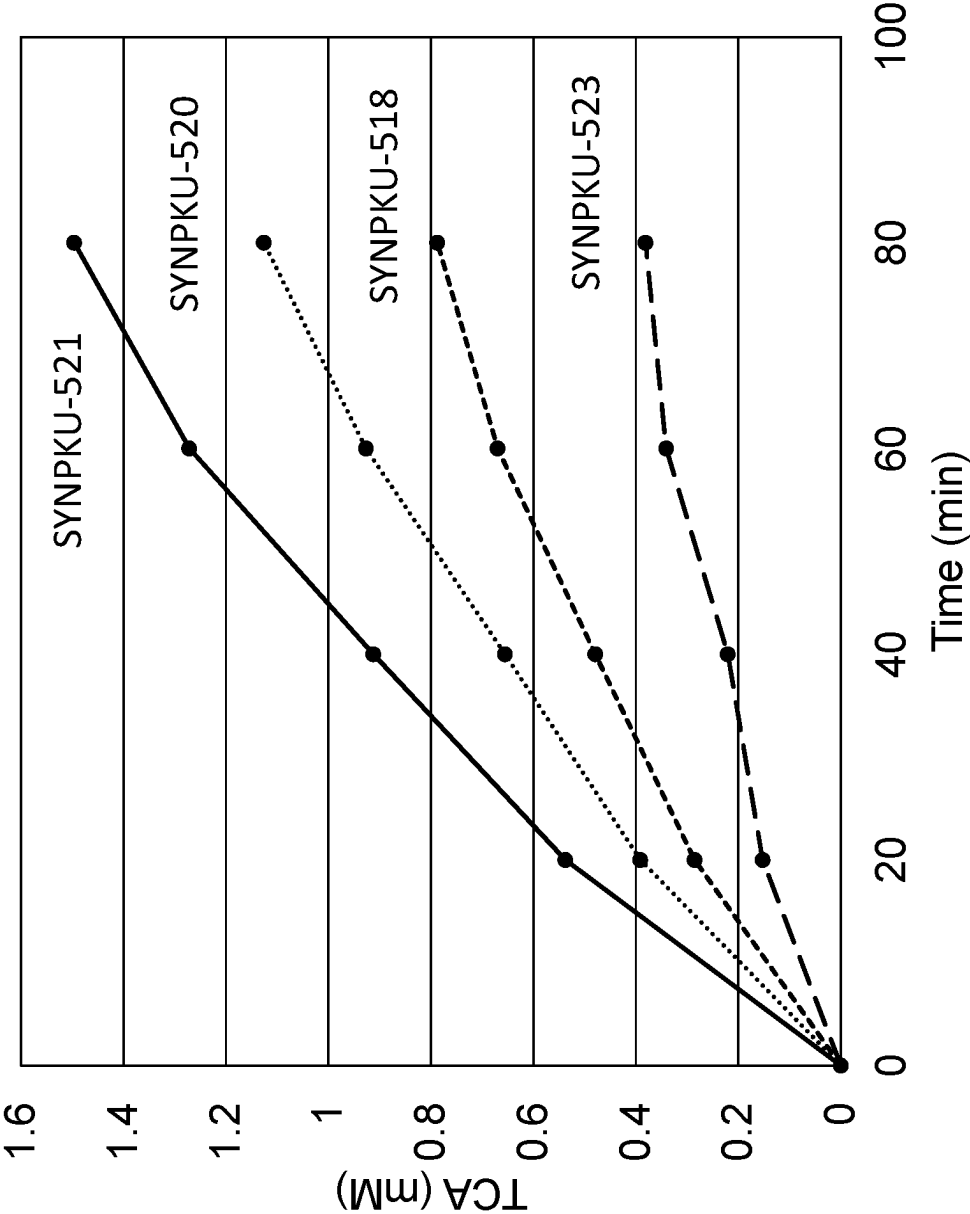


Fig. 22

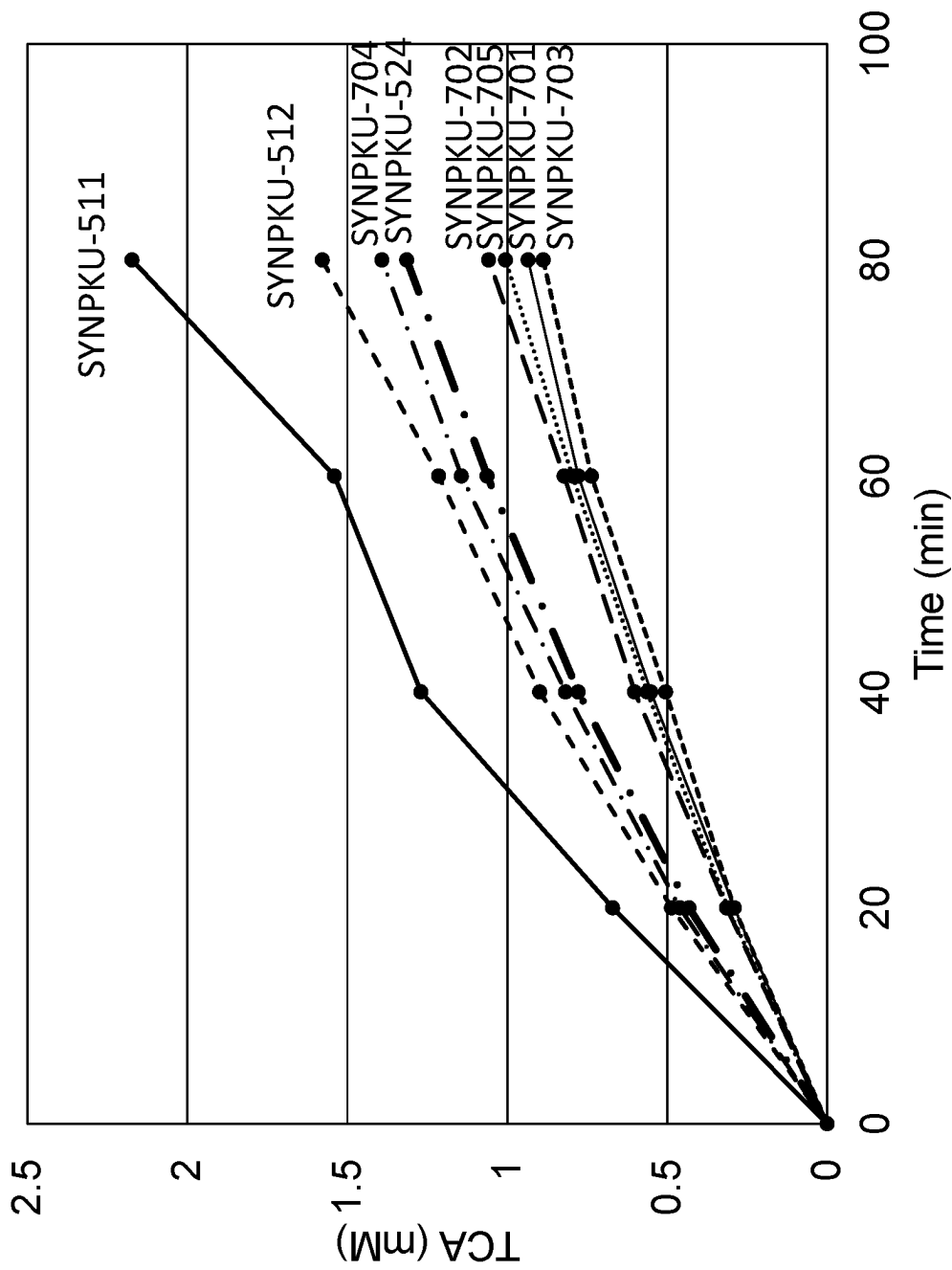


Fig. 23

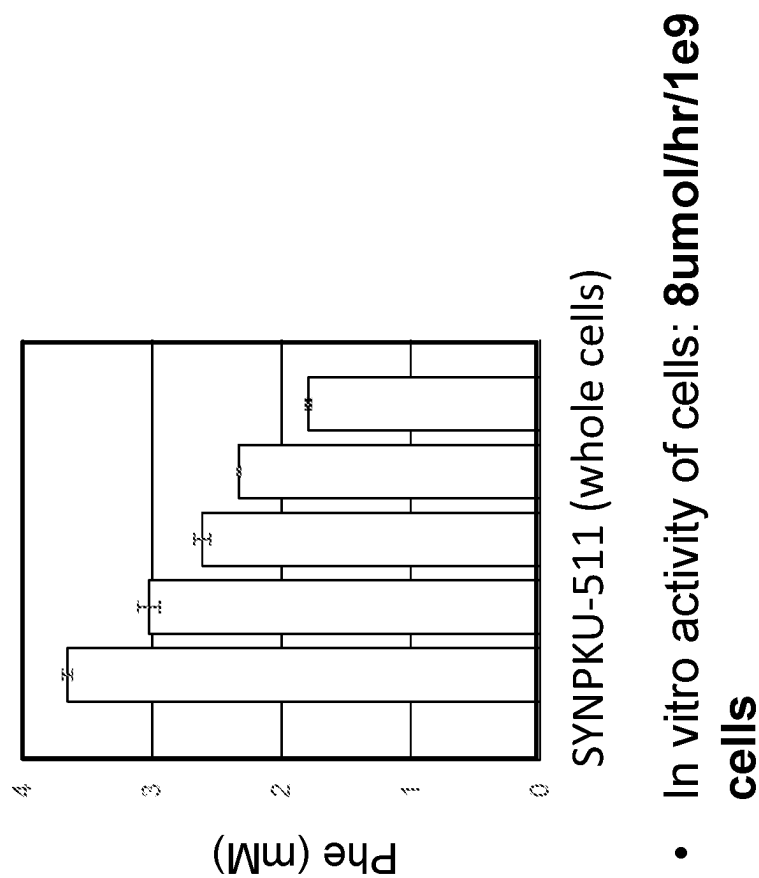


Fig. 24

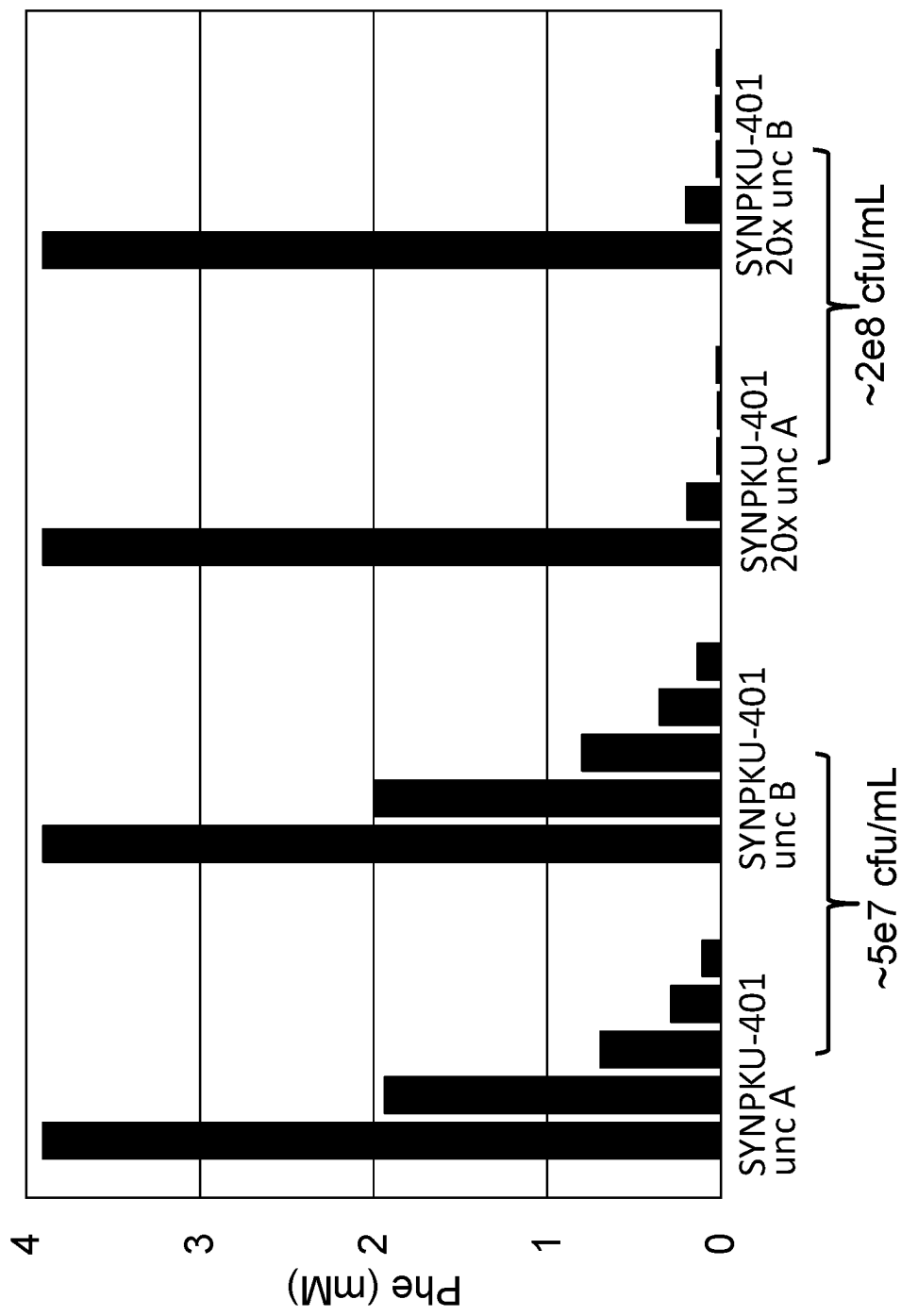


Fig. 25A

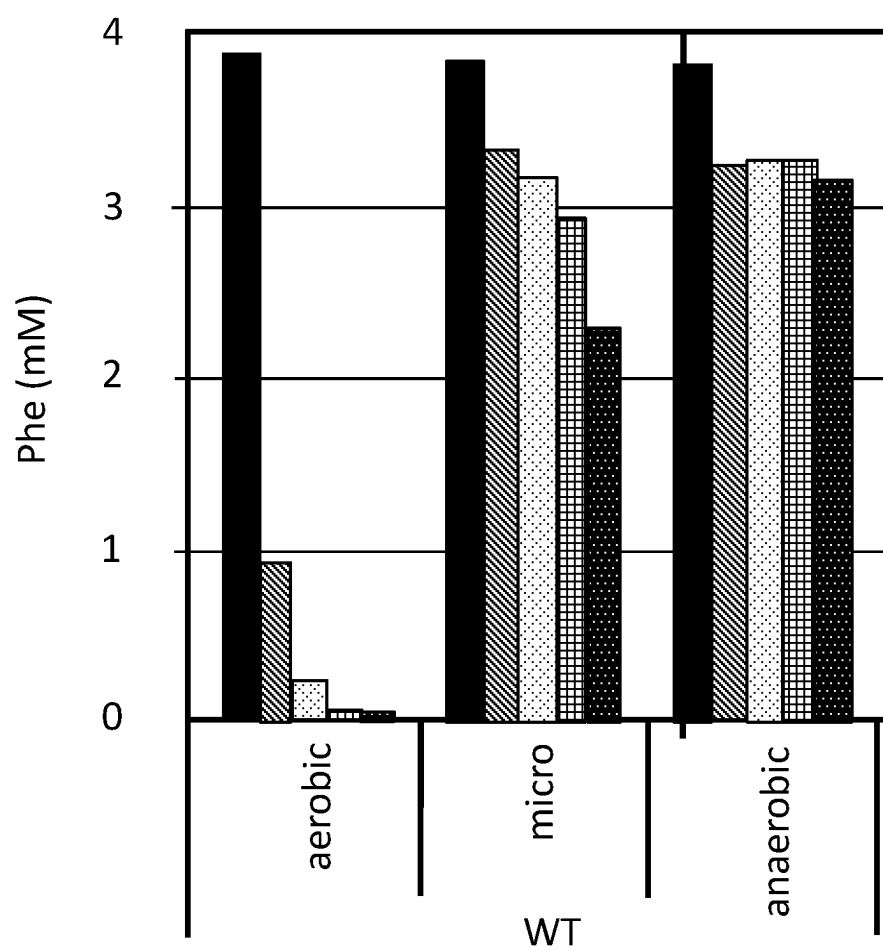


Fig. 25B

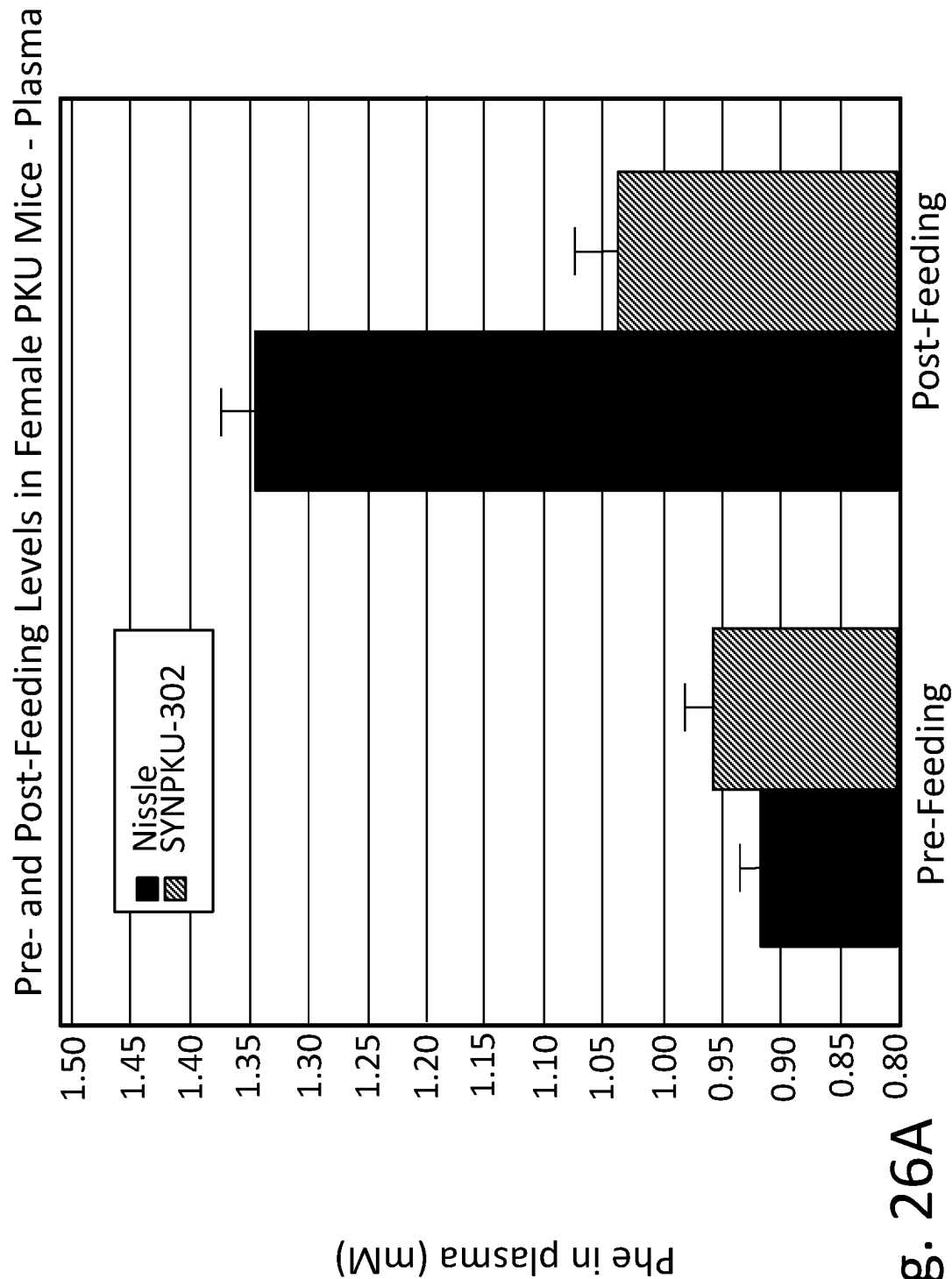


Fig. 26A

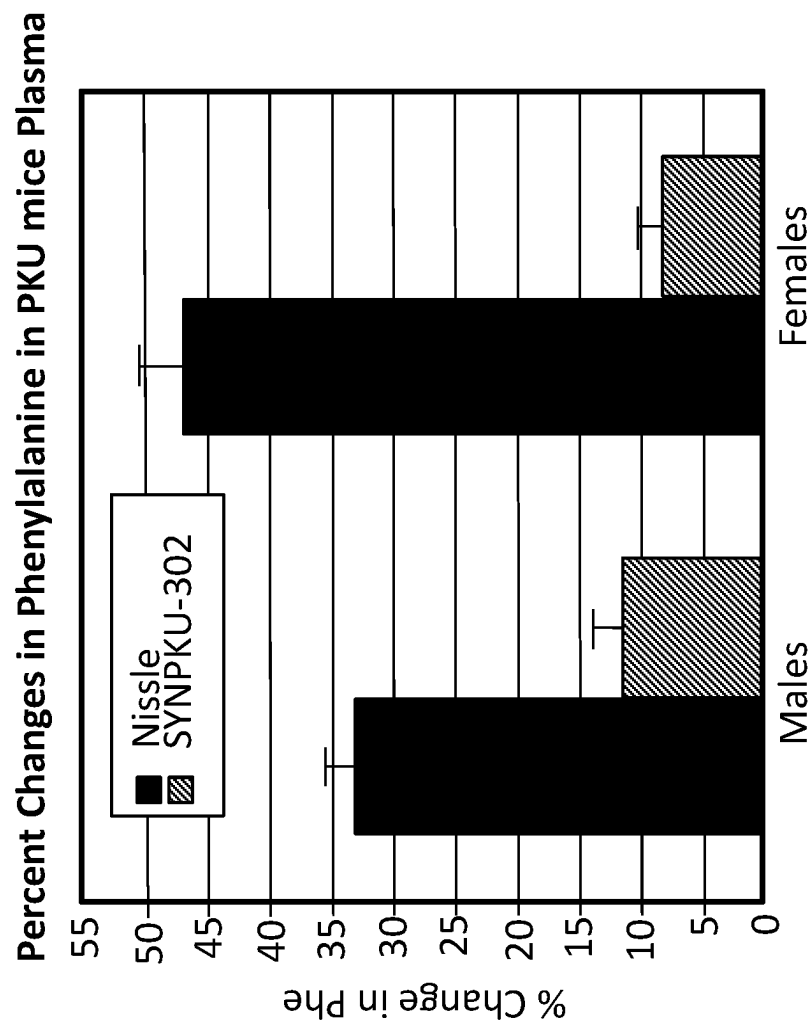


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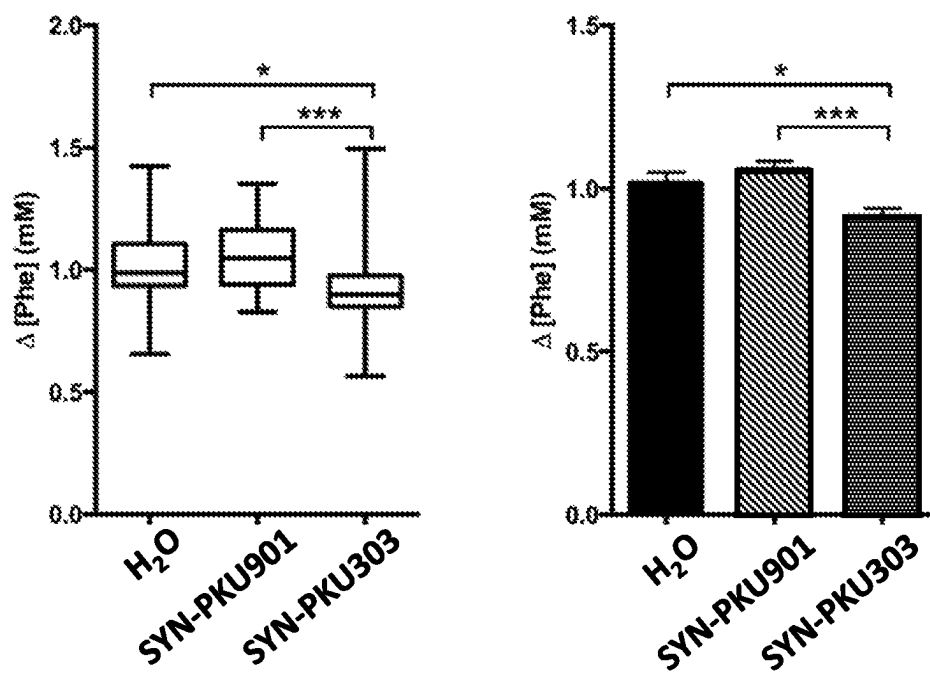


Fig. 27A

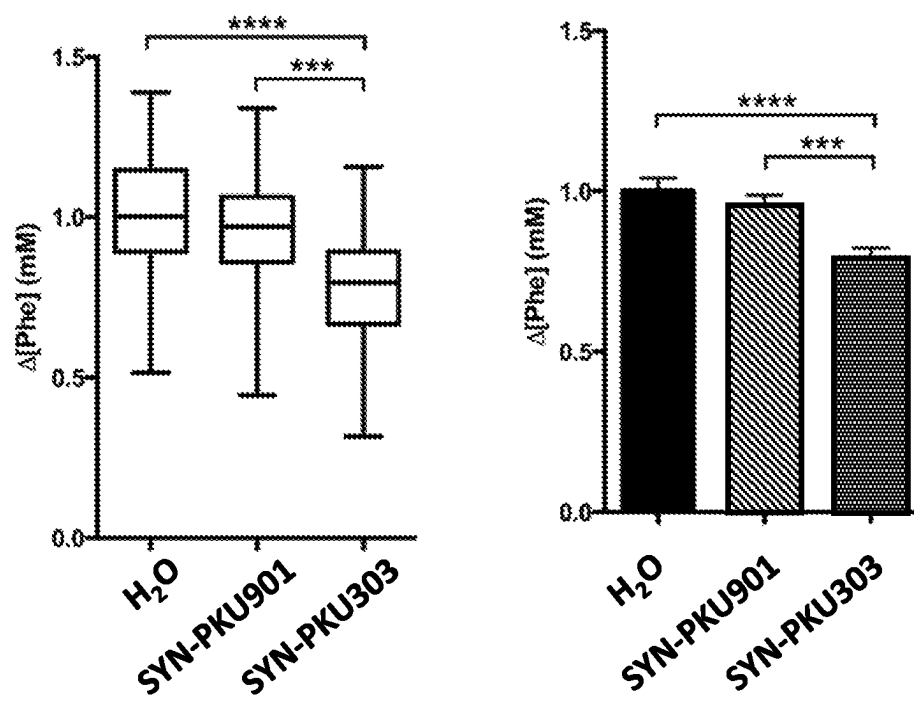


Fig. 27B

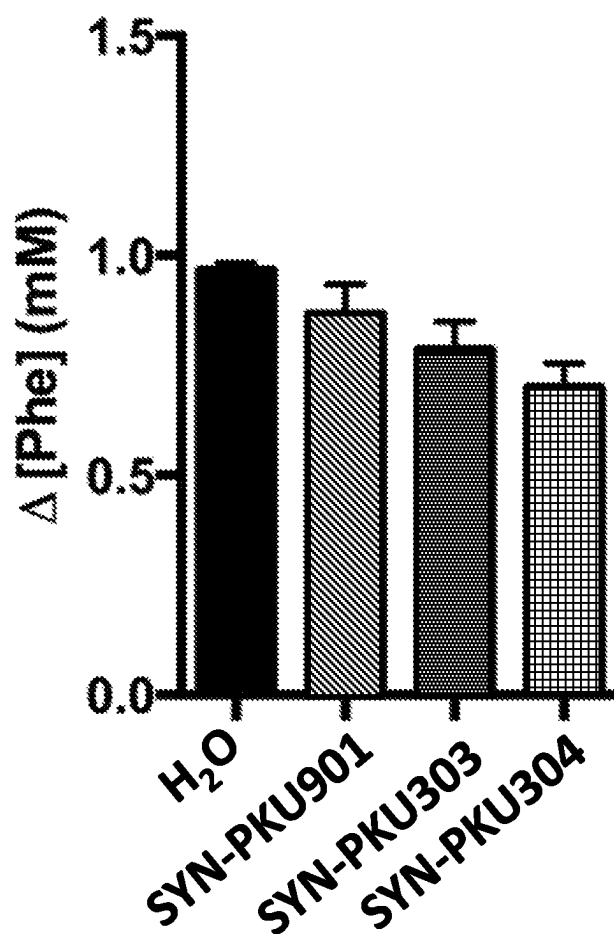


Fig. 28

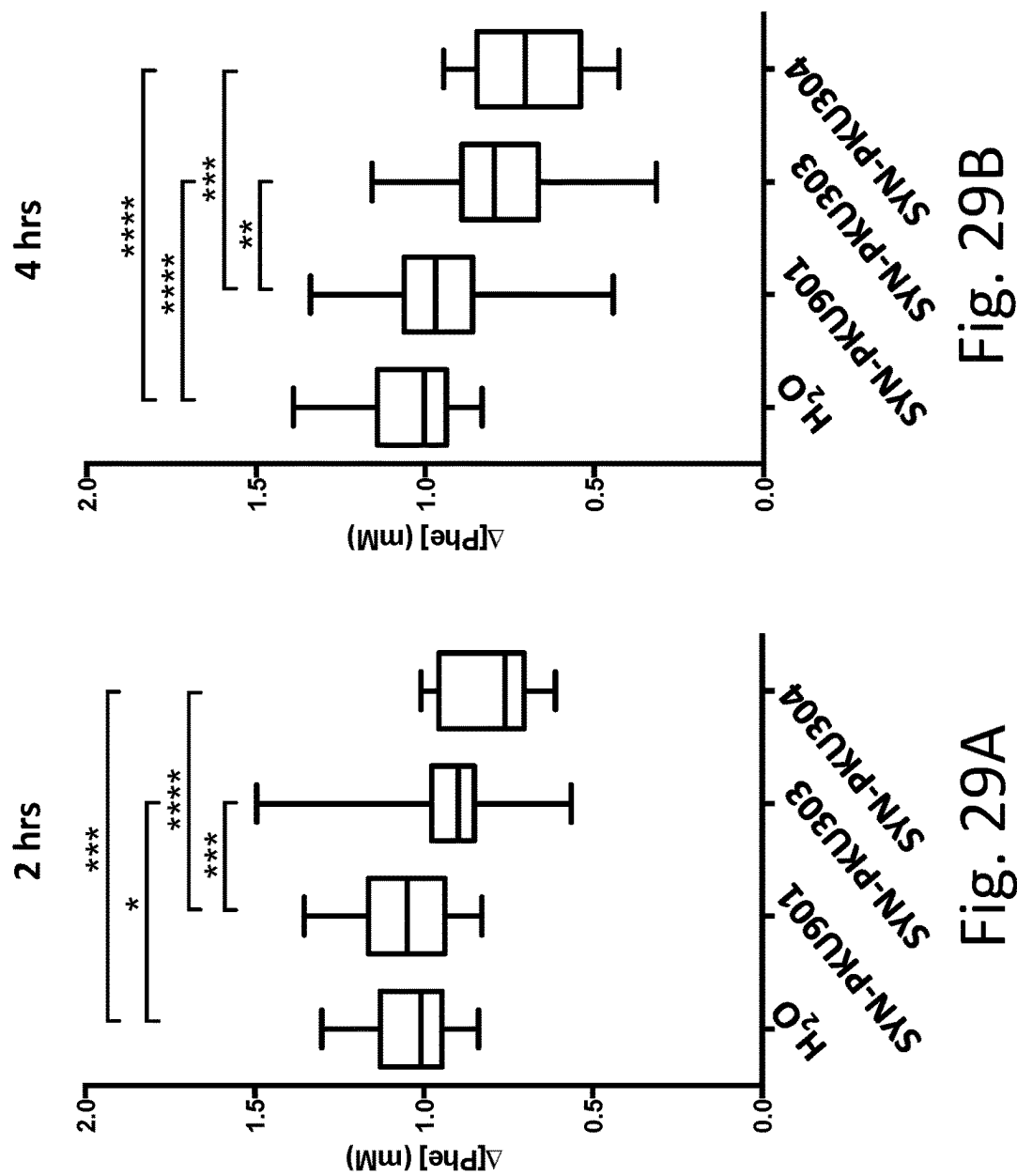


Fig. 29A

Fig. 29B

P-values: ANOVA with Tukey pairwise comparison
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

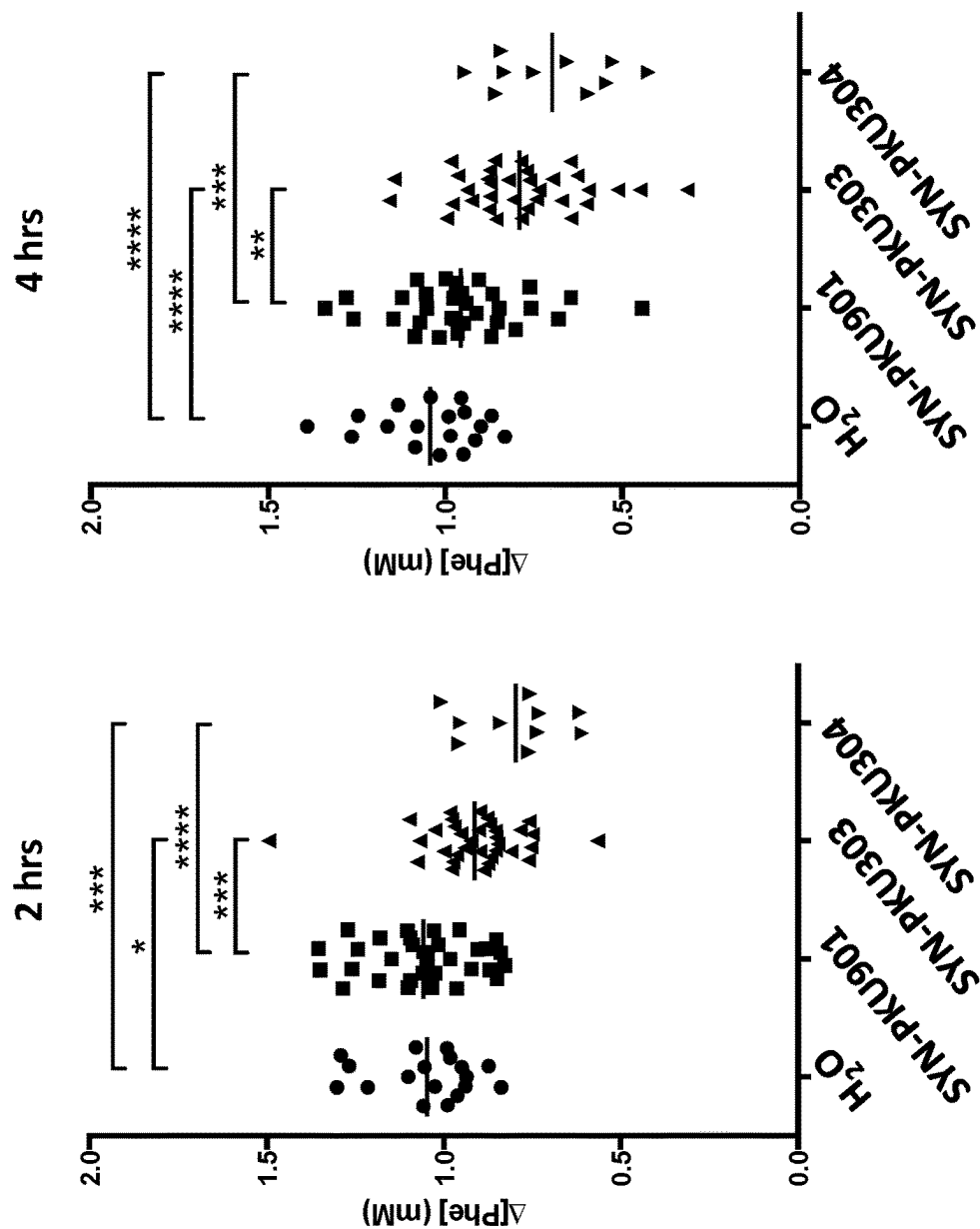


Fig. 29D

Fig. 29C

P-values: ANOVA with Tukey pairwise comparison
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

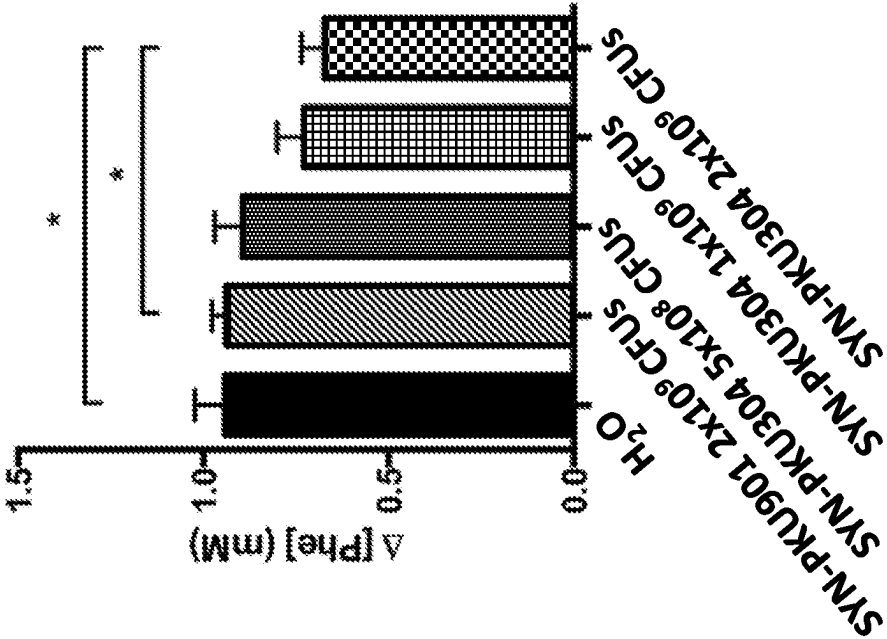


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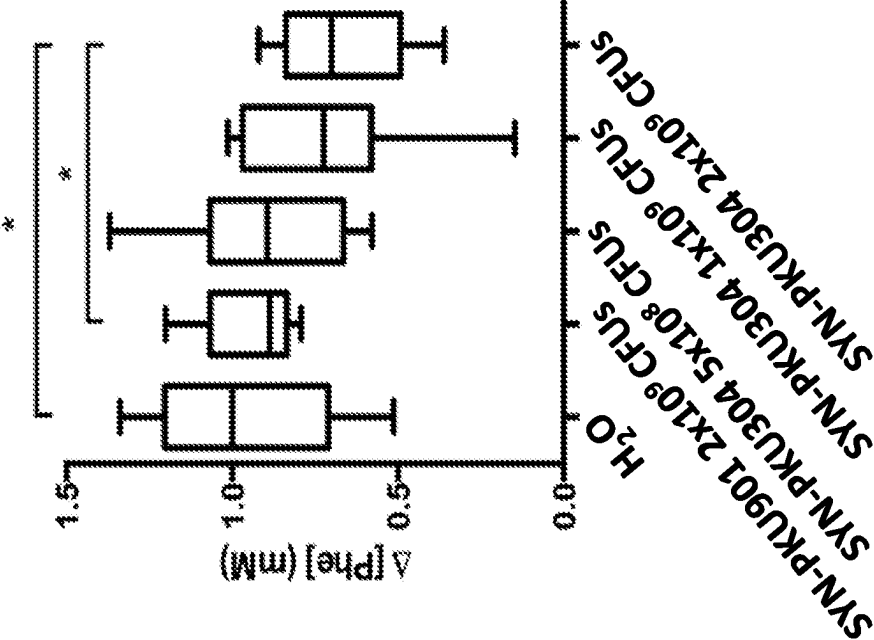
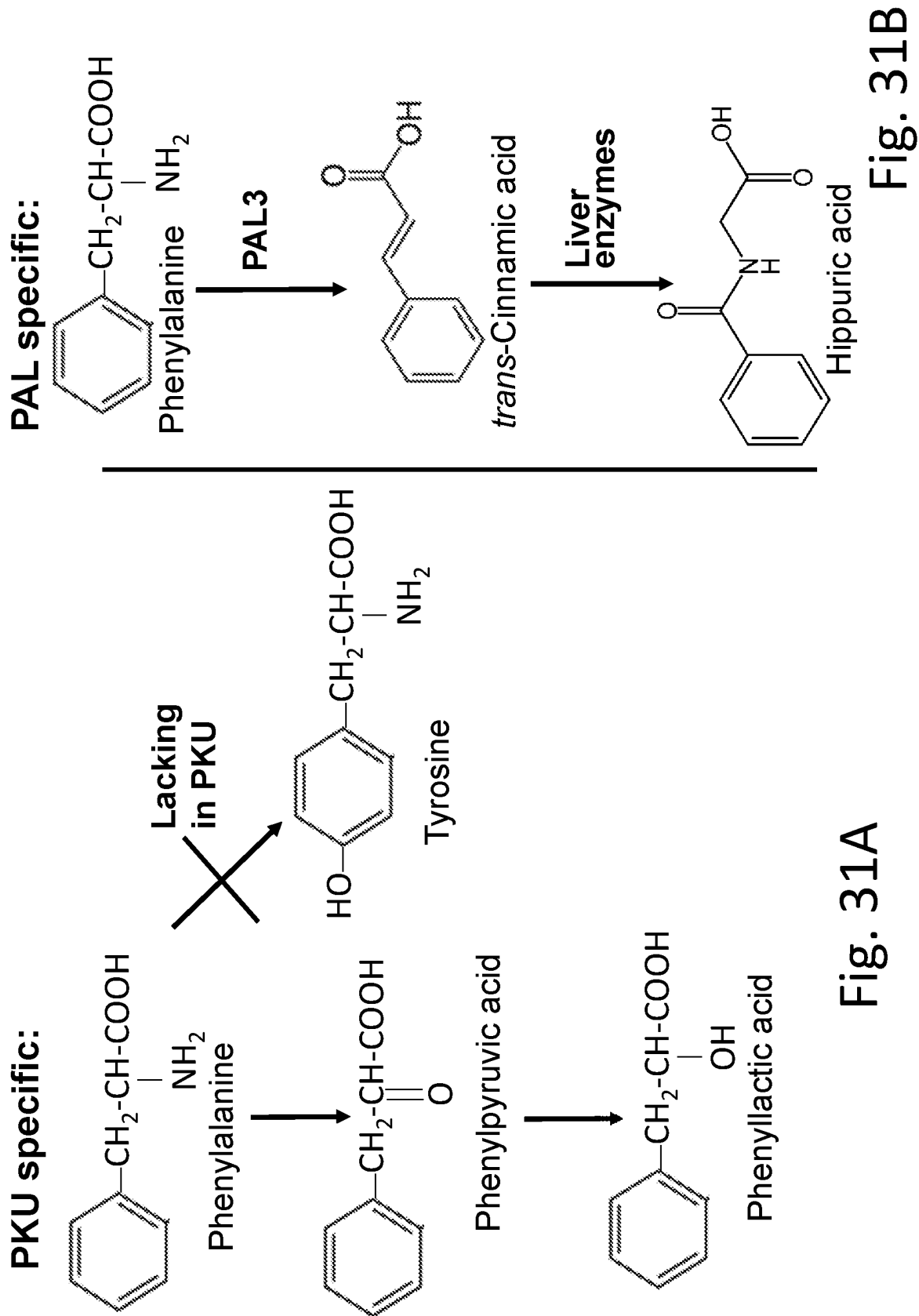


Fig. 30B



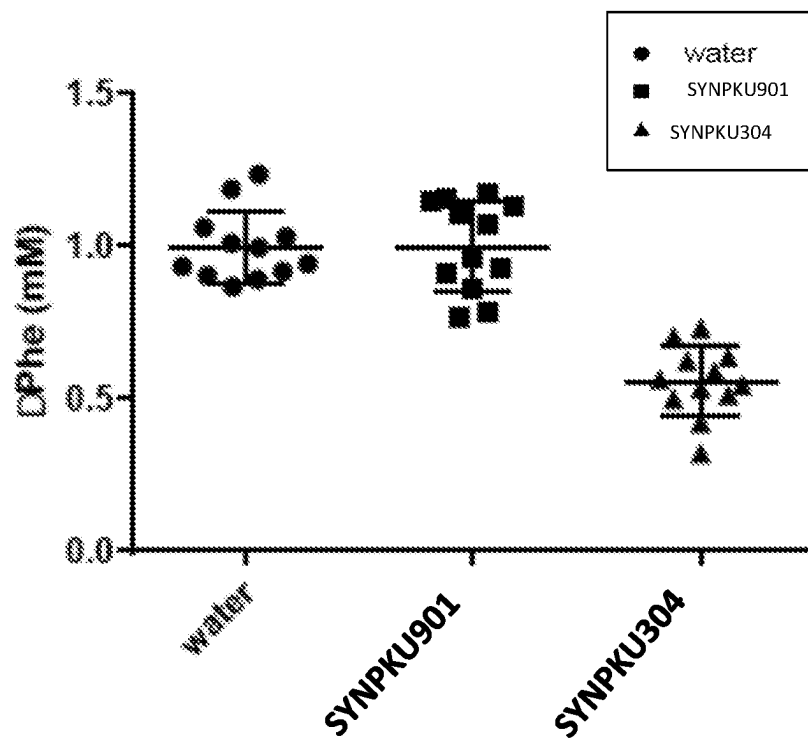


Fig. 32A

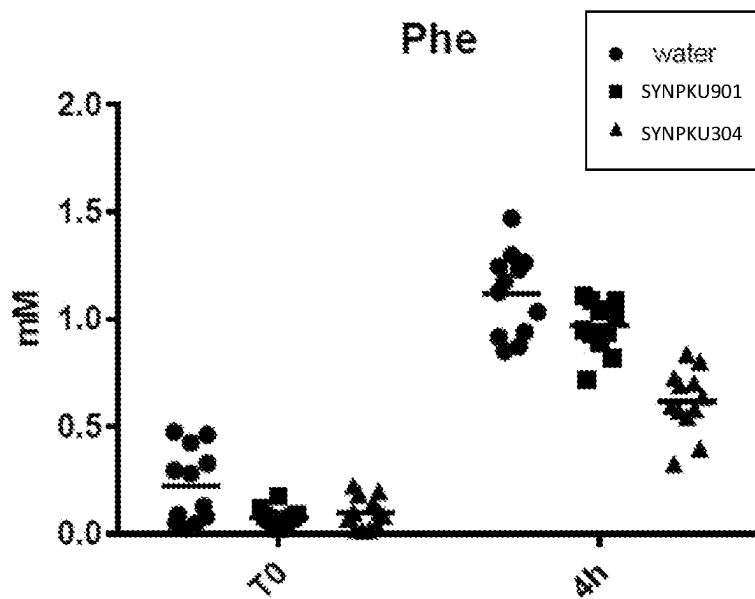


Fig. 32B

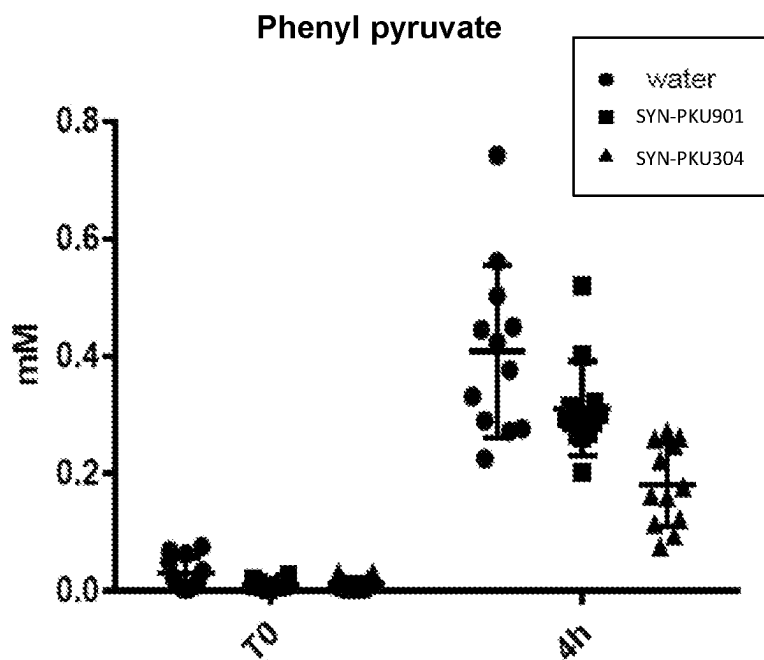


Fig.32C

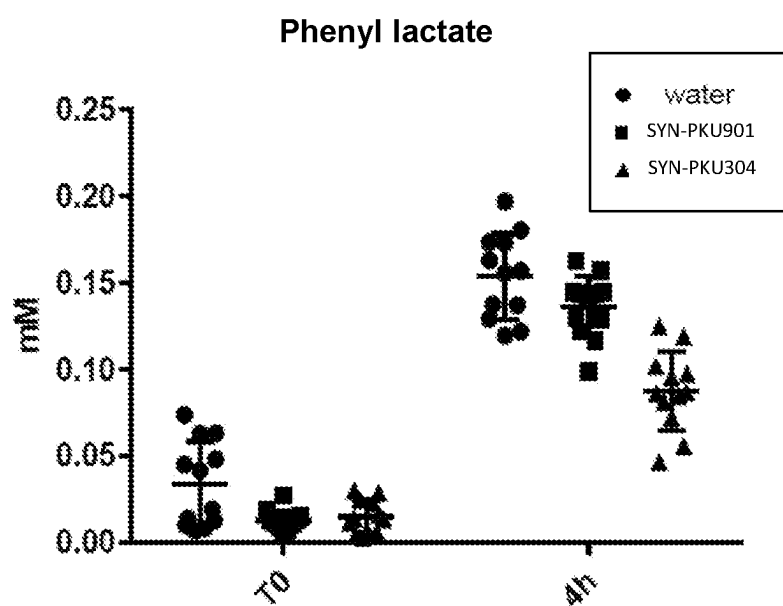


Fig.32D

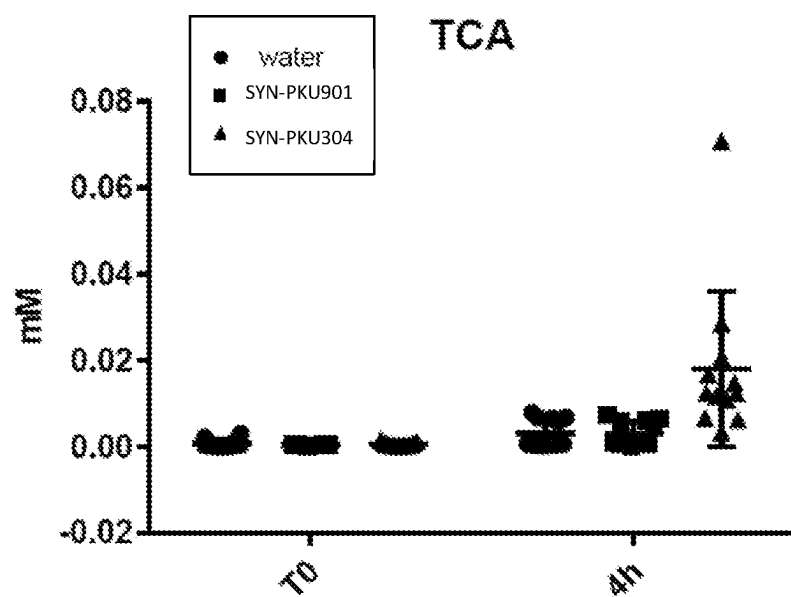


Fig. 32E

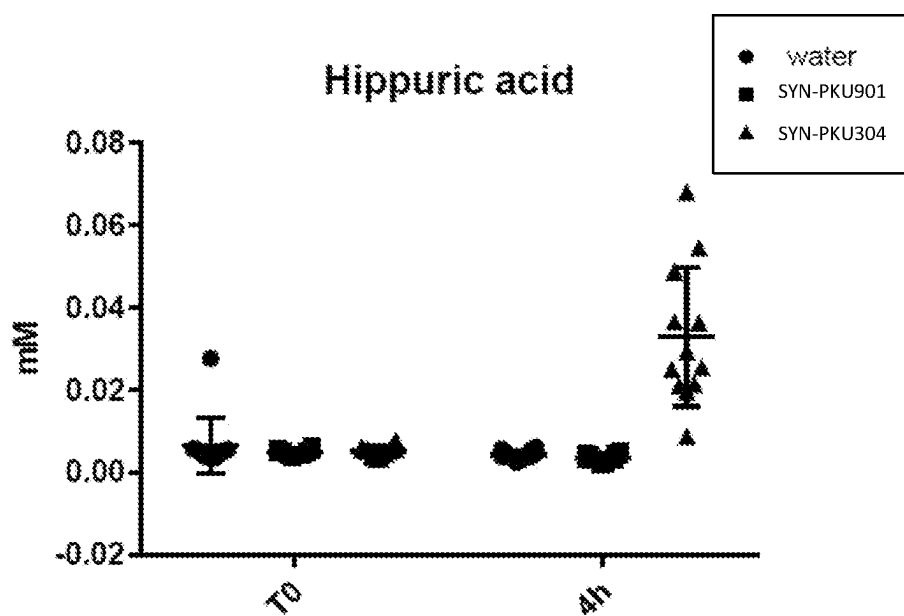


Fig. 32F

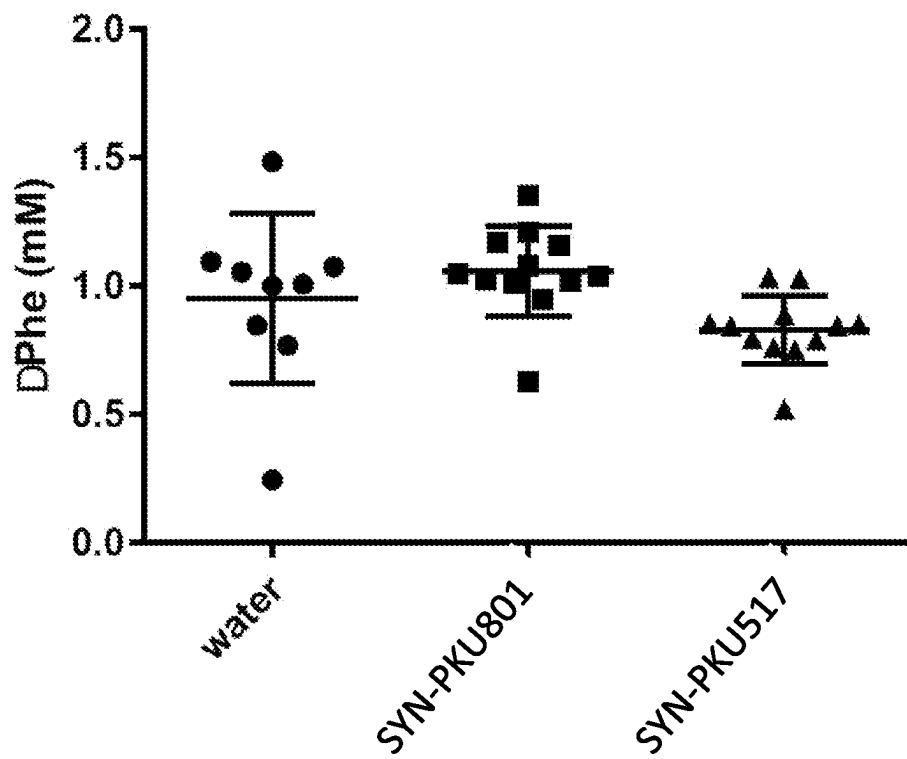


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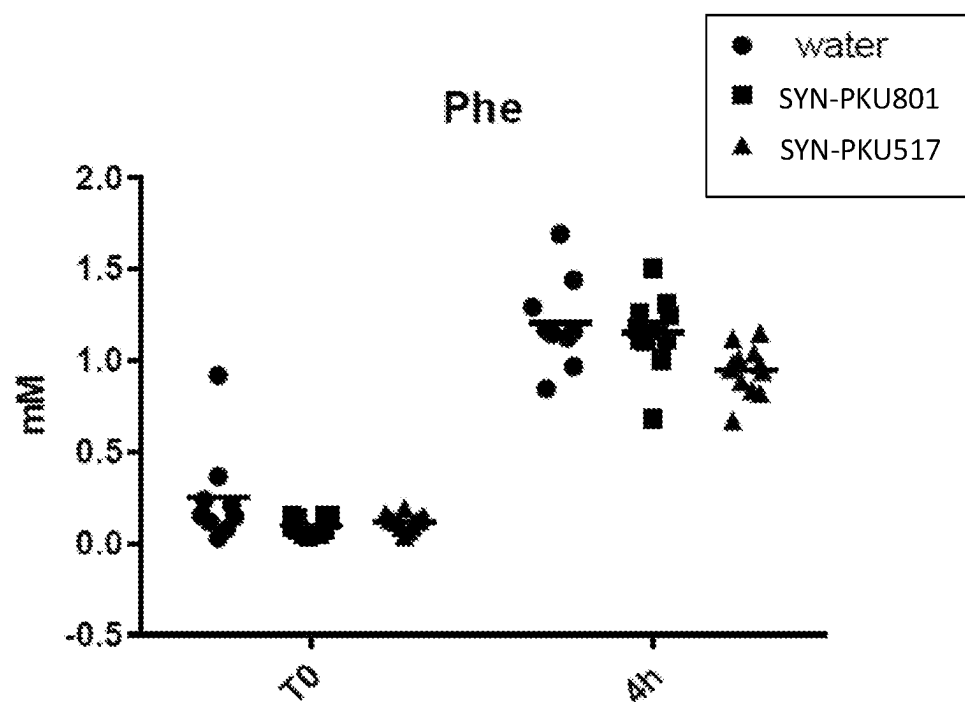


Fig. 33B

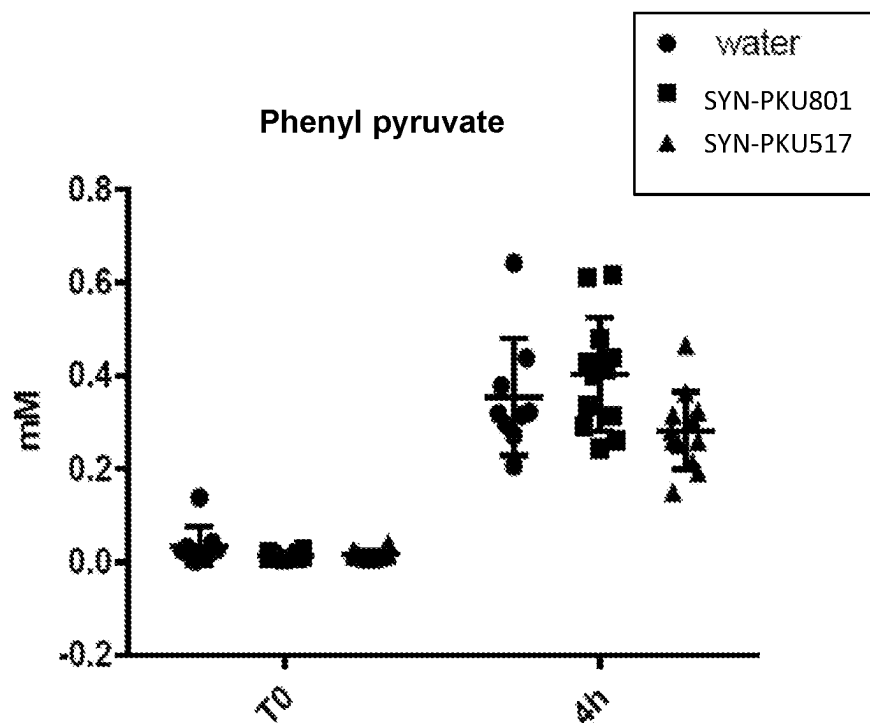


Fig. 33C

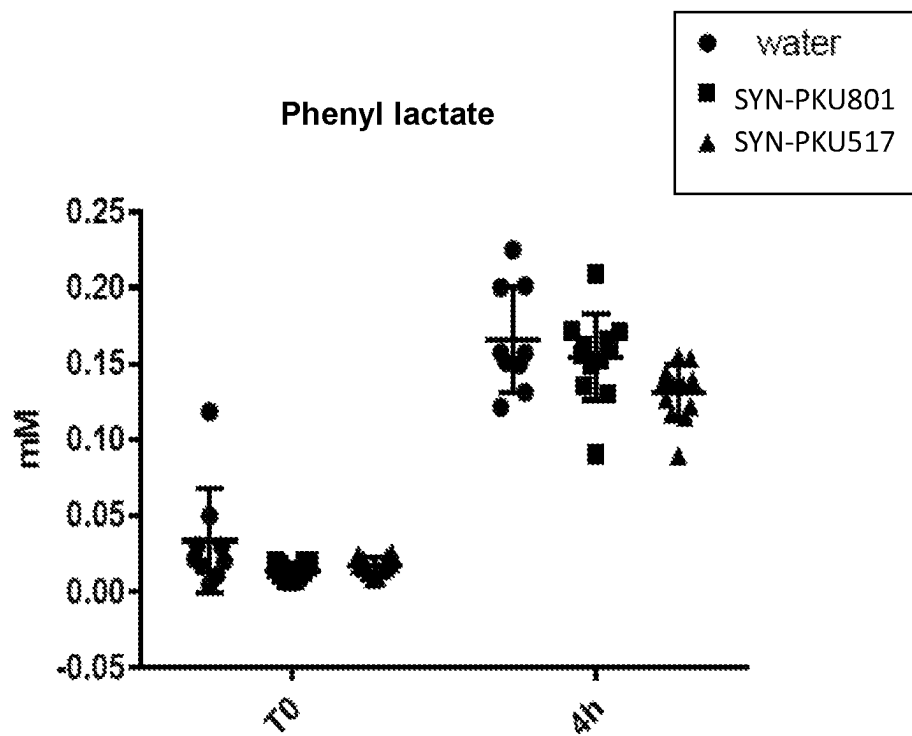


Fig. 33D

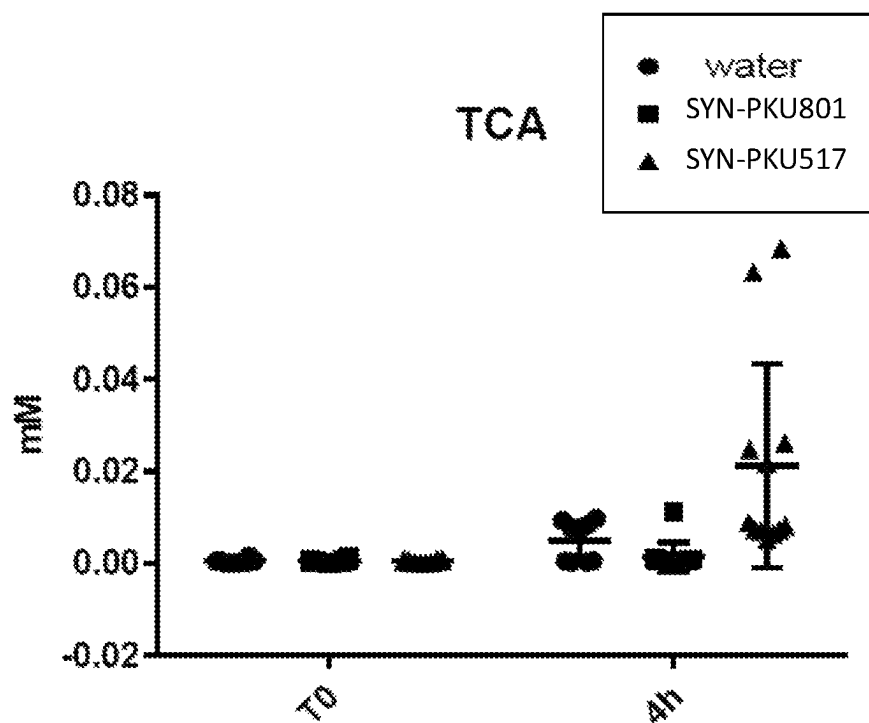


Fig. 33E

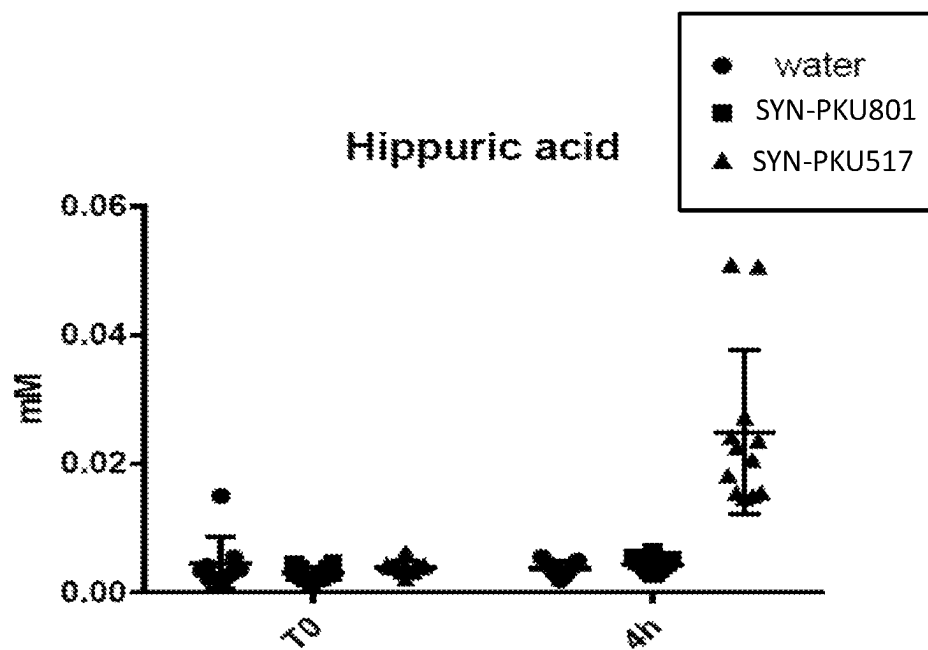


Fig. 33F

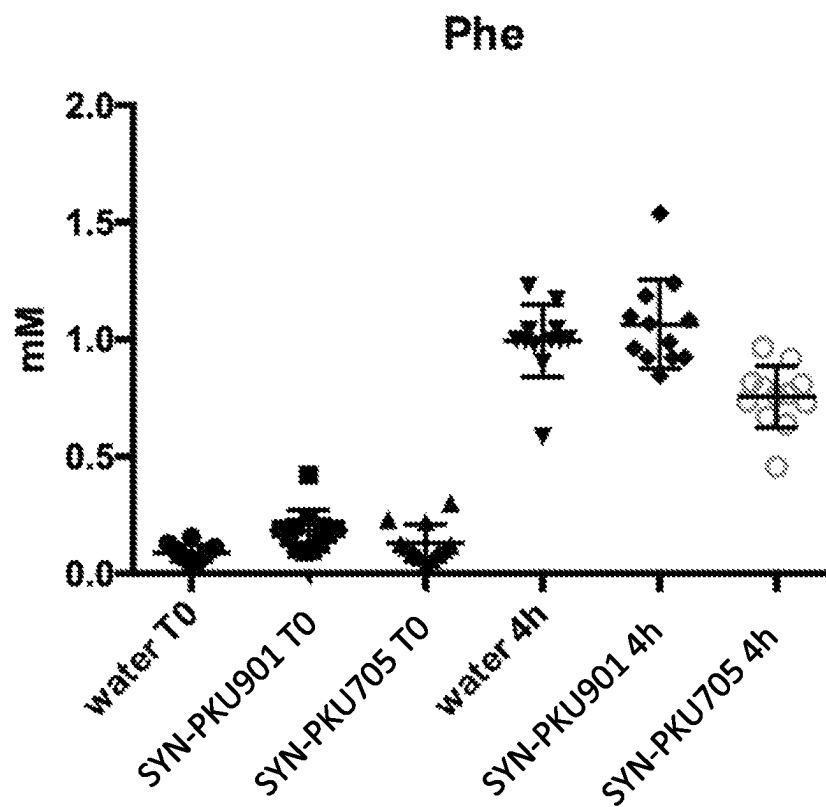


Fig. 34A

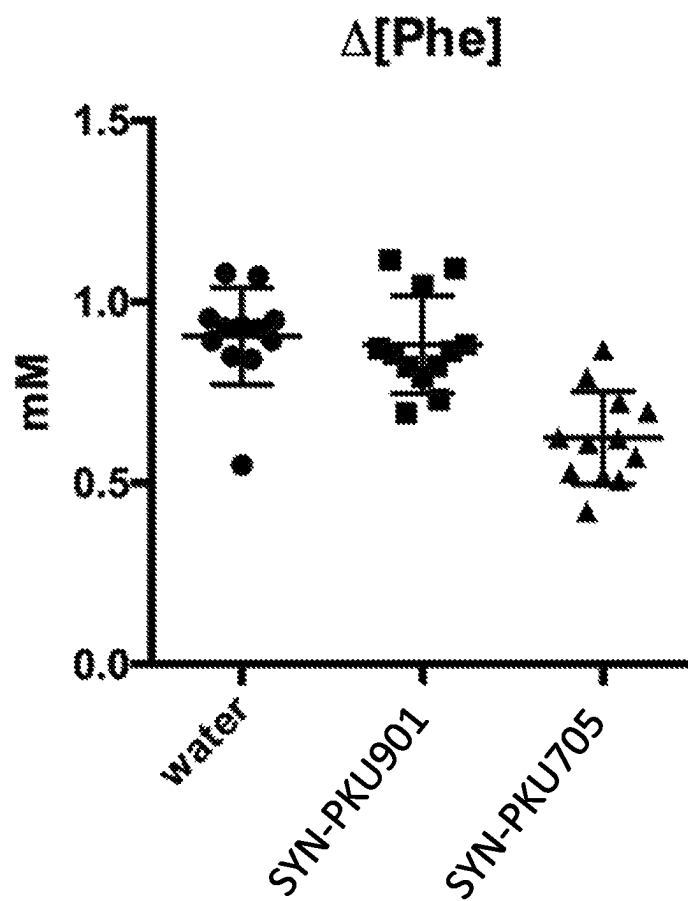


Fig. 34B

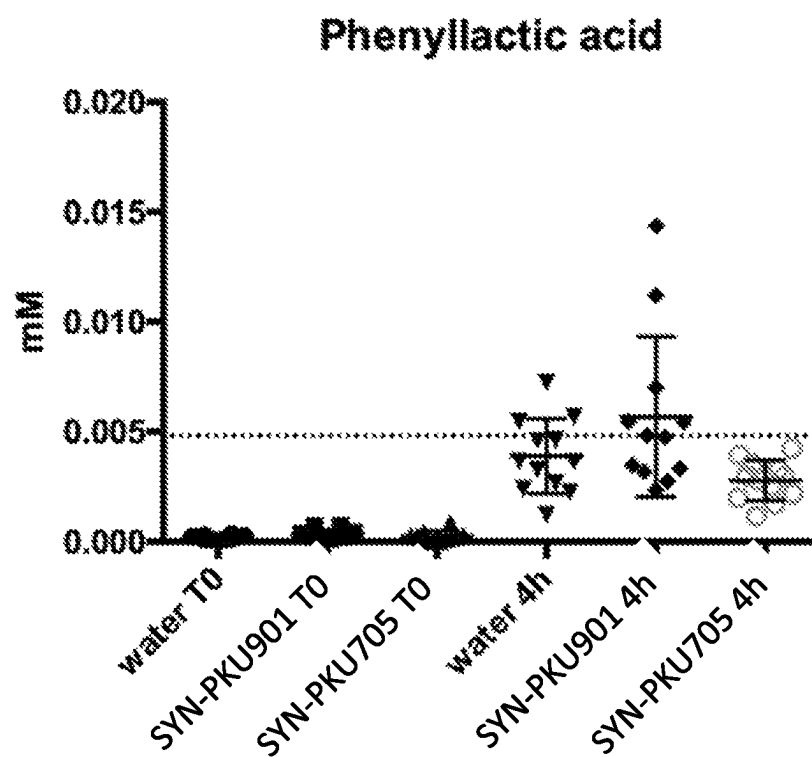


Fig. 34C

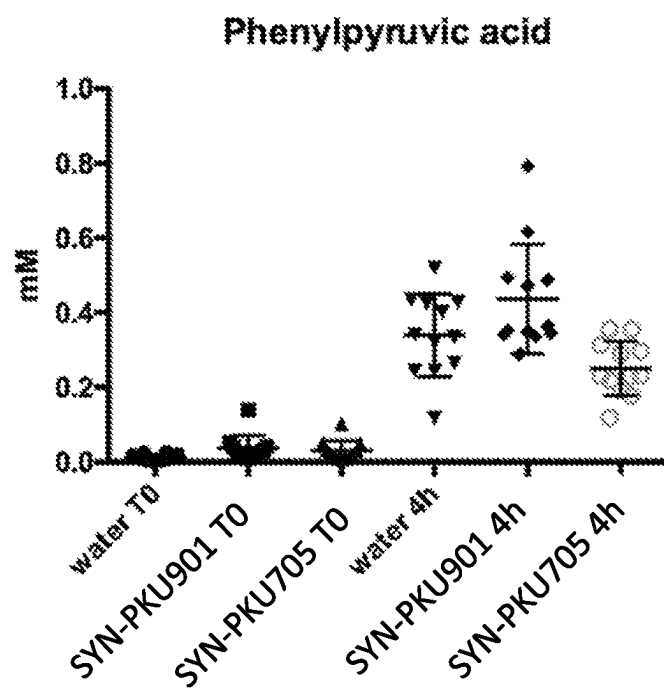


Fig. 34D

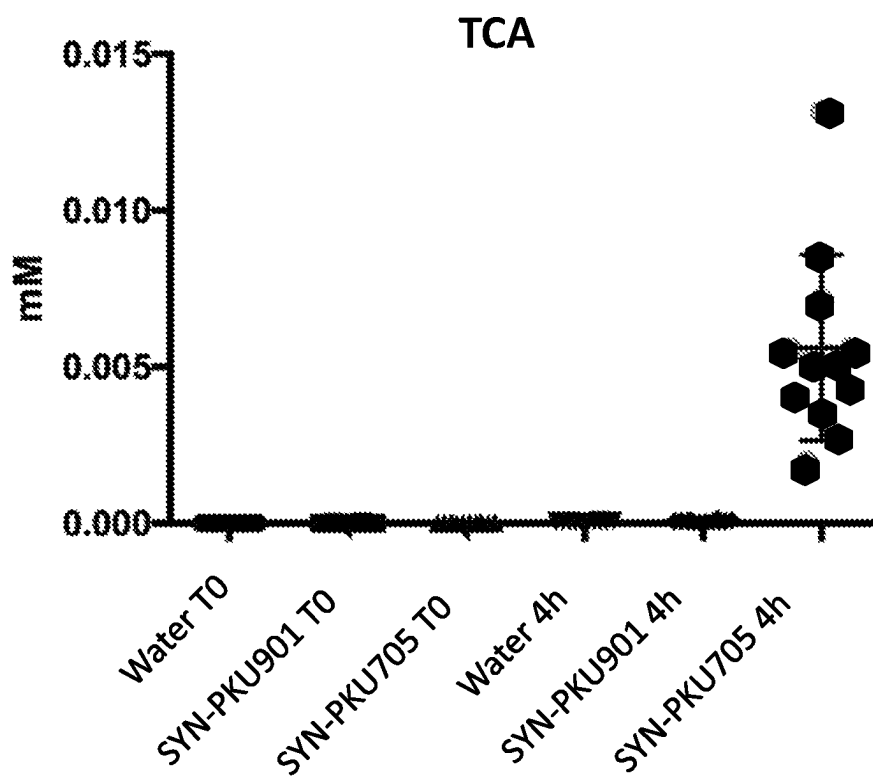


Fig. 34E

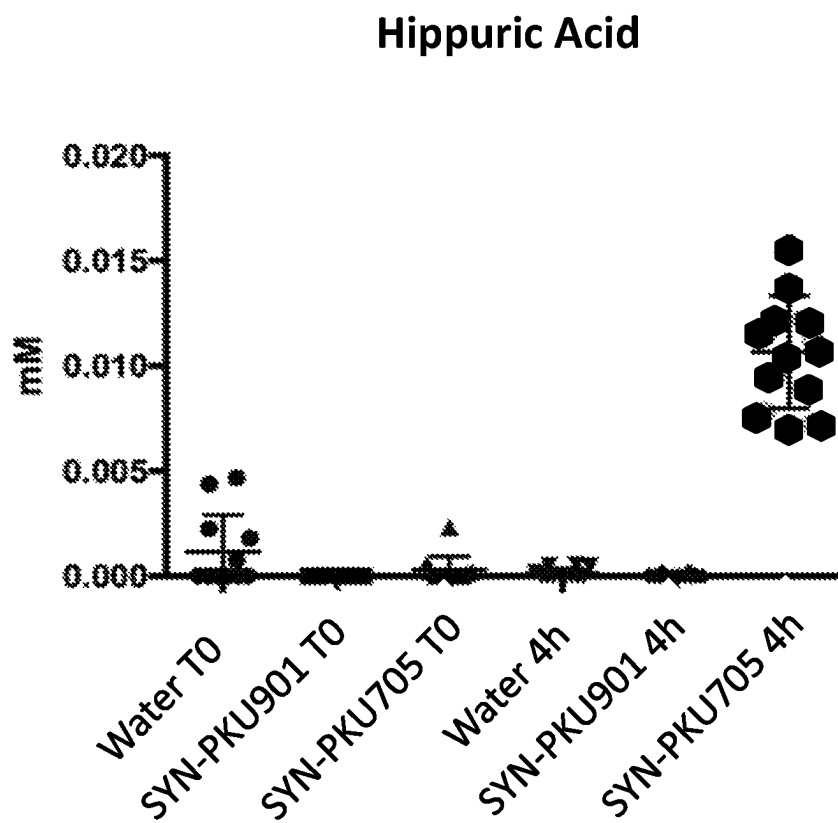
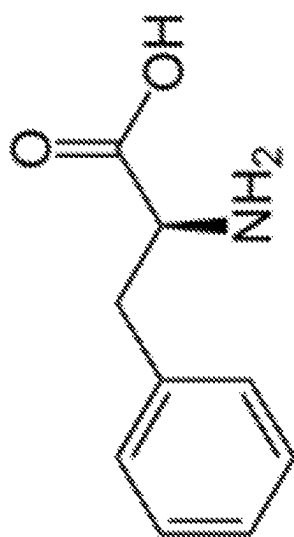
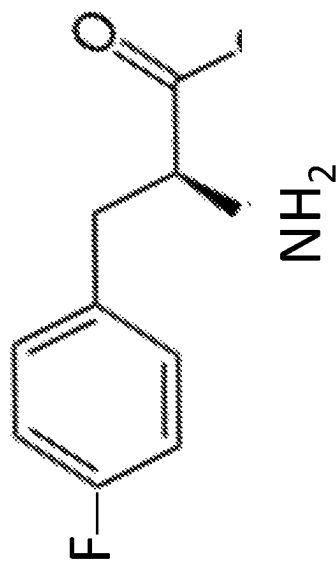


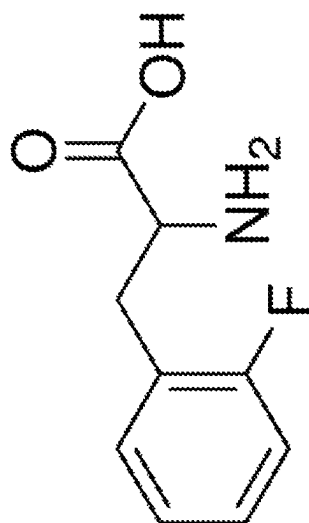
Fig. 34F



phenylalanine



p-fluoro-DL-phenylalanine



o-fluoro-DL-phenylalanine

Fig. 35

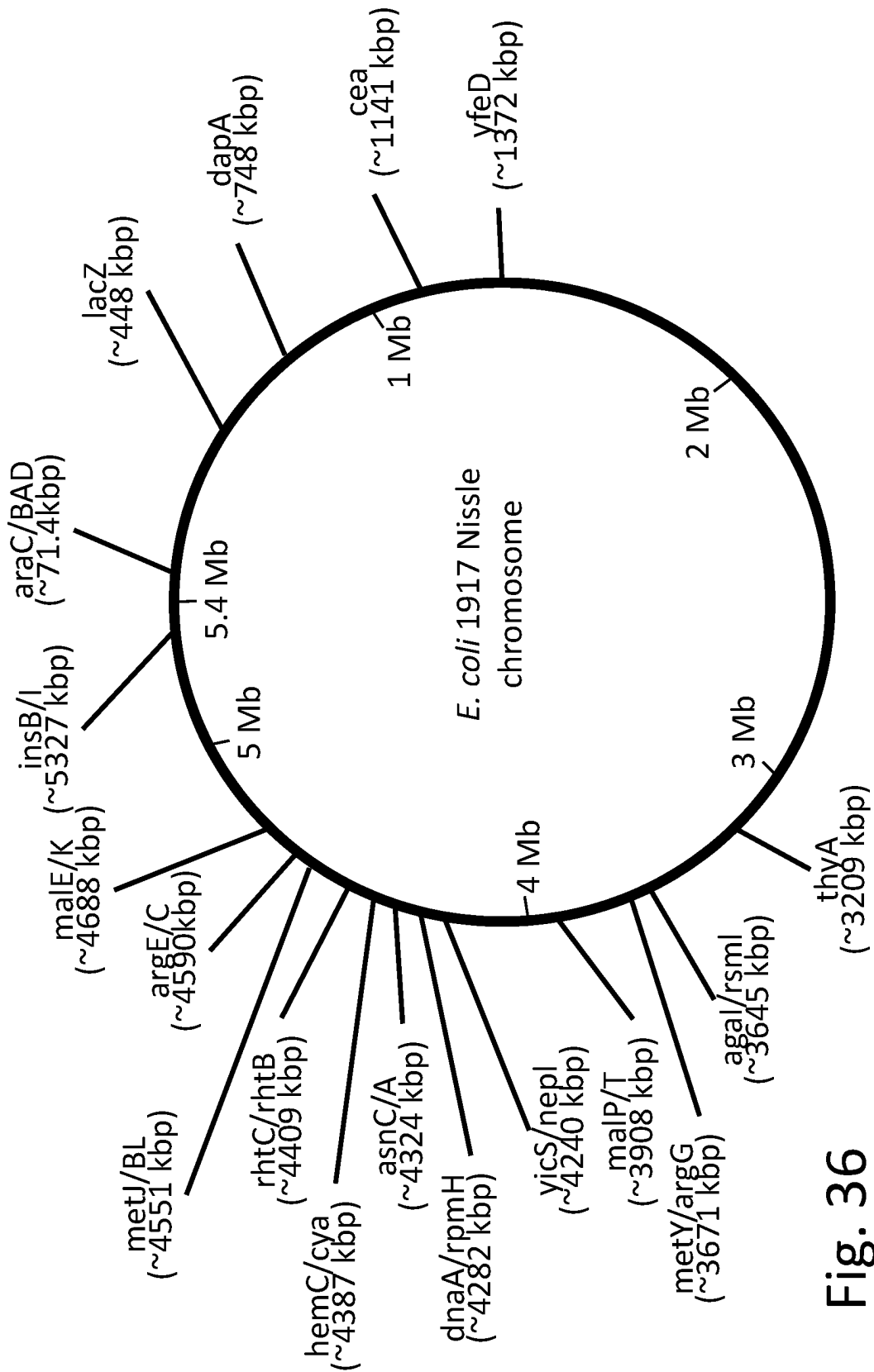
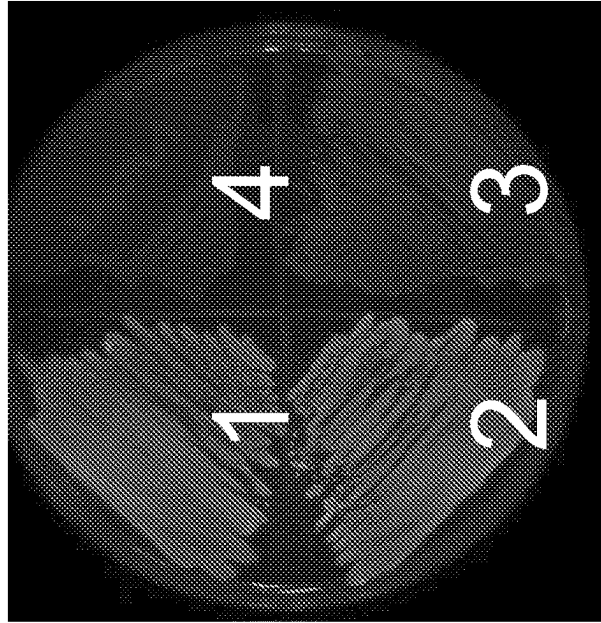


Fig. 36



Brightness of constitutive RFP integrated in
three locations:

1. AraB/C
2. MalE/K
3. MetY/ArgG
4. Nissle (non-fluorescent)

Fig. 37

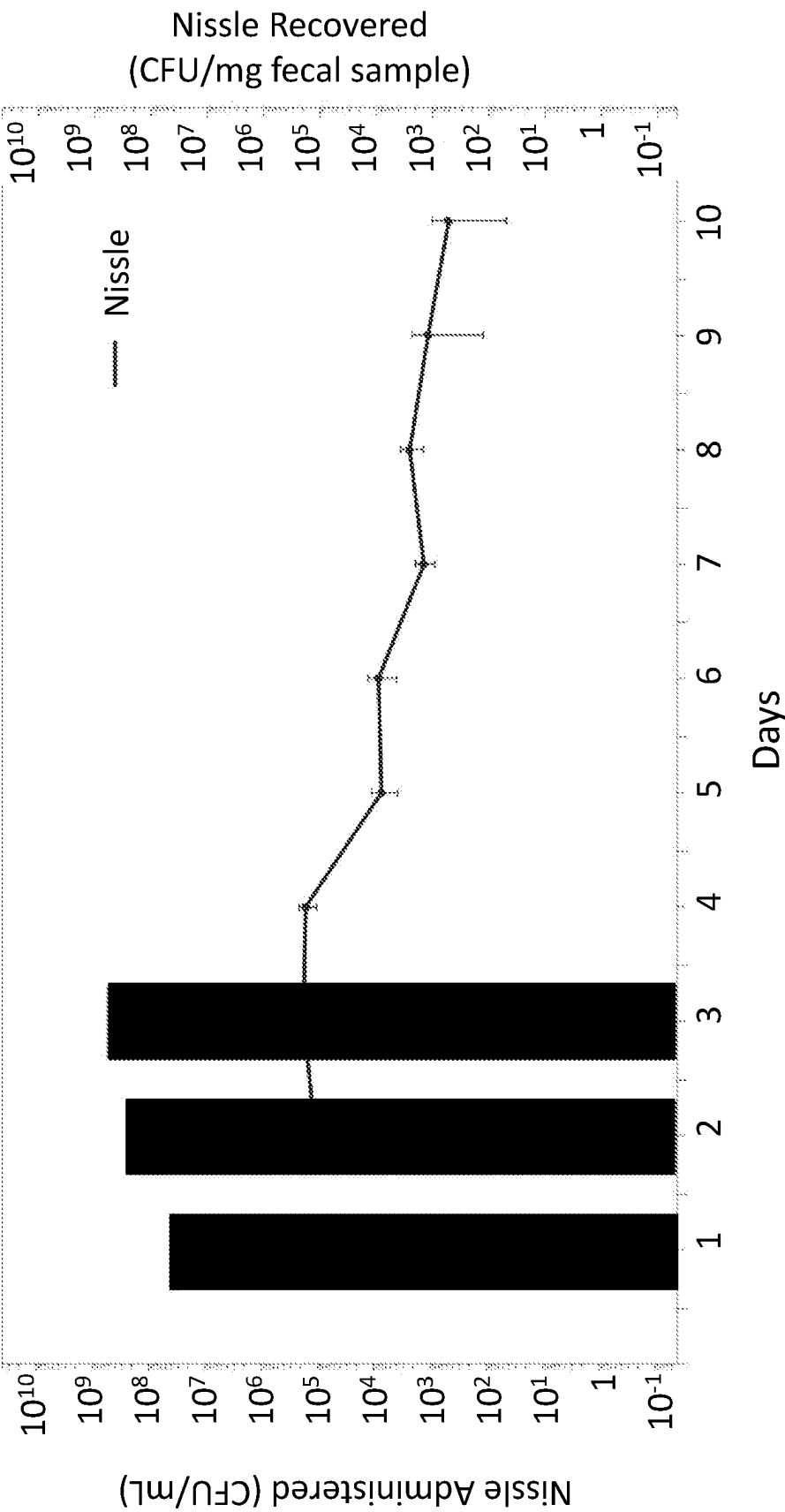


Fig. 38

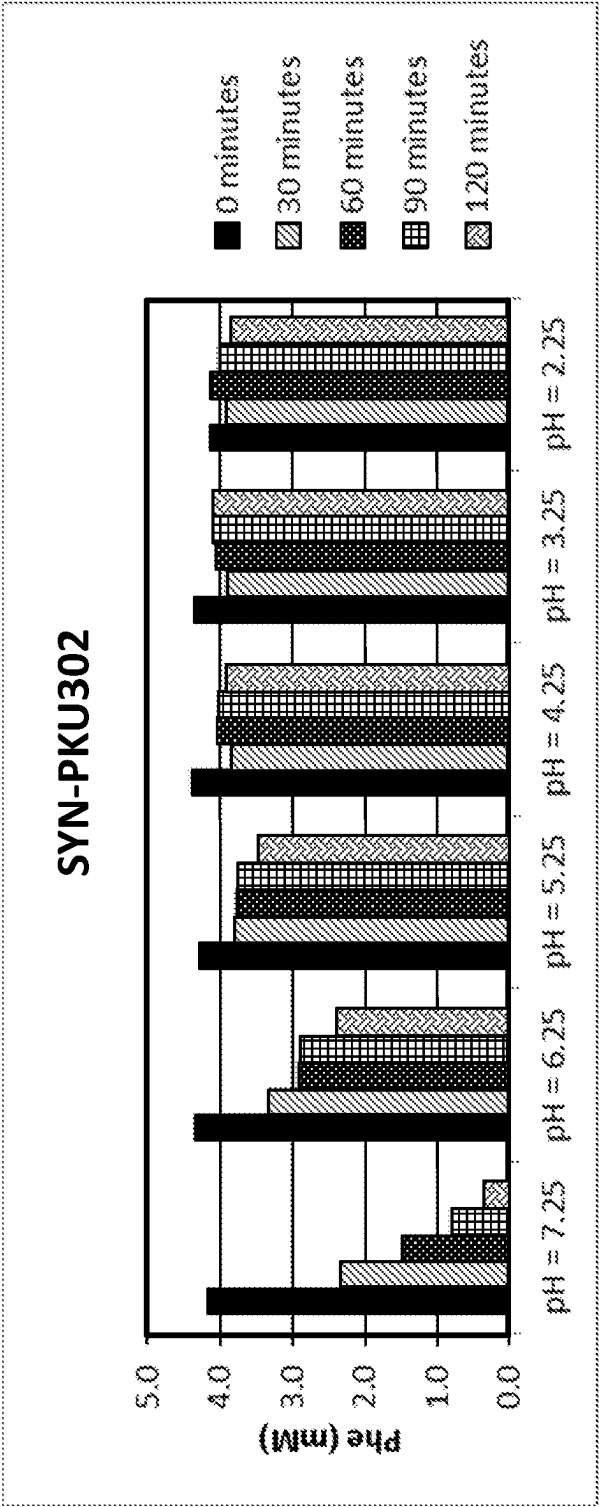


Fig. 39A

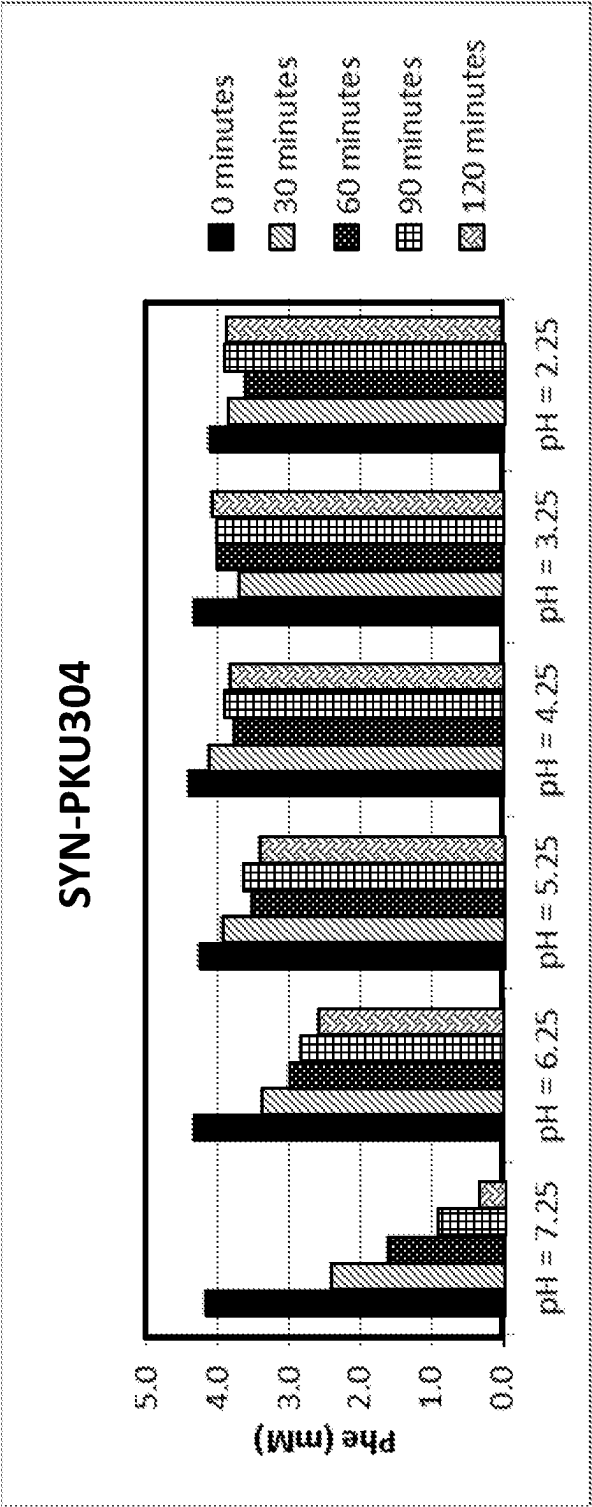


Fig. 39B

Synthetic Biotics: Single Product; Multiple MoAs

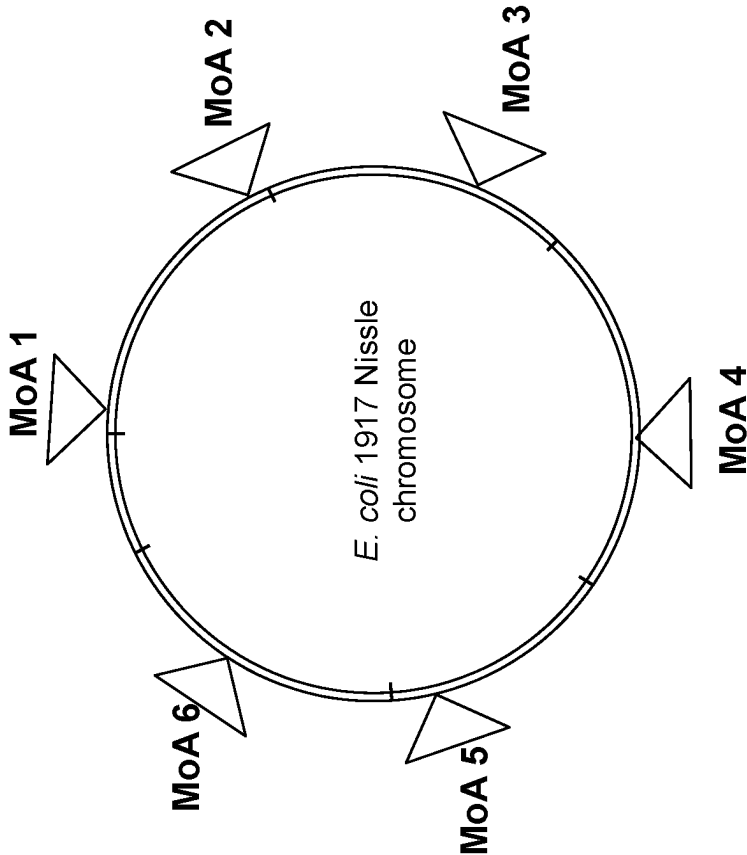


Fig. 40

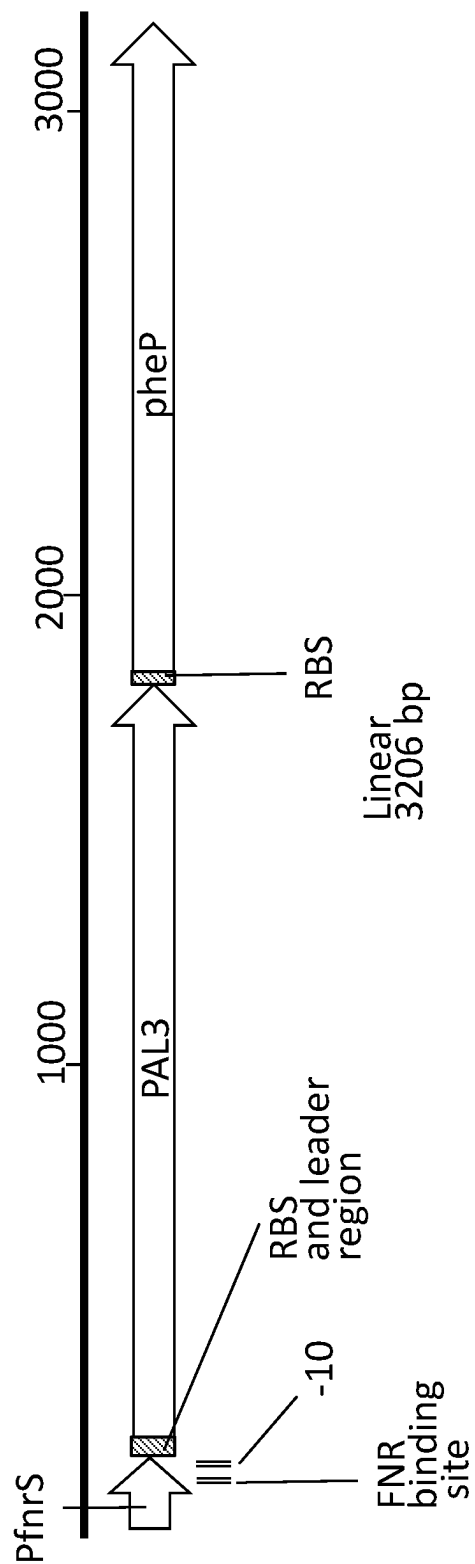


Fig. 41

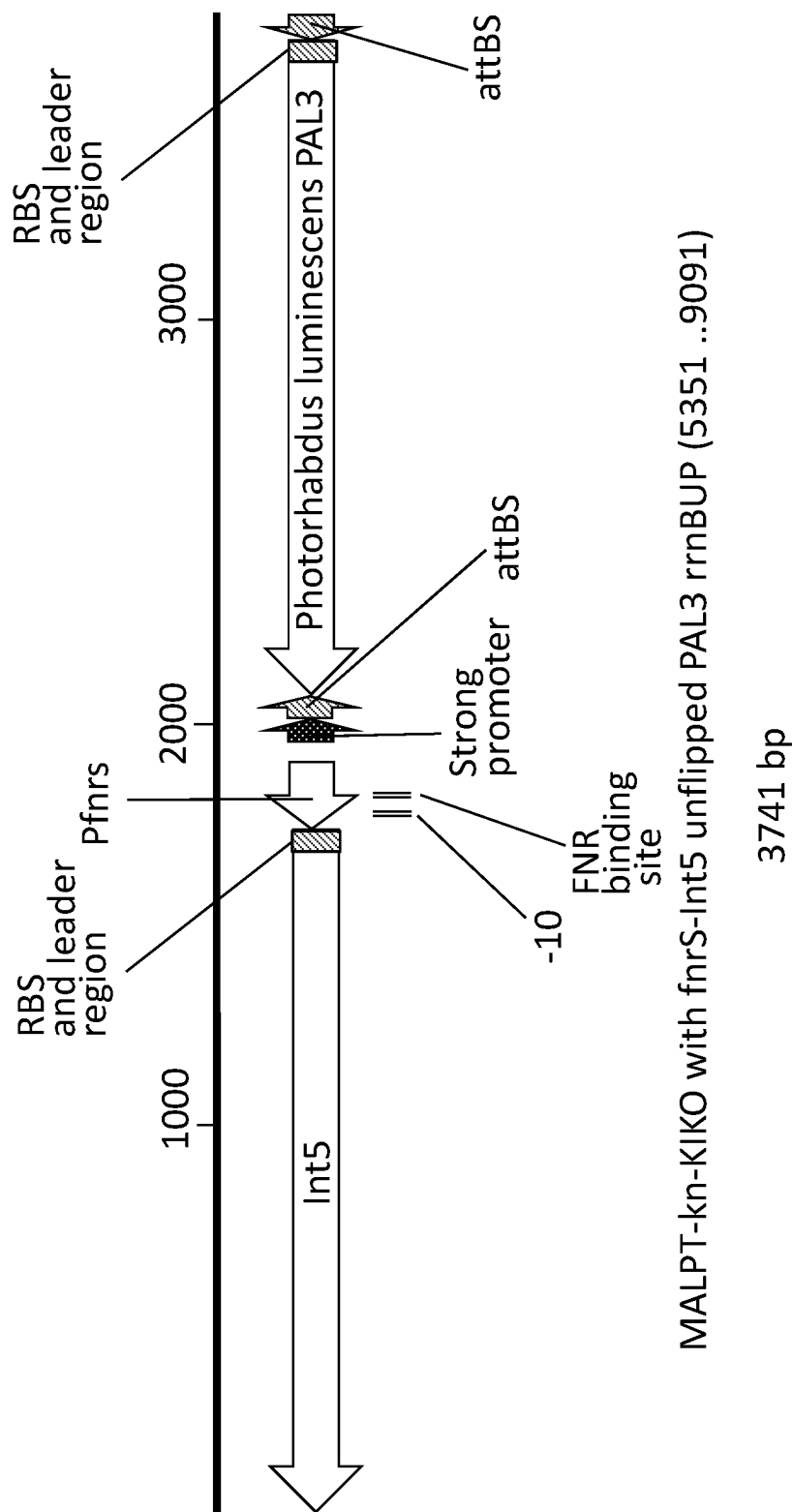


Fig. 42A

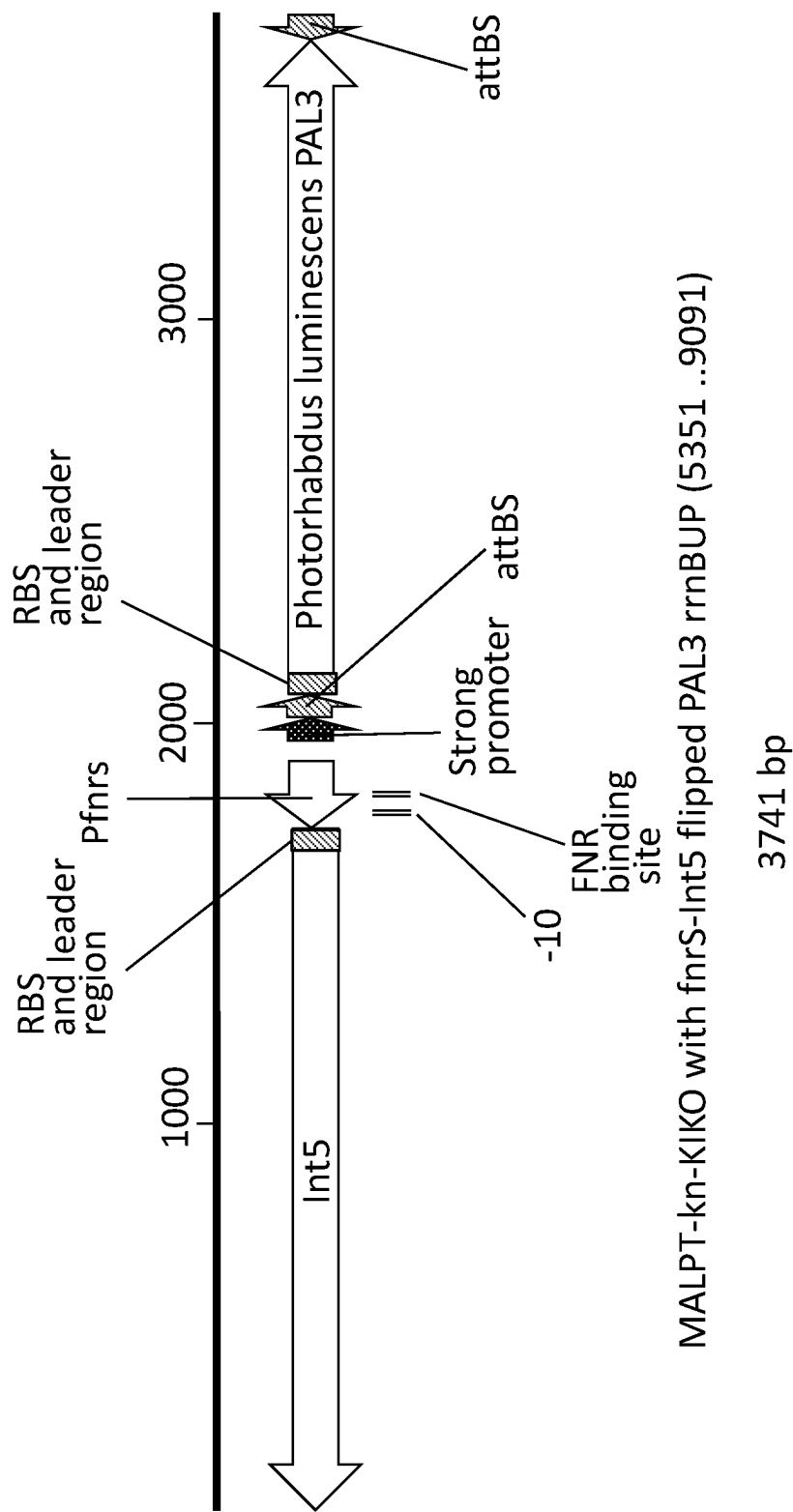
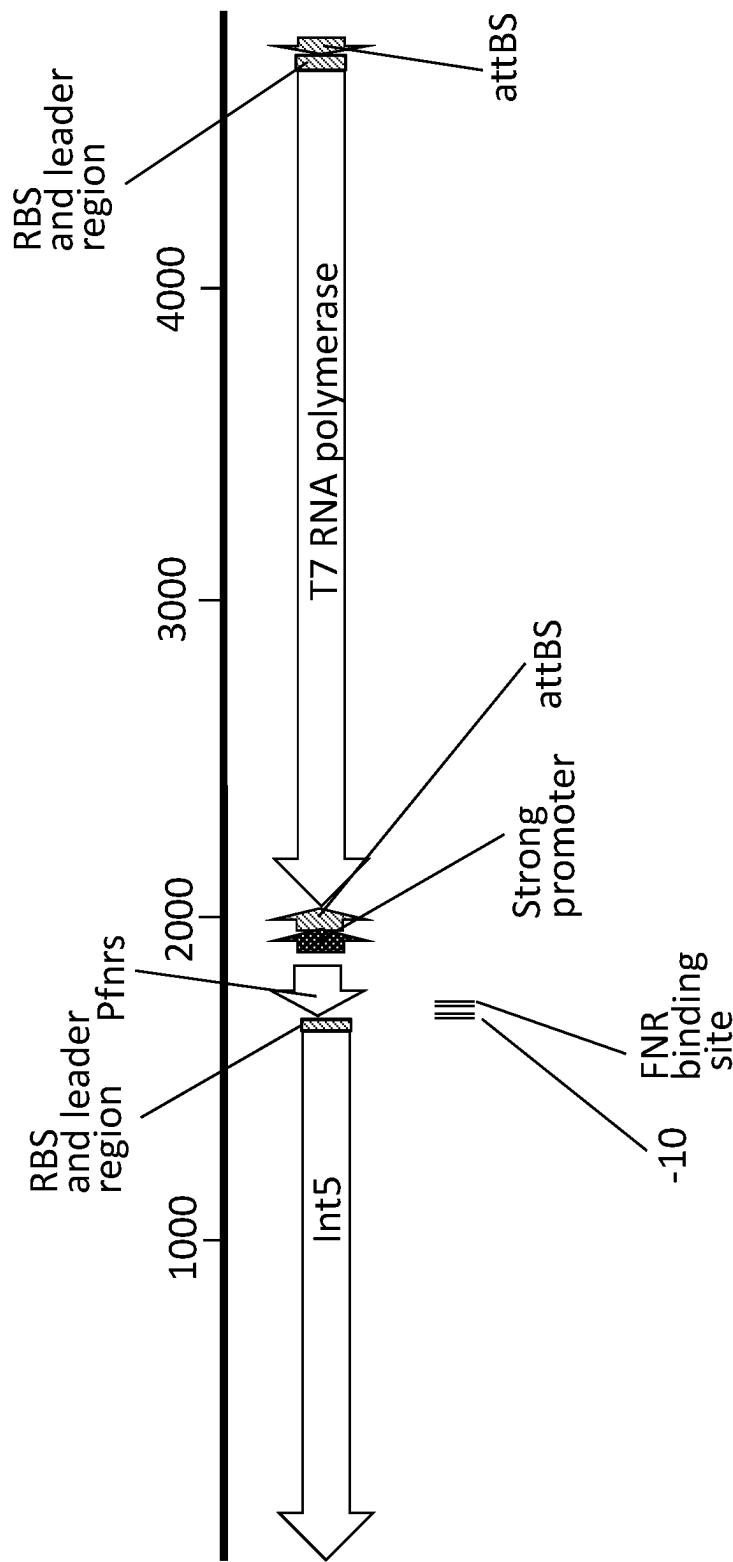


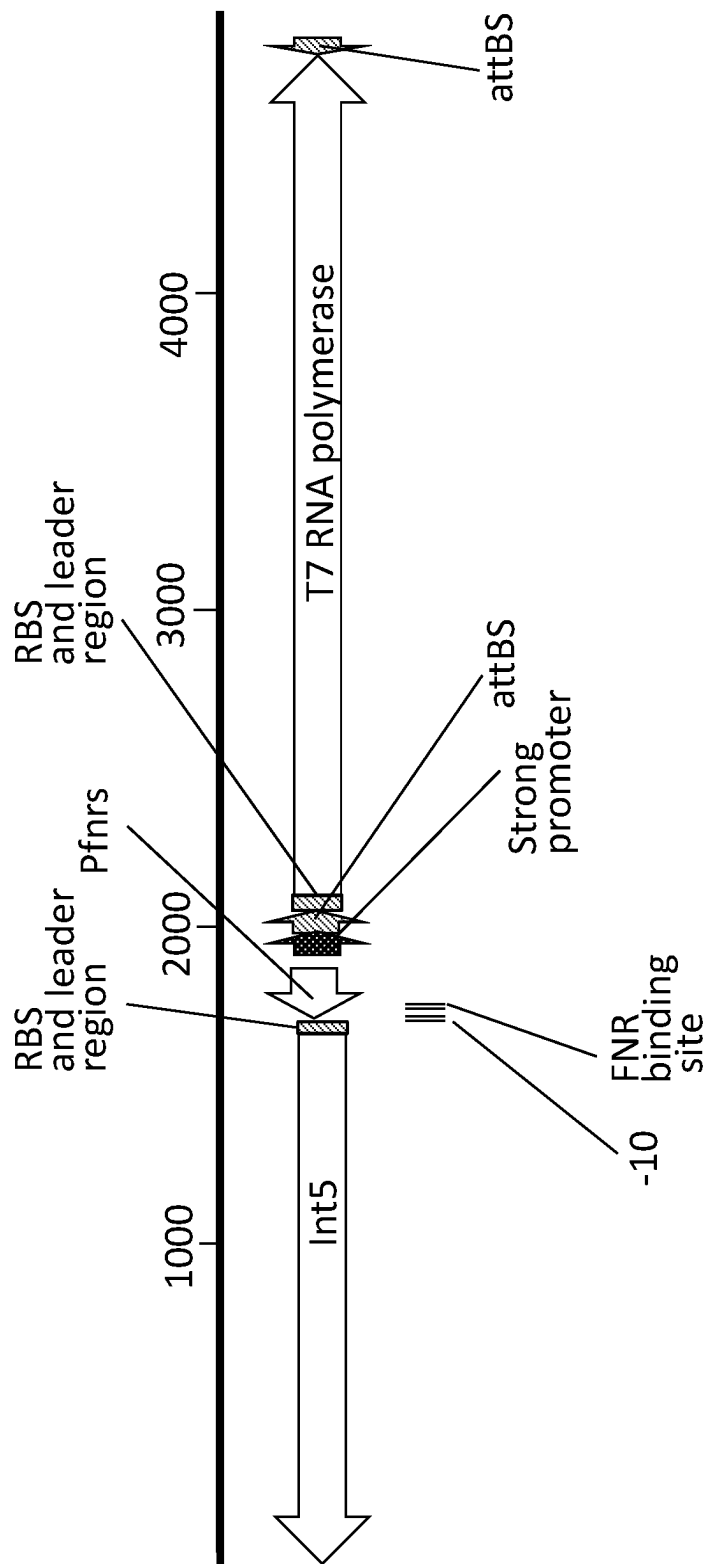
Fig. 42B



MALPT-kn-KIKO with fnrS-Int5 unflipped T7 polymerase rrnBUP (5351 ..9091)

4794 bp

Fig. 43A



MALPT-kn-KIKO with fnrS-Int5 flipped T7 polymerase rrnBUP (5351 ..9091)
4794 bp

Fig. 43B

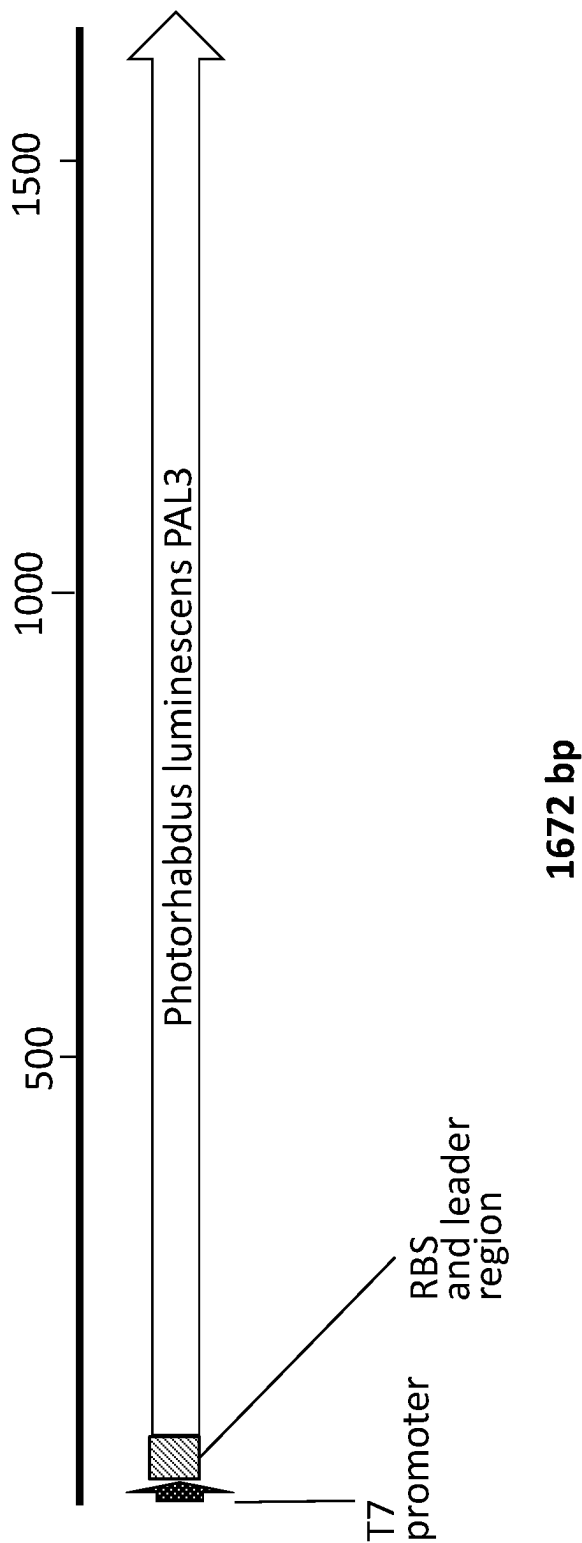


Fig. 43C

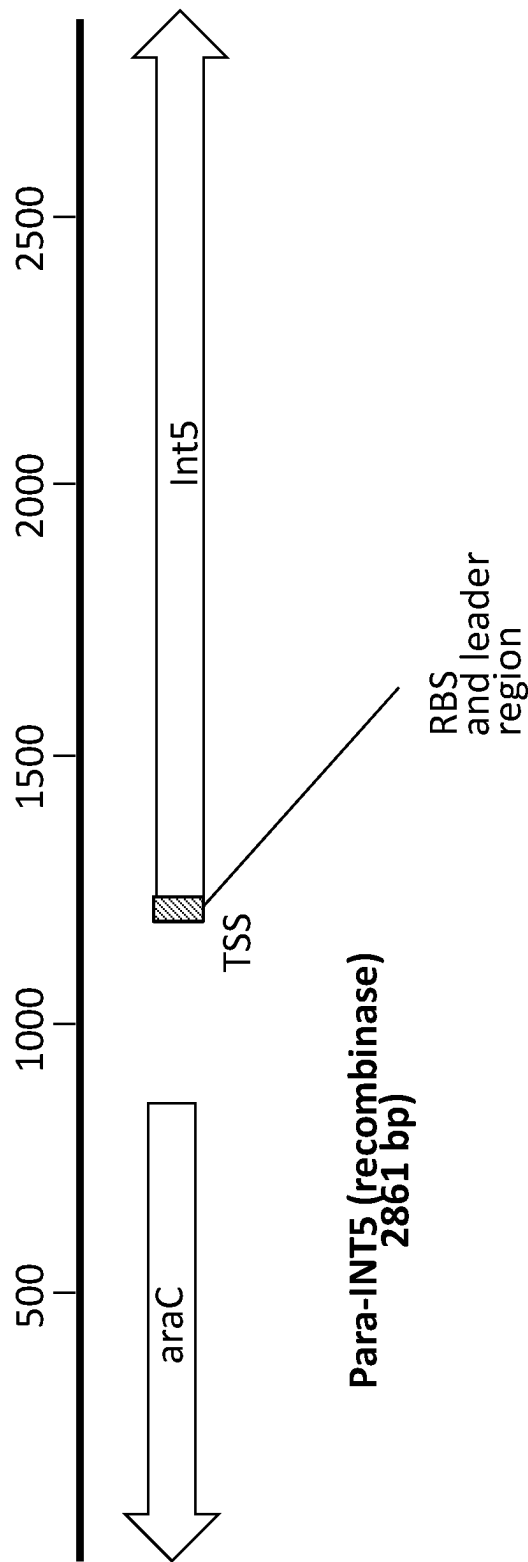


Fig. 44A

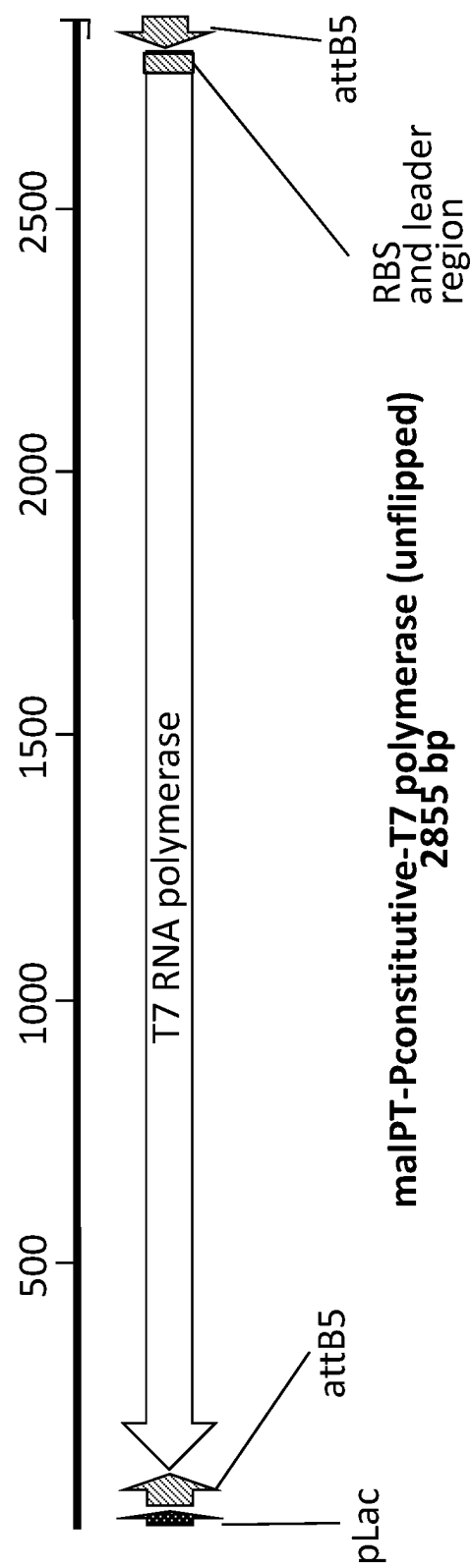


Fig. 44B

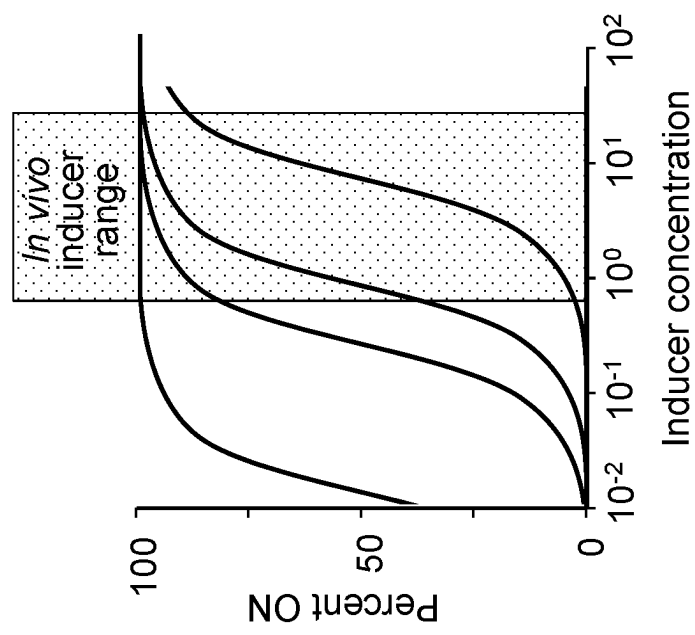


Fig. 45B

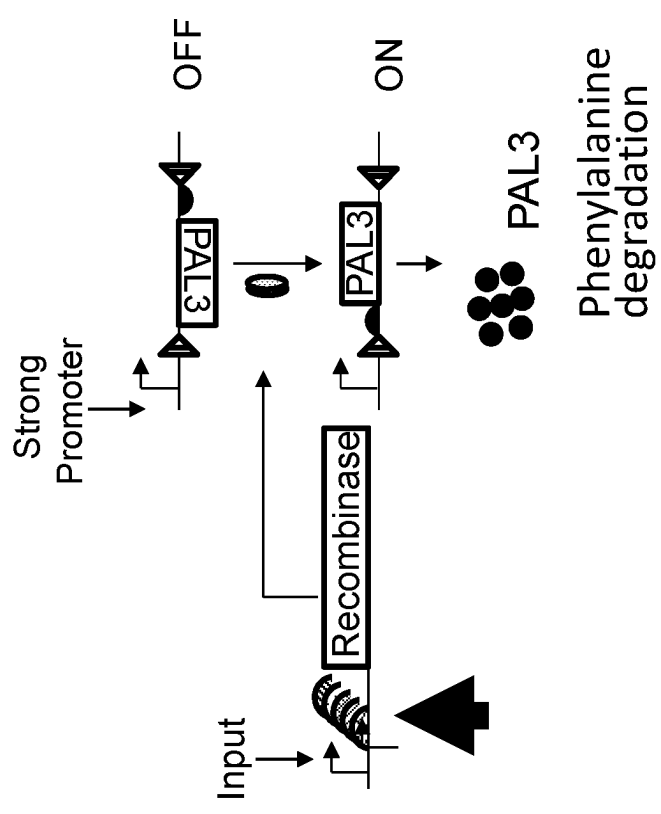


Fig. 45A

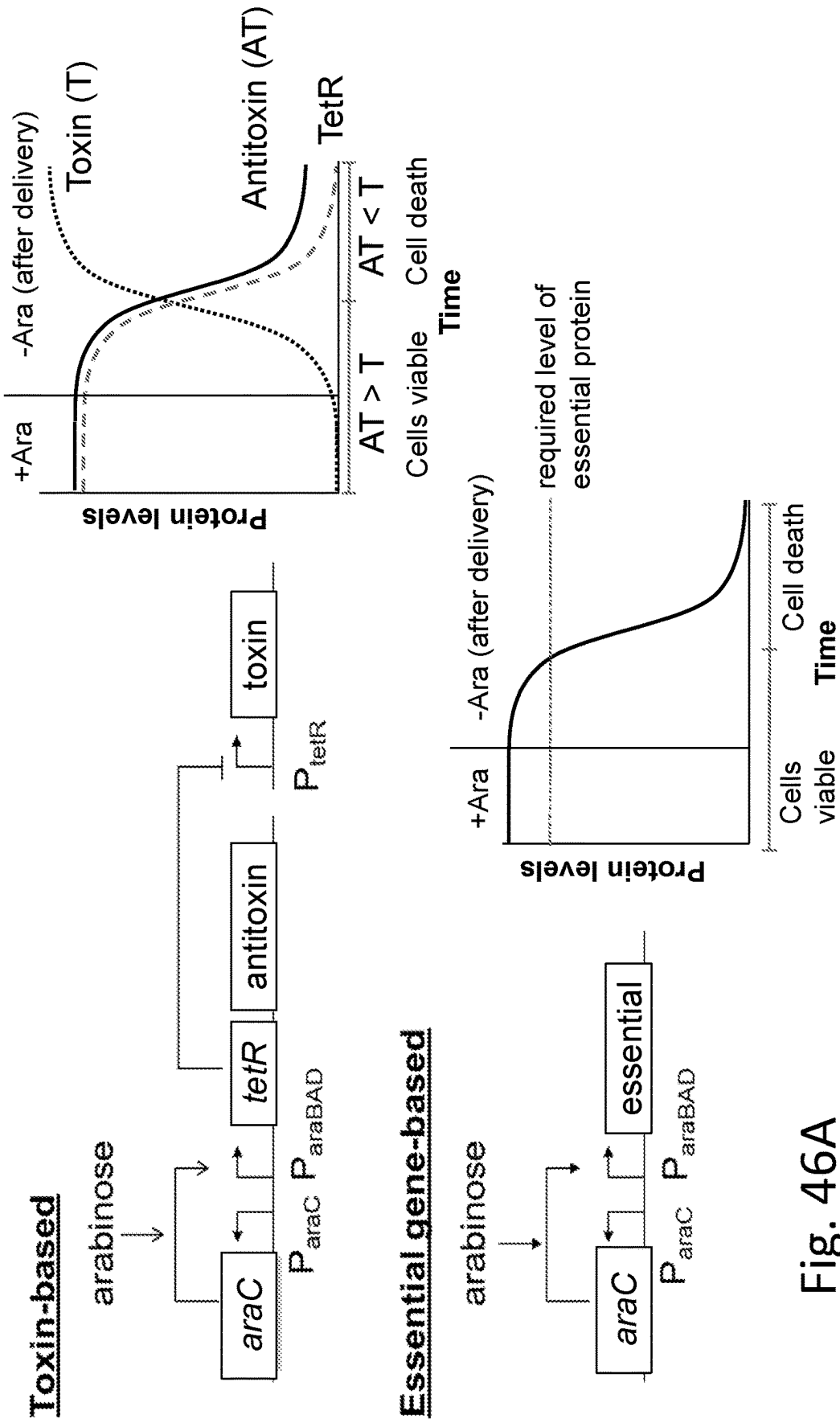


Fig. 46A

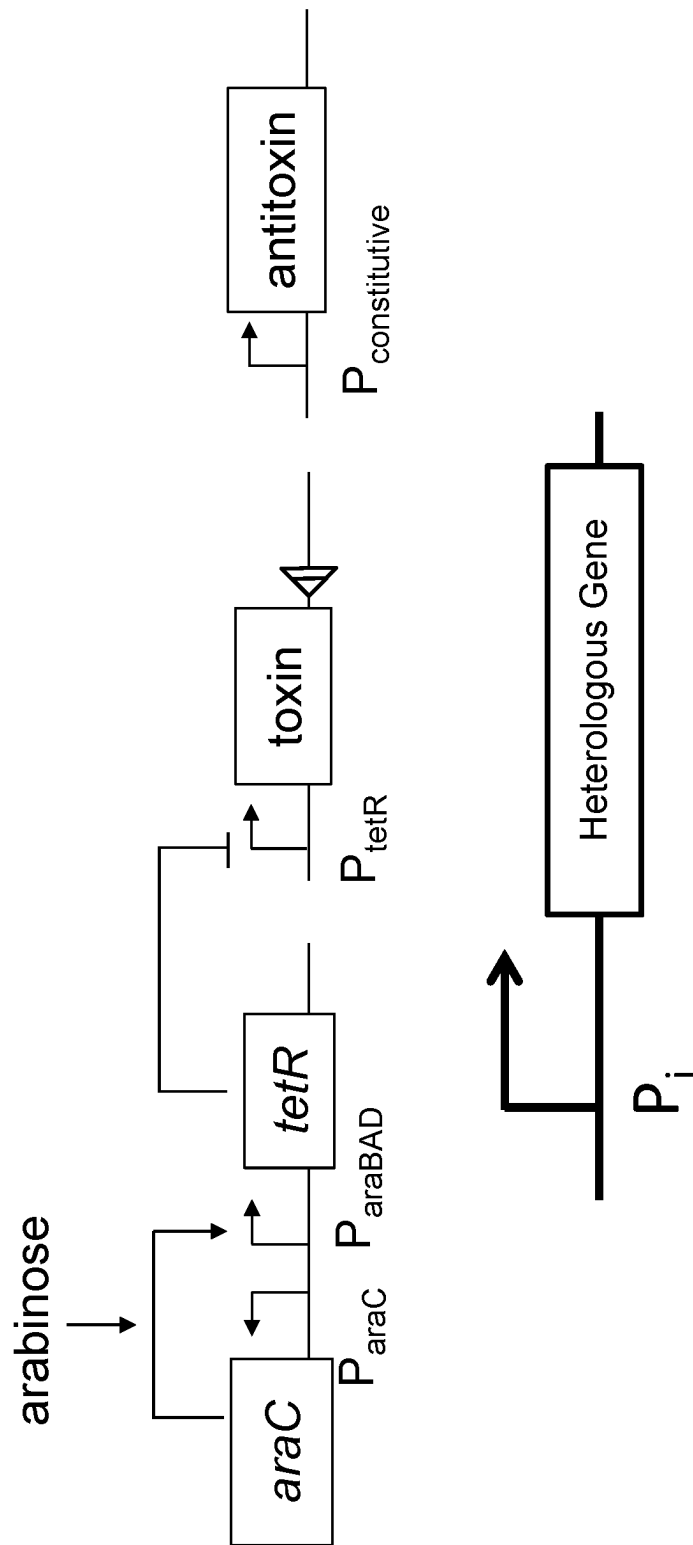


Fig. 46B

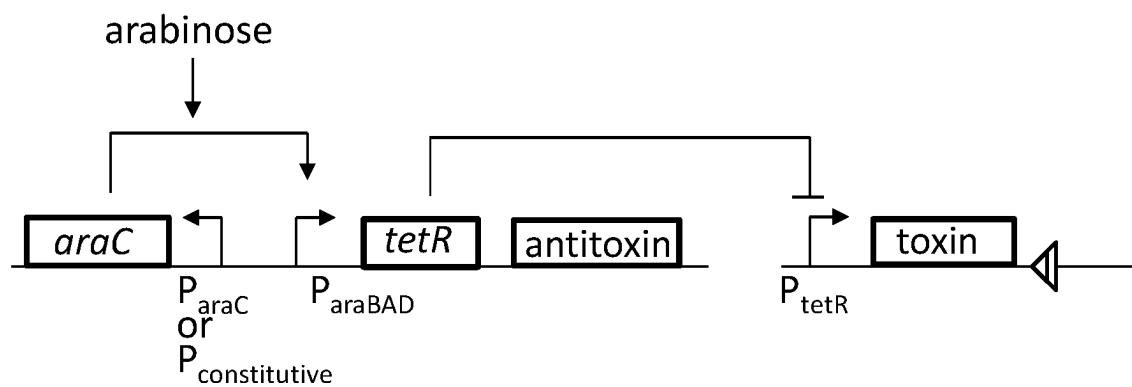


Fig. 46C

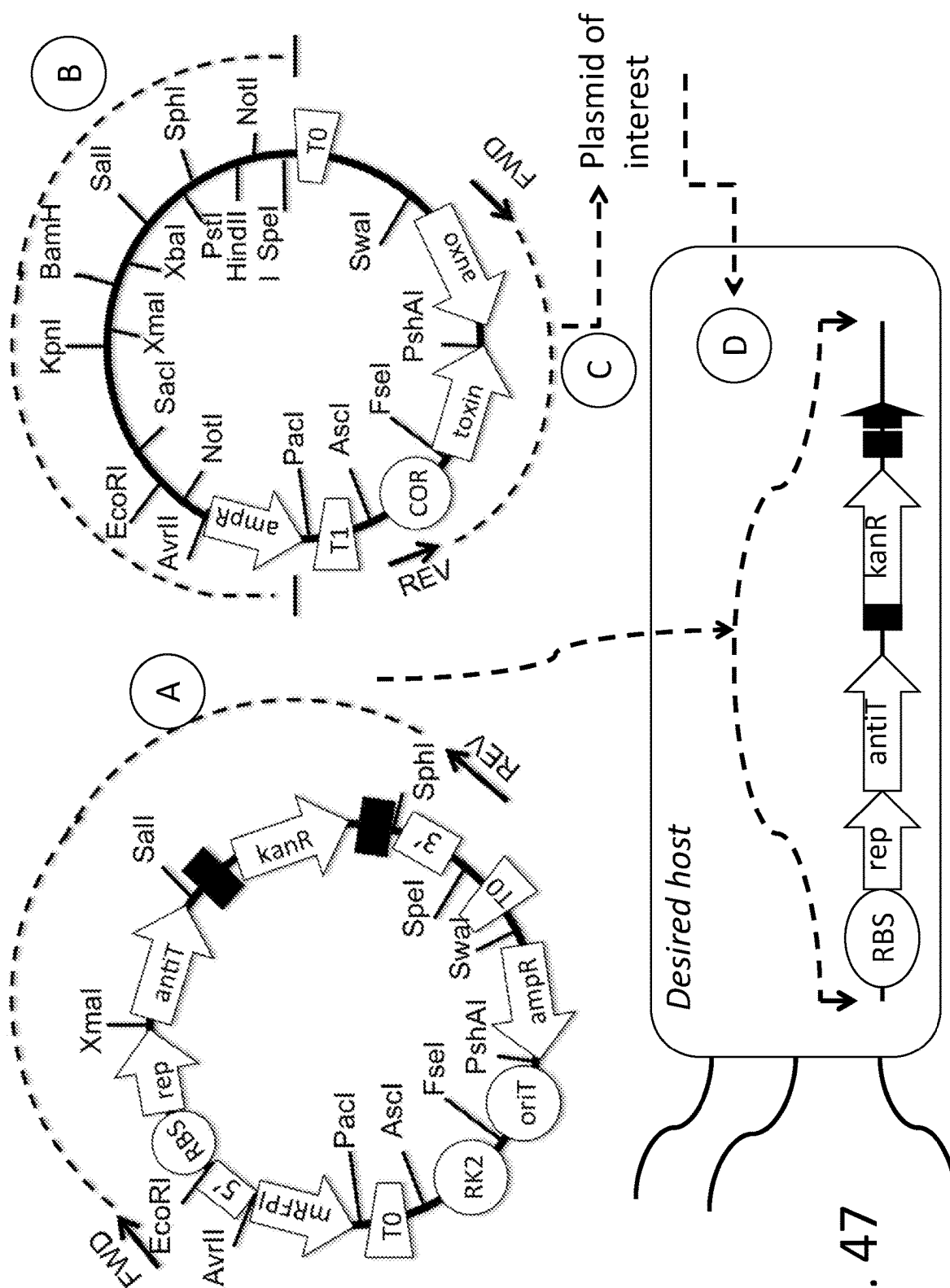


Fig. 47

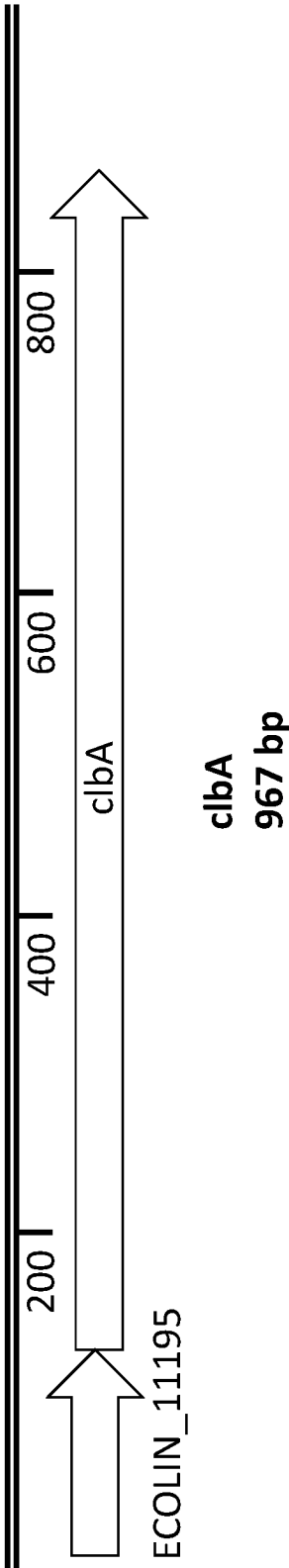


Fig. 48A

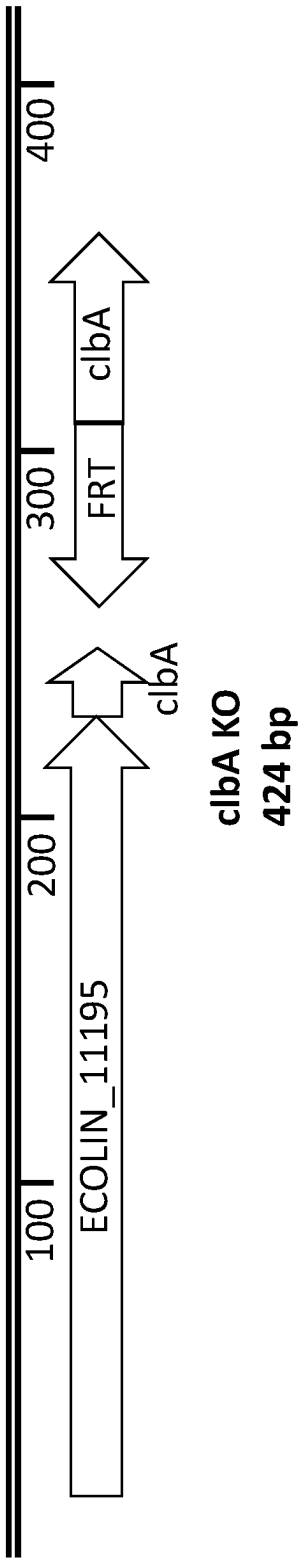


Fig. 48B

Wild-type clbA (SEQ ID NO: 64)
caaatatcacataatctttaacatatcaataaaacacagtaaagtgttcatgtgaaaaacatcaaaaca taaaatacaagctcggaatcacgaatcacgctatacacattgctaacaggaatgagattatctctaaa tgaggattgatataattaattggacatactagttttttcatcaaacagtagagataaacttcctt cactatctcaatgaggaaagaaataaaacgctatgatcagtttcattttgtgagtataaagaact ctatattttaagccgtatcctgctcaaaaacagcactaaaaagatatcaaacctgatgtctcattac aatcatggcaatttagtacgtgcaaatatggcaaacattttatagtttttccctcagttggcaaaa aagattttttttaaccctttcccatactatagatacagtagccgttgctattagttctcactgcga gcttggtgtcgataattgaacaaaataagagatttagacaactcttattctgaaatatcagtcagcatt ttttactccacaggaagctactaacatagtttccacttccctcgttatgaaggtcaattacttttt tggaaaaatgtggacgctcaaaagaagcttacatcaaaatcgcaggtaaaggccctatcttttaggact ggattgtattgaaatttcattttaacaaaataaaaaactaaacttcaaaaatatagaggttcacctgttt atttctctcaatggaaaaatatgttaactcattttctcgcattagcctctccactcatcacccctaaa ataactattgagctattttcctatgcagtcaccaactttatcacccagcactatcagctaatcattc gtcaaatgggcagaaattgaaatcgccacggataatctagacacttctgagccgtcgataaatga ttttcatattccgtcggtggtgtaagtatcccgcataaatcgtgccattcacatttag
clbA knockout (SEQ ID NO: 65)
ggatgggggaaacatggataagttcaaaagaaaaaacccgttatctctgctgaaagacaagta ttgcgcatgctggcacaaggtgatgagtactctcaaaatcacataaatcttaacataatcaataaa cacagtaaaagtttcatgtgaaaaacatcaaaacataaaatacaagctcggaatcacgaatcacgcta tacacattgctaacaggaatgagattatctaaatgaggatttgatGTGTAGGCTGGAGCTGCTTCG AAGTTCCTATACTTCTAGAGAAATAGGAACTTCGGAATAGGAACCTTCGGAATAGGAACTAAGGAG GATATTCATATGtcgtcaaatgggcagaaattgaaatcgccacggataaatctagacacttctgagcc gtcgataaatattgatttttcataattccgtcgtggtg

Fig. 49

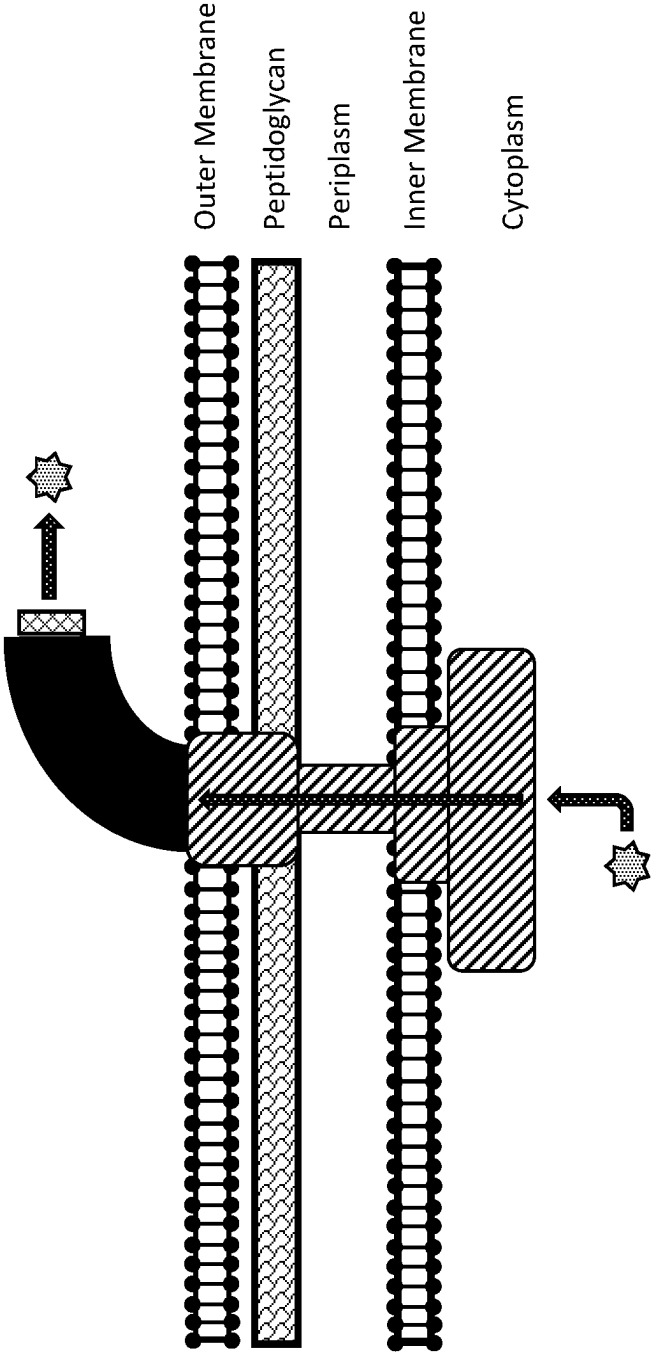


Fig. 50

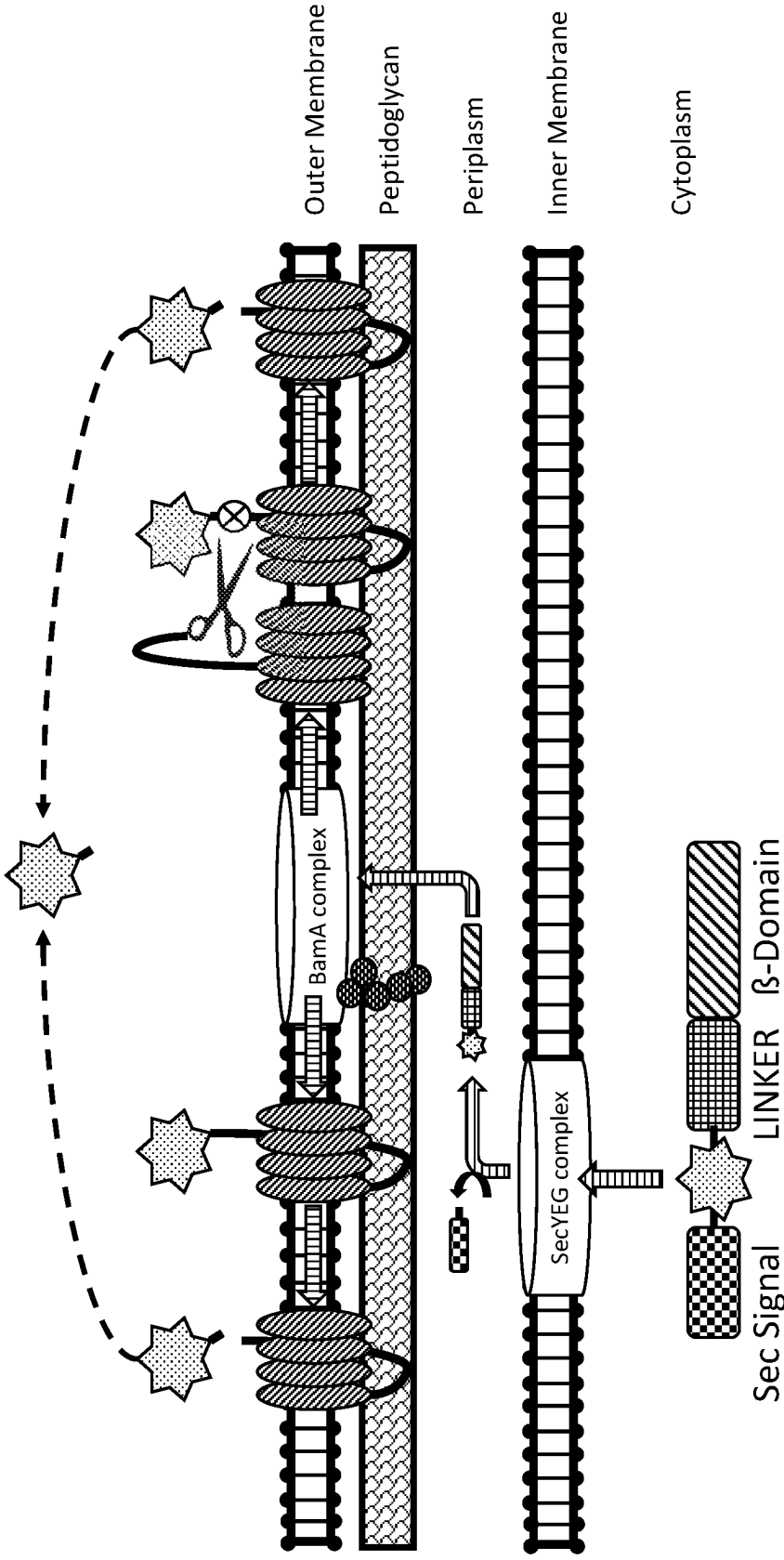


Fig. 51

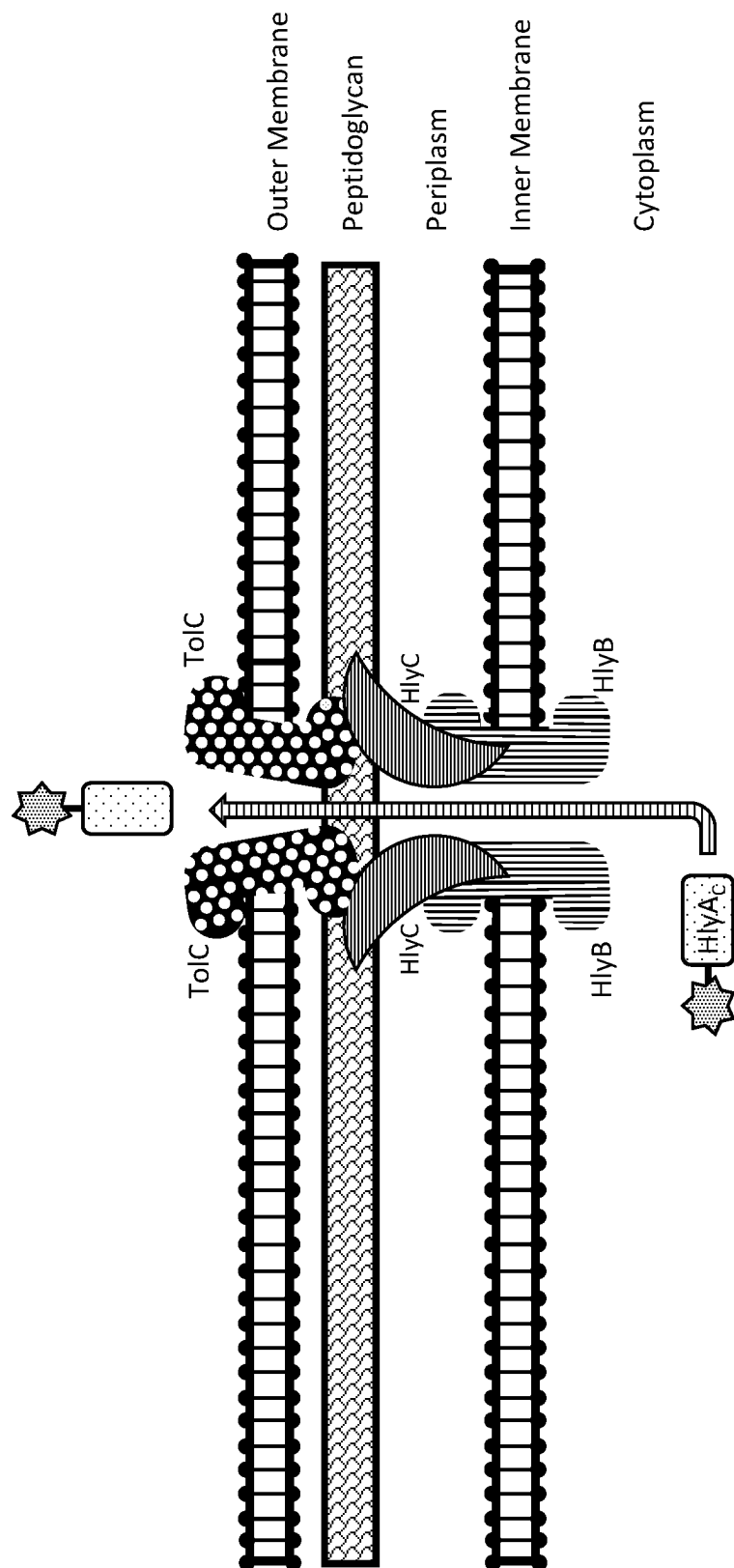


Fig. 52

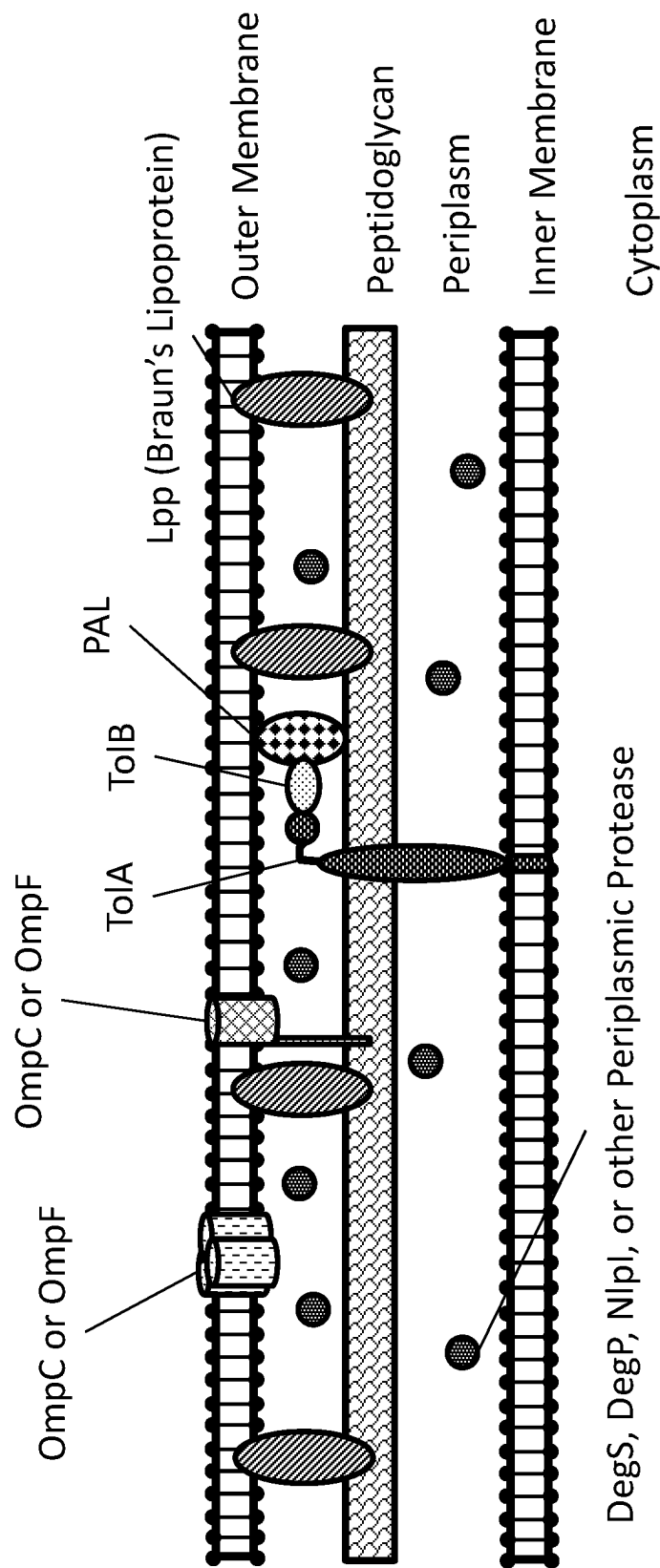


Fig. 53

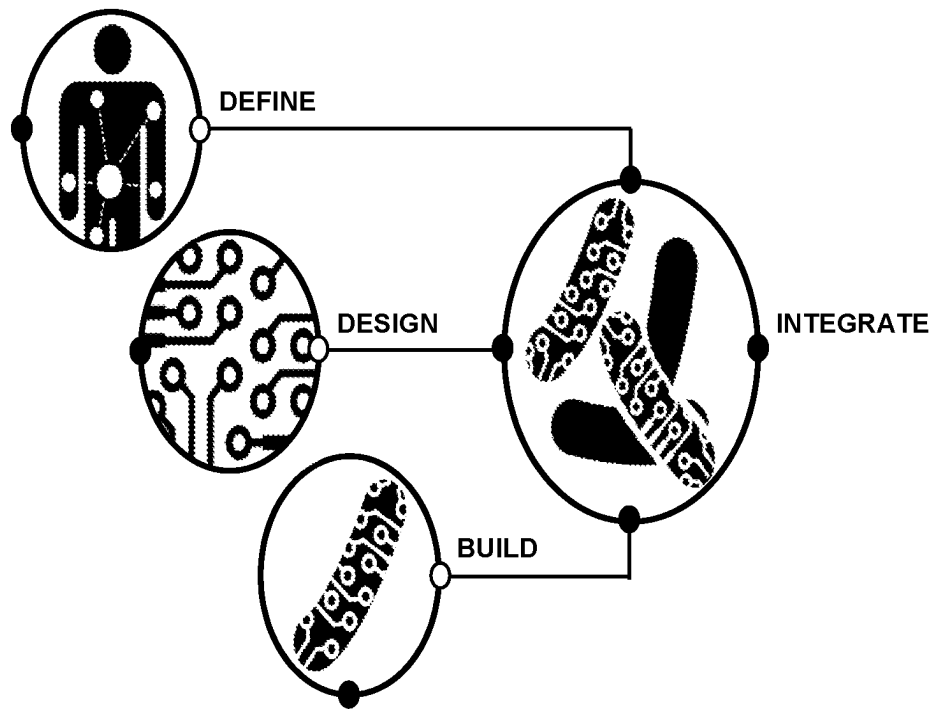
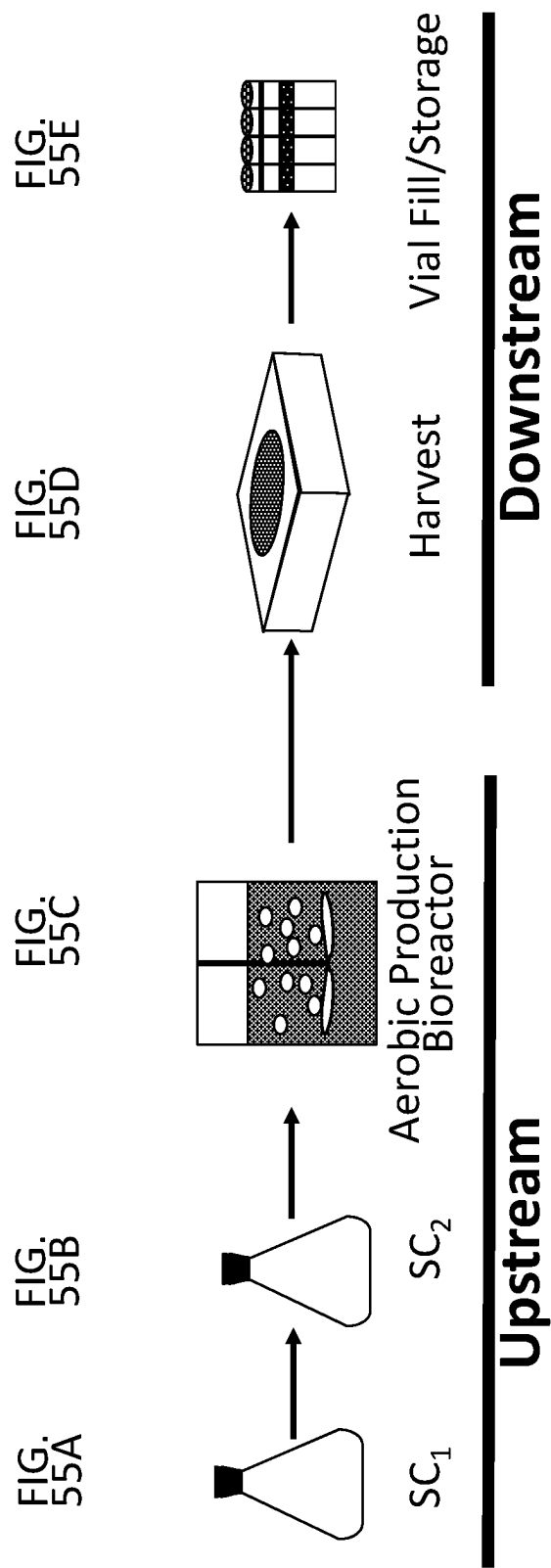


Fig. 54



BACTERIA ENGINEERED TO REDUCE HYPERPHENYLALANINEMIA

The present application is a Division of U.S. patent application Ser. No. 15/154,934, filed May 13, 2016, issued as U.S. Pat. No. 10,610,546 on April 7, 202, which claims the benefit of priority to U.S. Provisional Patent Application No. 62/161,137, filed May 13, 2015, and U.S. Provisional Patent Application No. 62/256,052, filed Nov. 16, 2015, the contents of which are hereby incorporated by reference herein in their entirety.

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jun. 23, 2016, is named 12671_0009-00000_SL.txt and is 207,920 bytes in size.

This disclosure relates to compositions and therapeutic methods for reducing hyperphenylalaninemia. In certain aspects, the disclosure relates to genetically engineered bacteria that are capable of reducing hyperphenylalaninemia in a mammal. In certain aspects, the compositions and methods disclosed herein may be used for treating diseases associated with hyperphenylalaninemia, e.g., phenylketonuria.

Phenylalanine is an essential amino acid primarily found in dietary protein. Typically, a small amount is utilized for protein synthesis, and the remainder is hydroxylated to tyrosine in an enzymatic pathway that requires phenylalanine hydroxylase (PAH) and the cofactor tetrahydrobiopterin. Hyperphenylalaninemia is a group of diseases associated with excess levels of phenylalanine, which can be toxic and cause brain damage. Primary hyperphenylalaninemia is caused by deficiencies in PAH activity that result from mutations in the PAH gene and/or a block in cofactor metabolism.

Phenylketonuria (PKU) is a severe form of hyperphenylalaninemia caused by mutations in the PAH gene. PKU is an autosomal recessive genetic disease that ranks as the most common inborn error of metabolism worldwide (1 in 3,000 births), and affects approximately 13,000 patients in the United States. More than 400 different PAH gene mutations have been identified (Hoeks et al., 2009). Current PKU therapies require substantially modified diets consisting of protein restriction. Treatment from birth generally reduces brain damage and mental retardation (Hoeks et al., 2009; Sarkissian et al., 1999). However, the protein-restricted diet must be carefully monitored, and essential amino acids as well as vitamins must be supplemented in the diet. Furthermore, access to low protein foods is a challenge as they are more costly than their higher protein, nonmodified counterparts (Vockley et al., 2014).

In children with PKU, growth retardation is common on a low-phenylalanine diet (Dobbelaere et al., 2003). In adulthood, new problems such as osteoporosis, maternal PKU, and vitamin deficiencies may occur (Hoeks et al., 2009). Excess levels of phenylalanine in the blood, which can freely penetrate the blood-brain barrier, can also lead to neurological impairment, behavioral problems (e.g., irritability, fatigue), and/or physical symptoms (e.g., convulsions, skin rashes, musty body odor). International guidelines recommend lifelong dietary phenylalanine restriction, which is widely regarded as difficult and unrealistic (Sarkissian et al., 1999), and “continued efforts are needed to overcome the biggest challenge to living with PKU—lifelong adherence to the low-phe diet” (Macleod et al., 2010).

In a subset of patients with residual PAH activity, oral administration of the cofactor tetrahydrobiopterin (also

referred to as THB, BH4, Kuvan, or sapropterin) may be used together with dietary restriction to lower blood phenylalanine levels. However, cofactor therapy is costly and only suitable for mild forms of phenylketonuria. The annual cost of Kuvan, for example, may be as much as \$57,000 per patient. Additionally, the side effects of Kuvan can include gastritis and severe allergic reactions (e.g., wheezing, lightheadedness, nausea, flushing of the skin).

The enzyme phenylalanine ammonia lyase (PAL) is capable of metabolizing phenylalanine to non-toxic levels of ammonia and transcinnamic acid. Unlike PAH, PAL does not require THB cofactor activity in order to metabolize phenylalanine. Studies of oral enzyme therapy using PAL have been conducted, but “human and even the animal studies were not continued because PAL was not available in sufficient amounts at reasonable cost” (Sarkissian et al., 1999). A pegylated form of recombinant PAL (PEG-PAL) is also in development as an injectable form of treatment. However, most subjects dosed with PEG-PAL have suffered from injection site reactions and/or developed antibodies to this therapeutic enzyme (Longo et al., 2014). Thus, there is significant unmet need for effective, reliable, and/or long-term treatment for diseases associated with hyperphenylalaninemia, including PKU.

L-amino acid deaminase (LAAD) catalyzes oxidative deamination of phenylalanine to generate phenylpyruvate, and trace amounts of ammonia and hydrogen peroxide. Phenylpyruvic acid (PPA) is widely used in the pharmaceutical, food, and chemical industries, and PPA is the starting material for the synthesis of D-phenylalanine, a raw intermediate in the production of many chiral drugs and food additives. LAAD has therefore been studied in the context of industrial PPA production (Hou et al. 2015, Appl Microbiol Biotechnol. 2015 October; 99(20):8391-402; “Production of phenylpyruvic acid from L-phenylalanine using an L-amino acid deaminase from *Proteus mirabilis*: comparison of enzymatic and whole-cell biotransformation approaches”). Phenylpyruvate is unable to cross the blood brain barrier (Steele, Fed Proc. 1986 June; 45(7):2060-4; “Blood-brain barrier transport of the alpha-keto acid analogs of amino acids,” indicating that this conversion is useful in controlling the neurological phenotypes of PKU).

In some embodiments, the disclosure provides genetically engineered bacteria that encode and express a phenylalanine metabolizing enzyme (PME). In some embodiments, the disclosure provides genetically engineered bacteria that encode and express phenylalanine ammonia lyase and/or phenylalanine hydroxylase and/or L-amino acid deaminase and are capable of reducing hyperphenylalaninemia.

In certain embodiments, the genetically engineered bacteria are non-pathogenic and may be introduced into the gut in order to reduce toxic levels of phenylalanine. In certain embodiments, the phenylalanine ammonia lyase and/or phenylalanine hydroxylase and/or L-amino acid deaminase is stably produced by the genetically engineered bacteria, and/or the genetically engineered bacteria are stably maintained in vivo and/or in vitro. In certain embodiments, the genetically engineered bacteria further comprise a phenylalanine transporter gene to increase their uptake of phenylalanine. The invention also provides pharmaceutical compositions comprising the genetically engineered bacteria, and methods of modulating and treating disorders associated with hyperphenylalaninemia.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 depicts a synthetic biotic for treating phenylketonuria (PKU) and disorders characterized by hyperphenylalaninemia.

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FIG. 2A depicts a schematic of phenylalanine hydroxylase action in phenylketonuria (PKU). FIG. 2B depicts a schematic of phenylalanine hydroxylase (PAH) action. FIG. 2C depicts a schematic of phenylalanine ammonia lyase (PAL) action. FIG. 2D depicts a schematic of L-amino acid deaminase (LAAD; e.g., from *Proteus mirabilis*) action.

FIG. 3 depicts a synthetic biotic for treating phenylketonuria (PKU) and disorders characterized by hyperphenylalaninemia.

FIG. 4 depicts a synthetic biotic for treating phenylketonuria (PKU) and disorders characterized by hyperphenylalaninemia.

FIG. 5 depicts a synthetic biotic for treating phenylketonuria (PKU) and disorders characterized by hyperphenylalaninemia.

FIG. 6 depicts the gene organization of an exemplary construct comprising a gene encoding PAL3 and a Tet promoter sequence on a high-copy plasmid e.g., as comprised in SYN-PKU202, SYN-PKU303.

FIG. 7 depicts the gene organization of an exemplary construct comprising a gene encoding PAL3 and an FNR promoter sequence on a low-copy plasmid, e.g., as comprised in SYN-PKU304, SYN-PKU307, SYN-PKU305, SYN-PKU306.

FIG. 8 depicts the gene organization of an exemplary construct comprising a gene encoding PAL3 and a Tet promoter sequence on a low-copy plasmid, e.g., SYN-PKU302, SYN-PKU201.

FIG. 9 depicts the gene organization of an exemplary construct, e.g., comprised in SYN-PKU401, comprising a cloned LAAD gene under the control of a Tet promoter sequence and a Tet repressor gene.

FIG. 10 depicts a schematic representation of the construction of *apheP* knock-in strain, wherein recombineering is used to insert a second copy of *pheP* into the Nissle *lacZ* gene.

FIG. 11 depicts the gene organization of an exemplary construct comprising a gene encoding *PheP*, a gene encoding *TetR*, and a tet promoter sequence for chromosomal insertion e.g., as for example comprised in SYN-PKU203, SYN-PKU401, SYN-PKU402, SYN-PKU302, and SYN-PKU303.

FIG. 12A depicts the gene organization of an exemplary construct, comprising a cloned PAL3 gene under the control of an FNR promoter sequence, on a low-copy, kanamycin-resistant plasmid (pSC101 origin of replication). Under anaerobic conditions, PAL3 degrades phenylalanine to non-toxic trans-cinnamate. FIG. 12B depicts an additional copy of the endogenous *E. coli* high affinity phenylalanine transporter, *pheP*, driven by the *PfmrS* promoter and inserted into the *lacZ* locus on the Nissle chromosome.

FIGS. 13A, 13B, and 13C depict schematic diagrams of non-limiting embodiments of the disclosure. FIG. 13A depicts phenylalanine degradation components integrated into the *E. coli* Nissle chromosome. In some embodiments, engineered plasmid-free bacterial strains are used to prevent plasmid conjugation in vivo. In some embodiments, multiple insertions of the PAL gene result in increased copy number and/or increased phenylalanine degradation activity. In some embodiments, a copy of the endogenous *E. coli* high affinity phenylalanine transporter, *pheP*, is driven by the *PfmrS* promoter and is inserted into the *lacZ* locus. FIG. 13B depicts a schematic diagram of one non-limiting embodiment of the disclosure, wherein the *E. coli* Nissle chromosome is engineered to contain four copies of *PfmrS*-PAL inserted at four different insertion sites across the genome (male/K, *yicS*/*nepI*, *agal*/*rsmI*, and *cea*), and one copy of a

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phenylalanine transporter gene inserted at a different insertion site (*lacZ*). In this embodiment, the PAL gene is PAL3 derived from *P. luminescens*, and the phenylalanine transporter gene is *pheP* derived from *E. coli*. In one embodiment, the strain is SYN-PKU511. FIG. 13C depicts a schematic diagram of one preferred embodiment of the disclosure, wherein the *E. coli* Nissle chromosome is engineered to contain five copies of PAL under the control of an oxygen level-dependent promoter (e.g., *PfmrS*-PAL3) inserted at different integration sites on the chromosome (male/K, *yicS*/*nepI*, *malP*/*T*, *agal*/*rsmI*, and *cea*), and one copy of a phenylalanine transporter gene under the control of an oxygen level-dependent promoter (e.g., *PfmrS*-*pheP*) inserted at a different integration site on the chromosome (*lacZ*). The genome is further engineered to include a *thyA* auxotrophy, in which the *thyA* gene is deleted and/or replaced with an unrelated gene, as well as a kanamycin resistance gene.

FIG. 14 depicts the gene organization of a non-limiting exemplary construct comprising a gene encoding *araC* and a gene encoding LAAD from *Proteus mirabilis* and an arabinose inducible promoter (ParaBAD) sequence for chromosomal insertion into the endogenous arabinose operon for chromosomal integration, e.g., as comprised in SYN-PKU705.

FIG. 15A depicts phenylalanine concentrations in samples comprising bacteria expressing PAL1 or on low-copy (LC; SYN-PKU101) or high-copy (HC; SYN-PKU102) plasmids or PAL3 on low-copy (LC; SYN-PKU201) or high-copy (HC; SYN-PKU202) plasmids, induced with anhydrous tetracycline (ATC), and then grown in culture medium supplemented with 4 mM (660,000 ng/mL) of phenylalanine. Samples were removed at 0 hrs, 4 hrs, and 23 hrs. Phenylalanine concentrations were determined by mass spectrometry. FIG. 15B depicts cinnamate levels in samples at 4 hrs and 23 hrs post-induction. In PAL3-expressing strains, the PAL3 gene is derived from *Photobacterium luminescens*, an enterobacterium in the same taxonomic subdivision as *Escherichia coli*.

FIG. 16A depicts phenylalanine concentrations in samples comprising bacteria expressing PAL1 or PAL3 on low-copy (LC) or high-copy (HC) plasmids, or further comprising a copy of *pheP* driven by the Tet promoter integrated into the chromosome. Bacteria were induced with ATC, and then grown in culture medium supplemented with 4 mM (660,000 ng/mL) of phenylalanine to an OD₆₀₀ of 2.0. Samples were removed at 0 hrs, 2 hrs, and 4 hrs post-induction and phenylalanine concentrations were determined by mass spectrometry. Notably, the additional copy of *pheP* permitted the degradation of phenylalanine (4 mM) in 4 hrs. FIG. 16B depicts cinnamate levels in samples at 2 hrs and 4 hrs post-induction. In some embodiments, cinnamate may be used as an alternative biomarker for strain activity. *PheP* overexpression improves phenylalanine metabolism in engineered bacteria. Strains analyzed in this data set are SYN-PKU101, SYN-PKU102, SYN-PKU202, SYN-PKU201, SYN-PKU401, SYN-PKU402, SYN-PKU203, SYN-PKU302, SYN-PKU303.

FIGS. 17A and 17B depict the state of one non-limiting embodiment of the PAL construct under non-inducing (FIG. 17A) and inducing (FIG. 17B) conditions. FIG. 17A depicts relatively low PAL and *PheP* production under aerobic conditions due to oxygen (O₂) preventing FNR from dimerizing and activating PAL and/or *pheP* gene expression. FIG. 17B depicts up-regulated PAL and *PheP* production under anaerobic conditions due to FNR dimerizing and inducing FNR promoter-mediated expression of PAL and

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pheP (squiggle above “PAL” and “pheP”). Arrows adjacent to a single rectangle, or a cluster of rectangles, depict the promoter responsible for driving transcription (in the direction of the arrow) of such gene(s). Arrows above each rectangle depict the expression product of each gene.

FIG. 18 depicts β -galactosidase levels in samples comprising bacteria harboring a low-copy plasmid expressing lacZ from an FNR-responsive promoter selected from the exemplary FNR promoters shown Table 3 (P_{fnr1-5}). Different FNR-responsive promoters were used to create a library of anaerobic-inducible reporters with a variety of expression levels and dynamic ranges. These promoters included strong ribosome binding sites. Bacterial cultures were grown in either aerobic (+O₂) or anaerobic conditions (−O₂). Samples were removed at 4 hrs and the promoter activity based on β -galactosidase levels was analyzed by performing standard β -galactosidase colorimetric assays.

FIG. 19A depicts a schematic representation of the lacZ gene under the control of an exemplary FNR promoter (P_{fmrS}). LacZ encodes the β -galactosidase enzyme and is a common reporter gene in bacteria. FIG. 19B depicts FNR promoter activity as a function of β -galactosidase activity in SYN-PKU904. SYN-PKU904, an engineered bacterial strain harboring a low-copy *fmrS-lacZ* fusion gene, was grown in the presence or absence of oxygen. Values for standard β -galactosidase colorimetric assays are expressed in Miller units (Miller, 1972). These data suggest that the *fmrS* promoter begins to drive high-level gene expression within 1 hr. under anaerobic conditions. FIG. 19C depicts the growth of bacterial cell cultures expressing lacZ over time, both in the presence and absence of oxygen.

FIGS. 20A and 20B depict phenylalanine levels produced under aerobic (FIG. 20A) or anaerobic conditions (FIG. 20B) in samples of wild-type Nissle, samples of bacteria comprising a low-copy plasmid expressing PAL3 from the Tet promoter or exemplary FNR promoters, or further comprising a copy of pheP driven by the Tet promoter and integrated into the chromosome. Samples were incubated in culture medium supplemented with ATC and 4 mM (660,000 ng/mL) of phenylalanine. Samples were removed at 0 hrs, 2 hrs, 4 hrs, and 24 hrs. Phenylalanine concentration was determined by mass spectrometry. These data suggest that the FNR-responsive *fmrS* promoter is as effective at activating PAL3 expression as a tetracycline-inducible promoter under anaerobic conditions.

FIG. 21 depicts phenylalanine concentrations in cultures of synthetic probiotic strains, with and without an additional copy of pheP inserted on the chromosome. After 1.5 hrs of growth, cultures were placed in Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂. After 4 hrs of induction, bacteria were resuspended in assay buffer containing 4 mM phenylalanine. Aliquots were removed from cell assays every 30 min for 3 hrs for phenylalanine quantification by mass spectrometry. Phenylalanine degradation rates in strains comprising an additional copy of pheP (SYN-PKU304 and SYN-PKU305; left) were higher than strains lacking an additional copy of pheP (SYN-PKU308 and SYN-PKU307; right).

FIG. 22 depicts trans-cinnamate concentrations (PAL activity) for strains comprising single PAL3 insertions at various locations on the chromosome.

FIG. 23 depicts trans-cinnamate concentrations (PAL activity) for strains comprising multiple PAL3 insertions at various locations on the chromosome.

FIG. 24 depicts phenylalanine concentrations in cultures of synthetic probiotic strain SYN-PKU511 over time. After 2.5 hrs of growth, cultures were placed in Coy anaerobic

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chamber supplying 90% N₂, 5% CO₂, and 5% H₂. After 3.5 hrs of induction in phenylalanine containing medium, whole cell extracts were prepared every 30 min for 3 hrs and phenylalanine was quantified by mass spectrometry. SYN-PKU511 comprises 5 integrated copies of an anaerobically (FNR) controlled gene encoding phenylalanine ammonia lyase (PAL) at 5 chromosomal locations and an anaerobically controlled gene encoding a high affinity Phe transporter (pheP) integrated in the lacZ locus.

FIGS. 25A and 25B depict phenylalanine concentrations in cultures of a synthetic probiotic strain, SYN-PKU401, which comprises a high copy pUC57-plasmid with LAAD driven by a Tet inducible promoter, cells were grown in flasks shaking at 37 C, and induced with TCA at early log phase for a duration of 2 hours. Cells were spun down and re-suspended in assay buffer containing phenylalanine. Cells were measured at various cell concentrations and at varying oxygen levels. Cells were either incubated aerobically (1 ml) in a 14 ml culture tube, shaking at 250 rpm. For microaerobic conditions, cells (1 ml) were incubated in a 1.7 ml conical tube without shaking. Cells were incubated anaerobically in a Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂. Aliquots were removed from cell assays every 30 min for 2 hrs for phenylalanine quantification by mass spectrometry. FIG. 25A depicts phenylalanine concentrations under aerobic conditions using two cell densities. A and B are duplicates under the same experimental conditions. The activity in aerobic conditions is ~50 umol/hr./1e9 cells. FIG. 25B depicts phenylalanine concentrations of aerobically, microaerobically, or anaerobically grown cells.

FIG. 26A shows phenylalanine concentrations before and after feeding in an in vivo mouse model of PKU. At the beginning of the study, homozygous BTBR-Pah^{enu2} mice were given water supplemented with 100 micrograms/mL ATC and 5% sucrose. Mice were fasted by removing chow overnight (10 hrs), and blood samples were collected by mandibular bleeding the next morning in order to determine baseline phenylalanine levels. Mice were given chow again, gavaged with 100 microliters (5×10⁹ CFU) of bacteria (SYN-PKU302 or control Nissle) after 1 hr., and allowed to feed for another 2 hrs. Serum phenylalanine concentrations were determined 2 hrs post-gavage. FIG. 26B shows the percent (%) change in blood phenylalanine concentrations before and after feeding as a male or female group average (p<0.01).

FIGS. 27A and 27B depict blood phenylalanine concentrations relative to baseline following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with 200 μ L of H₂O (n=30), SYN-PKU901 (n=33), or SYN-PKU303 (n=34) at 30 and 90 minutes post-phenylalanine injection (0.1 mg/gram of average group body weight). FIGS. 27A and 27B show blood phenylalanine concentrations at 2 hrs and 4 hrs post-phenylalanine injection, respectively. These data indicate that oral administration of the engineered probiotic strain SYN-PKU303 significantly reduces blood phenylalanine levels in mice, compared to mice administered mock treatment (H₂O) or the parental strain (SYN-PKU901) (*, p<0.05; ***, p<0.001; ****, p<0.00001). SYN-PKU303 is capable of intercepting enterorecirculating phenylalanine.

FIG. 28 depicts blood phenylalanine concentrations relative to baseline following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with 200 μ L of H₂O (n=30), SYN-PKU901 (n=33), SYN-PKU303 (n=34), or SYN-PKU304 (n=34) at 30 and 90 minutes post-phenylalanine injection (0.1 mg/gram of average group body weight). Blood phenylalanine concentra-

tions post phenylalanine injection indicate that SYN-PKU304 (low copy plasmid containing *fmrS*-PAL) is at least as effective as SYN-PKU303 (high copy plasmid containing Tet-PAL) in reducing circulating Phe levels in the entero-recirculation model.

FIGS. 29A and 29B depict blood phenylalanine concentrations relative to baseline following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with H₂O, SYN-PKU901, SYN-PKU303, or SYN-PKU304 at 30 and 90 minutes post-phenylalanine injection (0.1 mg/gram of average group body weight). FIGS. 29A and 29B show blood phenylalanine concentrations at 2 hrs and 4 hrs post-phenylalanine injection, respectively. These data indicate that oral administration of engineered probiotic strains SYN-PKU303 and SYN-PKU304 significantly reduces blood phenylalanine levels in mice compared to mice administered mock treatment (H₂O) or the parental strain (SYN-PKU901) (*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$). FIGS. 29C and 29D depict scatter plots of the data shown in FIGS. 29A and 29B.

FIGS. 30A and 30B depict blood phenylalanine concentrations relative to baseline following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with 200 μ L of H₂O (n=12), 200 μ L of SYN-PKU901 (n=12), or 100, 200, or 400 μ L of SYN-PKU304 (n=12 in each dose group) at 30 and 90 minutes post-phenylalanine injection (0.1 mg/gram of average group body weight). FIGS. 30A and 30B show a dose-dependent decrease in blood phenylalanine levels in SYN-PKU304-treated mice compared to mice administered mock treatment (H₂O) or the parental strain (SYN-PKU901) (*30% decrease; $p < 0.05$). This experiment represents one of eight studies of this same design, and each one shows that SYN-PKU304 is capable of intercepting enterorecirculating phenylalanine.

FIGS. 31A and 31B depicts a schematic of PKU specific and PAL specific phenylalanine metabolites. FIG. 31A depicts a schematic of the conversion of phenylalanine to phenylpyruvic acid and phenyllactic acid in the absence of functional PAH. FIG. 31B depicts a schematic of the conversion of phenylalanine to trans-cinnamic acid by PAL3, which is further metabolized to hippuric acid by liver enzymes. These metabolites can be detected by mass spectrometry as described in Examples 24-26 or by other means.

FIGS. 32A, 32B, 32C, 32D, 32E, and 32F depict blood phenylalanine concentrations relative to baseline and concentrations of phenylalanine (FIG. 32A), and absolute values of phenylalanine and PKU specific and PAL specific metabolites (FIGS. 32B, 32C, 32D, 32E, and 32F) following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with a total of 800 μ L of H₂O (n=12), SYN-PKU901 (n=12), or 800 μ L of SYN-PKU304 (n=12) (2.9×10^8 cfu/mouse) at 30 and 90 minutes post-phenylalanine injection. FIG. 32A depicts blood phenylalanine concentrations relative to baseline; total metabolic activity for SYN-PKU304 was calculated as 81.2 μ mol/hr. and the total reduction in Δ phe was 45% relative to SYN-PKU901 ($P < 0.05$). FIG. 32B depicts the blood phenylalanine concentration at 0 and 4 hours post phenylalanine injection. FIG. 32C depicts the blood phenylpyruvate concentration at 0 and 4 hours post phenylalanine injection. FIG. 32D depicts the blood phenyllactate concentration at 0 and 4 hours post phenylalanine injection. FIG. 32E depicts the blood t-cinnamic acid concentration at 0 and 4 hours post phenylalanine injection. FIG. 32F depicts the blood hippuric acid concentration at 0 and 4 hours post phenylalanine injection.

FIGS. 33A, 33B, 33C, 33D, 33E, and 33F depict blood phenylalanine concentrations relative to baseline and concentrations of phenylalanine (FIG. 33A), and absolute values of phenylalanine and PKU specific and PAL specific metabolites (FIGS. 33B, 33C, 33D, 33E, and 33F) following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with a total of 800 μ L of H₂O (n=9), SYN-PKU801 (n=12), or 800 μ L of SYN-PKU517 (n=12) (3.6×10^8 cfu/mouse) at 30 and 90 minutes post-phenylalanine injection. FIG. 33A depicts blood phenylalanine concentrations relative to baseline; total metabolic activity for SYN-PKU517 was calculated as 39.6 μ mol/hr. and the total reduction in Δ phe was 17% relative to SYN-PKU801 ($P < 0.05$). FIG. 33B depicts the blood phenylalanine concentration at 0 and 4 hours post phenylalanine injection. FIG. 33C depicts the blood phenylpyruvate concentration at 0 and 4 hours post phenylalanine injection. FIG. 33D depicts the blood phenyllactate concentration at 0 and 4 hours post phenylalanine injection. FIG. 33E depicts the blood t-cinnamic acid concentration at 0 and 4 hours post phenylalanine injection. FIG. 33F depicts the blood hippuric acid concentration at 0 and 4 hours post phenylalanine injection.

FIGS. 34A, 34B, 34C, 34D, 34E, and 34F depict blood phenylalanine concentrations relative to baseline and concentrations of phenylalanine (FIG. 34A), and absolute values of phenylalanine and PKU specific and PAL specific metabolites (FIGS. 34B, 34C, 34D, 34E, and 34F) following subcutaneous phenylalanine challenge in an in vivo mouse model of PKU. Mice were orally gavaged with a total of 800 μ L of H₂O (n=12), SYN-PKU901 (n=12), or 800 μ L of SYN-PKU705 (n=12) (3.6×10^8 cfu/mouse) at 30 and 90 minutes post-phenylalanine injection. FIG. 34A depicts blood phenylalanine concentrations relative to baseline; total metabolic activity for SYN-PKU705 was calculated as 133.2 μ mol/hr. and the total reduction in Δ phe was 30% relative to SYN-PKU901 ($P < 0.05$). FIG. 34B depicts the blood phenylalanine concentration at 0 and 4 hours post phenylalanine injection. FIG. 34C depicts the blood phenylpyruvate concentration at 0 and 4 hours post phenylalanine injection. FIG. 34D depicts the blood phenyllactate concentration at 0 and 4 hours post phenylalanine injection. FIG. 34E depicts the blood t-cinnamic acid concentration at 0 and 4 hours post phenylalanine injection. FIG. 34F depicts the blood hippuric acid concentration at 0 and 4 hours post phenylalanine injection.

FIG. 35 depicts phenylalanine and 2 toxic analogs, p-fluoro-DL-phenylalanine, and o-fluoro-DL-phenylalanine, which are useful for an untargeted approach to select PAL enzymes with increased activity. P-fluoro-DL-phenylalanine, and o-fluoro-DL-phenylalanine are incorporated into cellular protein in the place of phenylalanine, resulting in cell death. Since these compounds are readily taken up by PheP, and can act as a substrate for PAL as shown below, they can be employed in genetic selection and screening for the identification of strains with improved Phe consumption activity. Mutations allowing more efficient PAL metabolism may prevent the incorporation of the phenylalanine analog into cellular protein, therefore allowing growth under higher concentrations of the analog.

FIG. 36 depicts a map of exemplary integration sites within the *E. coli* 1917 Nissle chromosome. These sites indicate regions where circuit components may be inserted into the chromosome without interfering with essential gene expression. Backslashes (/) are used to show that the insertion will occur between divergently or convergently expressed genes. Insertions within biosynthetic genes, such

as *thyA*, can be useful for creating nutrient auxotrophies. In some embodiments, an individual circuit component is inserted into more than one of the indicated sites.

FIG. 37 depicts three bacterial strains which constitutively express red fluorescent protein (RFP). In strains 1-3, the *rfp* gene has been inserted into different sites within the bacterial chromosome, and results in varying degrees of brightness under fluorescent light. Unmodified *E. coli* Nissle (strain 4) is non-fluorescent.

FIG. 38 depicts a graph of Nissle residence in vivo. Streptomycin-resistant Nissle was administered to mice via oral gavage without antibiotic pre-treatment. Fecal pellets from 6 total mice were monitored post-administration to determine the amount of administered Nissle still residing within the mouse gastrointestinal tract. The bars represent the number of bacteria administered to the mice. The line represents the number of Nissle recovered from the fecal samples each day for 10 consecutive days.

FIGS. 39A and 39B depict phenylalanine concentrations in SYN-PKU302 cultures over time. After 1.5 hrs of growth, ATC was added to cultures of SYN-PKU302, and SYN-PKU304 cultures were placed in Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂. After 4 hrs of induction, bacteria were resuspended in assay buffer containing 4 mM phenylalanine and at different pH (pH range 7.25-2.25). Aliquots were removed from cell assays every 30 min for 2 hrs for phenylalanine quantification by mass spectrometry. Phenylalanine degradation rates decreased as pH of the assay buffer decreased in both strains, SYN-PKU302 (FIG. 39A) and SYN-PKU304 (FIG. 39B).

FIG. 40 depicts an exemplary schematic of the *E. coli* 1917 Nissle chromosome comprising multiple mechanisms of action (MoAs).

FIG. 41 depicts the gene organization of an exemplary construct in which the PAL3 and *pheP* genes are co-transcribed under the control of an exemplary FNR promoter (*P_{fmrS}*).

FIGS. 42A and 42B depict the gene organization of an exemplary construct in which the *Int5* recombinase gene is operably linked to an exemplary FNR promoter (*P_{fmrS}*), and the PAL3 gene is operably linked to a strong constitutive promoter. FIG. 42A depicts a schematic diagram of the PAL3 gene, flanked by *Int5* sites, in the OFF orientation (3' to 5'). When *Int5* gene expression is activated under anaerobic conditions, recombinatorial flipping of PAL3 to the ON orientation (5' to 3'; FIG. 42B) leads to the production of PAL3 and to phenylalanine metabolism. Any strong constitutive promoter sequence may be used.

FIGS. 43A, 43B, and 43C depict the gene organization of an exemplary construct in which the *Int5* recombinase gene is operably linked to an FNR promoter (*P_{fmrS}*), and the gene encoding T7 RNA polymerase is flanked by recombinase sites and operably linked to a strong constitutive promoter. FIG. 43A depicts a schematic diagram of the T7 RNA polymerase gene, flanked by *Int5* sites, in the OFF orientation. When *Int5* gene expression is activated under anaerobic conditions, the T7 RNA polymerase gene is flipped to the ON orientation (FIG. 43B). In engineered bacterial strains comprising a copy of PAL3 under the control of a T7-driven promoter (*P_{T7}*; FIG. 43C), T7 RNA polymerase expression leads to the production of PAL3 and to phenylalanine metabolism.

FIGS. 44A and 44B depict the gene organization of an exemplary construct in which the *Int5* recombinase gene is operably linked to an ParaBAD promoter (ParaBAD), and

the gene encoding T7 RNA polymerase is flanked by recombinase sites and operably linked to a strong constitutive promoter.

FIG. 45A depicts a schematic of a recombinase-based switch to activate PAL3 expression using different inducible promoters and ribosome binding sites. Recombinase expression causes recombinatorial flipping of the PAL3 gene to the ON orientation, leading to the production of PAL3 and to the degradation of phenylalanine. In some embodiments, recombinase-based switches are tuned to respond to specific levels of an inducer. FIG. 45B depicts the relationship between the concentration of an inducer and the percentage of PAL3-containing constructs in the ON orientation. The shaded area shows the predicted efficacy range of the inducer in vivo.

FIG. 46A depicts another non-limiting embodiment of the disclosure, wherein the expression of a heterologous gene is activated by an exogenous environmental signal. In the absence of arabinose, the AraC transcription factor adopts a conformation that represses transcription. In the presence of arabinose, the AraC transcription factor undergoes a conformational change that allows it to bind to and activate the ParaBAD promoter (ParaBAD), which induces expression of the Tet repressor (TetR) and an anti-toxin. The anti-toxin builds up in the recombinant bacterial cell, while TetR prevents expression of a toxin (which is under the control of a promoter having a TetR binding site). However, when arabinose is not present, both the anti-toxin and TetR are not expressed. Since TetR is not present to repress expression of the toxin, the toxin is expressed and kills the cell. FIG. 46A also depicts another non-limiting embodiment of the disclosure, wherein the expression of an essential gene not found in the recombinant bacteria is activated by an exogenous environmental signal. In the absence of arabinose, the AraC transcription factor adopts a conformation that represses transcription of the essential gene under the control of the araBAD promoter and the bacterial cell cannot survive. In the presence of arabinose, the AraC transcription factor undergoes a conformational change that allows it to bind to and activate the araBAD promoter, which induces expression of the essential gene and maintains viability of the bacterial cell.

FIG. 46B depicts a non-limiting embodiment of the disclosure, where an anti-toxin is expressed from a constitutive promoter, and expression of a heterologous gene is activated by an exogenous environmental signal. In the absence of arabinose, the AraC transcription factor adopts a conformation that represses transcription. In the presence of arabinose, the AraC transcription factor undergoes a conformational change that allows it to bind to and activate the araBAD promoter, which induces expression of TetR, thus preventing expression of a toxin. However, when arabinose is not present, TetR is not expressed, and the toxin is expressed, eventually overcoming the anti-toxin and killing the cell. The constitutive promoter regulating expression of the anti-toxin should be a weaker promoter than the promoter driving expression of the toxin. The araC gene is under the control of a constitutive promoter in this circuit.

FIG. 46C depicts another non-limiting embodiment of the disclosure, wherein the expression of a heterologous gene is activated by an exogenous environmental signal. In the absence of arabinose, the AraC transcription factor adopts a conformation that represses transcription. In the presence of arabinose, the AraC transcription factor undergoes a conformational change that allows it to bind to and activate the araBAD promoter, which induces expression of the Tet repressor (TetR) and an anti-toxin. The anti-toxin builds up

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in the recombinant bacterial cell, while TetR prevents expression of a toxin (which is under the control of a promoter having a TetR binding site). However, when arabinose is not present, both the anti-toxin and TetR are not expressed. Since TetR is not present to repress expression of the toxin, the toxin is expressed and kills the cell. The *araC* gene is either under the control of a constitutive promoter or an inducible promoter (e.g., *AraC* promoter) in this circuit.

FIG. 47 depicts the use of GeneGuards as an engineered safety component. All engineered DNA is present on a plasmid which can be conditionally destroyed. See, e.g., Wright et al., 2015.

FIG. 48A depicts a schematic diagram of a wild-type *clbA* construct. FIG. 48B depicts a schematic diagram of a *clbA* knockout construct.

FIG. 49 depicts exemplary sequences of a wild-type *clbA* construct and a *clbA* knockout construct.

FIG. 50 depicts a schematic of a secretion system based on the flagellar type III secretion in which a modified flagellum is used to secrete a therapeutic peptide of interest by recombinantly fusing the peptide to an N-terminal flagellar secretion signal of a native flagellar component so that the intracellularly expressed chimeric peptide can be mobilized across the inner and outer membranes into the surrounding host environment.

FIG. 51 depicts a schematic of a type V secretion system for the extracellular production of recombinant proteins in which a therapeutic peptide (star) can be fused to an N-terminal secretion signal, a linker and the beta-domain of an auto-secreter. In this system, the N-terminal signal sequence directs the protein to the SecA-YEG machinery, which moves the protein across the inner membrane into the periplasm, followed by subsequent cleavage of the signal sequence. The beta-domain is recruited to the Bam complex where the beta-domain is folded and inserted into the outer membrane as a beta-barrel structure. The therapeutic peptide is then threaded through the hollow pore of the beta-barrel structure ahead of the linker sequence. The therapeutic peptide is freed from the linker system by an autocatalytic cleavage or by targeting of a membrane-associated peptidase (scissors) to a complementary protease cut site in the linker.

FIG. 52 depicts a schematic of a type I secretion system, which translocates a passenger peptide directly from the cytoplasm to the extracellular space using HlyB (an ATP-binding cassette secretor; HlyD (a membrane fusion protein); and TolC (an outer membrane protein) which form a channel through both the inner and outer membranes. The secretion signal-containing C-terminal portion of HlyA is fused to the C-terminal portion of a therapeutic peptide (star) to mediate secretion of this peptide.

FIG. 53 depicts a schematic of the outer and inner membranes of a gram-negative bacterium, and several deletion targets for generating a leaky or destabilized outer membrane, thereby facilitating the translocation of a therapeutic polypeptides to the extracellular space, e.g., therapeutic polypeptides of eukaryotic origin containing disulfide bonds. Deactivating mutations of one or more genes encoding a protein that tethers the outer membrane to the peptidoglycan skeleton, e.g., *lpp*, *ompC*, *ompA*, *ompF*, *tolA*, *tolB*, *pal*, and/or one or more genes encoding a periplasmic protease, e.g., *degS*, *degP*, *nlpl*, generates a leaky phenotype. Combinations of mutations may synergistically enhance the leaky phenotype.

FIG. 54 depicts a schematic of non-limiting processes for designing and producing the genetically engineered bacteria of the present disclosure.

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FIGS. 55A, B, C, D, and E depict a schematic of non-limiting manufacturing processes for upstream and downstream production of the genetically engineered bacteria of the present disclosure. FIG. 55A depicts the parameters for starter culture 1 (SC1): loop full—glycerol stock, duration overnight, temperature 37° C., shaking at 250 rpm. FIG. 55B depicts the parameters for starter culture 2 (SC2): 1/100 dilution from SC1, duration 1.5 hours, temperature 37° C., shaking at 250 rpm. FIG. 55C depicts the parameters for the production bioreactor: inoculum—SC2, temperature 37° C., pH set point 7.00, pH dead band 0.05, dissolved oxygen set point 50%, dissolved oxygen cascade agitation/gas FLO, agitation limits 300-1200 rpm, gas FLO limits 0.5-20 standard liters per minute, duration 24 hours. FIG. 55D depicts the parameters for harvest: centrifugation at speed 4000 rpm and duration 30 minutes, wash 1×10% glycerol/PBS, centrifugation, re-suspension 10% glycerol/PBS. FIG. 55E depicts the parameters for vial fill/storage: 1-2 mL aliquots, -80° C.

DESCRIPTION OF EMBODIMENTS

The present disclosure includes genetically engineered bacteria, pharmaceutical compositions thereof, and methods of modulating and treating disorders associated with hyperphenylalaninemia. In some embodiments, the genetically engineered bacteria comprise a gene encoding non-native phenylalanine ammonia lyase (PAL) and are capable of processing and reducing phenylalanine in a mammal. Thus, the genetically engineered bacteria and pharmaceutical compositions comprising those bacteria may be used to metabolize phenylalanine in the body into non-toxic molecules in order to treat and/or prevent conditions associated with hyperphenylalaninemia, including PKU. In certain aspects, the compositions comprising the genetically engineered bacteria may be used in the methods of the disclosure to treat and/or prevent disorders associated with hyperphenylalaninemia.

In order that the disclosure may be more readily understood, certain terms are first defined. These definitions should be read in light of the remainder of the disclosure and as understood by a person of ordinary skill in the art. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by a person of ordinary skill in the art. Additional definitions are set forth throughout the detailed description.

“Hyperphenylalaninemia,” “hyperphenylalaninemic,” and “excess phenylalanine” are used interchangeably herein to refer to increased or abnormally high concentrations of phenylalanine in the body. In some embodiments, a diagnostic signal of hyperphenylalaninemia is a blood phenylalanine level of at least 2 mg/dL, at least 4 mg/dL, at least 6 mg/dL, at least 8 mg/dL, at least 10 mg/dL, at least 12 mg/dL, at least 14 mg/dL, at least 16 mg/dL, at least 18 mg/dL, at least 20 mg/dL, or at least 25 mg/dL. As used herein, diseases associated with hyperphenylalaninemia include, but are not limited to, phenylketonuria, classical or typical phenylketonuria, atypical phenylketonuria, permanent mild hyperphenylalaninemia, nonphenylketonuric hyperphenylalaninemia, phenylalanine hydroxylase deficiency, cofactor deficiency, dihydropteridine reductase deficiency, tetrahydropterin synthase deficiency, and Segawa's disease. Affected individuals can suffer progressive and irreversible neurological deficits, mental retardation, encephalopathy, epilepsy, eczema, reduced growth, microcephaly, tremor, limb spasticity, and/or hypopigmentation

(Leonard 2006). Hyperphenylalaninemia can also be secondary to other conditions, e.g., liver diseases.

“Phenylalanine ammonia lyase” and “PAL” are used to refer to a phenylalanine metabolizing enzyme (PME) that converts or processes phenylalanine to trans-cinnamic acid and ammonia. Trans-cinnamic acid has low toxicity and is converted by liver enzymes in mammals to hippuric acid, which is secreted in the urine. PAL may be substituted for the enzyme PAH to metabolize excess phenylalanine. PAL enzyme activity does not require THB cofactor activity. In some embodiments, PAL is encoded by a PAL gene derived from a prokaryotic species. In alternate embodiments, PAL is encoded by a PAL gene derived from a eukaryotic species. In some embodiments, PAL is encoded by a PAL gene derived from a bacterial species, including but not limited to, *Achromobacter xylosoxidans*, *Pseudomonas aeruginosa*, *Photobacterium luminescens*, *Anabaena variabilis*, and *Agrobacterium tumefaciens*. In some embodiments, PAL is encoded by a PAL gene derived from *Anabaena variabilis* and referred to as “PAL1” herein (Moffitt et al., 2007). In some embodiments, PAL is encoded by a PAL gene derived from *Photobacterium luminescens* and referred to as “PAL3” herein (Williams et al., 2005). In some embodiments, PAL is encoded by a PAL gene derived from a yeast species, e.g., *Rhodospiridium toruloides* (Gilbert et al., 1985). In some embodiments, PAL is encoded by a PAL gene derived from a plant species, e.g., *Arabidopsis thaliana* (Wanner et al., 1995). Any suitable nucleotide and amino acid sequences of PAL, or functional fragments thereof, may be used.

“Phenylalanine hydroxylase” and “PAH” are used to refer to an enzyme that catalyzes the hydroxylation of the aromatic side chain of phenylalanine to create tyrosine in the human body in conjunction with the cofactor tetrahydrobiopterin. The human gene encoding PAH is located on the long (q) arm of chromosome 12 between positions 22 and 24.2. The amino acid sequence of PAH is highly conserved among mammals. Nucleic acid sequences for human and mammalian PAH are well known and widely available. The full-length human cDNA sequence for PAH was reported in 1985 (Kwok et al. 1985). Active fragments of PAH are also well known (e.g., Kobe et al. 1997).

“L-Amino acid Deaminase” and “LAAD” are used to refer to an enzyme that catalyzes the stereospecific oxidative deamination of L-amino acids to generate their respective keto acids, ammonia, and hydrogen peroxide. For example, LAAD catalyzes the conversion of phenylalanine to phenylpyruvate. Multiple LAAD enzymes are known in the art, many of which are derived from bacteria, such as *Proteus*, *Providencia*, and *Morganella*, or venom. LAAD is characterized by fast reaction rate of phenylalanine degradation (Hou et al., Appl Microbiol Technol. 2015 October; 99(20): 8391-402; “Production of phenylpyruvic acid from L-phenylalanine using an L-amino acid deaminase from *Proteus mirabilis*: comparison of enzymatic and whole-cell biotransformation approaches”). Most eukaryotic and prokaryotic L-amino acid deaminases are extracellular; however, *Proteus* species LAAD are localized to the plasma membrane (inner membrane), facing outward into the periplasmic space, in which the enzymatic activity resides. As a consequence of this localization, phenylalanine transport through the inner membrane into the cytoplasm is not required for *Proteus* LAAD mediated phenylalanine degradation. Phenylalanine is readily taken up through the outer membrane into the periplasm without a transporter, eliminating the need for a transporter to improve substrate availability.

In some embodiments, the genetically engineered bacteria comprise a LAAD gene derived from a bacterial species,

including but not limited to, *Proteus*, *Providencia*, and *Morganella* bacteria. In some embodiments, the bacterial species is *Proteus mirabilis*. In some embodiments, the bacterial species is *Proteus vulgaris*. In some embodiments, the LAAD encoded by the genetically engineered bacteria is localized to the plasma membrane, facing into the periplasmic space and with the catalytic activity occurring in the periplasmic space.

“Phenylalanine metabolizing enzyme” or “PME” are used to refer to an enzyme which is able to degrade phenylalanine. Any phenylalanine metabolizing enzyme known in the art may be encoded by the genetically engineered bacteria. PME includes, but are not limited to, phenylalanine hydroxylase (PAH), phenylalanine ammonia lyase (PAL), aminotransferase, L-amino acid deaminase (L-AAD), and phenylalanine dehydrogenases.

Reactions with phenylalanine hydroxylases, phenylalanine dehydrogenases or aminotransferases require cofactors, while L-AAD and PAL do not require any additional cofactors. In some embodiments, the PME encoded by the genetically engineered bacteria requires a cofactor. In some embodiments, this cofactor is provided concurrently or sequentially with the administration of the genetically engineered bacteria. In other embodiments, the genetically engineered bacteria can produce the cofactor. In some embodiments, the genetically engineered bacteria encode a phenylalanine hydroxylase. In some embodiments, the genetically engineered bacteria encode a phenylalanine dehydrogenase. In some embodiments, the genetically engineered bacteria encode an aminotransferase. In some embodiments, the PME encoded by the genetically engineered bacteria does not require a cofactor. Without wishing to be bound by theory, the lack of need for a cofactor means that the rate of phenylalanine degradation by the enzyme is dependent on the availability of the substrate and is not limited by the availability of the cofactor. In some embodiments, the PME produced by the genetically engineered bacteria is PAL. In some embodiments, the PME produced by the genetically engineered bacteria is LAAD. In some embodiments, the genetically engineered bacteria encode combinations of PMEs.

In some embodiments, the catalytic activity of the PME is dependent on oxygen levels. In some embodiments, the PME is catalytically active under microaerobic conditions. As a non-limiting example, LAAD catalytic activity is dependent on oxygen. In some embodiments, LAAD is active under low oxygen conditions, such as microaerobic conditions. In some embodiments, of the invention, the PME functions at very low levels of oxygen or in the absence of oxygen, e.g. as found in the colon. As a non-limiting example, PAL activity is not dependent on the presence of oxygen.

In certain embodiments, new or improved PMEs can be identified according to methods known in the art or described herein, and are encoded by the genetically engineered bacteria. In some embodiments, the enzyme encoded by the genetically engineered bacteria is a wild type enzyme isolated from a viral, prokaryotic or eukaryotic organism. In some embodiments, the enzyme sequence has been further modified or mutated to increase one or more specific properties of the enzyme, such as stability or catalytic activity.

“Phenylalanine metabolite” refers to a metabolite that is generated as a result of the degradation of phenylalanine. The metabolite may be generated directly from phenylalanine, by the enzyme using phenylalanine as a substrate, or indirectly by a different enzyme downstream in the metabolic pathway, which acts on a phenylalanine metabolite

substrate. In some embodiments, phenylalanine metabolites are produced by the genetically engineered bacteria encoding a PME.

In some embodiments, the phenylalanine metabolite results directly or indirectly from PAH activity, e.g., from PAH produced by the genetically engineered bacteria. In some embodiments, the metabolite is tyrosine. In some embodiments, the phenylalanine metabolite accumulates in the blood or the urine of a PKU patient, due to defective PAH activity. Non-limiting examples of such PKU metabolites are phenylpyruvic acid and phenyl-lactic acid. Other examples include phenylacetate, phenylethylamine, and phenylacetyl glutamine.

In some embodiments, the phenylalanine metabolite results directly or indirectly from PAL action, e.g., from PAL produced by the genetically engineered bacteria. Non-limiting examples of such PAL metabolites are trans-cinnamic acid and hippuric acid. In some embodiments, the phenylalanine metabolite results directly or indirectly from LAAD action, e.g., from LAAD produced by the genetically engineered bacteria. Examples of such LAAD metabolites are phenylpyruvate and phenyllactic acid.

"Phenylalanine transporter" is used to refer to a membrane transport protein that is capable of transporting phenylalanine into bacterial cells (see, e.g., Pi et al., 1991). In *Escherichia coli*, the pheP gene encodes a high affinity phenylalanine-specific permease responsible for phenylalanine transport (Pi et al., 1998). In some embodiments, the phenylalanine transporter is encoded by apheP gene derived from a bacterial species, including but not limited to, *Acinetobacter calcoaceticus*, *Salmonella enterica*, and *Escherichia coli*. Other phenylalanine transporters include Aagenal amino acid permease, encoded by the aroP gene, transports three aromatic amino acids, including phenylalanine, with high affinity, and is thought, together with PheP, responsible for the lion share of phenylalanine import. Additionally, a low level of phenylalanine transport activity has been traced to the activity of the LIV-I/LS system, which is a branched-chain amino acid transporter consisting of two periplasmic binding proteins, the LIV-binding protein (LIV-I system) and LS-binding protein (LS system), and membrane components, LivHMGF. In some embodiments, the phenylalanine transporter is encoded by a aroP gene derived from a bacterial species. In some embodiments, the phenylalanine transporter is encoded by LIV-binding protein and LS-binding protein and LivHMGF genes derived from a bacterial species. In some embodiments, the genetically engineered bacteria comprise more than one type of phenylalanine transporter, selected from pheP, aroP, and the LIV-I/LS system.

"Phenylalanine" and "Phe" are used to refer to an amino acid with the formula $C_6H_5CH_2CH(NH_2)COOH$. Phenylalanine is a precursor for tyrosine, dopamine, norepinephrine, and epinephrine. L-phenylalanine is an essential amino acid and the form of phenylalanine primarily found in dietary protein; the stereoisomer D-phenylalanine is found in lower amounts in dietary protein; DL-phenylalanine is a combination of both forms. Phenylalanine may refer to one or more of L-phenylalanine, D-phenylalanine, and DL-phenylalanine.

"Operably linked" refers a nucleic acid sequence, e.g., a gene encoding PAL, that is joined to a regulatory region sequence in a manner which allows expression of the nucleic acid sequence, e.g., acts in cis. A regulatory region is a nucleic acid that can direct transcription of a gene of interest and may comprise promoter sequences, enhancer sequences, response elements, protein recognition sites, inducible ele-

ments, promoter control elements, protein binding sequences, 5' and 3' untranslated regions, transcriptional start sites, termination sequences, polyadenylation sequences, and introns.

An "inducible promoter" refers to a regulatory region that is operably linked to one or more genes, wherein expression of the gene(s) is increased in the presence of an inducer of said regulatory region.

A "directly inducible promoter" refers to a regulatory region, wherein the regulatory region is operably linked to a gene encoding a phenylalanine-metabolizing enzyme, e.g., PAL; in the presence of an inducer of said regulatory region, the phenylalanine-metabolizing enzyme is expressed. An "indirectly inducible promoter" refers to a regulatory system comprising two or more regulatory regions, for example, a first regulatory region that is operably linked to a gene encoding a first molecule, e.g., a transcriptional regulator, which is capable of regulating a second regulatory region that is operably linked to a gene encoding a phenylalanine-metabolizing enzyme. In the presence of an inducer of the first regulatory region, the second regulatory region may be activated or repressed, thereby activating or repressing expression of the phenylalanine-metabolizing enzyme. Both a directly inducible promoter and an indirectly inducible promoter are encompassed by "inducible promoter."

"Exogenous environmental conditions" refer to settings or circumstances under which the promoter described above is directly or indirectly induced. In some embodiments, the exogenous environmental conditions are specific to the gut of a mammal. In some embodiments, the exogenous environmental conditions are specific to the upper gastrointestinal tract of a mammal. In some embodiments, the exogenous environmental conditions are specific to the lower gastrointestinal tract of a mammal. In some embodiments, the exogenous environmental conditions are specific to the small intestine of a mammal. In some embodiments, exogenous environmental conditions refer to the presence of molecules or metabolites that are specific to the mammalian gut in a healthy or disease state, e.g., propionate. In some embodiments, the exogenous environmental conditions are low-oxygen, microaerobic, or anaerobic conditions, such as the environment of the mammalian gut.

"Exogenous environmental condition(s)" refer to setting(s) or circumstance(s) under which the promoter described herein is induced. The phrase "exogenous environmental conditions" is meant to refer to the environmental conditions external to the engineered microorganism, but endogenous or native to the host subject environment. Thus, "exogenous" and "endogenous" may be used interchangeably to refer to environmental conditions in which the environmental conditions are endogenous to a mammalian body, but external or exogenous to an intact microorganism cell. In some embodiments, the exogenous environmental conditions are specific to the gut of a mammal. In some embodiments, the exogenous environmental conditions are specific to the upper gastrointestinal tract of a mammal. In some embodiments, the exogenous environmental conditions are specific to the lower gastrointestinal tract of a mammal. In some embodiments, the exogenous environmental conditions are specific to the small intestine of a mammal. In some embodiments, the exogenous environmental conditions are low-oxygen, microaerobic, or anaerobic conditions, such as the environment of the mammalian gut. In some embodiments, exogenous environmental conditions are molecules or metabolites that are specific to the mammalian gut, e.g., propionate. In some embodiments, the exogenous environmental condition is a tissue-specific or

disease-specific metabolite or molecule(s). In some embodiments, the exogenous environmental condition is a low-pH environment. In some embodiments, the genetically engineered microorganism of the disclosure comprises a pH-dependent promoter. In some embodiments, the genetically engineered microorganism of the disclosure comprises an oxygen level-dependent promoter. In some aspects, bacteria have evolved transcription factors that are capable of sensing oxygen levels. Different signaling pathways may be triggered by different oxygen levels and occur with different kinetics.

An “oxygen level-dependent promoter” or “oxygen level-dependent regulatory region” refers to a nucleic acid sequence to which one or more oxygen level-sensing transcription factors is capable of binding, wherein the binding and/or activation of the corresponding transcription factor activates downstream gene expression.

Examples of oxygen level-dependent transcription factors include, but are not limited to, FNR, ANR, and DNR. Corresponding FNR-responsive promoters, ANR-responsive promoters, and DNR-responsive promoters are known in the art (see, e.g., Castiglione et al., 2009; Eiglmeier et al., 1989; Galimand et al., 1991; Hasegawa et al., 1998; Hoeren et al., 1993; Salmon et al., 2003). Non-limiting examples are shown in Table 1.

In a non-limiting example, a promoter (PfnrS) was derived from the *E. coli* Nissle fumarate and nitrate reductase gene S (fnrS) that is known to be highly expressed under conditions of low or no environmental oxygen (Durand and Storz, 2010; Boysen et al., 2010). The PfnrS promoter is activated under anaerobic conditions by the global transcriptional regulator FNR that is naturally found in Nissle. Under anaerobic conditions, FNR forms a dimer and binds to specific sequences in the promoters of specific genes under its control, thereby activating their expression. However, under aerobic conditions, oxygen reacts with iron-sulfur clusters in FNR dimers and converts them to an inactive form. In this way, the PfnrS inducible promoter is adopted to modulate the expression of proteins or RNA. PfnrS is used interchangeably in this application as FNRS, fnrS, FNR, P-FNRS promoter and other such related designations to indicate the promoter PfnrS.

TABLE 1

Examples of transcription factors and responsive genes and regulatory regions	
Transcription factor	Examples of responsive genes, promoters, and/or regulatory regions:
FNR	nirB, ydfZ, pdhR, focA, ndH, hlyE, narK, narX, narG, yfiD, tdcD
ANR	arcDABC
DNR	norB, norC

As used herein, a “non-native” nucleic acid sequence refers to a nucleic acid sequence not normally present in a bacterium, e.g., an extra copy of an endogenous sequence, or a heterologous sequence such as a sequence from a different species, strain, or substrain of bacteria, or a sequence that is modified and/or mutated as compared to the unmodified sequence from bacteria of the same subtype. In some embodiments, the non-native nucleic acid sequence is a synthetic, non-naturally occurring sequence (see, e.g., Purcell et al., 2013). The non-native nucleic acid sequence may be a regulatory region, a promoter, a gene, and/or one or

more genes in a gene cassette. In some embodiments, “non-native” refers to two or more nucleic acid sequences that are not found in the same relationship to each other in nature. The non-native nucleic acid sequence may be present on a plasmid or chromosome. In addition, multiple copies of any regulatory region, promoter, gene, and/or gene cassette may be present in the bacterium, wherein one or more copies of the regulatory region, promoter, gene, and/or gene cassette may be mutated or otherwise altered as described herein. In some embodiments, the genetically engineered bacteria are engineered to comprise multiple copies of the same regulatory region, promoter, gene, and/or gene cassette in order to enhance copy number or to comprise multiple different components of a gene cassette performing multiple different functions. In some embodiments, the genetically engineered bacteria of the invention comprise a gene encoding a phenylalanine-metabolizing enzyme that is operably linked to a directly or indirectly inducible promoter that is not associated with said gene in nature, e.g., an FNR promoter operably linked to a gene encoding PAL or a ParaBAD promoter operably linked to LAAD.

“Constitutive promoter” refers to a promoter that is capable of facilitating continuous transcription of a coding sequence or gene under its control and/or to which it is operably linked. Constitutive promoters and variants are well known in the art and include, but are not limited to, BBa_J23100, a constitutive *Escherichia coli* σ^S promoter (e.g., an osmY promoter (International Genetically Engineered Machine (iGEM) Registry of Standard Biological Parts Name BBa_J45992; BBa_J45993)), a constitutive *Escherichia coli* σ^{32} promoter (e.g., htpG heat shock promoter (BBa_J45504)), a constitutive *Escherichia coli* σ^{70} promoter (e.g., lacq promoter (BBa_J54200; BBa_J56015), *E. coli* CreABCD phosphate sensing operon promoter (BBa_J64951), GlnRS promoter (BBa_K088007), lacZ promoter (BBa_K119000; BBa_K119001); M13K07 gene I promoter (BBa_M13101); M13K07 gene II promoter (BBa_M13102), M13K07 gene III promoter (BBa_M13103), M13K07 gene IV promoter (BBa_M13104), M13K07 gene V promoter (BBa_M13105), M13K07 gene VI promoter (BBa_M13106), M13K07 gene VIII promoter (BBa_M13108), M13110 (BBa_M13110)), a constitutive *Bacillus subtilis* σ^A promoter (e.g., promoter veg (BBa_K143013), promoter 43 (BBa_K143013), PliaG (BBa_K823000), P_{lepA} (BBa_K823002), P_{veg} (BBa_K823003)), a constitutive *Bacillus subtilis* σ^B promoter (e.g., promoter ctc (BBa_K143010), promoter gsiB (BBa_K143011)), a *Salmonella* promoter (e.g., Pspv2 from *Salmonella* (BBa_K112706), Pspv from *Salmonella* (BBa_K112707)), a bacteriophage T7 promoter (e.g., T7 promoter (BBa_I712074; BBa_I719005; BBa_J34814; BBa_J64997; BBa_K113010; BBa_K113011; BBa_K113012; BBa_R0085; BBa_R0180; BBa_R0181; BBa_R0182; BBa_R0183; BBa_Z0251; BBa_Z0252; BBa_Z0253)), a bacteriophage SP6 promoter (e.g., SP6 promoter (BBa_J64998)), and functional fragments thereof.

“Gut” refers to the organs, glands, tracts, and systems that are responsible for the transfer and digestion of food, absorption of nutrients, and excretion of waste. In humans, the gut comprises the gastrointestinal (GI) tract, which starts at the mouth and ends at the anus, and additionally comprises the esophagus, stomach, small intestine, and large intestine. The gut also comprises accessory organs and glands, such as the spleen, liver, gallbladder, and pancreas. The upper gastrointestinal tract comprises the esophagus, stomach, and duodenum of the small intestine. The lower

gastrointestinal tract comprises the remainder of the small intestine, i.e., the jejunum and ileum, and all of the large intestine, i.e., the cecum, colon, rectum, and anal canal. Bacteria can be found throughout the gut, e.g., in the gastrointestinal tract, and particularly in the intestines.

As used herein, the term “gene sequence” is meant to refer to a genetic sequence, e.g., a nucleic acid sequence. The gene sequence or genetic sequence is meant to include a complete gene sequence or a partial gene sequence. The gene sequence or genetic sequence is meant to include sequence that encodes a protein or polypeptide and is also meant to include genetic sequence that does not encode a protein or polypeptide, e.g., a regulatory sequence, leader sequence, signal sequence, or other non-protein coding sequence.

“Microorganism” refers to an organism or microbe of microscopic, submicroscopic, or ultramicroscopic size that typically consists of a single cell. Examples of microorganisms include bacteria, viruses, parasites, fungi, certain algae, and protozoa. In some aspects, the microorganism is engineered (“engineered microorganism”) to produce one or more therapeutic molecules or proteins of interest. In certain aspects, the microorganism is engineered to take up and catabolize certain metabolites or other compounds from its environment, e.g., the gut. In certain aspects, the microorganism is engineered to synthesize certain beneficial metabolites or other compounds (synthetic or naturally occurring) and release them into its environment. In certain embodiments, the engineered microorganism is an engineered bacterium. In certain embodiments, the engineered microorganism is an engineered virus.

“Non-pathogenic bacteria” refer to bacteria that are not capable of causing disease or harmful responses in a host. In some embodiments, non-pathogenic bacteria are Gram-negative bacteria. In some embodiments, non-pathogenic bacteria are Gram-positive bacteria. In some embodiments, non-pathogenic bacteria are commensal bacteria, which are present in the indigenous microbiota of the gut. Examples of non-pathogenic bacteria include, but are not limited to, *Bacillus*, *Bacteroides*, *Bifidobacterium*, *Brevibacterium*, *Clostridium*, *Enterococcus*, *Escherichia*, *Lactobacillus*, *Lactococcus*, *Saccharomyces*, and *Staphylococcus*, e.g., *Bacillus coagulans*, *Bacillus subtilis*, *Bacteroides fragilis*, *Bacteroides subtilis*, *Bacteroides thetaiotaomicron*, *Bifidobacterium bifidum*, *Bifidobacterium infantis*, *Bifidobacterium lactis*, *Bifidobacterium longum*, *Clostridium butyricum*, *Enterococcus faecium*, *Escherichia coli*, *Lactobacillus acidophilus*, *Lactobacillus bulgaricus*, *Lactobacillus casei*, *Lactobacillus johnsonii*, *Lactobacillus paracasei*, *Lactobacillus plantarum*, *Lactobacillus reuteri*, *Lactobacillus rhamnosus*, *Lactococcus lactis*, and *Saccharomyces boulardii* (Sonnenborn et al., 2009; Dinleyici et al., 2014; U.S. Pat. Nos. 6,835,376; 6,203,797; 5,589,168; 7,731,976). Naturally pathogenic bacteria may be genetically engineered to provide reduce or eliminate pathogenicity.

“Probiotic” is used to refer to live, non-pathogenic microorganisms, e.g., bacteria, which can confer health benefits to a host organism that contains an appropriate amount of the microorganism. In some embodiments, the host organism is a mammal. In some embodiments, the host organism is a human. Some species, strains, and/or subtypes of non-pathogenic bacteria are currently recognized as probiotic. Examples of probiotic bacteria include, but are not limited to, *Bifidobacterium*, *Escherichia*, *Lactobacillus*, and *Saccharomyces*, e.g., *Bifidobacterium bifidum*, *Enterococcus faecium*, *Escherichia coli*, *Escherichia coli* strain Nissle, *Lactobacillus acidophilus*, *Lactobacillus bulgaricus*,

Lactobacillus paracasei, *Lactobacillus plantarum*, and *Saccharomyces boulardii* (Dinleyici et al., 2014; U.S. Pat. Nos. 5,589,168; 6,203,797; 6,835,376). The probiotic may be a variant or a mutant strain of bacterium (Arthur et al., 2012; Cuevas-Ramos et al., 2010; Olier et al., 2012; Nougayrede et al., 2006). Non-pathogenic bacteria may be genetically engineered to enhance or improve desired biological properties, e.g., survivability. Non-pathogenic bacteria may be genetically engineered to provide probiotic properties. Probiotic bacteria may be genetically engineered to enhance or improve probiotic properties.

As used herein, “stably maintained” or “stable” bacterium is used to refer to a bacterial host cell carrying non-native genetic material, e.g., a PAL gene, which is incorporated into the host genome or propagated on a self-replicating extra-chromosomal plasmid, such that the non-native genetic material is retained, expressed, and/or propagated. The stable bacterium is capable of survival and/or growth in vitro, e.g., in medium, and/or in vivo, e.g., in the gut. For example, the stable bacterium may be a genetically modified bacterium comprising a PAL gene, in which the plasmid or chromosome carrying the PAL gene is stably maintained in the host cell, such that PAL can be expressed in the host cell, and the host cell is capable of survival and/or growth in vitro and/or in vivo. In some embodiments, copy number affects the stability of expression of the non-native genetic material, e.g., a PAL gene or a PAH gene. In some embodiments, copy number affects the level of expression of the non-native genetic material, e.g., a PAL gene or a PAH gene.

As used herein, the terms “modulate” and “treat” and their cognates refer to an amelioration of a disease, disorder, and/or condition, or at least one discernible symptom thereof. In another embodiment, “modulate” and “treat” refer to an amelioration of at least one measurable physical parameter, not necessarily discernible by the patient. In another embodiment, “modulate” and “treat” refer to inhibiting the progression of a disease, disorder, and/or condition, either physically (e.g., stabilization of a discernible symptom), physiologically (e.g., stabilization of a physical parameter), or both. In another embodiment, “modulate” and “treat” refer to slowing the progression or reversing the progression of a disease, disorder, and/or condition. As used herein, “prevent” and its cognates refer to delaying the onset or reducing the risk of acquiring a given disease, disorder and/or condition or a symptom associated with such disease, disorder, and/or condition.

Those in need of treatment may include individuals already having a particular medical disease, as well as those at risk of having, or who may ultimately acquire the disease. The need for treatment is assessed, for example, by the presence of one or more risk factors associated with the development of a disease, the presence or progression of a disease, or likely receptiveness to treatment of a subject having the disease. Primary hyperphenylalaninemia, e.g., PKU, is caused by inborn genetic mutations for which there are no known cures. Hyperphenylalaninemia can also be secondary to other conditions, e.g., liver diseases. Treating hyperphenylalaninemia may encompass reducing or eliminating excess phenylalanine and/or associated symptoms, and does not necessarily encompass the elimination of the underlying disease.

As used herein a “pharmaceutical composition” refers to a preparation of genetically engineered bacteria of the invention with other components such as a physiologically suitable carrier and/or excipient.

The phrases “physiologically acceptable carrier” and “pharmaceutically acceptable carrier” which may be used

interchangeably refer to a carrier or a diluent that does not cause significant irritation to an organism and does not abrogate the biological activity and properties of the administered bacterial compound. An adjuvant is included under these phrases.

The term "excipient" refers to an inert substance added to a pharmaceutical composition to further facilitate administration of an active ingredient. Examples include, but are not limited to, calcium bicarbonate, calcium phosphate, various sugars and types of starch, cellulose derivatives, gelatin, vegetable oils, polyethylene glycols, and surfactants, including, for example, polysorbate 20.

The terms "therapeutically effective dose" and "therapeutically effective amount" are used to refer to an amount of a compound that results in prevention, delay of onset of symptoms, or amelioration of symptoms of a condition, e.g., hyperphenylalaninemia. A therapeutically effective amount may, for example, be sufficient to treat, prevent, reduce the severity, delay the onset, and/or reduce the risk of occurrence of one or more symptoms of a disease or condition associated with excess phenylalanine levels. A therapeutically effective amount, as well as a therapeutically effective frequency of administration, can be determined by methods known in the art and discussed below.

As used herein, the term "polypeptide" includes "polypeptide" as well as "polypeptides," and refers to a molecule composed of amino acid monomers linearly linked by amide bonds (i.e., peptide bonds). The term "polypeptide" refers to any chain or chains of two or more amino acids, and does not refer to a specific length of the product. Thus, "peptides," "dipeptides," "tripeptides," "oligopeptides," "protein," "amino acid chain," or any other term used to refer to a chain or chains of two or more amino acids, are included within the definition of "polypeptide," and the term "polypeptide" may be used instead of, or interchangeably with any of these terms. The term "dipeptide" refers to a peptide of two linked amino acids. The term "tripeptide" refers to a peptide of three linked amino acids. The term "polypeptide" is also intended to refer to the products of post-expression modifications of the polypeptide, including but not limited to glycosylation, acetylation, phosphorylation, amidation, derivatization, proteolytic cleavage, or modification by non-naturally occurring amino acids. A polypeptide may be derived from a natural biological source or produced by recombinant technology. In other embodiments, the polypeptide is produced by the genetically engineered bacteria or virus of the current invention. A polypeptide of the invention may be of a size of about 3 or more, 5 or more, 10 or more, 20 or more, 25 or more, 50 or more, 75 or more, 100 or more, 200 or more, 500 or more, 1,000 or more, or 2,000 or more amino acids. Polypeptides may have a defined three-dimensional structure, although they do not necessarily have such structure. Polypeptides with a defined three-dimensional structure are referred to as folded, and polypeptides, which do not possess a defined three-dimensional structure, but rather can adopt a large number of different conformations, are referred to as unfolded. The term "peptide" or "polypeptide" may refer to an amino acid sequence that corresponds to a protein or a portion of a protein or may refer to an amino acid sequence that corresponds with non-protein sequence, e.g., a sequence selected from a regulatory peptide sequence, leader peptide sequence, signal peptide sequence, linker peptide sequence, and other peptide sequence.

An "isolated" polypeptide or a fragment, variant, or derivative thereof refers to a polypeptide that is not in its natural milieu. No particular level of purification is required. Recombinantly produced polypeptides and proteins

expressed in host cells, including but not limited to bacterial or mammalian cells, are considered isolated for purposes of the invention, as are native or recombinant polypeptides which have been separated, fractionated, or partially or substantially purified by any suitable technique. Recombinant peptides, polypeptides or proteins refer to peptides, polypeptides or proteins produced by recombinant DNA techniques, i.e. produced from cells, microbial or mammalian, transformed by an exogenous recombinant DNA expression construct encoding the polypeptide. Proteins or peptides expressed in most bacterial cultures will typically be free of glycan. Fragments, derivatives, analogs or variants of the foregoing polypeptides, and any combination thereof are also included as polypeptides. The terms "fragment," "variant," "derivative" and "analog" include polypeptides having an amino acid sequence sufficiently similar to the amino acid sequence of the original peptide and include any polypeptides, which retain at least one or more properties of the corresponding original polypeptide. Fragments of polypeptides of the present invention include proteolytic fragments, as well as deletion fragments. Fragments also include specific antibody or bioactive fragments or immunologically active fragments derived from any polypeptides described herein. Variants may occur naturally or be non-naturally occurring. Non-naturally occurring variants may be produced using mutagenesis methods known in the art. Variant polypeptides may comprise conservative or non-conservative amino acid substitutions, deletions or additions.

Polypeptides also include fusion proteins. As used herein, the term "variant" includes a fusion protein, which comprises a sequence of the original peptide or sufficiently similar to the original peptide. As used herein, the term "fusion protein" refers to a chimeric protein comprising amino acid sequences of two or more different proteins. Typically, fusion proteins result from well known in vitro recombination techniques. Fusion proteins may have a similar structural function (but not necessarily to the same extent), and/or similar regulatory function (but not necessarily to the same extent), and/or similar biochemical function (but not necessarily to the same extent) and/or immunological activity (but not necessarily to the same extent) as the individual original proteins which are the components of the fusion proteins. "Derivatives" include but are not limited to peptides, which contain one or more naturally occurring amino acid derivatives of the twenty standard amino acids, "Similarity" between two peptides is determined by comparing the amino acid sequence of one peptide to the sequence of a second peptide. An amino acid of one peptide is similar to the corresponding amino acid of a second peptide if it is identical or a conservative amino acid substitution. Conservative substitutions include those described in Dayhoff, M. O., ed., *The Atlas of Protein Sequence and Structure* 5, National Biomedical Research Foundation, Washington, D.C. (1978), and in Argos, EMBO J. 8 (1989), 779-785. For example, amino acids belonging to one of the following groups represent conservative changes or substitutions: -Ala, Pro, Gly, Gln, Asn, Ser, Thr; -Cys, Ser, Tyr, Thr; -Val, Ile, Leu, Met, Ala, Phe; -Lys, Arg, His; -Phe, Tyr, Trp, His; and -Asp, Glu.

As used herein, the term "sufficiently similar" means a first amino acid sequence that contains a sufficient or minimum number of identical or equivalent amino acid residues relative to a second amino acid sequence such that the first and second amino acid sequences have a common structural domain and/or common functional activity. For example, amino acid sequences that comprise a common structural domain that is at least about 45%, at least about 50%, at least

about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 100%, identical are defined herein as sufficiently similar. Preferably, variants will be sufficiently similar to the amino acid sequence of the peptides of the invention. Such variants generally retain the functional activity of the peptides of the present invention. Variants include peptides that differ in amino acid sequence from the native and wt peptide, respectively, by way of one or more amino acid deletion(s), addition(s), and/or substitution(s). These may be naturally occurring variants as well as artificially designed ones.

As used herein the term “linker”, “linker peptide” or “peptide linkers” or “linker” refers to synthetic or non-native or non-naturally-occurring amino acid sequences that connect or link two polypeptide sequences, e.g., that link two polypeptide domains. As used herein the term “synthetic” refers to amino acid sequences that are not naturally occurring. Exemplary linkers are described herein. Additional exemplary linkers are provided in US 20140079701, the contents of which are herein incorporated by reference in its entirety.

As used herein the term “codon-optimized sequence” refers to a sequence, which was modified from an existing coding sequence, or designed, for example, to improve translation in an expression host cell or organism of a transcript RNA molecule transcribed from the coding sequence, or to improve transcription of a coding sequence. Codon optimization includes, but is not limited to, processes including selecting codons for the coding sequence to suit the codon preference of the expression host organism.

Many organisms display a bias or preference for use of particular codons to code for insertion of a particular amino acid in a growing polypeptide chain. Codon preference or codon bias, differences in codon usage between organisms, is allowed by the degeneracy of the genetic code, and is well documented among many organisms. Codon bias often correlates with the efficiency of translation of messenger RNA (mRNA), which is in turn believed to be dependent on, inter alia, the properties of the codons being translated and the availability of particular transfer RNA (tRNA) molecules. The predominance of selected tRNAs in a cell is generally a reflection of the codons used most frequently in peptide synthesis. Accordingly, genes can be tailored for optimal gene expression in a given organism based on codon optimization.

As used herein, the terms “secretion system” or “secretion protein” refers to a native or non-native secretion mechanism capable of secreting or exporting the protein(s) of interest or therapeutic protein(s) from the microbial, e.g., bacterial cytoplasm. The secretion system may comprise a single protein or may comprise two or more proteins assembled in a complex e.g., HlyBD. Non-limiting examples of secretion systems for gram negative bacteria include the modified type III flagellar, type I (e.g., hemolysin secretion system), type II, type IV, type V, type VI, and type VII secretion systems, resistance-nodulation-division (RND) multi-drug efflux pumps, various single membrane secretion systems. Non-limiting examples of secretion systems for gram positive bacteria include Sec and TAT secretion systems. In some embodiments, the proteins of interest include a “secretion tag” of either RNA or peptide origin to direct the protein(s) of interest or therapeutic protein(s) to specific secretion systems. In some embodiments, the secre-

tion system is able to remove this tag before secreting the protein(s) of interest from the engineered bacteria. For example, in Type V auto-secretion-mediated secretion the N-terminal peptide secretion tag is removed upon translocation of the “passenger” peptide from the cytoplasm into the periplasmic compartment by the native Sec system. Further, once the auto-secretor is translocated across the outer membrane the C-terminal secretion tag can be removed by either an autocatalytic or protease-catalyzed e.g., OmpT cleavage thereby releasing the protein(s) of interest into the extracellular milieu.]]

As used herein, the term “transporter” is meant to refer to a mechanism, e.g., protein or proteins, for importing a molecule, e.g., amino acid, toxin, metabolite, substrate, etc. into the microorganism from the extracellular milieu.

The articles “a” and “an,” as used herein, should be understood to mean “at least one,” unless clearly indicated to the contrary.

The phrase “and/or,” when used between elements in a list, is intended to mean either (1) that only a single listed element is present, or (2) that more than one element of the list is present. For example, “A, B, and/or C” indicates that the selection may be A alone; B alone; C alone; A and B; A and C; B and C; or A, B, and C. The phrase “and/or” may be used interchangeably with “at least one of” or “one or more of” the elements in a list.

Bacteria

The genetically engineered bacteria of the invention are capable of reducing excess phenylalanine. In some embodiments, the genetically engineered bacteria are non-pathogenic bacteria. In some embodiments, the genetically engineered bacteria are commensal bacteria. In some embodiments, the genetically engineered bacteria are probiotic bacteria. In some embodiments, the genetically engineered bacteria are naturally pathogenic bacteria that are modified or mutated to reduce or eliminate pathogenicity. In some embodiments, non-pathogenic bacteria are Gram-negative bacteria. In some embodiments, non-pathogenic bacteria are Gram-positive bacteria. Exemplary bacteria include, but are not limited to, *Bacillus*, *Bacteroides*, *Bifidobacterium*, *Brevibacterium*, *Clostridium*, *Enterococcus*, *Escherichia coli*, *Lactobacillus*, *Lactococcus*, *Saccharomyces*, and *Staphylococcus*, e.g., *Bacillus coagulans*, *Bacillus subtilis*, *Bacteroides fragilis*, *Bacteroides subtilis*, *Bacteroides thetaiotaomicron*, *Bifidobacterium bifidum*, *Bifidobacterium infantis*, *Bifidobacterium lactis*, *Bifidobacterium longum*, *Clostridium butyricum*, *Enterococcus faecium*, *Lactobacillus acidophilus*, *Lactobacillus bulgaricus*, *Lactobacillus casei*, *Lactobacillus johnsonii*, *Lactobacillus paracasei*, *Lactobacillus plantarum*, *Lactobacillus reuteri*, *Lactobacillus rhamnosus*, *Lactococcus lactis*, and *Saccharomyces boulardii*. In certain embodiments, the genetically engineered bacteria are selected from the group consisting of *Bacteroides fragilis*, *Bacteroides thetaiotaomicron*, *Bacteroides subtilis*, *Bifidobacterium bifidum*, *Bifidobacterium infantis*, *Bifidobacterium lactis*, *Clostridium butyricum*, *Escherichia coli* Nissle, *Lactobacillus acidophilus*, *Lactobacillus plantarum*, *Lactobacillus reuteri*, and *Lactococcus lactis*.

In some embodiments, the genetically engineered bacteria are *Escherichia coli* strain Nissle 1917 (*E. coli* Nissle), a Gram-negative bacterium of the Enterobacteriaceae family that has evolved into one of the best characterized probiotics (Ukena et al., 2007). The strain is characterized by its complete harmlessness (Schultz, 2008), and has GRAS (generally recognized as safe) status (Reister et al., 2014, emphasis added). Genomic sequencing confirmed that *E.*

E. coli Nissle lacks prominent virulence factors (e.g., *E. coli* α -hemolysin, P-fimbrial adhesins) (Schultz, 2008). In addition, it has been shown that *E. coli* Nissle does not carry pathogenic adhesion factors, does not produce any enterotoxins or cytotoxins, is not invasive, and is not uropathogenic (Sonnenborn et al., 2009). As early as in 1917, *E. coli* Nissle was packaged into medicinal capsules, called Mutaflor, for therapeutic use. It is commonly accepted that *E. coli* Nissle's therapeutic efficacy and safety have convincingly been proven (Ukena et al., 2007).

One of ordinary skill in the art would appreciate that the genetic modifications disclosed herein may be adapted for other species, strains, and subtypes of bacteria. Furthermore, genes from one or more different species can be introduced into one another, e.g., the PAL gene from *Rhodospiridium toruloides* can be expressed in *Escherichia coli* (Sarkissian et al., 1999), and it is known that prokaryotic and eukaryotic phenylalanine ammonia lyases share sequence homology (Xiang and Moore, 2005).

Unmodified *E. coli* Nissle and the genetically engineered bacteria of the invention may be destroyed, e.g., by defense factors in the gut or blood serum (Sonnenborn et al., 2009) or by activation of a kill switch, several hours or days after administration. Thus, the genetically engineered bacteria may require continued administration. In some embodiments, the residence time is calculated for a human subject. Residence time in vivo may be calculated for the genetically engineered bacteria of the invention (see, e.g., FIG. 38).

In some embodiments, the genetically engineered bacteria of the invention comprise a gene encoding PAL, wherein the PAL gene is operably linked to a directly or indirectly inducible promoter. In some embodiments, the bacteria comprise a non-native PAL gene. In some embodiments, the bacteria comprise additional copies of a native PAL gene. In some embodiments, the promoter is not associated with the PAL gene in nature. In some embodiments, the genetically engineered bacteria of the invention comprise a gene encoding PAH, wherein the PAH gene is operably linked to a directly or indirectly inducible promoter. In some embodiments, the bacteria comprise a non-native PAH gene. In some embodiments, the bacteria comprise additional copies of a native PAH gene. In some embodiments, the promoter is not associated with the PAH gene in nature.

The genetically engineered bacteria further comprise a gene encoding a phenylalanine transporter (PheP). In certain embodiments, the bacteria comprise additional copies of a native gene encoding a phenylalanine transporter, wherein the phenylalanine transporter gene is operably linked to a directly or indirectly inducible promoter. In alternate embodiments, the bacteria comprise a gene encoding a non-native phenylalanine transporter, wherein the phenylalanine transporter gene is operably linked to a directly or indirectly inducible promoter. Both embodiments are encompassed by the term "non-native" phenylalanine transporter. In some embodiments, the promoter is not associated with the pheP gene in nature. In some embodiments, the same promoter controls expression of PheP and PAL or PAH.

In some embodiments, the promoter that is operably linked to PAL, PAH, and/or pheP is directly or indirectly induced by exogenous environmental conditions. In some embodiments, the promoter is directly or indirectly induced by exogenous environmental conditions specific to the gut of a mammal. In some embodiments, the promoter is directly or indirectly induced by exogenous environmental conditions specific to the small intestine of a mammal. In some embodiments, the promoter is directly or indirectly induced

by low-oxygen or anaerobic conditions such as the environment of the mammalian gut. In some embodiments, the promoter is directly or indirectly induced by the presence of molecules or metabolites that are specific to the gut of a mammal, e.g., propionate. In some embodiments, the promoter is directly or indirectly induced by exposure to tetracycline. In some embodiments, the promoter is directly or indirectly induced by a molecule that is co-administered with the genetically engineered bacteria of the invention.

10 Reducing Hyperphenylalaninemia

The genetically engineered bacteria of the invention comprise a gene encoding a phenylalanine-metabolizing enzyme (PME) and are capable of reducing hyperphenylalaninemia.

Examples of phenylalanine metabolizing enzymes include, but are not limited to, phenylalanine hydroxylase (PAH), phenylalanine ammonia lyase (PAL), aminotransferases, L-amino acid deaminase (L-AAD), and phenylalanine dehydrogenases. Reactions with phenylalanine hydroxylases, phenylalanine dehydrogenases or aminotransferases require cofactors, while L-AAD and PAL do not require any extra cofactor. Without wishing to be bound by theory, the lack of need for a cofactor means that phenylalanine degradation by the enzyme encoded by the genetically engineered bacteria is dependent on the availability of the substrate and is not limited by the availability of the cofactor.

Phenylalanine ammonia lyase (PAL; EC 4.3.1.24) is an enzyme that catalyzes a reaction converting L-phenylalanine to ammonia and trans-cinnamic acid. Phenylalanine ammonia lyase is specific for L-Phe, and to a lesser extent, L-Tyrosine. The reaction catalyzed by PAL is the spontaneous, non-oxidative deamination of L-phenylalanine to yield trans-cinnamic acid and ammonia. Unlike the mammalian enzyme (PAH), PAL is a monomer and requires no cofactors (MacDonald et al., Biochem Cell Biol 2007; 85:273-82. A modern view of phenylalanine ammonia lyase). In microorganisms, it has a catabolic role, allowing them to utilize L-phenylalanine (L-Phe) as a sole source of carbon and nitrogen. In one embodiment, the genetically engineered bacteria of the invention comprise a PAL gene. PAL is capable of converting phenylalanine to non-toxic levels of trans-cinnamic acid and ammonia. Trans-cinnamic acid (TCA) can further be converted to TCA metabolites benzoic and hippuric acids (Sarkissian et al., J Mass Spectrom. 2007 June; 42(6):811-7; Quantitation of phenylalanine and its trans-cinnamic, benzoic and hippuric acid metabolites in biological fluids in a single GC-MS analysis). PAL enzyme activity does not require THB cofactor activity.

In some embodiments, PAL is encoded by a PAL gene derived from a bacterial species, including but not limited to, *Achromobacter xylosoxidans*, *Pseudomonas aeruginosa*, *Photobacterium luminescens*, *Anabaena variabilis*, and *Agrobacterium tumefaciens*. In some embodiments, the bacterial species is *Photobacterium luminescens*. In some embodiments, the bacterial species is *Anabaena variabilis*. In some embodiments, PAL is encoded by a PAL gene derived from a eukaryotic species, e.g., a yeast species, a plant species. Multiple distinct PAL proteins are known in the art. The genetically engineered bacteria convert more phenylalanine when the PAL gene is expressed than unmodified bacteria of the same bacterial subtype under the same conditions. Thus, the genetically engineered bacteria comprising PAL may be used to metabolize phenylalanine in the body into non-toxic molecules in order to treat conditions associated with hyperphenylalaninemia, including PKU. In some embodiments, the genetically engineered bacteria express *Anabaena variabilis* PAL ("PAL1"). In some embodiments, the genetically

engineered bacteria express *Photorhabdus luminescens* PAL ("PAL3"). Non-limiting examples of PAL sequences of interest are shown in Table 2.

LAAD catalyzes the stereospecific oxidative, i.e., oxygen consuming, deamination of L-amino acids to α -keto acids along with the production of ammonia and hydrogen peroxide via an imino acid intermediate. L-AADs are found in snake venoms, and in many bacteria (Bifulco et al. 2013), specifically in the cytomembranes of the *Proteus*, *Providencia*, and *Morganella* bacteria. L-AADs (EC 1.4.3.2) are flavoenzymes with a dimeric structure. Each subunit contains a non-covalently-bound flavin adenine dinucleotide (FAD) cofactor and do not require any external cofactors. *Proteus mirabilis* contains two types of L-AADs (Duerre and Chakrabarty 1975). One has broad substrate specificity and catalyzes the oxidation of aliphatic and aromatic L-amino acids to keto acids, typically L-phenylalanine (GenBank: U35383.1) (Baek et al., Journal of Basic Microbiology 2011, 51, 129-135; "Expression and characterization of a second L-amino acid deaminase isolated from *Proteus mirabilis* in *Escherichia coli*"). The other type acts mainly on basic L-amino acids (GenBank: EU669819.1). LAADs from bacterial, fungal, and plant sources appear to be involved in the utilization of L-amino acids (i.e., ammonia produced by the enzymatic activity) as a nitrogen source. Most eukaryotic and prokaryotic L-amino acid deaminases are extracellularly secreted, with the exception of from *Proteus* species LAADs, which are membrane-bound. In *Proteus mirabilis*, L-AADs have been reported to be located in the plasma membrane, facing outward into the periplasmic space, in which the enzymatic activity resides (Pelmont J et al., (1972) "L-amino acid oxidases of *Proteus mirabilis*: general properties" Biochimie 54: 1359-1374).

In one embodiment, the genetically engineered bacteria of the invention comprise a LAAD gene. LAAD is capable of converting phenylalanine to non-toxic levels of phenylpyruvate, which can also further be degraded, e.g., by liver enzymes, to phenyllactate. Phenylpyruvate cannot cross the blood brain barrier, which allows LAAD to reduce the levels of phenylalanine in the brain without allowing the accumulation of another potentially toxic metabolite. In some embodiments, LAAD is encoded by a LAAD gene derived from a bacterial species, including but not limited to, *Proteus*, *Providencia*, and *Morganella* bacteria. In some embodiments, the bacterial species is *Proteus mirabilis*. In some embodiments, the bacterial species is *Proteus vulgaris*.

In some embodiments, the genetically engineered bacteria express *Proteus mirabilis* LAAD enzyme GenBank: U35383.1. Non-limiting examples of LAAD sequences of interest are shown in Table 2. In some embodiments, the LAAD enzyme is derived from snake venom. According to the invention, genetically engineered bacteria convert more phenylalanine when the LAAD gene is expressed than unmodified bacteria of the same bacterial subtype under the same conditions. Thus, the genetically engineered bacteria comprising LAAD may be used to metabolize phenylalanine in the body into non-toxic molecules in order to treat conditions associated with hyperphenylalaninemia, including PKU.

In some embodiments, the genetically engineered bacteria encode a wild type enzyme as it occurs in nature. In some embodiments, the genetically engineered bacteria encode an enzyme which comprises mutations relative to the wild type sequence. In some embodiments, the mutations increase stability of the enzyme. In some embodiments, the mutations increase the catalytic activity of the enzyme. In some embodiments, the genetically engineered bacteria comprise a gene encoding one or more of the proteins listed in Table 2. In some embodiments, the genetically engineered bacteria comprise gene sequence(s) encoding one or more of the polypeptides comprising sequence of any of SEQ ID Nos: 1-8. In some embodiments, the genetically engineered bacteria comprise gene sequence(s) encoding a polypeptide having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity with any of the sequences of SEQ ID Nos: 1-8. In some embodiments, the genetically engineered bacteria encode one or more enzymes from Table 2, which comprise a mutation. In some embodiments, the genetically engineered bacteria comprise a gene encoding wild type PAH. In some embodiments, the genetically engineered bacteria encode a mutated PAH with increased stability and/or activity. In some embodiments, the genetically engineered bacteria comprise a gene encoding wild type PAL. In some embodiments, the genetically engineered bacteria encode a mutated PAL with increased stability and/or activity. In some embodiments, the genetically engineered bacteria comprise a gene encoding wild type LAAD. In some embodiments, the genetically engineered bacteria encode a mutated LAAD with increased stability and/or activity. Methods for screening for enzymes with desirable properties are known in the art and described herein.

TABLE 2

Sequences of Phenylalanine Metabolizing Enzymes		
Description	Sequence	SEQ ID NO
Phenylalanine ammonia-lyase (<i>Anabaena variabilis</i>) Acc. No.: Q3M5Z3.1	MKTLTSAQSKTSSQQFSFTGNSSANVIIGNQKLTIN DVARVARNGTLVSLTNNTDILQGIQASCDYINNAV ESGEPIYGVTSFGFGGMANVAISREQASELQTNLVW FLKTGAGNKLPLADVRAAMLLRANSHMRGASGIR LELIKRMEIFLNAGVTPVYVEFGSIGASGDLVPLSYI TGSLLGLDPSFKVDFNGKEMDAPTALRQLNLSPLTL LPKEGLAMMNGTSVMTGIAANCVDYDTQILTAIM GVHALDIQALNGTNSFPHFIHNSKPHPGQLWAAD QMISLLANSQVLVRDELDDGKHDYRDHELIQDRYSLR CLPQYLGPIVDGISQIAKQIEIINSVTDNPLIDVDNQ ASYHGGNFGQYVGMGMHLRYIIGLLAKHLDV QIALLASPEFSNGLPSSLGNRERKVNMLKGLQIC GNSIMPLLTFFYGNISADRFPTHAEQFNQINISQGYT SATLARRSVDIFQNYVAIALMFGVQAVDLRTYKKT GHYDARACLSPATERLYSAVRHVVGQKPTSDRPIYI WNDNEQGLDEHIARISADIAAGGVIVQAVQDILPCL H	SEQ ID NO: 1

TABLE 2-continued

Sequences of Phenylalanine Metabolizing Enzymes		
Description	Sequence	SEQ ID NO
histidine ammonia- lyase <i>Anabaena</i> <i>variabilis</i> ATCC 294131 (Acc. NO. ABA23593.1)	MKTLQAQSKTSSQQFSFTGNSSANVIIGNQKLTIN DVARVARNGTLVSLTNNTDILQGIQASCDYINNAV ESGEPIYGVTSQGGMANVAISREQASELQTNLVW FLKTGAGNKLPLADVRAAMLLRANSHMRGASGIR LELIKRMEIFLNAGVTPYVYEFSGSIGAGDLVPLSYI TGSLIGLDPSFKVDFNGKEMDAPTALRQLNLSPLTL LPKEGLAMMNGTSVMTGIAANCYDVTQILTAIM GVHALDIQALNGTNQSFHPFIHNSKPHPGQLWAAD QMISLLANSQLVRLDELGKHIDYRDHELIQDRYSLR CLPQYLGPIVDGISQIAKQIEIENSVDNPLDIDVDNQ ASYHGGNFGQYVGMGMDHLRYIIGLLAKHLDV QIALLASPEFSNGLPPSLLGNRERKVMNGLKGLQIC GNSIMPLLTFYGNISADRFPTHAEQFNQINSQGYT SATLARRSVDIFQNYVAIALMFGVQAVDLRTYKKT GHYDARACLSPATERLYSAVRHVVGQKPTSDRPYI WNDNEQGLDEHIARISADIAAGGVIVQAVQDILPCL H	SEQ ID NO: 2
histidine ammonia- lyase [<i>Photorhabdus</i> <i>luminescens</i>] (WP_011146 484)	MKAKDVQPTIIINKNGLISLEDIYDIAIKQKKVEISTE ITELLTHGREKLEKLSGSEVIYGINTEGFGGNANLV VPFEKIAEHQQNLLTFLSAGTDYMSKPCIKASQFT MLLSVCKGWSATRPIVAQAIVDHINHDIVPLVPRYG SVGASGDLIPLSYIARALCGIGKVVYMGAEIDAAEA IKRAGLTPLSLKAKEGLALINGTRVMSGISAITVIKL EKLFKASISAIALAVEALLASHEHYDARIQQVKNHP GQNAVASALRNLAGSTQVNLSSGVKEQANKACR HQEITQLNDTLQEVYSIRCAPQVLGIVPESLATARKI LEREVI SANDNPLIDPENGDLVHGGNFMGYVART MDALKLDIALIANHLHAIVALMMDNRFSGPLNSL SPTPGMYQGFKGVLQSQTALVAAIRHDCASGIHT LATEQYNQDIVSLGLHAAQDVLEMEQKLRNIVSMT ILVVCQAIHLRGNISEIAPETAKFYHAVREISSPLITD RALDEDIIRIADAIINDQLPLPEIMLEE	SEQ ID NO: 3
Histidine ammonia lyase (<i>Photorhabdus</i> <i>luminescens</i>) Acc. NO: CAE15566	MKQLTIYPGKLTLDLRLQVYLQPVKITLDSQIFPAIE RSVECVNAILAENRTAYGINTGFGLLASTRIEDNL EKLQRSLVVSHAAGVGKALDDNMTRLIMVLKINSL SRGYSGIRLAVIQALIALVNAEYIYPHIPCKGSVGASG DLAPLAHMSLLLLGEGQARYQGEWLPAKEALAKA NLQPTITLAAKEGLALLNGTQVSTAFALRGLFEAEDL LAAAIVCGLSVEAALGSRKPFDAHVHVVRGQQGQ IDVAALYRHVLEESSELSDSHINCPKVQDPYSLRCQ PQVMGACLTQLRHAADVILTANAVSDNPLVFAEQ GEVISGGNFHAEPVAMASDNLALVLAEIGALSERRI ALLMDSHMSQLPPLVENGVSNGFMIAQVTAAL ASENKALAHPASVDSLPTSANQEDHVSMAAPAAGR LWEMAENTRGILAEIWLACQGIDFRNGLKSSPILE EARVILRAKVDDYDQDRFFAPDIDAAVKLLAEQHL SSLLPSGQILQRKNR	SEQ ID NO: 4
amino acid deaminase (<i>Proteus</i> <i>mirabilis</i>) Acc. No: ACD36582	MAISRRKFI LGGTVVAVAAGAGVLTPLMTREGRFV PGTFRHGFVEGTGGPLPKQDDVVVIGAGILGIMTAI NLAERGLSVTIVEKGNIAEQSSRFYQQAISYKMPD ETFLHLHLGKHRWREMNKVGIDTTYRTQGRVEV PLDEEDLENVRKWIDAKSKDVGS DIPRTKMI EGAE LKQRLRGATTDWKIA GFEEDSGSFDPEVATFVMAE YAKKMGIKIFTNCAARGLTQAGVISDVVTEKGPIK TSRVVVGGSRLFMQNLNVDPVTLPAYQSQQLI SAAPNAPGGNVALPGGIFFRDQADGTATSPRVIVA PVVKESFTYGYKYLPLALPDPFVHISLNEQLINSFM QSTHWDLNEESPFEKYRDMTALPDLPELNASLEKL KKEFPAPKES TLIDQWSGAMAIAPDENPIISDVKEYP GLVINTATGWGMTESPVSAEITADLLLGKKPVLDA KPFSLYRF	SEQ ID NO: 5
amino acid deaminase [<i>Proteus</i> <i>mirabilis</i> HI4320] Acc. No.: AAA86752.1	MNISRRKLLLGVAAGVLAGGAALVPMVRRDGKF VEAKSRASFVEGTQGALPKEADVVIAGAGIGIMTA INLAERGMSVTILEKGQIAGEQSGRAYSQIISVQTSF EIFPLHHYGKILWRGMNEKIGADTSYRTQGRVEAL ADEKALDKAQAWIKTAKEAAGFDTPLNTRI KEGEE LSNRLVGAQTPWTVAAFEEDSGSVDPETGTPALAR YAKQIGVKIYTNCAVRGIETAGGKISDVVSEKGAIK TSQVVLAGGIWSRLFMGNMGIDIPTLNIVLSQQRV SGVPGAPRGNVHLPNGIHFRQADGTAYAVAPRIFTS	SEQ ID NO: 6

TABLE 2-continued

Sequences of Phenylalanine Metabolizing Enzymes		
Description	Sequence	SEQ ID NO
	SIVKDSFLLGPKFMHLLGGGELPLEFSIGEDLFNSFK MPTSWNLDEKTPFEQFRVATATQNTQHLDAVFQR MKTEFPVFEKSEVVERNGAVVSPTFDELPIISEVKE YPLGLVINTATVWGMTEGPAAGEVTADIVMGKKPVI DPTFFSLDRFKK	
L-AAD from <i>Proteus</i> <i>vulgaris</i> ; (Acc. No. BAA90864)	MAISRRKFIIGTTVVAVAGAGILTPMLTREGRFVP GTPRHGFVEGTEGALPKQADV VVVGAGILGIMTAI NLVERGLSVVIVEKGNIAEQSSRFYQAI SYKMPD ETFLHLHLGKHRWREMNKVGIDTTYRTQGRVEV PLDEEDLVNVRKWIDERSKNVGS DIPKTRIIIEGAEL NQRLRGATTDWKIAGFEEDSGSPDDEVATFVMAEY AKKMGVRIYTQCAARGLETQAGVISDVVTEKGAIK TSQVVVAGGVWSRLFMQNLNVDPVTLPAYQSQQ ISGSPTAPGGNVALPGGIFFREQADGTATSPRVIVA PVVKESFTYGYKYLPLALPDFFVHISLNEQLINSFM QSTHWNLEDEVSPFEQFRNM TALPDLPELNASLEKL KAEPFAFKESKLIDQWSGAMAIAPDENPIISEVKEYP GLVINTATGWGMTESPVSAELTADLLLGKKPVLD KPFSLYRF	SEQ ID NO: 7
Phenylalanine hydroxylase [<i>Homo</i> <i>sapiens</i>] (Acc. No. AAH26251)	MSTAVLENPGLGRKLSDFGQETSIEDNCNNGAIS LIFSLKEEVGALAKVLRLEENDVNLTHIESRPSRLK KDEYEFFTHLDRSLPALTNI IKILRHDIGATVHEL RDKKKDTVPWFPRTIQELDRFANQILSYGAELDAD HPGFKDPVYRARRKQFADIAYNRHHGQPIPRVEYM EEGKKTWGTVPFKTLKSLYKTHACEYVNHIFP LLEK YCGFHEDNIPQLEDVSQFLQCTGFRLRPVAGLLSS RDFLGGLAFRVFHCCTQYIRHGSKPMYTPEDICHE LGHVPLFSDRSFAQFSQEI GLASLGAPDEYIEKLATI YWFVTVEGLCKQGDSIKAYGAGLLSSFGELQYCLS EKP KLLPLELEKTAIQNYTVTEFQPLYVAESFNDA KEKVRNFAATIPRPFVRYDPYQRIEVL DNTQQLK ILADSINSEIGILCSALQKIK	SEQ ID NO: 8

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The PME, e.g., PAL, LAAD, or PAH, gene may be present on a plasmid or chromosome in the genetically engineered bacteria. In some embodiments, the PME gene is expressed under the control of a constitutive promoter. In some embodiments, the PME gene is expressed under the control of a promoter that is directly or indirectly induced by exogenous environmental conditions, as described herein. In some embodiments, the PME gene is expressed under the control of a promoter that is directly or indirectly induced by exogenous environmental conditions, such as in the presence of molecules or metabolites specific to the gut of a mammal. In one embodiment, the PME gene is expressed under the control of a promoter that is directly or indirectly induced by low-oxygen, microaerobic, or anaerobic conditions, wherein expression of the PME gene, e.g., the PAL gene, is activated under low-oxygen or anaerobic environments, such as the environment of the mammalian gut.

In one embodiment, the genetically engineered bacteria encode a PAL gene which is directly or indirectly induced by low-oxygen or anaerobic conditions, such as the mammalian gut. In one embodiment, the genetically engineered bacteria encode a LAAD gene which is directly or indirectly induced by oxygenated, low oxygen, or microaerobic conditions, such as conditions found in the proximal intestine, including but not limited to the stomach, duodenum, and ileum. In other embodiments, the genetically engineered bacteria encode a PME gene which is directly or indirectly induced by an environmental factor that is naturally present in a mammalian gut. In other embodiments, the genetically engineered bacteria encode a PME gene which is directly or indirectly induced by an environmental factor that is not

naturally present in a mammalian gut, e.g., arabinose. In other embodiments, the genetically engineered bacteria encode a PME gene which is directly or indirectly induced by an environmental factor that is naturally present in a mammalian gut under inflammatory conditions.

Bacteria have evolved transcription factors that are capable of sensing oxygen levels. Different signaling pathways may be triggered by different oxygen levels and occur with different kinetics. An oxygen level-dependent promoter is a nucleic acid sequence to which one or more oxygen level-sensing transcription factors is capable of binding, wherein the binding and/or activation of the corresponding transcription factor activates downstream gene expression. In one embodiment, the PME gene is expressed under the control of an oxygen level-dependent promoter. In a more specific aspect, the PAL gene is under the control of an oxygen level-dependent promoter that is activated under low-oxygen or anaerobic environments, such as the environment of the mammalian gut.

In certain embodiments, the genetically engineered bacteria comprise a PME, e.g., PAL, expressed under the control of the fumarate and nitrate reductase regulator (FNR) promoter. In *E. coli*, FNR is a major transcriptional activator that controls the switch from aerobic to anaerobic metabolism (Unden et al., 1997). In the anaerobic state, FNR dimerizes into an active DNA binding protein that activates hundreds of genes responsible for adapting to anaerobic growth. In the aerobic state, FNR is prevented from dimerizing by oxygen and is inactive. In some embodiments, multiple distinct FNR nucleic acid sequences are inserted in the genetically engineered bacteria. In alternate embodi-

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ments, the genetically engineered bacteria comprise a PME, e.g., PAL, expressed under the control of an alternate oxygen level-dependent promoter, e.g., an ANR promoter (Ray et al., 1997), a DNR promoter (Trunk et al., 2010). In some embodiments, phenylalanine metabolism is particularly activated in a low-oxygen or anaerobic environment, such as in the gut.

In *P. aeruginosa*, the anaerobic regulation of arginine deiminase and nitrate reduction (ANR) transcriptional regulator is “required for the expression of physiological functions which are inducible under oxygen-limiting or anaerobic conditions” (Winteler et al., 1996; Sawers 1991). *P. aeruginosa* ANR is homologous with *E. coli* FNR, and “the consensus FNR site (TTGAT---ATCAA) (SEQ ID NO: 66) was recognized efficiently by ANR and FNR” (Winteler et al., 1996). Like FNR, in the anaerobic state, ANR activates numerous genes responsible for adapting to anaerobic growth. In the aerobic state, ANR is inactive. *Pseudomonas fluorescens*, *Pseudomonas putida*, *Pseudomonas syringae*, and *Pseudomonas mendocina* all have functional analogs of ANR (Zimmermann et al., 1991). Promoters that are regulated by ANR are known in the art, e.g., the promoter of the arcDABC operon (see, e.g., Hasegawa et al., 1998).

The FNR family also includes the dissimilatory nitrate respiration regulator (DNR) (Arai et al., 1995), a transcriptional regulator which is required in conjunction with ANR for “anaerobic nitrate respiration of *Pseudomonas aeruginosa*” (Hasegawa et al., 1998). For certain genes, the FNR-binding motifs “are probably recognized only by DNR”

(Hasegawa et al., 1998). Any suitable transcriptional regulator that is controlled by exogenous environmental conditions and corresponding regulatory region may be used. Non-limiting examples include ArcA/B, ResD/E, NreA/B/C, and AirSR, and others are known in the art.

FNR promoter sequences are known in the art, and any suitable FNR promoter sequence(s) may be used in the genetically engineered bacteria of the invention. Any suitable FNR promoter(s) may be combined with any suitable PAL. Non-limiting FNR promoter sequences are provided in Table 3, and non-limiting PAL sequences are also provided herein. In some embodiments, the genetically engineered bacteria of the invention comprise one or more of: SEQ ID NO: 9, SEQ ID NO: 10, nirB1 promoter (SEQ ID NO: 11), nirB2 promoter (SEQ ID NO: 12), nirB3 promoter (SEQ ID NO: 13), ydfZ promoter (SEQ ID NO: 14), nirB promoter fused to a strong ribosome binding site (SEQ ID NO: 15), ydfZ promoter fused to a strong ribosome binding site (SEQ ID NO: 16), fnrS, an anaerobically induced small RNA gene (fnrS1 promoter SEQ ID NO: 9 or fnrS2 promoter SEQ ID NO: 17), nirB promoter fused to a crp binding site (SEQ ID NO: 18), and fnrS fused to a crp binding site (SEQ ID NO: 19).

In some embodiments, genetically engineered bacteria comprise a nucleic acid sequence that is at least about 80%, at least about 85%, at least about 90%, at least about 95%, or at least about 99% homologous to the DNA sequence of SEQ ID NO: 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 or a functional fragment thereof.

TABLE 3

FNR Sequences	
SEQ ID NO	FNR-responsive regulatory region Sequence
SEQ ID NO: 9	ATCCCCATCACTCTTGATGGAGATCAATTCCCCAAGCTGCTAGAGCGTTA CCTTGCCCTTAAACATTAGCAATGTCGATTATCAGAGGGCCGACAGGCT CCCACAGGAGAAAACCG
SEQ ID NO: 10	CTCTTGATCGTTATCAATTCCCACGCTGTTTCAGAGCGTTACCTTGCCCT TAAACATTAGCAATGTCGATTATCAGAGGGCCGACAGGCTCCACAGGA GAAAACCG
nirB1 SEQ ID NO: 11	GTCAGCATAACACCTGACCTCTCATTAAATGTTTCATGCCGGGCGGCACT ATCGTCGTCGGCCCTTTCTCTCTTACTCTGCTACGTACATCTATTTCT ATAAATCCGTTCAATTTGTCTGTTTTTGCACAAACATGAAATATCAGAC AATTCGCTGACTTAAGAAAATTTATACAAATCAGCAATATACCCCTTAAG GAGTATATAAAGGTGAATTTGATTACATCAATAAGCGGGTTGCTGAAT CGTTAAGGTAGGCGGTAATAGAAAAGAAATCGAGGCAAAA
nirB2 SEQ ID NO: 12	CGGCCGATCGTTGAACATAGCGGTCGCGAGGCGGCACTGCTTACAGCAA ACGGTCTGTACGCTGTCGTCCTTGTGATGTGCTTCTGTTAGGTTTCGTC AGCCGTCACCGTCAGCATAACACCTGACCTCTCATTAAATGCTCATGCCC GGACGGCACTATCGTCGTCGGCCCTTTCTCTCTTCCCGCTACGTGC ATCTATTTCTATAAACCCGCTCATTTTGTCTATTTTGCACAAACATGA AATATCAGACAATTCGCTGACTTAAGAAAATTTATACAAATCAGCAATAT ACCCATTAAGGAGTATATAAAGGTGAATTTGATTACATCAATAAGCGGG GTTGCTGAATCGTTAAGGTAGGCGGTAATAGAAAAGAAATCGAGGCAAAA atgtttgtttaactttaagaaggagatatcat
nirB3 SEQ ID NO: 13	GTCAGCATAACACCTGACCTCTCATTAAATGCTCATGCCGGGCGGCACT ATCGTCGTCGGCCCTTTCTCTCTTCCCCCGCTACGTGATCTATTTCT ATAAACCGCTCATTTTGTCTATTTTTCACAAACATGAAATATCAGAC AATTCGCTGACTTAAGAAAATTTATACAAATCAGCAATATACCCCTTAAG GAGTATATAAAGGTGAATTTGATTACATCAATAAGCGGGTTGCTGAAT CGTTAAGGTAGGCGGTAATAGAAAAGAAATCGAGGCAAAA
ydfZ SEQ ID NO: 14	ATTTCTCTCATCCCATCCGGGTGAGAGTCTTTTCCCCGACTTATGGC TCATGCGATGCATCAAAAAGATGTGAGCTTGATCAAAAACAAAAATATT TCACTCGACAGGAGTATTTATATTGCGCCCGTTACGTGGGCTTCGACTGT AATCAGAAAAGGAGAAAACACCT
nirB + RBS SEQ ID NO: 15	GTCAGCATAACACCTGACCTCTCATTAAATGTTTCATGCCGGGCGGCACT ATCGTCGTCGGCCCTTTCTCTCTTACTCTGCTACGTACATCTATTTCT

TABLE 3-continued

FNR Sequences	
SEQ ID NO	FNR-responsive regulatory region Sequence
	ATAATCCGTTCAATTTGCTGTTTTTTGCACAAACATGAAATATCAGAC AATTCGTGACTTAAGAAAATTTATACAAATCAGCAATATACCCCTTAAG GAGTATATAAAGGTGAATTTGATTACATCAATAAGCGGGTTGCTGAAT CGTTAAGGATCCCTCTAGAAATAATTTGTTTAACTTTAAGAAGGAGATA TACAT
ydfZ + RBS SEQ ID NO: 16	CATTTCCTCTCATCCCATCCGGGTGAGAGTCTTTCCCCGACTTATGG CTCATGCATGCATCAAAAAGATGTGAGCTTGATCAAAAACAAAAATAT TTCACTCGACAGGAGTATTTATATTGCGCCCGGATCCCTCTAGAAATAAT TTTGTTTAACTTTAAGAAGGAGATATACAT
fnrS1 SEQ ID NO: 17	AGTTGTTCTTATTGGTGGTGTGCTTTATGGTTGCATCGTAGTAAATGGT TGTAACAAAAGCAATTTTCCGGCTGTCTGTATACAAAACGCCGTAAAG TTTGAGCGAAGTCAATAAACTCTCTACCCATTGAGGCAATATCTCTCTT GGATCCCTCTAGAAATAATTTGTTTAACTTTAAGAAGGAGATATACAT
fnrS2 SEQ ID NO: 18	AGTTGTTCTTATTGGTGGTGTGCTTTATGGTTGCATCGTAGTAAATGGT TGTAACAAAAGCAATTTTCCGGCTGTCTGTATACAAAACGCCGTAAAG TTTGAGCGAAGTCAATAAACTCTCTACCCATTGAGGCAATATCTCTCTT GGATCCAAAGTGAATCTAGAAATAATTTGTTTAACTTTAAGAAGGAGA TATACAT
nirB + crp SEQ ID NO: 19	TCGTCTTTGTGATGTGCTTCTGTTAGGTTTCGTGAGCCGTCACCGTCAG CATAACACCCCTGACCTCTCATTAAATGCTCATGCCGACGGCACTATCGT CGTCCGGCCTTTCTCTCTTCCCGCTACGTGCATCTATTTCTATAAA CCCGCTCATTTGTCTATTTTTCGACAAACATGAAATATCAGACAAATC CGTGACTTAAGAAAATTTATACAAATCAGCAATATACCCATTAAAGGAGTA TATAAAGGTGAATTTGATTACATCAATAAGCGGGTTGCTGAATCGTTA AGGTAGaaatgtgatctagttcacatttGCGGTAATAGAAAAGAAATCGA GGCAAAAatggttgtttaactttaagaaggagatatatacat
fnrS + crp SEQ ID NO: 20	AGTTGTTCTTATTGGTGGTGTGCTTTATGGTTGCATCGTAGTAAATGGT TGTAACAAAAGCAATTTTCCGGCTGTCTGTATACAAAACGCCGTAAAG TTTGAGCGAAGTCAATAAACTCTCTACCCATTGAGGCAATATCTCTCaa atgtgatctagttcacatttGTTGTTAACTTTAAGAAGGAGATATACAT

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In other embodiments, a PME, e.g., PAL, is expressed under the control of an oxygen level-dependent promoter fused to a binding site for a transcriptional activator, e.g., CRP. CRP (cyclic AMP receptor protein or catabolite activator protein or CAP) plays a major regulatory role in bacteria by repressing genes responsible for the uptake, metabolism, and assimilation of less favorable carbon sources when rapidly metabolizable carbohydrates, such as glucose, are present (Wu et al., 2015). This preference for glucose has been termed glucose repression, as well as carbon catabolite repression (Deutscher, 2008; Görke and Stülke, 2008). In some embodiments, PME, e.g., PAL, expression is controlled by an oxygen level-dependent promoter fused to a CRP binding site. In some embodiments, PAL expression is controlled by an FNR promoter fused to a CRP binding site. In these embodiments, cyclic AMP binds to CRP when no glucose is present in the environment. This binding causes a conformational change in CRP, and allows CRP to bind tightly to its binding site. CRP binding then activates transcription of the PME gene, e.g., PAL gene, by recruiting RNA polymerase to the FNR promoter via direct protein-protein interactions. In the presence of glucose, cyclic AMP does not bind to CRP and a PME, e.g., PAL, gene transcription is repressed. In some embodiments, an oxygen level-dependent promoter (e.g., an FNR promoter) fused to a binding site for a transcriptional activator is used to ensure that a PME, e.g., PAL, is not expressed under anaerobic conditions when sufficient amounts of glucose are present, e.g., by adding glucose to growth media in vitro.

In another embodiment, a PME, e.g., LAAD, is expressed under the control of an inducible promoter fused to a binding

site for a transcriptional activator, e.g., CRP, such that expression is repressed in the presence of glucose.

In some embodiments, LAAD is not under the control of an FNRs promoter. LAAD requires oxygen to catalyze the degradation of phenylalanine to phenylpyruvate. Therefore, it would not be desirable to induce LAAD expression under strictly anaerobic conditions where it would be minimally active (FIG. 25).

In some embodiments, a PME, e.g., PAL or LAAD, is expressed under the control of an inducible promoter that is responsive to specific molecules or metabolites in the environment, e.g., the mammalian gut. For example, the short-chain fatty acid propionate is a major microbial fermentation metabolite localized to the gut (Hosseini et al., 2011). In one embodiment, PAL gene expression is under the control of a propionate-inducible promoter. In a more specific embodiment, PME gene expression is under the control of a propionate-inducible promoter that is activated by the presence of propionate in the mammalian gut. Any molecule or metabolite found in the mammalian gut, in a healthy and/or disease state, may be used to induce PME gene expression. Non-limiting examples include propionate, bilirubin, aspartate aminotransferase, alanine aminotransferase, blood coagulation factors II, VII, IX, and X, alkaline phosphatase, gamma glutamyl transferase, hepatitis antigens and antibodies, alpha fetoprotein, anti-mitochondrial, smooth muscle, and anti-nuclear antibodies, iron, transferrin, ferritin, copper, ceruloplasmin, ammonia, and manganese. In alternate embodiments, PME, e.g., PAL and/or LAAD, gene expression is under the control of a ParaBAD promoter, which is activated in the presence of the sugar arabinose. In one

embodiment, LAAD expression is under the control of the ParaBAD promoter. In one embodiment, expression of LAAD occurs under aerobic or microaerobic conditions.

In some embodiments, the PAL gene is expressed under the control of a promoter that is induced by exposure to tetracycline. In some embodiments, gene expression is further optimized by methods known in the art, e.g., by optimizing ribosomal binding sites, manipulating transcriptional regulators, and/or increasing mRNA stability.

In some embodiments, the genetically engineered bacteria comprise a stably maintained plasmid or chromosome carrying the PAL gene, such that PAL can be expressed in the host cell, and the host cell is capable of survival and/or growth in vitro, e.g., in medium, and/or in vivo, e.g., in the gut. In some embodiments, the genetically engineered bacteria comprise two or more distinct PAL genes. In some embodiments, the genetically engineered bacteria comprise multiple copies of the same PAL gene. In some embodiments, the PAL gene is present on a plasmid and operably linked to a directly or indirectly inducible promoter. In some embodiments, the PAL gene is present on a plasmid and operably linked to a promoter that is induced under low-oxygen or anaerobic conditions. In some embodiments, the PAL gene is present on a chromosome and operably linked to a directly or indirectly inducible promoter. In some embodiments, the PAL gene is present in the chromosome and operably linked to a promoter that is induced under low-oxygen or anaerobic conditions. In some embodiments, the PAL gene is present on a plasmid and operably linked to a promoter that is induced by exposure to tetracycline.

In some embodiments, the genetically engineered bacteria comprise a stably maintained plasmid or chromosome carrying the LAAD gene, such that LAAD can be expressed in the host cell, and the host cell is capable of survival and/or growth in vitro, e.g., in medium, and/or in vivo, e.g., in the gut. In some embodiments, the genetically engineered bacteria comprise two or more distinct LAAD genes. In some embodiments, the genetically engineered bacteria comprise multiple copies of the same LAAD gene. In some embodiments, the LAAD gene is present on a plasmid and operably linked to a directly or indirectly inducible promoter. In some embodiments, the LAAD gene is present on a plasmid and operably linked to a promoter that is inducible, e.g., by arabinose or tetracycline. In some embodiments, the LAAD gene is present on a chromosome and operably linked to a directly or indirectly inducible promoter. In some embodiments, the LAAD gene is present in the chromosome and operably linked to a promoter that is induced, e.g., by arabinose. In some embodiments, the LAAD gene is present on a plasmid and operably linked to a promoter that is induced by exposure to tetracycline.

In some embodiments, the genetically engineered bacteria comprise an oxygen-level dependent transcriptional regulator, e.g., FNR, ANR, or DNR, and corresponding promoter from a different bacterial species. The non-native oxygen-level dependent transcriptional regulator and promoter increase the transcription of genes operably linked to said promoter, e.g., PAL, in a low-oxygen or anaerobic environment, as compared to the native transcriptional regulator and promoter in the bacteria under the same conditions. In certain embodiments, the non-native oxygen-level dependent transcriptional regulator is an FNR protein from *N. gonorrhoeae* (see, e.g., Isabella et al., 2011). In some

alternate embodiments, the corresponding wild-type transcriptional regulator is deleted or mutated to reduce or eliminate wild-type activity.

In some embodiments, the genetically engineered bacteria comprise a wild-type oxygen-level dependent transcriptional regulator, e.g., FNR, ANR, or DNR, and corresponding promoter that is mutated relative to the wild-type promoter from bacteria of the same subtype. The mutated promoter enhances binding to the wild-type transcriptional regulator and increases the transcription of genes operably linked to said promoter, e.g., PAL, in a low-oxygen or anaerobic environment, as compared to the wild-type promoter under the same conditions. In some embodiments, the genetically engineered bacteria comprise a wild-type oxygen-level dependent promoter, e.g., FNR, ANR, or DNR promoter, and corresponding transcriptional regulator that is mutated relative to the wild-type transcriptional regulator from bacteria of the same subtype. The mutated transcriptional regulator enhances binding to the wild-type promoter and increases the transcription of genes operably linked to said promoter, e.g., PAL, in a low-oxygen or anaerobic environment, as compared to the wild-type transcriptional regulator under the same conditions. In certain embodiments, the mutant oxygen-level dependent transcriptional regulator is an FNR protein comprising amino acid substitutions that enhance dimerization and FNR activity (see, e.g., Moore et al., 2006).

In some embodiments, the genetically engineered bacteria of the invention comprise multiple copies of the endogenous gene encoding the oxygen level-sensing transcriptional regulator, e.g., the FNR gene. In some embodiments, the gene encoding the oxygen level-sensing transcriptional regulator is present on a plasmid. In some embodiments, the gene encoding the oxygen level-sensing transcriptional regulator and the gene encoding PAL are present on different plasmids. In some embodiments, the gene encoding the oxygen level-sensing transcriptional regulator and the gene encoding PAL are present on the same plasmid. In some embodiments, the gene encoding the oxygen level-sensing transcriptional regulator is present on a chromosome. In some embodiments, the gene encoding the oxygen level-sensing transcriptional regulator and the gene encoding PAL are present on different chromosomes. In some embodiments, the gene encoding the oxygen level-sensing transcriptional regulator and the gene encoding PAL are present on the same chromosome. In some instances, it may be advantageous to express the oxygen level-sensing transcriptional regulator under the control of an inducible promoter in order to enhance expression stability. In some embodiments, expression of the transcriptional regulator is controlled by a different promoter than the promoter that controls expression of the gene encoding the phenylalanine-metabolizing enzyme. In some embodiments, expression of the transcriptional regulator is controlled by the same promoter that controls expression of the phenylalanine-metabolizing enzyme. In some embodiments, the transcriptional regulator and the phenylalanine-metabolizing enzyme are divergently transcribed from a promoter region.

In some embodiments, the genetically engineered bacteria of the invention produce PAL under exogenous environmental conditions, such as the low-oxygen environment of the mammalian gut, to reduce blood phenylalanine by at least about 1.5-fold, at least about 2-fold, at least about 3-fold, at least about 4-fold, at least about 5-fold, at least about 6-fold, at least about 7-fold, at least about 8-fold, at least about 9-fold, at least about 10-fold, at least about 15-fold, at least about 20-fold, at least about 30-fold, at least about 40-fold,

or at least about 50-fold as compared to unmodified bacteria of the same subtype under the same conditions. Certain unmodified bacteria will not have appreciable levels of phenylalanine processing. In embodiments using genetically modified forms of these bacteria, PAL-mediated processing of phenylalanine will be appreciable under exogenous environmental conditions. Phenylalanine may be measured by methods known in the art, e.g., blood sampling and mass spectrometry. In some embodiments, cinnamate is measured by methods known in the art to assess PAL activity. Cinnamate production is directly correlated with phenylalanine degradation, and in some embodiments, that cinnamate may be used as an alternative biomarker for strain activity (FIG. 16B). Cinnamate can be further degraded to hippuric acid by liver enzymes; both can be measured as described in Example 24-26. In some embodiments, PAL expression is measured by methods known in the art to assess PAL activity.

In some embodiments, the genetically engineered bacteria of the invention produce LAAD, to reduce blood phenylalanine by at least about 1.5-fold, at least about 2-fold, at least about 3-fold, at least about 4-fold, at least about 5-fold, at least about 6-fold, at least about 7-fold, at least about 8-fold, at least about 9-fold, at least about 10-fold, at least about 15-fold, at least about 20-fold, at least about 30-fold, at least about 40-fold, or at least about 50-fold as compared to unmodified bacteria of the same subtype under the same conditions. Certain unmodified bacteria will not have appreciable levels of phenylalanine processing. In embodiments using genetically modified forms of these bacteria, LAAD-mediated processing of phenylalanine will be appreciable under exogenous environmental conditions. Phenylalanine may be measured by methods known in the art, e.g., blood sampling and mass spectrometry. Pyruvic acid and phenylpyruvate, the LAAD generated degradation products can be measured using mass spectrometry as described in Examples 24-26, and can be used as an additional readout of LAAD activity.

In some embodiments, the PME, e.g., PAL, LAAD, and/or PAH, is expressed on a low-copy plasmid. In some embodiments, the low-copy plasmid may be useful for increasing stability of expression. In some embodiments, the low-copy plasmid may be useful for decreasing leaky expression under non-inducing conditions. In some embodiments, the PME, e.g., PAL, LAAD, and/or PAH, is expressed on a high-copy plasmid. In some embodiments, the high-copy plasmid may be useful for increasing the PME, e.g., PAL, LAAD, and/or PAH, expression, thereby increasing the metabolism of phenylalanine and reducing hyperphenylalaninemia. In some embodiments, a genetically engineered bacterium comprising a the PME, e.g., PAL, LAAD, and/or PAH, expressed on a high-copy plasmid does not increase phenylalanine metabolism or decrease phenylalanine levels as compared to a genetically engineered bacterium comprising the same PME, e.g., PAL, LAAD, and/or PAH, expressed on a low-copy plasmid in the absence of heterologous pheP and additional copies of a native pheP. Genetically engineered bacteria comprising the same the PME gene, e.g., PAL, LAAD, and/or PAH gene on high and low copy plasmids were generated. For example, either PAL1 or PAL3 on a high-copy plasmid and a low-copy plasmid were generated, and each metabolized and reduced phenylalanine to similar levels (FIG. 15). Thus, in some embodiments, the rate-limiting step of phenylalanine metabolism is phenylalanine availability (see, e.g., FIG. 16). In these embodiments, it may be advantageous to increase phenylalanine transport into the cell, thereby enhancing

phenylalanine metabolism. In conjunction with pheP, even low-copy PAL plasmids are capable of almost completely eliminating Phe from a test sample (see, e.g., FIG. 16A). Furthermore, in some embodiments, that incorporate pheP, there may be additional advantages to using a low-copy PAL-expressing plasmid in conjunction in order to enhance the stability of PAL expression while maintaining high phenylalanine metabolism, and to reduce negative selection pressure on the transformed bacterium. In alternate embodiments, the phenylalanine transporter is used in conjunction with the high-copy plasmid.

In some embodiments, a transporter may not increase phenylalanine degradation. For example, *Proteus mirabilis* LAAD is localized to the plasma membrane, with the enzymatic catalysis occurring in the periplasm. Phenylalanine can readily traverse the outer membrane without the need of a transporter. Therefore, in embodiments, in which the genetically engineered bacteria express LAAD, a transporter may not be needed or improve phenylalanine metabolism.

In some embodiments, the PME, e.g., PAL, LAAD, and/or PAH, gene is expressed on a chromosome. In some embodiments, expression from the chromosome may be useful for increasing stability of expression of the PME. In some embodiments, the PME gene, e.g., PAL, LAAD, and/or PAH gene(s), is integrated into the bacterial chromosome at one or more integration sites in the genetically engineered bacteria. In some embodiments, the PME gene, e.g., PAL, LAAD, and/or PAH gene(s) is inserted into the bacterial genome at one or more of the following insertion sites in *E. coli* Nissle: malE/K, insB/I, araC/BAD, lacZ, agaI/rsmI, thyA, and malP/T. Any suitable insertion site may be used (see, e.g., FIG. 36). The insertion site may be anywhere in the genome, e.g., in a gene required for survival and/or growth, such as thyA (to create an auxotroph); in an active area of the genome, such as near the site of genome replication; and/or in between divergent promoters in order to reduce the risk of unintended transcription, such as between AraB and AraC of the arabinose operon. In some embodiments, more than one copy, e.g., two, three, four, five, six, seven, eight, nine, ten or more copies of the PME gene, e.g., PAL, PAH, and/or LAAD is integrated into the bacterial chromosome at one or more integration sites in the genetically engineered bacteria. The more than one copy of a PME gene may be more than one copy of the same PME gene or more than one copy of different PME genes.

Exemplary constructs are shown in 4-13 below. Table 4 shows the sequence of an exemplary construct comprising a gene encoding PheP and an FNR promoter sequence for chromosomal insertion (SEQ ID NO: 21), with the pheP sequence underlined and the FNR promoter sequence bolded. Table 5 shows the sequence of an exemplary construct comprising a gene encoding PAL1 and an FNR promoter sequence on a high-copy plasmid (SEQ ID NO: 22), with the PAL1 sequence underlined and the FNR promoter sequence bolded. Table 6 shows the sequence of an exemplary construct comprising a gene encoding PAL3 and an FNR promoter sequence on a high-copy plasmid (SEQ ID NO: 23), with the PAL3 sequence underlined and the FNR promoter sequence bolded. Table 7 shows the sequence of an exemplary construct comprising a gene encoding PAL1 and a Tet promoter sequence on a high-copy plasmid (SEQ ID NO: 24), with the PAL1 sequence underlined and the Tet promoter sequence bolded. Table 8 shows the sequence of an exemplary construct comprising a gene encoding PAL3 and a Tet promoter sequence on a high-copy plasmid (SEQ ID NO: 25), with the PAL3 sequence underlined and the Tet

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promoter sequence **bolded**. Table 9 shows the sequence of an exemplary construct comprising a gene encoding PAL1 and an FNR promoter sequence on a low-copy plasmid (SEQ ID NO: 26), with the PAL1 sequence underlined and the FNR promoter sequence **bolded**. Table 10 shows the sequence of an exemplary construct comprising a gene encoding PAL3 and an FNR promoter sequence on a low-copy plasmid (SEQ ID NO: 27), with the PAL3 sequence underlined and the FNR promoter sequence **bolded**. Table 11 shows the sequence of an exemplary construct comprising a gene encoding PAL1 and a Tet promoter sequence on a low-copy plasmid (SEQ ID NO: 28), with the PAL1 sequence underlined and the Tet promoter sequence **bolded**. Table 12 shows the sequence of an exemplary construct comprising a gene encoding PAL3 and a Tet promoter sequence on a low-copy plasmid (SEQ ID NO: 29), with the PAL3 sequence underlined and the Tet promoter sequence **bolded**. Table 13 shows the sequence of an exemplary construct comprising a gene encoding PheP, a gene coding TetR, and a Tet promoter sequence for chromosomal insertion (SEQ ID NO: 30), with the pheP sequence underlined, the TetR sequence **boxed**, and the FNR promoter sequence **bolded**.

TABLE 4

Nucleotide sequences of FNR promoter-PheP construct (SEQ ID NO: 21)	
CTCTTGATCGTTATCAATTCCACGCTGTTTCAGAGCGTTACCTTGCCCTT	
AAACATTAGCAATGTCGATTTATCAGAGGGCCGACAGGCTCCACAGGAGA	
AAACCGATGAAAAACGCGTCAACCGTATCGGAAGATACTGCGTCGAATCAA	
GAGCCGACGCTTCATCGCGGATTACATAACCGTCATATTCAACTGATTGCG	
TTGGGTGGCGCAATTGGTACTGGTCTGTTTCTTGGCATTGGCCCGGCGATT	
CAGATGGCGGGTCCGGCTGATTGCTGGGCTACGCGCTCGCCGGGATCATC	
GCTTTCCTGATTATGCGCCAGCTTGGCGAAATGGTGGTTAGGAGCCGGTA	
TCCGGTTCAATTTGCCCACTTTGCCATATAAACTAGGGACCGTTTGGCGGC	
TTCTCTCTGGCTGGAACCTACTGGGTAATGTTCTGCTGGTGGGAATGGCA	
GAGCTGACCGCTGCGGGCATCTATATGCAGTACTGGTTCCCGGATGTTCCA	
ACGTGGATTGGGGTGCAGCTTCTTTATTATCATCAACGCCGTTAACCTG	
GTGAACGTGCGCTTATATGGCGAAACCGAGTTCTGGTTTGGCTTGATTAA	
GTGCTGGCAATCATCGGTATGATCGGCTTGGCCTGTGGCTGCTGTTTTCT	
GGTCACGGCGGCGAGAAAGCCAGTATCGACAACCTCTGGCGCTACGGTGGT	
TTCTTCGCCACCGGCTGGAATGGGCTGATTTTGTGCTGGCGGTAATTATG	
TTCTCCTTCGGCGGTCTGGAGCTGATTGGGATTACTGCCGCTGAAGCGCGC	
GATCCGGAAGAAAGCATTCAGAAAGCGGTAATCAGGTGGTGTATCGCATC	
CTGCTGTTTTACATCGGTTCACTGGTGGTTTTACTGGCGCTCTATCCGTGG	
GTGGAAGTGAAATCCAACAGTAGCCGTTTGTGATGATTTCCATAATCTC	
GACAGCAACGTGGTAGCTTCTGCGCTGAACTTCGTCACTTCTGGTAGCATCG	
CTGTCACTGTATAACAGCGGGTTTTACTTAACAGCCGATGCTGTTTGGC	
CTTCTGTGACAGGTAATGCGCCGAAGTTTGGTACTCGCGTCAGCCGTCGC	
GGTGTGCCGATTAACTCGCTGATGCTTCCCGAGCGATCACTTCGCTGGTG	
GTGTTAATCAACTATCTGCTGCCGCAAAAGCGTTGGTCTGCTGATGGCG	

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TABLE 4-continued

Nucleotide sequences of FNR promoter-PheP construct (SEQ ID NO: 21)	
CTGGTGGTAGCAACGCTGCTGTTGAACTGGATTATGATCTGTCTGGCGCAT	5
CTGCGTTTTCTGTCAGCGATGCGACGTCAGGGCGGTGAAACACAGTTTAAG	
GCGCTGCTCTATCCGTTCCGGCAACTATCTCTGCATTGCCTTCTCCGGCATG	
ATTTTGCTGCTGATGTGCACGATGGATGATATGCGCTTGTGACGATCCTG	10
CTGCCGGTGTGGATTGTATTCCTGTTTATGGCATTAAACCGCTGCGTCGG	
AAATAA	15

TABLE 5

Nucleotide sequences of FNR promoter-PAL1 construct, high-copy (SEQ ID NO: 22)	
CTCTTGATCGTTATCAATTCCACGCTGTTTCAGAGCGTTACCTTGCCCTT	20
AAACATTAGCAATGTCGATTTATCAGAGGGCCGACAGGCTCCACAGGAGA	
AAACCGATGAAAACACTATCACAGGCCCAATCTAAACTTCTTCACAGCAA	25
TTCAGCTTTACCGGAACTCGTCTGCGAATGTAATTATCGGCAATCAAAAG	
CTGACCATTATGATGTAGTCTGCGTTGCCCGGAATGGCACTTTGGTGTCA	
CTGACGAACAATACCGACATTCGCAAGGTATTCAGCTAGCTGCGATTAT	30
ATCAATAACGCCGTTGAATCTGGCGAGCCAATCTACGGGGTAACAAGCGGT	
TTGGTGGGATGGCGAACGTTGCCATTAGCCGTGAACAGCGAGCGAACTT	
CAGACCAACCTCGTTTGGTTCCTAAAGACAGGAGCTGGTAATAAGTTACCT	35
CTGGCTGACGTAAGAGCCGCGATGCTGCTTCGCGCTAATAGTCACATGCGC	
GGCGCCAGTGGTATCCGCTTGTAGCTTATCAAGAGGATGGAATCTTCCCTC	
AACGCGGGTGTACACCATATGTTTATGAGTTTGGTAGTATCGGAGCCAGT	40
GGTGTCTTGTTCCTGAGTTATATTACGGGTTTATTGATTGGTTTAGAC	
CCGTCCCTTTAAAGTGGATTTTAAACGGGAAAGAAATGAGACGCCCCGACCGCT	
TTACGACAGCTTAATCTGAGCCCACTTACTTTGCTCCCTAAAGAGGCTCTT	45
GCCATGATGAATGGCACCTCTGTGATGACTGGAATTGCCGCAATTGTGTG	
TATGACACGCAGATCCTAACGGCCATTGCCATGGGTGTTACGCGTTGGAC	
ATTCAAGCCCTGAATGGTACAAACAGTCTGTTTCACTCCGTTTATCCATAAT	50
TCAAAACCCCATCCGGGACAGCTTTGGGCTGCTGATCAGATGATCTCACTC	
CTGGCCAATAGTCAACTGGTTCGGGACGAGCTCGACGGCAACATGATTAT	
CGCGATCATGAGCTCATCCAGGACCGGTATCACTTCGTGTCTCCACAA	55
TACCTGGGGCTATCGTTGATGGTATATCTCAAATGCGAAGCAAAATGAA	
ATTGAGATCAATAGCGTAACCGACAACCCGCTTATCGATGTTGATAATCAG	
GCCTCTTATACGGTGGCAATTTTCTGGGCCAGTATGTTGGTATGGGGATG	
GATCACCTGCGGTACTATATTGGGCTTCTGGCTAAACATCTTGATGTGCAG	60
ATTGCCCTTATTAGCTTCACCAGAATTTTCAAATGGAGTCCGCCATCATTG	
CTCGGTAAACAGAGAAAGGAAAGTAAATATGGGCCCTTAAGGGCTTCAGATA	
TGTGGTAACTCAATCATGCCCTCTGACCTTTTATGGGAACCAATTGCT	65

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TABLE 5-continued

Nucleotide sequences of FNR promoter-PAL1 construct, high-copy (SEQ ID NO: 22)	
<u>GATCGTTTTCCGACACATGCTGAACAGTTTAACCAAAACATTAAC</u>	
<u>GGCTATACATCCGCGACGTTAGCGCGTCGGTCCGTGGATATCTTCCAGAAT</u>	
<u>TATGTTGCTATCGCTCTGATGTTCCGGCTACAGGCCGTTGATTTGCGCACT</u>	
<u>TATAAAAAACCGGTCACTACGATGCTCGGGCTTGCCGTGCGCTGCCACC</u>	
<u>GAGCGGCTTTATAGCGCGTACGTGATGTTGGGTGAGAAACCGACGTCG</u>	
<u>GACCGCCCTATATTTGGAATGATAATGAACAAGGCTGGATGAACACATC</u>	
<u>GCCCGGATATCTGCCGATATTGCCCGCGAGGTGTCATCGTCCAGGCGGTA</u>	
<u>CAAGACATACTTCCTTGCTGCATTAAGCTTGGCGTAATCATGGTCATAGC</u>	
TGTTTCCTGTGTGAAATTGTTATCCGCTCACAATTCACACAACATACGAG	
CCGGAAGCATAAAGTGTAAGCCTGGGGTGCCTAATGAGTGAGCTAACTCA	
CATTAATTGCGTTGCGCTCACTGCCCGCTTCCAGTCGGGAAACCTGTCGT	
GCCAGCTGCATTAATGAATCGCCAACGCGCGGGGAGAGGCGGTTTGCCTA	
TTGGGCGCTCTTCCGCTTCTCGCTCACTGACTCGCTCGCTCGGTCGTTTC	
GGCTCGCGCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCA	
CAGAATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAA	
AGGCCAGGAACCGTAAAGAGCGCGGTTGCTGGCGTTTTTCCATAGGCTCC	
GCCCCCTGACGAGCATCAGGAAATCGACGCTCAAGTCAGAGGTGGCGAA	
ACCCGACAGGACTATAAGATACAGGCGTTTTCCCTGGAAGCTCCCTCG	
TGCGCTCTCTGTTCCGACCTGCGGCTTACCGGATACCTGTCCGCTTTC	
TCCCTTGGGAAGCGTGGCGCTTCTCATAGCTCACGCTGTAGGTATCTCA	
GTTGCGGTGTAGTCTGCTTCCGCTCAAGCTGGGCTGTGTGCACGAACCCCG	
TTCAGCCCCGACCGCTGCGCTTATCCGGTAACATATCGTCTTGAGTCCAACC	
CGGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTA	
GCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTA	
ACTACGGCTACACTAGAAGAAGTATTTGGTATCTGCGCTCTGTGAAGC	
CAGTTACCTTCGGAAGAGTGTGGTAGCTCTTGATCCGGCAACAAACCA	
CCGCTGGTAGCGGTGGTTTTTTTGTGTTGCAAGCAGCAGATTACGCGCAGAA	
AAAAAGGATCTCAAGAAGATCTTTGATCTTTTCTACGGGCTCTGACGCTC	
AGTGAACGAAACTACGTTAAGGGATTTTGGTCATGAGATTATCAAAAA	
GGATCTTCACCTAGATCTTTTAAATTAAAAATGAAGTTTAAATCAATCT	
AAAGTATATATGAGTAACTTGGTCTGACAGTTACCAATGCTTAATCAGTG	
AGGCACCTATCTCAGCGATCTGTCTATTTCTGTTTCCATAGTTGCTGAC	
TCCCGCTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCA	
GTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTATCAG	
CAATAAACAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCTGCAACTT	
TATCCGCCTCCATCCAGTCTATTAATTGTTGCCGGAAGCTAGAGTAAGTA	
GTTGCCAGTTAATAGTTTGCAGCAAGTGTGCTGCTACAGGCATCG	
TGGTGTACGCTCGCTTGGTATGGCTTCATTACGCTCCGGTTCCTCAAC	

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TABLE 5-continued

Nucleotide sequences of FNR promoter-PAL1 construct, high-copy (SEQ ID NO: 22)	
<u>GATCAAGGCGAGTTACATGATCCCCATGTTGTGCAAAAAAGCGGTTAGCT</u>	
<u>CCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGAGTGTATCAC</u>	
<u>TCATGGTTATGGCAGCACTGCATAATTCTTCTTACTGTCATGCCATCCGTAA</u>	
<u>GATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGT</u>	
<u>GTATGCGCGACCGAGTTGCTCTTGCCCGCGTCAATACGGGATAATACCG</u>	
<u>CGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAAAGTTCTTCGG</u>	
<u>GGCGAAAACCTCTCAAGGATCTTACCGTGTGAGATCCAGTTCGATGTAAC</u>	
<u>CCACTCGTGCACCCAAGTATCTTCAGCATCTTTTACTTTACCAGCGTTTT</u>	
<u>CTGGGTGAGCAAAAAAGGAAGGCAAAATGCCGCAAAAAAGGAATAAGGG</u>	
<u>CGACACGGAATGTTGAATACTCATACTCTTCTTTTCAATATTATTGAA</u>	
<u>GCATTTATCAGGTTATTGTCTCATGAGCGGATACATATTTGAATGTATTT</u>	
<u>AGAAAAATAAACAAATAGGGGTTCCGCGCACATTTCCCGAAAAGTGCCAC</u>	
<u>CTGACGTCTAAGAAACATTATTATCATGACATTAACCTATAAAAAATAGGC</u>	
<u>GTATCACGAGGCCCTTTCGTCTCGCGCGTTTCGGTGATGACGGTGAAAAAC</u>	
<u>TCTGACACATGCAGCTCCCGAGACGGTCACAGCTTGTCTGTAAGCGGATG</u>	
<u>CCGGGAGCAGACAAGCCCGTCAGGCGCGTCAGCGGGTGTGGCGGGTGTC</u>	
<u>GGGGTGGCTTAACTATGCGGCATCAGAGCAGATTGTACTGAGAGTGACAC</u>	
<u>ATATGCGGTGTGAAATACCGCACAGATGCGTAAGGAGAAAAATACCGCATCA</u>	
<u>GGCGCCATTTCGCCATTTCAGGCTGCGCAACTGTGGGAAGGGCGATCGGTGC</u>	
<u>GGGCTCTTCTGCTATTACGCCAGCTGGCGAAAGGGGATGTGCTGCAAGGC</u>	
<u>GATTAAGTTGGGTAACGCCAGGGTTTTTCCAGTCACGACGTT</u>	

TABLE 6

Nucleotide sequences of FNR promoter-PAL3 construct, high-copy (SEQ ID NO: 23)	
<u>CTCTTGATCGTTATCAATTCCCACGCTGTTTCAGAGCGTTACCTTGCCCTT</u>	
<u>AAACATTAGCAATGTCGATTTATCAGAGGGCCGACAGGCTCCCACAGGAGA</u>	
<u>AAACCGATGAAAGCTAAAGATGTTTCAGCCAACCATTTATTATTAATAAAAA</u>	
<u>GGCCTTATCTCTTTGGAAGATATCTATGACATTGCGATAAAACAAAAAAA</u>	
<u>GTAGAAATATCAACGAGATCACTGAACCTTTTACGCGATGGTCGTGAAAA</u>	
<u>TTAGAGGAAAAATTAAATTGAGGAGGTTATATATGAATCAATACAGGA</u>	
<u>TTTGAGGGAATGCCAATTTAGTTGTGCCATTTGAGAAAAATCGCAGAGCAT</u>	
<u>CAGCAAAATCTGTTAACTTTTCTTCTGCTGGTACTGGGGACTATATGTCC</u>	
<u>AAACCTTGATTAAGAGCGTCACAATTTACTATGTTACTTTCTGTTTGCAAA</u>	
<u>GGTGGTCTGCAACCAGACCAATTGTGCTCAAGCAATTGTTGATCATATT</u>	
<u>AATCATGACATTGTTCTCTGGTTCCTCGCTATGGCTCAGTGGGTGCAAGC</u>	
<u>GGTGATTTAATTCCTTTATCTTATATGACAGGATGTTGTTGATCGGC</u>	
<u>AAAGTTTATTATATGGGCGCAGAAATTGACGCTGCTGAAGCAATTAAACGT</u>	
<u>GCAGGGTTGACACCATTTATCGTTAAAGCCAAAGAAGTCTTGCTCTGATT</u>	

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TABLE 6-continued

Nucleotide sequences of FNR promoter-PAL3 construct, high-copy (SEQ ID NO: 23)	
<u>AACGGCACCCGGTAAATGTCAGGAATCAGTGCAATCACCGTCATTAACTG</u>	
<u>GAAAAACTATTTAAAGCCTCAATTTCTGCGATTGCCCTTGCTGTTGAAGCA</u>	
<u>TTACTTGCACTCATGAACATTATGATGCCCGGATTCAACAAGTAAAAAT</u>	
<u>CATCCTGGTCAAAACGCGGTGGCAAGTGCAATTGCGTAATTTATTGGCAGGT</u>	
<u>TCAACGCAAGTTAATCTATTATCTGGGGTTAAAGAACAAGCCAATAAAGCT</u>	
<u>TGTCGTCAATCAAGAAATACCCAATAAATGATACCTTACAGGAAGTTTAT</u>	
<u>TCAATTGCGTGTGCACCACAAGTATTAGGTATAGTGCCAGAATCTTTAGCT</u>	
<u>ACCGCTCGGAAAAATATTGGAACGGGAAGTTATCTCAGCTAATGATAATCCA</u>	
<u>TTGATAGATCCAGAAAAATGGCGATGTTCTACACGGTGAAATTTTATGGGG</u>	
<u>CAATATGTCGCCCCGAACAATGGATGCATTAAACATGGATATTGCTTTAATT</u>	
<u>GCCAATCATCTTCACGCCATTGTGGCTCTTATGATGGATAACCGTTTCTCT</u>	
<u>CGTGGATTACCTAATTCAGTCCGACACCCGGCATGTATCAAGGTTTTT</u>	
<u>AAAGGCGTCCAACCTTCTCAAACCGCTTTAGTTGCTGCAATTGCGCATGAT</u>	
<u>TGTGCTGCATCAGGTATTACATACCTCGCCACAGAACAATACAATCAAGAT</u>	
<u>ATTGTCAGTTTAGTCTGCATGCCGCTCAAGATGTTTTAGAGATGGAGCAG</u>	
<u>AAATTACGCAATATTGTTTCAATGACAATTCTGGTAGTTTGTGAGGCCATT</u>	
<u>CATCTTCGCGGCAATATTAGTGAAATTGCGCCTGAAACTGTAAATTTTAC</u>	
<u>CATGCAGTACGCGAAATCAGTTCCTCTTGATCACTGATCGTGGTGGAT</u>	
<u>GAAGATATAATCCGCATTGCGGATGCAATTATTAATGATCAACTTCCTCTG</u>	
<u>CCAGAAATCATGCTGGAAGAATAAGCTTGGCGTAATCATGGTCATAGCTGT</u>	
<u>TTCTGTGTGAAATGTTATCCGCTCACAATCCACACAACATACGAGCCG</u>	
<u>GAAGCATAAAGTGTAAAGCCTGGGTGCGCTAATGAGTGAGCTAATCAGAT</u>	
<u>TAATTGCGTTCGCGTCACTGCCCCGCTTCCAGTCGGGAAACCTGTCGTGCC</u>	
<u>AGCTGCATTAATGAATCGGCCAACGCGCGGGAGAGGCGGTTTGCCTATTG</u>	
<u>GGCGCTCTTCGCTTCCCTGCTCACTGACTGCTGCGCTCGGTGCTTCGGC</u>	
<u>TGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAAACGTTATCCACAG</u>	
<u>AATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGG</u>	
<u>CCAGGAACCGTAAAAGGCCGCGTTCGCTGGCGTTTTTCCATAGGCTCCGCC</u>	
<u>CCCCTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAACC</u>	
<u>CGACAGGACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTCGTGC</u>	
<u>GCTCTCCTGTTCCGACCTGCGCTTACCGGATACCTGTCGCCCTTTCTCC</u>	
<u>CTTCGGGAAGCGTGGCGCTTTCTCATAGCTCAGCTGTAGGTATCTCAGTT</u>	
<u>CGGTGTAGGTGTTGCTCCAAGCTGGGCTGTGTGCACGAACCCCCGTTTC</u>	
<u>AGCCCGACCGCTGCGCTTATCCGGTAACCTATCGTCTTGTAGTCCAAACCGG</u>	
<u>TAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCA</u>	
<u>GAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACT</u>	
<u>ACGGCTACACTAGAAGACAGTATTGTGATCTGCGCTCTGTGAAGCCAG</u>	
<u>TTACCTTCGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAAACAAACCACCG</u>	

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TABLE 6-continued

Nucleotide sequences of FNR promoter-PAL3 construct, high-copy (SEQ ID NO: 23)	
<u>CTGGTAGCGGTGGTTTTTTTGTGTTGCAAGCAGCAGATTACGCGCAGAAAAA</u>	
<u>AAGGATCTCAAGAAGATCCTTTGATCTTTTCTACGGGTCTGACGCTCAGT</u>	
<u>GGAACGAAAACTCACGTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGA</u>	
<u>TCTTCACCTAGATCCTTTTAAATTAATAATGAAGTTTTAAATCAATCTAAA</u>	
<u>GTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGG</u>	
<u>CACCTATCTCAGCGATCTGTCTATTTTGGTTCATCCATAGTTGCCTGACTCC</u>	
<u>CCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTG</u>	
<u>CTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTTATCAGCAA</u>	
<u>TAAACCGAGCCAGCCGAAGGGCCGAGCGCAGAGTGGTCTGCAACTTTAT</u>	
<u>CCGCTCCATCCAGTCTATTAATGTTGCCGGGAAGCTAGAGTAAGTAGTT</u>	
<u>CGCCAGTTAATAGTTTGCGCAACGTTGTTGCCATTGCTACAGGCATCGTGG</u>	
<u>TGTCACGCTCGTCGTTTGGTATGGCTTCATTCAGCTCCGGTTCCCAACGAT</u>	
<u>CAAGCGAGTTACATGATCCCCATGTTGTGCAAAAAAGCGGTTAGCTCCT</u>	
<u>TCGGTCTCCGATCGTTGTGAGAAGTAAGTTGGCCGCGAGTTTATCACTCA</u>	
<u>TGGTTATGGCAGCACTGCATAATTCTCTTACTGTGATGCCATCCGTAAGAT</u>	
<u>GCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTA</u>	
<u>TGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGC</u>	
<u>CACATAGCAGAACTTTAAAGTGCTCATATTGGAACAGTTCTTTCGGGGC</u>	
<u>GAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCA</u>	
<u>CTCGTGACCCCACTGATCTTCAGCATCTTTTACTTTTACCAGCGTTTCTG</u>	
<u>GGTGAGCAAAAAAGGAAGGCAAAATGCCGCAAAAAAGGGAATAAGGGCGA</u>	
<u>CACGAAATGTTGAATACTCATACTCTTCCTTTTCAATATTATTGAAGCA</u>	
<u>TTTATCAGGGTTATTGTCTCATGAGCGGATACATATTGAATGTATTAGA</u>	
<u>AAAATAAACAAATAGGGGTTCCGCGCACATTTCCCCGAAAAGTGCCACCTG</u>	
<u>ACGTCTAAGAAACCATTTATTTATCATGACATTAACCTATAAAAAATAGGCGTA</u>	
<u>TCACGAGGCCCTTTCGTCTCGCGCGTTTCGGTGATGACGGTGAAAACCTCT</u>	
<u>GACACATGCAGCTCCCGGAGACGGTCACAGCTTGTCTGTAAGCGGATGCCG</u>	
<u>GGAGCAGACAAGCCCGTCAGGGCGCGTCAGCGGGTGTGGCGGGTGTGCGG</u>	
<u>GCTGGCTTAACATATGCGGCATCAGAGCAGATTGTACTGAGAGTGACACCATA</u>	
<u>TGCGGTGTGAAATACCGCACAGATGCGTAAGGAGAAAAATACCGCATCAGGC</u>	
<u>GCCATTGCGCATTCAGGCTGCGCAACTGTTGGGAAGGGCGATCGGTGCGGG</u>	
<u>CCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGATGTGCTGCAAGGCGAT</u>	
<u>TAAGTTGGGTAAACGCCAGGGTTTTCCAGTCACGACGTT</u>	

TABLE 7

Nucleotide sequences of Tet promoter-PAL1 construct, high-copy (SEQ ID NO: 24)	
CTCTAGAAATAATTTTGTTTAACTTTAAGAAGGAGATATACATATGAAAACACTATCACA	
GGCCCAATCTAAACTTCTTCACAGCAATTCAGCTTTACCGGAACTCGTCTGCGAATGT	
AATTATCGGCAATCAAAGCTGACCATTAATGATGTAGCTCGCGTTGCCCGAATGGCAC	
TTTGGTGTCACTGACGAACAATACCGACATTCTGCAAGGTATTCAAGCTAGCTGCGATTA	
TATCAATAACGCCGTTGAATCTGGCGAGCCAATCTACGGGGTAACAAGCGGTTTGGTGG	
GATGGCGAACGTTGCCATTAGCCGTGAACAGGCGAGCGAACTTCAGACCAACCTCGTTTG	
GTTCCCTAAGACAGGAGCTGGTAATAAGTTACCTCTGGCTGACGTAAGAGCCGCGATGCT	
GCTTCGCGCTAATAGTCACATGCGCGGCGCCAGTGGTATCCGTCTTGAGCTTATCAAGAG	
GATGGAAATCTTCTCAACGCGGGTGTACACCATATGTTTATGAGTTGGTAGTATCGG	
AGCCAGTGGTGATCTTGTTCCTTGAGTTATATTACGGGTTCAATTGATTGGTTTAGACCC	
GTCTTTTAAAGTGGATTTTAAACGGGAAAGAAATGGACGCCCCGACCGCTTTACGACAGCT	
TAATCTGAGCCCACTTACTTTGCTCCCTAAGAAGGTCTTGCCATGATGAATGGCACCTC	
TGTGATGACTGGAATTGCCGCGAATTGTGTGTATGACACGCAGATCCTAACGGCCATTGC	
CATGGGTGTTACGCGTGGACATTCAAGCCCTGAATGGTACAAACCAGTCGTTTCATCC	
GTTTATCCATAATTCAAACCCCATCCGGGACAGCTTTGGGCTGCTGATCAGATGATCTC	
ACTCTGGCCAATAGTCAACTGGTTCGGGACGAGCTCGACGGCAACATGATTATCGCGA	
TCATGAGCTCATCCAGGACCGGTATTCCTTCGTTGTCTCCACAATACCTGGGGCTAT	
CGTTGATGTTATATCTCAAATTGCGAAGCAAATTGAAATTGAGATCAATAGCGTAACCGA	
CAACCCGCTTATCGATGTTGATAATCAGGCCTCTTATCAGGTGGCAATTTCTGGGCCA	
GTATGTTGGTATGGGGATGGATCACCTGCGTACTATATTGGGCTTCTGGCTAAACATCT	
TGATGTGCAGATTGCCTTATTAGCTTCACCAGAATTTCAAATGGACTGCCGCCATCATT	
GCTCGGTAACAGAGAAAGGAAAGTAAATATGGGCCTTAAGGCCTTCAGATATGTGGTAA	
CTCAATCATGCCCCCTCTGACCTTTTATGGGAACTCAATTGCTGATCGTTTCCGACACA	
TGCTGAACAGTTTAAACAAAACATTAATCAGGGCTATACATCCGCGACGTTAGCGCG	
TCGGTCCGTGGATATCTTCCAGAATTATGTTGCTATCGCTCTGATGTTCCGCGTACAGGC	
CGTTGATTGCGCACTTATAAAAAACCGGTCACTACGATGCTCGGGCTTGCTGTGCGC	
TGCCACCGAGCGGCTTTATAGCGCCGTACGTCATGTTGTGGGTGAGAAACCGACGTCGGA	
CCGCCCCATATTTGGAATGATAATGAACAAGGGCTGGATGAACACATCGCCCCGATATC	
TGCCGATATTGCCGCCGGAGGTGTCATCGTCCAGGCGGTACAAGACATACTTCCTTGCCT	
GCATTAAGCTTGCGTAATCATGGTCATAGCTGTTTCTGTGTGAAATTGTTATCCGCTC	
ACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTAAGCCTGGGGTGCCTAATGA	
GTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCCGCTTCCAGTCGGGAAACCTG	
TCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGAGAGGCGGTTTGCATATTGGG	
CGCTCTTCGCTTCTCGCTCACTGACTCGCTGCGCTCGGTGCTTCGCTGCGCGAGCG	
GTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGGATAACGCAGGA	
AAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAAAAAGGCCGCTTGCTG	
GCGTTTTTCCATAGGCTCCGCCCCCTGACGAGCATCAAAAAATCGACGCTCAAGTCAG	
AGGTGGCGAAACCCGACAGGACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTC	
GTGCGCTCTCTGTTCCGACCTGCGCTTACCGGATACCTGTCCGCTTTCTCCCTTCG	

TABLE 7-continued

Nucleotide sequences of Tet promoter-PAL1 construct, high-copy (SEQ ID NO: 24)
GGAAGCGTGGCGCTTTCTCATAGCTACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTT CGCTCCAAGCTGGGCTGTGTGCACGAACCCCCGTTACGCCGACCGCTGCGCCTTATCC GGTAACTATCGTCTTGAGTCCAACCCGGTAAGACACGACTTATCGCCACTGGCAGCAGCC ACTGGTAACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGG TGGCCTAACTACGGCTACACTAGAAGAACAGTATTTGGTATCTGCGCTCTGCTGAAGCCA GTTACCTTCGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAAACAAACCACCGCTGGTAGC GGTGGTTTTTTTGGTTTGAAGCAGCAGATTACGCGCAGAAAAAAGGATCTCAAGAAGAT CCTTTGATCTTTTCTACGGGGTCTGACGCTCAGTGAACGAAAACTCACGTTAAGGGATT TTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGT TTAAATCAATCTAAGTATATATGAGTAACTTGGTCTGACAGTTACCAATGCTTAATC AGTGAGGCACCTATCTCAGCGATCTGTCTATTTCTGTTTCCATAGTTGCTGACTCCCC GTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAATGATA CCGCGAGACCCACGCTACCGGCTCCAGATTTATCAGCAATAAACAGCCAGCCGGAAGG GCCGAGCGCAGAAGTGGTCTGCAACTTTATCCGCTCCATCCAGTCTATTAATTGTTGC CGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCACAACGTTGTTGCCATTGCT ACAGGCATCGTGGTGTACGCTCGTCGTTTGGTATGGCTTCATTACGCTCCGGTTCCCAA CGATCAAGCGCAGTTACATGATCCCCATGTTGTGCAAAAAAGCGGTTAGCTCCTTCGGT CCTCCGATCGTTGTGAGAAGTAAGTTGGCCGAGTGTTATCACTCATGGTTATGGCAGCA CTGCATAATTCTCTTACTGTCTGTCATCCGTAAGATGCTTTTCTGTGACTGGTGAAGTAC TCAACCAAGTCATTCTGAGAATAGTGATGCGGCGACCGAGTTGCTCTTGCCCGCGTCA ATACGGGATAATACCGGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAACCGT TCTTCGGGGCGAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCC ACTCGTGACCCAAGTATCTTACGATCTTTTACTTTACCGAGCGTTTCTGGGTGAGCA AAAAACAGGAAGGCAAAATGCCGAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATA CTCATACTCTTCTTTTCAATATTATTGAAGCATTATCAGGGTTATTGTCTCATGAGC GGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACATTTCCC CGAAAAGTGCCACCTGACGCTCTAAGAAACCATTATTATCATGACATTAACTATAAAAAAT AGCGCTATCACGAGGCCCTTTCTGCTCGCGCGTTTCGGTGATGACGGTGAAACCTCTGA CACATGCAGCTCCCGAGACGGTCAAGCTTGTCTGTAAGCGGATGCCGGGAGCAGACAA GCCCCGTAGGGCGCGTCAGCGGGTGTGGCGGGTGTGGGGCTGGCTTAACATATGCGGCA TCAGAGCAGATTGTACTGAGAGTGACCATATGCGGTGTGAAATACCGCACAGATGCGTA AGGAGAAAAATACCGCATCAGCGCCATTTCGCCATTACGGCTGCGCAACTGTTGGGAAGGG CGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGATGTGCTGCAAGG CGATTAAGTTGGGTAAAGCAGGGTTTTCCAGTCACGACGTTGTAAACGACGGCCAGT GAATTCGTTAAGACCACTTTTACATTTAAGTTGTTTTCTAATCCGCATATGATCAATT CAAGGCCGAATAAGAAGGCTGGCTCTGCACCTTGGTGATCAATAAATTCGATAGCTTGTC GTAATAATGGCGGCATACTATCAGTAGTAGGTGTTCCCTTCTTCTTTAGCGACTTGAT GCTCTTGATCTTCCAATACGCAACCTAAAGTAAATGCCCCACAGCGCTGAGTGATATA

TABLE 7-continued

Nucleotide sequences of Tet promoter-PAL1 construct, high-copy (SEQ ID NO: 24)

ATGCATTCTCTAGTGAAAAACCTTGTTGGCATAAAAAGGCTAATTGATTTTCGAGAGTTT
 CATACTGTTTTTCTGTAGGCCGTGTACCTAAATGTACTTTTGCTCCATCGCGATGACTTA
 GTAAAGCACATCTAAACCTTTTAGCGTTATTACGTAAAAATCTTGCCAGCTTTCCCTT
 CTAAAGGGCAAAGTGAGTATGGTGCTATCTAACATCTCAATGGCTAAGGCGTCGAGCA
 AAGCCCGCTTATTTTTTACATGCCAATACAATGTAGGCTGCTCTACACCTAGCTTCTGGG
 CGAGTTTACGGGTTGTTAAACCTTCGATTCGACCTCATTAAGCAGCTCTAATGCGCTGT
 TAATCACTTTACTTTTATCTAATCTAGACATCATTAATCCTAATTTTT**GTTGACACTCT**
ATCATTGATAGAGTTATTTTACCACCTCCCTATCAGTGATAGAGAAAAGTGAA

TABLE 8

Nucleotide sequences of Tet promoter-PAL3, high-copy construct (SEQ ID NO: 25)

CTCTAGAAATAATTTGTTTAACTTTAAGAAGGAGATATACATATGAAAGCTAAAGATGT
TCAGCCAACCATTATTATTAATAAAAAATGGCCTTATCTCTTTGGAAGATATCTATGACAT
TGCGATAAAACAAAAAAGTAGAAATATCAACGGAGATCACTGAACTTTGGACGCATGG
TCGTGAAAAATTAGAGGAAAAATTAAATTCAGGAGAGGTTATATATGGAATCAATACAGG
ATTTGGAGGGAATGCCAATTTAGTTGTGCCATTTGAGAAAATCGCAGAGCATCAGCAAAA
TCGTGTTAACTTTCTTTCTGCTGGTACTGGGGACTATATGTCCAACCTTGTATTAAAGC
GTCACAATTTACTATGTTACTTTCTGTTTGCAAAGGTTGGTCTGCAACCAGACCAATTGT
CGCTCAAGCAATTGTTGATCATATTAATCATGACATTGTTCTCTGGTTCCTCGTATGG
CTCAGTGGGTGCAAGCGGTGATTTAATTCCTTTATCTTATATTGCACGAGCATTATGTGG
TATCGGCAAAGTTTATTATATGGGCGCAGAAATTGACGCTGCTGAAGCAATTAAACGTGC
AGGGTTGACACCATTATCGTTAAAAGCCAAAGAAGGCTTGCTCTGATTAACGGCACCCG
GGTAATGTCAGGAATCAGTGCAATCACCGTCATTAACTGGAAAACTATTTAAAGCCTC
AATTTCTGCGATTGCCCTTGCTGTTGAAGCATTACTTGCATCTCATGAACATTATGATGC
CCGGATTCAACAAGTAAAAATCATCCTGGTCAAAACGCGGTGGCAAGTGCATTGCGTAA
TTTATTGGCAGGTTCAACGCAGGTTAATCTATTATCTGGGGTTAAAGAACCAAGCAATAA
AGCTTGTCGTATCAAGAAATTACCCAATAAATGATACCTTACAGGAAGTTTATTCAAT
TCGCTGTGCACCACAAGTATTAGGTATAGTGCCAGAATCTTAGCTACCGCTCGGAAAAA
ATTGGAACGGGAAGTTATCTCAGCTAATGATAATCCATTGATAGATCCAGAAAATGGCGA
TGTTCTACACGGTGGAATTTTATGGGGCAATATGTGCGCCGAACAATGGATGCATTAA
ACTGGATATTGCTTTAATTGCCAATCATCTTCACGCCATTGTGGCTCTTATGATGGATAA
CCGTTTCTCTCGTGGATTACCTAATTCAGTCCGACACCCGGCATGTATCAAGGTTT
TAAAGGCGTCCAACCTTCTCAAACCGCTTTAGTTGCTGCAATTGCGCATGATTGTGCTGC
ATCAGGTATTATACCCCTCGCCACAGAACAAATACAATCAAGATATTGTGAGTTTAGGTCT
GCATGCCGCTCAAGATGTTTTAGAGATGGAGCAGAAATTACGCAATATTGTTTCAATGAC
AATTCGTGAGTTTGTGAGGCCATTATCTTCGCGGCAATATTAGTGAAATTGCGCCTGA
AACTGCTAAATTTTACCATGCAGTACGCGAAATCAGTTCTCCTTTGATCACTGATCGTGC
GTTGGATGAAGATATAATCCGATTGCGGATGCAATTATTAATGATCAACTTCCTCTGCC

TABLE 8-continued

Nucleotide sequences of Tet promoter-PAL3,
high-copy construct (SEQ ID NO: 25)

AGAAATCATGCTGGAAGAATAAGCTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGA
AATTGTTATCCGCTCACAAATCCACACAACATACGAGCCGGAAGCATAAAGTGTAAGCC
TGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTC
CAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGC
GGTTTGCGTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTGTT
CGGCTGCGGCGAGCGGTATCAGCTCACTCAAGGCGGTAAATACGGTTATCCACAGAATCA
GGGGATAACGCGAGGAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAA
AAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCTGACGAGCATCACAAAAAT
CGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGGCGTTTCCC
CCTGGAAGCTCCCTCGTGCCTCTCCTGTTCCGACCTGCCGCTTACCGGATACCTGTCC
GCCTTTCCTCCTTCGGGAAGCGTGGCGCTTTCATAGCTCACGCTGATAGTATCTCAGT
TCGGTGTAGGTGTTTCGCTCCAAGCTGGGCTGTGTGCACGAACCCCCGTTTCAGCCCGAC
CGCTGCGCCTTATCCGGTAACATATCGTCTTGAGTCCAACCCGGTAAGACACGACTTATCG
CCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACA
GAGTTCCTTGAAGTGGTGGCCTAACTACGGCTACACTAGAAGAACAGTATTTGGTATCTGC
GCTCTGCTGAAGCCAGTTACCTTCGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAACAA
ACCACCGCTGGTAGCGGTGGTTTTTTGTTTGAAGCAGCAGATTACGCGCAGAAAAAA
GGATCTCAAGAAGATCCTTTGATCTTTTCTACGGGTCTGACGCTCAGTGGAACGAAAAAC
TCACGTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTA
AATTAATAATGAAGTTTTAAATCAATCTAAAGTATATATGAGTAACTTGGTCTGACAGT
TACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTTCGTTTCATCCATA
GTTGCTGACTCCCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCC
AGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAAC
CAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAG
TCTATTAATTGTTGCCGGAAGCTAGAGTAAGTAGTTGCCAGTTAATAGTTTGCGCAAC
GTTGTTGCCATTGCTACAGGCATCGTGGTGTACGCTCGTCTGTTGGTATGGCTTCATTC
AGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCATGTTGTGCAAAAAGCG
GTTAGCTCCTTCGGTCCTCCGATCGTTGTGAGAAGTAAGTTGGCCGCGAGTGTATCACTC
ATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAGATGCTTTTCT
GTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCGGCGACCGAGTTGC
TCTTGCCCGCGCTCAATACGGGATAATACCGCGCCACATAGCAGAACTTTAAAGTGCTC
ATCATTGGAAAACGTTCTTCGGGGCGAAAACCTCAAGGATCTTACCGCTGTTGAGATCC
AGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTCAGCATCTTTACTTTTACCAGC
GTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAAGGGAATAAGGGCGACA
CGAAATGTTGAATACTCATACTCTTCCTTTTCAATATTATTGAAGCATTATCAGGGT
TATTGTCTCATGAGCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTT
CCGCGCACATTTCCCGAAAAGTGCCACCTGACGCTAAGAAACCATTATTATCATGACA
TTAACCTATAAAAATAGGCGTATCACGAGGCCCTTTCGTCTCGCGGTTTCGGTGATGAC

TABLE 8-continued

Nucleotide sequences of Tet promoter-PAL3, high-copy construct (SEQ ID NO: 25)
GGTGAACCTCTGACACATGCAGCTCCCGAGACGGTCACAGCTTGTCTGTAAGCGGAT
GCCGGGAGCAGACAAGCCCGTCAGGGCGCGTCAGCGGGTGTGGCGGGTGTGCGGGCTGG
CTTAACATATGCGGCATCAGAGCAGATTGTAAGTGTGAGAGTGACCATATGCGGTGTGAAATA
CCGCACAGATGCGTAAGGAGAAAAATACCGCATCAGGCGCCATTTCGCCATTCAGGCTGCGC
AACTGTTGGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGG
GGATGTGCTGCAAGGCGATTAAAGTTGGGTAAAGCCAGGGTTTTCCAGTCACGACGTTGT
AAAACGACGCGCAGTGAATTTCGTTAAGACCCACTTTCACATTTAAGTTGTTTTCTAATC
CGCATATGATCAATTCAAGGCCGAATAAGAAGGCTGGCTCTGCACCTTGGTGATCAAATA
ATTGATAGCTTGTGTAATAATGGCGGCATACTATCAGTAGTAGGTGTTCCCTTTCTT
CTTTAGCGACTTGATGCTCTTGATCTTCCAATACGCAACCTAAAGTAAATGCCCCACAG
CGCTGAGTGATATAATGCATTCTCTAGTGAAAAACCTTGTGGCATAAAAAGGCTAATT
GATTTTCGAGAGTTTCATACTGTTTTCTGTAGGCGGTACCTAAATGTACTTTTGCTC
CATCGCGATGACTTAGTAAGCACATCTAAACTTTTAGCGTTATTACGTAAAAATCTT
GCCAGCTTTCCTCTTAAGGGCAAAAGTGAGTATGGTGCTATCTAACATCTCAATGG
CTAAGGCGTCGAGCAAGCCGCTTATTTTTTACATGCCAATACAATGTAGGCTGCTCTA
CACCTAGCTTCTGGCGAGTTTACGGGTTGTTAAACCTTCGATTCCGACCTCATTAGCA
GCTCTAATGCGCTGTTAATCACTTTACTTTTATCTAATCTAGACATCATTAAATCCTAAT
TTTTGTTGACACTCTATCATTGATAGAGTTATTTTACCCTCCCTATCAGTGATAGAGAA
AAGTGAA

TABLE 9

Nucleotide sequences of FNR promoter-PAL1 construct, low-copy (SEQ ID NO: 26)
CTCTTGATCGTTATCAATTCACGCTGTTTCAGAGCGTTACCTTGCCCTTAACATTAG
CAATGTCGATTTATCAGAGGGCGACAGGCTCCACAGGAGAAAAAGGATGAAAACATA
TCACAGGCCCAATCTAAAACCTCTTCACAGCAATTCAGCTTTACCGGGAACCTGCTGCG
AATGTAATTATCGGCAATCAAAGCTGACCATTAAATGATGTAGCTCGCGTTGCCCGGAAT
GGCACTTTGGTGCTACTGACGAACAATACCGACATTCTGCAAGGTATTCAAGCTAGCTGC
GATTATATCAATAACGCCGTTGAATCTGGCGAGCAATCTACGGGGTAACAAGCGGTTTT
GGTGGGATGGCGAAGCTTGCCATTAGCCGTGAACAGGCGAGCGAATTCAGACCAACCTC
GTTTGGTTCTTAAAGACAGGAGCTGGTAATAAGTTACCTCTGGCTGACGTAAGAGCCGCG
ATGCTGCTTCGCGCTAATAGTCACATGCGCGGCGCCAGTGGTATCCGTCTTGAGCTTATC
AAGAGGATGGAATCTTCTCAACGCGGGTGTACACCATATGTTTATGAGTTTGGTAGT
ATCGGAGCCAGTGGTGATCTTGTTCCTGAGTTATATTACGGGTTTATGATTGGTTTA
GACCCGTCCTTTAAAGTGGATTTTAAAGGAAAGAAATGGACGCCCCGACCGCTTTACGA
CAGCTTAATCTGAGCCACTTACTTTGCTCCCTAAAGAAGGTCTTGCCATGATGAATGGC
ACCTCTGTGATGACTGGAATTGCGCGAATTGTTGTGTATGACACGCAGATCCTAACGGCC
ATTGCCATGGGTGTTACGCGTTGGACATTCAAGCCCTGAATGGTACAAACAGTCGTTT
CATCCGTTTATCCATAATTCAAACCCCATCCGGACAGCTTTGGGCTGCTGATCAGATG

TABLE 9-continued

Nucleotide sequences of FNR promoter-PAL1 construct, low-copy (SEQ ID NO: 26)	
<u>ATCTCACTCCTGGCCAATAGTCAACTGGTTCGGGACGAGCTCGACGGCAAACATGATTAT</u>	
<u>CGCGATCATGAGCTCATCCAGGACCGGTATTCAC TTCGTTGTCTCCACAATACCTGGGG</u>	
<u>CCTATCGTTGATGGTATATCTCAAATTGCGAAGCAAATTGAAATTGAGATCAATAGCGTA</u>	
<u>ACCGACAACCCGCTTATCGATGTTGATAATCAGGCCTCTTATCACGGTGGCAATTTCTG</u>	
<u>GGCCAGTATGTTGGTATGGGGATGGATCACCTGCGGTACTATATTGGGCTTCTGGCTAAA</u>	
<u>CATCTTGATGTGCAGATTGCCCTTATTAGCTTCACCAGAATTTTCAAATGGACTGCCGCCA</u>	
<u>TCATTGCTCGGTAACAGAGAAAGGAAAGTAAATATGGGCCCTTAAGGGCCTTCAGATATGT</u>	
<u>GGTAACTCAATCATGCCCTCCTGACCTTTTATGGGAACTCAATTGCTGATCGTTTCCG</u>	
<u>ACACATGCTGAACAGTTTAACCAAAACATTAACCTACAGGGCTATACATCCGCGACGTTA</u>	
<u>GCGCGTCGGTCCGTGGATATCTCCAGAATTATGTTGCTATCGCTCTGATGTTGCGCGTA</u>	
<u>CAGGCCGTTGATTGCGCACTTATAAAAAACCGGTCACTACGATGCTCGGGCTTGCCTG</u>	
<u>TCGCCTGCCACCGAGCGGCTTTATAGCGCCGTACGTCATGTTGTGGGTGAGAAACCGACG</u>	
<u>TCGGACCGCCCTATATTTGGAATGATAATGAACAAGGGCTGGATGAACACATCGCCCGG</u>	
<u>ATATCTGCCGATATTGCCCGCGGAGGTGTCATCGTCCAGGCGGTACAAGACATACTTCCT</u>	
<u>TGCCTGCATTAAAGCTTGGCGTAATCATGGTCATAGCTGTTTCCGTGTGAAATTGTTATC</u>	
<u>CGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTAAGCCTGGGGTGCCT</u>	
<u>AATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCGCTTTCCAGTCGGGAA</u>	
<u>ACCTGTCTGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGCGGTTTGCCTA</u>	
<u>TTGGGCGCTCTTCCGCTTCTCGCTCACTGACTCGCTGCGCTCGGTCGTTGCGCTGCGGC</u>	
<u>GAGCGGTATCAGCTCACTCAAAGGCGGTAGTACGGGTTTGTGTCGCGCAAACGGGCTGT</u>	
<u>TCTGGTGTGCTAGTTTGTATCAGAATCGCAGATCCGGCTTCAGGTTTCCCGCTGAAA</u>	
<u>GCGCTATTTCTCCAGAATTGCCATGATTTTTCACCGGAGGCGTCACTGGCTCCCG</u>	
<u>TGTTGTGCGCAGCTTTGATTGATAAGCAGCATCGCCTGTTTCAGGCTGTCTATGTGTGA</u>	
<u>CTGTTGAGCTGTAACAAGTTGTCTCAGGTGTTCAATTTTCATGTTCTAGTTGCTTTGTTTT</u>	
<u>ACTGTTTTACCTGTCTTATTAGGTGTTACATGCTGTTTCATCTGTTACATTGTCGATCTG</u>	
<u>TTTCATGGTGAACAGCTTTAAATGCACCAAAAACCTCGTAAAAGCTCTGATGTATCTATCTT</u>	
<u>TTTTACACCGTTTTTCATCTGTGCATATGGACAGTTTTCCCTTTGATATCTAACGGTGAAC</u>	
<u>AGTTGTTCTACTTTTGTGTTAGTCTTGATGCTTCACTGATAGATACAAGAGCCATAAG</u>	
<u>AACCTCAGATCCTTCCGTATTTAGCCAGTATGTTCTCTAGTGTGGTTCGTTGTTTTGCG</u>	
<u>TGAGCCATGAGAACGAACCATTGAGATCATGCTTACTTTGCATGTCACTCAAAAATTTTG</u>	
<u>CCTCAAACTGGTGAGCTGAATTTTTCAGTTAAAGCATCGTGTAGTGTTCCTTAGTCTC</u>	
<u>CGTTACGTAGGTAGGAATCTGATGTAATGGTTGTTGGTATTTTGTACCATTCATTTTTA</u>	
<u>TCTGGTGTCTCTCAAGTTCGGTTACGAGATCCATTTGTCTATCTAGTTCAACTTGAAAA</u>	
<u>TCAACGTATCAGTCGGCGGCCTCGCTTATCAACCACCAATTTTCATATTGCTGTAAGTGT</u>	
<u>TTAAATCTTTACTTATTGGTTTCAAACCCATTGGTTAAGCCTTTTAACTCATGGTAGT</u>	
<u>TATTTTCAAGCATTAAACATGAACCTTAAATTCATCAAGGCTAATCTCTATATTTGCCTGT</u>	
<u>GAGTTTCTTTTGTGTTAGTCTTTTAATAACCACTCATAAATCCTCATAGAGTATTTGT</u>	
<u>TTTCAAAAGACTTAACATGTTCCAGATTATTTTATGAATTTTTTAACTGGAAAAGAT</u>	

TABLE 9-continued

Nucleotide sequences of FNR promoter-PAL1 construct, low-copy (SEQ ID NO: 26)
AAGGCAATATCTCTTCACTAAAACTAATTCTAATTTTCGCTTGAGAACTTGGCATAGT TTGTCCACTGGAAATCTCAAAGCCTTTAACCAAAGGATTCCTGATTTCCACAGTTCTCG TCATCAGCTCTCTGGTTGCTTTAGCTAATACACCATAAGCATTTCCTTACTGATGTTCA TCATCTGAGCGTATTGGTTATAAGTGAACGATACCGTCCGTTCTTTCCTTGTAGGGTTT CAATCGTGGGGTTGAGTAGTGCCACACAGCATAAAATTAGCTTGGTTTCATGCTCCGTTA AGTCATAGCGACTAATCGCTAGTTCATTTGCTTTGAAAACAATAATTCAGACATACATC TCAATTGGTCTAGGTGATTTTAACTACTATACCAATTGAGATGGGCTAGTCAATGATAAT TACTAGTCCTTTTCCTTTGAGTTGTGGGTATCTGTAAATTCGCTAGACCTTTGCTGGAA AACTTGTAATTTCTGCTAGACCTCTGTAAATTCGCTAGACCTTTGTGTGTTTTTTTG TTTATATTCAAGTGGTTATAATTTATAGAATAAGAAAGAATAAAAAAGATAAAAAGAA TAGATCCCAGCCCTGTGTATAACTCACTACTTTAGTCAGTCCGCAGTATTACAAAAGGA TGTCGCAAAACGCTGTTTGCTCCTCTACAAAACAGACCTTAAACCCCTAAAGGCTTAAGTA GCACCTCGCAAGCTCGGGCAAAATCGCTGAATATTCCTTTTGTCTCCGACCATCAGGCAC CTGAGTCGCTGTCTTTTCGTGACATTGAGTTCGCTGCGCTCACGGCTCTGGCAGTGAAT GGGGGTAAATGGCACTACAGGCGCCTTTTATGGATTGATGCAAGGAACTACCCATAATA CAAGAAAAGCCGCTCACGGGCTTCTCAGGGCGTTTATGGCGGGTCTGCTATGTGGTGCT ATCTGACTTTTGTGTTTCAGCAGTTCTGCGCTCTGATTTTCCAGTCTGACCACTTCGG ATTATCCCGTGACAGGTGATTGAGTGGCTAATGCACCCAGTAAGCAGCGGTATCATC AACAGGCTTACCCGCTTACTGTCTTTTCTACGGGTCTGACGCTCAGTGGAACGAAAAC TCACGTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTA AATTAATAATGAAGTTTAAATCAATCTAAAGTATATATGAGTAACTTGGTCTGACAGT TACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTTCGTTTATCCATA GTTGCCGTGACTCCCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCC AGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAAC CAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCTGCAACTTTATCCGCTCCATCCAG TCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAAC GTTGTTGCCATTGCTACAGGCATCGTGGTGTACGCTCGTCTGTTGGTATGGCTTCATTC AGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCATGTTGTGCAAAAAGCG GTTAGCTCCTTCGGTCCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGAGTGTATCACTC ATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAGATGCTTTTCT GTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGATGCGGCGACCGAGTTGC TCTTGCCCCGCGTCAATACGGGATAATACCGCGCCACATAGCAGAACTTTAAAAGTGCTC ATCATTTGAAAACGTTCTTCGGGGCGAAAACCTCAAGGATCTTACCGCTGTTGAGATCC AGTTCGATGTAACCCACTCGTGACCCCACTGATCTTCAGCATCTTTTACTTTCACCAGC GTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAAGGGAATAAGGGCGACA CGAAATGTTGAATACTCATACTCTTCCTTTTCAATATTATTGAAGCATTATCAGGGT TATTGTCTCATGAGCGGATACATATTTGAATGTATTAGAAAAATAACAATAGGGGTT CCGCGCACATTTCCCGAAAAGTGCCACCTGACGCTCTAAGAAACCATTTATCATGACA TTAACCTATAAAAATAGGCGTATCACGAGGCCCTTTCGTCTCGCGGTTTCGGTGATGAC

TABLE 9-continued

Nucleotide sequences of FNR promoter-PAL1 construct, low-copy (SEQ ID NO: 26)
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GGTGAAAACCTCTGACACATGCAGCTCCCGGAGACGGTCACAGCTTGTCTGTAAGCGGAT
GCCGGGAGCAGACAAGCCCGTCAGGGCGCGTCAGCGGGTGTGGCGGGTGTGGGGCTGG
CTTAAGTATGCGGCATCAGAGCAGATTGTACTGAGAGTGACCATATGCGGTGTGAAATA
CCGCACAGATGCGTAAGGAGAAAAATACCGCATCAGGCGCCATTGCGCATTCAGGCTGCGC
AACTGTTGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGG
GGATGTGCTGCAAGGCGATTAAAGTTGGGTAAAGCCAGGGTTTTCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAATTCG

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TABLE 10

Nucleotide sequences of FNR promoter-PAL3 construct, low-copy (SEQ ID NO: 27)
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CTCTTGATCGTTATCAATTCCACGCTGTTTCAGAGCGTTACCTTGCCCTTAAACATTAG
CAATGTCGATTATTCAGAGGGCCGACAGGCTCCACAGGAGAAAAACCGATGAAAGCTAAA
GATGTTGAGCCCAACCATTTATTATTAATAAAAAATGGCCTTATCTCTTGGAAGATATCTAT
GACATTGCGATAAAAAAAGTAGAAATATCAACGGAGATCACTGAACCTTTTGACG
CATGGTCGTGAAAAATTAGAGGAAAAATTAAATTCAGGAGAGGTTATATATGGAATCAAT
ACAGGATTGAGGGGAATGCCAATTTAGTTGTGCCATTGAGAAAAATCGCAGAGCATCAG
CAAAATCTGTTAACTTTTCTTCTGCTGCTACTGGGGACTATATGTCCAAACCTTGTATT
AAAGCGTCACAATTTACTATGTTACTTTCTGTTTGCAAAGGTTGGTCTGCAACCAGACCA
ATTGTCGCTCAAGCAATTGTTGATCATATTAATCATGACATTGTTCTCTGTTCTCTCGC
TATGGCTCAGTGGGTGCAAGCGGTGATTTAATTCCTTTATCTTATATTGCACGAGCATTA
TGTTGATCGGCAAGTTTATATATGCGCGCAGAAATGACGCTGCTGAGCAATTAA
CGTGCAGGGTTGACACCATTTATCGTTAAAAGCCAAAGAAGGCTTGTCTCTGATTAAACGGC
ACCCGGGTAATGTCAGGAATCAGTGCAATCACCGTCATTAACTGGAAAACTATTTAA
GCCTCAATTTCTGCGATTGCCCTTGCTGTTGAAGCATTACTGTCATCTCATGAACATTAT
GATGCCCGGATTCAACAAGTAAAAAATCATCCTGGTCAAAACGCGGTGGCAAGTGCATTG
CGTAATTTATTGGCAGGTTCAACGCAGGTTAATCTATTATCTGGGGTTAAAGAACAAGCC
AATAAAGCTTGTGTCATCAAGAAATTACCCAATAAATGATACCTTACAGGAAGTTTAT
TCAATTGCTGTGCACCAAGTATTAGGTATAGTGCCAGAATCTTAGCTACCGCTCGG
AAAATATTGGAACGGGAAGTTATCTCAGCTAATGATAATCCATTGATAGATCCAGAAAAAT
GGCGATGTTCTACAGGTGGAATTTTATGGGGCAATATGTCGCCCGAACAATGGATGCA
TTAAACTGGATATTGCTTTAATTGCCAATCATCTTCACGCCATTGTGGCTCTTATGATG
GATAACCGTTTCTCTCGTGGATTACCTAATTCACGTGAGTCCGACACCCGGCATGTATCAA
GGTTTTAAAGCGTCCAACCTTCTCAAACCGCTTTAGTTGCTGCAATTCGCCATGATTGT
GCTGCATCAGGTATTCATACCCTCGCCACAGAACAATAAATCAAGATATTGTCAGTTTA
GGTCTGCATGCCGCTCAAGATGTTTTAGAGATGGAGCAGAAATTACGCAATATTGTTTCA
ATGACAATTTCTGGTAGTTTGTGAGGCCATTATCTTCGCGGCAATATTAGTGAAATTGCG
CCTGAAACTGCTAAATTTTACCATGCAGTACGCGAAATCAGTTCTCCTTTGATCACTGAT

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TABLE 10-continued

Nucleotide sequences of FNR promoter-PAL3 construct, low-copy (SEQ ID NO: 27)
CGTGCGTTGGATGAAGATATAATCCGCATTGCGGATGCAATTATTAATGATCAACTTCCT CTGCCAGAAATCATGCTGGAAGAATAAGCTTGGCGTAATCATGGTCATAGCTGTTTCCTG TGTGAAATTGTTATCCGCTCACAATCCACACAACATACGAGCCGGAAGCATAAAGTGTA AAGCCTGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCCG CTTTCAGTCGGGAAACCTGTCTGTCAGCTGCATTAATGAATCGGCCAACGCGCGGGGA GAGGCGGTTTTCGTATTGGGCGCTCTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGG TCGTTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAGTACGGGTTTGTCTGC CCGCAAACGGGCTGTTCTGGTGTGCTAGTTTGTATCAGAATCGCAGATCCGGCTTCAG GTTTGCCGGCTGAAAGCGCTATTTCTCCAGAATTGCCATGATTTTTTCCACGCGGAGG CGTCACTGGCTCCCGTGTGTGCGCAGCTTTGATTGATAAGCAGCATCGCCTGTTTCAG GCTGTCTATGTGTGACTGTTGAGCTGTAACAAGTTGTCTCAGGTGTTCAATTCATGTTT TAGTTGCTTTGTTTTACTGGTTTCACCTGTTCTATTAGGTGTTACATGCTGTTTCATCTGT TACATTGTGATCTGTTTATGTTGAACAGCTTTAAATGCACCAAAAACCTCGTAAAGCTC TGATGTATCTATCTTTTTTACACCGTTTTTCATCTGTGCATATGGACAGTTTTCCCTTTGA TATCTAACGGTGAACAGTTGTTCTACTTTTGTGTTAGTCTTGATGCTTCACTGATAGA TACAAGAGCCATAAGAACCTCAGATCCTTCGGTATTAGCCAGTATGTTCTCTAGTGTGG TTGTTGTTTTTTCGCTGAGCCATGAGAACGAACCATTGAGATCATGCTTACTTTGCATGT CACTCAAAAATTTTGCTCAAACTGGTGAGCTGAATTTTGCAGTTAAAGCATCGTGTA GTGTTTTTCTTAGTCCGTACGTAGGTAGGAATCTGATGTAATGGTTGTTGGTATTTTGT CACCATTCAATTTTATCTGGTTGTTCTCAAGTTCGGTTACGAGATCCATTTGTCTATCTA GTTCAACTTGGAATAACAGTATCAGTCGGGCGGCTCGCTTATCAACCACCAATTTCA TATTGCTGTAAGTGTTTAAATCTTTACTTATGGTTTCAAAACCCATTGGTTAAGCCTTT TAAACTCATGGTAGTTATTTTCAAGCATTAACTGAACCTTAAATTCATCAAGGCTAATCT CTATATTTGCCTTGTGAGTTTTCTTTTGTGTTAGTCTTTTAAATAACCACTCATAAATCC TCATAGAGTATTTGTTTTCAAAAGACTTAACATGTTCCAGATTATTTTTATGAATTTTT TTAACTGGAAAAGATAAGGCAATATCTCTTCACTAAAACTAATTCATTTTTTCGCTTG AGAACTTGGCATAGTTTGTCCACTGGAATCTCAAAGCCTTTAACCAGGATTTCCTGA TTTCCACAGTTCTCGTCATCAGCTCTCTGGTTGCTTTAGCTAATACACCATAAGCATTTT CCCTACTGATGTTTCATCATCTGAGCGTATTGGTTATAAGTGAACGATACCGTCCGTTCTT TCCTTGTAGGGTTTTCAATCGTGGGTTGAGTAGTGCCACACAGCATAAAATTAGCTTGG TTTCATGCTCCGTTAAGTCATAGCGACTAATCGCTAGTTCATTTGCTTTGAAAACAATA ATTTCAGACATACATCTCAATTGGTCTAGGTGATTTTAACTACTATACCAATTGAGATGGG CTAGTCAATGATAATTACTAGTCCTTTTCTTTGAGTTGTGGGTATCTGTAAATTCGTCT AGACCTTTGCTGGAAGACTTGTAAATCTGCTAGACCTCTGTAAATTCGCTAGACCTT TGTGTGTTTTTTTTGTTTATATTCAAGTGGTTATAATTTATAGAATAAAGAAAGATAAA AAAAGATAAAAAGAATAGATCCAGCCCTGTGTATAACTCACTACTTTAGTCAGTTCCGC AGTATTACAAAAGGATGTCGCAACGCTGTTTGCTCCTCTACAAAACAGACCTTAAACCC CTAAAGGCTTAAGTAGCACCTCGCAAGCTCGGGCAAATCGCTGAATATTCCTTTTGTCT CCGACCATCAGGCACCTGAGTCGCTGTCTTTTTCGTGACATTAGTTCGCTGCGCTCAG

TABLE 10-continued

Nucleotide sequences of FNR promoter-PAL3 construct, low-copy (SEQ ID NO: 27)
GCTCTGGCAGTGAATGGGGTAAATGGCACTACAGGCGCCTTTTATGGATTCATGCAAGG
AAACTACCCATAATACAAGAAAAGCCCGTCACGGGCTTCTCAGGGCGTTTATGGCGGGT
CTGCTATGTGGTGCTATCTGACTTTTGTCTTTCAGCAGTTCCTGCCCTCTGATTTTCCA
GTCTGACCACTTCGATTATCCCGTGACAGGTCATTGAGCTGGCTAATGCACCCAGTAA
GGCAGCGGTATCATCAACAGGCTTACCCGTCTTACTGTCTTTCTACGGGGTCTGACGCT
CAGTGGAAACGAAAACCTACGTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTC
ACCTAGATCCTTTTAAATTAAAAATGAAGTTTAAATCAATCTAAAGTATATATGAGTAA
ACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTA
TTTCGTTTCATCCATAGTTGCCTGACTCCCCGTCTGTGTAGATAACTACGATACGGGAGGGC
TTACCATCTGGCCCCAGTGTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGAT
TTATCAGCAATAAACAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCTTCAACTTTA
TCCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTT
AATAGTTTTCGCGAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTACGCTCGTCGTTT
GGTATGGCTTCATTGAGCTCCGGTCCCAACGATCAAGGCGAGTTACATGATCCCCCATG
TTGTGCAAAAAGCGGTTAGCTCCTTCGGTCTCCGATCGTTGTCAGAAGTAAGTTGGCC
GCAGTGTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCC
GTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATG
CGGCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCAGA
ACTTTAAAGTGCTCATATTGGAACGTTCTTCGGGGCGAAAACCTCTCAAGGATCTTA
CCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCAACCACTGATCTTCAGCATCT
TTTACTTTCACAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAAG
GGAATAAGGGCGACACGGAATGTTGAATACTCATACTTCTCCTTTTCAATATTATTGA
AGCATTTATCAGGGTTATTGTCTCATGAGCGGATACATATTGAATGTATTAGAAAAAT
AAACAAATAGGGGTTCCGCGCACATTTCCCGAAAAGTGCCACCTGACGTCTAAGAAACC
ATTATTATCATGACATTAACTTATAAAAATAGGCGTATCACGAGGCCCTTTCGTCTCGCG
CGTTTCGGTGATGACGGTGAAAACCTCTGACACATGCAGCTCCCGGAGACGGTCACAGCT
TGTCTGTAAGCGGATGCCGGGAGCAGACAAGCCCGTCAGGGCGCGTCAGCGGGTGTGGC
GGGTGTGGGGCTGGCTTAACTATGCGGCATCAGAGCAGATTGTACTGAGAGTGCAACCAT
ATGCGGTGTGAAATACCGCACAGATGCGTAAGGAGAAAATACCGCATCAGGCGCCATTCTG
CCATTGAGGCTGCGCAACTGTTGGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGC
CAGCTGGCGAAAGGGGGATGTGCTGCAAGGCGATTAAAGTTGGGTAAACGCCAGGGTTTCC
CAGTCACGACGTTGTAAAACGACGGCCAGTGAATTCG

TABLE 11

Nucleotide sequences of Tet promoter-PAL1 construct, low-copy (SEQ ID NO: 28)
ACCACTCCCTATCAGTGATAGAGAAAAGTGAAGTCTAGAATAATTTGTTAACTTTAA
GAAGGAGATATACATATGAAAACACTATCACAGGCCCAATCTAAACTTCTTCACAGCAA

TABLE 11-continued

Nucleotide sequences of Tet promoter-PAL1 construct, low-copy (SEQ ID NO: 28)
<u>TTCAGCTTTACCGGGAACTCGTCTGCGAATGTAATTATCGGCAATCAAAAGCTGACCATT</u>
<u>AATGATGTAGCTCGCGTTGCCCGGAATGGCACTTTGGTGTCACTGACGAACAATACCGAC</u>
<u>ATTCTGCAAGGTATTCAAGCTAGCTGCGATTATATCAATAACGCCGTTGAATCTGGCGAG</u>
<u>CCAATCTACGGGGTAACAAGCGTTTGGTGGGATGGCGAACGTTGCCATTAGCCGTGAA</u>
<u>CAGGCGAGCGAACTTCAGACCAACCTCGTTTGGTTCCTAAAGACAGGAGCTGGTAATAAG</u>
<u>TTACCTCTGGCTGACGTAAGAGCCGCGATGCTGCTTCGCGCTAATAGTCACATGCGCGGC</u>
<u>GCCAGTGGTATCCGTCTTGAGCTTATCAAGAGGATGGAATCTTCCTCAACGCGGGTGTC</u>
<u>ACACCATATGTTTATGAGTTTGGTAGTATCGGAGCCAGTGGTGATCTTGTTCCCTGAGT</u>
<u>TATATTACGGGTTCAATTGATTGGTTTAGACCCGTCCTTTAAAGTGGATTTTAAACGGGAAA</u>
<u>GAAATGGACGCCCCGACCGCTTACGACAGCTTAATCTGAGCCCACTTACTTTGCTCCCT</u>
<u>AAAGAAGGTCTTGCCATGATGAATGGCACCTCTGTGATGACTGGAATTGCCGCGAATTGT</u>
<u>GTGTATGACACGCAGATCCTAACGGCCATTGCCATGGGTGTTACGCGTTGGACATTCAA</u>
<u>GCCCTGAATGGTACAAACAGTCGTTTCATCCGTTTATCCATAATTCAAAACCCCATCCG</u>
<u>GGACAGCTTTGGGCTGCTGATCAGATGATCTCACTCCTGGCCAATAGTCAACTGGTTCGG</u>
<u>GACGAGCTCGACGGCAACATGATTATCGCGATCATGAGCTCATCCAGGACCGGTATTCA</u>
<u>CTTCGTTGTCTCCACAATACCTGGGGCCTATCGTTGATGGTATATCTCAAATTGCGAAG</u>
<u>CAAATTGAAATTGAGATCAATAGCGTAACCGACAACCCGCTTATCGATGTTGATAATCAG</u>
<u>GCCTCTTATCAGGTGGCAATTTCTGGGCCAGTATGTTGGTATGGGATGGATCACCTG</u>
<u>CGGTACTATATTGGGCTTCTGGCTAAACATCTTGATGTGAGATTGCCTTATTAGCTTCA</u>
<u>CCAGAATTTTCAAATGGACTGCCGCCATCATTGCTCGGTAACAGAGAAAGGAAAGTAAAT</u>
<u>ATGGGCCCTTAAGGGCCTTCAGATATGTGGTAACTCAATCATGCCCTCCTGACCTTTTAT</u>
<u>GGGAACCTCAATTGCTGATCGTTTCCGACACATGCTGAACAGTTTAAACAAAACATTAAAC</u>
<u>TCACAGGGCTATACATCCGCGAGCTTAGCGCGTCGGTCCGTGGATATCTCCAGAATTAT</u>
<u>GTGCTATCGCTCTGATGTTTCGGCGTACAGGCCGTTGATTTGCGCACTTATAAAAAACC</u>
<u>GGTCACTACGATGCTCGGGCTTGCCCTGTCGCTGCCACCGAGCGGCTTTATAGCGCGTA</u>
<u>CGTCATGTTGTGGGTGAGAAACCGACGTCGGACCGCCCTATATTGGAATGATAATGAA</u>
<u>CAAGGGCTGGATGAACACATCGCCCGGATATCTGCCGATATTGCCGCGGAGGTGTCATC</u>
<u>GTCCAGGCGGTACAAGACATACTTCCTTGCCCTGCATTAAGCTTGGCGTAATCATGGTCAT</u>
<u>AGCTGTTTCCGTGTGAAATTGTTATCCGCTCACAATTCCACACAACATACGAGCCGGAA</u>
<u>GCATAAAGTGTAAGCCTGGGGTGCTAATGAGTGAGCTAACTCACATTAATTGCGTTGC</u>
<u>GCTCACTGCCCCTTTCCAGTCGGGAAACCTGTCGTGCCAGTGCATTAATGAATCGGCC</u>
<u>AACGCGCGGGGAGAGGCGGTTTGCCTATTGGGCGCTCTTCCGCTTCTCGCTCACTGACT</u>
<u>CGCTGCGCTCGGTCGTTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAGTAC</u>
<u>GGGTTTGTGCCCCGAAACGGGCTGTTCTGGTGTGCTAGTTTGTATCAGAATCGCAG</u>
<u>ATCCGGCTTCAGGTTTGCCGGCTGAAAGCGCTATTTCTCCAGAATTGCCATGATTTTTT</u>
<u>CCCCACGGGAGGCGTCACTGGCTCCCGTGTGTGCGGAGCTTTGATTGATAAGCAGCAT</u>
<u>CGCTGTTTTCAGGCTGTCTATGTGTGACTGTTGAGCTGTAACAAGTTGTCTCAGGTGTTT</u>
<u>AATTTTCATGTTCTAGTTGCTTTGTTTACTGGTTTCACCTGTTCTATTAGGTGTTACATG</u>
<u>CTGTTTCATCTGTTACATTGTCGATCTGTTTCATGGTGAACAGCTTTAAATGCACCAAAAC</u>

TABLE 11-continued

Nucleotide sequences of Tet promoter-PAL1 construct, low-copy (SEQ ID NO: 28)
TCGTAAAAGCTCTGATGTATCTATCTTTTTTACACCGTTTTTCATCTGTGCATATGGACAG TTTTCCCTTTGATATCTAACGGTGAACAGTTGTTCTACTTTTGTTTGTAGTCTTGATGC TTCCTGATAGATACAAGAGCCATAAGAACCTCAGATCCTCCGTATTTAGCCAGTATGT TCTCTAGTGTGGTTCGTTGTTTTTGCCTGAGCCATGAGAACGAACATTGAGATCATGCT TACTTTGCATGTCACTCAAAAATTTGCCTCAAACTGGTGAGCTGAATTTTGCAGTTA AAGCATCGTGTAGTGTTCCTTAGTCCGTTACGTAGGTAGGAATCTGATGTAATGGTTG TTGGTATTTTGTCAACATTCATTTTATCTGGTTGTTCTCAAGTTCGGTTACGAGATCCA TTTGTCTATCTAGTTCAACTTGGAATCAACGTATCAGTCGGGCGGCCTCGCTTATCAA CCACCAATTTTATATTGCTGTAAGTGTAAATCTTTACTTATTGGTTTCAAAACCCATT GGTTAAGCCTTTTAACTCATGGTAGTTATTTTCAAGCATTAACTGAACCTAAATTCAT CAAGCTAATCTCTATATTGCTTGTGAGTTTCTTTTGTGTAGTTCTTTAATAACC ACTCATAAATCCTCATAGAGTATTTGTTTTCAAAGACTTAACATGTTCCAGATTATATT TTATGAATTTTTTAACTGGAAGATAAGGCAATATCTCTTCACTAACTAATTTCTA ATTTTTCGCTTGAGAACTTGGCATAGTTTGTCCACTGGAAATCTCAAAGCCTTTAACCA AAGGATTCCTGATTTCCACAGTTCTCGTCATCAGCTCTCTGGTTGCTTTAGCTAATACAC CATAAGCATTTCCTTACTGATGTTTCATCATCTGAGCGTATTGGTTATAAGTGAACGATA CCGTCCGTCTTTCTTGTAGGTTTTCAATCGTGGGTTGAGTAGTGCCACACAGCATA AAATTAGCTTGGTTTCATGCTCCGTTAAGTCATAGCGACTAATCGCTAGTTTATTGCTT TGAAACAACCTAATTCAGACATACATCTCAATTGGTCTAGGTGATTTTAACTACATACC AATTGAGATGGGCTAGTCAATGATAATTACTAGTCTTTTCTTTGAGTTGTGGGTATCT GTAAATTCGCTAGACCTTTGCTGGAAACTTGTAATTCGCTAGACCTCTGTAAATT CCGTAGACCTTTGTGTGTTTTTTTTGTTTATATTCAAGTGGTTATAATTTATAGAATAA AGAAAGATAAAAAAGATAAAAAAGATAGATCCAGCCCTGTGTATAACTCACTACTTT AGTCAGTTCCGCAGTATTACAAAAGGATGTCGAAACGCTGTTTGCTCCTCTACAAAACA GACCTTAAACCCCTAAAGGCTTAAGTAGCACCTCGCAAGCTCGGGCAAATCGCTGAATA TTCCTTTTGTCTCCGACCATCAGGCACCTGAGTCGCTGTCTTTTTCGTGACATTGAGTTC GCTGCGCTCACGGCTCTGGCAGTGAATGGGGTAAATGGCACTACAGGCGCCTTTATGG ATTATGCAAGGAACTACCCATAATACAAGAAAGCCGTCACGGGCTTCTCAGGGCGT TTTATGGCGGCTGCTATGTGGTGCTATCTGACTTTTGTGTTGAGCAGTTCTTGCCC TCTGATTTTCCAGTCTGACCACTTCGGATTATCCCGTGACAGGTCATTGAGCTGGCTAA TGCACCCAGTAAGGCAGCGGTATCATCAACAGGCTTACCGTCTTACTGTCTTTTCTACG GGGTCTGACGCTCAGTGAACGAAACTCACGTTAAGGGATTTTGGTCATGAGATTATCA AAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTAAATCAATCTAAAGT ATATATGAGTAACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGGCACCTATCTCA GCGATCTGTCTATTTCGTTTATCCATAGTTGCCTGACTCCCGTCGTGTAGATAACTACG ATACGGGAGGGCTTACCATCTGGCCCCAGTGTGCAATGATACCGCGAGACCCACGCTCA CCGGTCCAGATTTATCAGCAATAAACCAGCCGGAAGGGCCGAGCGCAGAAGTGGT CCTGCAACTTTATCCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGT

TABLE 11-continued

Nucleotide sequences of Tet promoter-PAL1 construct, low-copy (SEQ ID NO: 28)
AGTTCGCCAGTTAATAGTTTGGCGAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTCA CGCTCGTCGTTTGGTATGGCTTCATTACAGCTCCGGTTCCCAACGATCAAGGCGAGTTACA TGATCCCCCATGTTGTGCAAAAAGCGGTTAGCTCCTTCGGTCTCCGATCGTTGTGAGA AGTAAGTTGGCCGAGTGTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACT GTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGA GAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGCGCTCAATACGGGATAATACCGCG CCACATAGCAGAACTTTAAAAGTGCTCATCTTGAAACGTTCTTCGGGGCGAAAACCTC TCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGA TCTTCAGCATCTTTTACTTTTACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAAT GCCGCAAAAAGGGAATAAGGGCGACACGGAAATGTGAATACTCATACTCTTCCTTTTT CAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGAGCGGATACATATTTGAATGT ATTTAGAAAAATAACAAATAGGGGTTCCGCGCACATTTCCCGAAAAGTGCCACCTGAC GTCTAAGAAACCATTATTATCATGACATTAACTATAAAAATAGGCGTATCAGAGGCC TTTCGTCTCGCGCGTTTCGGTGATGACGGTGAAAACCTCTGACACATGCAGCTCCCGGAG ACGGTCACAGCTTGTCTGTAAGCGGATGCCGGGAGCAGACAAGCCCGTCAGGGCGCGTCA GCGGGTGTGGCGGGTGTCGGGGCTGGCTTAATATGCGGCATCAGAGCAGATTGTACTG AGAGTGCAACCATATGCGGTGTGAAATACCGCACAGATGCGTAAGGAGAAAAATACCGCATC AGCGGCCATTGCCATTACAGCTGCGCAACTGTTGGGAAGGGCGATCGGTGCGGGCCTCT TCGCTATTACGCCAGCTGGCGAAAGGGGATGTGCTGCAAGCGATTAAAGTTGGGTAACG CCAGGGTTTTCCAGTCACGACGTTGTAAAACGACGGCCAGTGAATTGTTAAGACCCAC TTTCACATTTAAGTTGTTTTTCTAATCCGCATATGATCAATTCAAGGCCGAATAAGAAGG CTGGCTCTGCACCTTGGTGATCAAATAATTCGATAGCTTGTCGTAATAATGGCGCATAC TATCAGTAGTAGGTGTTCCCTTTCTTTTAGCGACTTGATGCTCTTGATCTTCCAATA CGCAACCTAAAGTAAATGCCCCACAGCGCTGAGTGCATATAATGCATTCTCTAGTGAAA AACCTTGTGGCATAAAAAGGCTAATTGATTTTCGAGAGTTTCATACTGTTTTTCTGTAG GCCGTGTACCTAAATGTACTTTTGCTCCATCGCGATGACTTAGTAAAGCACATCTAAAAC TTTTAGCGTTATTACGTAAAAATCTTGCCAGCTTTCCCTTCTAAAGGGCAAAAGTGAG TATGGTGCCTATCTAACATCTCAATGGCTAAGGCGTCGAGCAAAGCCCGCTATTTTTTA CATGCCAATACAATGATAGGCTGCTCTACACCTAGCTTCTGGGCGAGTTTACGGGTGTGA AACCTTCGATTCCGACCTCATTAAGCAGCTCTAATGCGCTGTTAATCACTTTACTTTTAT CTAATCTAGACATCATTAATTCCTAATTTTTGTTGACACTCTATCATTGATAGAGTTATT TT

TABLE 12

Nucleotide sequences of Tet promoter-PAL3 construct, low-copy (SEQ ID NO: 29)
<u>ACCACCTCCCTATCAGTGATAGAGAAAAGTGA</u> <u>ACTCTAGAAATAATTTGTTTA</u> <u>ACTTTAA</u> <u>GAAGGAGATATACATATGAAAGCTAAAGATGTT</u> <u>CAGCCAACCATTATTATTA</u> <u>ATAAAAAAT</u> <u>GGCCTTATCTCTTTGGAAGATATCTATGACATTGCGATAAAACAAAAAAGTAGAAATA</u>

TABLE 12-continued

Nucleotide sequences of Tet promoter-PAL3 construct, low-copy (SEQ ID NO: 29)	
<u>TCAACGGAGATCACTGAACTTTTGACGCATGGTCGTGAAAAATTAGAGGAAAAATTAAT</u>	
<u>TCAGGAGAGGTTATATATGGAATCAATACAGGATTTGGAGGGAATGCCAATTTAGTTGTG</u>	
<u>CCATTTGAGAAAAATCGCAGAGCATCAGCAAAATCTGTTAACTTTTCTTTCTGCTGGTACT</u>	
<u>GGGGACTATATGTCCAACCTTGTATTAAAGCGTCACAATTTACTATGTTACTTTCTGTT</u>	
<u>TGCAAAGGTTGGTCTGCAACCAGACCAATTGTCGCTCAAGCAATTGTTGATCATATTAAT</u>	
<u>CATGACATTGTTCTCTGGTTCCTCGCTATGGCTCAGTGGGTGCAAGCGGTGATTTAATT</u>	
<u>CCTTTATCTTATATGACAGCATTATGTGGTATCGGCAAGTTTATTATATGGGCGCA</u>	
<u>GAAATTGACGCTGCTGAAGCAATTAAACGTGCAGGGTTGACACCATTATCGTTAAAAGCC</u>	
<u>AAAGAAGGTCTTGCTCTGATTAACGGCACCCGGGTAATGTCAGGAATCAGTGCAATCACC</u>	
<u>GTCATTAAACTGGA AAAACTATTTAAAGCCTCAATTTCTGCGATTGCCCTTGCTGTTGAA</u>	
<u>GCATTACTTGCATCTCATGAACATTATGATGCCCGGATTCAACAAGTAAAAAATCATCCT</u>	
<u>GGTCAAAACGCGGTGGCAAGTGCATTGCGTAATTTATTGGCAGGTTCAACGCAGTTAAT</u>	
<u>CTATTATCTGGGGTTAAAGAACAGCCAATAAAGCTTGTCGTCATCAAGAAATTACCCAA</u>	
<u>CTAAATGATACCTTACAGGAAGTTTATTCAATTCTGCTGTGCCACCAAGTATTAGGTATA</u>	
<u>GTGCCAGAATCTTTAGCTACCGCTCGGAAAAATTTGGAACGGGAAGTTATCTCAGCTAAT</u>	
<u>GATAATCCATTGATAGATCCAGAAAATGGCGATGTTCTACACGGTGGAATTTTATGGGG</u>	
<u>CAATATGTCGCCCCGAACAATGGATGCATTAAACTGGATATTGCTTTAATTGCCAATCAT</u>	
<u>CTTCACGCCATTGTGGCTCTTATGATGGATAACCGTTTCTCTCGTGGATTACCTAATTCA</u>	
<u>CTGAGTCCGACACCCGGCATGTATCAAGGTTTAAAGGCGTCCAACCTTCTCAAACCGT</u>	
<u>TTAGTTGCTGCAATTCCGCATGATTGTGCTGCATCAGGTATTCATACCTCGCCACAGAA</u>	
<u>CAATACAATCAAGATATTGTCAGTTTAGGTCTGCATGCCGCTCAAGATGTTTTAGAGATG</u>	
<u>GAGCAGAAATTACGCAATATTGTTTCAATGACAATCTGGTAGTTTGTGAGGCCATTTCAT</u>	
<u>CTTCGCGGCAATATTAGTGAAATTGCGCCTGAACTGCTAAATTTACCATGCAGTACGC</u>	
<u>GAAATCAGTTCTCCTTTGATCACTGATCGTGCCTGGATGAAGATATAATCCGCATTGCG</u>	
<u>GATGCAATTATTAATGATCAACTTCCTCTGCCAGAAATCATGCTGGAAGAATAAGCTTGG</u>	
<u>CGTAATCATGGTCATAGCTGTTTCTGTGTGAAATTGTTATCCGCTCACAATTCACACA</u>	
<u>ACATACGAGCCGAAGCATAAAGTGTAAGCCTGGGGTGCCTAATGAGTGAGCTAACTCA</u>	
<u>CATTAATTGCGTTGCGCTCACTGCCCGCTTTCAGTCGGGAAACCTGTCGTGCCAGCTGC</u>	
<u>ATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCCTATTGGGCGCTCTTCCGCTT</u>	
<u>CCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCCGCTGCGGCGAGCGGTATCAGCTCACT</u>	
<u>CAAAGGCGGTAGTACGGGTTTTGCTGCCCGCAAACGGGCTGTTCTGGTGTGCTAGTTTG</u>	
<u>TTATCAGAATCGCAGATCCGGCTTCAGGTTTGCCGGCTGAAAGCGCTATTTCTTCCAGAA</u>	
<u>TTGCCATGATTTTTTCCCCACGGGAGGCGTCACTGGCTCCCGTGTGTCGGCAGCTTTGA</u>	
<u>TTGATAAGCAGCATCGCCTGTTTCAGGCTGTCTATGTGTGACTGTTGAGCTGTAAACAG</u>	
<u>TTGTCTCAGGTGTTCAATTTTCATGTTCTAGTTGCTTTGTTTTACTGGTTTACCTGTTCT</u>	
<u>ATTAGGTGTTACATGCTGTTTCATCTGTTACATTGTGATCTGTTTCATGGTGAACAGCTTT</u>	
<u>AAATGCACCAAAACTCGTAAAAGCTCTGATGTATCTATCTTTTTTACACCGTTTTTCATC</u>	
<u>TGTGCATATGGACAGTTTTCCCTTTGATATCTAACGGTGAACAGTTGTTCTACTTTTGT</u>	

TABLE 12-continued

Nucleotide sequences of Tet promoter-PAL3 construct, low-copy (SEQ ID NO: 29)
TGTTAGTCTTGATGCTTCACTGATAGATACAAGAGCCATAAGAACCTCAGATCCTCCGT ATTTAGCCAGTATGTTCTCTAGTGTGGTTCGTTGTTTTGCGTGAGCCATGAGAACGAAC CATTGAGATCATGCTTACTTTGCATGTCACTCAAAAATTTTGCCTCAAACTGGTGAGCT GAATTTTTCAGTTAAAGCATCGTGTAGTGTCTTTCTTAGTCCGTTACGTAGGTAGGAAT CTGATGTAATGGTGTGGTATTTTGTCAACCATTCATTTTATCTGGTGTCTCAAGTT CGGTTACGAGATCCATTTGTCTATCTAGTTCACCTTGAAAAATCAACGTATCAGTCGGGC GGCCTCGCTTATCAACCACCAATTTCATATTGCTGTAAGTGTAAATCTTTACTTATTG GTTTCAAAACCCATTGGTTAAGCCTTTTAACTCATGGTAGTTATTTCAAGCATTAAACA TGAACCTAAATTCATCAAGGCTAATCTCTATATTGCTTGTGAGTTTCTTTTGTGTTA GTTCTTTTAATAACCACTATAAATCCTCATAGAGTATTTGTTTTCAAAAGACTTAACAT GTTCCAGATTATATTTTATGAATTTTTTAACTGGAAAAGATAAGGCAATATCTCTCAC TAAAACTAATTTCTAATTTTTCGCTTGAGAACTTGCATAGTTTGTCCACTGAAAACTCT CAAAGCCTTTAACCAGGATTCTCTGATTTCCACAGTTCTCGTCATCAGCTCTCGGTTG CTTTAGCTAATACACCATAAGCATTTCCTTACTGATGTTTCATCATCTGAGCGTATTGGT TATAAGTGAACGATACCGTCCGTTCTTTCTTGTAGGGTTTTCAATCGTGGGGTTGAGTA GTGCCACACAGCATAAAATTAGCTTGGTTTCATGCTCCGTTAAGTCATAGCGACTAATCG CTAGTTCATTTGCTTTGAAAACTAATTCAGACATACATCTCAATTGGTCTAGGTGAT TTTAATCACTATACCAATTGAGATGGGCTAGTCAATGATAATTACTAGTCTTTTCTCTT GAGTTGTGGGTATCTGTAAATCTGCTAGACCTTTGCTGAAAACTTGTAATCTGCTA GACCCTCTGTAATTCGCTAGACCTTTGTGTGTTTTTTTTGTTTATATCAAGTGGTTA TAATTTATAGAATAAAGAAAGATAAAAAAGATAAAAGAATAGATCCAGCCCTGTGT ATAACTACTACTTTAGTCAGTTCCGCAGTATTACAAAAGGATGTCGCAACGCTGTTTG CTCCTCTACAAAACAGACCTTAAACCCCTAAGGCTTAAGTAGCACCCCTCGCAAGCTCGG GCAAATCGCTGAATATTCCTTTTGTCTCCGACCATCAGGCACCTGAGTCGCTGTCTTTT CGTGACATTCAGTTTCGCTGCGCTCACGGCTCTGGCAGTGAATGGGGTAAATGGCACTAC AGGCGCCTTTTATGGATTTCATGCAAGGAACTACCCATAATACAAGAAAGCCCGTCACG GGCTTCTCAGGGCGTTTATGGCGGTCTGCTATGTGGTGTATCTGACTTTTGTGTT CAGCAGTTCCTGCCCTCTGATTTCCAGTCTGACCACTTCGGATTATCCCGTGACAGGTC ATTGAGACTGGCTAATGCACCCAGTAAGGCAGCGGTATCATCAACAGGCTTACCCGTCTT ACTGTCTTTTCTACGGGTCTGACGCTCAGTGAACGAAACTCACGTTAAGGGATTTTG GTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAATAATGAAGTTT AAATCAATCTAAAGTATATATGAGTAACTTGGTCTGACAGTTACCAATGCTTAATCAGT GAGGCACCTATCTCAGCGATCTGTCTATTTGTTTCATCATAGTTGCTGACTCCCCGTC GTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGTGCAATGATACCG CGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAGCCGAAGGGCC GAGCGCAGAAGTGGTCTGCAACTTTATCCGCCTCCATCCAGTCTATTAATGTGCGCG GAAGCTAGAGTAAGTAGTTTCGCCAGTTAATAGTTTTCGCAACGTTGTTGCCATTGCTACA GGCATCGTGGTGTACGCTCGTCTTGGTATGGCTTCATTCAGCTCCGTTCCCAACGA TCAAGCGAGTTACATGATCCCCATGTTGTGCAAAAAGCGGTTAGTCTCTTCGTCCT

TABLE 12-continued

Nucleotide sequences of Tet promoter-PAL3 construct, low-copy (SEQ ID NO: 29)
CCGATCGTTGTCAGAAGTAAGTTGGCCGCGAGTGTATCACTCATGGTTATGGCAGCACTG CATAATTCCTTACTGTGTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCA ACCAAGTCATTCTGAGAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATA CGGGATAATACCGCGCCACATAGCAGAACTTTAAAAGTGCTCATCATTTGGAAAACGTTCT TCGGGGCGAAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACT CGTGCACCCACTGATCTTCAGCATCTTTACTTTTACCAGCGTTTCTGGGTGAGCAAAA ACAGGAAGGCAAAATGCCGCAAAAAGGGAATAAGGGCGACACGGAATGTTGAATACTC ATACTCTTCTTTTCAATATTATTGAAGCATTATCAGGGTTATTGTCTCATGAGCGGA TACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACATTTCCCCGA AAAGTGCCACCTGACGCTCTAAGAAACATTATTATCATGACATTAACCTATAAAAATAGG CGTATCACGAGGCCCTTTCGTCTCGCGCGTTTCGGTGATGACGGTGAAAACCTCTGACAC ATGCAGCTCCCGGAGACGGTCACAGCTTGTCTGTAAGCGGATGCCGGGAGCAGACAAGCC CGTCAGGGCGCGTCAGCGGGTGTGGCGGGTGTCGGGGCTGGCTTAATATGCGGCATCA GAGCAGATTGTACTGAGAGTGACCATATGCGGTGTGAAATACCGCACAGATGCGTAAGG AGAAAAATACCGCATCAGGCGCCATTGCCATTAGGGCTGCGCAACTGTTGGGAAGGGCGA TCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGATGTGCTGCAAGGCGA TTAAGTTGGGTAAAGCCAGGGTTTTCCAGTCACGACGTTGTAAAACGACGGCCAGTGAA TTCGTTAAGACCCACTTTCACATTTAAGTTGTTTTCTAATCCGCATATGATCAATTCAA GGCCGAATAAGAAGGCTGGCTCTGCACCTTGGTGATCAAATAATTCGATAGCTTGTGTA ATAATGGCGGCATACTATCAGTAGTAGGTGTTTCCCTTCTTCTTTAGCGACTTGATGCT CTTGATCTTCCAATACGCAACCTAAAGTAAATGCCCCACAGCGCTGAGTGATATAATG CATTCTCTAGTGAAAAACCTTGTGGCATAAAAAGGCTAATTGATTTTCGAGAGTTTCAT ACTGTTTTTCTGTAGGCCGTGTACCTAAATGTACTTTTGCTCCATCGCGATGACTTAGTA AAGCACATCTAAACCTTTTAGCGTTATTACGTAAAAATCTTGCCAGCTTCCCCTTCTA AAGGGCAAAAGTGAGTATGGTGCCTATCTAACATCTCAATGGCTAAGGCGTCGAGCAAAG CCCGCTTATTTTTTACATGCCAATACAATGTAGGCTGCTCTACACCTAGCTTCTGGGCGA GTTTACGGGTGTTAAACCTTCGATTCCGACCTCATTAAAGCAGCTCTAATGCGCTGTTAA TCACTTTACTTTTATCTAATCTAGACATCATTAAATCCTAATTTTTGTTGACACTCTATC ATTGATAGAGTTATTTT

TABLE 13

Nucleotide sequences of TetR-PheP construct, low-copy (SEQ ID NO: 30)
ccagtggaattcgcttaagaccactttcacatttaagttgtttttctaatccgcatatgat caattcaaggccgaataagaaggctggctctgcaccttggtgatcaaataatcgatagc ttgtcgtaataatggcgccatactatcagtagtaggtgtttccctttcttcttagcgac ttgatgctcttgatcttccaatacgcacctaagtaaaatgccccacagcgctgagtgct

TABLE 13-continued

Nucleotide sequences of TetR-PheP construct, low-copy (SEQ ID NO: 30)	
<u>atataatgcattctctagtgaaaaaccttggtggcataaaaaggctaattgattttcgag</u>	
<u>agtttcatactgtttttctgtaggccgtgtacctaagtacttttgcctccatcgcgatg</u>	
<u>acttagtaaagcacatctaaaacttttagcggtattacgtaaaaaatcttgccagctttc</u>	
<u>cccttctaaagggcaaaagtgagtatggcgctatctaacaatctcaatggctaaggcgctc</u>	
<u>gagcaaaagcccgcttatttttacatgccaatacaatgtaggctgctctacacctagctt</u>	
<u>ctggcgagtttacgggtgttaaaccttcgattccgacctcattaagcagctcctaatgc</u>	
<u>gctgttaatcactttacttttatctaatactagacat</u> cattaattcctaattttt gttgac	
actctatcattgatagagttat tttaccactccctat cagtgatagagaaa agtgaactc	
tagaaataattttgtttaactttaagaaggagataacat ATGAAAAACGCGTCAACCGT	
ATCGGAAGATACTGCGTCGAATCAAGAGCCGACGCTTCATCGCGGATTACATAACCGTCA	
TATTCACATGATTGCGTTGGGTGGCGCAATTGGTACTGGTCTGTTCTTGGCATTGGCCC	
GGCGATTAGATGGCGGGTCCGGCTGTATTGCTGGGCTACGGCGTCGCCGGGATCATCGC	
TTTCCTGATTATGCGCCAGCTTGGCGAAATGGTGGTTGAGGAGCCGGTATCCGGTTCATT	
TGCCCCACTTTGCCTATAAATACTGGGGACCGTTTGCGGGCTTCCTCTCGGCTGGAATA	
CTGGGTAATGTTTCGTGCTGGTGGGAATGGCAGAGCTGACCGCTCGGGCATCTATATGCA	
GTA CTGGTTCCCGATGTTCCAACGTGGATTGGGCTGCCGCTTCTTTATTATCATCAA	
CGCCGTTAACCTGGTGAACGTGCGCTTATATGGCGAAACCGAGTTCTGGTTTGCCTTGAT	
TAAAGTGTGGCAATCATCGGTATGATCGGCTTTGGCCTGTGGCTGCTGTTTCTGGTCA	
CGGCGGCGAGAAAGCCAGTATCGACAACCTCTGGCGCTACGGTGGTTTCTTCGCCACCGG	
CTGGAATGGGCTGATTTTGTGCTGGCGTAATTATGTTCTCCTTCGGCGGTCTGGAGCT	
GATTGGGATTACTGCGCTGAAGCGCGCATCCGAAAAAGCATTCCAAAAGCGGTAAA	
TCAGGTGGTGTATCGCATCCTGCTGTTTACATCGGTTCACTGGTGGTTTACTGGCGCT	
CTATCCGTGGGTGGAAGTGAAATCCAACAGTAGCCCGTTTGTGATGATTTCCATAATCT	
CGACAGCAACGTGGTAGCTTCTGCGCTGAACTTCGTCACTCTGGTAGCATCGCTGTCAGT	
GTATAACAGCGGGGTTTACTCTAACAGCCGATGCTGTTTGGCCTTCTGTGCAGGGTAA	
TGCGCCGAAGTTTTTGACTCGCGTCAGCCGTCGCGGTGTCCGATTAACTCGCTGATGCT	
TTCCGGAGCGATCACTTCGCTGGTGGTGTAAATCAACTATCTGCTGCCGCAAAAAGCGTT	
TGGTCTGCTGATGGCGCTGGTGGTAGCAACGCTGCTGTTGAAGTGGATTATGATCTGCT	
GGCGCATCTGCGTTTTCTGTGACGATGCGACGTCAGGGCGTGAAACACAGTTTAAGGC	
GCTGCTCTATCCGTTTCGGCAACTATCTCTGCATTGCCTTCCTCGGCATGATTTTCTGCT	
GATGTGCACGATGGATGATATGCGCTTGTGACGATCCTGCTGCCGGTGTGGATTGTATT	
CTGTTTTATGGCATTAAACGCTGCGTCGGAAATAA	

In some embodiments, the genetically engineered bacteria contain gene sequence(s) comprising one or more sequence(s) of any of SEQ ID Nos: 21-30. In some embodiments, the genetically engineered bacteria contain gene sequence(s) comprising one or more sequence(s) having at least 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity with any of the sequences of SEQ ID Nos: 21-30.

Phenylalanine Transport

Each of PAL1 and PAL3 was expressed on a high-copy plasmid and a low-copy plasmid in genetically engineered *E. coli* Nissle. Surprisingly, each construct metabolized and reduced phenylalanine to similar levels (FIG. 15), and the rate-limiting step of phenylalanine metabolism was phenylalanine availability (FIG. 16). Thus, in some embodiments, it is advantageous to increase phenylalanine transport into the cell, thereby enhancing phenylalanine metabolism. Unexpectedly, even low-copy PAL plasmids are capable of almost completely eliminating Phe from a test sample when expressed in conjunction with pheP (FIG. 16A). Furthermore, there may be additional advantages to using a low-copy PAL-expressing plasmid in conjunction with pheP in order to enhance the stability of PAL expression while maintaining high phenylalanine metabolism, and to reduce negative selection pressure on the transformed bacterium. In alternate embodiments, the phenylalanine transporter is used in conjunction with the high-copy plasmid.

The genetically engineered bacteria further comprise a gene encoding a phenylalanine transporter. Phenylalanine transporters may be expressed or modified in the genetically engineered bacteria of the invention in order to enhance phenylalanine transport into the cell.

PheP is a membrane transport protein that is capable of transporting phenylalanine into bacterial cells (see, e.g., Pi et al., 1991). In some embodiments, the native pheP gene in the genetically modified bacteria of the invention is not modified. In some embodiments, the genetically engineered bacteria of the invention comprise multiple copies of the native pheP gene. In some embodiments, the genetically engineered bacteria of the invention comprise multiple copies of a non-native pheP gene. In some embodiments, the genetically engineered bacteria of the invention comprise a pheP gene that is controlled by its native promoter, an inducible promoter, a promoter that is stronger than the native promoter, e.g., the GlnRS promoter or the P(Bla) promoter, or a constitutive promoter. In some embodiments, expression of the pheP gene is controlled by a different promoter than the promoter that controls expression of the gene encoding the phenylalanine-metabolizing enzyme and/or the transcriptional regulator. In some embodiments, expression of the pheP gene is controlled by the same promoter that controls expression of the phenylalanine-metabolizing enzyme and/or the transcriptional regulator. In some embodiments, the pheP gene and the phenylalanine-metabolizing enzyme and/or the transcriptional regulator are divergently transcribed from a promoter region. In some embodiments, expression of each of the genes encoding PheP, the phenylalanine-metabolizing enzyme, and the transcriptional regulator is controlled by a different promoter. In some embodiments, expression of the genes encoding PheP, the phenylalanine-metabolizing enzyme, and the transcriptional regulator is controlled by the same promoter.

In some embodiments, the native pheP gene in the genetically modified bacteria is not modified, and one or more additional copies of the native pheP gene are inserted into the genome under the control of the same inducible pro-

moter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter. In alternate embodiments, the native pheP gene is not modified, and a copy of a non-native pheP gene from a different bacterial species is inserted into the genome under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter.

In some embodiments, the native pheP gene in the genetically modified bacteria is not modified, and one or more additional copies of the native pheP gene are present in the bacteria on a plasmid and under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of the PME, or a constitutive promoter. In alternate embodiments, the native pheP gene is not modified, and a copy of a non-native pheP gene from a different bacterial species is present in the bacteria on a plasmid and under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter.

In some embodiments, the native pheP gene is mutagenized, mutants exhibiting increased phenylalanine transport are selected, and the mutagenized pheP gene is isolated and inserted into the genetically engineered bacteria (see, e.g., Pi et al., 1996; Pi et al., 1998). The phenylalanine transporter modifications described herein may be present on a plasmid or chromosome.

In some embodiments, the genetically engineered bacterium is *E. coli* Nissle, and the native pheP gene in *E. coli* Nissle is not modified; one or more additional copies the native *E. coli* Nissle pheP genes are inserted into the *E. coli* Nissle genome under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter. In an alternate embodiment, the native pheP gene in *E. coli* Nissle is not modified, and a copy of a non-native pheP gene from a different bacterium is inserted into the *E. coli* Nissle genome under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter. In some embodiments, the genetically engineered bacterium is *E. coli* Nissle, and the native pheP gene in *E. coli* Nissle is not modified; one or more additional copies the native *E. coli* Nissle pheP genes are present in the bacterium on a plasmid and under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter. In an alternate embodiment, the native pheP gene in *E. coli* Nissle is not modified, and a copy of a non-native pheP gene from a different bacterium, are present in the bacterium on a plasmid and under the control of the same inducible promoter that controls expression of PAL, e.g., the FNR promoter, or a different inducible promoter than the one that controls expression of PAL, or a constitutive promoter.

It has been reported that *Escherichia coli* has five distinct transport systems (AroP, Mtr, PheP, TnaB, and TyrP) for the accumulation of aromatic amino acids. A general amino acid permease, encoded by the aroP gene, transports three aromatic amino acids, including phenylalanine, with high affinity, and is thought, together with PheP, responsible for the

lion share of phenylalanine import. Additionally, a low level of accumulation of phenylalanine was observed in an aromatic amino acid transporter-deficient *E. coli* strain (Δ aroP Δ pheP Δ mtr Δ tna Δ tyrP), and was traced to the activity of the LIV-I/LS system, which is a branched-chain amino acid transporter consisting of two periplasmic binding proteins, the LIV-binding protein (LIV-I system) and LS-binding protein (LS system), and membrane components, LivHMGF (Koyanagi et al., and references therein; Identification of the LIV-I/LS System as the Third Phenylalanine Transporter in *Escherichia coli* K-12).

In some embodiments, the genetically engineered bacteria comprise an aroP gene. In some embodiments, the genetically engineered bacterium is *E. coli* Nissle, and the native aroP gene in *E. coli* Nissle is not modified; one or more additional copies of the native *E. coli* Nissle aroP genes are present in the bacterium on a plasmid or in the chromosome and under the control of the same inducible promoter that controls expression of the PME, e.g., the FNR promoter, or the araBAD promoter, a different inducible promoter than the one that controls expression of the PME, or a constitutive promoter. In an alternate embodiment, the native aroP gene in *E. coli* Nissle is not modified, and a copy of a non-native aroP gene from a different bacterium, are present in the bacterium on a plasmid or in the chromosome and under the control of the same inducible promoter that controls expression of the PME, e.g., the FNR promoter or the AraBAD promoter, or a different inducible promoter than the one that controls expression of the PME, or a constitutive promoter.

In other embodiments, the genetically engineered bacteria comprise AroP and PheP, under the control of the same or different inducible or constitutive promoters.

In some embodiments, the pheP gene is expressed on a chromosome. In some embodiments, expression from the chromosome may be useful for increasing stability of expression of pheP. In some embodiments, the pheP gene is integrated into the bacterial chromosome at one or more integration sites in the genetically engineered bacteria. In some embodiments, the pheP gene is inserted into the bacterial genome at one or more of the following insertion sites in *E. coli* Nissle: male/K, insB/I, araC/BAD, lacZ, agal/rsmI, thyA, and malP/T. Any suitable insertion site may be used (see, e.g., FIG. 36). The insertion site may be anywhere in the genome, e.g., in a gene required for survival and/or growth, such as thyA (to create an auxotroph); in an active area of the genome, such as near the site of genome replication; and/or in between divergent promoters in order to reduce the risk of unintended transcription, such as between AraB and AraC of the arabinose operon.

In some embodiments, the genetically engineered bacterium comprises multiple mechanisms of action and/or one or more auxotrophies. In certain embodiments, the bacteria are genetically engineered to comprise five copies of PAL under the control of an oxygen level-dependent promoter (e.g., P_{fms} -PAL3) inserted at different integration sites on the chromosome (e.g., male/K, yicS nepI, malP/T, agal/rsmI, and cea), and one copy of a phenylalanine transporter gene under the control of an oxygen level-dependent promoter (e.g., P_{fms} -pheP) inserted at a different integration site on the chromosome (e.g., lacZ). In a more specific aspect, the bacteria are genetically engineered to further include a kanamycin resistance gene, and a thyA auxotrophy, in which the thyA gene is deleted and/or replaced with an unrelated gene.

Multiple Mechanisms of Action

In some embodiments, the bacteria are genetically engineered to include multiple mechanisms of action (MoAs),

e.g., circuits producing multiple copies of the same product (e.g., to enhance copy number) or circuits performing multiple different functions. Examples of insertion sites include, but are not limited to, male/K, yicS nepI, insB/I, araC/BAD, lacZ, agal/rsmI, thyA, malP/T, dapA, and cea, and others shown in FIG. 36. For example, the genetically engineered bacteria may include four copies of PAL inserted at four different insertion sites, e.g., male/K, insB/I, araC/BAD, and lacZ. The genetically engineered bacteria may also include four copies of PAL inserted at four different insertion sites, e.g., male/K, yicS nepI, agal/rsmI, and cea, and one copy of a phenylalanine transporter gene inserted at a different insertion site, e.g., lacZ (FIG. 13B). Alternatively, the genetically engineered bacteria may include three copies of PAL inserted at three different insertion sites, e.g., male/K, insB/I, and lacZ, and three copies of a phenylalanine transporter gene inserted at three different insertion sites, e.g., dapA, cea, and araC/BAD.

In some embodiments, the genetically engineered bacteria comprise one or more of (1) PAL, PAH, LAAD for degradation of phenylalanine, in wild type or in a mutated form (for increased stability or metabolic activity) (2) transporter PheP or AroP for uptake of phenylalanine, in wild type or in mutated form (for increased stability or metabolic activity) (3) PAL, PAH, LAAD, and/or PheP for secretion and extracellular phenylalanine degradation, (4) components of secretion machinery, as described herein (5) Auxotrophy, e.g., deltaThyA (6) antibiotic resistance, including but not limited to, kanamycin or chloramphenicol resistance (7) mutations/deletions in genes involved in oxygen metabolism, as described herein and (8) mutations/deletions in genes of the endogenous Nissle phenylalanine synthesis pathway (e.g., deltaPheA for Phe auxotrophy).

In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, (e.g., under the control of a Pfnr promoter) and one or more copies of PAL1 (e.g. under the control of a Pfnr promoter). In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, (e.g., under the control of a Pfnr promoter) and one or more copies of PAL1 (e.g. under the control of a Pfnr promoter); and further comprises one or more copies of a phenylalanine transporter (e.g., PheP and/or AroP, e.g., under the control of a Pfnr promoter). In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, (e.g., under the control of a Pfnr promoter) and one or more copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, (e.g., under the control of a Pfnr promoter) and one or more copies of LAAD (e.g., under the control of the ParaBAD promoter); and further comprises one or more copies of a phenylalanine transporter (e.g., PheP and/or AroP, e.g., under the control of a Pfnr promoter). In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, (e.g., under the control of a Pfnr promoter) and one or more copies of PAH. In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, (e.g., under the control of a Pfnr promoter) and one or more copies of PAH; and further comprises one or more copies of a phenylalanine transporter (e.g., PheP and/or AroP, e.g., under the control of a Pfnr promoter). In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL1, (e.g., under the control of a Pfnr promoter) and one or more copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL1, (e.g., under the control of

or more copies of a phenylalanine transporter (e.g., PheP and/or AroP, e.g., under the control of a P_{fir} promoter). PME and/or transporters may be integrated into any of the insertion sites described herein. Alternatively, PME and/or transporters may be comprised on low or high copy plasmids. PME and/or transporters may be integrated into any of the insertion sites described herein in combination with PME and/or transporters that are comprised on low or high copy plasmids.

In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, e.g. (e.g., under the control of a P_{fir} promoter), one or more copies of PAL1, e.g. (e.g., under the control of a P_{fir} promoter), one or more copies of LAAD (e.g., under the control of the ParaBAD promoter), and one or more copies of PAH. In one embodiment, the genetically engineered bacteria comprise one or more copies of PAL3, e.g. (e.g., under the control of a P_{fir} promoter), one or more copies of PAL1, e.g. (e.g., under the control of a P_{fir} promoter), one or more copies of LAAD (e.g., under the control of the ParaBAD promoter), and one or more copies of PAH; and further comprise one or more copies of a phenylalanine transporter (e.g., PheP and/or AroP, e.g., under the control of a P_{fir} promoter). PME and transporters may be integrated into any of the insertion sites described herein. Alternatively, PMEs and/or transporters may be comprised on low or high copy plasmids.

In one embodiment, the genetically engineered bacteria comprise one copy of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fmr} promoter) one copy of PheP (e.g., under the control of a P_{fmr} promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise one copy of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fmr} promoter) two copies of PheP (e.g., under the control of a P_{fmr} promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise one copy of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fmr} promoter) one copy of PheP (e.g., under the control of a P_{fmr} promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise one copy of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fmr} promoter) two copies of PheP (e.g., under the control of a P_{fmr} promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). PME's and transporters may be integrated into any of the insertion sites described herein. Alternatively, located PME's and/or transporters may be comprised on low or high copy plasmids.

In one embodiment, the genetically engineered bacteria comprise two copies of PAL (e.g., PAL1 or PAL3, e.g., under the control of a P_{fir} promoter) one copy of PheP (e.g., under the control of a P_{fir} promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise two copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fir} promoter) two copies of PheP (e.g., under the control of a P_{fir} promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise two copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fir} promoter) one copy of PheP (e.g., under the control of a P_{fir} promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise two copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a P_{fir} promoter) two copies of PheP (e.g., under the control of a

Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter).

In one embodiment, the genetically engineered bacteria comprise three copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) one copy of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise three copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) two copies of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise three copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) one copy of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise three copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) two copies of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise three copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter), three copies of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise three copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter), three copies of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter).

In one embodiment, the genetically engineered bacteria comprise four copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) one copy of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise four copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) two copies of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise four copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) one copy of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise four copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) two copies of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter).

In one embodiment, the genetically engineered bacteria comprise five copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) one copy of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise five copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) two copies of PheP (e.g., under the control of a Pfmr promoter), and one copy of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise five copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) one copy of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter). In one embodiment,

the genetically engineered bacteria comprise five copies of PAL, (e.g., PAL1 or PAL3, e.g., under the control of a Pfmr promoter) two copies of PheP (e.g., under the control of a Pfmr promoter), and two copies of LAAD (e.g., under the control of the ParaBAD promoter).

In one embodiment, the genetically engineered bacteria comprise one or more PME for metabolizing phenylalanine in combination with one or more PME for secretion. In one embodiment, the genetically engineered bacteria comprise one or more PME for metabolizing phenylalanine and a phenylalanine transporter in combination with one or more PME for secretion. In one embodiment, the genetically engineered bacteria comprise one or more PME for metabolizing phenylalanine and a phenylalanine transporter in combination with one or more PME for secretion, and also include an auxotrophy and/or an antibiotic resistance. Secretion systems described herein are utilized to secrete the PME in the genetically engineered bacteria with multiple mechanisms of action.

In one embodiment, the genetically engineered bacteria comprise two additional copies of PheP (in addition to the wild type gene). This provides redundancy, in case one of the PheP genes acquires a mutation. In one embodiment, the PheP genes are inserted at lacZ and agal/rsml. In one embodiment, the two copies of PheP are under the control of the PfmrS promoter. In one embodiment, the genetically engineered bacteria comprise three copies of PAL3. In one embodiment, the genetically engineered bacteria comprise three copies of PAL3, inserted at maleK, malPT, yicS/nepl. In one embodiment, the expression of the three copies of PAL3 is under the control of the PfmrS promoter. In one embodiment, the genetically engineered bacteria comprise one or more copies of LAAD. In one embodiment, the genetically engineered bacteria comprise one copy of LAAD, inserted in the arabinose operon. In one embodiment, LAAD is under the control of the endogenous ParaBAD promoter. In one embodiment, the genetically engineered bacteria comprise an auxotrophy, e.g., deltaThyA. In one embodiment, the genetically engineered bacteria comprise an antibiotic resistance. In one embodiment, the genetically engineered bacteria comprise an antibiotic resistance and an auxotrophy, e.g., deltaThyA. In one embodiment, the genetically engineered bacteria do not comprise an auxotrophy, e.g., deltaThyA. In one embodiment, the genetically engineered bacteria do not comprise an antibiotic resistance. In one embodiment, the genetically engineered bacteria comprise neither an antibiotic resistance nor an auxotrophy, e.g., deltaThyA.

In one embodiment, the genetically engineered bacteria comprise three copies of PAL, e.g., PAL3, 2 copies of PheP (in addition to the endogenous PheP), and one copy of LAAD. In one embodiment, the genetically engineered bacteria comprise three copies of PAL, e.g., PAL3, 2 copies of PheP (in addition to the endogenous PheP), and one copy of LAAD, and an auxotrophy, e.g., deltaThyA. In one embodiment, the genetically engineered bacteria comprise three copies of PAL, 2 copies of PheP (in addition to the endogenous PheP), and one copy of LAAD, and an antibiotic resistance gene. In one embodiment, the genetically engineered bacteria comprise three copies of PAL, 2 copies of PheP (in addition to the endogenous PheP), and one copy of LAAD, and an antibiotic resistance gene and an auxotrophy, e.g., deltaThyA.

In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a PfmrS promoter), 2 copies of PheP (each under control of a PfmrS promoter), and one copy of LAAD (under the control of the

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endogenous ParaBAD promoter). In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter), 2 copies of PheP (each under control of a P_fnrS promoter), and one copy of LAAD (under the control of the endogenous ParaBAD promoter), and an antibiotic resistance. In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter), 2 copies of PheP (each under control of a P_fnrS promoter), and one copy of LAAD (under the control of the endogenous ParaBAD promoter), and an auxotrophy, e.g., delta ThyA. In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter), 2 copies of PheP (each under control of a P_fnrS promoter), and one copy of LAAD (under the control of the endogenous ParaBAD promoter), and an antibiotic resistance and an auxotrophy, e.g., deltaThyA.

In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter and inserted at the malEK, malPT, and yicS/nepI sites), 2 copies of PheP (each under control of a P_fnrS promoter and inserted at the LacZ and agal/rsml sites), and one copy of LAAD (under the control of the endogenous ParaBAD promoter, and inserted in the endogenous arabinose operon). In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter and inserted at the malEK, malPT, and yicS/nepI sites), 2 copies of PheP (each under control of a P_fnrS promoter and inserted at the LacZ and agal/rsml sites), and one copy of LAAD (under the control of the endogenous ParaBAD promoter, and inserted in the endogenous arabinose operon), and further comprise an antibiotic resistance. In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter and inserted at the malEK, malPT, and yicS/nepI sites), 2 copies of PheP (each under control of a P_fnrS promoter and inserted at the LacZ and agal/rsml sites), and one copy of LAAD (under the control of the endogenous ParaBAD promoter, and inserted in the endogenous arabinose operon) and further comprise an auxotrophy, e.g., deltaThyA. In one embodiment, the genetically engineered bacteria comprise three copies of PAL (each under control of a P_fnrS promoter and inserted at the malEK, malPT, and yicS/nepI sites), 2 copies of PheP (each under control of a P_fnrS promoter and inserted at the LacZ and agal/rsml sites), and one copy of LAAD (under the control of the endogenous ParaBAD promoter, and inserted in the endogenous arabinose operon), and further comprise an antibiotic resistance and an auxotrophy, e.g., deltaThyA.

In one embodiment, the genetically engineered bacteria are SYN-PKU705. In one embodiment, SYN-PKU705 further comprises an antibiotic resistance. In one embodiment, SYN-PKU705 further comprises an auxotrophy, e.g., deltaThyA. In one embodiment, SYN-PKU705 further comprises an antibiotic resistance and auxotrophy, e.g., deltaThyA.

Table 14 contains non-limiting examples of the genetically engineered bacteria of the disclosure. In certain embodiments, the genetically engineered bacteria of Table 14 further contain a PME for secretion.

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TABLE 14

Non-limiting Examples of Embodiments of the Disclosure	
Strain Name	Genotype
Plasmid -based strains	
SYN-PKU101	Low copy pSC101-Ptet::PAL1, ampicillin resistant
SYN-PKU102	High copy pColE1-Ptet::PAL1, ampicillin resistant,
SYN-PKU201	Low copy pSC101-Ptet::PAL3, ampicillin resistant
SYN-PKU202	High copy pColE1-Ptet::PAL3, ampicillin resistant,
SYN-PKU203	lacZ::Ptet-pheP::cam
SYN-PKU401	Low copy pSC101-Ptet::PAL1, ampicillin resistant, chromosomal lacZ::Ptet-pheP::cam
SYN-PKU402	High copy pColE1-Ptet::PAL1, ampicillin resistant, chromosomal lacZ::Ptet-pheP::cam
SYN-PKU302	Low Copy pSC101-Ptet::PAL3, ampicillin resistant; chromosomal lacZ::Ptet-pheP::cam
SYN-PKU303	High copy pColE1-Ptet::PAL3, ampicillin resistant, chromosomal lacZ::Ptet-pheP::cam
SYN-PKU304	Low Copy pSC101-P _f nrS::PAL3, ampicillin resistant; chromosomal lacZ::P _f nrS-pheP::cam
SYN-PKU305	Low Copy pSC101-P _f nrS::PAL3, kanamycin resistant; chromosomal lacZ::P _f nrS-pheP::cam
SYN-PKU306	Low Copy pSC101-P _f nrS::PAL3, kanamycin resistant; thyA
SYN-PKU307	Low Copy pSC101-P _f nrS::PAL3, ampicillin resistant;
SYN-PKU308	Low Copy pSC101-P _f nrS::PAL3, kanamycin resistant;
SYN-PKU401	High Copy pUC57-Ptet::LAAD; kanamycin resistant
Integrated strains	
SYN-PKU501	malPT::P _f nrS::PAL3::kan
SYN-PKU502	malPT::P _f nrS::PAL3::kan; bicistronic lacZ::P _f nrS::PAL3-pheP::cam
SYN-PKU503	malEK::P _f nrS::PAL3::cam
SYN-PKU504	agal/rsml::P _f nrS::PAL3
SYN-PKU505	cea::P _f nrS::PAL3
SYN-PKU506	malEK::P _f nrS::PAL3; agal/rsml::P _f nrS::PAL3; cea::P _f nrS::PAL3
SYN-PKU507	malEK::P _f nrS::PAL3; agal/rsml::P _f nrS::PAL3; cea::P _f nrS::PAL3; lacZ::P _f nr-pheP::cam
SYN-PKU508	malEK::P _f nrS::PAL3; pheA auxotroph
SYN-PKU509	malEK::P _f nrS::PAL3; agal/rsml::P _f nrS::PAL3; cea::P _f nrS::PAL3; lacZ::P _f nr-pheP::cam
SYN-PKU601	malPT::P _f nrS-INT5::kan, rrnBUP-[PAL3]; lacZ::P _f nr-pheP::cam (recombinase based strain)

TABLE 14-continued

Non-limiting Examples of Embodiments of the Disclosure	
Strain Name	Genotype
SYN-PKU510	malEK::PfurS::PAL3; agaI/rsmI::PfurS::PAL3; cea::PfurS::PAL3;
SYN-PKU511	malEK::PfurS::PAL3; agaI/rsmI::PfurS::PAL3; cea::PfurS::PAL3; yicS/nepI::PfurS-PAL3::kan; malPT::PfurS::PAL3; lacZ::Pfur-pheP; AthyA lacZ::Pfur-pheP::cam
SYN-PKU204	
SYN-PKU512	malEK::PfurS::PAL3; agaI/rsmI::PfurS::PAL3; cea::PfurS::PAL3; malPT::PfurS::PAL3; lacZ::Pfur-pheP::cam; AthyA
SYN-PKU513	malEK::PfurS::PAL3; agaI/rsmI::PfurS::PAL3; cea::PfurS::PAL3; lacZ::Pfur-pheP; AthyA
SYN-PKU514	malEK::PfurS::PAL3; agaI/rsmI::PfurS::PAL3; cea::PfurS::PAL3; malPT::PfurS::PAL3; AthyA
SYN-PKU515	malEK::PfurS::PAL3; agaI/rsmI::PfurS::PAL3; cea::PfurS::PAL3; AthyA
SYN-PKU516	agaI/rsmI::PfurS::PAL3::kan
SYN-PKU517	malEK::PfurS::PAL3::cam; malPT::PfurS::PAL3::kan; lacZ::PfurS-pheP; AthyA
SYN-PKU518	malEK-PfurS::PAL3::cam; PfurS-pheP::kan
SYN-PKU519	ParaBC-PAL3::cam; PfurS-pheP::kan
SYN-PKU520	agaI/rsmI::PfurS::PAL3::kan; PfurS-PheP::cam
SYN-PKU801	ΔargR; thyA::cam
SYN-PKU701	ParaBC-LAAD::cam; malEK-PfurS-PAL3; malPT::PfurS-PAL3::kan; PfurS-pheP yicS/nepI::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam
SYN-PKU521	cea::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam
SYN-PKU522	malPT::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam
SYN-PKU523	malPT::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam
SYN-PKU524	malEK::PfurS::PAL3; malPT::PfurS::PAL3; lacZ::Pfur-pheP
SYN-PKU702	malEK::PfurS::PAL3; lacZ::Pfur-pheP; Para::LAAD
SYN-PKU703	malEK::PfurS::PAL3; malPT::PfurS-PAL3; lacZ::Pfur-pheP; agaI/rsmI::PfurS-pheP; Para::LAAD
SYN-PKU704	malEK::PfurS::PAL3; malPT::PfurS::PAL3; yicS/nepI::PfurS-PAL3; lacZ::Pfur-pheP; Para::LAAD
SYN-PKU705	malEK::PfurS::PAL3; malPT::PfurS::PAL3; yicS/nepI::PfurS-PAL3::kan; lacZ::Pfur-pheP; agaI/rsmI::PfurS-pheP Para::LAAD
SYN-PKU602	malEK::PT7::PAL3; Para::INT5::cam (recombinase); lacZ::Pfur-pheP; malPT::Pconstitutive::T7 polymerase (unflipped);

TABLE 14-continued

Non-limiting Examples of Embodiments of the Disclosure	
Strain Name	Genotype
SYN-PKU901	Nissle with streptomycin resistance
5	
10	Secretion
15	In some embodiments, the genetically engineered bacteria further comprise a native secretion mechanism (e.g., gram positive bacteria) or non-native secretion mechanism (e.g., gram negative bacteria) that is capable of secreting the protein(s) of interest or therapeutic protein(s), e.g., PAH, PAL or LAAD, from the bacterial cytoplasm. Many bacteria have evolved sophisticated secretion systems to transport substrates across the bacterial cell envelope. Substrates, such as small molecules, proteins, and DNA, may be released into the extracellular space or periplasm (such as the gut lumen or other space), injected into a target cell, or associated with the bacterial membrane.
20	In Gram-negative bacteria, secretion machineries may span one or both of the inner and outer membranes. In some embodiments, the genetically engineered bacteria further comprise a non-native double membrane-spanning secretion system. Double membrane-spanning secretion systems include, but are not limited to, the type I secretion system (T1SS), the type II secretion system (T2SS), the type III secretion system (T3SS), the type IV secretion system (T4SS), the type VI secretion system (T6SS), and the resistance-nodulation-division (RND) family of multi-drug efflux pumps (Pugsley 1993; Gerlach et al., 2007; Collinson et al., 2015; Costa et al., 2015; Reeves et al., 2015; WO2014138324A1, incorporated herein by reference). Examples of such secretion systems are shown in FIGS. 3-6. Mycobacteria, which have a Gram-negative-like cell envelope, may also encode a type VII secretion system (T7SS) (Stanley et al., 2003). With the exception of the T2SS, double membrane-spanning secretions generally transport substrates from the bacterial cytoplasm directly into the extracellular space or into the target cell. In contrast, the T2SS and secretion systems that span only the outer membrane may use a two-step mechanism, wherein substrates are first translocated to the periplasm by inner membrane-spanning transporters, and then transferred to the outer membrane or secreted into the extracellular space. Outer membrane-spanning secretion systems include, but are not limited to, the type V secretion or autotransporter system (T5SS), the curli secretion system, and the chaperone-usher pathway for pili assembly (Saier, 2006; Costa et al., 2015).
25	In some embodiments, the genetically engineered bacteria of the invention further comprise a type III or a type III-like secretion system (T3SS) from <i>Shigella</i> , <i>Salmonella</i> , <i>E. coli</i> , <i>Bivrio</i> , <i>Burkholderia</i> , <i>Yersinia</i> , <i>Chlamydia</i> , or <i>Pseudomonas</i> . The T3SS is capable of transporting a protein from the bacterial cytoplasm to the host cytoplasm through a needle complex. The T3SS may be modified to secrete the molecule from the bacterial cytoplasm, but not inject the molecule into the host cytoplasm. Thus, the molecule is secreted into the gut lumen or other extracellular space. In some embodiments, the genetically engineered bacteria comprise said modified T3SS and are capable of secreting the protein(s) of interest or therapeutic protein(s) from the bacterial cytoplasm. In some embodiments, the secreted molecule, such as a heterologous protein or peptide, e.g., the protein of interest or therapeutic protein e.g., PAH, PAL or LAAD, comprises
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a type III secretion sequence that allows the protein(s) of interest or therapeutic protein(s) to be secreted from the bacteria.

In some embodiments, a flagellar type III secretion pathway is used to secrete the molecule of interest, e.g., PAH, PAL or LAAD. In some embodiments, an incomplete flagellum is used to secrete a therapeutic peptide of interest, e.g., PAH, PAL or LAAD, by recombinantly fusing the peptide to an N-terminal flagellar secretion signal of a native flagellar component. In this manner, the intracellularly expressed chimeric peptide can be mobilized across the inner and outer membranes into the surrounding host environment.

In some embodiments, a Type V Autotransporter Secretion System is used to secrete the therapeutic peptide, e.g., PAH, PAL or LAAD. Due to the simplicity of the machinery and capacity to handle relatively large protein fluxes, the Type V secretion system is attractive for the extracellular production of recombinant proteins. As shown in FIG. 10, a therapeutic peptide (star) can be fused to an N-terminal secretion signal, a linker, and the beta-domain of an autotransporter. The N-terminal signal sequence directs the protein to the SecA-YEG machinery which moves the protein across the inner membrane into the periplasm, followed by subsequent cleavage of the signal sequence. The Beta-domain is recruited to the Bam complex ('Beta-barrel assembly machinery') where the beta-domain is folded and inserted into the outer membrane as a beta-barrel structure. The therapeutic peptide, e.g., PAH, PAL or LAAD, is threaded through the hollow pore of the beta-barrel structure ahead of the linker sequence. Once exposed to the extracellular environment, the therapeutic peptide, e.g., PAH, PAL or LAAD, can be freed from the linker system by an autocatalytic cleavage (left side of Bam complex) or by targeting of a membrane-associated peptidase (black scissors; right side of Bam complex) to a complimentary protease cut site in the linker. Thus, in some embodiments, the secreted molecule, such as a heterologous protein or peptide, e.g., the protein of interest or therapeutic protein, comprises an N-terminal secretion signal, a linker, and beta-domain of an autotransporter so as to allow the molecule to be secreted from the bacteria.

In some embodiments, a Hemolysin-based Secretion System is used to secrete the molecule of interest, e.g., e.g., PAH, PAL or LAAD. Type I Secretion systems offer the advantage of translocating their passenger peptide directly from the cytoplasm to the extracellular space, obviating the two-step process of other secretion types. FIG. 11 shows the alpha-hemolysin (HlyA) of uropathogenic *Escherichia coli*. This pathway uses HlyB, an ATP-binding cassette transporter; HlyD, a membrane fusion protein; and TolC, an outer membrane protein. The assembly of these three proteins forms a channel through both the inner and outer membranes. Natively, this channel is used to secrete HlyA, however, to secrete the therapeutic peptide of the present disclosure, the secretion signal-containing C-terminal portion of HlyA is fused to the C-terminal portion of a therapeutic peptide (star) to mediate secretion of this peptide.

In alternate embodiments, the genetically engineered bacteria further comprise a non-native single membrane-spanning secretion system. Single membrane-spanning exporters may act as a component of a secretion system, or may export substrates independently. Such exporters include, but are not limited to, ATP-binding cassette translocases, flagellum/virulence-related translocases, conjugation-related translocases, the general secretory system (e.g., the SecYEG complex in *E. coli*), the accessory secretory system in

mycobacteria and several types of Gram-positive bacteria (e.g., *Bacillus anthracis*, *Lactobacillus johnsonii*, *Corynebacterium glutamicum*, *Streptococcus gordonii*, *Staphylococcus aureus*), and the twin-arginine translocation (TAT) system (Saier, 2006; Rigel and Braunstein, 2008; Albinak et al., 2013). It is known that the general secretory and TAT systems can both export substrates with cleavable N-terminal signal peptides into the periplasm, and have been explored in the context of biopharmaceutical production. The TAT system may offer particular advantages, however, in that it is able to transport folded substrates, thus eliminating the potential for premature or incorrect folding. In certain embodiments, the genetically engineered bacteria comprise a TAT or a TAT-like system and are capable of secreting the protein(s) of interest or therapeutic protein(s), e.g., PAH, PAL or LAAD, from the bacterial cytoplasm. One of ordinary skill in the art would appreciate that the secretion systems disclosed herein may be modified to act in different species, strains, and subtypes of bacteria, and/or adapted to deliver different payloads.

In order to translocate a protein, e.g., therapeutic polypeptide, e.g., PAH, PAL or LAAD, to the extracellular space, the polypeptide must first be translated intracellularly, mobilized across the inner membrane and finally mobilized across the outer membrane. Many effector proteins (e.g., therapeutic polypeptides)—particularly those of eukaryotic origin—contain disulphide bonds to stabilize the tertiary and quaternary structures. While these bonds are capable of correctly forming in the oxidizing periplasmic compartment with the help of periplasmic chaperones, in order to translocate the polypeptide across the outer membrane the disulphide bonds must be reduced and the protein unfolded again.

One way to secrete properly folded proteins in gram-negative bacteria—particularly those requiring disulphide bonds—is to target the periplasm in a bacterium with a destabilized outer membrane. In this manner the protein is mobilized into the oxidizing environment and allowed to fold properly. In contrast to orchestrated extracellular secretion systems, the protein is then able to escape the periplasmic space in a correctly folded form by membrane leakage. These "leaky" gram-negative mutants are therefore capable of secreting bioactive, properly disulphide-bonded polypeptides. In some embodiments, the genetically engineered bacteria have a "leaky" or de-stabilized outer membrane. Destabilizing the bacterial outer membrane to induce leakiness can be accomplished by deleting or mutagenizing genes responsible for tethering the outer membrane to the rigid peptidoglycan skeleton, including for example, lpp, ompC, ompA, ompF, tolA, tolB, pal, degS, degP, and nlpI. Lpp is the most abundant polypeptide in the bacterial cell existing at ~500,000 copies per cell and functions as the primary 'staple' of the bacterial cell wall to the peptidoglycan. Silhavy, T. J., Kahne, D. & Walker, S. The bacterial cell envelope. *Cold Spring Harb Perspect Biol* 2, a000414 (2010). TolA-PAL and OmpA complexes function similarly to Lpp and are other deletion targets to generate a leaky phenotype. Additionally, leaky phenotypes have been observed when periplasmic proteases are deactivated. The periplasm is very densely packed with protein and therefore encode several periplasmic proteins to facilitate protein turnover. Removal of periplasmic proteases such as degS, degP or nlpI can induce leaky phenotypes by promoting an excessive build-up of periplasmic protein. Mutation of the proteases can also preserve the effector polypeptide by preventing targeted degradation by these proteases. Moreover, a combination of these mutations may synergistically

enhance the leaky phenotype of the cell without major sacrifices in cell viability. Thus, in some embodiments, the engineered bacteria have one or more deleted or mutated membrane genes. In some embodiments, the engineered bacteria have a deleted or mutated lpp gene. In some embodiments, the engineered bacteria have one or more deleted or mutated gene(s), selected from ompA, ompA, and ompF genes. In some embodiments, the engineered bacteria have one or more deleted or mutated gene(s), selected from tolA, tolB, and pal genes. In some embodiments, the engineered bacteria have one or more deleted or mutated periplasmic protease genes. In some embodiments, the engineered bacteria have one or more deleted or mutated periplasmic protease genes selected from degS, degP, and nlpI. In some embodiments, the engineered bacteria have one or more deleted or mutated gene(s), selected from lpp, ompA, ompA, ompF, tolA, tolB, pal, degS, degP, and nlpI genes.

To minimize disturbances to cell viability, the leaky phenotype can be made inducible by placing one or more membrane or periplasmic protease genes, e.g., selected from lpp, ompA, ompA, ompF, tolA, tolB, pal, degS, degP, and nlpI, under the control of an inducible promoter. For example, expression of lpp or other cell wall stability protein or periplasmic protease can be repressed in conditions where the therapeutic polypeptide needs to be delivered (secreted). For instance, under inducing conditions a transcriptional repressor protein or a designed antisense RNA can be expressed which reduces transcription or translation of a target membrane or periplasmic protease gene. Conversely, overexpression of certain peptides can result in a destabi-

lized phenotype, e.g., overexpression of colicins or the third topological domain of TolA, wherein peptide overexpression can be induced in conditions in which the therapeutic polypeptide needs to be delivered (secreted). These sorts of strategies would decouple the fragile, leaky phenotypes from biomass production. Thus, in some embodiments, the engineered bacteria have one or more membrane and/or periplasmic protease genes under the control of an inducible promoter.

Table 15 and Table 16 list secretion systems for Gram positive bacteria and Gram negative bacteria. These can be used to secrete polypeptides, proteins of interest or therapeutic protein(s) from the engineered bacteria, which are reviewed in Milton H. Saier, Jr. Microbe/Volume 1, Number 9, 2006 "Protein Secretion Systems in Gram-Negative Bacteria Gram-negative bacteria possess many protein secretion-membrane insertion systems that apparently evolved independently", the contents of which is herein incorporated by reference in its entirety.

TABLE 15

Secretion systems for gram positive bacteria	
Bacterial Strain	Relevant Secretion System
<i>C. novyi</i> -NT (Gram+)	Sec pathway Twin-arginine (TAT) pathway
<i>C. butyricum</i> (Gram+)	Sec pathway Twin-arginine (TAT) pathway
<i>Listeria monocytogenes</i> (Gram+)	Sec pathway Twin-arginine (TAT) pathway

TABLE 16

Secretion Systems for Gram negative bacteria Protein secretory pathways (SP) in gram-negative bacteria and their descendants							
Type (Abbreviation)	Name	TC# ²	Bacteria	Archaea	Eukarya	# Proteins/ Energy System	Source
IMPS - Gram-negative bacterial inner membrane channel-forming translocases							
ABC (SIP)	ATP binding cassette translocase	3.A.1	+	+	+	3-4	ATP
SEC (IISP)	General secretory translocase	3.A.5	+	+	+	~12	GTP OR ATP + PMF
Fla/Path (IIISP)	Flagellum/virulence-related translocase	3.A.6	+	-	-	>10	ATP
Conj (IVSP)	Conjugation-related translocase	3.A.7	+	-	-	>10	ATP
Tat (IISP)	Twin-arginine targeting translocase	2.A.64	+	+	+	2-4	PMF (chloroplasts)
Oxal (YidC)	Cytochrome oxidase biogenesis family	2.A.9	+	+	+	1	None or PMF (mitochondria chloroplasts)
MscL	Large conductance mechanosensitive channel family	1.A.22	+	+	+	1	None

TABLE 16-continued

Secretion Systems for Gram negative bacteria Protein secretory pathways (SP) in gram-negative bacteria and their descendants							
Type (Abbreviation)	Name	TC# ²	Bacteria	Archaea	Eukarya	# Proteins/ System	Energy Source
Holins	Holin functional superfamily	1.E.1*21	+	-	-	1	None
Eukaryotic Organelles							
MPT	Mitochondria 1 protein translocase	3.A.B	-	-	+	>20	ATP
CEPT	Chloroplast envelope protein translocase	3.A.9	(+)	-	+	≥3	GTP
Bcl-2	Eukaryotic Bcl-2 family (programmed cell death)	1.A.21	-	-	+	1?	None
Gram-negative bacterial outer membrane channel-forming translocases							
MTB (IISP)	Main terminal branch of the general secretory translocase	3.A.15	+ ^b	-	-	~14	ATP; PMF
FUP AT-1	Fimbrial	1.B.11	+ ^b	-	-	1	None
	usher protein	1.B.12	+ ^b	-	-	1	None
AT-2 OMF (ISP)	Autotransporter- 1	1.B.40	+ ^b	-	-	1	None
	2	1.B.17	+ ^b	-	+(?)	1	None
TPS		1.B.20	+	-	+	1	None
Secretin (IISP and IISP)		1.B.22	+ ^b	-	-	1	None
OmpIP	Outer membrane insertion porin	1.B.33	+	-	+	≥4	None ?

In some embodiments, the genetically engineered bacterial comprise a native or non-native secretion system described herein for the secretion of a PME, e.g., PAH, PAL and/or LAAD. In some embodiments, the secretion system is selected from the modified type III flagellar, type I (e.g., hemolysin secretion system), type II, type IV, type V, type VI, and type VII secretion systems, resistance-nodulation-division (RND) multi-drug efflux pumps, a single membrane secretion system, Sec and, TAT secretion systems.

In some embodiments, the PMEs secreted by the genetically engineered bacteria are modified to increase resistance to proteases. For example, in some embodiments, the one or more PME administered is modified as described in Sakissian et al., 2011, Mol Genet Metab. 2011 November; 104(3): 249-254, the contents of which is herein incorporated by reference in its entirety. In some embodiments, the secreted PAL is Av-p.C503S/p.C565S/p.F18A PAL. In some embodiments, the secreted PAL is PEG-Av-p.C503S/p.C565S/p.F18A PAL.

In some embodiments, the one or more PMEs for secretion are under the control of an inducible promoter, as described herein. In one example, the one or more PMEs are under the control of the FNR promoter and are produced and secreted under anaerobic conditions. In some embodiments, the PMEs for secretion are under the control of the ParaBAD

promoter. In some embodiments, the PMEs for secretion are under the control of a constitutive promoter.

In some embodiments in which the one or more PMEs are secreted or exported from the microorganism, the engineered microorganism comprises gene sequence(s) that includes a secretion tag. In some embodiments, the PME(s) include a "secretion tag" of either RNA or peptide origin to direct the PME(s) to specific secretion systems. For example, a secretion tag for the Type I Hemolysin secretion system is encoded in the C-terminal 53 amino acids of the alpha hemolysin protein (HlyA). HlyA secretion signal.

HlyB inserts into inner membrane to form a pore, HlyD aligns HlyB with TolC (outer membrane pore) thereby forming a channel through inner and outer membrane. The C-terminal secretion tag can be removed by either an autocatalytic or protease-catalyzed e.g., OmpT cleavage thereby releasing the PME(s) into the extracellular milieu.

The Type V Auto-secretion System utilizes an N-terminal Sec-dependent peptide tag (inner membrane) and C-terminal tag (outer-membrane). This uses Sec-system to get from cytoplasm to periplasm. C-terminal tag then inserts into the outer membrane forming a pore through which the "passenger protein" threads through. Once across the outer membrane, the passenger (anti-cancer molecule) is released from the membrane-embedded C-terminal tag by either an auto-

catalytic, intein-like mechanism or via a membrane-bound protease (i.e., OmpT). The N-terminal tag is removed by the Sec system. Thus, in some embodiments, the secretion system is able to remove this tag before secreting the PME(s), e.g., PAL, PAH, and/or LAAD from the engineered bacteria. In the Type V auto-secretion-mediated secretion the N-terminal peptide secretion tag is removed upon translocation of the “passenger” peptide from the cytoplasm into the periplasmic compartment by the native Sec system. Further, once the auto-secretor is translocated across the outer membrane the C-terminal secretion tag can be removed by either an autocatalytic or protease-catalyzed e.g., OmpT cleavage thereby releasing the anti-cancer molecule(s) into the extracellular milieu.

In the Flagellar modified Type III Secretion, the tag is encoded in 5'untranslated region of the mRNA and thus there is no peptide tag to cleave/remove. This modified system does not contain the “syringe” portion and instead uses the basal body of the flagella structure as the pore to translocate across both membranes and out through the forming flagella. If the *fliC/fliD* genes (encoding the flagella “tail”/whip) are disrupted the flagella cannot fully form and this promotes overall secretion. In some embodiments, the tail portion can be removed entirely. In the Type III traditional secretion system, the basal body closely resembles the flagella, however, instead of a “tail”/whip, the traditional T3SS has a syringe to inject the passenger proteins into host cells. The secretion tag is encoded by an N-terminal peptide (lengths vary and there are several different tags, see PCT/US14/020972). The N-terminal tag is not removed from the polypeptides in this secretion system.

In some embodiments the PME contains expressed as fusion protein with the 53 amino acids of the C termini of alpha-hemolysin (hlyA) of *E. coli* CFT073 (C terminal secretion tag).

Oxygen Consuming Enzymes

LAAD catalytic activity is dependent on oxygen, and therefore may not be active in anaerobic environments in the intestine, e.g., the colon. Oxygen is present in more proximal compartments of the GI tract.

The oxygen tension as measured in healthy mice is shown in Table 17. He et al., Proc Natl Acad Sci USA. 1999 Apr. 13; 96(8):4586-91; “Noninvasive measurement of anatomic structure and intraluminal oxygenation in the gastrointestinal tract of living mice with spatial and spectral EPR imaging”, the contents of which is herein incorporated by reference in its entirety. A marked oxygen gradient from the proximal to the distal GI tract. As noted by He et al., the observed oxygen gradient seen along the GI tract can be explained by a combination of processes. Without wishing to be bound by theory, food, when swallowed, is initially equilibrated with the oxygen tension of ambient room air. On passage to the stomach and later the small intestine, the oxygen levels may fall as oxygen diffuses across the mucosal membrane. A gradual process of equilibration with the capillary levels of oxygen (i.e., 5-10 torr; ref. 9) may occur. On passage to the colon, with its heavy bacterial colonization, further decreases in oxygenation occur. Finally, the lumen of the distal colon displays marked hypoxia, as expected, based on the abundance of anaerobic bacteria at this site.

TABLE 17

Oxygen Tension in Gastrointestinal Tract Compartments	
Compartment	Oxygen Tension
Ambient Air	159 Torr
stomach	~60 torr
duodenum and first part of jejunum	(~30 torr); ~20% oxygen in ambient air
ileum	(~10 torr); ~6% oxygen in ambient air
colon	(<2 torr)

As shown in FIG. 25B, LAAD activity is retained in microaerobic conditions, albeit at lower levels than under aerobic conditions (FIG. 25A and FIG. 25B). LAAD therefore may be active in the more proximal areas of the intestine, such as stomach, duodenum, jejunum, and ileum. It is contemplated as part of this disclosure that LAAD expressed by the genetically engineered bacteria may advantageously be active in a different compartment than PAL, which may be expressed in the colon if under the control of an FNR promoter. In one embodiment, the genetically engineered bacteria express two enzymes, which have different oxygen requirements and/or are induced under different oxygen conditions, such that an PME is expressed and active throughout the entire gastrointestinal system. For example, the first enzyme, e.g., LAAD, which is dependent on the presence of oxygen, is expressed in one or more of stomach, duodenum and ileum under the control of a constitutive or inducible promoter (such as ParaBAD), and the second enzyme, e.g., PAL, is expressed in the colon under the control of an FNR promoter.

Several strategies can be employed to further increase LAAD activity under oxygen limiting conditions. For example, the activity of other enzymes that consume large amounts of oxygen can be reduced or extinguished. One such enzyme is NADH dehydrogenase. *E. coli* has two NADH dehydrogenases; nuo and ndh2, and is has been shown that knock out of both of these enzymes reduces oxygen consumption by 80%. In some embodiments, additional measures are taken to conserve limiting oxygen, i.e., to allow LAAD to function under lower exogenous oxygen conditions in the genetically engineered bacteria expressing LAAD. In some embodiments, the genetically engineered bacteria further comprise a mutation in one or more genes involved in oxygen consumption. In some embodiments, one or both *E. coli* NADH dehydrogenases are knocked out. In some embodiments, the knocked out NADH dehydrogenase is nuo. In some embodiments the knocked out NADH dehydrogenase is ndh2. In some embodiments nuo and ndh2 are knocked out. Other enzymes involved in *E. coli* oxygen metabolism may also be knocked out, including enzymes in the respiratory chain, such as cydB (a subunit of high affinity terminal oxidase), cydD (an enzyme required to make cytochrome D), and cyoABC (subunits of low affinity cytochrome oxidase). In some embodiments, the genetically engineered bacteria harbor a knock out mutation/deletion in one or more genes selected from cydB, cydD, and cyoABC.

In one embodiment, the one or more PME encoded by the genetically engineered bacteria are expressed and show activity in the stomach. In one embodiment, the one or more PME encoded by the genetically engineered bacteria are expressed and show activity in the duodenum. In one embodiment, the one or more PME encoded by the genetically engineered bacteria are expressed and show activity in the jejunum. In one embodiment, the one or more PME encoded by the genetically engineered bacteria are expressed and show activity in the ileum. In one embodi-

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ment, the one or more PME encoded by the genetically engineered bacteria are expressed and show activity in the colon.

Essential Genes and Auxotrophs

As used herein, the term “essential gene” refers to a gene that is necessary for cell growth and/or survival. Bacterial essential genes are well known to one of ordinary skill in the art, and can be identified by directed deletion of genes and/or random mutagenesis and screening (see, e.g., Zhang and Lin, “DEG 5.0, a database of essential genes in both prokaryotes and eukaryotes,” Nucl Acids Res, 2009; 37:D455-D458 and Gerdes et al., “Essential genes on metabolic maps,” Curr Opin Biotechnol, 2006; 17(5):448-456, the entire contents of each of which are expressly incorporated herein by reference).

An “essential gene” may be dependent on the circumstances and environment in which an organism lives. For example, a mutation of, modification of, or excision of an essential gene may result in the genetically engineered bacteria of the disclosure becoming an auxotroph. An auxotrophic modification is intended to cause bacteria to die in the absence of an exogenously added nutrient essential for survival or growth because they lack the gene(s) necessary to produce that essential nutrient. In some embodiments, any of the genetically engineered bacteria described herein also comprise a deletion or mutation in a gene required for cell survival and/or growth. In one embodiment, the essential gene is a DNA synthesis gene, for example, thyA. In another embodiment, the essential gene is a cell wall synthesis gene, for example, dapA. In yet another embodiment, the essential gene is an amino acid gene, for example, serA or MetA. Any gene required for cell survival and/or growth may be targeted, including but not limited to, cysE, glnA, ilvD, leuB, lysA, serA, metA, glyA, hisB, ilvA, pheA, proA, thrC, trpC, tyrA, thyA, uraA, dapA, dapB, dapD, dapE, dapF, flhD, metB, metC, proAB, and thi1, as long as the corresponding wild-type gene product is not produced in the bacteria. Table 18 lists exemplary bacterial genes which may be disrupted or deleted to produce an auxotrophic strain. These include, but are not limited to, genes required for oligonucleotide synthesis, amino acid synthesis, and cell wall synthesis.

TABLE 18

Non-limiting Examples of Bacterial Genes Useful for Generation of an Auxotroph		
Amino Acid	Oligonucleotide	Cell Wall
cysE	thyA	dapA
glnA	uraA	dapB
ilvD		dapD
leuB		dapE
lysA		dapF
serA		
metA		
glyA		
hisB		
ilvA		
pheA		
proA		
thrC		
trpC		
tyrA		

Table 19 shows the survival of various amino acid auxotrophs in the mouse gut, as detected 24 hrs and 48 hrs post-gavage. These auxotrophs were generated using BW25113, a non-Nissle strain of *E. coli*.

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TABLE 19

Survival of amino acid auxotrophs in the mouse gut				
Gene	AA Auxotroph	Pre-Gavage	24 hours	48 hours
argA	Arginine	Present	Present	Absent
cysE	Cysteine	Present	Present	Absent
glnA	Glutamine	Present	Present	Absent
glyA	Glycine	Present	Present	Absent
hisB	Histidine	Present	Present	Present
ilvA	Isoleucine	Present	Present	Absent
leuB	Leucine	Present	Present	Absent
lysA	Lysine	Present	Present	Absent
metA	Methionine	Present	Present	Present
pheA	Phenylalanine	Present	Present	Present
proA	Proline	Present	Present	Absent
serA	Serine	Present	Present	Present
thrC	Threonine	Present	Present	Present
trpC	Tryptophan	Present	Present	Present
tyrA	Tyrosine	Present	Present	Present
ilvD	Valine/ Isoleucine/ Leucine	Present	Present	Absent
thyA	Thiamine	Present	Absent	Absent
uraA	Uracil	Present	Absent	Absent
flhD	FlhD	Present	Present	Present

For example, thymine is a nucleic acid that is required for bacterial cell growth; in its absence, bacteria undergo cell death. The thyA gene encodes thymidylate synthetase, an enzyme that catalyzes the first step in thymine synthesis by converting dUMP to dTMP (Sat et al., 2003). In some embodiments, the bacterial cell of the disclosure is a thyA auxotroph in which the thyA gene is deleted and/or replaced with an unrelated gene. A thyA auxotroph can grow only when sufficient amounts of thymine are present, e.g., by adding thymine to growth media in vitro, or in the presence of high thymine levels found naturally in the human gut in vivo. In some embodiments, the bacterial cell of the disclosure is auxotrophic in a gene that is complemented when the bacterium is present in the mammalian gut. Without sufficient amounts of thymine, the thyA auxotroph dies. In some embodiments, the auxotrophic modification is used to ensure that the bacterial cell does not survive in the absence of the auxotrophic gene product (e.g., outside of the gut).

Diaminopimelic acid (DAP) is an amino acid synthesized within the lysine biosynthetic pathway and is required for bacterial cell wall growth (Meadow et al., 1959; Clarkson et al., 1971). In some embodiments, any of the genetically engineered bacteria described herein is a dapD auxotroph in which dapD is deleted and/or replaced with an unrelated gene. A dapD auxotroph can grow only when sufficient amounts of DAP are present, e.g., by adding DAP to growth media in vitro, or in the presence of high DAP levels found naturally in the human gut in vivo. Without sufficient amounts of DAP, the dapD auxotroph dies. In some embodiments, the auxotrophic modification is used to ensure that the bacterial cell does not survive in the absence of the auxotrophic gene product (e.g., outside of the gut).

In other embodiments, the genetically engineered bacterium of the present disclosure is a uraA auxotroph in which uraA is deleted and/or replaced with an unrelated gene. The uraA gene codes for UraA, a membrane-bound transporter that facilitates the uptake and subsequent metabolism of the pyrimidine uracil (Andersen et al., 1995). A uraA auxotroph can grow only when sufficient amounts of uracil are present, e.g., by adding uracil to growth media in vitro, or in the presence of high uracil levels found naturally in the human gut in vivo. Without sufficient amounts of uracil, the uraA

auxotroph dies. In some embodiments, auxotrophic modifications are used to ensure that the bacteria do not survive in the absence of the auxotrophic gene product (e.g., outside of the gut).

In complex communities, it is possible for bacteria to share DNA. In very rare circumstances, an auxotrophic bacterial strain may receive DNA from a non-auxotrophic strain, which repairs the genomic deletion and permanently rescues the auxotroph. Therefore, engineering a bacterial strain with more than one auxotroph may greatly decrease the probability that DNA transfer will occur enough times to rescue the auxotrophy. In some embodiments, the genetically engineered bacteria of the invention comprise a deletion or mutation in two or more genes required for cell survival and/or growth.

Other examples of essential genes include, but are not limited to, *yhbV*, *yagG*, *hemB*, *secD*, *secF*, *ribD*, *ribE*, *thiL*, *dxs*, *ispA*, *dnaX*, *adk*, *hemH*, *lpxH*, *cysS*, *fold*, *rplT*, *infC*, *thrS*, *nadE*, *gapA*, *yeaZ*, *aspS*, *argS*, *pgsA*, *yefM*, *metG*, *folE*, *yejM*, *gyrA*, *nrdA*, *nrdB*, *folC*, *accD*, *fabB*, *gltX*, *ligA*, *zipA*, *dapE*, *dapA*, *der*, *hisS*, *ispG*, *suhB*, *tadA*, *acpS*, *era*, *rnc*, *ftsB*, *eno*, *pyrG*, *chpR*, *lgt*, *JbaA*, *pgk*, *yqgD*, *metK*, *yqgF*, *plsC*, *ygiT*, *pare*, *ribB*, *cca*, *ygiD*, *tdcF*, *yraL*, *yihA*, *ftsN*, *murI*, *murB*, *birA*, *secE*, *nusG*, *rplJ*, *rplL*, *rpoB*, *rpoC*, *ubiA*, *plsB*, *lexA*, *dnaB*, *ssb*, *alsK*, *groS*, *psd*, *orn*, *yjeE*, *rpsR*, *chpS*, *ppa*, *valS*, *yjgP*, *yjgQ*, *dnaC*, *ribF*, *ispA*, *ispH*, *dapB*, *folA*, *imp*, *yabQ*, *ftsL*, *ftsI*, *murE*, *murF*, *mraY*, *murD*, *ftsW*, *murG*, *murC*, *ftsQ*, *ftsA*, *ftsZ*, *lpxC*, *secM*, *secA*, *can*, *folK*, *hemL*, *yadR*, *dapD*, *map*, *rpsB*, *infB*, *nusA*, *ftsH*, *obgE*, *rpmA*, *rplU*, *ispB*, *murA*, *yrbB*, *yrbK*, *yhbN*, *rpsI*, *rplM*, *degS*, *mreD*, *mreC*, *mreB*, *accB*, *accC*, *yrdC*, *def*, *fnt*, *rplQ*, *rpoA*, *rpsD*, *rpsK*, *rpsM*, *entD*, *mrdB*, *mrdA*, *nadD*, *hlePB*, *rpoE*, *psaA*, *yfiO*, *rplS*, *trmD*, *rpsP*, *fliH*, *grpE*, *yfjB*, *csrA*, *ispF*, *ispD*, *rplW*, *rplD*, *rplC*, *rpsJ*, *fusA*, *rpsG*, *rpsL*, *trpS*, *yrfF*, *asd*, *rpoH*, *ftsX*, *ftsE*, *ftsY*, *frr*, *dxr*, *ispU*, *rfaK*, *kdtA*, *coaD*, *rpmB*, *dfp*, *dut*, *gmK*, *spot*, *gyrB*, *dnaN*, *dnaA*, *rpmH*, *rpmA*, *yidC*, *tnaB*, *glmS*, *glmU*, *wzyE*, *hemD*, *hemC*, *yigP*, *ubiB*, *ubiD*, *hemG*, *secY*, *rplO*, *rpmD*, *rpsE*, *rplR*, *rplF*, *rpsH*, *rpsN*, *rplE*, *rplX*, *rplN*, *rpsQ*, *rpmC*, *rplP*, *rpsC*, *rplV*, *rpsS*, *rplB*, *cdsA*, *yaeL*, *yaeT*, *lpxD*, *fabZ*, *lpxA*, *lpxB*, *dnaE*, *accA*, *tilS*, *proS*, *yafF*, *tsf*, *pyrH*, *ola*, *rlpB*, *leuS*, *lnt*, *glnS*, *fldA*, *cydA*, *infA*, *cydC*, *ftsK*, *lolA*, *serS*, *rpsA*, *msbA*, *lpxK*, *kdsB*, *mukF*, *mukE*, *mukB*, *asnS*, *fabA*, *mviN*, *rne*, *yceQ*, *fabD*, *fabG*, *acpP*, *tmk*, *holB*, *lolC*, *lolD*, *lolE*, *purB*, *ymK*, *minE*, *mind*, *pth*, *rsa*, *ispE*, *lolB*, *hemA*, *prfA*, *prmC*, *kdsA*, *topA*, *ribA*, *fabI*, *racR*, *dicA*, *ydfB*, *tyrS*, *ribC*, *ydlL*, *pheT*, *pheS*, *yhhQ*, *bcsB*, *glyQ*, *yibJ*, and *gpsA*. Other essential genes are known to those of ordinary skill in the art.

In some embodiments, the genetically engineered bacterium of the present disclosure is a synthetic ligand-dependent essential gene (SLiDE) bacterial cell. SLiDE bacterial cells are synthetic auxotrophs with a mutation in one or more essential genes that only grow in the presence of a particular ligand (see Lopez and Anderson, "Synthetic Auxotrophs with Ligand-Dependent Essential Genes for a BL21 (DE3) Biosafety Strain," *ACS Synth Biol* 2015; 4(12):1279-1286, the entire contents of which are expressly incorporated herein by reference).

In some embodiments, the SLiDE bacterial cell comprises a mutation in an essential gene. In some embodiments, the essential gene is selected from the group consisting of *pheS*, *dnaN*, *tyrS*, *metG*, and *adk*. In some embodiments, the essential gene is *dnaN* comprising one or more of the following mutations: H191N, R240C, I317S, F319V, L340T, V347I, and S345C. In some embodiments, the essential gene is *dnaN* comprising the mutations H191N, R240C, I317S, F319V, L340T, V347I, and S345C. In some

embodiments, the essential gene is *pheS* comprising one or more of the following mutations: F125G, P183T, P184A, R186A, and I188L. In some embodiments, the essential gene is *pheS* comprising the mutations F125G, P183T, P184A, R186A, and I188L. In some embodiments, the essential gene is *tyrS* comprising one or more of the following mutations: L36V, C38A, and F40G. In some embodiments, the essential gene is *tyrS* comprising the mutations L36V, C38A, and F40G. In some embodiments, the essential gene is *metG* comprising one or more of the following mutations: E45Q, N47R, I49G, and A51C. In some embodiments, the essential gene is *metG* comprising the mutations E45Q, N47R, I49G, and A51C. In some embodiments, the essential gene is *adk* comprising one or more of the following mutations: I4L, L5I, and L6G. In some embodiments, the essential gene is *adk* comprising the mutations I4L, L5I, and L6G.

In some embodiments, the genetically engineered bacterium is complemented by a ligand. In some embodiments, the ligand is selected from the group consisting of benzothiazole, indole, 2-aminobenzothiazole, indole-3-butyric acid, indole-3-acetic acid, and L-histidine methyl ester. For example, bacterial cells comprising mutations in *metG* (E45Q, N47R, I49G, and A51C) are complemented by benzothiazole, indole, 2-aminobenzothiazole, indole-3-butyric acid, indole-3-acetic acid, or L-histidine methyl ester. Bacterial cells comprising mutations in *dnaN* (H191N, R240C, I317S, F319V, L340T, V347I, and S345C) are complemented by benzothiazole, indole, or 2-aminobenzothiazole. Bacterial cells comprising mutations in *pheS* (F125G, P183T, P184A, R186A, and I188L) are complemented by benzothiazole or 2-aminobenzothiazole. Bacterial cells comprising mutations in *tyrS* (L36V, C38A, and F40G) are complemented by benzothiazole or 2-aminobenzothiazole. Bacterial cells comprising mutations in *adk* (I4L, L5I and L6G) are complemented by benzothiazole or indole.

In some embodiments, the genetically engineered bacterium comprises more than one mutant essential gene that renders it auxotrophic to a ligand. In some embodiments, the bacterial cell comprises mutations in two essential genes. For example, in some embodiments, the bacterial cell comprises mutations in *tyrS* (L36V, C38A, and F40G) and *metG* (E45Q, N47R, I49G, and A51C). In other embodiments, the bacterial cell comprises mutations in three essential genes. For example, in some embodiments, the bacterial cell comprises mutations in *tyrS* (L36V, C38A, and F40G), *metG* (E45Q, N47R, I49G, and A51C), and *pheS* (F125G, P183T, P184A, R186A, and I188L).

In some embodiments, the genetically engineered bacterium is a conditional auxotroph whose essential gene(s) is replaced using the arabinose system shown in FIGS. 43-47.

In some embodiments, the genetically engineered bacterium of the disclosure is an auxotroph and also comprises kill switch circuitry, such as any of the kill switch components and systems described herein. For example, the genetically engineered bacteria may comprise a deletion or mutation in an essential gene required for cell survival and/or growth, for example, in a DNA synthesis gene, for example, *thyA*, cell wall synthesis gene, for example, *dapA* and/or an amino acid gene, for example, *serA* or *MetA* and may also comprise a toxin gene that is regulated by one or more transcriptional activators that are expressed in response to an environmental condition(s) and/or signal(s) (such as the described arabinose system) or regulated by one or more recombinases that are expressed upon sensing an exogenous environmental condition(s) and/or signal(s) (such as the recombinase systems described herein). Other embodiments

are described in Wright et al., "GeneGuard: A Modular Plasmid System Designed for Biosafety," ACS Synth Biol, 2015; 4(3):307-316, the entire contents of which are expressly incorporated herein by reference). In some embodiments, the genetically engineered bacterium of the disclosure is an auxotroph and also comprises kill switch circuitry, such as any of the kill switch components and systems described herein, as well as another biosecurity system, such as a conditional origin of replication (Wright et al., 2015).

The addition of a Phe-auxotrophy may also have utility for increasing the rate of phenylalanine degradation. For example, the deletion of the pheA gene confers phenylalanine auxotrophy. By turning off endogenous bacterial phenylalanine production, this may drive increased uptake from the environment and also result in increased degradation of phenylalanine taken up from the environment.

Genetic Regulatory Circuits

In some embodiments, the genetically engineered bacteria comprise multilayered genetic regulatory circuits for expressing the constructs described herein (see, e.g., U.S. Provisional Application No. 62/184,811, incorporated herein by reference in its entirety). The genetic regulatory circuits are useful to screen for mutant bacteria that produce a phenylalanine-metabolizing enzyme or rescue an auxotroph. In certain embodiments, the invention provides methods for selecting genetically engineered bacteria that produce one or more genes of interest.

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a T7 polymerase-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a first gene encoding a T7 polymerase, wherein the first gene is operably linked to an FNR-responsive promoter; a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme, wherein the second gene or gene cassette is operably linked to a T7 promoter that is induced by the T7 polymerase; and a third gene encoding an inhibitory factor, lysY, that is capable of inhibiting the T7 polymerase. In the presence of oxygen, FNR does not bind the FNR-responsive promoter, and the phenylalanine-metabolizing enzyme is not expressed. LysY is expressed constitutively (P-lac constitutive) and further inhibits T7 polymerase. In the absence of oxygen, FNR dimerizes and binds to the FNR-responsive promoter, T7 polymerase is expressed at a level sufficient to overcome lysY inhibition, and the phenylalanine-metabolizing enzyme is expressed. In some embodiments, the lysY gene is operably linked to an additional FNR binding site. In the absence of oxygen, FNR dimerizes to activate T7 polymerase expression as described above, and also inhibits lysY expression.

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a protease-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a first gene encoding an mf-lon protease, wherein the first gene is operably linked to an FNR-responsive promoter; a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme operably linked to a Tet regulatory region (TetO); and a third gene encoding an mf-lon degradation signal linked to a Tet repressor (TetR), wherein the TetR is capable of binding to the Tet regulatory region and repressing expression of the second gene or gene cassette. The mf-lon protease is capable of recognizing the mf-lon degradation signal and degrading the TetR. In the presence

of oxygen, FNR does not bind the FNR-responsive promoter, the repressor is not degraded, and the phenylalanine-metabolizing enzyme is not expressed. In the absence of oxygen, FNR dimerizes and binds the FNR-responsive promoter, thereby inducing expression of the mf-lon protease. The mf-lon protease recognizes the mf-lon degradation signal and degrades the TetR, and the phenylalanine-metabolizing enzyme is expressed.

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a repressor-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a first gene encoding a first repressor, wherein the first gene is operably linked to an FNR-responsive promoter; a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme operably linked to a first regulatory region comprising a constitutive promoter; and a third gene encoding a second repressor, wherein the second repressor is capable of binding to the first regulatory region and repressing expression of the second gene or gene cassette. The third gene is operably linked to a second regulatory region comprising a constitutive promoter, wherein the first repressor is capable of binding to the second regulatory region and inhibiting expression of the second repressor. In the presence of oxygen, FNR does not bind the FNR-responsive promoter, the first repressor is not expressed, the second repressor is expressed, and the phenylalanine-metabolizing enzyme is not expressed. In the absence of oxygen, FNR dimerizes and binds the FNR-responsive promoter, the first repressor is expressed, the second repressor is not expressed, and the phenylalanine-metabolizing enzyme is expressed.

Examples of repressors useful in these embodiments include, but are not limited to, ArgR, TetR, ArsR, AscG, LacI, CscR, DeoR, DgoR, FruR, GalR, GatR, CI, LexA, RarR, QacR, and PtxS (US20030166191).

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a regulatory RNA-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a first gene encoding a regulatory RNA, wherein the first gene is operably linked to an FNR-responsive promoter, and a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme. The second gene or gene cassette is operably linked to a constitutive promoter and further linked to a nucleotide sequence capable of producing an mRNA hairpin that inhibits translation of the phenylalanine-metabolizing enzyme. The regulatory RNA is capable of eliminating the mRNA hairpin and inducing translation via the ribosomal binding site. In the presence of oxygen, FNR does not bind the FNR-responsive promoter, the regulatory RNA is not expressed, and the mRNA hairpin prevents the phenylalanine-metabolizing enzyme from being translated. In the absence of oxygen, FNR dimerizes and binds the FNR-responsive promoter, the regulatory RNA is expressed, the mRNA hairpin is eliminated, and the phenylalanine-metabolizing enzyme is expressed.

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a CRISPR-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a Cas9 protein; a first gene encoding a CRISPR guide RNA, wherein the first gene is operably linked to an FNR-responsive promoter; a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme, wherein the second

gene or gene cassette is operably linked to a regulatory region comprising a constitutive promoter; and a third gene encoding a repressor operably linked to a constitutive promoter, wherein the repressor is capable of binding to the regulatory region and repressing expression of the second gene or gene cassette. The third gene is further linked to a CRISPR target sequence that is capable of binding to the CRISPR guide RNA, wherein said binding to the CRISPR guide RNA induces cleavage by the Cas9 protein and inhibits expression of the repressor. In the presence of oxygen, FNR does not bind the FNR-responsive promoter, the guide RNA is not expressed, the repressor is expressed, and the phenylalanine-metabolizing enzyme is not expressed. In the absence of oxygen, FNR dimerizes and binds the FNR-responsive promoter, the guide RNA is expressed, the repressor is not expressed, and the phenylalanine-metabolizing enzyme is expressed.

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a recombinase-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a first gene encoding a recombinase, wherein the first gene is operably linked to an FNR-responsive promoter, and a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme operably linked to a constitutive promoter. The second gene or gene cassette is inverted in orientation (3' to 5') and flanked by recombinase binding sites, and the recombinase is capable of binding to the recombinase binding sites to induce expression of the second gene or gene cassette by reverting its orientation (5' to 3'). In the presence of oxygen, FNR does not bind the FNR-responsive promoter, the recombinase is not expressed, the gene or gene cassette remains in the 3' to 5' orientation, and no functional phenylalanine-metabolizing enzyme is produced. In the absence of oxygen, FNR dimerizes and binds the FNR-responsive promoter, the recombinase is expressed, the gene or gene cassette is reverted to the 5' to 3' orientation, and a functional phenylalanine-metabolizing enzyme is produced (see, e.g., FIG. 42).

In some embodiments, the invention provides genetically engineered bacteria comprising a gene or gene cassette for producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and a polymerase- and recombinase-regulated genetic regulatory circuit. For example, the genetically engineered bacteria comprise a first gene encoding a recombinase, wherein the first gene is operably linked to an FNR-responsive promoter; a second gene or gene cassette for producing a phenylalanine-metabolizing enzyme operably linked to a T7 promoter; a third gene encoding a T7 polymerase, wherein the T7 polymerase is capable of binding to the T7 promoter and inducing expression of the phenylalanine-metabolizing enzyme. The third gene encoding the T7 polymerase is inverted in orientation (3' to 5') and flanked by recombinase binding sites, and the recombinase is capable of binding to the recombinase binding sites to induce expression of the T7 polymerase gene by reverting its orientation (5' to 3'). In the presence of oxygen, FNR does not bind the FNR-responsive promoter, the recombinase is not expressed, the T7 polymerase gene remains in the 3' to 5' orientation, and the phenylalanine-metabolizing enzyme is not expressed. In the absence of oxygen, FNR dimerizes and binds the FNR-responsive promoter, the recombinase is expressed, the T7 polymerase gene is reverted to the 5' to 3' orientation, and the phenylalanine-metabolizing enzyme is expressed (see, e.g., FIG. 43).

Synthetic gene circuits expressed on plasmids may function well in the short term but lose ability and/or function in the long term (Danino et al., 2015). In some embodiments, the genetically engineered bacteria comprise stable circuits for expressing genes of interest over prolonged periods. In some embodiments, the genetically engineered bacteria are capable of producing a phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and further comprise a toxin-anti-toxin system that simultaneously produces a toxin (hok) and a short-lived anti-toxin (sok), wherein loss of the plasmid causes the cell to be killed by the long-lived toxin (Danino et al., 2015). In some embodiments, the genetically engineered bacteria further comprise *alp7* from *B. subtilis* plasmid pL20 and produces filaments that are capable of pushing plasmids to the poles of the cells in order to ensure equal segregation during cell division (Danino et al., 2015).

Host-Plasmid Mutual Dependency

In some embodiments, the genetically engineered bacteria of the invention also comprise a plasmid that has been modified to create a host-plasmid mutual dependency. In certain embodiments, the mutually dependent host-plasmid platform is GeneGuard (Wright et al., 2015). In some embodiments, the GeneGuard plasmid comprises (i) a conditional origin of replication, in which the requisite replication initiator protein is provided in trans; (ii) an auxotrophic modification that is rescued by the host via genomic translocation and is also compatible for use in rich media; and/or (iii) a nucleic acid sequence which encodes a broad-spectrum toxin. The toxin gene may be used to select against plasmid spread by making the plasmid DNA itself disadvantageous for strains not expressing the anti-toxin (e.g., a wild-type bacterium). In some embodiments, the GeneGuard plasmid is stable for at least 100 generations without antibiotic selection. In some embodiments, the GeneGuard plasmid does not disrupt growth of the host. The GeneGuard plasmid is used to greatly reduce unintentional plasmid propagation in the genetically engineered bacteria of the invention.

The mutually dependent host-plasmid platform may be used alone or in combination with other biosafety mechanisms, such as those described herein (e.g., kill switches, auxotrophies). In some embodiments, the genetically engineered bacteria comprise a GeneGuard plasmid. In other embodiments, the genetically engineered bacteria comprise a GeneGuard plasmid and/or one or more kill switches. In other embodiments, the genetically engineered bacteria comprise a GeneGuard plasmid and/or one or more auxotrophies. In still other embodiments, the genetically engineered bacteria comprise a GeneGuard plasmid, one or more kill switches, and/or one or more auxotrophies.

Kill Switch

In some embodiments, the genetically engineered bacteria of the invention also comprise a kill switch (see, e.g., U.S. Provisional Application Nos. 62/183,935 and 62/263,329, incorporated herein by reference in their entireties). The kill switch is intended to actively kill genetically engineered bacteria in response to external stimuli. As opposed to an auxotrophic mutation where bacteria die because they lack an essential nutrient for survival, the kill switch is triggered by a particular factor in the environment that induces the production of toxic molecules within the microbe that cause cell death.

Bacteria comprising kill switches have been engineered for in vitro research purposes, e.g., to limit the spread of a biofuel-producing microorganism outside of a laboratory environment. Bacteria engineered for in vivo administration to treat a disease may also be programmed to die at a specific

time after the expression and delivery of a heterologous gene or genes, for example, a phenylalanine-metabolizing enzyme, or after the subject has experienced the therapeutic effect. For example, in some embodiments, the kill switch is activated to kill the bacteria after a period of time following oxygen level-dependent expression of the phenylalanine-metabolizing enzyme (e.g., PAL or PAH) and/or the phenylalanine transporter gene. In some embodiments, the kill switch is activated in a delayed fashion following oxygen level-dependent expression of the phenylalanine-metabolizing enzyme and/or phenylalanine transporter gene. Alternatively, the bacteria may be engineered to die after the bacterium has spread outside of a disease site. Specifically, it may be useful to prevent long-term colonization of subjects by the microorganism, spread of the microorganism outside the area of interest (for example, outside the gut) within the subject, or spread of the microorganism outside of the subject into the environment (for example, spread to the environment through the stool of the subject). Examples of such toxins that can be used in kill switches include, but are not limited to, bacteriocins, lysins, and other molecules that cause cell death by lysing cell membranes, degrading cellular DNA, or other mechanisms. Such toxins can be used individually or in combination. The switches that control their production can be based on, for example, transcriptional activation (toggle switches; see, e.g., Gardner et al., 2000), translation (riboregulators), or DNA recombination (recombinase-based switches), and can sense environmental stimuli such as anaerobiosis or reactive oxygen species. These switches can be activated by a single environmental factor or may require several activators in AND, OR, NAND and NOR logic configurations to induce cell death. For example, an AND riboregulator switch is activated by tetracycline, isopropyl 3-D-1-thiogalactopyranoside (IPTG), and arabinose to induce the expression of lysins, which permeabilize the cell membrane and kill the cell. IPTG induces the expression of the endolysin and holin mRNAs, which are then derepressed by the addition of arabinose and tetracycline. All three inducers must be present to cause cell death. Examples of kill switches are known in the art (Callura et al., 2010).

Kill switches can be designed such that a toxin is produced in response to an environmental condition or external signal (e.g., the bacteria is killed in response to an external cue) or, alternatively designed such that a toxin is produced once an environmental condition no longer exists or an external signal is ceased.

Thus, in some embodiments, the genetically engineered bacteria of the disclosure are further programmed to die after sensing an exogenous environmental signal, for example, in a low-oxygen environment. In some embodiments, the genetically engineered bacteria of the present disclosure comprise one or more genes encoding one or more recombinase(s), whose expression is induced in response to an environmental condition or signal and causes one or more recombination events that ultimately leads to the expression of a toxin which kills the cell. In some embodiments, the at least one recombination event is the flipping of an inverted heterologous gene encoding a bacterial toxin which is then constitutively expressed after it is flipped by the first recombinase. In one embodiment, constitutive expression of the bacterial toxin kills the genetically engineered bacterium. In these types of kill switch systems once the engineered bacterial cell senses the exogenous environmental condition and expresses the heterologous gene of interest, the recombinant bacterial cell is no longer viable.

In another embodiment in which the genetically engineered bacteria of the present disclosure express one or more recombinase(s) in response to an environmental condition or signal causing at least one recombination event, the genetically engineered bacterium further expresses a heterologous gene encoding an anti-toxin in response to an exogenous environmental condition or signal. In one embodiment, the at least one recombination event is flipping of an inverted heterologous gene encoding a bacterial toxin by a first recombinase. In one embodiment, the inverted heterologous gene encoding the bacterial toxin is located between a first forward recombinase recognition sequence and a first reverse recombinase recognition sequence. In one embodiment, the heterologous gene encoding the bacterial toxin is constitutively expressed after it is flipped by the first recombinase. In one embodiment, the anti-toxin inhibits the activity of the toxin, thereby delaying death of the genetically engineered bacterium. In one embodiment, the genetically engineered bacterium is killed by the bacterial toxin when the heterologous gene encoding the anti-toxin is no longer expressed when the exogenous environmental condition is no longer present.

In another embodiment, the at least one recombination event is flipping of an inverted heterologous gene encoding a second recombinase by a first recombinase, followed by the flipping of an inverted heterologous gene encoding a bacterial toxin by the second recombinase. In one embodiment, the inverted heterologous gene encoding the second recombinase is located between a first forward recombinase recognition sequence and a first reverse recombinase recognition sequence. In one embodiment, the inverted heterologous gene encoding the bacterial toxin is located between a second forward recombinase recognition sequence and a second reverse recombinase recognition sequence. In one embodiment, the heterologous gene encoding the second recombinase is constitutively expressed after it is flipped by the first recombinase. In one embodiment, the heterologous gene encoding the bacterial toxin is constitutively expressed after it is flipped by the second recombinase. In one embodiment, the genetically engineered bacterium is killed by the bacterial toxin. In one embodiment, the genetically engineered bacterium further expresses a heterologous gene encoding an anti-toxin in response to the exogenous environmental condition. In one embodiment, the anti-toxin inhibits the activity of the toxin when the exogenous environmental condition is present, thereby delaying death of the genetically engineered bacterium. In one embodiment, the genetically engineered bacterium is killed by the bacterial toxin when the heterologous gene encoding the anti-toxin is no longer expressed when the exogenous environmental condition is no longer present.

In one embodiment, the at least one recombination event is flipping of an inverted heterologous gene encoding a second recombinase by a first recombinase, followed by flipping of an inverted heterologous gene encoding a third recombinase by the second recombinase, followed by flipping of an inverted heterologous gene encoding a bacterial toxin by the third recombinase.

In one embodiment, the at least one recombination event is flipping of an inverted heterologous gene encoding a first excision enzyme by a first recombinase. In one embodiment, the inverted heterologous gene encoding the first excision enzyme is located between a first forward recombinase recognition sequence and a first reverse recombinase recognition sequence. In one embodiment, the heterologous gene encoding the first excision enzyme is constitutively expressed after it is flipped by the first recombinase. In one

embodiment, the first excision enzyme excises a first essential gene. In one embodiment, the programmed recombinant bacterial cell is not viable after the first essential gene is excised.

In one embodiment, the first recombinase further flips an inverted heterologous gene encoding a second excision enzyme. In one embodiment, the inverted heterologous gene encoding the second excision enzyme is located between a second forward recombinase recognition sequence and a second reverse recombinase recognition sequence. In one embodiment, the heterologous gene encoding the second excision enzyme is constitutively expressed after it is flipped by the first recombinase. In one embodiment, the genetically engineered bacterium dies or is no longer viable when the first essential gene and the second essential gene are both excised. In one embodiment, the genetically engineered bacterium dies or is no longer viable when either the first essential gene is excised or the second essential gene is excised by the first recombinase.

In one embodiment, the genetically engineered bacterium dies after the at least one recombination event occurs. In another embodiment, the genetically engineered bacterium is no longer viable after the at least one recombination event occurs.

In any of these embodiment, the recombinase can be a recombinase selected from the group consisting of: BxbI, PhiC31, TP901, BxbI, PhiC31, TP901, HK022, HP1, R4, Int1, Int2, Int3, Int4, Int5, Int6, Int7, Int8, Int9, Int10, Int11, Int12, Int13, Int14, Int15, Int16, Int17, Int18, Int19, Int20, Int21, Int22, Int23, Int24, Int25, Int26, Int27, Int28, Int29, Int30, Int31, Int32, Int33, and Int34, or a biologically active fragment thereof.

In the above-described kill switch circuits, a toxin is produced in the presence of an environmental factor or signal. In another aspect of kill switch circuitry, a toxin may be repressed in the presence of an environmental factor (i.e., not produced) and then produced once the environmental condition or external signal is no longer present. Such kill switches are called repression-based kill switches and represent systems in which the bacterial cells are viable only in the presence of an external factor or signal, such as arabinose or other sugar. Exemplary kill switch designs in which the toxin is repressed in the presence of an external factor or signal (and activated once the external signal is removed) is shown in FIGS. 43-47. The disclosure provides recombinant bacterial cells which express one or more heterologous gene(s) upon sensing arabinose or other sugar in the exogenous environment. In this aspect, the recombinant bacterial cells contain the *araC* gene, which encodes the AraC transcription factor, as well as one or more genes under the control of the *araBAD* promoter (ParaBAD). In the absence of arabinose, the AraC transcription factor adopts a conformation that represses transcription of genes under the control of the *araBAD* promoter. In the presence of arabinose, the AraC transcription factor undergoes a conformational change that allows it to bind to and activate the *araBAD* promoter, which induces expression of the desired gene, for example TetR, which represses expression of a toxin gene. In this embodiment, the toxin gene is repressed in the presence of arabinose or other sugar. In an environment where arabinose is not present, the TetR gene is not activated and the toxin is expressed, thereby killing the bacteria. The arabinose system can also be used to express an essential gene, in which the essential gene is only expressed in the presence of arabinose or other sugar and is not expressed when arabinose or other sugar is absent from the environment.

Thus, in some embodiments, in which one or more heterologous gene(s) are expressed upon sensing arabinose in the exogenous environment, the one or more heterologous genes are directly or indirectly under the control of the *araBAD* promoter. In some embodiments, the expressed heterologous gene is selected from one or more of the following: a heterologous therapeutic gene, a heterologous gene encoding an anti-toxin, a heterologous gene encoding a repressor protein or polypeptide, for example, a TetR repressor, a heterologous gene encoding an essential protein not found in the bacterial cell, and/or a heterologous encoding a regulatory protein or polypeptide.

Arabinose inducible promoters are known in the art, including *P_{araC}*, *P_{araB}*, *P_{araC}*, and ParaBAD. In one embodiment, the arabinose inducible promoter is from *E. coli*. In some embodiments, the *P_{araC}* promoter and the ParaBAD promoter operate as a bidirectional promoter, with the ParaBAD promoter controlling expression of a heterologous gene(s) in one direction, and the *P_{araC}* (in close proximity to, and on the opposite strand from the ParaBAD promoter), controlling expression of a heterologous gene(s) in the other direction. In the presence of arabinose, transcription of both heterologous genes from both promoters is induced. However, in the absence of arabinose, transcription of both heterologous genes from both promoters is not induced.

In one exemplary embodiment of the disclosure, the genetically engineered bacteria of the present disclosure contain a kill switch having at least the following sequences: a ParaBAD promoter operably linked to a heterologous gene encoding a tetracycline repressor (TetR) protein, a *P_{araC}* promoter operably linked to a heterologous gene encoding the AraC transcription factor, and a heterologous gene encoding a bacterial toxin operably linked to a promoter which is repressed by the TetR protein. In the presence of arabinose, the AraC transcription factor activates the ParaBAD promoter, which activates transcription of the TetR protein which, in turn, represses transcription of the toxin. In the absence of arabinose, however, AraC suppresses transcription from the ParaBAD promoter and no TetR protein is expressed. In this case, expression of the heterologous toxin gene is activated, and the toxin is expressed. The toxin builds up in the recombinant bacterial cell, and the recombinant bacterial cell is killed. In one embodiment, the *araC* gene encoding the AraC transcription factor is under the control of a constitutive promoter and is therefore constitutively expressed.

In one embodiment of the disclosure, the genetically engineered bacterium further comprises an anti-toxin under the control of a constitutive promoter. In this situation, in the presence of arabinose, the toxin is not expressed due to repression by TetR protein, and the anti-toxin protein builds-up in the cell. However, in the absence of arabinose, TetR protein is not expressed, and expression of the toxin is induced. The toxin begins to build-up within the recombinant bacterial cell. The recombinant bacterial cell is no longer viable once the toxin protein is present at either equal or greater amounts than that of the anti-toxin protein in the cell, and the recombinant bacterial cell will be killed by the toxin.

In another embodiment of the disclosure, the genetically engineered bacterium further comprises an anti-toxin under the control of the ParaBAD promoter. In this situation, in the presence of arabinose, TetR and the anti-toxin are expressed, the anti-toxin builds up in the cell, and the toxin is not expressed due to repression by TetR protein. However, in the absence of arabinose, both the TetR protein and the anti-toxin are not expressed, and expression of the toxin is

induced. The toxin begins to build-up within the recombinant bacterial cell. The recombinant bacterial cell is no longer viable once the toxin protein is expressed, and the recombinant bacterial cell will be killed by the toxin.

In another exemplary embodiment of the disclosure, the genetically engineered bacteria of the present disclosure contain a kill switch having at least the following sequences: a ParaBAD promoter operably linked to a heterologous gene encoding an essential polypeptide not found in the recombinant bacterial cell (and required for survival), and a P_{araC} promoter operably linked to a heterologous gene encoding the AraC transcription factor. In the presence of arabinose, the AraC transcription factor activates the ParaBAD promoter, which activates transcription of the heterologous gene encoding the essential polypeptide, allowing the recombinant bacterial cell to survive. In the absence of arabinose, however, AraC suppresses transcription from the ParaBAD promoter and the essential protein required for survival is not expressed. In this case, the recombinant bacterial cell dies in the absence of arabinose. In some embodiments, the sequence of ParaBAD promoter operably linked to a heterologous gene encoding an essential polypeptide not found in the recombinant bacterial cell can be present in the bacterial cell in conjunction with the TetR/toxin kill switch system described directly above. In some embodiments, the sequence of ParaBAD promoter operably linked to a heterologous gene encoding an essential polypeptide not found in the recombinant bacterial cell can be present in the bacterial cell in conjunction with the TetR/toxin/anti-toxin kill switch system described directly above.

In yet other embodiments, the bacteria may comprise a plasmid stability system with a plasmid that produces both a short-lived anti-toxin and a long-lived toxin. In this system, the bacterial cell produces equal amounts of toxin and anti-toxin to neutralize the toxin. However, if/when the cell loses the plasmid, the short-lived anti-toxin begins to decay. When the anti-toxin decays completely the cell dies as a result of the longer-lived toxin killing it.

In some embodiments, the engineered bacteria of the present disclosure further comprise the gene(s) encoding the components of any of the above-described kill switch circuits.

In any of the above-described embodiments, the bacterial toxin is selected from the group consisting of a lysin, Hok, Fst, TisB, LdrD, Kid, SymE, MazF, FlmA, Ibs, XCV2162, dinJ, CcdB, MazF, ParE, YafO, Zeta, hicB, relB, yhaV, yoeB, chpBK, hipA, microcin B, microcin B17, microcin C, microcin C7-C51, microcin J25, microcin ColV, microcin 24, microcin L, microcin D93, microcin L, microcin E492, microcin H47, microcin 147, microcin M, colicin A, colicin E1, colicin K, colicin N, colicin U, colicin B, colicin Ia, colicin Ib, colicin 1b, colicin10, colicin S4, colicin Y, colicin E2, colicin E7, colicin E8, colicin E9, colicin E3, colicin E4, colicin E6, colicin E5, colicin D, colicin M, and cloacin DF13, or a biologically active fragment thereof.

In any of the above-described embodiments, the anti-toxin is selected from the group consisting of an anti-lysin, Sok, RNAII, IstR, RdlD, Kis, SymR, MazE, FlmB, Sib, ptaRNA1, yafQ, CcdA, MazE, ParD, yafN, Epsilon, HicA, relE, prfF, yefM, chpBI, hipB, MccE, MccE^{CTD}, MccF, Cai, ImmE1, Cki, Cni, Cui, Cbi, Iia, Imm, Cfi, Im10, Csi, Cyi, Im2, Im7, Im8, Im9, Im3, Im4, ImmE6, cloacin immunity protein (Cim), ImmE5, ImmD, and Cmi, or a biologically active fragment thereof.

In one embodiment, the bacterial toxin is bactericidal to the genetically engineered bacterium. In one embodiment, the bacterial toxin is bacteriostatic to the genetically engineered bacterium.

In some embodiments, the genetically engineered bacterium provided herein is an auxotroph. In one embodiment, the genetically engineered bacterium is an auxotroph selected from a *cysE*, *glnA*, *ilvD*, *leuB*, *lysA*, *serA*, *metA*, *glyA*, *hisB*, *ilvA*, *pheA*, *proA*, *thrC*, *trpC*, *tyrA*, *thyA*, *uraA*, *dapA*, *dapB*, *dapD*, *dapE*, *dapF*, *flhD*, *metB*, *metC*, *proAB*, and *thiI* auxotroph. In some embodiments, the engineered bacteria have more than one auxotrophy, for example, they may be a Δ thyA and Δ dapA auxotroph.

In some embodiments, the genetically engineered bacterium provided herein further comprises a kill switch circuit, such as any of the kill switch circuits provided herein. For example, in some embodiments, the genetically engineered bacteria further comprise one or more genes encoding one or more recombinase(s) under the control of an inducible promoter and an inverted toxin sequence. In some embodiments, the genetically engineered bacteria further comprise one or more genes encoding an anti-toxin. In some embodiments, the engineered bacteria further comprise one or more genes encoding one or more recombinase(s) under the control of an inducible promoter and one or more inverted excision genes, wherein the excision gene(s) encode an enzyme that deletes an essential gene. In some embodiments, the genetically engineered bacteria further comprise one or more genes encoding an anti-toxin. In some embodiments, the engineered bacteria further comprise one or more genes encoding a toxin under the control of a promoter having a TetR repressor binding site and a gene encoding the TetR under the control of an inducible promoter that is induced by arabinose, such as ParaBAD. In some embodiments, the genetically engineered bacteria further comprise one or more genes encoding an anti-toxin.

In some embodiments, the genetically engineered bacterium is an auxotroph comprising a gene encoding a phenylalanine-metabolizing enzyme and further comprises a kill switch circuit, such as any of the kill switch circuits described herein.

In some embodiments, of the above described genetically engineered bacteria, the gene or gene cassette for producing the phenylalanine-metabolizing enzyme is present on a plasmid in the bacterium and operatively linked on the plasmid to the promoter that is induced under low-oxygen or anaerobic conditions. In other embodiments, the gene or gene cassette for producing the phenylalanine-metabolizing enzyme is present in the bacterial chromosome and is operatively linked in the chromosome to the promoter that is induced under low-oxygen or anaerobic conditions.

Pharmaceutical Compositions and Formulations

Pharmaceutical compositions comprising the genetically engineered bacteria of the invention may be used to treat, manage, ameliorate, and/or prevent diseases associated with hyperphenylalaninemia, e.g., PKU. Pharmaceutical compositions of the invention comprising one or more genetically engineered bacteria, alone or in combination with prophylactic agents, therapeutic agents, and/or pharmaceutically acceptable carriers are provided. In certain embodiments, the pharmaceutical composition comprises one species, strain, or subtype of bacteria that are engineered to comprise the genetic modifications described herein. In alternate embodiments, the pharmaceutical composition comprises two or more species, strains, and/or subtypes of bacteria that are each engineered to comprise the genetic modifications described herein.

The pharmaceutical compositions described herein may be formulated in a conventional manner using one or more physiologically acceptable carriers comprising excipients and auxiliaries, which facilitate processing of the active ingredients into compositions for pharmaceutical use. Methods of formulating pharmaceutical compositions are known in the art (see, e.g., "Remington's Pharmaceutical Sciences," Mack Publishing Co., Easton, PA). In some embodiments, the pharmaceutical compositions are subjected to tableting, lyophilizing, direct compression, conventional mixing, dissolving, granulating, levigating, emulsifying, encapsulating, entrapping, or spray drying to form tablets, granulates, nanoparticles, nanocapsules, microcapsules, microtablets, pellets, or powders, which may be enterically coated or uncoated. Appropriate formulation depends on the route of administration.

The genetically engineered bacteria described herein may be formulated into pharmaceutical compositions in any suitable dosage form (e.g., liquids, capsules, sachet, hard capsules, soft capsules, tablets, enteric coated tablets, suspension powders, granules, or matrix sustained release formations for oral administration) and for any suitable type of administration (e.g., oral, topical, injectable, immediate-release, pulsatile-release, delayed-release, or sustained release). Suitable dosage amounts for the genetically engineered bacteria may range from about 10^5 to 10^{12} bacteria, e.g., approximately 10^5 bacteria, approximately 10^6 bacteria, approximately 10^7 bacteria, approximately 10^8 bacteria, approximately 10^9 bacteria, approximately 10^{10} bacteria, approximately 10^{11} bacteria, or approximately 10^{12} bacteria. The composition may be administered once or more daily, weekly, or monthly. The composition may be administered before, during, or following a meal. In one embodiment, the pharmaceutical composition is administered before the subject eats a meal. In one embodiment, the pharmaceutical composition is administered currently with a meal. In one embodiment, the pharmaceutical composition is administered after the subject eats a meal.

The genetically engineered bacteria may be formulated into pharmaceutical compositions comprising one or more pharmaceutically acceptable carriers, thickeners, diluents, buffers, buffering agents, surface active agents, neutral or cationic lipids, lipid complexes, liposomes, penetration enhancers, carrier compounds, and other pharmaceutically acceptable carriers or agents. For example, the pharmaceutical composition may include, but is not limited to, the addition of calcium bicarbonate, sodium bicarbonate, calcium phosphate, various sugars and types of starch, cellulose derivatives, gelatin, vegetable oils, polyethylene glycols, and surfactants, including, for example, polysorbate 20. In some embodiments, the genetically engineered bacteria of the invention may be formulated in a solution of sodium bicarbonate, e.g., 1 molar solution of sodium bicarbonate (to buffer an acidic cellular environment, such as the stomach, for example). The genetically engineered bacteria may be administered and formulated as neutral or salt forms. Pharmaceutically acceptable salts include those formed with anions such as those derived from hydrochloric, phosphoric, acetic, oxalic, tartaric acids, etc., and those formed with cations such as those derived from sodium, potassium, ammonium, calcium, ferric hydroxides, isopropylamine, triethylamine, 2-ethylamino ethanol, histidine, procaine, etc.

The genetically engineered bacteria disclosed herein may be administered topically and formulated in the form of an ointment, cream, transdermal patch, lotion, gel, shampoo, spray, aerosol, solution, emulsion, or other form well-known to one of skill in the art. See, e.g., "Remington's Pharmaceutical Sciences," Mack Publishing Co., Easton, PA. In an embodiment, for non-sprayable topical dosage forms, viscous to semi-solid or solid forms comprising a carrier or one or more excipients compatible with topical application and having a dynamic viscosity greater than water are employed.

Suitable formulations include, but are not limited to, solutions, suspensions, emulsions, creams, ointments, powders, liniments, salves, etc., which may be sterilized or mixed with auxiliary agents (e.g., preservatives, stabilizers, wetting agents, buffers, or salts) for influencing various properties, e.g., osmotic pressure. Other suitable topical dosage forms include sprayable aerosol preparations wherein the active ingredient in combination with a solid or liquid inert carrier, is packaged in a mixture with a pressurized volatile (e.g., a gaseous propellant, such as freon) or in a squeeze bottle. Moisturizers or humectants can also be added to pharmaceutical compositions and dosage forms. Examples of such additional ingredients are well known in the art. In one embodiment, the pharmaceutical composition comprising the recombinant bacteria of the invention may be formulated as a hygiene product. For example, the hygiene product may be an antibacterial formulation, or a fermentation product such as a fermentation broth. Hygiene products may be, for example, shampoos, conditioners, creams, pastes, lotions, and lip balms.

The genetically engineered bacteria disclosed herein may be administered orally and formulated as tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspensions, etc. Pharmacological compositions for oral use can be made using a solid excipient, optionally grinding the resulting mixture, and processing the mixture of granules, after adding suitable auxiliaries if desired, to obtain tablets or dragee cores. Suitable excipients include, but are not limited to, fillers such as sugars, including lactose, sucrose, mannitol, or sorbitol; cellulose compositions such as maize starch, wheat starch, rice starch, potato starch, gelatin, gum tragacanth, methyl cellulose, hydroxypropylmethyl-cellulose, sodium carbomethylcellulose; and/or physiologically acceptable polymers such as polyvinylpyrrolidone (PVP) or polyethylene glycol (PEG). Disintegrating agents may also be added, such as cross-linked polyvinylpyrrolidone, agar, alginic acid or a salt thereof such as sodium alginate.

Tablets or capsules can be prepared by conventional means with pharmaceutically acceptable excipients such as binding agents (e.g., pregelatinised maize starch, polyvinylpyrrolidone, hydroxypropyl methylcellulose, carboxymethylcellulose, polyethylene glycol, sucrose, glucose, sorbitol, starch, gum, kaolin, and tragacanth); fillers (e.g., lactose, microcrystalline cellulose, or calcium hydrogen phosphate); lubricants (e.g., calcium, aluminum, zinc, stearic acid, polyethylene glycol, sodium lauryl sulfate, starch, sodium benzoate, L-leucine, magnesium stearate, talc, or silica); disintegrants (e.g., starch, potato starch, sodium starch glycolate, sugars, cellulose derivatives, silica powders); or wetting agents (e.g., sodium lauryl sulphate). The tablets may be coated by methods well known in the art. A coating shell may be present, and common membranes include, but are not limited to, polylactide, polyglycolic acid, polyanhydride, other biodegradable polymers, alginate-polylysine-alginate (APA), alginate-polymethylene-co-guanidine-alginate (A-PMCG-A), hydroxymethylacrylate-methyl methacrylate (HEMA-MMA), multilayered HEMA-MMA-MAA, polyacrylonitrilevinylchloride (PAN-PVC), acrylonitrile/sodium methallylsulfonate (AN-69), polyethylene glycol/poly pentamethylcyclopentasiloxane/polydimethylsiloxane (PEG/PD5/PDMS), poly N,N-dimethyl acrylamide (PDMAAm), siliceous encapsulates, cellulose sulphate/sodium alginate/polymethylene-co-guanidine (CS/A/PMCG), cellulose acetate phthalate, calcium alginate, k-carrageenan-locust bean gum gel beads, gellan-xanthan beads, poly(lactide-co-glycolides), carrageenan, starch poly-anhydrides, starch polymethacrylates, polyamino acids, and enteric coating polymers.

In some embodiments, the genetically engineered bacteria are enterically coated for release into the gut or a particular region of the gut, for example, the large intestine. The typical pH profile from the stomach to the colon is about 1-4

(stomach), 5.5-6 (duodenum), 7.3-8.0 (ileum), and 5.5-6.5 (colon). In some diseases, the pH profile may be modified. In some embodiments, the coating is degraded in specific pH environments in order to specify the site of release. In some embodiments, at least two coatings are used. In some

embodiments, the outside coating and the inside coating are degraded at different pH levels. Liquid preparations for oral administration may take the form of solutions, syrups, suspensions, or a dry product for constitution with water or other suitable vehicle before use. Such liquid preparations may be prepared by conventional means with pharmaceutically acceptable agents such as suspending agents (e.g., sorbitol syrup, cellulose derivatives, or hydrogenated edible fats); emulsifying agents (e.g., lecithin or acacia); non-aqueous vehicles (e.g., almond oil, oily esters, ethyl alcohol, or fractionated vegetable oils); and preservatives (e.g., methyl or propyl-p-hydroxybenzoates or sorbic acid). The preparations may also contain buffer salts, flavoring, coloring, and sweetening agents as appropriate. Preparations for oral administration may be suitably formulated for slow release, controlled release, or sustained release of the genetically engineered bacteria described herein.

In one embodiment, the genetically engineered bacteria of the disclosure may be formulated in a composition suitable for administration to pediatric subjects. As is well known in the art, children differ from adults in many aspects, including different rates of gastric emptying, pH, gastrointestinal permeability, etc. (Ivanovska et al., 2014). Moreover, pediatric formulation acceptability and preferences, such as route of administration and taste attributes, are critical for achieving acceptable pediatric compliance. Thus, in one embodiment, the composition suitable for administration to pediatric subjects may include easy-to-swallow or dissolvable dosage forms, or more palatable compositions, such as compositions with added flavors, sweeteners, or taste blockers. In one embodiment, a composition suitable for administration to pediatric subjects may also be suitable for administration to adults.

In one embodiment, the composition suitable for administration to pediatric subjects may include a solution, syrup, suspension, elixir, powder for reconstitution as suspension or solution, dispersible/effervescent tablet, chewable tablet, gummy candy, lollipop, freezer pop, troche, chewing gum, oral thin strip, orally disintegrating tablet, sachet, soft gelatin capsule, sprinkle oral powder, or granules. In one embodiment, the composition is a gummy candy, which is made from a gelatin base, giving the candy elasticity, desired chewy consistency, and longer shelf-life. In some embodiments, the gummy candy may also comprise sweeteners or flavors.

In one embodiment, the composition suitable for administration to pediatric subjects may include a flavor. As used herein, "flavor" is a substance (liquid or solid) that provides a distinct taste and aroma to the formulation. Flavors also help to improve the palatability of the formulation. Flavors include, but are not limited to, strawberry, vanilla, lemon, grape, bubble gum, and cherry.

In certain embodiments, the genetically engineered bacteria may be orally administered, for example, with an inert diluent or an assimilable edible carrier. The compound may also be enclosed in a hard or soft shell gelatin capsule, compressed into tablets, or incorporated directly into the subject's diet. For oral therapeutic administration, the compounds may be incorporated with excipients and used in the form of ingestible tablets, buccal tablets, troches, capsules, elixirs, suspensions, syrups, wafers, and the like. To administer a compound by other than parenteral administration, it may be necessary to coat the compound with, or co-administer the compound with, a material to prevent its inactivation.

In another embodiment, the pharmaceutical composition comprising the recombinant bacteria of the invention may be a comestible product, for example, a food product. In one embodiment, the food product is milk, concentrated milk, fermented milk (yogurt, sour milk, frozen yogurt, lactic acid bacteria-fermented beverages), milk powder, ice cream, cream cheeses, dry cheeses, soybean milk, fermented soybean milk, vegetable-fruit juices, fruit juices, sports drinks, confectionery, candies, infant foods (such as infant cakes), nutritional food products, animal feeds, or dietary supplements. In one embodiment, the food product is a fermented food, such as a fermented dairy product. In one embodiment, the fermented dairy product is yogurt. In another embodiment, the fermented dairy product is cheese, milk, cream, ice cream, milk shake, or kefir. In another embodiment, the recombinant bacteria of the invention are combined in a preparation containing other live bacterial cells intended to serve as probiotics. In another embodiment, the food product is a beverage. In one embodiment, the beverage is a fruit juice-based beverage or a beverage containing plant or herbal extracts. In another embodiment, the food product is a jelly or a pudding. Other food products suitable for administration of the recombinant bacteria of the invention are well known in the art. See, e.g., US 2015/0359894 and US 2015/0238545, the entire contents of each of which are expressly incorporated herein by reference. In yet another embodiment, the pharmaceutical composition of the invention is injected into, sprayed onto, or sprinkled onto a food product, such as bread, yogurt, or cheese.

In some embodiments, the composition is formulated for intrainestinal administration, intrajejunal administration, intraduodenal administration, intraileal administration, gastric shunt administration, or intracolonic administration, via nanoparticles, nanocapsules, microcapsules, or microtablets, which are enterically coated or uncoated. The pharmaceutical compositions may also be formulated in rectal compositions such as suppositories or retention enemas, using, e.g., conventional suppository bases such as cocoa butter or other glycerides. The compositions may be suspensions, solutions, or emulsions in oily or aqueous vehicles, and may contain suspending, stabilizing and/or dispersing agents.

The genetically engineered bacteria described herein may be administered intranasally, formulated in an aerosol form, spray, mist, or in the form of drops, and conveniently delivered in the form of an aerosol spray presentation from pressurized packs or a nebuliser, with the use of a suitable propellant (e.g., dichlorodifluoromethane, trichlorofluoromethane, dichlorotetrafluoroethane, carbon dioxide or other suitable gas). Pressurized aerosol dosage units may be determined by providing a valve to deliver a metered amount. Capsules and cartridges (e.g., of gelatin) for use in an inhaler or insufflator may be formulated containing a powder mix of the compound and a suitable powder base such as lactose or starch.

The genetically engineered bacteria may be administered and formulated as depot preparations. Such long acting formulations may be administered by implantation or by injection, including intravenous injection, subcutaneous injection, local injection, direct injection, or infusion. For example, the compositions may be formulated with suitable polymeric or hydrophobic materials (e.g., as an emulsion in an acceptable oil) or ion exchange resins, or as sparingly soluble derivatives (e.g., as a sparingly soluble salt).

In some embodiments, disclosed herein are pharmaceutically acceptable compositions in single dosage forms. Single dosage forms may be in a liquid or a solid form. Single dosage forms may be administered directly to a patient without modification or may be diluted or reconstituted prior to administration. In certain embodiments, a single dosage form may be administered in bolus form, e.g., single injection, single oral dose, including an oral dose that comprises multiple tablets, capsule, pills, etc. In alternate

embodiments, a single dosage form may be administered over a period of time, e.g., by infusion.

Single dosage forms of the pharmaceutical composition may be prepared by portioning the pharmaceutical composition into smaller aliquots, single dose containers, single dose liquid forms, or single dose solid forms, such as tablets, granulates, nanoparticles, nanocapsules, microcapsules, microtablets, pellets, or powders, which may be enterically coated or uncoated. A single dose in a solid form may be reconstituted by adding liquid, typically sterile water or saline solution, prior to administration to a patient.

In other embodiments, the composition can be delivered in a controlled release or sustained release system. In one embodiment, a pump may be used to achieve controlled or sustained release. In another embodiment, polymeric materials can be used to achieve controlled or sustained release of the therapies of the present disclosure (see, e.g., U.S. Pat. No. 5,989,463). Examples of polymers used in sustained release formulations include, but are not limited to, poly(2-hydroxy ethyl methacrylate), poly(methyl methacrylate), poly(acrylic acid), poly(ethylene-co-vinyl acetate), poly(methacrylic acid), polyglycolides (PLG), polyanhydrides, poly(N-vinyl pyrrolidone), poly(vinyl alcohol), polyacrylamide, poly(ethylene glycol), polylactides (PLA), poly(lactide-co-glycolides) (PLGA), and polyorthoesters. The polymer used in a sustained release formulation may be inert, free of leachable impurities, stable on storage, sterile, and biodegradable. In some embodiments, a controlled or sustained release system can be placed in proximity of the prophylactic or therapeutic target, thus requiring only a fraction of the systemic dose. Any suitable technique known to one of skill in the art may be used.

Dosage regimens may be adjusted to provide a therapeutic response. Dosing can depend on several factors, including severity and responsiveness of the disease, route of administration, time course of treatment (days to months to years), and time to amelioration of the disease. For example, a single bolus may be administered at one time, several divided doses may be administered over a predetermined period of time, or the dose may be reduced or increased as indicated by the therapeutic situation. The specification for the dosage is dictated by the unique characteristics of the active compound and the particular therapeutic effect to be achieved. Dosage values may vary with the type and severity of the condition to be alleviated. For any particular subject, specific dosage regimens may be adjusted over time according to the individual need and the professional judgment of the treating clinician. Toxicity and therapeutic efficacy of compounds provided herein can be determined by standard pharmaceutical procedures in cell culture or animal models. For example, LD₅₀, ED₅₀, EC₅₀, and IC₅₀ may be determined, and the dose ratio between toxic and therapeutic effects (LD₅₀/ED₅₀) may be calculated as the therapeutic index. Compositions that exhibit toxic side effects may be used, with careful modifications to minimize potential damage to reduce side effects. Dosing may be estimated initially from cell culture assays and animal models. The data obtained from in vitro and in vivo assays and animal studies can be used in formulating a range of dosage for use in humans.

The ingredients are supplied either separately or mixed together in unit dosage form, for example, as a dry lyophilized powder or water-free concentrate in a hermetically sealed container such as an ampoule or sachet indicating the quantity of active agent. If the mode of administration is by injection, an ampoule of sterile water for injection or saline can be provided so that the ingredients may be mixed prior to administration.

The pharmaceutical compositions may be packaged in a hermetically sealed container such as an ampoule or sachet indicating the quantity of the agent. In one embodiment, one or more of the pharmaceutical compositions is supplied as a

dry sterilized lyophilized powder or water-free concentrate in a hermetically sealed container and can be reconstituted (e.g., with water or saline) to the appropriate concentration for administration to a subject. In an embodiment, one or more of the prophylactic or therapeutic agents or pharmaceutical compositions is supplied as a dry sterile lyophilized powder in a hermetically sealed container stored between 2° C. and 8° C. and administered within 1 hour, within 3 hours, within 5 hours, within 6 hours, within 12 hours, within 24 hours, within 48 hours, within 72 hours, or within one week after being reconstituted. Cryoprotectants can be included for a lyophilized dosage form, principally 0-10% sucrose (optimally 0.5-1.0%). Other suitable cryoprotectants include trehalose and lactose. Other suitable bulking agents include glycine and arginine, either of which can be included at a concentration of 0-0.05%, and polysorbate-80 (optimally included at a concentration of 0.005-0.01%). Additional surfactants include but are not limited to polysorbate 20 and BRIJ surfactants. The pharmaceutical composition may be prepared as an injectable solution and can further comprise an agent useful as an adjuvant, such as those used to increase absorption or dispersion, e.g., hyaluronidase.

Methods of Treatment

Another aspect of the invention provides methods of treating a disease associated with hyperphenylalaninemia or symptom(s) associated with hyperphenylalaninemia. In some embodiments, the disease is selected from the group consisting of: phenylketonuria, classical or typical phenylketonuria, atypical phenylketonuria, permanent mild hyperphenylalaninemia, nonphenylketonuric hyperphenylalaninemia, phenylalanine hydroxylase deficiency, cofactor deficiency, dihydropteridine reductase deficiency, tetrahydropterin synthase deficiency, and Segawa's disease. In some embodiments, hyperphenylalaninemia is secondary to other conditions, e.g., liver diseases. In some embodiments, the invention provides methods for reducing, ameliorating, or eliminating one or more symptom(s) associated with these diseases, including but not limited to neurological deficits, mental retardation, encephalopathy, epilepsy, eczema, reduced growth, microcephaly, tremor, limb spasticity, and/or hypopigmentation. In some embodiments, the subject to be treated is a human patient.

In certain embodiments, the genetically engineered bacteria are capable of metabolizing phenylalanine in the diet in order to treat a disease or disorder associated with hyperphenylalaninemia, e.g., PKU. In some embodiments, the genetically engineered bacteria are delivered simultaneously with dietary protein. In other embodiments, the genetically engineered bacteria are not delivered simultaneously with dietary protein. Studies have shown that pancreatic and other glandular secretions into the intestine contain high levels of proteins, enzymes, and polypeptides, and that the amino acids produced as a result of their catabolism are reabsorbed back into the blood in a process known as "enterorecirculation" (Chang, 2007; Sarkissian et al., 1999). Thus, high intestinal levels of phenylalanine may be partially independent of food intake, and are available for breakdown by PAL. In some embodiments, the genetically engineered bacteria and dietary protein are delivered after a period of fasting or phenylalanine-restricted dieting. In these embodiments, a patient suffering from hyperphenylalaninemia may be able to resume a substantially normal diet, or a diet that is less restrictive than a phenylalanine-free diet. In some embodiments, the genetically engineered bacteria may be capable of metabolizing phenylalanine from additional sources, e.g., the blood, in order to treat a disease associated with hyperphenylalaninemia, e.g., PKU. In these embodiments, the genetically engineered bacteria need not be delivered simultaneously with dietary protein, and a phenylalanine gradient is generated, e.g., from blood to gut, and the genetically engineered bacteria metabolize phenylalanine and reduce phenylalaninemia.

The method may comprise preparing a pharmaceutical composition with at least one genetically engineered species, strain, or subtype of bacteria described herein, and administering the pharmaceutical composition to a subject in a therapeutically effective amount. In some embodiments, the genetically engineered bacteria of the invention are administered orally, e.g., in a liquid suspension. In some embodiments, the genetically engineered bacteria of the invention are lyophilized in a gel cap and administered orally. In some embodiments, the genetically engineered bacteria of the invention are administered via a feeding tube or gastric shunt. In some embodiments, the genetically engineered bacteria of the invention are administered rectally, e.g., by enema. In some embodiments, the genetically engineered bacteria of the invention are administered topically, intraintestinally, intrajejunally, intraduodenally, intraileally, and/or intracolically.

In certain embodiments, the pharmaceutical composition described herein is administered to reduce phenylalanine levels in a subject. In some embodiments, the methods of the present disclosure reduce the phenylalanine levels in a subject by at least about 10%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 75%, 80%, 85%, 90%, 95%, or more as compared to levels in an untreated or control subject. In some embodiments, reduction is measured by comparing the phenylalanine level in a subject before and after administration of the pharmaceutical composition. In some embodiments, the method of treating or ameliorating hyperphenylalaninemia allows one or more symptoms of the condition or disorder to improve by at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or more.

Before, during, and after the administration of the pharmaceutical composition, phenylalanine levels in the subject may be measured in a biological sample, such as blood, serum, plasma, urine, peritoneal fluid, cerebrospinal fluid, fecal matter, intestinal mucosal scrapings, a sample collected from a tissue, and/or a sample collected from the contents of one or more of the following: the stomach, duodenum, jejunum, ileum, cecum, colon, rectum, and anal canal. In some embodiments, the methods may include administration of the compositions of the invention to reduce phenylalanine. In some embodiments, the methods may include administration of the compositions of the invention to reduce phenylalanine to undetectable levels in a subject. In some embodiments, the methods may include administration of the compositions of the invention to reduce phenylalanine concentrations to undetectable levels, or to less than about 1%, 2%, 5%, 10%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 75%, or 80% of the subject's phenylalanine levels prior to treatment.

In certain embodiments, the genetically engineered bacteria are *E. coli* Nissle. The genetically engineered bacteria may be destroyed, e.g., by defense factors in the gut or blood serum (Sonnenbom et al., 2009) or by activation of a kill switch, several hours or days after administration. Thus, the pharmaceutical composition comprising the genetically engineered bacteria may be re-administered at a therapeutically effective dose and frequency. Length of Nissle residence in vivo in mice is shown in FIG. 38. In alternate embodiments, the genetically engineered bacteria are not destroyed within hours or days after administration and may propagate and colonize the gut.

The methods of the invention may comprise administration of the pharmaceutical composition alone or in combination with one or more additional therapeutic agents. In some embodiments, the pharmaceutical composition is administered in conjunction with the cofactor tetrahydrobiopterin (e.g., Kuvan/sapropterin), large neutral amino acids (e.g., tyrosine, tryptophan), glycomacropeptides, a probiotic (e.g., VSL3), an enzyme (e.g., pegylated-PAL), and/or other agents used in the treatment of phenylketonuria (Al Hafid and Christodoulou, 2015).

In some embodiments, the genetically engineered bacteria are administered in combination with one or more recombinantly produced PME enzymes, e.g. recombinant PAL, LAAD or PAH. In some embodiments, the recombinant enzymes are further formulated for improved stability and/or delivery. In some embodiments, the one or more PME enzyme administered in combination with the genetically engineered bacteria is pegylated. In some embodiments, the one or more PME enzyme administered in combination with the genetically engineered bacteria is delivered as a fusion protein. A non-limiting example of such a fusion protein is a fusion between a PME and a transduction domain for uptake into cells. A non-limiting example of such transduction domain or cell penetrating peptide is the TAT peptide. In some embodiments, the one or more PME enzyme administered in combination with the genetically engineered bacteria is formulated in a nanoparticle. A non-limiting example of such a nanoparticle is a dextran sulfate/chitosan PME nanoparticle. In some embodiments, the one or more PME enzyme administered in combination with the genetically engineered bacteria is delivered as a PME microsphere. A non-limiting example of such a microsphere is a barium alginate PME microsphere. In some embodiments, the one or more PME enzyme administered in combination with the genetically engineered bacteria is delivered as amorphous silica PME particles.

In some embodiments, the genetically engineered bacteria are administered in combination with PAL. In some embodiments, the genetically engineered bacteria are administered in combination with PAH. In some embodiments, the genetically engineered bacteria are administered in combination with LAAD. In some embodiments, the genetically engineered bacteria are administered in combination with PAL and PAH. In some embodiments, the genetically engineered bacteria are administered in combination with PAL and LAAD. In some embodiments, the genetically engineered bacteria are administered in combination with PAH and LAAD. In some embodiments, the genetically engineered bacteria are administered in combination with PAL, PAH, and LAAD.

In some embodiments, the genetically engineered bacteria are administered in combination with pegylated PAL. In some embodiments, the genetically engineered bacteria are administered in combination with pegylated PAH. In some embodiments, the genetically engineered bacteria are administered in combination with pegylated LAAD. In some embodiments, the genetically engineered bacteria are administered in combination with a PAL fusion protein, e.g., a cell penetrating peptide. In some embodiments, the genetically engineered bacteria are administered in combination with a PAH fusion protein, e.g., a cell penetrating peptide. In some embodiments, the genetically engineered bacteria are administered in combination with a LAAD fusion protein, e.g., a cell penetrating peptide. In some embodiments, the genetically engineered bacteria are administered in combination with PAL-nanoparticles. In some embodiments, the genetically engineered bacteria are administered in combination with PAH-nanoparticles. In some embodiments, the genetically engineered bacteria are administered in combination with LAAD nanoparticles. In some embodiments, the genetically engineered bacteria are administered in combination with PAL-microspheres. In some embodiments, the genetically engineered bacteria are administered in combination with PAH-microspheres. In some embodiments, the genetically engineered bacteria are administered in combination with LAAD-microspheres. In some embodiments, the genetically engineered bacteria are administered in combination with PAL-silica particles. In some embodiments, the genetically engineered bacteria are administered in combination with PAH-silica particles. In some embodiments, the genetically engineered bacteria are administered in combination with LAAD-silica particles.

In some embodiments, a recombinant enzyme replacement therapy or substitution therapy, e.g. PAL, PAH, and/or LAAD is administered without the genetically engineered bacteria.

In some embodiments, the one or more PME administered is PAL. In some embodiments, PAL is modified as described in Sakissian et al., 2011, Mol Genet Metab. 2011 November; 104(3): 249-254, the contents of which is herein incorporated by reference in its entirety. In some embodiments, the PAL is Av-p.C503S/p.C565S/p.F18A PAL. In some embodiments, the PAL is PEG-Av-p.C503S/p.C565S/p.F18A PAL.

In some embodiments, the PAL is PEGylated. In one embodiment, the pegylated PAL is from *Anabaena variabilis*. In one embodiment, the pegylated PAL is from *Photorhabdus luminescens*. In some embodiments, the one or more PME administered is PAH. In one embodiment, PAH is human PAH. In some embodiments, the one or more PME administered is LAAD. In one embodiment, the LAAD protein administered is derived from *Proteus mirabilis*. In some embodiments, the one or more PME administered in combination with PAL and PAH. In some embodiments, the one or more PME administered is PAL and LAAD. In some embodiments, the one or more PME administered is PAH and LAAD. In some embodiments, the one or more PME administered is PAL, PAH, and LAAD.

In some embodiments, the recombinant enzymes are further formulated for improved stability and/or delivery. In some embodiments, the one or more PME enzyme administered is peggylated. In some embodiments, the one or more PME enzyme administered is delivered as a fusion protein. A non-limiting example of such a fusion protein is a fusion between a PME and a transduction domain for uptake into cells. A non-limiting example of such transduction domain or cell penetrating peptide is the TAT peptide. In some embodiments, the one or more PME enzyme administered is formulated in a nanoparticle. A non-limiting example of such a nanoparticle is a dextran sulfate/chitosan PME nanoparticle. In some embodiments, the one or more PME enzyme administered is delivered as a PME microsphere. A non-limiting example of such a microsphere is a barium alginate PME microsphere. In some embodiments, the one or more PME enzyme administered is delivered as amorphous silica PME particles.

In some embodiments, pegylated PAL is administered. In some embodiments, pegylated LAAD is administered. In some embodiments peggylated LAAD from *Proteus mirabilis* is administered. In some embodiments, pegylated PAH is administered.

In one embodiment, a PAL fusion protein, e.g., with a cell penetrating peptide, is administered. In one embodiment, a LAAD fusion protein, e.g., with a cell penetrating peptide, is administered. In one embodiment, a PAH fusion protein, e.g., with a cell penetrating peptide, is administered. In some embodiments, PAL-nanoparticles are administered. In some embodiments, PAH-nanoparticles are administered. In some embodiments, LAAD-nanoparticles are administered. In some embodiments, PAL-microspheres are administered. In some embodiments, PAH-microspheres are administered. In some embodiments, LAAD-microspheres are administered. In some embodiments, PAL-silica particles are administered. In some embodiments, PAH-silica particles are administered. In some embodiments, LAAD-silica particles are administered.

In some embodiments the PME, e.g., PAH, PAL, and/or LAAD is formulated with aprotinin, e.g., 40 mg/ml aprotinin.

In some embodiments the PMEs are delivered as gene therapy. In some embodiments, a CRISPR technology is used. In some embodiments a gene therapy vector is used to deliver the one or more PME, e.g., PAL, LAAD, and/or PAH. Gene therapy vectors are known in the art and include, but are not limited to, retroviral vectors, adenoviral vectors, adeno-associated viral vectors. Alternatively, formulated or naked PME gene DNA or RNA can be delivered.

An important consideration in the selection of the one or more additional therapeutic agents is that the agent(s) should be compatible with the genetically engineered bacteria of the invention, e.g., the agent(s) must not interfere with or kill the bacteria. In some embodiments, the pharmaceutical composition is administered with food. In alternate embodiments, the pharmaceutical composition is administered before or after eating food. The pharmaceutical composition may be administered in combination with one or more dietary modifications, e.g., low-phenylalanine diet. The dosage of the pharmaceutical composition and the frequency of administration may be selected based on the severity of the symptoms and the progression of the disease. The appropriate therapeutically effective dose and/or frequency of administration can be selected by a treating clinician.

The methods of the invention also include kits comprising the pharmaceutical composition described herein. The kit can include one or more other elements including, but not limited to: instructions for use; other reagents, e.g., a label, an additional therapeutic agent; devices or materials for measuring phenylalanine levels, or levels of other molecules or metabolites associated with hyperphenylalaninemia, in a subject; devices or other materials for preparing the pharmaceutical composition of the invention for administration; and devices or other materials for administration to a subject. Instructions for use can include guidance for therapeutic application, such as suggested dosages and/or modes of administration, e.g., in a patient with hyperphenylalaninemia. The kit can further contain at least one additional therapeutic agent, and/or one or more additional genetically engineered bacterial strains of the invention, formulated as appropriate, in one or more separate pharmaceutical preparations.

In some embodiments, the kit is used for administration of the pharmaceutical composition to a subject. In some embodiments, the kit is used for administration of the pharmaceutical composition, alone or in combination with one or more additional therapeutic agents, to a subject. In some embodiments, the kit is used for measuring phenylalanine levels (e.g., blood phenylalanine levels) in a subject before, during, or after administration of the pharmaceutical composition to the subject. In certain embodiments, the kit is used for administration and/or re-administration of the pharmaceutical composition, alone or in combination with one or more additional therapeutic agents, when blood phenylalanine levels are increased or abnormally high. In some embodiments, a diagnostic signal of hyperphenylalaninemia is a blood phenylalanine level of at least 2 mg/dL, at least 4 mg/dL, at least 6 mg/dL, at least 8 mg/dL, at least 10 mg/dL, at least 12 mg/dL, at least 14 mg/dL, at least 16 mg/dL, at least 18 mg/dL, at least 20 mg/dL, or at least 25 mg/dL.

Table 20 shows non-limiting examples of target degradation rates, based on levels of phenylalanine on average in classical PKU patients.

TABLE 20

	Target Degradation Rates								
	Age (years)								
	0-6 months	7-12 months	1-3	4-8	9-13	14-18 (M)	14-18 (F)	>18 (M)	>18 (F)
RDA	9.1	11	13	19	34	52	46	56	46
Protein (g/d)									
Daily PHE	428	517	611	893	1598	2444	2162	2632	2162
(mg)- Healthy subject									
(1 g protein = 47 mg PHE)									
Daily PHE	250	250	250	250	250	250	250	250	250
tolerance (mg)									
(Classical PKU)									
Target Reduction	178	267	361	643	1348	2194	1912	2382	1912
(mg)									
Target Reduction	1.08	1.62	2.19	3.89	8.16	13.28	11.57	14.42	11.57
(mmol)									
Target degradation rate	0.15	0.22	0.3	0.54	1.13	1.84	1.61	2	1.61
($\mu\text{mol}/10^9$ CFUs/hr)									
(based on 3.10^{11} CFUs/day dose)									
assuming all dose functioning for 24 hours									
Target degradation rate 2 hrs transit time	0.6	0.9	1.21	2.16	4.53	7.38	6.43	8.01	6.43
($\mu\text{mol}/10^9$ CFUs/hr)									
assuming 2 hour transit time per dose									
Target degradation rate 6 hrs transit time	0.2	0.3	0.4	0.72	1.51	2.46	2.14	2.67	2.14
($\mu\text{mol}/10^9$ CFUs/hr)									
assuming 6 hour transit time per dose									

In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.15 to about 8.01 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.15 to about 2 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.6 to about 8.01 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.2 to about 2.67 $\mu\text{mol}/10^9$ CFUs/hr.

In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.15 to about 0.6 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.22 to about 0.9 $\mu\text{mol}/10^9$ CFUs/hr. In some embodi-

ments, the genetically engineered bacteria achieve a target degradation rate of about 0.3 to about 1.21 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.54 to about 2.16 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 1.13 to about 4.53 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 1.84 to about 7.38 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 1.61 to about 6.43 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 2 to about 8.01 $\mu\text{mol}/10^9$ CFUs/hr.

In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 0.1 to about 1

$\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 1 to about 2 $\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 2 to about 3 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 3 to about 4 $\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 4 to about 5 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 5 to about 6 $\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 6 to about 7 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of about 7 to about 8 $\text{mol}/10^9$ CFUs/hr.

In some embodiments, the genetically engineered bacteria achieve a target reduction rate of less than 0.15 $\mu\text{mol}/10^9$ CFUs/hr. In some embodiments, the genetically engineered bacteria achieve a target degradation rate of greater than 8.01 $\mu\text{mol}/10^9$ CFUs/hr.

In some embodiments, the genetically engineered bacteria achieve a target reduction of between about 178 mg and 2382 mg. In some embodiments, the genetically engineered bacteria achieve a target reduction of 1.08 mmol to 14.42 mmol. In some embodiments, the reduction is less than 1.08 mmol. In some embodiments, the reduction is greater than 14.42 mmol.

In some embodiments, target reduction and target degradation rates are based on classical PKU phenylalanine levels. In some embodiments, the target reduction and target degradation rates are based on phenylalanine levels observed in mild PKU. In some embodiments, target reduction and target degradation rates are based on phenylalanine levels observed in mild hyperphenylalaninemia.

Treatment In Vivo

The genetically engineered bacteria of the invention may be evaluated in vivo, e.g., in an animal model. Any suitable animal model of a disease or condition associated with hyperphenylalaninemia may be used (see, e.g., Sarkissian et al., 1999). In some embodiments, the animal model is a mouse model of PKU. In certain embodiments, the mouse model of PKU is an PAH mutant BTBR mouse (BTBR-Pah^{enu2}, Jackson Laboratories). In these embodiments, the mouse model contains a chemically (ENU)-induced homozygous missense mutation (T835C) in exon 7 of the Pah gene, which results in a phenylalanine to serine substitution at amino acid 263 (F263S). This residue is located in the active site of the PAH enzyme, as shown by crystal structure analysis, and results in the complete loss of PAH activity. On normal diets, these mutant mice demonstrate a 10- to 20-fold increase in serum phenylalanine levels compared to unaffected controls. The genetically engineered bacteria of the invention may be administered to the animal, e.g., by oral gavage, and treatment efficacy is determined, e.g., by measuring blood phenylalanine and/or cinnamate before and after treatment. In animal models, it is noted that residence time of the genetically engineered bacteria within the GI tract may be shorter than residence time in humans. The animal may be sacrificed, and tissue samples may be collected and analyzed.

In some embodiments, pharmacokinetics and pharmacodynamic studies may be conducted in non-human primates to determine any potential toxicities arising from administration of the genetically engineered bacteria. the pharmacokinetics and pharmacodynamics of the genetically engi-

neered bacteria. Non-limiting examples of such studies are described in Examples 30 and 31.

In some embodiments, the genetically engineered bacteria expressing LAAD can be specifically detected in the feces and differentiated from other *E. coli* strains. A Phenylalanine Deaminase Test "Phenylalanine Agar Slant" can be used for this purpose. Phenylalanine agar used to determine whether the microbe can use phenylalanine and convert it to phenyl pyruvate. When the test chemicals are added to the tube containing the sample on the phenylalanine agar, phenylpyruvate is converted to a green compound, indicating a positive test. Wild type *E. coli* does not produce phenylpyruvate, since they do not encode an enzyme, which can produce phenylpyruvate from phenylalanine, allowing differentiation from other *E. coli* strains. The genetically engineered bacteria can be differentiated from other bacterial species which are able to produce phenylpyruvate by PCR-based tests known in the art. For example, species specific sequences can be amplified. For example, universal PCR that amplifies conserved regions in various bacteria is ideal to detect any pathogen in screening of specimens. For this purpose, the conserved region of the 16S rRNA gene can be used as a target gene for the universal PCR; the 16S rRNA gene contains species-specific regions by which a large number of bacterial species can be differentiated.

In some embodiments, the Phenylalanine Deaminase Test can be used to detect the genetically engineered bacteria in a feces sample. In some embodiments, PCR-based tests can be conducted to differentiate the genetically engineered bacteria from other bacterial species.

Screening Methods

In some embodiments, of the disclosure a genetically engineered strain may be improved upon by using screening and selection methods, e.g., to increase PME enzymatic activity or to increase the ability of a strain to take up phenylalanine. In some embodiments, the screen serves to generate a bacterial strain with improved PME activity. In some embodiments, the screen serves to generate a bacterial strain which has improved phenylalanine uptake ability. In some embodiments, the screen may identify a bacterial strain with both improved PME activity and enhanced substrate import. Non-limiting examples of methods of screening which can be used are described herein.

Generation of Bacterial Strains with Enhance Ability to Transport Biomolecules

Due to their ease of culture, short generation times, very high population densities and small genomes, microbes can be evolved to unique phenotypes in abbreviated timescales. Adaptive laboratory evolution (ALE) is the process of passaging microbes under selective pressure to evolve a strain with a preferred phenotype. Most commonly, this is applied to increase utilization of carbon/energy sources or adapting a strain to environmental stresses (e.g., temperature, pH), whereby mutant strains more capable of growth on the carbon substrate or under stress will outcompete the less adapted strains in the population and will eventually come to dominate the population.

This same process can be extended to any essential metabolite by creating an auxotroph. An auxotroph is a strain incapable of synthesizing an essential metabolite and must therefore have the metabolite provided in the media to grow. In this scenario, by making an auxotroph and passaging it on decreasing amounts of the metabolite, the resulting dominant strains should be more capable of obtaining and incorporating this essential metabolite.

For example, if the biosynthetic pathway for producing an amino acid is disrupted a strain capable of high-affinity

capture of said amino acid can be evolved via ALE. First, the strain is grown in varying concentrations of the auxotrophic amino acid, until a minimum concentration to support growth is established. The strain is then passaged at that concentration, and diluted into lowering concentrations of the amino acid at regular intervals. Over time, cells that are most competitive for the amino acid—at growth-limiting concentrations—will come to dominate the population. These strains will likely have mutations in their amino acid-transporters resulting in increased ability to import the essential and limiting amino acid.

Similarly, by using an auxotroph that cannot use an upstream metabolite to form an amino acid, a strain can be evolved that not only can more efficiently import the upstream metabolite, but also convert the metabolite into the essential downstream metabolite. These strains will also evolve mutations to increase import of the upstream metabolite, but may also contain mutations which increase expression or reaction kinetics of downstream enzymes, or that reduce competitive substrate utilization pathways.

In the previous examples, a metabolite innate to the microbe was made essential via mutational auxotrophy and selection was applied with growth-limiting supplementation of the endogenous metabolite. However, phenotypes capable of consuming non-native compounds can be evolved by tying their consumption to the production of an essential compound. For example, if a gene from a different organism is isolated which can produce an essential compound or a precursor to an essential compound this gene can be recombinantly introduced and expressed in the heterologous host. This new host strain will now have the ability to synthesize an essential nutrient from a previously non-metabolizable substrate. Hereby, a similar ALE process can be applied by creating an auxotroph incapable of converting an immediately downstream metabolite and selecting in growth-limiting amounts of the non-native compound with concurrent expression of the recombinant enzyme. This will result in mutations in the transport of the non-native substrate, expression and activity of the heterologous enzyme and expression and activity of downstream native enzymes. It should be emphasized that the key requirement in this process is the ability to tether the consumption of the non-native metabolite to the production of a metabolite essential to growth.

Once the basis of the selection mechanism is established and minimum levels of supplementation have been established, the actual ALE experimentation can proceed. Throughout this process several parameters must be vigilantly monitored. It is important that the cultures are maintained in an exponential growth phase and not allowed to reach saturation/stationary phase. This means that growth rates must be checked during each passaging and subsequent dilutions adjusted accordingly. If growth rate improves to such a degree that dilutions become large, then the concentration of auxotrophic supplementation should be decreased such that growth rate is slowed, selection pressure is increased and dilutions are not so severe as to heavily bias subpopulations during passaging. In addition, at regular intervals cells should be diluted, grown on solid media and individual clones tested to confirm growth rate phenotypes observed in the ALE cultures.

Predicting when to halt the stop the ALE experiment also requires vigilance. As the success of directing evolution is tied directly to the number of mutations “screened” throughout the experiment and mutations are generally a function of errors during DNA replication, the cumulative cell divisions (CCD) acts as a proxy for total mutants which have been

screened. Previous studies have shown that beneficial phenotypes for growth on different carbon sources can be isolated in about 10^{11-2} CCD¹. This rate can be accelerated by the addition of chemical mutagens to the cultures—such as N-methyl-N-nitro-N-nitrosoguanidine (NTG)—which causes increased DNA replication errors. However, when continued passaging leads to marginal or no improvement in growth rate the population has converged to some fitness maximum and the ALE experiment can be halted.

At the conclusion of the ALE experiment, the cells should be diluted, isolated on solid media and assayed for growth phenotypes matching that of the culture flask. Best performers from those selected are then prepped for genomic DNA and sent for whole genome sequencing. Sequencing with reveal mutations occurring around the genome capable of providing improved phenotypes, but will also contain silent mutations (those which provide no benefit but do not detract from desired phenotype). In cultures evolved in the presence of NTG or other chemical mutagen, there will be significantly more silent, background mutations. If satisfied with the best performing strain in its current state, the user can proceed to application with that strain. Otherwise the contributing mutations can be deconvoluted from the evolved strain by reintroducing the mutations to the parent strain by genome engineering techniques. See Lee, D.-H., Feist, A. M., Barrett, C. L. & Palsson, B. Ø. Cumulative Number of Cell Divisions as a Meaningful Timescale for Adaptive Laboratory Evolution of *Escherichia coli*. *PLoS ONE* 6, e26172 (2011).

In some embodiments, the ALE method can be used to identify genetically engineered bacteria with improved phenylalanine uptake.

Specific Screen to Improve PME Activity

Screens using genetic selection are conducted to improve phenylalanine consumption in the genetically engineered bacteria. Toxic phenylalanine analogs exert their mechanism of action (MOA) by being incorporated into cellular protein, causing cell death. These compounds, such as paralog p-fluoro-DL-phenylalanine and ortholog o-fluoro-DL-phenylalanine have utility in an untargeted approach to select PAL enzymes with increased activity. Assuming that these toxic compounds can be metabolized by PAL into a non-toxic metabolite, rather than being incorporated into cellular protein, genetically engineered bacteria which have improved phenylalanine degradation activity can tolerate higher levels of these compounds, and can be screened for and selected on this basis.

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EXAMPLES

The following examples provide illustrative embodiments of the disclosure. One of ordinary skill in the art will recognize the numerous modifications and variations that may be performed without altering the spirit or scope of the disclosure. Such modifications and variations are encompassed within the scope of the disclosure. The Examples do not in any way limit the disclosure.

Example 1. Construction of PAL Plasmids

To facilitate inducible production of PAL in *Escherichia coli* Nissle, the PAL gene of *Anabaena variabilis* ("PAL1") or *Photobacterium luminescens* ("PAL3"), as well as transcriptional and translational elements, were synthesized (Gen9, Cambridge, MA) and cloned into vector pBR322. The PAL gene was placed under the control of an inducible promoter. Low-copy and high-copy plasmids were generated for each of PAL1 and PAL3 under the control of an inducible FNR promoter or a Tet promoter. Exemplary FNR promoters are shown in Table 3. Organization and nucleotide sequences of these constructs are shown in FIGS. 6-9. However, as noted above, other promoters may be used to drive expression of the PAL gene, other PAL genes may be used, and other phenylalanine metabolism-regulating genes may be used.

Example 2. Transforming *E. coli*

Each of the plasmids described herein was transformed into *E. coli* Nissle for the studies described herein according to the following steps. All tubes, solutions, and cuvettes were pre-chilled to 4° C. An overnight culture of *E. coli* Nissle was diluted 1:100 in 5 mL of lysogeny broth (LB) containing ampicillin and grown until it reached an OD₆₀₀ of 0.4-0.6. The *E. coli* cells were then centrifuged at 2,000 rpm for 5 min at 4° C., the supernatant was removed, and the cells were resuspended in 1 mL of 4° C. water. The *E. coli* were again centrifuged at 2,000 rpm for 5 min at 4° C., the supernatant was removed, and the cells were resuspended in 0.5 mL of 4° C. water. The *E. coli* were again centrifuged at 2,000 rpm for 5 min at 4° C., the supernatant was removed, and the cells were finally resuspended in 0.1 mL of 4° C. water. The electroporator was set to 2.5 kV. Plasmid (0.5 µg) was added to the cells, mixed by pipetting, and pipetted into a sterile, chilled cuvette. The dry cuvette was placed into the sample chamber, and the electric pulse was applied. One mL of room-temperature SOC media was added immediately, and the mixture was transferred to a culture tube and

incubated at 37° C. for 1 hr. The cells were spread out on an LB plate containing ampicillin and incubated overnight.

Example 3. Comparison of Phenylalanine Metabolism Between High-Copy and Low Copy Plasmids Expressing PAL1 and PAL2

Genetically engineered bacteria comprising the same PAL gene, either PAL3 on a low-copy plasmid or high copy plasmid (SYN-PKU101 and SYN-PKU102) or PAL3 on a low-copy plasmid or a high copy plasmid (SYN-PKU201 and SYN-PKU202) were assayed for phenylalanine metabolism in vitro.

Engineered bacteria were induced with anhydrous tetracycline (ATC), and then grown in culture medium supplemented with 4 mM (660,000 ng/mL) of phenylalanine for 2 hours. Samples were removed at 0 hrs, 4 hrs, and 23 hrs, and phenylalanine (FIG. 15A) and trans-cinnamic acid (TCA) (FIG. 15B) concentrations were determined by mass spectrometry as described in Examples 24-26.

High copy plasmids and low copy plasmid strains were found to metabolize and reduce phenylalanine to similar levels (FIG. 15). A greater reduction in phenylalanine levels and increase in TCA levels was observed in the strains expressing PAL3.

Example 4. Phenylalanine Transporter—Integration of PheP into the Bacterial Chromosome

In some embodiments, it may be advantageous to increase phenylalanine transport into the cell, thereby enhancing phenylalanine metabolism. Therefore, a second copy of the native high affinity phenylalanine transporter, PheP, driven by an inducible promoter, was inserted into the Nissle genome through homologous recombination. Organization of the construct is shown in FIG. 11. The pheP gene was placed downstream of the P_{tet} promoter, and the tetracycline repressor, TetR, was divergently transcribed (see, e.g., FIG. 11). This sequence was synthesized by Genewiz (Cambridge, MA). To create a vector capable of integrating the synthesized TetR-PheP construct into the chromosome, Gibson assembly was first used to add 1000 bp sequences of DNA homologous to the Nissle lacZ locus into the R6K origin plasmid pKD3. This targets DNA cloned between these homology arms to be integrated into the lacZ locus in the Nissle genome (FIG. 10). Gibson assembly was used to clone the TetR-PheP fragment between these arms. PCR was used to amplify the region from this plasmid containing the entire sequence of the homology arms, as well as the pheP sequence between them. This PCR fragment was used to transform electrocompetent Nissle-pKD46, a strain that contains a temperature-sensitive plasmid encoding the lambda red recombinase genes. After transformation, cells were grown for 2 hrs before plating on chloramphenicol at 20 µg/mL at 37° C. Growth at 37° C. cures the pKD46 plasmid. Transformants containing anhydrous tetracycline (ATC)-inducible pheP were lac-minus (lac-) and chloramphenicol resistant.

Example 5. Effect of the Phenylalanine Transporter on Phenylalanine Degradation

To determine the effect of the phenylalanine transporter on phenylalanine degradation, phenylalanine degradation and trans-cinnamate accumulation achieved by genetically engineered bacteria expressing PAL1 or PAL3 on low-copy (LC) or high-

copy (HC) plasmids in the presence or absence of a copy of pheP driven by the Tet promoter integrated into the chromosome was assessed.

For in vitro studies, all incubations were performed at 37°

C. Cultures of *E. coli* Nissle transformed with a plasmid comprising the PAL gene driven by the Tet promoter were grown overnight and then diluted 1:100 in LB. The cells were grown with shaking (200 rpm) to early log phase. Anhydrous tetracycline (ATC) was added to cultures at a concentration of 100 ng/mL to induce expression of PAL, and bacteria were grown for another 2 hrs. Bacteria were then pelleted, washed, and resuspended in minimal media, and supplemented with 4 mM phenylalanine. Aliquots were removed at 0 hrs, 2 hrs, and 4 hrs for phenylalanine quantification (FIG. 16A), and at 2 hrs and 4 hrs for cinnamate quantification (FIG. 16B), by mass spectrometry, as described in Examples 24-26. As shown in FIG. 16, expression of pheP in conjunction with PAL significantly enhances the degradation of phenylalanine as compared to PAL alone or pheP alone. Notably, the additional copy of pheP permitted the complete degradation of phenylalanine (4 mM) in 4 hrs (FIG. 16A). FIG. 16B depicts cinnamate levels in samples at 2 hrs and 4 hrs post-induction. Since cinnamate production is directly correlated with phenylalanine degradation, these data suggest that phenylalanine disappearance is due to phenylalanine catabolism, and that cinnamate may be used as an alternative biomarker for strain activity. PheP overexpression improves phenylalanine metabolism in engineered bacteria.

In conclusion, in conjunction with pheP, even low-copy PAL-expressing plasmids are capable of almost completely eliminating phenylalanine from a test sample (FIGS. 16A and 16B). Furthermore, without wishing to be bound by theory, in some embodiments, that incorporate pheP, there may be additional advantages to using a low-copy PAL-expressing plasmid in conjunction in order to enhance the stability of PAL expression while maintaining high phenylalanine metabolism, and to reduce negative selection pressure on the transformed bacterium. In alternate embodiments, the phenylalanine transporter is used in conjunction with a high-copy PAL-expressing plasmid.

Example 6. FNR Promoter Activity

In order to measure the promoter activity of different FNR promoters, the lacZ gene, as well as transcriptional and translational elements, were synthesized (Gen9, Cambridge, MA) and cloned into vector pBR322. The lacZ gene was placed under the control of any of the exemplary FNR promoter sequences disclosed in Table 3. The nucleotide sequences of these constructs are shown in Tables 21-28 (SEQ ID NOs 31-38). However, as noted above, the lacZ gene may be driven by other inducible promoters in order to analyze activities of those promoters, and other genes may be used in place of the lacZ gene as a readout for promoter activity. Alternatively, beta-galactosidase may be used as a reporter, exemplary results are shown in FIG. 18.

Table 21 shows the nucleotide sequence of an exemplary construct comprising a gene encoding lacZ, and an exemplary FNR promoter, P_{fnr1} (SEQ ID NO: 3). The construct comprises a translational fusion of the Nissle nirB1 gene and the lacZ gene, in which the translational fusions are fused in frame to the 8th codon of the lacZ coding region. The P_{fnr1} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The lacZ sequence is underlined upper case. ATG site is bolded upper

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case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 22 shows the nucleotide sequence of an exemplary construct comprising a gene encoding lacZ, and an exemplary FNR promoter, P_{fmr2} (SEQ ID NO: 6). The construct comprises a translational fusion of the Nissle ydfZ gene and the lacZ gene, in which the translational fusions are fused in frame to the 8th codon of the lacZ coding region. The P_{fmr2} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The lacZ sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 23 shows the nucleotide sequence of an exemplary construct comprising a gene encoding lacZ, and an exemplary FNR promoter, P_{fmr3} (SEQ ID NO: 7). The construct comprises a transcriptional fusion of the Nissle nirB gene and the lacZ gene, in which the transcriptional fusions use only the promoter region fused to a strong ribosomal binding site. The P_{fmr3} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The lacZ sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 24 shows the nucleotide sequence of an exemplary construct comprising a gene encoding lacZ, and an exemplary FNR promoter, P_{fmr4} (SEQ ID NO: 8). The construct comprises a transcriptional fusion of the Nissle ydfZ gene and the lacZ gene. The P_{fmr4} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The lacZ sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 25 shows the nucleotide sequence of an exemplary construct comprising a gene encoding lacZ, and an exemplary FNR promoter, P_{fmrS} (SEQ ID NO: 9). The construct comprises a transcriptional fusion of the anaerobically induced small RNA gene, fnrS1, fused to lacZ. The P_{fmrS} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The lacZ sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 26 shows the nucleotide sequence of an exemplary construct comprising a gene encoding PAL3, and an exemplary FNR promoter, P_{fmr3} (SEQ ID NO: 7). The construct comprises a transcriptional fusion of the Nissle nirB gene and the PAL3 gene, in which the transcriptional fusions use only the promoter region fused to a strong ribosomal binding site. The P_{fmr3} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The PAL3 sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 27 shows the nucleotide sequences of an exemplary construct comprising a gene encoding PAL3, and an exemplary FNR promoter, P_{fmr4} (SEQ ID NO: 8). The construct comprises a transcriptional fusion of the Nissle ydfZ gene and the PAL3 gene. The P_{fmr4} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The PAL3 sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

Table 28 shows the nucleotide sequences of an exemplary construct comprising a gene encoding PAL3, and an exemplary FNR promoter, P_{fmrS} (SEQ ID NO: 9). The construct comprises a transcriptional fusion of the anaerobically

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induced small RNA gene, fnrS1, fused to PAL3. The P_{fmrS} sequence is bolded lower case, and the predicted ribosome binding site within the promoter is underlined. The PAL3 sequence is underlined upper case. ATG site is bolded upper case, and the cloning sites used to synthesize the construct are shown in regular upper case.

TABLE 21

Nucleotide sequences of Pfnr1-lacZ construct, low-copy (SEQ ID NO: 31)

GGTACC**gtcagc**ataaacaccctgacctctcattaatgttcatgcccgg
 ggggcactatcgctcgccgcttttctctcttactctgtctacgtac
 atctattttctataaatccggttcaatttctgtctgttttttgcacaaacat
 gaaatatcagacaattccgtgacttaagaaattatatacaaatcagca
 atatacccccctaaggagtataataaaggatgaatttgatttacatcaata
 agcgggggttgcgtgaatcgttaaggtaggcggtatagaaaagaaatcg
 agggcaaa**ATG**agcaagtcagactcgcaattatGGATCTCTGGCCG
 TCGTATTACAACGTCGTGACTGGGAAACCCCTGGCGTTACCCAACTTA
 ATCGCCTTGCAGCACATCCCCCTTTCGCCAGCTGGCGTAATAGCGAAG
 AGGCCCGCACCGATCGCCCTTCCCAACAGTTGCGCAGCCTGAATGGCG
 AATGGCGCTTTGCCTGGTTTCCGGCACCAGAACGGTGCCGGAAGCT
 GGCTGGAGTGCAGTCTTCTGACGCCGATACTGTCTGTCTCCCTCAA
 ACTGCGAGATGCACGGTTACGATGCGCTATCTACACCAACGTGACCT
 ATCCCATACGGTCAATCCGCGTTTGTTCGCCGAGAAATCCGACAG
 GTTGTACTCGCTCACATTTAATATTGATGAAGCTGGCTACAGGAAG
 GCCAGACGCAATTATTTTGTATGGCGTTAACTCGCGTTTCATCTGT
 GGTGCAACGGCGCTGGGTGGTTACGCCAGGACAGCCGTTTGCCGT
 CTGAATTTGACCTGAGCGCATTTTACGCGCCGAGAAACCGCCTCG
 CGGTGATGGTGTCTGCGTGGAGTGACGGCAGTTATCTGGAAGATCAGG
 ATATGTGGCGATGAGCGCATTTTCCGTGACGTCTCGTTGCTGCATA
 AACCACACCGCAATCAGCGATTCCAAGTTACCACTCTCTTTAATG
 ATGATTTAGCCGCGCGTACTGGAGGAGAGAGTTACAGATGACGGCG
 AGCTGCGCGATGAATGCGGGTGACGGTTTCTTTGTGGCAGGGTGAAA
 CGCAGGTGCGCAGCGGCACCGCGCTTTCGGCGGTGAAATTATCGATG
 AGCGTGGCGGTTATGCCGATCGCGTCACACTACGCCTGAACGTTGAAA
 ATCCGGAACGTGGAGCGCCGAAATCCGAATCTCTATCGTGCAGTGG
 TTGAAGTGCACACCGCCGACGGCAGCGTGATTGAAGCAGAAGCCTGCG
 ACGTCGGTTTCCGCGAGGTGCGGATTGAAATGGTCTGCTGCTGTA
 ACGGCAAGCCGTTGCTGATTGCGGGGTTAACCGTCACGAGCATCATC
 CTCTGCATGGTCAGGTGATGGATGAGCAGCATGGTGAGGATATCC
 TGCTGATGAAGCAGAACTTTAACGCCGTGCGCTGTTTCGATTATC
 CGAACCATCCGCTGTGGTACACGCTGTGCGACCGCTACGGCCTGTATG
 TGGTGGATGAAGCCAATATTGAAACCCACGGCATGGTGCCAATGAATC
 GTCTGACCGATGATCCGCGTGGCTACCCGCGATGAGCGAACCGGTAA

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TABLE 21-continued

Nucleotide sequences of Pfnr1-lacZ construct, low-copy (SEQ ID NO: 31)	
<u>CGCGGATGGTGCAGCGGATCGTAATCACCCGAGTGTGATCATCTGGT</u>	
<u>CGCTGGGGAATGAATCAGGCCACGGCGCTAATCACGACGCGCTGTATC</u>	
<u>GCTGGATCAAATCTGTCGATCCTTCCCGCCGGTACAGTATGAAGGCG</u>	
<u>GCGGAGCCGACACCACGGCCACCGATATTATTGGCCGATGTACGCGC</u>	
<u>GCGTGGATGAAGACAGCCCTTCCCGCGGTGCCGAAATGGTCCATCA</u>	
<u>AAAAATGGCTTTTCGTCGCTGGAGAAATGCGCCCGTGTATCCTTTGCG</u>	
<u>AATATGCCACGCGATGGGTAACAGTCTTGGCGGCTTCGCTAAATACT</u>	
<u>GGCAGGCGTTTCGTAGTACCCCGTTTACAGGGCGGCTTCGTCTGGG</u>	
<u>ACTGGGTGGATCAGTCGTGATTAAATATGATGAAACGGCAACCCGT</u>	
<u>GGTCGGCTTACGGCGGTGATTTTGGCGATACGCCGAACGATCGCCAGT</u>	
<u>TCTGTATGAACGGTCTGGTCTTTGCGGACCGCACGCCGATCCGGCGC</u>	
<u>TGACGGAAGCAAAACACCAACAGCAGTATTTCAGTTCGGTTTATCCG</u>	
<u>GGCGAACCATCGAAGTGACGAGCAATACCTGTTCCGTATAGCGATA</u>	
<u>ACGAGTTCCTGCATGGATGGTGGCACTGGATGGCAAGCCGCTGGCAA</u>	
<u>GCGGTGAAGTGCTCTGGATGTTGGCCCGCAAGGTAAGCAGTTGATTG</u>	
<u>AACTGCCTGAAGTGCCTGAGCGCGGAGAGCGCCGGACAACCTCTGGCTAA</u>	
<u>CGGTACGCGTAGTGCAACCAACGCGACCGCATGGTCAGAAGCCGGAC</u>	
<u>ACATCAGCGCCTGGCAGCAATGGCGTCTGGCGGAAACCTCAGCGTGA</u>	
<u>CACCTCCCTCCGCGTCCCACGCCATCCCTCAACTGACCACGAGCGGAA</u>	
<u>CGGATTTTTCATCAGAGTGGGTAATAAGCGTTGGAATTTAAACGCC</u>	
<u>AGTCAGGCTTTCTTTACAGATGTGGATTGGCGATGAAAAACAACCTGC</u>	
<u>TGACCCCGCTGCGCGATCAGTTCACCCGTCGCCGCTGGATAACGACA</u>	
<u>TTGGCGTAAGTGAAGCGACCCGCATTGACCCTAACGCCCTGGGTGGAAC</u>	
<u>GCTGGAAGCGCGCGGCCATTACAGGCGGAAGCGCGTGTGTGAGT</u>	
<u>GCACGGCAGATACACTTGCCGACGCGGTGCTGATTACAACGCCCCACG</u>	
<u>CGTGCGAGCATCAGGGGAAACCTTATTATCAGCCGAAACCTACC</u>	
<u>GGATTGATGGGCACGGTGAGATGGTCATCAATGTGGATGTTGCGGTGG</u>	
<u>CAAGCGATACACCGCATCCGGCGGGATTGGCCTGACCTGCCAGCTGG</u>	
<u>CGCAGGTCTCAGAGCGGTAACCTGGCTCGGCCTGGGCGCGAAGAAA</u>	
<u>ACTATCCCGACCGCCTTACTGCAGCCTGTTTGGACCGCTGGGATCTGC</u>	
<u>CATTGTGACAGATGTATACCCCGTACGTCTTCCCGAGCGAAACGGTC</u>	
<u>TGCGCTGCGGGACGCGCAATTGAATTATGGCCACACAGTGGCGCG</u>	
<u>GCGACTTCCAGTTCAACATCAGCCGCTACAGCCAACAACAACCTGATGG</u>	
<u>AAACGAGCATCGCCATCTGCTGCACGCGGAAGAAGGCACATGGCTGA</u>	
<u>ATATCGACGGTTTCCATATGGGGATTGGTGGCGACGACTCCTGGAGCC</u>	
<u>CGTCAGTATCGCGGAATTCCAGCTGAGCGCGGTGCTACCATTACC</u>	
<u>AGTTGGTCTGGTGTCAAAAATAA</u>	

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TABLE 22

Nucleotide sequences of Pfnr2-lacZ construct, low-copy (SEQ ID NO: 32)	
<u>GGTACCcatttctctcatcccatccgggtgagagtctttttcccccg</u>	5
<u>aottatggctcatgcatgcataaaaaagatgtgagcttgatcaaaaa</u>	
<u>caaaaaatatttctactcgacaggagtatttatattgcccgggttaagt</u>	
<u>gggcttcgactgtaaatcagaaggagaaaaacacctATGacgacctac</u>	10
<u>gatcgGGATCCTCTGGCCGTCGTATTACAACGTCGTGACTGGGAAAAAC</u>	
<u>CCTGGCGTTACCCAACTTAATCGCCTTGGCGCACATCCCCCTTTTCGCC</u>	
<u>AGCTGGCGTAATAGCGAAGAGGCCCGCACCGATCGCCCTTCCCAACAG</u>	15
<u>TTGCGCAGCCTGAATGGCGAATGGCGCTTTGCCTGGTTTCCGGCACCA</u>	
<u>GAAGCGGTGCCGGAAGCTGGCTGGAGTGCATCTTCTGACGCCGAT</u>	
<u>ACTGTCGTCTGCCCTCAAACCTGGCAGATGCACGGTTACGATGCGCCT</u>	20
<u>ATCTACACCAACGTGACCTATCCCATTTACGGTCAATCCGCCGTTTGT</u>	
<u>CCCGCGGAGATCCGACAGGTTGTACTCGCTCACATTTAATATTGAT</u>	
<u>GAAAGCTGGCTACAGGAAGGCCAGACGCAATTATTTTGTATGGCGTT</u>	25
<u>AACTCGGCGTTTCATCTGTGGTGCAACGGCGCTGGGTGCGTTACGGC</u>	
<u>CAGGACAGCCGTTTGCCGCTGAATTTGACCTGAGCGCATTTTACGC</u>	
<u>GCCGGAGAAAACCGCCTCGCGGTGATGGTGTCTGCGCTGGAGTGACGGC</u>	30
<u>AGTTATCTGGAAGATCAGGATATGTGGCGGATGAGCGGCATTTTCCGT</u>	
<u>GACGTCTCGTTGCTGCATAAACCGACCACGCAAAATCAGCGATTTCCAA</u>	
<u>GTTACCACTCTCTTAAATGATGATTTAGCCGCGCGGTACTGGAGGCA</u>	35
<u>GAAGTTCAGATGTACGGCGAGCTGCGCGATGAAGTGGGGTGACGGTT</u>	
<u>TCTTTGTGGCAGGGTGAAACGCAGGTGCGCAGCGCACCGCGCCTTTC</u>	
<u>GGCGGTGAAATTATCGATGAGCGTGGCGGTTATGCCGATCGCGTCACA</u>	40
<u>CTACGCCTGAACGTTGAAAAATCCGGAACGTGGAGCGCCGAAATCCCG</u>	
<u>AATCTCTATCGTGCAGTGGTTGAAGTGCACACCGCCGACGGCACGCTG</u>	
<u>ATTGAAGCAGAAGCCTGCGACGTGGTTTCCGCGAGGTGCGGATTGAA</u>	45
<u>AATGGTCTGCTGCTGCTGAACGGCAAGCCGTTGCTGATTGCGCGCGTT</u>	
<u>AACCGTCACGAGCATCATCTCTGCATGGTCAGGTCAATGATGAGCAG</u>	
<u>ACGATGGTGCAGGATATCCTGCTGATGAAGCAGAACAACCTTTAACGCC</u>	
<u>GTGCGCTGTTTCGATTATCCGAACCATCCGCTGTGTACACGCTGTGC</u>	50
<u>GACCGCTACGGCCTGTATGTGGTGGATGAAGCCAATATTGAAACCCAC</u>	
<u>GGCATGGTGCCAATGAATCGTCTGACCGATGATCCGCGCTGGCTACCC</u>	
<u>GCGATGAGCGAACGCGTAACGCGGATGGTGCAGCGCGATCGTAATCAC</u>	55
<u>CCGAGTGTGATCATCTGGTCGCTGGGGAATGAATCAGGCCACGGCGCT</u>	
<u>AATCACGACGCGCTGTATCGCTGGATCAAATCTGTCGATCCTTCCCGC</u>	
<u>CCGGTACAGTATGAAGGCGGCGGAGCCGACACCACGGCCACCGATATT</u>	60
<u>ATTTGCCGATGTACGCGCGGTGGATGAAGACCAGCCCTTCCGGCGG</u>	
<u>GTGCCGAAATGGTCCATCAAAAAATGGCTTTTCGCTGCTGGAGAAATG</u>	
<u>CGCCCGCTGATCCTTTGCGAATATGCCACGCGATGGGTAAACAGTCTT</u>	65
<u>GGCGGCTTCGCTAAATACTGGCAGGCGTTTCGTAGTACCCCGCTTTA</u>	

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TABLE 22-continued

Nucleotide sequences of Pfnr2-lacZ construct, low-copy (SEQ ID NO: 32)	
CAGGGCGGCTTCGCTCGGACTGGGTGGATCAGTCGCTGATTAAATAT	
GATGAAAACGGCAACCCGTGGTCGGCTTACGGCGGTGATTTGGCGAT	
ACGCCGAACGATCGCCAGTTCTGTATGAACGGTCTGGTCTTTGCCGAC	
CGCACGCCGATCCGGCGCTGACGGAAGCAAACACCAACAGCAGTAT	
TTCCAGTTCGGTTTATCCGGGCGAACCATCGAAGTGACCAGCGAATAC	
CTGTTCCGTCATAGCGATAACGAGTTCCTGCACTGGATGGTGGCACTG	
GATGGCAAGCCGCTGGCAAGCGGTGAAGTGCTCTGGATGTTGGCCCCG	
CAAGGTAAGCAGTTGATTGAACTGCCTGAACTGCGCAGCCGGAGAGC	
GCCGGACAACCTCTGGCTAACGCTACGCGTAGTGCAACCAACCGCGACC	
GCATGGTCAGAAGCCGGACACATCAGCGCTGGCAGCAATGGCGTCTG	
GCGGAAAACCTCAGCGTGACACTCCCTCCGCGTCCCACGCCATCCCT	
CAACTGACCACCAGCGGAACGATTTTTGTCATCGAGCTGGTAATAAG	
CGTTGGCAATTTAACCGCCAGTCAGGCTTTCTTTCACAGATGTGGATT	
GGCGATGAAAAACAACCTGCTGACCCCGCTGCGCGATCAGTTCACCCGT	
GCGCCGCTGGATAACGACATTGGCGTAAGTGAAGCGACCCGATTGAC	
CCTAACGCTCGGTGCAACGCTGGAAGCGCGCGGCCATTACCAGGCC	
GAAGCGCGTGTGTCAGTGACCGGAGATACACTTGCCGACGCGGTG	
CTGATTACAACCGCCACGCGTGGCAGCATCAGGGGAAAACCTTATTT	
ATCAGCCGAAAACCTACCGGATTGATGGGCAGGTGAGATGGTCATC	
AATGTGATGTTGCGGTGGCAAGCGATACCGCATCCGGCGCGGATT	
GGCCTGACCTGCCAGCTGGCGCAGGTCTCAGAGCGGGTAACTGGCTC	
GGCTTGGGGCCGCAAGAAAATATCCCGACCGCTTACTGACGCTGT	
TTTGACCGCTGGGATCTGCCATTGTGAGACATGTATACCCGTACGTC	
TTCCCGAGCGAAAACGGTCTGCGCTGCGGGACGCGCAATTGAATTAT	
GGCCACACCACTGGCGCGGCGACTTCCAGTTCAACATCAGCCGCTAC	
AGCCAAACAACCTGATGGAACAGCCATCGCCATCTGCTGCACGCG	
GAAGAAGGCATGGCTGAATATCGACGGTTTCCATATGGGGATTGGT	
GGCGACGACTCCTGGAGCCCGTCAGTATCGGCGGAATTCCAGCTGAGC	
GCCGGTCGCTACCATTACAGTTGGTCTGGTGTCAAAAATAA	

TABLE 23

Nucleotide sequences of Pfnr3-lacZ construct, low-copy (SEQ ID NO: 33)	
GGTACGctcagcataaacacccctgacctctcattaatgttcatgcgg	
gcggaactatcgctcgccggccttttctctcttactctgctacgtac	
atctatctctataaactcggttcaatttctgtgttttttgcacaaacat	
gaaatatcagacaatttcgtgacttaagaaaattatacaaatcagca	
atatacccttaaggagtataaaaggatgaatttgatttacatcaata	

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TABLE 23-continued

Nucleotide sequences of Pfnr3-lacZ construct, low-copy (SEQ ID NO: 33)	
5	<u>agcggggttgctgaatcggttaaggATCCctctagaaataattttgttt</u>
	<u>aactttaagaaggagatatacatATG</u> ACTATGATTACGGATTCTCTGG
	<u>CCGTCGTATTACAACGTCGTGACTGGGAAAACCTGGCGTTACCCAAC</u>
10	<u>TTAATCGCCTTGCGGCACATCCCCCTTTCGCCAGCTGGCGTAATAGCG</u>
	<u>AAGAGGCCCCGCACCGATCGCCCTTCCCAACAGTTGCGCAGCCTGAATG</u>
	<u>GCGAATGGCGCTTTGCCTGGTTTCCGGCACAGAAGCGGTGCCGAAA</u>
15	<u>GCTGGCTGGAGTGCATCTTCCTGACGCCGATACTGTCTGTCTCCCCCT</u>
	<u>CAAACCTGGCAGATGCACGGTTACGATGCGCCTATCTACACCAACGTGA</u>
	<u>CCTATCCCATTAACGGTCAATCCGCCGTTTGTTCGCCGAGAAATCCGA</u>
20	<u>CAGGTTGTTACTCGCTCACATTTAATATTGATGAAAGCTGGCTACAGG</u>
	<u>AAGGCCAGACGCGAATTATTTTGTATGGCGTTAACTCGCGTTTCATC</u>
	<u>TGTGGTGCAACGGGCGCTGGGTGCGTTACGGCCAGGACAGCGTTTGC</u>
25	<u>CGTCTGAATTTGACCTGAGCGCATTTTACGCGCCGGAGAAAACCGCC</u>
	<u>TCGCGGTGATGGTGTGCGCTGGAGTGACGGCAGTTATCTGGAAGATC</u>
	<u>AGGATATGTGGCGGATGAGCGGCATTTTCCGTGACGCTCTCGTTGCTGC</u>
	<u>ATAAACCGACCACGCAATCAGCGATTTCCAAGTTACCACTCTCTTTA</u>
30	<u>ATGATGATTTTCCGCGCGGTACTGGAGGCAGAAATTCAGATGTACG</u>
	<u>GCGAGCTGCGGATGAACTGCGGGTGACGGTTTCTTTGTGGCAGGGTG</u>
	<u>AAACGCAAGTCGCCAGCGGCACCGCGCTTTCGGCGGTGAAATTATCG</u>
35	<u>ATGAGCGTGGCGGTTATGCCGATCGCGTCACACTACGCCTGAACGTTG</u>
	<u>AAAATCCGGAACGTGGAGCGCGAAATCCCGAATCTCTATCGTGACG</u>
	<u>TGGTTGAACTGCACACCGCGACGGCACGCTGATTGAAGCAGAAGCCT</u>
40	<u>GCGACGTCGGTTTCCGCGAGGTGCGGATTGAAAATGGTCTGCTGCTGC</u>
	<u>TGAACGGCAAGCCGTTGCTGATTTCGCGCGTTAACCGTCACGAGCATC</u>
	<u>ATCCTCTGCATGGTCAGGTGATGGATGAGCAGACGATGGTGACGATA</u>
45	<u>TCCTGCTGATGAAGCAGAACAACTTTAACGCCGTGCGCTGTTTCGATT</u>
	<u>ATCCGAACCATCCGCTGTGGTACACGCTGTGCGACCGCTACGGCCTGT</u>
	<u>ATGTGGTGGATGAAGCCAATATTGAAACCCACGCATGGTGCCAATGA</u>
50	<u>ATCGTCTGACCGATGATCCGCGCTGGCTACCCGCGATGAGCGAACCGG</u>
	<u>TAACGCGGATGGTGCAGCGCGATCGTAATCACCCGAGTGTGATCATCT</u>
	<u>GGTCGCTGGGGAATGAATCAGGCCACGGCGCTAATCACGACGCGCTGT</u>
55	<u>ATCGCTGGATCAAATCTGTGATCCTTCCCGCCGGTACAGTATGAAG</u>
	<u>GCGCGGAGCCGACACCACGGCCACCGATATTATTTGCCCGATGTACG</u>
	<u>CGCGCGTGGATGAAGACCAGCCCTTCCCGCGGTGCCGAAATGGTCCA</u>
60	<u>TCAAAAAATGGCTTTCGCTGCTGGAGAAATGCGCCCGCTGATCCTTT</u>
	<u>GCGAATATGCCACGCGATGGGTAAAGTCTTGGCGGCTTCGCTAAAT</u>
	<u>ACTGCGAGGCGTTTCGTCAGTACCCCGTTTACAGGGCGGCTTCGTC</u>
	<u>GGGACTGGGTGGATCAGTCGCTGATTAAATATGATGAAAACGGCAACC</u>
65	<u>CGTGGTCGGCTTACGGCGGTGATTTTGGCGATACGCCGAACGATCGCC</u>

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TABLE 23-continued

Nucleotide sequences of Pfnr3-lacZ construct, low-copy (SEQ ID NO: 33)	
<u>AGTTCTGTATGAACGGTCTGGTCTTTGCCGACCGCACGCCGCATCCGG</u>	
<u>CGCTGACGGAAGCAAAACACCAACAGCAGTATTTCCAGTTCCGTTTAT</u>	
<u>CCGGGCGAACCATCGAAGTGACCAGCGAATACCTGTTCCGTCATAGCG</u>	
<u>ATAACGAGTTCCTGCACTGGATGGTGGCACTGGATGGCAAGCCGCTGG</u>	
<u>CAAGCGGTGAAGTGCTCTGGATGTTGGCCCGCAAGTAAGCAGTTGA</u>	
<u>TTGAACTGCCTGAAGTCCGCGACGCCGAGAGCGCCGGACAACCTCTGGC</u>	
<u>TAAACGCTACGCGTAGTGCAACCAACGCGACCGCATGGTCAGAAGCCG</u>	
<u>GACACATCAGCGCCTGGCAGCAATGGCGTCTGGCGGAAAACCTCAGCG</u>	
<u>TGACACTCCCCCTCCGCGTCCCACGCCATCCCTCAACTGACCACCAGCG</u>	
<u>GAACGGATTTTTCATCGAGCTGGGTAATAAGCGTTGGCAATTTAACC</u>	
<u>GCCAGTCAGGCTTTCTTTACAGATGTGGATTGGCGATGAAAAACAAC</u>	
<u>TGCTGACCCCGCTGCGCGATCAGTTACCCCGTGCGCCGCTGGATAACG</u>	
<u>ACATTGGCGTAAGTGAAGCGACCCCGCATGACCTTAACGCCTGGGTGCG</u>	
<u>AACGCTGGAAGGCGCGGCCATTACCAGGCCGAAGCGCGCTTGTGTC</u>	
<u>AGTGCAAGCGAGATACACTTGCCGACGCGGTGCTGATTACAACGCCCC</u>	
<u>ACGCGTGGCAGCATCAGGGGAAAACCTTATTTATCAGCGGAAAACCT</u>	
<u>ACCGGATTGATGGGCACGGTGAGATGGTCATCAATGTGGATGTTGCGG</u>	
<u>TGGCAAGCGATACACCGCATCCGCGCGGATTGGCCTGACCTGCCAGC</u>	
<u>TGGCGCAGGTCTCAGAGCGGGTAAACTGGCTCGGCCTGGGGCCGCAAG</u>	
<u>AAAACATATCCGACCGCCTTACTGCAGCCTGTTTTGACCGCTGGGATC</u>	
<u>TGCCATTGTGACAGATGTATACCCCGTACGTCTTCCGAGCGAAAAACG</u>	
<u>GTCTGCGCTGCGGGACGCGCAATTGAATTATGGCCACACCACTGGC</u>	
<u>GCGGCGACTTCCAGTTCAACATCAGCCGCTACAGCCAACAACAACATGA</u>	
<u>TGAAACACAGCCATCGCCATCTGCTGCACGCGGAAGAAGGCACATGGC</u>	
<u>TGAATATCGACGGTTTCCATATGGGGATTGGTGGCGACGACTCTGGGA</u>	
<u>GCCCGTCAGTATCGCGGAATTCCAGCTGAGCGCCGGTCGCTACCAAT</u>	
<u>ACCAGTTGGTCTGGTGTCAAAAATAA</u>	

TABLE 24

Nucleotide sequences of Pfnr4-lacZ construct, low-copy (SEQ ID NO: 34)	
<u>GGTACCCatttctctctccatcccatccgggtgagagtcttttcccccgac</u>	
<u>ttatggctcatgcatgcatcaaaaagatgtgagcttgatcaaaaacaaa</u>	
<u>aaatattttcactgcagaggagtatttatattgccccGGATCCctctaga</u>	
<u>aaataattttgtttaactttaagaaggagatatatacatATGACTATGATTAC</u>	
<u>GGATTCTCTGGCCGTCGTATTACAACGTCGTGACTGGGAAAACCTGGCG</u>	
<u>TTACCCAACTTAATCGCCTTGCGGCACATCCCCCTTTCGCCAGCTGGCGT</u>	
<u>AATAGCGAAGAGGCCCGCACCGATCGCCCTTCCCAACAGTTGCGCAGCCT</u>	

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TABLE 24-continued

Nucleotide sequences of Pfnr4-lacZ construct, low-copy (SEQ ID NO: 34)	
5	<u>GAATGGCGAATGGCGCTTTGCCTGGTTCCGGCACCAAGAAGCGGTGCCGG</u>
	<u>AAAGCTGGCTGGAGTGCGATCTTCTGACGCCGATACTGTCGTCGTCGCC</u>
	<u>TCAAACGGCAGATGCACGGTTACGATGCGCCTATCTACACCAACGTGAC</u>
10	<u>CTATCCCATACGGTCAATCCGCCGTTTGTTCGCCGCGAGAATCCGACAG</u>
	<u>GTTGTTACTCGCTCACATTTAATATTGATGAAAGCTGGCTACAGGAAGGC</u>
	<u>CAGACGCGAATTATTTTTGATGGCGTTAACTCGCGCTTTCATCTGTGGTG</u>
15	<u>CAACGGGCGCTGGGTGCGTTACGGCCAGGACAGCCGTTTGGCGTCTGAAT</u>
	<u>TTGACCTGAGCGCATTTTTACGCGCCGAGAAAAACGCCCTCGCGGTGATG</u>
	<u>GTGCTGCGCTGGAGTGACGGCAGTTATCTGGAAGATCAGGATATGTGGCG</u>
20	<u>GATGAGCGGCATTTTCCGTGACGTCTCGTTGCTGCATAAACCGACACGCG</u>
	<u>AAATCAGCGATTTCCAAGTTACCACTCTCTTTAATGATGATTTCAGCCCG</u>
	<u>GCGGTACTGGAGGCGAAGTTGAGATGTACGGCGAGCTGCGCGATGAACT</u>
	<u>GCGGGTGACGGTTTCTTTGTGGCAGGGTGAAACGCGAGTCCGACGCGCA</u>
25	<u>CCGCGCCTTTCCGCGGTGAAATTATCGATGAGCGTGGCGTTATGCCGAT</u>
	<u>CGCGTCACACTACGCCTGAACGTTGAAAATCCGGAACGTGGAGCGCCGA</u>
	<u>AATCCCCGAATCTCTATCGTGAGTGGTTGAACTGCACACCGCCGACGCGCA</u>
30	<u>CGCTGATTGAAGCAGAAGCCTGCGACGTCGGTTTCCGCGAGGTGCGGATT</u>
	<u>GAAAATGGTCTGCTGCTGCTGAACGGCAAGCCGTGCTGATTCCGCGCGT</u>
	<u>TAACCGTCAAGAGCATCATCTCTGATGGTCAGGTCATGGATGAGCAGA</u>
35	<u>CGATGGTGCAGGATATCTGCTGATGAAGCAGAACAACCTTTAACGCCGTG</u>
	<u>CGCTGTTTCGCTATTATCCGAACCATCCGCTGTGGTACACGCTGTGCGACCG</u>
	<u>CTACGGCCTGTATGTGGTGGATGAAGCCAATATTGAAACCCACGGCATGG</u>
40	<u>TGCCAATGAATCGTCTGACCGATGATCCGCGCTGGCTACCCGCGATGAGC</u>
	<u>GAACGCGTAACGCGGATGGTGCAGCGCGATCGTAATCACCCGAGTGTGAT</u>
	<u>CATCTGGTTCGCTGGGGAATGAATCAGGCCACGGCGCTAATCACGACGCGC</u>
45	<u>TGTATCGCTGGATCAAATCTGTCGATCCTTCCGCCCGGTACAGTATGAA</u>
	<u>GGCGGCGGAGCGACACACCGCCACCGATATTATTGCCCCGATGTACGC</u>
	<u>GCGCGTGGATGAAGACAGCCCTTCCCGCGGTGCCGAAATGGTCCATCA</u>
50	<u>AAAAATGGCTTTCGCTGCCTGGAGAAATGCGCCCGCTGATCCTTTGCGAA</u>
	<u>TATGCCACGCGATGGGTAAACAGTCTTGGCGGCTTCGCTAAATACTGGCA</u>
	<u>GGCGTTTCGTCAGTACCCCCGTTTACAGGGCGGCTTCGCTGGGACTGGG</u>
55	<u>TGGATCAGTCGCTGATTAAATATGATGAAAACGGCAACCCGTGGTCCGCT</u>
	<u>TACGGCGGTGATTTTGGCGATACGCCGAACGATCGCCAGTTCTGTATGAA</u>
	<u>CGGTCTGGTCTTTGCCGACGACCGCGCATCCGGCGCTGACGGAAGCAA</u>
	<u>AACACCAACAGCAGTATTTCCAGTTCGGTTTATCCGGGCGAACCATCGAA</u>
60	<u>GTGACCAGCGAATACCTGTTCCGTCATAGCGATAACGAGTTCCTGCACTG</u>
	<u>GATGGTGGCACTGGATGGCAAGCCGCTGGCAAGCGGTGAAGTGCTCTGG</u>
	<u>ATGTTGGCCCGCAAGGTAAGCAGTTGATTGAACTGCCTGAACTGCCGCGAG</u>
65	<u>CCGGAGAGCGCCGACAACTCTGGCTAACGGTACGCGTAGTGCAACCAAA</u>

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TABLE 24-continued

Nucleotide sequences of Pfnr4-lacZ construct, low-copy (SEQ ID NO: 34)	
<u>CGCGACCGCATGGTCAGAAAGCCGACACATCAGCGCCTGGCAGCAATGGC</u>	
<u>GTCTGGCGGAAAACCTCAGCGTGACACTCCCCCTCCGCGTCCCACGCCATC</u>	
<u>CCTCAACTGACCACCAGCGGAACGGATTTTTCATCGAGCTGGGTAATAA</u>	
<u>GCGTTGGCAATTTAACCGCCAGTCAGGCTTTCTTTTACAGATGTGGATTG</u>	
<u>GCGATGAAAAACAAGTGTGACCCCGCTGCGCGATCAGTTTACCCGTGCG</u>	
<u>CCGCTGGATAACGACATTGGCGTAAGTGAAGCGACCCGATTGACCCATA</u>	
<u>CGCTGGGTGCAAGCGTGAAGCGCGCGGCCATTACAGGCCGAAGCGG</u>	
<u>CGTTGTTGAGTGACGCGAGATACACTTGCCGACGCGGTGCTGATTACA</u>	
<u>ACCGCCACGCGTGGCAGCATCAGGGGAAAACCTTATTTATCAGCCGGAA</u>	
<u>AACCTACCGGATTGATGGGCACGGTGAGATGGTCATCAATGTGGATGTTG</u>	
<u>CGGTGGCAAGCAGATACCCGCATCCGGCGCGGATTGGCCTGACCTGCCAG</u>	
<u>CTGGCGCAGGTCTCAGAGCGGGTAAACTGGCTCGGCCTGGGCGCGCAAGA</u>	
<u>AAACTATCCCGACCGCCTTACTGACGCTGTTTTGACCGCTGGGATCTGC</u>	
<u>CATTGTGAGACATGTATACCCCGTACGTCTTCCGAGCGAAAACGGTCTG</u>	
<u>CGCTGCGGGACGCGCAATTGAATTATGGCCACACAGTGGCGCGGCGA</u>	
<u>CTTCCAGTTCAACATCAGCCGTACAGCCAACAACAAGTATGGAACCA</u>	
<u>GCCATCGCCATCTGCTGACGCGGAAGAAGGCACATGGCTGAATATCGAC</u>	
<u>GGTTTCCATATGGGATTGGTGGCAGGACTCCTGGAGCCCGTCAGTATC</u>	
<u>GGCGGAATTCCAGCTGAGCGCGGTGCTACCATACAGTTGGTCTGGT</u>	
<u>GTCAAAAATAA</u>	

TABLE 25

Nucleotide sequences of Pfnrs-lacZ construct, low-copy (SEQ ID NO: 35)	
<u>GGTACAggtgtgttcttattggtggtgttgetttatggttgatcgtagta</u>	
<u>aatggtgtgaacaaaagcaatttttccgctgtctgtatacaaaaacgoc</u>	
<u>gtaaagtttgagogaagtcaataaaactctctacccattcagggaatatc</u>	
<u>tctcttGGATCCctctagaataaattttgtttaaactttaagaaggagata</u>	
<u>tacatATGCTATGATTACGGATTCTCTGGCCGTCGTATTACAACGTCGTG</u>	
<u>ACTGGAAAAACCTGGCGTTACCCAACTTAATCGCCTTGCGGCACATCCC</u>	
<u>CCTTTCGCCAGCTGGCGTAATAGCGAAGAGGCCCGACCGATCGCCCTTC</u>	
<u>CCAACAGTTGCGCAGCCTGAATGGCGAATGGCGCTTTGCCTGGTTTCCGG</u>	
<u>CACGAGAAGCGGTGCCGAAAGCTGGCTGGAGTGGCATCTTCCTGACGCC</u>	
<u>GATACTGTCGTGCTCCCTCAAAGTGGCAGATGCACGGTTACGATGCGCC</u>	
<u>TATCTACACCAACGTGACCTATCCATTACGGTCAATCCGCGTTTGTTT</u>	
<u>CCGCGGAGAATCCGACAGGTGTTTACTCGCTCACATTTAATATTGATGAA</u>	
<u>AGCTGGCTACAGGAAGGCCAGACGCAATTATTTTGTGTCGTTAACTC</u>	
<u>GGCGTTTCATCTGTGGTGCAACGGGCGCTGGGTGCGTTACGGCCAGGACA</u>	

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TABLE 25-continued

Nucleotide sequences of Pfnrs-lacZ construct, low-copy (SEQ ID NO: 35)	
5	<u>GCCGTTTGCCGTCTGAATTGACCTGAGCGCATTTTACGCGCGGAGAA</u>
	<u>AACCGCCTCGCGGTGATGGTGTGCTGCGCTGGAGTGACGGCAGTTATCTGGA</u>
	<u>AGATCAGGATATGTGGCGGATGAGCGGCATTTCCGTGACGTCTCGTTGC</u>
10	<u>TGCATAAACCGACCAACGCAATCAGCGATTTCCAAGTTACCACTCTCTTT</u>
	<u>AATGATGATTTTACGCGCGCGGTACTGGAGGCAGAAGTTAGATGTACGG</u>
	<u>CGAGCTGCGCGATGAACGCGGGTGACGGTTTCTTTGTGCGAGGGTGAA</u>
15	<u>CGCAGGTGCGCAGCGGCACCGCGCTTTCCGCGGTGAAATTATCGATGAG</u>
	<u>CGTGGCGGTTATGCCGATCGCGTCACACTACGCGCTGAACGTTGAAAAATCC</u>
	<u>GGAACTGTGGAGCGCGAAATCCGAAATCTCTATCGTCAGTGGTTGAAC</u>
20	<u>TGCACACCGCCGACGGCACGCTGATTGAAGCAGAAGCTGCGACGTCGGT</u>
	<u>TTCCGCGAGGTGCGGATTGAAAAATGGTCTGCTGCTGCTGAACGGCAAGCC</u>
	<u>GTTGCTGATTGCGCGCGTTAACCGTCACGAGCATCATCTCTGCATGGTC</u>
	<u>AGGTGATGGATGAGCAGACGATGGTGCAGGATATCCTGCTGATGAAGCAG</u>
25	<u>AACAACCTTTAACCGCGTGCCTGTTTCGCATTATCCGAACCATCCGCTGTG</u>
	<u>GTACACGCTGTGCGACCGCTACGGCTGTATGTGGTGGATGAAGCCAATA</u>
	<u>TTGAAACCCACGGCATGGTGCCAATGAATCGTCTGACCGATGATCCGCGC</u>
30	<u>TGGCTACCCCGCATGAGCGAACGCGTAACGCGGATGGTGCAGCGCGATCG</u>
	<u>TAATACCCCGAGTGTGATCATCTGGTCTGGGAATGAATCAGGCCACG</u>
	<u>GCGCTAATCAGCAGCGCTGTATCGCTGGATCAAATCTGTGATCCTTCC</u>
35	<u>CGCCCGGTACAGTATGAAGGCGGCGGAGCCGACACCGGCCACCGATAT</u>
	<u>TATTTGCCCGATGTACGCGCGGTGGATGAAGACCAGCCCTTCCCGCGCG</u>
	<u>TGCCGAAATGGTCCATCAAAAAATGGCTTTGCTGCCTGGAGAAATGCGC</u>
40	<u>CCGCTGATCCTTTGCGAATATGCCACGCGATGGGTAAAGTCTTGGCGG</u>
	<u>CTTCGCTAAATACTGGCAGGCGTTTCGTGATACCCCGTTTACAGGGCG</u>
	<u>GCTTCGTCTGGGACTGGGTGGATCAGTCGCTGATTAAATATGATGAAAA</u>
45	<u>GGCAACCCGTGGTGGCTTACGGCGGTGATTTGGCGATACGCCGAACGA</u>
	<u>TCGCCAGTCTGTATGAACGGTCTGGTCTTTGCCGACCGCACGCGCATC</u>
	<u>CGGCGCTGACGGAAGCAAAACACCAACAGCAGTATTTCAGTTCCGTTTA</u>
50	<u>TCCGGGCGAACCATCGAAGTGACCGCAATACTGTTCCGTCATAGCGA</u>
	<u>TAACGAGTTCCTGCACTGGATGGTGGCACTGGATGGCAAGCCGCTGGCAA</u>
	<u>GCGGTGAAGTGCCCTGGATGTTGGCCGCAAGGTAAGCAGTTGATTGAA</u>
55	<u>CTGCCTGAACTGCGCAGCCGAGAGCGCCGGACAACCTCTGGCTAACGGT</u>
	<u>ACGCGTAGTGCAACCAACGCGACCGCATGGTCAGAAGCCGGACACATCA</u>
	<u>GCGCTGGCAGCAATGGCGTCTGGCGGAAAACCTCAGCGTGACACTCCCC</u>
	<u>TCCGCGTCCCACGCCATCCCTCAACTGACCACAGCGGAACGATTTTTG</u>
60	<u>CATCGAGCTGGGTAAAGCGTTGGCAATTTAACCGCCAGTCAGGCTTTC</u>
	<u>TTTCACAGATGTGGATTGGCGATGAAAAACAAGTCTGACCCCGCTGCGC</u>
	<u>GATCAGTTTACCCGTGCGCGCTGGATAACGACATTGGCGTAAGTGAAGC</u>
65	<u>GACCCGATTGACCTAACGCTGGGTGCAACGCTGGAAGGCGCGGGCC</u>

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TABLE 25-continued

Nucleotide sequences of Pfnrs-lacZ construct, low-copy (SEQ ID NO: 35)
ATTACCAGGCCGAAGCGCGTGTGTCAGTGCACGGCAGATACACTTGCC
GACGCGGTGCTGATTACAACCGCCACGCGTGGCAGCATCAGGGGAAAAAC
CTTATTTATCAGCCGAAAAACCTACCGGATTGATGGGCACGGTGAGATGG
TCATCAATGTGGATGTTGCGGTGGCAAGCGATACACCGCATCCGGCGCGG
ATTGGCCTGACCTGCCAGCTGGCGCAGGTCTCAGAGCGGGTAACTGGCT
CGGCCTGGGGCCGCAAGAAAACTATCCCGACCGCCTTACTGCAGCCTGTT
TTGACCGCTGGGATCTGCCATTGTGCAGACATGTATACCCGCTACGCTCTC
CCGAGCGAAAAACGGTCTGCGCTGCGGGACGCGGAATTGAATTATGGCCC
ACACCAGTGGCGCGCGACTTCCAGTTCAACATCAGCCGCTACAGCCAAC
AACAACCTGATGGAACAGCCATCGCCATCTGCTGCACGCGGAAGAAGGC
ACATGGCTGAATATCGACGGTTTCCATATGGGGATTGGTGGCGACGACTC
CTGGAGCCCGTCAGTATCGGCGGAATTCCAGCTGAGCGCCGGTCGCTACC
ATTACCAGTTGGTCTGGTGTCAAAAATAA

TABLE 26

Nucleotide sequences of Pfnr3-PAL3 construct, low-copy (SEQ ID NO: 36)
GGTACCgtcagcataaacacctgacotctcattaattgttcatgcgggc
ggaactatcgctcgccgcttttctctcttactctgctacgtacatct
atttctataaatccgttcaatttgtctgttttttgcacaaacatgaaata
tcagacaattccgtgacttaagaaaatttatacaaatcagcaatatccc
cttaaggagtatataaagggtgaatttgatttacatcaataagcggggttg
ctgaatcgtaaGGATCCctctagaataaattttgtttaactttaagaag
gagatatacatATGAAAGCTAAAGATGTTACGCCAACCATTTATTAAAT
AAAAATGGCCTTATCTCTTTGGAAGATATCTATGACATTGCGATAAAACA
AAAAAAGTAGAAATATCAACGGAGATCACTGAACTTTTGACGCATGGTC
GTGAAAAATTAGAGGAAAAATTAAATTGAGGAGAGGTATATATGGAATC
AATACAGGATTGAGGGAATGCCAATTAGTTGTGCCATTGAGAAAAAT
CGCAGAGCATCAGCAAAATCTGTTAACTTTTCTTCTGCTGGTACTGGGG
ACTATATGTCCAAACCTTGTATTAAAGCGTCACAATTTACTATGTTACTT
TCTGTTTGCAAGGTTGGTCTGCAACCAGACCAATTGTCGCTCAAGCAAT
TGTTGATCATATTAATCATGACATTGTTCTCTGGTTCCTCGCTATGGCT
CAGTGGGTGCAAGCGGTGATTTAATTCCTTTATCTTATATGACGAGCA
TTATGTGGTATCGGCAAGTTTATTATATGGGCGCAGAAATTGACGCTGC
TGAAGCAATTAAACGTGACGGGTGACACCATTATCGTTAAAGCCAAAG
AAGGTCTTGCTCTGATTAAACGCCACCCGGTAATGTCAGGAATCAGTGCA
ATCACCGTCATTAAACTGGAAAACTATTTAAAGCCTCAATTTCTGCGAT
TGCCCTTGCTGTTGAAGCATTACTTGCATCTCATGAACATTATGATGCC

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TABLE 26-continued

Nucleotide sequences of Pfnr3-PAL3 construct, low-copy (SEQ ID NO: 36)
5 GGATTCAACAAGTAAAAAATCATCCTGGTCAAACCGGGTGGCAAGTGCA
TTGCGTAATTTATTGGCAGGTTCAACGCAGGTTAATCTATTATCTGGGGT
TAAAGAACAAGCCAATAAAGCTTGTCTCATCAAGAAATTACCCAATAA
10 ATGATACCTTACAGGAAGTTTATTCAATTCTGCTGTGCACCACAAGTATTA
GGTATAGTGCCAGAATCTTTAGCTACCGCTCGGAAAAATATTGGAACGGGA
AGTTATCTCAGCTAATGATAATCCATTGATAGATCCAGAAAATGGCGATG
15 TTCTACACGGTGGAAATTTTATGGGGCAATATGTCGCCGAACAATGGAT
GCATTAAAACTGGATATTGCTTTAATTGCCAATCATCTTCACGCCATTGT
GGCTCTTATGATGGATAACCGTTTCTCTCGTGATTACCTAATTCACCTGA
20 GTCCGACACCCGGCATGTATCAAGGTTTAAAGGCGTCCAACCTTTCTCAA
ACCGCTTTAGTTGCTGCAATTCGCCATGATTGTGCTGCATCAGGTATTCA
TACCTCGCCACAGAACAATAACAATCAAGATATTGTGAGTTTAGGCTGTC
25 ATGCCGCTCAAGATGTTTTAGAGATGGAGCAGAAATTACGCAATATTGTT
TCAATGACAATTCTGGTAGTTTGTGAGCCATTTCATCTTCGCGCAATAT
TAGTGAAATTGCGCTGAAACTGCTAAATTTTACCATGCAGTACGCGAAA
30 TCAGTTCTCCTTTGATCACTGATCGTGCCTGGATGAAGATATAATCCGC
ATTGCGGATGCAATTATTAATGATCAACTTCTCTGCCAGAAATCATGCT
GGAAGAATAA

TABLE 27

Nucleotide sequences of Pfnr4-PAL3 construct, low-copy (SEQ ID NO: 37)
40 GGTACCcatttctctctatcccatccgggtgagagctcttttccccgcac
ttatggctcatgcatgcatcaaaaagatgtgagcttgatcaaaaacaaa
aaatatcttactcgacaggagtatttatattgcccggatCCctctaga
45 aataattttgtttaactttaagaaggagatatacatATGAAAGCTAAAGA
TGTTGAGCCAAACCATTTATTAATAAAAAATGGCCTTATCTCTTTGGAAG
ATATCTATGACATTGCGATAAAACAAAAAAGTAGAAATATCAACGGAG
50 ATCACTGAACTTTTGACGCATGGTCTGAAAAATTAGAGGAAAAATTAA
TTGAGGAGGTTATATATGGAATCAATACAGGATTTGAGGGAATGCCA
ATTTAGTTGTGCCATTTGAGAAAAATCGCAGAGCATCAGCAAAATCTGTTA
55 ACTTTTCTTTCTGCTGGTACTGGGGACTATATGTCCAAACCTTGTATTAA
AGCGTCACAATTTACTATGTTACTTTCTGTTTGCAAGGTTGGTCTGCAA
CCAGACCAATTGTCGCTCAAGCAATTGTTGATCATATTAATCATGACATT
GTCCTCTGGTTCCTCGCTATGGCTCAGTGGGTGCAAGCGGTGATTTAAT
60 TCCTTTATCTTATATGACGAGCATTATGTGGTATCGGCAAGTTTATT
ATATGGGCGCAGAAATTGACGCTGCTGAAGCAATTAAACGTGACGGGTTG
ACACCATTTATCGTTAAAGCCAAAGAAGGCTTGCTCTGATTAAACGGCAC
65 CCGGTAATGTGAGGAATCAGTGCAATCACCGTCATTAACTGGAATAAC

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TABLE 27-continued

Nucleotide sequences of Pfnr4-PAL3 construct, low-copy (SEQ ID NO: 37)	
<u>TATTTAAAGCCTCAATTTCTGCGATTGCCCTTGCTGTTGAAGCATTACTT</u>	
<u>GCATCTCATGAACATTATGATGCCCGGATTCAACAAGTAAAAATCATCC</u>	
<u>TGGTCAAAACGCGGTGGCAAGTGCATTGCGTAATTTATTGGCAGGTTCAA</u>	
<u>CGCAGGTTAATCTATTATCTGGGGTTAAAGAACAAGCCAATAAAGCTTGT</u>	
<u>CGTCATCAAGAAATTACCAACTAAATGATACCTTACAGGAAGTTTATTC</u>	
<u>AATTCGCTGTGCACCACAAGTATTAGGTATAGTGCCAGAATCTTTAGCTA</u>	
<u>CCGCTCGGAAAATATTGGAACGGGAAGTTATCTCAGCTAATGATAATCCA</u>	
<u>TTGATAGATCCAGAAAATGGCGATGTTCTACACGGTGGAATTTTATGGG</u>	
<u>GCAATATGTCGCCCGAACAATGGATGCATTAAACTGGATATTGCTTTAA</u>	
<u>TTGCCAATCATCTTCACGCCATTGTGGCTCTTATGATGGATAACCGTTTC</u>	
<u>TCTCGTGGATTACCTAATTCAGTCCGACACCCGCATGTATCAAGG</u>	
<u>TTTTAAAGGCGTCCAACCTTCTCAAACCGCTTTAGTTGCTGCAATTCGCC</u>	
<u>ATGATTGTGCTGCATCAGGTATTATACCTCGCCACAGAACAAATACAAT</u>	
<u>CAAGATATTGTCAAGTTAGGTCTGCATGCCGCTCAAGATGTTTTAGAGAT</u>	
<u>GGAGCAGAAATTACGCAATATTGTTTCAATGACAATCTGGTAGTTTGTC</u>	
<u>AGGCCATTCTCTCGCGCAATATTAGTGAAATTGCGCCTGAAACTGCT</u>	
<u>AAATTTTACCATGCAGTACGCGAAATCAGTTCTCTCTTGATCACTGATCG</u>	
<u>TGCGTTGGATGAAGATATAATCCGATTGCGGATGCAATTATTAATGATC</u>	
<u>AACTTCTCTGCCAGAAATCATGCTGGAAGAATAA</u>	

TABLE 28

Nucleotide sequences of PfnrS-PAL3 construct, low-copy (SEQ ID NO: 38)	
<u>GGTACCagttgttcttatttggtggtgttgctttatggttgcatcgtagta</u>	
<u>aatggttgtaacaaaagaatttttccggctgtctgtatatacaaaaacgoc</u>	
<u>gtaaagtttgagcggaagtcaataaaactctctacccattcagggcaatato</u>	
<u>tctcttGGATCCctctagaataattttgtttaactttaagaaggagata</u>	
<u>tacatATGAAAGCTAAAGATGTTACGCCAACCATATTATTATATAAAAAAT</u>	
<u>GGCCTTATCTCTTGGAGATATCTATGACATTGCGATAAAACAAAAAA</u>	
<u>AGTAGAAATATCAACGGAGATCACTGAACCTTTTGACGCATGTCGTGAAA</u>	
<u>AATTAGAGGAAAAATTAAATTAGGAGAGGTTATATATGGAATCAATACA</u>	
<u>GGATTGGAGGAATGCCAATTTAGTTGTGCCATTGAGAAAATCGCAGA</u>	
<u>GCATCAGCAAAATCTGTTAACTTTTCTTCTGCTGGTACTGGGACTATA</u>	
<u>TGTCCAAACCTTGTATTAAAGCGTCACAATTTACTATGTTACTTTCTGTT</u>	
<u>TGCAAGGTTGGTCTGCAACAGACCAATTGTCGCTCAAGCAATTGTTGA</u>	
<u>TCATATTATCATGACATTGTTCTCTGGTTCCTCGCTATGGCTCAGTGG</u>	
<u>GTGCAAGCGGTGATTTAATTCCTTTATCTTATATGCACGAGCATTATGT</u>	
<u>GGTATCGGCAAGTTTATTATATGGCGCAGAAATTGACGCTGCTGAAGC</u>	

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TABLE 28-continued

Nucleotide sequences of PfnrS-PAL3 construct, low-copy (SEQ ID NO: 38)	
5	<u>AATTAACGTCAGGGTTGACACCATTATCGTTAAAAGCCAAAGAAGTCT</u>
	<u>TTGCTCTGATTAACGGCACCCGGGTAATGTGAGGAATCAGTGCAATCACC</u>
	<u>GTCATTAAACTGGA AAAACTATTTAAAGCCTCAATTTCTGCGATTGCCCT</u>
10	<u>TGCTGTTGAAGCATTACTTGATCTCATGAACATTATGATGCCCGGATTCT</u>
	<u>AACAAGTAAAAATCATCCTGGTCAAAACGCGGTGGCAAGTGCATTGCGT</u>
	<u>AATTTATTGGCAGGTTCAACGCAGGTTAATCTATTATCTGGGGTTAAAGA</u>
15	<u>ACAAGCCAATAAAGCTTGTGTCATCAAGAAATTACCAACTAAATGATA</u>
	<u>CCTTACAGGAAGTTTATCAATTGCTGTGCACCACAAGTATTAGGTATA</u>
	<u>GTGCCAGAATCTTTAGCTACCGCTCGGAAAATATTGGAACGGGAAGTTAT</u>
20	<u>CTCAGCTAATGATAATCCATTGATAGATCCAGAAAATGGCGATGTTCTAC</u>
	<u>ACGGTGGAAAATTTTATGGGGCAATATGTCGCCCGAACAATGGATGCATTA</u>
	<u>AACTGGATATTGCTTTAATTGCCAATCATCTTCACGCCATTGTGGCTCT</u>
	<u>TATGATGGATAACCGTTTCTCTCGTGGATTACCTAATTCAGTCCGCA</u>
25	<u>CACCCGGCATGTATCAAGGTTTAAAGGCGTCCAACCTTCTCAAACCGCT</u>
	<u>TTAGTTGCTGCAATTCGCCATGATTGTGCTGCATCAGGTATTATACCTCT</u>
	<u>CGCCACAGAACAAATACAATCAAGATATTGTGAGTTAGGTCTGCATGCCG</u>
30	<u>CTCAAGATGTTTTAGAGATGGAGCAGAAATTACGCAATATTGTTTCAATG</u>
	<u>ACAATTCTGGTAGTTTGTGAGCCATTATCTTCGCGCAATATTAGTGA</u>
	<u>AATTGCGCTGAACTGCTAAATTTTACCATGCAGTACGCGAAATCAGTT</u>
35	<u>CTCCTTTGATCACTGATCGTGGTGGATGAAGATATAATCCGCAATTGCG</u>
	<u>GATGCAATTATTAATGATCAACTTCTCTGCCAGAAATCATGCTGGAAGA</u>
	<u>ATAA</u>

Each of the plasmids was transformed into *E. coli* Nissle, as described above. Cultures of transformed *E. coli* Nissle were grown overnight and then diluted 1:200 in LB. The cells were grown with shaking at 250 rpm either aerobically or anaerobically in a Coy anaerobic chamber supplied with 90% N₂, 5% CO₂, and 5% H₂. After 4-6 hrs of incubation, samples were collected, and promoter activity was analyzed by performing β -galactosidase assays (Miller, 1972). As shown in FIG. 20, the activities of the FNR promoters were greatly enhanced under anaerobic conditions compared to aerobic conditions.

Example 7. Measuring the Activity of an FNR Promoter

To determine the kinetics of FNR promoter-driven gene expression, *E. coli* strains harboring a low-copy *fnrS-lacZ* fusion gene (FIG. 19A) were grown aerobically with shaking at 250 rpm. Cultures were split after 1 hr., and then incubated either aerobically or anaerobically in a Coy anaerobic chamber (supplying 90% N₂, 5% CO₂, and 5% H₂) at 37° C. Promoter activity was measured as a function of 0-galactosidase activity using a standard colorimetric assay (Miller, 1972). FIG. 19B demonstrates that the *fnrS* promoter begins to drive high-level gene expression within 1 hr. under anaerobic conditions. Growth curves of bacterial

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cell cultures expressing lacZ are shown in FIG. 19C, both in the presence and absence of oxygen.

Example 8. Production of PAL from FNR Promoter in Recombinant *E. coli*

Cultures of *E. coli* Nissle transformed with a plasmid comprising the PAL gene driven by any of the exemplary FNR promoters were grown overnight and then diluted 1:200 in LB. The bacterial cells may further comprise the pheP gene driven by the Tet promoter and incorporated into the chromosome. ATC was added to cultures at a concentration of 100 ng/mL to induce expression of pheP, and the cells were grown with shaking at 250 rpm either aerobically or anaerobically in a Coy anaerobic chamber supplied with 90% N₂, 5% CO₂, and 5% H₂. After 4 hrs of incubation, cells were pelleted down, washed, and resuspended in M9 minimal medium supplemented with 0.5% glucose and 4 mM phenylalanine. Aliquots were collected at 0 hrs, 2 hrs, 4 hrs, and 24 hrs for phenylalanine quantification (FIG. 20). As shown in FIG. 20B, the genetically engineered bacteria expressing PAL3 driven by the FNR promoter are more efficient at removing phenylalanine from culture medium under anaerobic conditions, compared to aerobic conditions (FIG. 20A). The expression of pheP in conjunction with PAL3 further decreased levels of phenylalanine.

Example 9. Phenylalanine Degradation in Recombinant *E. coli* with and without pheP Overexpression

The SYN-PKU304 and SYN-PKU305 strains contain low-copy plasmids harboring the PAL3 gene, and a copy of pheP integrated at the lacZ locus. The SYN-PKU308 and SYN-PKU307 strains also contain low-copy plasmids harboring the PAL3 gene, but lack a copy of pheP integrated at the lacZ locus. In all four strains, expression of PAL3 and pheP (when applicable) is controlled by an oxygen level-dependent promoter.

To determine rates of phenylalanine degradation in engineered *E. coli* Nissle with and without pheP on the chromosome, overnight cultures of SYN-PKU304 and SYN-PKU307 were diluted 1:100 in LB containing ampicillin, and overnight cultures of SYN-PKU308 and SYN-PKU305 were diluted 1:100 in LB containing kanamycin. All strains were grown for 1.5 hrs before cultures were placed in a Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂. After 4 hrs of induction, bacteria were pelleted, washed in PBS, and resuspended in 1 mL of assay buffer. Assay buffer contained M9 minimal media supplemented with 0.5% glucose, 8.4% sodium bicarbonate, and 4 mM of phenylalanine.

For the activity assay, starting counts of colony-forming units (cfu) were quantified using serial dilution and plating. Aliquots were removed from each cell assay every 30 min for 3 hrs for phenylalanine quantification by mass spectrometry. Specifically, 150 μ L of bacterial cells were pelleted and the supernatant was harvested for LC-MS analysis, with assay media without cells used as the zero-time point. FIG. 21 shows the observed phenylalanine degradation for strains with pheP on the chromosome (SYN-PKU304 and SYN-PKU305; left), as well as strains lacking pheP on the chromosome (SYN-PKU308 and SYN-PKU307; right). These data show that pheP overexpression is important in order to increase rates of phenylalanine degradation in synthetic probiotics.

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Example 10. Activity of Strains with Single and Multiple Chromosomal PAL3 Insertions

To assess the effect of insertion site and number of insertions on the activity of the genetically engineered bacteria, in vitro activity of strains with different single insertions of PAL3 at various chromosomal locations and with multiple PAL3 insertions was measured.

Cells were grown overnight in LB and diluted 1:100. After 1.5 hrs of growth, cultures were placed in Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂. After 4 hrs of induction, bacteria were resuspended in assay buffer containing 50 mM phenylalanine. Aliquots were removed from cell assays every 20 min for 1.5 hrs for trans-cinnamate quantification by absorbance at 290 nm. Results are shown in FIGS. 22 and 23 and Table 39 and Table 40. FIG. 22 depicts trans-cinnamate concentrations (PAL activity) for strains comprising single PAL3 insertions at various locations on the chromosome. FIG. 23 depicts trans-cinnamate concentrations (PAL activity) for strains comprising multiple PAL3 insertions at various locations on the chromosome.

TABLE 39

Activity of various strains comprising a single PAL3 chromosomal insertion at various sites		
Insertion:	Strain:	rate (μ mol/hr./ 1e9 cells):
agaI/rsmI	SYN-PKU520	1.97
yicS/nepI	SYN-PKU521	2.44
cea	SYN-PKU522	ND
malEK	SYN-PKU518	1.66
malPT	SYN-PKU523	0.47

TABLE 40

In vitro activity of various strains comprising one or more chromosomal PAL3 insertions		
Genotypes:	Strain	Rate (μ mol/hr./ 1e9 cells)
agaI:PAL, cea:PAL, matPT:PAL, malEK:PAL, lacZ:pheP, thyA-agaI:PAL, yicS:PAL, cea:PAL, matPT:PAL, malEK:PAL, lacZ:pheP, thyA-malPT:PAL, malEK:PAL, lacZ:pheP	SYN-PKU512	6.76
	SYN-PKU511	7.65
	SYN-PKU524	2.89
malEK:PAL, lacZ:pheP, ara-LAAD	SYN-PKU702	1.53
malPT:PAL, malEK:PAL, lacZ:pheP, ara-LAAD	SYN-PKU701	2.65
malPT:PAL, malEK:PAL, lacZ:pheP, agaI:pheP, ara-LAAD	SYN-PKU703	3.14
yicS:PAL, malPT:PAL, malEK:PAL, lacZ:pheP, ara-LAAD	SYN-PKU704	3.47
yicS:PAL, malPT:PAL, malEK:PAL, lacZ:pheP, agaI:pheP, ara-LAAD	SYN-PKU705	3.74

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Example 11. Activity of a Strain with Five Chromosomal Copies of PAL3

The activity of a strain SYN-PKU511, a strain comprising five integrated copies of an anaerobically (FNR) controlled PAL3 and an anaerobically controlled pheP integrated in the lacZ locus, was assessed.

The genetically engineered bacteria were grown overnight, diluted and allowed to grow for another 2.5 hours. Cultures were then placed in Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂. After 3.5 hrs of induction in phenylalanine containing medium (4 mM phenylalanine), whole cell extracts were prepared every 30 min for 3 hrs and phenylalanine was quantified by mass spectrometry. Results are shown in FIG. 24. The in vitro activity of the cells was 8 umol/hr./1e9 cells. Phenylalanine levels drop to about half of the original levels after 2 hours.

Example 12. Activity of a Strain Expressing LAAD

To assess whether LAAD expression can be used as an alternative, additional or complementary phenylalanine degradation means to PAL3, the ability of genetically engineered strain SYN-PKU401, which contains a high copy plasmid expressing LAAD driven by a Tet-inducible promoter, was measured at various cell concentrations and at varying oxygen levels.

Overnight cultures of SYN-PKU401 were diluted 1:100 and grown to early log phase before induction with ATC (100 ng/ml) for 2 hours. Cells were spun down and incubated as follows.

Cells (1 ml) were incubated aerobically in a 14 ml culture tube, shaking at 250 rpm (FIGS. 25 A and B). For microaerobic conditions, cells (1 ml) were incubated in a 1.7 ml conical tube without shaking. Cells were incubated anaerobically in a Coy anaerobic chamber supplying 90% N₂, 5% CO₂, and 5% H₂ (FIG. 25B). Aliquots were removed from cell assays every 30 min for 2 hrs for phenylalanine quantification by mass spectrometry, and results are shown in FIGS. 25A and 25B. FIG. 25A shows cell concentration dependent aerobic activity. The activity in aerobic conditions is ~50 umol/hr./1e9 cells, and some activity is retained under microaerobic conditions, which may allow for activity in environments with oxygen concentrations less than ambient air. The activity of SYN-PKU401 under microaerobic conditions is comparable to SYN-PKU304 under anaerobic conditions, however, activity seems to be dependent on cell density.

Table 41 and Table 42 contain LAAD constructs of interest. Table 41 shows the sequence of an exemplary construct comprising a gene encoding LAAD from *Proteus mirabilis* and a Tet repressor gene and a Tet promoter sequence and RBS and leader region, on a plasmid SEQ ID NO: 39, with the LAAD sequence underlined the TetR sequence in italics and the Tet promoter sequence bolded and the RBS and leader region underlined and italics. Table 42 shows the sequence of an exemplary construct comprising a gene encoding araC and a gene encoding LAAD from *Proteus mirabilis* and an arabinose inducible promoter (ParaBAD) sequence for chromosomal insertion into the endogenous arabinose operon (SEQ ID NO: 40), with the araC sequence underlined and the ParaBAD promoter sequence bolded and the LAAD sequence in italics and the RBS and leader region underlined and in italics.

In some embodiments, genetically engineered bacteria comprise a nucleic acid sequence that is at least about 80%, at least about 85%, at least about 90%, at least about 95%,

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or at least about 99% homologous to the DNA sequence of SEQ ID NO: 20-42, or a functional fragment thereof.

TABLE 41

LAAD driven by a Tet inducible promoter on a plasmid	
Nucleotide sequences of TetR-LAAD plasmid construct (SEQ ID NO: 39)	
5	Ttaagaccacactttcacatttaagttgtttttcctaaccgcatatgatca
10	attcaaggccgaataagaaggctggctctgcaccttggtgatcaataat
	tcgatagcttgcgtataatggcgccatactatcagtagtaggtgtttc
15	cctttcttcttttagcgacttgatgctcttgatcttccaatcgcacacct
	aagtaaaatgccccacagcgctgagtgcatataatgcattctctagttaa
	aaacctgttggcataaaaaggctaattgattttcgagagtttcatactg
20	ttttctgtaggcgtgtacctaataatgtacttttgcctccatcgogtagc
	ttagtaaacacatctaaaacttttagcgttattacgtaaaaactcttgc
	cagctttcccttctaaaggccaaaagtgagtaggtgcctatctaacat
	ctcaatggctaaaggcgtcgagcaaaagccgcttatttttacatgccaat
25	acaatgtaggctgctctacacctagcttctggcgagtttacgggtgtt
	aaacctcgattccgacctcattaagcagctcctaagcgctgttaataca
	tttacttttctctaatactagacatcattaattcctaattttgttgacac
30	tttatcattgatagagttattttaccactccctatcagtgatagagaaaa
	gtgaactctagaaaaaattttgtttaactttaagaaggagatatacatat
	gaacatttcaaggagaaagctacttttaggtgttggtgctcgccggcggtt
35	tagcaggtggtgcccgttttagttccaatgggtcgccgtgacggcaattt
	gtggaagctaaatcaagagcatcatttgttgagggtacgcaaggggctct
	tcctaaagaagcagatgtagtgattattggtgcccgtattcaagggatca
40	tgaccgctattaaccttgctgaacgtggtatgagtgctactatcttagaa
	aagggctcagattgcccgtgagcaatcaggccgtgcatcacagccaaattat
	tagttaccaaacatcgccagaaatcttccattacaccattatgggaaaa
45	tattatggcgtggcatgaatgagaaaaattggtgcccgtatcaccagttatcgt
	actcaaggtcgtgtagaagcgtggcagatgaaaaagcatttagataaagc
	tcaagcgtggatcaaaacagctaaagaagcggcaggttttgataccacct
50	taaatactcgcatcattaaagggtgaagagctatcaaatcgcttagtcggt
	gctcaaacgccatggactgttgctgcatttgaagaagattcaggctctgt
	tgatcctgaaacaggcacacctgcactcgctcgttatgccaaacaaatcg
55	gtgtgaaaaatttataccaactgtgcagtaagaggtattgaaactgcccgt
	ggtaaaatctctgatgtggtgagtgagaagggggcattaaaacgtctca
	agttgtactcgctgggggtatctggctcgcttatttatgggcaatatgg
	gtattgatatcccaacgctcaatgtatatctatcacacaacagctgtctca
60	gggggtcctggtgaccacgtggtaagtgtgcatctacctaattggtattca
	tttcgcgcaacaagcggatggtacttatgccgttgaccacgtatcttta
	cgagttcaatagtcaaaagatagcttctgctagggcctaaatttatgcac
65	ttattaggtggcggagagtaccggtggaattctctattggtgaagatct

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TABLE 41-continued

LAAD driven by a Tet inducible promoter on a plasmid Nucleotide sequences of TetR-LAAD plasmid construct (SEQ ID NO: 39)
<u>at</u> tttaattcattttaaattgcccacctcttgaatttagatgaaaaaacac
<u>cattcgaacaattccgagttgccacgggcaacacaaaatcgcacactta</u>
<u>gatgctgttttccaaagaatgaaaacagaattccagctatttgaaaaatc</u>
<u>agaagttgttgaaacgttggggtgccgttgtgagtgccaacatttgatgaat</u>
<u>tacctatcattttctgaggtcaagaataccaggcttagtgattaacacg</u>
<u>gcaacagtggtgggtatgacagaaggccggcgagcggtgaagtgaccgc</u>
<u>tgatattgtcatgggcaagaaacctgttattgatccaacgcgctttagtt</u>
<u>tggtacgttttaagaagtaa</u>

TABLE 42

LAAD sequence driven by the AraBAD promoter for insertion into the Ara operon Nucleotide sequences of AraC-ARABAD promoter-LAAD construct (SEQ ID NO: 40)
<u>Ttattcacaacctgcccctaactcgctcggactcgccccggtgcattttt</u>
<u>taaatactcgcgagaaatagagttgatcgctcaaaaccgacattgcgaccg</u>
<u>acgggtggcgatagggcatccgggtgggtgctcaaaagcagcttcgcctgact</u>
<u>gatgcgctgggtcctcgccagcttaatacgcctaactcctactgctggc</u>
<u>ggaaacaaatgcgacagacgcgacggcgacaggcagacatgctgtgcgacg</u>
<u>ctggcgatatacaaaattactgtctgccagggtgatcgctgatgtactgaca</u>
<u>agcctcgctgacccgattatccatcggtggatggagcgactcggttaatcg</u>
<u>cttccatgcgcgcgagtaacaattgctcaagcagatttatcgccagcaat</u>
<u>tccgaatagcgcccttcccttgtcggcattaatgatttgcacaaacag</u>
<u>gtcgctgaaatgcggctgggtgcgcttcatccgggcgaagaaaccggtat</u>
<u>tggaacaaatgcagggccagtttaagccattcatgccagtaggcgcgcgga</u>
<u>cgaaagtaaaaccactgggtgatccattcgtagcctccggtgacgaccc</u>
<u>gtagtgatgaatctctccaggcggaacagcaaaatatacccccgtcgcc</u>
<u>agacaaattctcgctccctgatttttaccacccccctgaccgcgaatgggtg</u>
<u>agattgagaatataacctttcattccacggcggtcggtcgataaaaaaac</u>
<u>gagataaccgttggcctcaatcggtgtaaaaccgccaccagatggggcgt</u>
<u>taaacgagtatccggcgacgaggggatcatttgcgcttcagccatactt</u>
<u>ttcatactcccgccattcagagaagaaaccaattgtccatattgcatcag</u>
<u>acattgccgtcactcggtcttttactggctcttctcgctaaccacaaccg</u>
<u>taaccccgcttattaaaagcattctgttaacaagcgggacaaaagccatg</u>
<u>acaaaaacgcgtacaaaaagtgctataatcacggcgagaaaagtcacat</u>
<u>tgattatttgcacggcgctcacactttgctatgccatagcatttttatcca</u>
<u>taagattagcggatccagcctgacgcttttttgcgaactctctactggt</u>
<u>tctccatAccctctagaaataattttgtttaactttaagaaggagatatac</u>

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TABLE 42-continued

LAAD sequence driven by the AraBAD promoter for insertion into the Ara operon Nucleotide sequences of AraC-ARABAD promoter-LAAD construct (SEQ ID NO: 40)
<u>at</u> atgaacatttcaaggagaaagctacttttaggtgttggtgctgcgggc
<u>gttttagcaggtggtgcggctttagttccaatggttcgcccgtgacggcaa</u>
<u>at</u> tttgtggaagctaaatcaagagcatcatttgttgaaggtagcgaagggg
<u>ctcttcctaagaagcagatgtagtgattattggtgcccgtattcaaggg</u>
<u>atcatgaccgctattaaaccttgctgaacgtggtagtgatgtcactatctt</u>
<u>ag</u> aaaaagggtcagattgcccgtgagcaatcaggccgtgcatacagccaaa
<u>ttattagttacaaaacatcgccagaaatcttccattacaccattatggg</u>
<u>aaaaatattatggcgtggcatgaatgagaaaaatgtgtgcggataccagtta</u>
<u>tc</u> gtagtcaaggctgtagaagcgctggcagatgaaaaagcatttagata
<u>aa</u> gctcaagcgtggatcaaaacagctaaagaagcggcaggttttgatata
<u>cc</u> attaaatactcgcatcattaaaggtagaagcgtatcaaatcgcttagt
<u>cg</u> gtgctcaaacgccatggactgttgccttgaagaagattcaggct
<u>ct</u> gttgatcctgaaacaggcacacgtgactcgctgcttattgcaaaacaa
<u>at</u> cggtgtgaaaaattataccaactgtgcagtaagaggatttgaactgc
<u>gg</u> gtggtaaaatctctgatgtggtgagtgagaaagggcgattaaacgt
<u>ct</u> caagttgtactcgctgggggtatctggtcgctttatttatgggcaat
<u>at</u> gggtattgatatacccaacgctcaatgtatatctatcacacaacgtgt
<u>ct</u> cagggttctcgtgacccacgtggtaattgtcatttacctaattggta
<u>tt</u> cattttccgcgaacaagcggatggtagtacttgccgttgaccacgtatc
<u>tt</u> tacgagttcaatagtagtcaagatagcttctgtagggcctaaatttat
<u>gc</u> acttattagtggtggcgagagttaccgttgaattctctattggtgaa
<u>at</u> ctatttaattcattttaaattgcccacgtcttggaatttagatgaaaaa
<u>ac</u> accattcgaacaattccgagttgccacggcaacacaaaaacgcaca
<u>ct</u> tagatgctgttttccaaagaatgaaacagaattcccgattttgaaa
<u>aa</u> tcagaagttgttgaaacgttgggggtgcccgttgtagtccaacatttgat
<u>ga</u> attacctatcatttctgaggtcaaagaataccaggcttagtgattaa
<u>ca</u> cggcaacagtggtgggtatgacagaaggccggcgagcggtgaaagtga
<u>cc</u> gctgatattgtcatgggcaagaaacctgttattgatccaacgccgttt
<u>ag</u> tttggatcggttttaagaagtaa

Example 13. Efficacy of PAL-Expressing Bacteria in a Mouse Model of PKU

For in vivo studies, BTBR-Pah^{emu2} mice were obtained from Jackson Laboratory and bred to homozygosity for use as a model of PKU. Bacteria harboring a low-copy pSC101 origin plasmid expressing PAL3 from the Tet promoter, as well as a copy of pheP driven by the Tet promoter integrated into the genome (SYN-PKU302), were grown. SYN-PKU1 was induced by ATC for 2 hrs prior to administration. Bacteria were resuspended in phosphate buffered saline (PBS) and 10⁹ ATC-induced SYN-PKU302 or control Nissle bacteria were administered to mice by oral gavage.

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At the beginning of the study, mice were given water that was supplemented with 100 micrograms/mL ATC and 5% sucrose. Mice were fasted by removing chow overnight (10 hrs), and blood samples were collected by mandibular bleeding the next morning in order to determine baseline phenylalanine levels. Blood samples were collected in heparinized tubes and spun at 2G for 20 min to produce plasma, which was then removed and stored at -80°C . Mice were given chow again, and were gavaged after 1 hr. with 100 μL (5×10^9 CFU) of bacteria that had previously been induced for 2 hrs with ATC. Mice were put back on chow for 2 hrs. Plasma samples were prepared as described above.

FIG. 26A shows phenylalanine levels before and after feeding, and FIG. 26B shows the percent (%) change in blood phenylalanine levels before and after feeding as a male or female group average ($p < 0.01$). As shown in FIG. 26, PKU mice treated with SYN-PKU1 exhibit a significantly reduced post-feeding rise in serum phenylalanine levels compared to controls.

Example 14. Efficacy of PAL-Expressing Bacteria Following Subcutaneous Phenylalanine Challenge

Streptomycin-resistant *E. coli* Nissle (SYN-PKU901) was grown from frozen stocks to a density of 10^{10} cells/mL. Bacteria containing a copy of pheP under the control of a Tet promoter integrated into the lacZ locus, as well as a high-copy plasmid expressing PAL3 under the control of a Tet promoter (SYN-PKU303) were grown to an A_{600} of 0.25 and then induced by ATC (100 ng/mL) for 4 hrs. Bacteria were centrifuged, washed, and resuspended in bicarbonate buffer at density of 1×10^{10} cells/mL before freezing at -80°C .

Beginning at least 3 days prior to the study (i.e., Days -6 to -3), homozygous BTBR-Pah^{enu2} mice (approx. 6-12 weeks of age) were maintained on phenylalanine-free chow and water that was supplemented with 0.5 grams/L phenylalanine. On Day 1, mice were randomized into treatment groups and blood samples were collected by sub-mandibular skin puncture to determine baseline phenylalanine levels. Mice were also weighed to determine the average weight for each group. Mice were then administered single dose of phenylalanine by subcutaneous injection at 0.1 mg per gram body weight, according to the average group weight. At 30 and 90 min post-injection, 200 μL of H_2O ($n=30$), SYN-PKU901 ($n=33$), or SYN-PKU303 ($n=34$) were administered to mice by oral gavage. Blood samples were collected at 2 hrs and 4 hrs following phenylalanine challenge, and phenylalanine levels in the blood were measured using mass spectrometry.

FIG. 27 shows phenylalanine blood concentrations relative to baseline concentrations at 2 hrs (FIG. 27A) and 4 hrs (FIG. 27B) post-phenylalanine injection. These data suggest that subcutaneous injection of phenylalanine causes hyperphenylalaninemia in homozygous enu2/enu2 mice, and that oral administration of SYN-PKU303 significantly reduces blood phenylalanine levels following phenylalanine challenge, compared to control groups ($p < 0.00001$ at 4 hrs). Moreover, these results confirm that the orally-administered engineered bacteria, and not the non-engineered Nissle parent, can significantly impact blood-phenylalanine levels independent of dietary exposure. Thus, a PKU-specific probiotic may not need to be co-administered in conjunction with diet.

Example 15. Dose-Response Activity of PAL-Expressing Bacteria on Systemic Phenylalanine

Streptomycin-resistant *E. coli* Nissle (SYN-PKU901) were grown from frozen stocks to a density of 10^{10} cells/mL.

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Bacteria containing a copy of pheP under the control of a P_{fmrS} promoter integrated into the lacZ locus, as well as a low-copy plasmid expressing PAL3 under the control of a P_{fmrS} promoter (SYN-PKU304) were grown to an A_{600} of 0.25 and then induced anaerobically by purging the bacterial fermenter with nitrogen for 4 hrs. Bacteria were centrifuged, washed, and resuspended in bicarbonate buffer at density of 5×10^9 cells/mL before freezing at -80°C .

Beginning at least 3 days prior to the study (i.e., Days -6 to -3), mice were maintained on phenylalanine-free chow and water that was supplemented with 0.5 grams/L phenylalanine. On Day 1, mice were randomized into treatment groups and blood samples were collected by sub-mandibular skin puncture to determine baseline phenylalanine levels. Mice were also weighed to determine the average weight for each group. Mice were then administered single dose of phenylalanine by subcutaneous injection at 0.1 mg per gram body weight, according to the average group weight. At 30 and 90 min post-injection, 200 μL of H_2O ($n=12$), 200 μL of SYN-PKU901 ($n=12$), or 100 μL , 200 μL , or 400 μL of SYN-PKU304 ($n=12$ in each dose group) were administered to mice by oral gavage. Blood samples were collected at 2 hrs and 4 hrs following phenylalanine challenge, and phenylalanine levels in the blood were measured using mass spectrometry.

FIG. 30 shows phenylalanine blood concentrations relative to baseline concentrations post-phenylalanine injection. These data demonstrate a dose-dependent decrease in blood phenylalanine levels in SYN-PKU304-treated mice compared to mock treatment (H_2O) or administration of the parental strain (SYN-PKU901), following subcutaneous injection of phenylalanine (*30% decrease; $p < 0.05$).

Example 16. Phenylalanine Degradation Activity In Vivo (PAL)

To compare the correlation between in vivo and in vitro phenylalanine activity, SYN-PKU304 (containing a low copy plasmid expressing PAL3 with a chromosomal insertion of P_{fmrS}-pheP at the LacZ locus, was compared to SYN-PKU901, a control Nissle strain with streptomycin resistance in vivo).

Beginning at least 3 days prior to the study (i.e., Days -6 to -3), homozygous BTBR-Pah^{enu2} mice (approx. 6-12 weeks of age) were maintained on phenylalanine-free chow and water that was supplemented with 0.5 grams/L phenylalanine. On Day 1, mice were randomized into treatment groups and blood samples were collected by sub-mandibular skin puncture to determine baseline phenylalanine levels. Mice were also weighed to determine the average weight for each group. Mice were then administered single dose of phenylalanine by subcutaneous injection at 0.1 mg per gram body weight, according to the average group weight. At 30 and 90 min post-injection, the bacteria were administered to mice by oral gavage.

To prepare the cells, cells were diluted 1:100 in LB (2 L), grown for 1.5 h aerobically, then shifted to the anaerobe chamber for 4 hours. Prior to administration, cells were concentrated 200 $\times$ and frozen (15% glycerol, 2 g/L glucose, in PBS). Cells were thawed on ice, and 4×10^8 cfu/mL and mixed 9:1 in 1M bicarbonate. Each mouse gavaged 800 μL total, or 2.9×10^8 cfu/mouse.

Blood samples were collected at 2 hrs and 4 hrs following phenylalanine challenge, and phenylalanine levels in the blood were measured using mass spectrometry, and the change in Phenylalanine concentration per hour was calculated. Results are shown in FIG. 32. The total metabolic

activity measured was 81.2 $\mu\text{mol/hr.}$ and the total reduction in change in phenylalanine was 45% ($P<0.05$). These same cells showed an in vitro activity of 2.8 $\mu\text{mol/hr./1e9 cells.}$

Additionally, various metabolites were measured to determine whether secondary metabolites can be used as an additional parameter to assess the rate of phenylalanine consumption of the engineered bacteria. When PAH activity is reduced in PKU, the accumulated phenylalanine is converted into PKU specific metabolites phenylpyruvate, which can be further converted into phenyllactic acid. In the presence of the genetically engineered bacteria, phenylalanine is converted by PAL to PAL specific metabolites trans-cinnamic acid, which then can be further converted by liver enzymes to hippuric acid (FIG. 32). Blood samples were analyzed for phenylpyruvate, phenyllactate, trans-cinnamic acid, and hippuric acid as described in Example 24-26. Results are shown in FIGS. 32C, 32D, 32E, and 32F and are consistent with the phenylalanine degradation shown in FIGS. 32A and 32B. For SYN-PKU304, PAL specific metabolites are detected at 4 hours, and moreover, lower levels of PKU specific metabolites are observed as compared to SYN-PKU901, indicating that PAL phenylalanine degradation may cause a shift away from PKU specific metabolites in favor or PAL specific metabolites.

Example 17. Phenylalanine Degradation Activity In Vivo (PAL)

SYN-PKU517 (comprising 2 chromosomal insertions of PAL (2 $\times$ fmrS-PAL (malEK, malPT)), and a chromosomal insertion of pheP (fmrS-pheP (lacZ)), thyA auxotrophy (kan/cm)) was compared to SYN-PKU901.

Mice were maintained, fed, and administered phenylalanine as described above. To prepare the bacterial cells for gavage, cells were diluted 1:100 in LB (2 L), grown for 1.5 h aerobically, then shifted to the anaerobe chamber for 4 hours. Prior to administration, cells were concentrated 200 $\times$ and frozen (15% glycerol, 2 g/L glucose, in PBS). Cells were thawed on ice, and 4e10 cfu/mL was mixed 9:1 in 1M bicarbonate. Each mouse gavaged 800 μL total, or 3.6e10 cfu/mouse.

As described above, blood samples were collected, and the change in phenylalanine concentration as compared to baseline was calculated. Results are shown in FIGS. 33A and 33B. The total metabolic activity measured was 39.6 $\mu\text{mol/hr.}$ and the total reduction in change in phenylalanine was 17% ($P<0.05$). These same cells showed an in vitro activity of 1.1 $\mu\text{mol/hr./1e9 cells.}$

Absolute levels of phenylalanine and of PKU and PAL metabolites are shown in FIGS. 33C, 33D, 33E, and 33F and are consistent with the phenylalanine degradation shown in FIGS. 33A and 33B. For SYN-PKU517, PAL specific metabolites were detected at 4 hours, and moreover, lower levels of PKU specific metabolites were observed as compared to SYN-PKU901, indicating that PAL phenylalanine degradation may cause a shift away from PKU specific metabolites in favor or PAL specific metabolites.

In some embodiments, urine is collected at predetermined time points, and analyzed for phenylalanine levels and levels of PAL and PKU metabolites.

Example 18. Phenylalanine Degradation Activity In Vivo (PAL)

SYN-PKU705 (comprising 3 chromosomal insertions of PAL (3 $\times$ fmrS-PAL (malEK, malPT, yicS/nepI)), and 2 chromosomal insertions of pheP (2 $\times$ fmrS-pheP (lacZ, agal/rsml)),

and LAAD (driven by the ParaBAD promoter integrated within the endogenous arabinose operon) was compared to SYN-PKU901.

Mice were maintained, fed, and administered phenylalanine as described above. To prepare the bacterial cells for gavage, cells were diluted 1:100 in LB (2 L), grown for 1.5 h aerobically, then shifted to the anaerobe chamber for 4 hours. Prior to administration, cells were concentrated 200 $\times$ and frozen (15% glycerol, 2 g/L glucose, in PBS). Cells were thawed on ice, and 5e10 cfu/mL was mixed 9:1 in 1M bicarbonate. Each mouse gavaged 800 μL total, or 3.6e10 cfu/mouse. Note: Though this strain contains the LAAD gene, it was not induced in this study.

As described above, blood samples were collected, and the change in phenylalanine concentration as compared to baseline was calculated. Results are shown in FIG. 34A. The total metabolic activity measured was 133.2 $\mu\text{mol/hr.}$ and the total reduction in change in phenylalanine was 30% ($P<0.05$). These same cells showed an in vitro activity of 3.7 $\mu\text{mol/hr./1e9 cells.}$

Absolute levels of phenylalanine and of PKU and PAL metabolites are shown in FIGS. 34C, 34D, 34E, and 34F and are consistent with the phenylalanine degradation shown in FIGS. 34A and 34B. PAL specific metabolites were detected at 4 hours, and moreover, lower levels of PKU specific metabolites were observed as compared to SYN-PKU901, indicating that PAL phenylalanine degradation may cause a shift away from PKU specific metabolites in favor or PAL specific metabolites. total metabolic activity measured activity was greater than the total metabolic activity measured of the PAL3 plasmid-based strain SYN-PKU304 and the total reduction in phenylalanine approached that of SYN-PKU304 (30% as compared to 45%).

In some embodiments, urine is collected at predetermined time points, and analyzed for phenylalanine levels and levels of PAL and PKU metabolites.

Example 19. Phenylalanine Degradation Activity In Vivo (PAL) LAAD

The suitability of *P. proteus* LAAD for phenylalanine degradation by the genetically engineered bacteria is further assessed in vivo. Bacterial strain SYN-PKU401 (comprising a high copy plasmid comprising LAAD driven by a Tet-inducible promoter is compared to SYN-PKU901.

Mice are maintained, fed, and administered phenylalanine as described above. To prepare the bacterial cells for gavage, cells are diluted 1:100 in LB (2 L), grown for 1.5 h aerobically, then ATC is added and the cells are grown for another 2 hours. Prior to administration, cells are concentrated 200 $\times$ and frozen for storage. Cells are thawed on ice, and resuspended. Cells are mixed 9:1 in 1M bicarbonate. Each mouse is gavaged four times with 800 μL total volume, or with a total of bacteria ranging from 2 $\times 10^9$ to 1 $\times 10^{10}$. Blood samples are collected from the mice described in the previous examples and are analyzed for phenylalanine, phenylpyruvate, phenyllactate, trans-cinnamic acid, and hippuric acid levels. Total reduction in phenylalanine and total metabolic activity are calculated.

Example 20. Effect of pH on Phenylalanine Degradation in Recombinant *E. coli*

To determine whether the rates of phenylalanine degradation in SYN-PKU304 and SYN-PKU302 are affected by low pH, overnight cultures of both strains were diluted 1:100 in LB and grown with shaking (250 rpm) at 37 $^{\circ}$ C. After 1.5

hrs of growth, ATC (100 ng/mL) was added to cultures of SYN-PKU302, and SYN-PKU304 cultures were placed in a Coy anaerobic chamber (supplying 90% N₂, 5% CO₂, and 5% H₂). After 4 hrs of induction, bacteria were pelleted, washed in PBS, and resuspended in assay buffer (M9 minimal media with 0.5% glucose, 8.4% sodium bicarbonate, and 4 mM Phe) to a concentration of 5e9 cells/mL. Assay buffer was prepared with incrementally decreasing values of pH, ranging from 7.25-2.25, using 1M HCl. Aliquots were removed from the cell assay every 30 min for 2 hrs for phenylalanine quantification by mass spectrometry. As shown in FIG. 39, phenylalanine degradation rates decreased as pH of the assay buffer decreased in both strains, SYN-PKU302 (FIG. 39A) and SYN-PKU304 (FIG. 39B).

Example 21. Degradation of Dipeptides and Tripeptides

Overnight strains of SYN-PKU304, and SYN-PKU705 were diluted 1:100 and grown to early log before shifting to anaerobic conditions for induction of PAL and pheP. One culture of SYN-PKU705 was also induced with arabinose to induce the LAAD protein. The focus of this study was to determine if PKU strains could degrade Phe when sequestered in the form of di and tripeptides. After strain induction Cells were spun down and resuspended in assay buffer containing M9 minimal media, 0.5% glucose, 50 mM MOPS, and 50 mM of Phe or Phe-containing di- or tripeptide. Supernatant samples were removed every 20 minutes for a total of 80 minutes, and supernatant was analyzed on a UV-Vis spectrophotometer to measure absorbance at 290 nm (the absorption peak for trans-cinnamic acid). Results are shown in Table 43 indicated that PKU strains were capable of degrading Phe rapidly even in the form of di- and tri-peptides.

TABLE 43

Dipeptide and Tripeptide Degradation Rates						
Rate (umol TCA produced/hr./1e9 cfu)						
	Phe	Phe-Val	Phe-Ala	Gly-Phe	Phe-Pro	Phe-Gly-Gly
SYN-PKU304	4.1	3.9	3.5	1.7	1.1	2.0
SYN-PKU705	6.9	5.8	5.0	4.1	1.3	4.5
SYN-PKU705 + ara	4.8	5.8	4.2	2.0	1.4	3.3

Example 22. Engineering Bacterial Strains Using Chromosomal Insertions

Bacterial strains, in which the pheP and/or PAL3 genes are integrated directly into the *E. coli* Nissle genome under the control of an FNR-responsive promoter, were constructed. The methods described below may be used for engineering bacterial strains comprising chromosomal insertions (e.g., SYN-PKU902 and/or any of the integrated strains listed in Table 14).

The SYN-PKU902 strain (lacZ::P_{fmrS}-PAL3-pheP) contains a copy of PAL3 and a copy of pheP integrated at the lacZ locus, with both genes operatively linked to a single fmrS promoter and co-transcribed in a bicistronic message (FIG. 41). Table 21 shows the sequence of an exemplary construct in which the PAL3 and pheP genes are co-transcribed under the control of an exemplary FNR promoter (SEQ ID NO: 31), with the FNR promoter sequence bolded,

the PAL3 sequence **boxed**, the pheP sequence underlined, and ribosomal binding sites **highlighted**.

To create a vector capable of integrating the P_{fmrS}-PAL3-pheP sequence into the chromosome, Gibson assembly was used to add 1000 bp sequences of DNA homologous to the Nissle lacZ locus to both sides of a flippase recombination target (FRT) site-flanked chloramphenicol resistance (cm^R) cassette on a knock-in knock-out (KIKO) plasmid. Gibson assembly was then used to clone the P_{fmrS}-PAL3-pheP DNA sequence between these homology arms, adjacent to the FRT-cm^R-FRT site. Successful insertion of the fragment was validated by sequencing. PCR was used to amplify the entire lacZ::FRT-cm^R-FRT::P_{fmrS}-PAL3-pheP::lacZ region. This knock-in PCR fragment was used to transform an electro-competent Nissle strain that contains a temperature-sensitive plasmid encoding the lambda red recombinase genes. After transformation, cells were grown for 2 hrs at 37° C. Growth at 37° C. cured the temperature-sensitive plasmid. Transformants with successful chromosomal integration of the fragment were selected on chloramphenicol at 20 µg/mL.

The SYN-PKU501 strain (malPT::P_{fmrS}-PAL3, lacZ::P_{fmrS}-pheP) contains a copy of PAL3 integrated at the malP/T locus, and a copy of pheP integrated at the lacZ locus, with both genes operatively linked to separate fmrS promoters (see Table 28; SEQ ID NO: 38). The SYN-PKU502 strain (malPT::P_{fmrS}-PAL3, lacZ::P_{fmrS}-PAL3-pheP) contains a copy of PAL3 integrated at the malP/T locus under the control of an fmrS promoter (see Table 28; SEQ ID NO: 38), as well as a PAL3-pheP construct integrated at the lacZ locus, wherein both genes at the lacZ locus are operatively linked to a single fmrS promoter and co-transcribed in a bicistronic message (see Table 21; SEQ ID NO: 31).

To create a vector capable of integrating the P_{fmrS}-PAL3 sequence (SEQ ID NO: 38) into the *E. coli* Nissle chromosome in SYN-PKU501 and SYN-PKU502, Gibson assembly was used to add 1000 bp sequences of DNA homologous to the Nissle malP and malT loci on either side of an FRT site-flanked kanamycin resistance (kn^R) cassette on a KIKO plasmid. Gibson assembly was then used to clone the P_{fmrS}-PAL3 DNA sequence between these homology arms, adjacent to the FRT-kn^R-FRT site. Successful insertion of the fragment was validated by sequencing. PCR was used to amplify the entire malP::FRT-kn^R-FRT::P_{fmrS}-PAL3::malT region. This knock-in PCR fragment was used to transform an electrocompetent Nissle strain already containing P_{fmrS}-pheP or bicistronic P_{fmrS}-PAL3-pheP in the lacZ locus, and expressing the lambda red recombinase genes. After transformation, cells were grown for 2 hrs at 37° C. Transformants with successful integration of the fragment were selected on kanamycin at 50 µg/mL. These same methods may be used to create a vector capable of integrating the P_{fmrS}-PAL3 sequence (SEQ ID NO: 38) at the malE/K insertion site in SYN-PKU506 and SYN-PKU507.

In some embodiments, recombinase-based switches may be used to activate PAL3 expression. The SYN-PKU601 strain (malPT::P_{fmrS}-Int5, rrmBUP-PAL3; lacZ::P_{fmrS}-pheP) contains the Int5 recombinase operably linked to a P_{fmrS} promoter, as well as a copy of PAL3 under the control of a strong constitutive promoter, integrated at the mal/T locus (FIG. 42). Table 45 shows the sequence of an exemplary P_{fmrS}-Int5, rrmBUP-PAL3 construct (SEQ ID NO: 42), wherein P_{fmrS}, Int5, and PAL3 are in reverse orientation. The

Int5 sequence is bolded, the P_{fmrS} sequence is **boxed**, the PAL3 sequence is underlined, and recombinase sites are

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bolded and underlined. Ribosomal binding sites are highlighted, and the *rrnBUP* constitutive promoter sequence is boxed. The UP element-containing *E. coli* *rrnBUP* promoter was selected to yield high PAL3 expression (Estrem et al., 1998), although any strong promoter may be used. SYN-PKU601 also contains a copy of *pheP* integrated at the *lacZ* locus.

To construct the SYN-PKU601 strain, the P_{fms} -driven *Int5* gene and the *rrnBUP*-driven, recombinase site-flanked PAL3 gene sequences were synthesized by Genewiz (Cambridge, MA). Gibson assembly was used to add 1000 bp sequences of DNA homologous to the Nissle *malP* and *malT* loci on either side of the P_{fms} -*Int5*, *rrnBUP*-PAL3 DNA sequence and to clone this sequence between the homology arms. Successful insertion of the fragment into a KIKO plasmid was validated by sequencing. PCR was used to amplify the entire P_{fms} -*Int5*, *rrnBUP*-PAL3 region. This knock-in PCR fragment was used to transform an electro-competent Nissle strain already containing P_{fms} -*pheP* in the

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lacZ locus, and expressing the lambda red recombinase genes. After transformation, cells were grown for 2 hrs at 37° C. Transformants with successful integration of the P_{fms} -PAL3 fragment at the *malPT* intergenic region were selected on kanamycin at 50 µg/mL. This strategy may also be used to construct a recombinase-based strain requiring T7 polymerase activity for PAL3 expression (FIG. 43). [Table 46 shows the sequence of an exemplary P_{fms} -*Int5*, *rrnBUP*-T7 construct (SEQ ID NO: 43), wherein P_{fms} , *Int5*, and the T7 polymerase gene are in reverse orientation. The *Int5* sequence is bolded, the P_{fms} sequence is boxed, the T7 polymerase sequence is underlined, and recombinase sites are bolded and underlined. Ribosomal binding sites are highlighted, and the *rrnBUP* constitutive promoter sequence is boxed. Table 44 shows the sequence of an exemplary P_{T7} -PAL3 construct, with the P_{T7} sequence highlighted, the ribosome binding site underlined, and the PAL3 sequence bolded.

TABLE 44

Nucleotide sequences of FNR promoter-PAL3- <i>pheP</i> construct (SEQ ID NO: 41)	
ggtaccAGTTGTTCTTATTGGTGGTGTGCTTTATGGTTGCATCGTAGTAAATGGTTGT	
AACAAAAGCAATTTTTCCGGCTGTCTGTATACAAAAACGCCGCAAGTTTGAGCGAAGT	
CAATAAACTCTCTACCCATTTCAGGGCAATATCTCTCTTggaatccaaagtgaactctaga	
aatcaattttgttttaactttaagaaggagatatatacatATGAAAGCTAAAGATGTTTCAGCG	
AACCATTATTATTAATAAAAAATGGCCTTATCTCTTTGGAAGATATCTATGACATTGCGA	
TAAAACAAAAAAAGTAGAAATATCAACGGAGATCACTGAACCTTTTGACGCATGGTCGT	
GAAAAATTAGAGGAAAAATTAAATTCAGGAGAGGTTATATATGGAATCAATACAGGATT	
TGGAGGGAATGCCAATTTAGTTGTGCCATTTGAGAAAAATCGCAGAGCATCAGCAAAATC	
TGTTAACTTTTCTTCTGCTGGTACTGGGGACTATATGTCCAAACCTTGTATTAAAGCG	
TCACAATTACTATGTTACTTTCTGTTTGCAAAGGTTGGTCTGCAACCAGACCAATTGT	
CGCTCAAGCAATTGTTGATCATATTAATCATGACATTGTTCTCTGGTTCCTCGCTATG	
GCTCAGTGGGTGCAAGCGGTGATTTAATTCCTTTTATCTTATATTGCACGAGCATTATGT	
GGTATCGGCAAAGTTTATTATATGGGCGCAGAAATTGACGCTGCTGAAGCAATTAAACG	
TGCAGGGTTGACACCATTATCGTTAAAGCCAAAGAAGGCTTGCTCTGATTAAACGCCA	
CCCGGTAATGTCAGGAATCAGTGCAATCACCGTCATTAAACTGGAAAACTATTTAAA	
GCCTCAATTTCTGCGATTGCCCTTGCTGTTGAAGCATTACTTGATCTCATGAACATTA	
TGATGCCCGGATTCAACAAGTAAAAATCATCCTGGTCAAAACGCGTGGCAAGTGCAT	
TGCGTAATTTATGGCAGGTTCAACGCAGGTTAATCTATTATCTGGGGTTAAAGAACA	
GCCAATAAAGCTTGTCGTCATCAAGAAATTACCAACTAAATGATACCTTACAGGAGGT	
TTATTCAATTGCTGTGCACCACAGTATTAGGTATAGTGCCAGAATCTTTAGTACCGC	
CTCGGAAATATTGGAACGGGAAGTTATCTCAGCTAATGATAATCCATTGATAGATCCA	
GAAATGGCGATGTTCTACACGGTGAAATTTTATGGGGCAATATGTGCGCCGAACAAT	

TABLE 44-continued

Nucleotide sequences of FNR promoter-PAL3-pheP
construct (SEQ ID NO: 41)

GGATGCATTAAACTGGATATTGCTTTAATTGCCAATCATCTTCACGCCATTGTGGCTC
TTATGATGGATAACCGTTTCTCTCGTGGATTACCTAATTCAGTCCGACACCCGGC
ATGTATCAAGGTTTAAAGGCGTCCAACCTTCTCAAACCGCTTAGTTGCTGCAATTCG
CCATGATTGTGCTGCATCAGGTATTCATACCCTCGCCACAGAACAATACAATCAAGATA
TTGTCAAGTTAGGTCTGCATGCCGCTCAAGATGTTTAGAGATGGAGCAGAAATTACGC
AATATTGTTTCAATGACAATTCCTGGTAGTTTGTCAAGGCCATTTCATCTTCGCGGCAATAT
TAGTGAAATTGCGCCTGAACTGCTAAATTTTACCATGCAGTACGCGAAATCAGTTCTC
CTTTGATCACTGATCGTGCCTGGATGAAGATATAATCCGATTGCGGATGCAATTATT
AATGATCAACTTCTCTGCGCAAAATCATGCTGGAAGAATAAaagaaggagatatacac
atgAAAAACGCGTCAACCGTATCGGAAGATACTGCGTCGAATCAAGAGCCGACGCTTCA
TCGCGGATTACATAACCGTCATATTCAACTGATTGCGTTGGGTGGCGCAATTGGTACTG
GTCTGTTTCTTGGCATTGGCCCGGCGATTAGATGGCGGGTCCGGCTGTATTGCTGGGC
TACGGCGTCGCGGGGATCATCGCTTTCCTGATTATGCGCCAGCTTGGCGAAATGGTGGT
TGAGGAGCCGCTATCCGTTTCAATTTGCCCACTTTCCTATAAAATACTGGGGACCGTTTG
CGGGCTTCCTCTCTGGCTGGAACACTGCGGTAATGTTCTGCTGGTGGGAATGGCAGAG
CTGACCGCTGCGGGCATCTATATGCAGTACTGGTTCCCGGATGTTCCAACGTGGATTG
GGCTGCCGCTTCTTTATTATCATCAACGCCGTTAACCTGGTGAACGTGCGCTTATATG
GCGAAACCGAGTTCTGGTTTGCCTTGAATTAAGTGTGGCAATCATCGGTATGATCGGC
TTTGGCCTGTGGTCTGCTGTTTTCTGGTCACGGCGCGAGAAAGCCAGTATCGACAACCT
CTGGCGCTACGGTGGTTTTCTTCGCCACCGGCTGGAATGGGCTGATTTTGTCTGCTGGCGG
TAATTATGTTCTCCTTCGGCGGTCTGGAGCTGATTGGGATTACTGCCGCTGAAGCGCGC
GATCCGGAAAAAGCATTCAAAAGCGGTAAATCAGGTGGTGTATCGCATCCTGCTGTT
TTACATCGGTTCACTGGTGGTTTTACTGGCGCTCTATCCGTGGGTGGAAGTGAATCCA
ACAGTAGCCCGTTTGTGATGATTTCCATAATCTCGACAGCAACGTGGTAGCTTCTGCG
CTGAACCTTCGTCTTCTGGTAGCATCGCTGTAGTGATAACAGCGGGGTTTACTCTAA
CAGCCGATGCTGTTTGGCCTTTCTGTGAGGGTAATGCGCCGAAGTTTTTGAATCGCG
TCAGCCGTGCGGTGTGCCGATTAACCTCGCTGATGCTTTCGGAGCGATCACTTCGCTG
GTGGTGTTAATCAACTATCTGTGCGCAAAAAGCGTTTGGTCTGCTGATGGCGCTGGT
GGTAGCAACGCTGCTGTGAAGTGGATTATGATCTGTCTGGCGCATCTGCGTTTTCTGTG
CAGCGATGCGACGTGAGGGCGTGAACACAGTTTAAGGCGTCTCTATCCGTTTCGGC
AACTATCTCTGCATTGCTTCTCGGCATGATTTTGTGCTGATGTGCACGATGGATGA
TATGCGCTTGTGAGCATCCTGCTGCCGTTGGATTGTATTCTGTTTATGGCATTTA
AAACGCTGCTCGGAAataa

TABLE 45

Nucleotide sequences of FNR promoter-Int5, rrnBUP-PAL3 construct (SEQ ID NO: 42)
ttaggtacgggctgccatttgattttaacgcgttcaccatcaaacggacgaccac gctggccttttgcaacccaatttcacgatgcaggtatcaataattgcattacgcag gtcgggggtgcaacgcagccacagttcttcataatcgctgctatcaacaatccagtaac atcaactgctgcgcttgcgctgcttctgctaactgcatctttggctgcctgcaggtgc tcagtgcctcttgatgatgcaggggcaaaaaactgttctgcggaccatcataaacacca ttctgacgatcagcagcagggcgaccagattttttcggttcacgaactgcggcttt tgcatacttttcacgttcgcttgctgcggatgggtcagtgctgcccagcagatctgcaa ctgcaataaacaacggatcatccggttcgcttgcgtgtaattttgctgcccaacgaaat gcaacatattcttcaacgcttttacgtgcaacataggtcggtgccggacaaccacctt cacactgctacgccaacaacgataaaccattaccgctatagctacagctaccaccacaac ccggacaacgcatacgcgctcagcagatgtttgcgacgggtatcatgatcgctacca tccagcggaacaccaaacaccatcttcaccttaacggctgcttttgcggcttcttgctc ttcacggtcaccagcggaggaccatgcataaacgctaacacgctttaccttcaccgttat aaaaggctcagacgagctgtttaccatcctgacgacctgtggtctgccaaccgcgat gcgggattctgaatcatatcacgcacggtaactgcaatccacggaccaccggctcgggct cggaatttcacgggtattcattgcatgtgcggctgctgcatagctcagacgatcggtaa ccggcagggtaaaaaccagacgggctgcttctgctttggctcagaccatcaggaccaccc gcatcttcacatctgctgcccagtttacgttcacatattcatcacctcttcacact aacggtaaccagaacaacacgcagaccatacgggtgcacgggcattaacccattcaccat ttacacgctgatgtgctttgggtatcacgaacacgcttgcgtcagttttctgcttcttcg cgtgcttcttctgcacgacgaatcagttcacgcgcatcagtttattggtgctatccag aaccggacgacgggtatcttcatcccaaccaaacgcagacgacgaggtatccatctt ccgggttcgataattttcagaattgcaccggcaccaccacgatcccaacgatccagacga taacaccacagtgcaaccaacttcacgcgtttccagggtttcagtgctttgctctgac atcacgtgctttacctttacgaaaacggcttgcgctaccaactctttccaaacatgac gaacctgcataccacgagtgctgcaactttacgaccagggtttcttgctgctgcaatg ctaatttcttggttaacgacgctgacctgcaccatttgacggcttttaactgctttgct ttacgacaaaaacagggtcaatcagacctgcaggatccggaccgggttcgggtggtcatac caggcatatgtatatctctcttcttaagtttaacaaaattattctagagttcactttg gatcc
AAGAGAGATATTGCCCTGAATGGGTAGAGAGTTTATTGACTTCGCTCAAACCTT
GCGGCGTTTTTGTATACAGACAGCCGGAATAATGCTTTTGTACCAACCTTTACTACG
ATGCAACCATAAAGCAACACCACCAATAAGAACAACtggtacg
CATGGCAGCTAGCCCTGCAGGGTGCACTCAGAAAATTATTTTAAATTTCCTCTTGTC
GGCCGGAATAACTCCCTATAATGCGCCACCAcagcgcggatcagggagtgagcggc
ctgggagcgtacacgctgtggctgcggctcgggtgcTTATTCTTCAGCATGATTTCTGG
CAGAGGAAGTGTGATCATTAAATAATTGCATCCGCAATGCGGATTATATCTTCATCCAACG
CACGATCAGTGATCAAAGGAGAACTGATTCGCGTACTGCATGGTAAAATTAGCAGTT

TABLE 45-continued

Nucleotide sequences of FNR promoter-Int5, rrnBUP-PAL3 construct (SEQ ID NO: 42)
TCAGGCGCAATTTCACTAATATTGCCGCGAAGATGAATGGCCTGACAACTACCAGAAT
TGTCATTGAAACAATATTGCGTAATTTCTGCTCCATCTCTAAAACATCTTGAGCGGCAT
GCAGACCTAAACTGACAATATCTTGATTGTATTGTTCTGTGGCGAGGGTATGAATACCT
GATGCAGCACAATCATGGCGAATTGCAGCAACTAAAGCGGTTTGAGAAAGTTGGACGCC
TTTAAACCTTGATACATGCCGGGTGTCGGACTCAGTGAATTAGGTAATCCACGAGAGA
AACGGTTATCCATCATAAGAGCCACAATGGCGTGAAGATGATTGGCAATTAAAGCAATA
TCCAGTTTAAATGCATCCATTGTCGGGCGACATATTGCCCCATAAAATTTCCACCGTG
TAGAACATCGCCATTTTCTGGATCTATCAATGGATTATCATTAGCTGAGATAACTTCCC
GTTCCAATATTTTCCGAGCGGTAGCTAAAGATTCTGGCACTATACCTAATACTTGTGGT
GCACAGCGAATTGAATAAACTTCCTGTAAGGTATCATTAGTTGGGTAATTTCTTGATG
ACGACAAGCTTTATTGGCTTGTCTTTAACCCAGATAATAGATTAACCTGCGTTGAAC
CTGCCAATAAATTACGCAATGCACTTGCCACCGCGTTTGGACAGGATGATTTTTTACT
TGTTGAATCCGGGCATCATAATGTTTCATGAGATGCAAGTAATGCTTCAACAGCAAGGGC
AATCGCAGAAATTGAGGCTTTAAATAGTTTTTCCAGTTAATGACGGTGATTGCACTGA
TTCTTGACATTACCCGGGTGCCGTTAATCAGAGCAAGACCTTCTTGGCTTTTAACGAT
AATGGTGTCAACCTGCACGTTTAATTGCTTCAGCAGCGTCAATTTCTGCGCCCATATA
ATAAACTTTGCCGATACCACATAATGCTCGTGCAATATAAGATAAAGGAATTAAATCAC
CGCTTGCACCCACTGAGCCATAGCGAGGAACAGAGGAACAATGTCATGATTAATATGA
TCAACAATTGCTTGAGCGACAATTGGTCTGGTTGCAGACCAACCTTGCAAAACAGAAAG
TAACATAGTAAATTGTGACGCTTTAATACAAGGTTTGACATATAGTCCCCAGTACCAG
CAGAAAGAAAAGTTAACAGATTTTGCTGATGCTCTGCGATTTTCTCAAATGGCACAAC
AAATTGGCATTCCCTCCAAATCCTGTATTGATTCCATATATAACCTCTCCTGAATTAA
TTTTTCCTCTAATTTTTCAGACCATGCGTCAAAAGTTCAGTGATCTCCGTTGATATTT
CTACTTTTTTTTGTATTCGCAATGTCATAGATATCTTCCAAGAGATAAGGCCATTT
TTATTAATAAATGGTTGGCTGAACATCTTTAGCTTTCAATgatatatctctcttctta
aaagllaaacaaaattatttttagagcagatcaggggtgcgcaagttgtcaacgctcccag
gagagttatcgacttgcggtattaggg

TABLE 46

Nucleotide sequences of FNR promoter-Int5, rrnBUP-T7 construct (SEQ ID NO: 43)
ttaggtacgggctgcccatttgatttttaacggttcacaccatcaaacggagaccac
gctggccttttgcaacccaaatttcacgatgcaggatcaataattgcattacgcatg
gtcggggttgcaocgagccacagttcttcataatcgctgctatcaacaatccagctaac
ataaactgctgcgcttgctgctgtttcgctaaactgcactcttggctgcctgcaggggtgc
tcagtgctctcttgatgcaggggcaaaaaactgttctgcggaccatcataaacacca
ttctgacgatcaocgagcaggcgaccagatttttttcggcttcacgaactgcggcttt
tgcatacttttcatcttcgcttgctgcggatgggtcagtgctgccagcgatctgcaa

TABLE 46-continued

Nucleotide sequences of FNR promoter-Int5, rrnBUP-T7
construct (SEQ ID NO: 43)

ctgcaataacaaacggatcatccggttcgcttgctgctaattttgctgcccacgaaat
gcaacatattcttcaacgcttttacgtgcaacataggtcggcgccgacacccactt
cacactgctacgccaacacgataaaccattaccgctatagctacagctaccaccacaac
cgggacaacgcatacagccgctcagcagatgtttggaagggtatcatgatcgctacca
tccagcggaaacaccaacacccatcttccactttaacggctgcttttgcggtctcttggtc
ttcatcggtcaccagcggaggaccatgcataacgctaacacggtttaccttcacggttat
aaaaggtcagacgaagctgtttaccatcctgacgaacctggtgtgccaacccgcataat
gcgggattctgaatcatatcacgcacggtaactgcaatccacggaccacggtcgggct
cggaatttcacgggtattcattgcatgtgcggtgcctgcatagctcagacgatcggttaa
cgggcagggtaaaaaacagacgggtgcttctgctttggtcagaccatcaggaccaccc
gcatcttcacatctgctgccagtttacgttcatcatattcatccctcttcatcact
aacggtaaccagaacaacacgcagaccatacgggtgcacgggcatttaaccattcaccat
tttcacgctgatgtgctttggtatcacgaacacggttcgctcagttttctgcttcttcg
cgtgcttctctgcaacgaatcagttccacgcgatacagtttattggtgctatccag
aacgggacgacgggtatcttcatcccaaccaaacgcagacgaagggataccatctt
cgggttcgataattttcagaattgcacgggcaccacacgatcccaacgatccagacga
taacaccacagtgccaacttcacgctttccagggtttcagtgctttgctctgato
atcagtgctttacctttacgaaaacggcttgctgctaccaacttctttccaaacatgac
gaacctgcataccacgagtgctgcaactttacgaccagggtttcttgctgctgcaatg
ctaatttcttggttacgaagctgacctgcaccatttgcaaggcttttaactgctttgct
tttacgacaaaacagggtcaatcagacctgcaggatccggacgggtttcggtggtcatac
caggcatatgtatatctctcttctttaaagtttaacaaaattatttctagagttcacttgc
gagggAAGAGAGATATTGCCCTGAATGGGTAGAGAGTTTATTGACTTCGCTCAAACTTT
GCGGCGTTTTTGTATACAGACAGCCGGAATAATTGCTTTTGTACAAACATTACTACG
ATGCAACCATAAAGCAACACCACCAATAAGAACAACCTggtacCGATATTCATATGGAC
CATGGCAGTAGCCCTGCAGGGTGCAGTCAGAAAATTATTTTAAATTTCTCTTGTTCG
GGCCGGAATAACTCCCTATAATGCGCCACCACgagcgcggatcaggagtggaaggc
ctgggagcgtacacgctgtggtgcggtcggtgcttacgcgaacgcgaagtccgactc
taagatgtcacggagggtcaagttacctttagccggaagtgtggcattttgtccaatt
gagactcgtgcaactgggtcagcgaactgggtcgtagaaatcagccagtagatcacaaagac
tcataatgtgtcaaccatagtttcgcgcactgctttgaacagggtcgcagcgtcagccgg
aatggtaccgaaggagtcgtgaatcagtgcaaaagattcgattccgtacttctcgtgtg
cccacactacagtcttacgaagggtggtaccgtcttggtgtgtacaaagtaggagcg
ataccagactcctgtttgtgtgcatcaatctcgctatctttgttggtgttaatggtagg
ctgtaagcgggaactgaccgaggaacatcagggttaagcgcgctgtaaataggcttctgtg
attcctgccacacagggaacacatcaggagttaccaatgcacagcgaacgcttgcga
agaatctctccagtccttcttatctttgacctcagcagccagcagcttagcagcagactt

TABLE 46-continued

Nucleotide sequences of FNR promoter-Int5, rrnBUP-T7
construct (SEQ ID NO: 43)

aagccagttcattgcttcaaccgcagctaccaccgtcacgctcacagattcccaaatca
gcttagccatgtatccagcagcctgattcggtgagtgaaacatcagacccttgccggaa
tcaatagctggctgaatgggtatcttccagcacttggtgacggaagccgaactctttgga
cccgtaagccagcgtcatgactgaacgcttagtcacactgaggaacaccgtaagcca
gccattgaccagccagtgcccttagtgcccagcttgactttctcagagatttcaccagtg
ttctcatcggtcacggtaactacttcgttatcggtccattgattgctgctgtag
aatctcgttgactttcttagcaacaatcccgtagatgtcctgaacgggttcactaggaa
gcaagttaaccgcgcgaccacctacctcatctcgagcatcgcgagagaagtgctggatg
ccagagcaagaccgcgtaaacgccagcggaagggagcagttatagctcaggccgtggtg
ctgtaccccgctactcaaagcagaacgcaaggagcagaacggagaatcttgcag
cccccaagtgttctccagtgagacttagcgcaagccatgatgttctcgtgggtttcc
tcaatgaacttgatgagctcagggaaacggaaccttatcgacaccgcacagtttgacc
gtggattttcagccagtagtaaccttccttaccgattgggtttacctttcgcagcgtaa
gcagtcctttgggtcatatcggtaccttgccgggtgaacattgacacagcgtaaacacga
ccgcgcagtcctatgttgaaggaaccagatggccttatggtagcaaaactattggc
ttgctcaagcatgaactcaaggctgatacggcgagacttgcgagccttgctccttcgggt
acacagcagcggcagcacggtttccacgcggtgagagcctcaggattcatgtcgatgtct
tccggtttcatcggtggttcttcacgctcaatcgagggatgtcctcgaccggacaatg
cttcacttggtgattacgttggcgaccgctaggactttcttggtagattttccatgcgg
tgttttgcgaatgttaatcgctttgtacacctcaggcatgtaaacgtcttcgtagcgc
atcagtgtctttcttactgtgagtacgcaccagcgccagaggacgacgaccttagccca
atagccaccaccagtaaatgccagtcacggcttaggaggaactacgcaaggttggaaca
tcgagagatgccagccagcgacctgcacgggttgcatagcctcagcgattcaggt
gcgagttcagatagctcagagtcctgacotactacgccagcattttggcgggtgaagt
aaccattccggttgactcaatgagcatctcgatgcagcgtaactcctacatgaatagat
cttccttatgccacgaagaccacgcctcgccaccgagtagacccttagagagcatgtca
gcctcgacaacttgcataaatgctttctttagacgtgccctacgcgcttggtgagttg
ttcctcaacgtttttcttgaagtgttagcttcaaggtcacggatcgaccgaagcgag
cctcgtcctcaatggcccgaccgattgcgcttgctacagcctgaacgggtgtattgtca
gcactgggttaggcaagccagagtggtcttaattggtgatgtacgctacggcttccggtt
gatttcttgaggaaactggaaggctgtcgggcgcttgccgcgcttagctttcacttcct
caaaccagtcgttgatgcgtgcaatcatcttagggagtagggtagtgtgagaggttg
gcggcagcggttatccgcaacctcaccagcttaagttgacgctcaaacatcttgccgaa
gcgtgcttcacccatctcgtaagactcatgctcaaggccaactgttcgcgagctaaac
gtcacccgtaaatggtcagccagagtggtgaacgggtagcagccagttcagatgtcagag
aagtcgttcttagcgatgttaatcggtgttcataatgacatctcctcttaagttaaac

TABLE 46-continued

Nucleotide sequences of FNR promoter-Int5, rrnBUP-T7
construct (SEQ ID NO: 43)

aaatctctctctctagagcagatcagggtgcgcaagttgtcaacgctcccaggagagttat
cgacttgcggtattaggg

TABLE 47

Nucleotide sequences of T7 promoter-PAL3 construct (SEQ ID NO: 44)

caatcagactccctatagggagaaagtgaactctagaaataattttgtttaactttaag
aaggagatatatacatATGAAAGCTAAAGATGTTGACCAACCATTATTATTAATAAAAAAT
GGCCTTATCTCTTTGGAAGATATCTATGACATTGCGATAAAACAAAAAAGTAGAAAT
ATCAACGGAGATCACTGAACTTTTGACGCATGGTCGTGAAAAATTAGAGGAAAAATTAA
ATTGAGGAGAGGTTATATATGGAATCAATACAGGATTTGGAGGGAATGCCAATTAGTT
GTGCCATTTGAGAAAAATCGCAGAGCATCAGCAAAATCTGTTAACTTTCTTTCTGCTGG
TACTGGGGACTATATGTCCAAACCTTGTATTAAAGCGTCACAATTTACTATGTTACTTT
CTGTTTGCAAAGGTTGGTCTGCAACCAGACCAATTGTCGCTCAAGCAATTGTTGATCAT
ATTAATCATGACATTGTTCTCTGTTCTCGCTATGGCTCAGTGGGTGCAAGCGGTGA
TTTAATTCCTTTATCTTATATTGCACGAGCATTATGTGGTATCGGCAAAGTTTATTATA
TGGGCGCAGAAATTGACGCTGCTGAAGCAATTAAACGTGCAGGGTTGACACCATTTATCG
TTAAAGCCAAAGAAGGTCTTGCTCTGATTAAACGGCACCCGGGTAATGTCAGGAATCAG
TGCAATCACCGTCATTAAACTGGAAAACTATTTAAAGCCTCAATTTCTGCGATTGCCC
TTGCTGTTGAAGCATTACTTGCACTCATGAACATTATGATGCCCGGATTCAACAAGTA
AAAAATCATCCTGGTCAAACGCGGTGGCAAGTGCAATGCGTAATTTATTGGCAGGTTT
AACGCAGGTTAATCTATTATCTGGGGTTAAAGAACAAAGCCAATAAAGCTTGTGTCATC
AAGAAATTACCCAATAAATGATACCTTACAGGAAGTTTATTCAATTCGCTGTGACCA
CAAGTATTAGGTATAGTGCCAGAATCTTTAGCTACCGCTCGGAAAAATATTGGAACGGGA
AGTTATCTCAGCTAATGATAATCCATTGATAGATCCAGAAAAATGGCGATGTTCTACAG
GTGGAATTTTATGGGGCAATATGTCGCCCAGCAATGGATGCATTAAACTGGATATT
GCTTTAATTGCCAATCATCTTCAGCCATTGTGGCTCTTATGATGGATAACCGTTTCTC
TCGTGGATTACCTAATTCACTGAGTCCGACACCCGGCATGTATCAAGGTTTAAAGGCG
TCCAACCTTCTCAAACCGCTTTAGTTGCTGCAATTCGCCATGATTGTGCTGCATCAGGT
ATTATACCTCGCCACAGAACAAATACAATCAAGATATTGTCAGTTTAGGTCTGCATGC
CGCTCAAGATGTTTTAGAGATGGAGCAGAAATTACGCAATATTGTTTCAATGACAATTC
TGGTAGTTTGTGAGCCATTCTTCGCGGCAATATTAGTGAATTCGCGCTGAACT
GCTAAATTTTACCATGCAGTACGCGAAATCAGTTCTCCTTTGATCACTGATCGTGCCTT
GGATGAAGATATAATCCGCATTGCGGATGCAATTATTAATGATCAACTTCCTCTGCCAG
AAATCATGCTGGAAGAATAA

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To construct the SYN-PKU602 strain comprising P_{ARA}-Int5 construct, P_{T7}-PAL3 construct, and P_{Lac}-T7 polymerase construct (FIG. 44), Gibson assembly was used essentially as described above.

Table 48 shows the sequence of an exemplary P_{ARA}-Int5 construct (SEQ ID NO: 45), for integration at the Ara locus. The Int5 sequence is **bolded**, the P_{ara} sequence containing TSS and RBS sites is underlined, and AraC sequence is in *italics*.

TABLE 48

Nucleotide Sequence of P _{ARA} -Int5 construct; SEQ ID NO: 45	
TTATTACAACTGCCCCCTAACTCGCTCGGACTCGCCCCGGTGCAATTTT	
TAAATACTCGCAGAGAAATAGAGTTGATCGTCAAAACCGACATTGCGACCG	
ACGGTGGCGATAGGCATCCGGTGGTGCTCAAAAGCAGCTTCGCCTGACT	
GATGCGCTGGTCTCGCGCCAGCTTAATACGCTAATCCCTAACTGCTGGC	
GGAACAAATGCGACAGACGCGACGGCGACAGGCAGACATGCTGTGCGACG	
CTGGCGATATCAAAATTACTGTCTGCCAGGTGATCGCTGATGTACTGACA	
AGCCTCGCTACCCGATTATCCATCGGTGGATGGAGCGACTCGTTAATCG	
CTTCCATGCGCCGAGTAACAATTGCTCAAGCAGATTTATCGCCAGCAAT	
TCCGAATAGCGCCCTTCCCTTGTCCGGCATTAAATGATTGCCCAAACAG	
GTGCTGAAATGCGGCTGGTGCGCTTCATCCGGGCGAAAGAAACCGGTAT	
TGGCAAAATATCGACGCCAGTTAAGCCATTATGCCAGTAGCGCGCGGA	
CGAAAGTAAACCCACTGGTGATACCATTCGTGAGCCTCCGGATGACGACC	
GTAGTGATGAATCTCTCCAGGCGGGAACAGCAAAATATCACCCGGTCGGC	
AGACAAATCTCGTCCCTGATTTTTCACCACCCCCTGACCGGAATGGTG	
AGATTGAGAATATAACCTTTCATTCCAGCGGTCCGTCGATAAAAAATC	
GAGATAACCGTTGGCCTCAATCGGCGTTAAACCCGCCACAGATGGGCGT	
TAAACGAGTATCCCGCAGCAGGGGATCATTTTGGCGTTTCAGCCATACTT	
<u>TTCTACTCCCGCCATTTCAGAGAAGAAACCAATTGTCATATGCATCAG</u>	
<u>ACATTGCCGTCCTGCTCTTTTACTGGTCTTCTCGCTAACCCAAACCGG</u>	
<u>TAACCCCGCTTATTAAGCATTCTGTAAACAAAGCGGACCAAGGCCATG</u>	
<u>ACAAAACGCGTAACAAAAGTGTCTATAATCACGGCAGAAAAGTCCACAT</u>	
<u>TGATTATTGACGCGCTCACACTTTGCTATGCCATAGCATTTTATCCA</u>	
<u>TAAGATTAGCGGATCCAGCCTGACGCTTTTTTCGCAACTCTCTACTGTT</u>	
<u>TCTCCATACCTCTAGAAATAATTTTGTTTAACTTTAAGAAGGAGATATAC</u>	
<u>ATATGCCTGGTATGACCACCGAAACCGGTCCGGATCCTGCAGGTCTGATT</u>	
GACCTGTTTTGTGCTAAAGCAAGCAGTTAAAGCCGTGCAAAATGGTGC	
AGGTGAGCGTCGTAACAAGAAATTAGCATTCGACACAAAGAAACCTGG	
GTCGTAAGTTGACGACCTGCTGGGTATGCAGGTTTCGTATGTTTGGAAA	
GAAAGTTGGTAGCGCAAGCCGTTTTTCGTAAAGTTAAAGCAGTGATGATCA	
GAGCAAGACCTGAAAGCCCTGAAAGCGGTGAAAGTTGGTGCACTGTGGT	
GTTATCGTCTGGATCGTTGGGATCGTGGTGGTGCAGGTCGAATTCGAAA	
ATTATCGAACCAGGATGGTATGCCTCGTCTGCTGTTTGGTTGGGA	
TGAAGATACCGGTCTGCGGTTCTGGATAGCACCAATAAAGCTGATCGCG	

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TABLE 48-continued

Nucleotide Sequence of P _{ARA} -Int5 construct; SEQ ID NO: 45	
GTGAAGTATCGTCTGCGAGAGAGCACGCGAAGAGCAGAAAACTG	5
AGCGAACGTGTTCTGATACCAAAGCACATCAGCGTGAAAAATGGTGAATG	
GGTTAATGCCCGTGACCGTATGGTCTGCGTGTGTTCTGGTTACCGTTA	10
GTGATGAAGAGGGTGATGAATATGATGAACGTAACTGGCAGCAGATGAT	
GAAGATGCGGGTGGTCTGATGGTCTGACCAAAGCAGAAAGCAGCCCTCT	
GGTTTTTACCCTGCCGTTACCGATCGTCTGAGCTATGAGGCACCGCAC	15
ATGCAATGAATACCCGTGAAATTCGAGCCCGACCGGTGGTCCGTGGATT	
GCAGTTACCGTGCCTGATGATGATTCAGAATCCGGCATATGCGGGTTGGCA	
GACCACAGGTGTCAGGATGGTAAACAGCGTCGCTGACCTTTTATAACG	20
GTGAAGGTAAACGTGTTAGCGTTATGATGGTCTCCGCTGGTGACCGAT	
GAAGAACAAGAGCCGCAAAAGCAGCCGTTAAAGGTGAAGATGGTGTGG	
TGTTCCGCTGGATGGTAGCGCATCATGATACCCGTCCGCAACATCTGCTGA	25
GCGGTCTGATGCGTTGTCCGGTTGTGGTGGTAGCTGTAGCTATAGCGGT	
AATGGTTATCGTTGTGGCGTAGCAGTGTGAAGGTGGTGTCCGGCACC	
GACCTATGTTGCACGTAAAGCGTTGAAGAATATGTTGCATTTTCGTTGGG	30
CAGCAAAATTAGCAGCAAGCGAACCGGATGATCCGTTTGTATTGTCAGTT	
GCAGATCGCTGGGCAGCACTGACCCATCCGCGAGGCAAGCGAAGATGAAAA	35
GTATGCAAAAGCCGAGTTTCGTGAAGCCGAAAAAATCTGGGTGCGCTGC	
TGCGTGATCGTCAGAATGGTGTATGATGGTCCGGCAGAACAGTTTTTT	
GCCCTGCATATCAAGAAGCACTGAGCACCTGCAGGCAGCAAGATGTC	40
AGTTAGCGAAAGCAGCGCAAGCGCAGCAGTTGATGTTAGCTGGATTGTTG	
ATAGCAGCGATTATGAAGAACTGTGGCTGCGTGAACCCCGACCATGCGT	
AATGCAATTATTGATACCTGCATCGATGAAATTTGGGTTGCAAAAGGCCA	45
GCGTGGTCTGCGTTTGTATGGTATGAACCGGTTAAATCAAATGGGCAG	
CCCGTACCTAA	50

Example 23. Generation of DeltaThyA

55 An auxotrophic mutation causes bacteria to die in the absence of an exogenously added nutrient essential for survival or growth because they lack the gene(s) necessary to produce that essential nutrient. In order to generate genetically engineered bacteria with an auxotrophic modification, the thyA, a gene essential for oligonucleotide synthesis was deleted. Deletion of the thyA gene in *E. coli* Nissle yields a strain that cannot form a colony on LB plates unless they are supplemented with thymidine.

65 A thyA::cam PCR fragment was amplified using 3 rounds of PCR as follows. Sequences of the primers used at a 100 um concentration are found in Table 49.

TABLE 49

Primer Sequences			
Name	Sequence	Description	SEQ ID NO
SR36	tagaactgatgcaaaaagtgctcgacgaaggcacacagaTGTGTAG GCTGGAGCTGCTTC	Round 1: binds on pKD3	SEQ ID NO: 46
SR38	gtttcgtaattagatagccaccggcgctttaatgcccggaCATATGAA TATCCTCCTTAG	Round 1: binds on pKD3	SEQ ID NO: 47
SR33	caacacgtacctgaggaacctatgaacagtatttagaactgatgcaaaaag	Round 2: binds to round 1 PCR product	SEQ ID NO: 48
SR34	cgcacactggcgctcggtctggcaggatgtttcgtaattagatagc	Round 2: binds to round 1 PCR product	SEQ ID NO: 49
SR43	atategtcgcagccacagcaacacgctttcctgagg	Round 3: binds to round 2 PCR product	SEQ ID NO: 50
SR44	aagaatttaacggaggggcaaaaaaacggacgcacactggcgctcggc	Round 3: binds to round 2 PCR product	SEQ ID NO: 51

For the first PCR round, 4×50 ul PCR reactions containing 1 ng pKD3 as template, 25 ul 2×phusion, 0.2 ul primer SR36 and SR38, and either 0, 0.2, 0.4 or 0.6 ul DMSO were brought up to 50 ul volume with nuclease free water and amplified under the following cycle conditions:

- step1: 98c for 30 s
- step2: 98c for 10 s
- step3: 55c for 15 s
- step4: 72c for 20 s
- repeat step 2-4 for 30 cycles
- step5: 72c for 5 min

Subsequently, 5 ul of each PCR reaction was run on an agarose gel to confirm PCR product of the appropriate size. The PCR product was purified from the remaining PCR reaction using a Zymoclean gel DNA recovery kit according to the manufacturer's instructions and eluted in 30 ul nuclease free water.

For the second round of PCR, 1 ul purified PCR product from round 1 was used as template, in 4×50 ul PCR reactions as described above except with 0.2 ul of primers SR33 and SR34. Cycle conditions were the same as noted above for the first PCR reaction. The PCR product run on an agarose gel to verify amplification, purified, and eluted in 30 ul as described above.

For the third round of PCR, 1 ul of purified PCR product from round 2 was used as template in 4×50 ul PCR reactions as described except with primer SR43 and SR44. Cycle conditions were the same as described for rounds 1 and 2. Amplification was verified, the PCR product purified, and eluted as described above. The concentration and purity was measured using a spectrophotometer. The resulting linear DNA fragment, which contains 92 bp homologous to upstream of thyA, the chloramphenicol cassette flanked by *frt* sites, and 98 bp homologous to downstream of the thyA gene, was transformed into a *E. coli* Nissle 1917 strain containing pKD46 grown for recombineering. Following electroporation, 1 ml SOC medium containing 3 mM thymidine was added, and cells were allowed to recover at 37 C for 2 h with shaking. Cells were then pelleted at 10,000×g for 1 minute, the supernatant was discarded, and the cell pellet was resuspended in 100 ul LB containing 3 mM thymidine and spread on LB agar plates containing 3 mM

thy and 20 ug/ml chloramphenicol. Cells were incubated at 37 C overnight. Colonies that appeared on LB plates were restreaked. +cam 20 ug/ml + or -thy 3 mM. (thyA auxotrophs will only grow in media supplemented with thy 3 mM).

Next, the antibiotic resistance was removed with pCP20 transformation. pCP20 has the yeast Flp recombinase gene, FLP, chloramphenicol and ampicillin resistant genes, and temperature sensitive replication. Bacteria were grown in LB media containing the selecting antibiotic at 37° C. until OD600=0.4-0.6. 1 mL of cells were washed as follows: cells were pelleted at 16,000×g for 1 minute. The supernatant was discarded and the pellet was resuspended in 1 mL ice-cold 10% glycerol. This wash step was repeated 3× times. The final pellet was resuspended in 70 ul ice-cold 10% glycerol. Next, cells were electroporated with 1 ng pCP20 plasmid DNA, and 1 mL SOC supplemented with 3 mM thymidine was immediately added to the cuvette. Cells were resuspended and transferred to a culture tube and grown at 30° C. for 1 hours. Cells were then pelleted at 10,000×g for 1 minute, the supernatant was discarded, and the cell pellet was resuspended in 100 ul LB containing 3 mM thymidine and spread on LB agar plates containing 3 mM thy and 100 ug/ml carbenicillin and grown at 30° C. for 16-24 hours. Next, transformants were colony purified non-selectively (no antibiotics) at 42° C.

To test the colony-purified transformants, a colony was picked from the 42° C. plate with a pipette tip and resuspended in 10 µL LB. 3 µL of the cell suspension was pipetted onto a set of 3 plates: Cam, (37° C.; tests for the presence/absence of CamR gene in the genome of the host strain), Amp, (30° C., tests for the presence/absence of AmpR from the pCP20 plasmid) and LB only (desired cells that have lost the chloramphenicol cassette and the pCP20 plasmid), 37° C. Colonies were considered cured if there is no growth in neither the Cam or Amp plate, picked, and re-streaked on an LB plate to get single colonies, and grown overnight at 37° C.

Example 24. Phenylalanine Quantification (Dansyl-Chloride Derivatization)

For in vitro and in vivo assays described herein, which assess the ability of the genetically engineered bacteria to

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degrade phenylalanine and which require quantification of phenylalanine levels in the sample, a dansyl-chloride derivatization protocol was employed as follows.

Sample Preparation

Phenylalanine standards (1000, 500, 250, 100, 20, 4 and 0.8 g/mL in water) were prepared. On ice, 10 μ L of sample was pipetted into a V-bottom polypropylene 96-well plate, and 190 μ L of 60% acetonitrile with 1 μ g/mL of L-Phenyl-d₅-alanine internal standard was added. The plate was heat sealed, mixed well, and centrifuged at 4000 rpm for 5 min. Next, 5 L of diluted samples were added to 95 μ L of derivatization mix (85 L 10 mM NaHCO₃ pH 9.7 and 10 μ L 10 mg/mL dansyl-chloride (diluted in acetonitrile)) in a V-bottom 96-well polypropylene plate, and the plate was heat-sealed and mixed well. The samples were incubated at 60° C. for 45 min for derivatization and then centrifuged at 4000 rpm for 5 minutes. Next, 20 μ L of the derivatized samples were added to 180 μ L of water with 0.1% formic acid in a round-bottom 96-well plate, plates were heat-sealed and mixed well.

LC-MS/MS Method

Phenylalanine was measured by liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) using a Thermo TSQ Quantum Max triple quadrupole mass spectrometer. HPLC Method details are described in Table 50 and Table 51. Tandem Mass Spectrometry details are described in Table 52.

TABLE 50

HPLC Method Details	
Column	Luna C18(2) column, 5 μ m (50 $\times$ 2.1 mm)
Mobile Phase A	100% H ₂ O, 0.1% Formic Acid
Mobile Phase B	100% ACN, 0.1% Formic Acid
Injection volume	10 μ L

TABLE 51

HPLC Method Details			
Total Time (min)	Flow Rate (μ L/min)	A %	B %
0	400	90	10
0.5	400	90	10
0.6	400	10	90
2	400	10	90
2.01	400	90	10
3	400	90	10

TABLE 52

Tandem Mass Spectrometry Details	
Ion Source	HESI-II
Polarity	Positive
SRM transitions	
L-Phenylalanine	399.1/170.1
L-Phenyl-d ₅ -alanine	404.1/170.1

Example 25 Trans-Cinnamic Acid Quantification (Trifluoroethylamine Derivatization)

For in vitro and in vivo assays described herein, which assess the ability of the genetically engineered bacteria to

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degrade phenylalanine and which require quantification of Trans-cinnamic acid levels in the sample, a trifluoroethylamine derivatization protocol was employed as follows.

Sample Preparation

Trans-cinnamic acid standard (500, 250, 100, 20, 4 and 0.8 g/mL in water) were prepared. On ice, 10 μ L of sample was pipetted into a V-bottom polypropylene 96-well plate. Next, 30 μ L of 80% acetonitrile with 2 μ g/mL of trans-cinnamic acid-d₇ internal standard was added, and the plate was heat sealed, mixed well, and centrifuged at 4000 rpm for 5 minutes. Next, 20 μ L of diluted samples were added to 180 μ L of 10 mM MES pH4, 20 mM N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide (EDC), 20 mM trifluoroethylamine in a round-bottom 96-well polypropylene plate. The plate was heat-sealed, mixed well, and samples were incubated at room temperature for 1 hour.

LC-MS/MS Method

Trans-cinnamic acid was measured by liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) using a Thermo TSQ Quantum Max triple quadrupole mass spectrometer. HPLC Method details are described in Table 53 and Table 54. Tandem Mass Spectrometry details are described in Table 55.

TABLE 53

HPLC Method Details	
Column	Thermo Aquasil C18 column, 5 μ m (50 $\times$ 2.1 mm)
Mobile Phase A	100% H ₂ O, 0.1% Formic Acid
Mobile Phase B	100% ACN, 0.1% Formic Acid
Injection volume	10 μ L

TABLE 54

HPLC Method Details			
Total Time (min)	Flow Rate (μ L/min)	A %	B %
0	500	100	0
1	500	100	0
2	500	10	90
4	500	10	90
4.01	500	100	0
5	500	100	0

TABLE 55

Tandem Mass Spectrometry Details	
Ion Source:	HESI-II
Polarity	Positive
SRM transitions	
Trans-cinnamic acid:	230.1/131.1
Trans-cinnamic acid-d ₇	237.1/137.2

Example 26. Phenylalanine, Trans-Cinnamic Acid, Phenylacetic Acid, Phenylpyruvic Acid, Phenyllactic Acid, Hippuric Acid and Benzoic Acid Quantification (2-Hydrazinoquinoline Derivatization)

For in vitro and in vivo assays described herein, which assess the ability of the genetically engineered bacteria to degrade phenylalanine and which require quantification of

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phenylalanine, trans-cinnamic acid, phenylacetic acid, phenylpyruvic acid, phenyllactic acid, hippuric acid, and benzoic acid levels in the sample, a 2-Hydrazinoquinoline derivatization protocol was employed as follows

Sample Preparation

Standard solutions containing 250, 100, 20, 4, 0.8, 0.16 and 0.032 $\mu\text{g/mL}$ of each standard in water were prepared. On ice, 10 μL of sample was pipetted into a V-bottom polypropylene 96-well plate, and 90 μL of the derivatizing solution containing 50 mM of 2-Hydrazinoquinoline (2-HQ), dipyrldyl disulfide, and triphenylphosphine in acetonitrile with 1 $\mu\text{g/mL}$ of L-Phenyl- d_5 -alanine, 1 $\mu\text{g/mL}$ of hippuric acid- d_5 and 0.25 $\mu\text{g/mL}$ trans-cinnamic acid- d_7 internal standards was added. The plate was heat-sealed, mixed well, and samples were incubated at 60° C. for 1 hour for derivatization, and then centrifuged at 4000 rpm for 5 min. In a round-bottom 96-well plate, 20 μL of the derivatized samples were added to 180 μL of water with 0.1% formic acid. Plates were heat-sealed and mixed well.

LC-MS/MS Method

Metabolites derivatized by 2-HQ were measured by liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) using a Thermo TSQ Quantum Max triple quadrupole mass spectrometer. HPLC details are described in Table 56 and Table 57. Tandem Mass Spectrometry details are described in Table 58.

TABLE 56

HPLC Method Details	
Column	Luna C18(2) column, 3 μm (150 $\times$ 2.1 mm)
Mobile Phase A	100% H_2O , 0.1% Formic Acid
Mobile Phase B	100% ACN, 0.1% Formic Acid
Injection volume	10 μL

TABLE 57

HPLC Method Details			
Total Time (min)	Flow Rate ($\mu\text{L}/\text{min}$)	A %	B %
0	500	90	10
0.5	500	90	10
2	500	10	90
4	500	10	90
4.01	500	90	10
4.25	500	90	10

TABLE 58

Tandem Mass Spectrometry Details	
Ion Source Polarity	HESI-II Positive
SRM transitions	
L-Phenylalanine:	307.1/186.1
L-Phenyld d_5 -alanine	312.1/186
Trans-cinnamic acid	290.05/131.1
Trans-cinnamic acid- d_7	297.05/138.1
Hippuric acid	321.1/160.1
Hippuric acid- d_5	326/160
Phenylacetic acid	278.05/160.1
Phenyllactic acid	308.05/144.1
Benzoic acid	264.05/105.1
Phenylpyruvate	306.05/260.1

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Example 27. Relative Efficacy of Chromosomal Insertion and Plasmid-Bearing Strains

To compare the rate of phenylalanine degradation between engineered bacterial strains with chromosomal insertions and those harboring plasmids, overnight cultures were diluted 1:100 in LB and grown with shaking (250 rpm) at 37° C. After 1.5 hrs of growth, cultures were placed in a Coy anaerobic chamber supplying 90% N_2 , 5% CO_2 , 5% H_2 . After 4 hrs of induction, bacteria were pelleted, washed in PBS, and resuspended in assay buffer (M9 minimal media with 0.5% glucose, 8.4% sodium bicarbonate, and 4 mM Phe). Rates of phenylalanine degradation (i.e., disappearance from the assay solution) or cinnamate accumulation from 30 to 90 min were normalized to 1e9 cells. Table 59 shows the normalized rates for all strains and describes genotypes and the activities of non-limiting examples of engineered plasmid-bearing strains and engineered strains comprising chromosomal insertions.

TABLE 59

Genotype and Activity of engineered plasmid-bearing strains and engineered strains comprising chromosomal insertions.				
Strain Name	Genotype	PAL Activity ($\mu\text{mol}/\text{hr}/10^9$ cells)	LAAD activity ($\mu\text{mol}/\text{hr}/10^9$ cells)	
Plasmid -based strains				
SYN-PKU101	Low copy pSC101-Ptet::PAL1, ampicillin resistant	ND	NA	
SYN-PKU102	High copy pColE1-Ptet::PAL1, ampicillin resistant,	ND	NA	
SYN-PKU201	Low copy pSC101-Ptet::PAL3, ampicillin resistant	ND	NA	
SYN-PKU202	High copy pColE1-Ptet::PAL3, ampicillin resistant,	ND	NA	
SYN-PKU203	lacZ::Ptet-pheP::cam	0	NA	
SYN-PKU401	Low copy pSC101-Ptet::PAL1, ampicillin resistant, chromosomal lacZ::Ptet-pheP::cam	1.1	NA	
SYN-PKU402	High copy pColE1-Ptet::PAL1, ampicillin resistant, chromosomal lacZ::Ptet-pheP::cam	0.8	NA	
SYN-PKU302	Low Copy pSC101-Ptet::PAL3, ampicillin resistant; chromosomal lacZ::Ptet-pheP::cam	2.2	NA	
SYN-PKU303	High copy pColE1-Ptet::PAL3, ampicillin resistant, chromosomal lacZ::Ptet-pheP::cam	7.1	NA	
SYN-PKU304	Low Copy pSC101-PfmrS::PAL3, ampicillin resistant; chromosomal lacZ::PfmrS-pheP::cam	3	NA	
SYN-PKU305	Low Copy pSC101-PfmrS::PAL3, kanamycin resistant; chromosomal lacZ::PfmrS-pheP::cam	3	NA	
SYN-PKU306	Low Copy pSC101-PfmrS::PAL3, kanamycin resistant; thyA	0.3	NA	
SYN-PKU307	Low Copy pSC101-PfmrS::PAL3, ampicillin resistant;	0.3	NA	
SYN-PKU308	Low Copy pSC101-PfmrS::PAL3, kanamycin resistant;	0.3	NA	
SYN-PKU401	High Copy pUC57-Ptet::LAAD; kanamycin resistant	NA	50 (* O_2), 0(- O_2)	

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TABLE 59-continued

Genotype and Activity of engineered plasmid-bearing strains and engineered strains comprising chromosomal insertions.			
Strain Name	Genotype	PAL Activity (umol/hr./ 10 ⁹ cells)	LAAD activity (umol/hr./ 10 ⁹ cells)
Integrated strains			
SYN-PKU501	malPT::PfurS::PAL3::kan	0.3	NA
SYN-PKU502	malPT::PfurS::PAL3::kan; bicistronic lacZ::PfurS::PAL3-pheP::cam	ND	NA
SYN-PKU503	malEK::PfurS::PAL3::cam	0.3	NA
SYN-PKU504	agal/rsmI::PfurS::PAL3	0.3	NA
SYN-PKU505	cea::PfurS::PAL3	0.3	NA
SYN-PKU506	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3	0.7	NA
SYN-PKU507	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; lacZ::Pfur-pheP::cam	5.2	NA
SYN-PKU508	malEK::PfurS::PAL3; pheA auxotroph	0.4	NA
SYN-PKU509	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; lacZ::Pfur-pheP::cam	4.9	NA
SYN-PKU601	malPT::PfurS-INT5::kan, rrnBUP - [PAL3]; lacZ::Pfur-pheP::cam (recombinase based strain)	0.9	NA
SYN-PKU510	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3	0.6	NA
SYN-PKU511	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; yicS/nepI::PfurS-PAL3::kan; malPT::PfurS::PKL3; lacZ::Pfur-pheP; AthyA	7.7	NA
SYN-PKU204	lacZ::Pfur-pheP::cam	ND	NA
SYN-PKU512	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; malPT::PfurS::PKL3; lacZ::Pfur-pheP::cam; AthyA	6.7	NA
SYN-PKU513	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; lacZ::Pfur-pheP; AthyA	4.9	NA
SYN-PKU514	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; malPT::PfurS::PAL3; AthyA	0.8	NA
SYN-PKU515	malEK::PfurS::PAL3; agal/rsmI::PfurS::PAL3; cea::PfurS::PAL3; AthyA	0.7	NA
SYN-PKU516	agal/rsmI::PfurS::PAL3::kan	0.3	NA
SYN-PKU517	malEK::PfurS::PAL3::cam; malPT::PfurS::PAL3::kan; lacZ::Pfur-pheP; AthyA	2.9	NA
SYN-PKU518	malEK-PfurS::PAL3::cam; PfurS::pheP::kan	1.7	NA
SYN-PKU519	ParaBC-PAL3::cam; PfurS-pheP::kan	1.3	NA
SYN-PKU520	agal/rsmI::PfurS::PAL3::kan; PfurS-PheP::cam	2.0	NA
SYN-PKU801	AargR; thyA::cam	ND	NA

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TABLE 59-continued

Genotype and Activity of engineered plasmid-bearing strains and engineered strains comprising chromosomal insertions.			
Strain Name	Genotype	PAL Activity (umol/hr./ 10 ⁹ cells)	LAAD activity (umol/hr./ 10 ⁹ cells)
SYN-PKU701	ParaBC-LAAD::cam; malEK-PfurS-PAL3; malPT::PfurS-PAL3::kan; PfurS-pheP yicS/nepI::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam	2.7	28 (*O ₂), 0 (-O ₂)
SYN-PKU521	cea::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam	2.4	NA
SYN-PKU522	malPT::PfurS-PAL3::kan; lacZ::Pfur-pheP::cam	ND	NA
SYN-PKU523	malEK::PfurS::PAL3; malPT::PfurS::PAL3; lacZ::Pfur-pheP	0.5	NA
SYN-PKU524	malEK::PfurS::PAL3; lacZ::Pfur-pheP	2.9	NA
SYN-PKU702	malEK::PfurS::PAL3; lacZ::Pfur-pheP; Para::LAAD	1.5	ND
SYN-PKU703	malEK::PfurS::PAL3; malPT::PfurS::PAL3; lacZ::Pfur-pheP; agal/rsmI::PfurS::pheP; Para::LAAD	3.1	ND
SYN-PKU704	malEK::PfurS::PAL3; malPT::PfurS::PAL3; yicS/nepI::PfurS-PAL3; lacZ::Pfur-pheP; Para::LAAD	3.5	ND
SYN-PKU705	malEK::PfurS::PAL3; malPT::PfurS::PAL3; yicS/nepI::PfurS-PAL3::kan; lacZ::Pfur-pheP; agal/rsmI::PfurS::pheP Para::LAAD	3.7	ND
SYN-PKU602	malEK::PT7::PAL3; Para::INT5::cam (recombinase); lacZ::Pfur-pheP; malPT::Pconstitutive::T7 polymerase (unflipped);	2.4	NA
SYN-PKU901	Nissle with streptomycin resistance	NA	NA

Example 28. Screening for Improved Phe Consumption

Screens using genetic selection are conducted to improve phenylalanine consumption in the genetically engineered bacteria. Toxic phenylalanine analogs exert their mechanism of action (MOA) by being incorporated into cellular protein, causing cell death. These compounds were evaluated for their utility in an untargeted approach to select PAL enzymes with increased activity. Assuming that these toxic compounds can be metabolized by PAL into a non-toxic metabolite, rather than being incorporated into cellular protein, genetically engineered bacteria which have improved phenylalanine degradation activity can tolerate higher levels of these compounds, and can be screened for and selected on this basis.

Various genetically engineered bacterial strains as well as control Nissle were treated with two analogs, p-fluoro-DL-minimum phenylalanine and o-fluoro-DL-phenylalanine (FIG. 35) at increasing concentrations. Minimum inhibitory concentration (MIC) was determined and the fold change relative to the wild type Nissle was determined. Results are shown in Table 60.

These results indicate that the para-analog appear to be taken up readily by pheP and are potentially a substrate of PAL, and that the ortholog appears to be taken up readily by pheP and is potentially a substrate of PAL. As a result, these compounds have utility for screening for PAL enzymes with greater activity.

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TABLE 60

MIC and Fold Change Relative to WT for various strains		
MIC (ug/mL)	fold change (WT)	Strain
para-fluoro-Phe		
1250	1	Wild Type Nissle
<2.4	>↓520X	SYN-PKU203 (Ptet::pheP chr.)
2500	↑2X	SYN-PKU202 (Ptet-PAL3 high copy)
19.5	↓64X	SYN-PKU302 (Ptet-PAL low copy + Ptet-pheP chr.)
39	↓32X	SYN-PKU303 (Ptet-PAL high copy + Ptet-pheP chr.)
ortho-fluoro-Phe		
62.5	1	Wild Type Nissle
1	↓64X	SYN-PKU203 (Ptet::pheP chr.)
250	↑4X	SYN-PKU202 (Ptet-PAL3 high copy)
31.3	↓2X	SYN-PKU302 (Ptet-PAL low copy + Ptet-pheP chr.)
15.6	↓4X	SYN-PKU303 (Ptet-PAL high copy + Ptet-pheP chr.)

Example 29. Repeat-Dose Pharmacokinetic and Pharmacodynamic Study of Genetically Engineered Bacteria Following Daily Nasogastric Gavage Dose Administration for 28-Days in Cynomolgus Monkeys (Non-GLP)

To evaluate any potential toxicities arising from administration of the genetically engineered bacteria or *E. coli* Nissle alone, the pharmacokinetics and pharmacodynamics of the genetically engineered bacteria and an *E. coli* Nissle are studied following daily nasogastric gavage (NG) dose administration for 28-days to female cynomolgus monkeys. Cynomolgus monkeys is selected because this species is closely related, both phylogenetically and physiologically, to humans and is a species commonly used for nonclinical toxicity evaluations. The genetically engineered bacteria are administered by nasal gastric gavage, consistent with the proposed route of administration in humans. Animals overall well-being (clinical observations), weight clinical pathology (serum chemistry, hematology, and coagulation) are tracked. Plasma is analyzed for ammonia levels, and fecal samples examined for bacterial load.

The genetically engineered strain comprises one or more copies of PAL3 integrated into the chromosome and one or more copies of PheP integrated into the chromosome, each of which are under the control of an FNRS promoter. In some embodiments, the genetically engineered strain also comprises one or more copies of LAAD, driven by an arabinose inducible promoter, e.g., ParaBAD. In some embodiments, the strain further comprises a auxotrophy mutation, e.g., deltaThyA. In some embodiments, the genetically engineered bacteria further comprise an antibiotic resistance, e.g., kanamycin. In some embodiments, the genetically engineered bacteria do not comprise an auxotrophy mutation. In some embodiments, the genetically engineered bacteria do not comprise an antibiotic resistance.

Materials, Animals and Dosing Regimen

The study is conducted in compliance with nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58; effective Jun. 20, 1979) and the OECD Principles on Good Laboratory

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Practice (C [97]186/Final; effective 1997). The animals are individually housed based on the recommendations set forth in the Guide for the Care and Use of Laboratory Animals (National Research Council 2011).

5 Animals used in the study are Female Purpose-bred, non-naive cynomolgus monkey (*Macaca fascicularis*) with 3 to 6 kg (at initial physical exam) 3 to 8 years (at initial physical exam) of age (SNBL USA stock, Origin: Cambodia).

10 For the duration of the study, animals are offered PMI LabDiet® Fiber-Plus® Monkey Diet 5049 biscuits twice daily. Animal are fasted for at least 2 hours prior to dose administration and fed within 1-hour post dose. Animals also are fasted as required by specific procedures (e.g., prior to blood draws for serum chemistry, fecal collection). The diet is routinely analyzed for contaminants and found to be within manufacturer's specifications. No contaminants are expected to be present at levels that would interfere with the outcome of the study. Food analysis records are maintained in the testing facility records.

15 Fresh drinking water is provided ad libitum to all animals. The water is routinely analyzed for contaminants. No contaminants are present at levels that would interfere with the outcome of the study. Animals are given fruits, vegetables, other dietary supplements, and cage enrichment devices throughout the course of the study.

20 Previously quarantined animals are acclimated to the study room for 7 days prior to initiation of dosing (day 1). The last dosing occurs on day 28. A stratified randomization scheme incorporating body weights is used to assign animals to study groups. Animals are assigned to groups and treated as indicated in Table 61.

TABLE 61

Group Assignments						
Test/ Control Group	Articles	Dose			Flu	
		Level (cfu/ Animal)	Conc. (cfu/ mL)	Volume (mL/ Animal)	Bicarb. Conc. (M)	Volume (mL/ Animal)
1	Control Article	0	0	10	0.36	5
2	<i>E. coli</i> Nissle	1×10^9	1×10^9	1	0.12	14
3	<i>E. coli</i> Nissle	1×10^{12}	1×10^{11}	10	0.36	5
4	Genetically engineered bacteria	1×10^9	1×10^9	1	0.12	14
5	Genetically engineered bacteria	1×10^{12}	1×10^{11}	10	0.36	5

55 Nissle control and genetically engineered bacterial stocks are prepared at 1×10^9 cfu/mL and 1×10^{11} cfu/mL in 15% glycerol in 1xPBS with 2.2% glucose and 3 mM thymidine and are kept at 86 to -60° C. (see Table 61). PBS made in 20% glycerol with sodium bicarbonate is used as a control vehicle. Carbonate concentration is 0.36M and 0.12M for sodium bicarbonate (see table XXX). On the day of each dosing, bacteria and vehicle control are removed from the freezer and put on ice and thawed and placed on ice until dosing.

60 Animals are dosed at 0, 1×10^9 , or 1×10^{12} cfu/animal. All animals are dosed via nasal gastric gavage (NG) followed by control/vehicle flush once daily for 28-days. The concentra-

tion of bicarbonate and volume for each group is specified in Table YYY. Vials are inverted at least 3 times prior to drawing the dose in the syringe. The dose site and dose time (end of flush time) is recorded.

Analysis

Overall Condition:

Clinical observations are performed twice daily beginning on the second day of acclimation for each animal. The first observation is in the AM, prior to room cleaning. The second observation is no sooner than 4 hours after the AM observation. During the dosing phase, the second observation is performed 4 hour (+10 minutes) post dose administration. Additional clinical observations are performed, as necessary.

Weight:

Each animal is weighed on Day -6, 1, 8, 15, 22, and 29 prior to the first feeding and also prior to dose administration. Additional body weights are taken as needed if necessary.

Blood Collection:

Blood is collected from a peripheral vein of restrained, conscious animals. Whenever possible, blood is collected via a single draw and then divided appropriately. Specimen collection frequency is summarized in Table 62.

TABLE 62

Specimen collection frequency					
Time Point	Hematology	Coagulation	Serum Chemistry	Plasma Sample (on ice)	Fecal sample (on ice)
Acclimation Week 1	1x	1x	1x	1x	1x
Dosing	Day 2 (Predose)	Day 2 (Predose)	Day 2 (Predose)	Days 2 and 7 (Predose)	Days 2 and 7
Dosing	Day 14 (Predose)	Day 14 (Predose)	Day 14 (Predose)	Day 14 (Predose)	Day 14-20
Dosing	—	—	—	—	Day 21-27—
Dosing	—	—	—	Day 28 (Predose)	Day 28-30
Dosing Weeks	Day 30	Day 30	Day 30	Day 30	Day 35, 40

— = Not applicable

x = Number of times procedure performed within the week

Hematology:

Approximately 1.3 mL of blood is tested in 2 mL K2EDTA tubes using an Advia automated analyzer. Parameters measured are White Blood Cells, Red Blood Cells, Hemoglobin, Hematocrit, Mean Corpuscular Volume, Mean Corpuscular Hemoglobin, Mean Corpuscular Hemoglobin Concentration, Red Cell Distribution Width, Platelets, Mean Platelet Volume, Differential leukocyte count (absolute); Neutrophils Absolute Lymphocytes Absolute Monocytes Absolute Eosinophils Absolute, Basophils Absolute Reticulocyte Percent, and Reticulocyte Absolute Count.

Coagulation:

Approximately 1.3 mL of blood is tested in 1.8 mL 3.2% sodium citrate tubes. The following Coagulation parameters are determined using a STACompact automated analyzer: Activated Partial Thromboplastin Time, Fibrinogen, and Prothrombin Time. Sodium citrate-treated plasma is stored at -60 to -86° C. prior to analysis and discarded after analysis.

Serum Chemistry:

Animals are fasted for 4 hours prior to removal of sample. The following parameters are tested in approximately 1 mL of blood in 4 mL serum separator tubes using a AU680 analyzer: Albumin, Alkaline Phosphatase, Alanine Amino-transferase Aspartate Aminotransferase, Total Bilirubin, Calcium, Total Cholesterol, Creatine Kinase, Creatinine, Glucose, Inorganic Phosphorus, Total Protein, Triglyceride, Sodium, Potassium, Chloride Globulin, Albumin/Globulin Ratio, Blood Urea Nitrogen, and Gamma Glutamyltransferase.

Residual serum is stored at -60 to -86° C. and disposed of prior to study finalization.

Plasma Samples:

Animals are fasted for 4 hours prior to removal of the sample. Blood samples are collected from the femoral vein at the target time points listed in Table YYY. After aliquotting the target volume of blood in the blood tube, approximately 0.05 mL of mineral oil is added covering the surface of blood. Tubes are not inverted and placed on a rack and wet ice. Blood sample collection dates and times were recorded. The minimum sample volume is 1 mL of blood collected in a 2 mL lithium heparin tube. Within 15 minutes of collection, the samples are centrifuged at 2 to 8° C. to obtain plasma. Plasma is transferred to a vial and stored at -60 to -86° C. Specimens are stored on dry ice prior to analysis. Analysis of specimens is conducted using a blood ammonia analyzer instrument.

Phenylalanine, trans-cinnamic acid, and hippuric acid is measured by liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS) using a Thermo TSQ Quantum Max triple quadrupole mass spectrometer.

Fecal Sample Collection:

Two fecal samples per animal are collected at the target time points listed in Table YYY. Sample collection dates and times are recorded. 50 mL falcon tube with approximately 5 mL PBS are used as the container (If feces is liquid, no PBS is added). To get the fecal sample weight, pre- and post-sampling weight of container was taken. Samples are collected from the bottom of the cage from each animal. To get fresh and un-contaminated samples, remaining food is removed and the cage pan was cleaned and squeegeed to remove debris and/or water before the collection. Sample is put on wet ice immediately after the collection. Samples are stored at -20 to -15° C. until analysis. Analysis of specimens is conducted using a PCR analytical method.

Example 30. 4-Week Toxicity Study in Cynomolgus Monkeys with a 4-Week Recovery (GLP)

To evaluate any potential toxicities arising from administration of the genetically engineered bacteria, the pharmacokinetics and pharmacodynamics of the genetically engineered bacteria is studied following daily nasogastric gavage (NG) dose administration for 28-days to female cynomolgus monkeys under GLP conditions.

The genetically engineered strain comprises one or more copies of PAL3 integrated into the chromosome and one or more copies of PheP integrated into the chromosome, each of which are under the control of an FNRS promoter. In some embodiments, the genetically engineered strain also comprises one or more copies of LAAD, driven by and arabinose inducible promoter, e.g., ParaBAD. In some embodiments, the strains further comprise a auxotrophy mutation, e.g., deltaThyA. In some embodiments, the genetically engineered bacteria further comprise an antibiotic resistance, e.g., kanamycin. In some embodiments, the

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genetically engineered bacteria do not comprise an auxotrophy mutation. In some embodiments, the genetically engineered bacteria do not comprise an antibiotic resistance.

The study is conducted in compliance with nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58; effective Jun. 20, 1979) and the OECD Principles on Good Laboratory Practice (C[97]186/Final; effective 1997). The animals are individually housed based on the recommendations set forth in the Guide for the Care and Use of Laboratory Animals (National Research Council 2011).

Animals are administered the genetically engineered bacteria or control vehicle essentially as described in Example 29, except that all materials are manufactured under GMP standards. Dosing is tabulated in Table 63. Additionally, animals are acclimated for 14 days and the dosing period is daily for 28 days followed by a recovery period of 28 days. Additionally, animals are euthanized at the end of the study to conduct histological analysis.

TABLE 63

Dosing Period and Regimen					
	ACCLIMATION	14 days			
	TEST ARTICLE	Daily			
	PREP				
	DOSING PERIOD	Daily for 28 days			
	RECOVERY PERIOD	28 days			
	REGULATIONS	FDA GLP			
NUMBER OF ANIMALS					
GROUP	TEST ARTICLE	DOSE LEVEL	DOSE ROUTE	MALES (♂)	FEMALES (♀)
1	Vehicle	0	NG	3 ^a + 2 ^b	3 ^a + 2 ^b
2	Genetically engineered bacteria	1 × 10 ⁹	NG	3 ^a	3 ^a
3	Genetically engineered bacteria	1 × 10 ¹⁰	NG	3 ^a	3 ^a
4	Genetically engineered bacteria	1 × 10 ¹¹	NG	3 ^a + 2 ^b	3 ^a + 2 ^b

^aTerminal Necropsy, Day 29

^bRecovery Necropsy, Day 56

Study Analysis is conducted as described in Table 64. Hematology, Coagulation, Serum Chemistry and Plasma Samples parameters are essentially as described in Example 30, and are analyzed using the methods described in Example 30. Collection and analysis of fecal samples is essentially conducted as described in Example

TABLE 64

Study Analysis	
PROCEDURE	TIME POINTS
DOSE	Day 1 and Day 28
CONCENTRATION ANALYSIS	
CLINICAL OBSERVATIONS	Twice Daily (cageside observations)
FOOD CONSUMPTION	Daily (qualitative)
BODY WEIGHT	Weekly
OPHTHALMOLOGY	Once during acclimation, Week 4, and Week 8

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TABLE 64-continued

Study Analysis	
PROCEDURE	TIME POINTS
ECGS/HR/BP	Once during acclimation, Week 4, and Week 8
HEMATOLOGY	Twice during acclimation, Day 2 (pre-dose), Day 15 (pre-dose), Day 29, Day 42, and Day 56
COAGULATION	Twice during acclimation, Day 2 (pre-dose), Day 15 (pre-dose), Day 29, Day 42, and Day 56
SERUM CHEMISTRY	Twice during acclimation, Day 2 (pre-dose), Day 15 (pre-dose), Day 29, Day 42, and Day 56
BODY (RECTAL) TEMPERATURE	Twice during acclimation (with at least 7 days between measurements); once weekly during dosing (~6 hrs post-dose), and Weeks 5 and 8
STOOL SAMPLE COLLECTION (BACTERIAL CULTURE)	Once during acclimation, prior to dosing on Days 2, 7, and 14, Day 29, Day 33, and Week 8
	Rectal/Fecal swabs are collected via cotton tip applicator; the cotton part of the swab is transferred to a tube with an appropriate broth/media and immediately put on wet ice.
	Fecal samples are stored at 2 to 8° C. until time of analysis.
CYTOKINE BLOOD COLLECTIONS	Once during acclimation, Days 1, 3, 7, 14 and 28 (6 hrs post-dose), and Day 56
ARCHIVE BLOOD SAMPLE COLLECTION (SAMPLES TO BE HELD FOR POSSIBLE ANALYSIS)	Once during acclimation, Days 1, 3, 7, 14 and 28 (6 hrs post-dose), and Day 56; Blood samples are processed to serum; samples are stored frozen.
NECROPSY & TISSUE COLLECTION	All animals (e.g., colon, intestine, cecum, liver, spleen)
ORGAN WEIGHTS	All animals
TISSUE COLLECTION FOR PK/PD ASSESSMENT	All animals
HISTOPATHOLOGY	Comparative (Anova/Bartlett's)
STATISTICAL ANALYSIS	

Example 31: Genetically Engineered Bacteria with HlyA Tag for Secretion of PME

Constructs for secretion of PME were generated as shown in Table 65. This sequences are subsequently tagged, e.g., with a HIS tag, e.g., inserted before the C terminal secretion sequence. *E. coli* are transformed with the constructs on a low-copy plasmid. Secreted PME are isolated from the media using affinity chromatography (His-Tag). PME molecular weight is confirmed by western blot. Activity of the purified enzyme is tested in an in vitro assay in a phenylalanine-containing buffer. Metabolites are measured over time as described in Examples 24-26.

TABLE 65

Secretion Sequences		
SEQ ID NO	Description	Sequence
SEQ ID NO: 52	HlyA Secretion tag	<u>LNPLINEISKIISAAGNFDVKEERAAASL</u> <u>LQLSGNASDFS YGRNSITLTASA</u>
SEQ ID NO: 53	PAL (upper case) expressed as fusion protein with the 53 amino acids of the C termini of alpha-hemolysin (hlyA) of <i>E. coli</i> CFT073 (lower case).	MKAKDVQPTIIINKNGLISLEDIYDIAIK QKKVEISTEITELLTHGREKLEEKLSNGE VIYGINTGFGGNANLVVPEKIAEHQONL LTFLSAGTGDYMSKPCIKASQFTMLLSVC KGWSATRPIVAQAIVDHIHNDIVPLVPRY GSVGASGDLIPLSYIARALCGIKVYVMG AEIDAAEAIKRAGLTPLSLKAKEGLALIN GTRVMSGISAITVIKLEKLPKASISAIAL AVEALLASHEHYDARIQQVKNHPGQNAVA SALRNLLAGSTQVNLLSGVKEQANKACRH QEITQLNDTLQEVYSIRCAPQVLGIVPES LATARKILEREVISANDNPLIDPENGDLV HGGNFMGQYVARTMDALKLDIALIANHLH AIVALMMDNRFSRGLPNSLSPTPGMYQGF KGVQLSQ TALVAAIRHDCASGIHTLATE QYNQDIVSLGLHAAQDVLEMEQKLRNIVS MTILVVQCAIHLRGNISEIAPETAKFYHA VREISSPLITDRALDEDIIRIADAIINDQ LPLPEIMLEE <u>lnplineiskiisaagnfdvkeeraaasl</u> <u>lqlsgnasdfs ygrnsitltasa*</u>
SEQ ID NO: 54	LAAD (uppercase) expressed as fusion protein with the 53 amino acids of the C termini of alpha- hemolysin (hlyA) of <i>E.</i> <i>coli</i> CFT073 (lower case)	MNISRRKLLLGVGAGVLGGAALVPMVR RDGKFVEAKSRASFVEGTQGALPKEADV IIGAGIQGINTAINLAERGMSVTILEKGQ IAGEQSGRAYSQIISYQTSPEIFPLHHYG KILWRGMNEKIGADTSYRTQGRVEALADE KALDKAQAWIKTAKAAGFDTPLNTRI I K GEELSNRLVGAQTPWTVAAPFEEDSGSVDP ETGTPALARYAKQIGVKIYTNCAVRGIET AGGKISDVVSEKGAIKTSQVVLAGGIWSR LFMGNMGIDIPTLNVYLSQQRVSGVPGAP RGNVHLFPNGIHFEQADGTAVAPRIFTS SIVKDSFLLGPKFMHLLGGGELPLEFSIG EDLFNSFKMPTSWNLDEKTPFEQFRVATA TQNTQHLDAVFQRMKTEFFVFEKSEVVER WGA VVSPTFDELPIISEVKEYPGLVINTA TVWGMTEGPAAAGEVTADIVMGKKPVIDPT PFS LDRFKK <u>lnplineiskiisaagnfdvkeeraaasl</u> <u>lqlsgnasdfs ygrnsitltasa</u>
SEQ ID NO: 55	HlyA secretion signal	CTTAATCCATTAATTAATGAATCAGCAA AATCATTT CAGCTGCAGGTAATTTTGATG TTAAAGAGGAAAGAGCTGCAGCTTCTTTA TTGCAGTTGTCCGGTAATGCCAGTGATTT TTCATATGGACGGAACCTCAATACTTTGA CAGCATCAGCATAA
SEQ ID NO: 56	LAAD (bold italics) driven by ParaBAD (underlined) with C terminal HlyA Secretion tag (bold)	<u>Acttttcatactcccgccattcagagaag</u> <u>aaaccaattgtccatattgcatcagacat</u> <u>tgccgctcactgcgctcttttactggctctt</u> <u>ctcgctaaccceaacccggtaaccccgctta</u> <u>ttaaaagcattctgtacaaagcgggacc</u> <u>aaagccatgacaaaaacgcgtaacaaaag</u> <u>tgtctataatcacggcgagaaaa</u> <u>atgaacatttcaaggagaaagctactttt</u> <u>aggtgttggtgctgcgggcgttttagcag</u> <u>gtggtgcggttttagttccaatggttcgc</u> <u>cgtgacggc aaa tttgtggaagctaaatc</u> <u>aagagcatcat tttgttgaaggtagcgaag</u> <u>gggctcttcttaaagaagcagatgttagtg</u> <u>attattggtgccggtattcaaggatcat</u> <u>gaccgctattaaccttgctgaacgttggt</u> <u>tgagtgctcactatcttagaaaagggtcag</u> <u>attgccggtgagcaatcaggccgtgcata</u> <u>cagccaaattattagttaccaaactcgc</u> <u>cagaaatcttccattacaccattatggg</u> <u>aaaaattatggcgtggcatgaatgagaa</u> <u>aattggtgcggataccagttatcgacttc</u> <u>aaggtcgtgtagaagcgtggcagatgaa</u> <u>aaagcattagataaagctcaagcgtggat</u> <u>caaacagctaaagcggcaggttttg</u>

TABLE 65-continued

Secretion Sequences		
SEQ ID NO	Description	Sequence
		atacaccattaaatactcgcatcattaaa ggtgaagagctatcaaatcgcttagtcgg tgctcaaacgccatggactgttgctgcat ttgaagaagattcaggctctgtgatcct gaaacaggcacacctgactcgctcgta tgccaaacaaatcggtgtgaaaatttata ccaactgtgcagtaagaggtattgaaact gcgggtggtaaaatctctgagtgtgag tgagaaagggcgattaaaacgtctcaag ttgtactcgctgggggtaactggctcggt ttatttatgggcaatatgggtattgatat cccaacgctcaatgtatactatcacaaac aacgtgtctcaggggttctgggtgcacca cgtggttaatgtgcatctacctaattggtat tcatctccgcgaacaagcggtggtactt atgcggttgaccacgtaactttacgagt tcaatagtcagaatagcttctgctagg gcctaaatttatgcactattagtggtgag gagagtaccgttggaattctctattgggt gaagatctatttaattcatttaaaatgcc gacctcttggaatttagatgaaaaaacac cattcgaaacaattccgagttgccacggca acacaaaaacgcacacacttagatgctgt ttccaaagaatgaaaacagaattccag tatttgaaaaatcagaagttgtgaaacgt tgggggtgccgttgtgagtcacaatttga tgaattacctatcattctctgaggtcaaa aatcccgaggttagtgattaacacggca acagtggtgggtatgacagaaggcccggt agcgggtgaaagtgaccgctgatattgtca tgggcaagaaacctgttattgatccaacg ccgttttagtttgatcgctttaaagaagta aCTTAATCCATTAAATTAATGAAATCAGCA AAATCATTTACAGCTGCAGGTAATTTGAT GTTAAAGAGGAAAGAGCTGCAGCTTCTTT ATTGCAGTTGTCCGTAATGCCAGTGATT TTTCATATGGACGGAACCAATAACTTTG ACAGCATCAGCATAA
SEQ ID NO: 57	PfnrS-PAL3 with C terminal secretion tag. PfnrS (bolded lower case), PAL3 sequence is underlined upper case C terminal secretion tag is bold uppercase	GGTACCagttgttcttattgggtggtgttg ctttatggttgcatcgtagtaaatgggttg taacaaaagcaattttccggctgtctgt atacaaaacgccgtaaagtttgagcgaa gtcaataaaactctctacccattcagggca atatctctcttGGATCCctctagaaataa ttttgtttaactttaagaaggagatatac atATGAAAGCTAAAGATGTTTCAGCCAACC ATTATTATTAAATAAAATGGCCTTATCTC TTTGGAAGATATCTATGACATTGCGATAA AACAAAAAAGTAGAATATCAACGGAG ATCACTGAACCTTTTGACGCATGGTCGTGA AAAATTAGAGGAAAAATTAAATTCAGGAG AGGTTATATATGGAATCAATACAGGATTT GGAGGGAATGCCAATTAGTTGTGCCATT TGAGAAAATCGCAGAGCATCAGCAAAATC TGTTAACTTTTCTTTCTGCTGGTACTGGG GACTATATGTCCAAACCTTGATTAAAGC GTCACAATTTACTATGTTACTTTCTGTTT GCAAAGGTTGGTCTGCAACCAGACCAATT GTCGCTCAAGCAATTGTTGATCATATTAA TCATGACATTGTTCTCTGGTTCCTCGCT ATGGCTCAGTGGGTGCAAGCGGTGATTTA ATTCTTTATCTTATATTGCACGAGCATT ATGTTGATTCGGCAAAGTTTATTATATGG GCGCAGAAATTGACGCTGCTGAAGCAATT AAACGTGCAGGGTTGACACCATTAATCGTT AAAAGCCAAAGAAAGTCTTGTCTGATTA ACGGCACCCGGGTAATGTCAGGAATCAGT GCAATCACCGTCATTAACTGGAAAAACT ATTTAAAGCCTCAATTTCTGCGATTGCCC TTGCTGTTGAAGCATTACTTGCATCTCAT GAACATTATGATGCCCCGATTCAACAAGT AAAAATCATCCTGGTCAAACGCGGTGG CAAGTGCATTGCGTAATTTATTGGCAGGT TCACGCGAGTTAATCTATTCTGGGGT TAAAGACAAGCCAATAAAGCTTGTGCTC

TABLE 65-continued

Secretion Sequences		
SEQ ID NO	Description	Sequence
		ATCAAGAAATTACCCAACTAAATGATACC TTACAGGAAGTTTATTCAATTCGCTGTGC ACCACAAGTATTAGGTATAGTGCCAGAAT CTTTAGCTACCGCTCGGAAATATTGGAA CGGGAAGTTATCTCAGCTAATGATAATCC ATTGATAGATCCAGAAAATGGCGATGTTT TACACGGTGGAAATTTTATGGGCAATAT GTCGCCCGAACAATGGATGCATTAAAACT GGATATTGCTTTAATTGCCAATCATCTTC ACGCCATTGTGGCTCTTATGATGGATAAC CGTTTCTCTCGTGGATTACCTAATTCAC GAGTCCGACACCCGGCATGTATCAAGGTT TTAAAGGCGTCCAACTTTCTCAAACCGCT TTAGTTGCTGCAATTCGCCATGATTGTGC TGCATCAGGTATTATACCCCTCGCCACAG AACAATACAATCAAGATATTGTCTAGTTTA GGTCTGCATGCCGCTCAAGATGTTTTAGA GATGGAGCAGAAAATTACGCAATATTGTTT CAATGACAATTCTGGTAGTTTGTCTAGGCC ATTCATCTTCGCGGCAATATTAGTGAAAT TCGCCTGAACTGCTAAATTTTACCATG CAGTACGCGAAATCAGTTCTCCTTTGATC ACTGATCGTGCCTGGATGAAGATATAAT CCGCATGCGGATGCAATTATTAATGATC AACTTCCTCTGCCAGAAATCATGCTGGAA GAATAACTTAATCCATTAAATGAAAT CAGCAAAATCATTTCAGCTGCAGGTAATT TTGATGTTAAAGAGGAAAGAGCTGCAGCT TCTTTATTGCAGTTGTCCGGTAATGCCAG TGATTTTTCATATGGACGGAACTCAATAA CTTTGACAGCATCAGCATAA

TABLE 66

HlyB and HlyD protein sequences		
SEQ ID NO:	HlyB protein	
58		MDSCHKIDYGLYALEILAQYHNVSVNPEE IKHRFDTDGTGLGLTSWLLAAKSLLEKVK QVKKTIIDRLNFISLPALVWREDGRHFILT KVSKEANRYLIFDLEQRNPRVLEQSEFEA LYQGHILIASRSSVTGKLAKFDFWTFIP AIIKYRKIFIETLVVSVFLQLFALITPLF PQVVMKVLVHRGFS TLNVTVALSVVVV FEIILSGLRTYIFAHSTSRIDVELGAKLF RHLLALPISYFESRRVGD TVARVRELDQI RNFLTQQALTSVLDLLFSFIFFAVMWYYS PKLTLVILFSLPCYAAWSVFISPILRRL DDKFSRNADNQSFVSVTAINTIKAMAV SPQMTNIWDKQLAGYVAAGFKVTVLATIG QQGIQLIQKTVMI INLWLGHLVISGDLIS IGQLIAFNMLAQIVAPVIRLAQIWQDPQ QVGISVTRLGDLNSPTESYHGKLLALPEI NGNITFRNIRFRYKPDSPVILDNINLSIK QGEVIGIVGRSGSGKSTLTKLIQRFYIPE NGQVLIDGHD LALADPNWLRQVGVVLQD NVLLNRSI IDNISLANPGMSVEKVIYAAK LAGAHDFISELREGYNTIVGQAGLSGG QRQRIAIARALVNNPKILIFDEATSALDY ESEHIIMRNMHKICKGRTVIIIAHRLSTV KNADRIIVMEKGKIVEQKHKELLSEPESE LYSYLYQLQSD
SEQ ID NO:	HlyD protein	
59		MKTWLMGFSEFLRYKLVWSETWKIRKQL DTPVREKDENEFLPAHLELIETPVSRRR LVAYFIMGFLVIAVILSVLGQVEIVATAN GKLTLSGRSKEIKPIENSIVKEIIVKEGE SVRKGDVLLKLTALGAEADTLKTQSSLLQ TRLEQTRYQILSRSEIENKLPKLDPDEP YFQNVSEEEVLRSLI KEQFSTWQNKY QKELNLDKKRAERLTILARINRYENLSRV EKSRLDDFRSLHKKQAIKHAVLEQENKY VEAANELRVYKSLQEQIESEILSAKEEYQ

TABLE 66-continued

HlyB and HlyD protein sequences	
	LVTQLFKNEILDKLRQTTDNIELLTLELE KNEERQQASVIRAPVSGKVQQLKVHTEGG VVTTAETLMVIVPEDDTLEVLTALVQNKDI GFINVGNQNAIKVEAFPYTRYGYLVGKVK NINLDAIEDQKLGVLVFNVIVSVEENDLST GNKHIPSSGMAVTAEIKTGMRSVISYLL SPLEESVTESLHER

Example 32: Genetically Engineered Bacteria
Comprising Additional Constructs

TABLE 67

Description	Sequence	SEQ ID NO
phenylalanine transporter [<i>Escherichia coli</i> str. K-12 substr. MG1655] Acc. No. NP_415108 (PheP)	MKNASTVSEDTASNQEPTLHRGLHNRHIQLIA LGGAIGTGLFLGIGPAIQMAGPAVLLGYGVAG IIAFLIMRQLGEMVVEEPVSGSFAHFAYKYWG PFAGFLSGWNVWVFLVGMALTAAGIYMQY WFPDVPWIIWAAAFIIINAVNLVNVRLYGET EFWFALIKVLAIIGMIGFGLWLLFSGHGGEKA SIDNLWRYGGFFATGWNGLILSLAVIMFSFGG LELIGITAAEARDPEKSIKAVNQVVRILLF YIGSLVLLALYPWVEVKSNSSPFVMIFHNLD SNVVASALNFVILVASLSVYNSGVYSNSRMLF GLSVQGNAPKFLTRVSRGVPINSLMLSGAIT SLVVLINYLLOKAFGLLMALVVATLLLNWIM ICLAHLRFRAAMRRQGRETFKALLYPPGNYL CIAFLGMILLMCTMDMRLSAILLPVWIVFL FMAFKTLRRK	60
aromatic amino acid transport protein AroP [<i>Escherichia coli</i> F11] Acc. NO: EDV65095	MEGQQHGEQLKRGLKNRHIQLIALGGAIGTGL FLGSASVIQSAGPGIILGYAIGFIAFLIMRQ LGEMVVEEPVAGSFSHFAYKYWGSFAGFASGW NYWVLYVLVAMAELTAVGKYIQFWYPEIPTV SAAVFFVVINAINLTNVKVFGESEFWFAIKV IAVVAMIIFGAWLLFSGNGGPQASVSNLWDQG GFLPHGFTGLVMMMAIIMFSFGGLELVGITAA EADNPEQSIKATNQVIYRILIFYIGSLAVLL SLMPWTRVTADTSPFVIFHELGDTFVANALN IVVLTAALSVYNSCVYCNRMFLGLAQGNAP KALASVDKRGVPVNTILVSALVTALCVLINYL APESAFGLLMALVVSALVINWAMISLAHMKFR RAKQEQGVVTRFPALLYPLGNVWCLLFMAAVL VIMLMTPGMAISVYLIPVWLIVLGIGYLFKEK TAKAVKAH	61
FNRS promoter (bold, lower case) -PheP (upper case underlined)	GGTACCagttgttctctattggtggtgtt tatggttgcatcgtagtaaatggttgtaacaa aagcaatttttccggtctgtctatacaaaaa cgccgtaagtttgagcgaagtcataaaactc tctacccattcagggcaatatctctcttGGAT CCctctagaaataattttgttttaactttaaga aggagataacatATGAAAAACGCGTCAACCG TATCGGAAGATACTGCGTCGAATCAAGAGCCG ACGCTTCATCGCGGATTACATAACCGTCATAT TCAACTGATTGCGTTGGGTGGCGCAATTGGTA CTGGTCTGTTTCTTGCCATGGCCCCGGCGATT CAGATGGCGGGTCCGGCTGATGCTGGGCTA CGGCGTCGCCGGGATCATCGCTTTCCTGATTA TGCGCCAGCTTGGCGAAATGGTGGTTGAGGAG CCGGTATCCGGTTCATTGCCCCACTTGCCTA TAAATACTGGGACCGTTTGCGGGCTTCCTCT CTGGCTGGAACACTGGGTAATGTTTCGTGCTG GTGGGAATGGCAGAGCTGACCGCTGCGGGCAT CTATATGCAGTACTGGTTCCCGGATGTTCCAA CGTGGATTGGGCTGCCGCTTCTTTATTATC ATCAACGCCGTTAACTGGTGAACGTGCGCTT ATATGGCGAAACCGAGTCTGGTTTGCCTTGA TTAAAGTGCTGGCAATCATCGGTATGATCGGC TTTGGCTGTGGCTGCTGTTTTCTGGTCACGG CGGCGAGAAAGCCAGTATCGACAACCTCTGGC GCTACGGTGGTTTCTCGCCACCGCTGGAAT GGGCTGATTTGTCTGCTGGCGTAATTATGTT	62

TABLE 67-continued

Description	Sequence	SEQ ID NO
	CTCCTTCGGCGGTCTGGAGCTGATTGGGATTA CTGCCGCTGAAGCGCGGATCCGGAAAAAGC ATTCCAAAAGCGGTAAATCAGGTGGTGATCG CATCCTGCTGTTTACATCGGTTCAGTGGTGG TTTACTGGCGCTCTATCCGTGGGTGGAAGTG AAATCCAACAGTAGCCCGTTTGTGATGATTTT CCATAATCTCGACAGCAACGTGGTAGCTTCTG CGCTGAACCTCGTCAATCTGGTAGCATCGCTG TCAGTGATAACAGCGGGTTTACTCTAACAG CCGCATGCTGTTTGGCCTTCTGTGCAGGGTA ATGCCGCCAAGTTTTTGACTCGCGTCAGCCGT CGCGGTGTGCCGATTAATCGCTGATGCTTTC CGGAGCGATCACTTCGCTGGTGGTGTTAATCA ACTATCTGCTGCCGCAAAAAGCGTTTGGTCTG CTGATGGCGCTGGTGGTAGCAACGCTGCTGTT GAACTGGATTATGATCTGTCTGGCGCATCTGC GTTTTCTGTCAGCGATGCGACGTCAGGGCGCT GAAACACAGTTTAAGGCGCTGCTCTATCCGTT CGGCAACTATCTCTGCATTGCCTTCCTCGGCA TGATTTTGCTGCTGATGTGCACGATGGATGAT ATGCGCTTGTGACGATCCTGCTGCCGGTGTG GATTGTATTCTGTTTATGGCATTTAAACGC TGCGTCGGAAATAA	
FNRS promoter (bold, lower case)-AroP (upper case underlined, codon optimized)	GGTACCagttgttctctattggtggtgttgctt tatggttgcatcgtagtaaatggttgtaacaa aagcaatttttccggtgtctgtatacaaaaa cgccgtaaaagtttgagcgaagtcaataaactc tctacccattcagggcaatatctctcttGGAT CCctctagaaataatttgtttaactttaaga aggagataatacatATGGAGGGGACAGCATG GGGAGCAACTGAAGCGCGGTTAAAAATCGT CACATTCAATTAATCGCGCTGGGCGGAGCAAT TGGTACGGGATTGTTCTGGGTTAGCGAGCG TCATCCAATCGGCAAGTCCAGGGATCATCTTG GGATATGCGATCGCAGGCTTTATCGCTTTTCT TATTATGCGCAATTAGGTGAGATGGTGGTCG AGGAGCCTGTAGCTGGCTCCTTCTCACATTTC GCGTACAAGTATTGGGGATCCTTTGCGGGATT TGCTTCTGGTTGGAATATTGGGTTCTTTATG TCCTGGTGGCCATGGCGGAGCTGACCGCGGTT GGAAAAATATCCAGTTCTGGTACCCCGAGAT CCCGACGTGGGTCTCAGCCGCGGTATTCTTTG TTGTTATCAATGCAATCAATTTAACCAACGTA AAAGTATTGGTGAAATGGAGTCTCGGTTCCG GATTATCAAAGTAATTGCCGTAGTTGCTATGA TTATTTTGGGCATGGTTGCTTTTCTCAGGA AATGGCGGACCACAAGCGTCGGTTTCAAACCT GTGGGATCAAGGGGATTTCCTGCCGCACGGAT TTACGGGCTTGGTGATGATGAGGCTATCATT ATGTTTCTTTCGGTGGTCTTGAATTAGTGGG TATTACCGCAGCAGAGGCAGATAATCCCGAAC AAAGCATCCAAAAGCTACTAACCAGTTATT TACCGTATCCTGATTTTTATATTGGTTCTCT GGCAGTCTGCTTTCCTAATGCCCTGGACAC GTGTAAACGCCGATACATCCCCTTTGTACTT ATCTTTCAGAACTGGGAGACAGTTCGTTCGC CAATGCATTAAACATGTTGTGCTGACAGCTG CCTTATCTGTGTATAATAGCTGCGTTTATTGC AATTACGTATGTTATTCGGGCTTGCTCAGCA GGGTAAACGCGCAAAGCGTTGGCCTCAGTAG ATAAGCGCGAGTGCTGTAAATACAATTTTG GTCAGCGCATTAGTCACGGCTCTTTCGCTTCT GATTAACTATCTGGCTCCTGAAAGCGCATTCG GATTACTTATGGCCCTGGTTGTTTCCGCCCTG GTTATCAATTGGGCAATGATTAGTTTGGCACA TATGAAGTTCGCGGTGCTAAACAAGAACAG GTGTCGTAACCTGTTTCCCTGCCTTATTGTAT CCGCTGGGGAATTGGGTATGCCTTCTTTTAT GGCCGAGTACTGGTAATTATGTTGATGACGC CCGGCATGGCTATTAGTGATACCTTATTCCG GTATGGTTAATCGTCTGGGTATCGGCTACTT ATTTAAAGAAAAACAGCAAAAGCCGTAAAGG CTCAT	63

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 66

<210> SEQ ID NO 1

<211> LENGTH: 567

<212> TYPE: PRT

<213> ORGANISM: *Anabaena variabilis*

<400> SEQUENCE: 1

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Met Lys Thr Leu Ser Gln Ala Gln Ser Lys Thr Ser Ser Gln Gln Phe
 1              5              10              15

Ser Phe Thr Gly Asn Ser Ser Ala Asn Val Ile Ile Gly Asn Gln Lys
      20              25              30

Leu Thr Ile Asn Asp Val Ala Arg Val Ala Arg Asn Gly Thr Leu Val
      35              40              45

Ser Leu Thr Asn Asn Thr Asp Ile Leu Gln Gly Ile Gln Ala Ser Cys
      50              55              60

Asp Tyr Ile Asn Asn Ala Val Glu Ser Gly Glu Pro Ile Tyr Gly Val
      65              70              75              80

Thr Ser Gly Phe Gly Gly Met Ala Asn Val Ala Ile Ser Arg Glu Gln
      85              90              95

Ala Ser Glu Leu Gln Thr Asn Leu Val Trp Phe Leu Lys Thr Gly Ala
      100             105             110

Gly Asn Lys Leu Pro Leu Ala Asp Val Arg Ala Ala Met Leu Leu Arg
      115             120             125

Ala Asn Ser His Met Arg Gly Ala Ser Gly Ile Arg Leu Glu Leu Ile
      130             135             140

Lys Arg Met Glu Ile Phe Leu Asn Ala Gly Val Thr Pro Tyr Val Tyr
      145             150             155             160

Glu Phe Gly Ser Ile Gly Ala Ser Gly Asp Leu Val Pro Leu Ser Tyr
      165             170             175

Ile Thr Gly Ser Leu Ile Gly Leu Asp Pro Ser Phe Lys Val Asp Phe
      180             185             190

Asn Gly Lys Glu Met Asp Ala Pro Thr Ala Leu Arg Gln Leu Asn Leu
      195             200             205

Ser Pro Leu Thr Leu Leu Pro Lys Glu Gly Leu Ala Met Met Asn Gly
      210             215             220

Thr Ser Val Met Thr Gly Ile Ala Ala Asn Cys Val Tyr Asp Thr Gln
      225             230             235             240

Ile Leu Thr Ala Ile Ala Met Gly Val His Ala Leu Asp Ile Gln Ala
      245             250             255

Leu Asn Gly Thr Asn Gln Ser Phe His Pro Phe Ile His Asn Ser Lys
      260             265             270

Pro His Pro Gly Gln Leu Trp Ala Ala Asp Gln Met Ile Ser Leu Leu
      275             280             285

Ala Asn Ser Gln Leu Val Arg Asp Glu Leu Asp Gly Lys His Asp Tyr
      290             295             300

Arg Asp His Glu Leu Ile Gln Asp Arg Tyr Ser Leu Arg Cys Leu Pro
      305             310             315             320

Gln Tyr Leu Gly Pro Ile Val Asp Gly Ile Ser Gln Ile Ala Lys Gln
      325             330             335

Ile Glu Ile Glu Ile Asn Ser Val Thr Asp Asn Pro Leu Ile Asp Val
      340             345             350

Asp Asn Gln Ala Ser Tyr His Gly Gly Asn Phe Leu Gly Gln Tyr Val
      355             360             365

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Gly Met Gly Met Asp His Leu Arg Tyr Tyr Ile Gly Leu Leu Ala Lys
 370 375 380
 His Leu Asp Val Gln Ile Ala Leu Leu Ala Ser Pro Glu Phe Ser Asn
 385 390 395 400
 Gly Leu Pro Pro Ser Leu Leu Gly Asn Arg Glu Arg Lys Val Asn Met
 405 410 415
 Gly Leu Lys Gly Leu Gln Ile Cys Gly Asn Ser Ile Met Pro Leu Leu
 420 425 430
 Thr Phe Tyr Gly Asn Ser Ile Ala Asp Arg Phe Pro Thr His Ala Glu
 435 440 445
 Gln Phe Asn Gln Asn Ile Asn Ser Gln Gly Tyr Thr Ser Ala Thr Leu
 450 455 460
 Ala Arg Arg Ser Val Asp Ile Phe Gln Asn Tyr Val Ala Ile Ala Leu
 465 470 475 480
 Met Phe Gly Val Gln Ala Val Asp Leu Arg Thr Tyr Lys Lys Thr Gly
 485 490 495
 His Tyr Asp Ala Arg Ala Cys Leu Ser Pro Ala Thr Glu Arg Leu Tyr
 500 505 510
 Ser Ala Val Arg His Val Val Gly Gln Lys Pro Thr Ser Asp Arg Pro
 515 520 525
 Tyr Ile Trp Asn Asp Asn Glu Gln Gly Leu Asp Glu His Ile Ala Arg
 530 535 540
 Ile Ser Ala Asp Ile Ala Ala Gly Gly Val Ile Val Gln Ala Val Gln
 545 550 555 560
 Asp Ile Leu Pro Cys Leu His
 565

<210> SEQ ID NO 2
 <211> LENGTH: 567
 <212> TYPE: PRT
 <213> ORGANISM: *Anabaena variabilis*

<400> SEQUENCE: 2

Met Lys Thr Leu Ser Gln Ala Gln Ser Lys Thr Ser Ser Gln Gln Phe
 1 5 10 15
 Ser Phe Thr Gly Asn Ser Ser Ala Asn Val Ile Ile Gly Asn Gln Lys
 20 25 30
 Leu Thr Ile Asn Asp Val Ala Arg Val Ala Arg Asn Gly Thr Leu Val
 35 40 45
 Ser Leu Thr Asn Asn Thr Asp Ile Leu Gln Gly Ile Gln Ala Ser Cys
 50 55 60
 Asp Tyr Ile Asn Asn Ala Val Glu Ser Gly Glu Pro Ile Tyr Gly Val
 65 70 75 80
 Thr Ser Gly Phe Gly Gly Met Ala Asn Val Ala Ile Ser Arg Glu Gln
 85 90 95
 Ala Ser Glu Leu Gln Thr Asn Leu Val Trp Phe Leu Lys Thr Gly Ala
 100 105 110
 Gly Asn Lys Leu Pro Leu Ala Asp Val Arg Ala Ala Met Leu Leu Arg
 115 120 125
 Ala Asn Ser His Met Arg Gly Ala Ser Gly Ile Arg Leu Glu Leu Ile
 130 135 140
 Lys Arg Met Glu Ile Phe Leu Asn Ala Gly Val Thr Pro Tyr Val Tyr
 145 150 155 160
 Glu Phe Gly Ser Ile Gly Ala Ser Gly Asp Leu Val Pro Leu Ser Tyr
 165 170 175

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Ile Thr Gly Ser Leu Ile Gly Leu Asp Pro Ser Phe Lys Val Asp Phe
      180                      185                      190

Asn Gly Lys Glu Met Asp Ala Pro Thr Ala Leu Arg Gln Leu Asn Leu
      195                      200                      205

Ser Pro Leu Thr Leu Leu Pro Lys Glu Gly Leu Ala Met Met Asn Gly
      210                      215                      220

Thr Ser Val Met Thr Gly Ile Ala Ala Asn Cys Val Tyr Asp Thr Gln
      225                      230                      235                      240

Ile Leu Thr Ala Ile Ala Met Gly Val His Ala Leu Asp Ile Gln Ala
      245                      250                      255

Leu Asn Gly Thr Asn Gln Ser Phe His Pro Phe Ile His Asn Ser Lys
      260                      265                      270

Pro His Pro Gly Gln Leu Trp Ala Ala Asp Gln Met Ile Ser Leu Leu
      275                      280                      285

Ala Asn Ser Gln Leu Val Arg Asp Glu Leu Asp Gly Lys His Asp Tyr
      290                      295                      300

Arg Asp His Glu Leu Ile Gln Asp Arg Tyr Ser Leu Arg Cys Leu Pro
      305                      310                      315                      320

Gln Tyr Leu Gly Pro Ile Val Asp Gly Ile Ser Gln Ile Ala Lys Gln
      325                      330                      335

Ile Glu Ile Glu Ile Asn Ser Val Thr Asp Asn Pro Leu Ile Asp Val
      340                      345                      350

Asp Asn Gln Ala Ser Tyr His Gly Gly Asn Phe Leu Gly Gln Tyr Val
      355                      360                      365

Gly Met Gly Met Asp His Leu Arg Tyr Tyr Ile Gly Leu Leu Ala Lys
      370                      375                      380

His Leu Asp Val Gln Ile Ala Leu Leu Ala Ser Pro Glu Phe Ser Asn
      385                      390                      395                      400

Gly Leu Pro Pro Ser Leu Leu Gly Asn Arg Glu Arg Lys Val Asn Met
      405                      410                      415

Gly Leu Lys Gly Leu Gln Ile Cys Gly Asn Ser Ile Met Pro Leu Leu
      420                      425                      430

Thr Phe Tyr Gly Asn Ser Ile Ala Asp Arg Phe Pro Thr His Ala Glu
      435                      440                      445

Gln Phe Asn Gln Asn Ile Asn Ser Gln Gly Tyr Thr Ser Ala Thr Leu
      450                      455                      460

Ala Arg Arg Ser Val Asp Ile Phe Gln Asn Tyr Val Ala Ile Ala Leu
      465                      470                      475                      480

Met Phe Gly Val Gln Ala Val Asp Leu Arg Thr Tyr Lys Lys Thr Gly
      485                      490                      495

His Tyr Asp Ala Arg Ala Cys Leu Ser Pro Ala Thr Glu Arg Leu Tyr
      500                      505                      510

Ser Ala Val Arg His Val Val Gly Gln Lys Pro Thr Ser Asp Arg Pro
      515                      520                      525

Tyr Ile Trp Asn Asp Asn Glu Gln Gly Leu Asp Glu His Ile Ala Arg
      530                      535                      540

Ile Ser Ala Asp Ile Ala Ala Gly Gly Val Ile Val Gln Ala Val Gln
      545                      550                      555                      560

Asp Ile Leu Pro Cys Leu His
      565

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<210> SEQ ID NO 3

<211> LENGTH: 532

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<212> TYPE: PRT

<213> ORGANISM: Photorhabdus luminescens

<400> SEQUENCE: 3

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Met Lys Ala Lys Asp Val Gln Pro Thr Ile Ile Ile Asn Lys Asn Gly
1      5      10      15

Leu Ile Ser Leu Glu Asp Ile Tyr Asp Ile Ala Ile Lys Gln Lys Lys
      20      25      30

Val Glu Ile Ser Thr Glu Ile Thr Glu Leu Leu Thr His Gly Arg Glu
      35      40      45

Lys Leu Glu Glu Lys Leu Asn Ser Gly Glu Val Ile Tyr Gly Ile Asn
50      55      60

Thr Gly Phe Gly Gly Asn Ala Asn Leu Val Val Pro Phe Glu Lys Ile
65      70      75      80

Ala Glu His Gln Gln Asn Leu Leu Thr Phe Leu Ser Ala Gly Thr Gly
      85      90      95

Asp Tyr Met Ser Lys Pro Cys Ile Lys Ala Ser Gln Phe Thr Met Leu
100     105     110

Leu Ser Val Cys Lys Gly Trp Ser Ala Thr Arg Pro Ile Val Ala Gln
115     120     125

Ala Ile Val Asp His Ile Asn His Asp Ile Val Pro Leu Val Pro Arg
130     135     140

Tyr Gly Ser Val Gly Ala Ser Gly Asp Leu Ile Pro Leu Ser Tyr Ile
145     150     155     160

Ala Arg Ala Leu Cys Gly Ile Gly Lys Val Tyr Tyr Met Gly Ala Glu
165     170     175

Ile Asp Ala Ala Glu Ala Ile Lys Arg Ala Gly Leu Thr Pro Leu Ser
180     185     190

Leu Lys Ala Lys Glu Gly Leu Ala Leu Ile Asn Gly Thr Arg Val Met
195     200     205

Ser Gly Ile Ser Ala Ile Thr Val Ile Lys Leu Glu Lys Leu Phe Lys
210     215     220

Ala Ser Ile Ser Ala Ile Ala Leu Ala Val Glu Ala Leu Leu Ala Ser
225     230     235     240

His Glu His Tyr Asp Ala Arg Ile Gln Gln Val Lys Asn His Pro Gly
245     250     255

Gln Asn Ala Val Ala Ser Ala Leu Arg Asn Leu Leu Ala Gly Ser Thr
260     265     270

Gln Val Asn Leu Leu Ser Gly Val Lys Glu Gln Ala Asn Lys Ala Cys
275     280     285

Arg His Gln Glu Ile Thr Gln Leu Asn Asp Thr Leu Gln Glu Val Tyr
290     295     300

Ser Ile Arg Cys Ala Pro Gln Val Leu Gly Ile Val Pro Glu Ser Leu
305     310     315     320

Ala Thr Ala Arg Lys Ile Leu Glu Arg Glu Val Ile Ser Ala Asn Asp
325     330     335

Asn Pro Leu Ile Asp Pro Glu Asn Gly Asp Val Leu His Gly Gly Asn
340     345     350

Phe Met Gly Gln Tyr Val Ala Arg Thr Met Asp Ala Leu Lys Leu Asp
355     360     365

Ile Ala Leu Ile Ala Asn His Leu His Ala Ile Val Ala Leu Met Met
370     375     380

Asp Asn Arg Phe Ser Arg Gly Leu Pro Asn Ser Leu Ser Pro Thr Pro
385     390     395     400

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Gly Met Tyr Gln Gly Phe Lys Gly Val Gln Leu Ser Gln Thr Ala Leu
 405 410 415
 Val Ala Ala Ile Arg His Asp Cys Ala Ala Ser Gly Ile His Thr Leu
 420 425 430
 Ala Thr Glu Gln Tyr Asn Gln Asp Ile Val Ser Leu Gly Leu His Ala
 435 440 445
 Ala Gln Asp Val Leu Glu Met Glu Gln Lys Leu Arg Asn Ile Val Ser
 450 455 460
 Met Thr Ile Leu Val Val Cys Gln Ala Ile His Leu Arg Gly Asn Ile
 465 470 475 480
 Ser Glu Ile Ala Pro Glu Thr Ala Lys Phe Tyr His Ala Val Arg Glu
 485 490 495
 Ile Ser Ser Pro Leu Ile Thr Asp Arg Ala Leu Asp Glu Asp Ile Ile
 500 505 510
 Arg Ile Ala Asp Ala Ile Ile Asn Asp Gln Leu Pro Leu Pro Glu Ile
 515 520 525
 Met Leu Glu Glu
 530

<210> SEQ ID NO 4

<211> LENGTH: 514

<212> TYPE: PRT

<213> ORGANISM: Photorhabdus luminescens

<400> SEQUENCE: 4

Met Lys Gln Leu Thr Ile Tyr Pro Gly Lys Leu Thr Leu Asp Glu Leu
 1 5 10 15
 Arg Gln Val Tyr Leu Gln Pro Val Lys Ile Thr Leu Asp Ser Gln Ile
 20 25 30
 Phe Pro Ala Ile Glu Arg Ser Val Glu Cys Val Asn Ala Ile Leu Ala
 35 40 45
 Glu Asn Arg Thr Ala Tyr Gly Ile Asn Thr Gly Phe Gly Leu Leu Ala
 50 55 60
 Ser Thr Arg Ile Glu Glu Asp Asn Leu Glu Lys Leu Gln Arg Ser Leu
 65 70 75 80
 Val Val Ser His Ala Ala Gly Val Gly Lys Ala Leu Asp Asp Asn Met
 85 90 95
 Thr Arg Leu Ile Met Val Leu Lys Ile Asn Ser Leu Ser Arg Gly Tyr
 100 105 110
 Ser Gly Ile Arg Leu Ala Val Ile Gln Ala Leu Ile Ala Leu Val Asn
 115 120 125
 Ala Glu Ile Tyr Pro His Ile Pro Cys Lys Gly Ser Val Gly Ala Ser
 130 135 140
 Gly Asp Leu Ala Pro Leu Ala His Met Ser Leu Leu Leu Gly Glu
 145 150 155 160
 Gly Gln Ala Arg Tyr Gln Gly Glu Trp Leu Pro Ala Lys Glu Ala Leu
 165 170 175
 Ala Lys Ala Asn Leu Gln Pro Ile Thr Leu Ala Ala Lys Glu Gly Leu
 180 185 190
 Ala Leu Leu Asn Gly Thr Gln Val Ser Thr Ala Phe Ala Leu Arg Gly
 195 200 205
 Leu Phe Glu Ala Glu Asp Leu Leu Ala Ala Ala Ile Val Cys Gly Ser
 210 215 220
 Leu Ser Val Glu Ala Ala Leu Gly Ser Arg Lys Pro Phe Asp Ala Arg

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225	230	235	240
Val His Val Val Arg Gly Gln Gln Gly Gln Ile Asp Val Ala Ala Leu	245	250	255
Tyr Arg His Val Leu Glu Glu Ser Ser Glu Leu Ser Asp Ser His Ile	260	265	270
Asn Cys Pro Lys Val Gln Asp Pro Tyr Ser Leu Arg Cys Gln Pro Gln	275	280	285
Val Met Gly Ala Cys Leu Thr Gln Leu Arg His Ala Ala Asp Val Ile	290	295	300
Leu Thr Glu Ala Asn Ala Val Ser Asp Asn Pro Leu Val Phe Ala Glu	305	310	315
Gln Gly Glu Val Ile Ser Gly Gly Asn Phe His Ala Glu Pro Val Ala	325	330	335
Met Ala Ser Asp Asn Leu Ala Leu Val Leu Ala Glu Ile Gly Ala Leu	340	345	350
Ser Glu Arg Arg Ile Ala Leu Leu Met Asp Ser His Met Ser Gln Leu	355	360	365
Pro Pro Phe Leu Val Glu Asn Gly Gly Val Asn Ser Gly Phe Met Ile	370	375	380
Ala Gln Val Thr Ala Ala Ala Leu Ala Ser Glu Asn Lys Ala Leu Ala	385	390	395
His Pro Ala Ser Val Asp Ser Leu Pro Thr Ser Ala Asn Gln Glu Asp	405	410	415
His Val Ser Met Ala Pro Ala Ala Gly Arg Arg Leu Trp Glu Met Ala	420	425	430
Glu Asn Thr Arg Gly Ile Leu Ala Ile Glu Trp Leu Ser Ala Cys Gln	435	440	445
Gly Ile Asp Phe Arg Asn Gly Leu Lys Ser Ser Pro Ile Leu Glu Glu	450	455	460
Ala Arg Val Ile Leu Arg Ala Lys Val Asp Tyr Tyr Asp Gln Asp Arg	465	470	475
Phe Phe Ala Pro Asp Ile Asp Ala Ala Val Lys Leu Leu Ala Glu Gln	485	490	495
His Leu Ser Ser Leu Leu Pro Ser Gly Gln Ile Leu Gln Arg Lys Asn	500	505	510

Asn Arg

<210> SEQ ID NO 5

<211> LENGTH: 471

<212> TYPE: PRT

<213> ORGANISM: Proteus mirabilis

<400> SEQUENCE: 5

Met Ala Ile Ser Arg Arg Lys Phe Ile Leu Gly Gly Thr Val Val Ala	1	5	10	15
Val Ala Ala Gly Ala Gly Val Leu Thr Pro Met Leu Thr Arg Glu Gly	20	25	30	
Arg Phe Val Pro Gly Thr Pro Arg His Gly Phe Val Glu Gly Thr Gly	35	40	45	
Gly Pro Leu Pro Lys Gln Asp Asp Val Val Val Ile Gly Ala Gly Ile	50	55	60	
Leu Gly Ile Met Thr Ala Ile Asn Leu Ala Glu Arg Gly Leu Ser Val	65	70	75	80
Thr Ile Val Glu Lys Gly Asn Ile Ala Gly Glu Gln Ser Ser Arg Phe				

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85					90					95					
Tyr	Gly	Gln	Ala	Ile	Ser	Tyr	Lys	Met	Pro	Asp	Glu	Thr	Phe	Leu	Leu
		100						105					110		
His	His	Leu	Gly	Lys	His	Arg	Trp	Arg	Glu	Met	Asn	Ala	Lys	Val	Gly
		115					120					125			
Ile	Asp	Thr	Thr	Tyr	Arg	Thr	Gln	Gly	Arg	Val	Glu	Val	Pro	Leu	Asp
	130					135					140				
Glu	Glu	Asp	Leu	Glu	Asn	Val	Arg	Lys	Trp	Ile	Asp	Ala	Lys	Ser	Lys
145						150					155				160
Asp	Val	Gly	Ser	Asp	Ile	Pro	Phe	Arg	Thr	Lys	Met	Ile	Glu	Gly	Ala
			165						170					175	
Glu	Leu	Lys	Gln	Arg	Leu	Arg	Gly	Ala	Thr	Thr	Asp	Trp	Lys	Ile	Ala
			180					185					190		
Gly	Phe	Glu	Glu	Asp	Ser	Gly	Ser	Phe	Asp	Pro	Glu	Val	Ala	Thr	Phe
		195					200					205			
Val	Met	Ala	Glu	Tyr	Ala	Lys	Lys	Met	Gly	Ile	Lys	Ile	Phe	Thr	Asn
	210					215					220				
Cys	Ala	Ala	Arg	Gly	Leu	Glu	Thr	Gln	Ala	Gly	Val	Ile	Ser	Asp	Val
225						230					235				240
Val	Thr	Glu	Lys	Gly	Pro	Ile	Lys	Thr	Ser	Arg	Val	Val	Val	Ala	Gly
			245						250					255	
Gly	Val	Gly	Ser	Arg	Leu	Phe	Met	Gln	Asn	Leu	Asn	Val	Asp	Val	Pro
			260					265					270		
Thr	Leu	Pro	Ala	Tyr	Gln	Ser	Gln	Gln	Leu	Ile	Ser	Ala	Ala	Pro	Asn
		275					280					285			
Ala	Pro	Gly	Gly	Asn	Val	Ala	Leu	Pro	Gly	Gly	Ile	Phe	Phe	Arg	Asp
	290					295					300				
Gln	Ala	Asp	Gly	Thr	Tyr	Ala	Thr	Ser	Pro	Arg	Val	Ile	Val	Ala	Pro
305						310					315				320
Val	Val	Lys	Glu	Ser	Phe	Thr	Tyr	Gly	Tyr	Lys	Tyr	Leu	Pro	Leu	Leu
			325						330					335	
Ala	Leu	Pro	Asp	Phe	Pro	Val	His	Ile	Ser	Leu	Asn	Glu	Gln	Leu	Ile
			340					345					350		
Asn	Ser	Phe	Met	Gln	Ser	Thr	His	Trp	Asp	Leu	Asn	Glu	Glu	Ser	Pro
		355					360					365			
Phe	Glu	Lys	Tyr	Arg	Asp	Met	Thr	Ala	Leu	Pro	Asp	Leu	Pro	Glu	Leu
	370					375					380				
Asn	Ala	Ser	Leu	Glu	Lys	Leu	Lys	Lys	Glu	Phe	Pro	Ala	Phe	Lys	Glu
385						390					395				400
Ser	Thr	Leu	Ile	Asp	Gln	Trp	Ser	Gly	Ala	Met	Ala	Ile	Ala	Pro	Asp
			405						410					415	
Glu	Asn	Pro	Ile	Ile	Ser	Asp	Val	Lys	Glu	Tyr	Pro	Gly	Leu	Val	Ile
		420						425					430		
Asn	Thr	Ala	Thr	Gly	Trp	Gly	Met	Thr	Glu	Ser	Pro	Val	Ser	Ala	Glu
		435					440					445			
Ile	Thr	Ala	Asp	Leu	Leu	Gly	Lys	Lys	Pro	Val	Leu	Asp	Ala	Lys	
	450					455					460				
Pro	Phe	Ser	Leu	Tyr	Arg	Phe									
465						470									

<210> SEQ ID NO 6

<211> LENGTH: 473

<212> TYPE: PRT

<213> ORGANISM: Proteus mirabilis

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<400> SEQUENCE: 6

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Met Asn Ile Ser Arg Arg Lys Leu Leu Leu Gly Val Gly Ala Ala Gly
 1           5           10
Val Leu Ala Gly Gly Ala Ala Leu Val Pro Met Val Arg Arg Asp Gly
 20           25           30
Lys Phe Val Glu Ala Lys Ser Arg Ala Ser Phe Val Glu Gly Thr Gln
 35           40           45
Gly Ala Leu Pro Lys Glu Ala Asp Val Val Ile Ile Gly Ala Gly Ile
 50           55           60
Gln Gly Ile Met Thr Ala Ile Asn Leu Ala Glu Arg Gly Met Ser Val
 65           70           75           80
Thr Ile Leu Glu Lys Gly Gln Ile Ala Gly Glu Gln Ser Gly Arg Ala
 85           90           95
Tyr Ser Gln Ile Ile Ser Tyr Gln Thr Ser Pro Glu Ile Phe Pro Leu
100           105           110
His His Tyr Gly Lys Ile Leu Trp Arg Gly Met Asn Glu Lys Ile Gly
115           120           125
Ala Asp Thr Ser Tyr Arg Thr Gln Gly Arg Val Glu Ala Leu Ala Asp
130           135           140
Glu Lys Ala Leu Asp Lys Ala Gln Ala Trp Ile Lys Thr Ala Lys Glu
145           150           155           160
Ala Ala Gly Phe Asp Thr Pro Leu Asn Thr Arg Ile Ile Lys Gly Glu
165           170           175
Glu Leu Ser Asn Arg Leu Val Gly Ala Gln Thr Pro Trp Thr Val Ala
180           185           190
Ala Phe Glu Glu Asp Ser Gly Ser Val Asp Pro Glu Thr Gly Thr Pro
195           200           205
Ala Leu Ala Arg Tyr Ala Lys Gln Ile Gly Val Lys Ile Tyr Thr Asn
210           215           220
Cys Ala Val Arg Gly Ile Glu Thr Ala Gly Gly Lys Ile Ser Asp Val
225           230           235           240
Val Ser Glu Lys Gly Ala Ile Lys Thr Ser Gln Val Val Leu Ala Gly
245           250           255
Gly Ile Trp Ser Arg Leu Phe Met Gly Asn Met Gly Ile Asp Ile Pro
260           265           270
Thr Leu Asn Val Tyr Leu Ser Gln Gln Arg Val Ser Gly Val Pro Gly
275           280           285
Ala Pro Arg Gly Asn Val His Leu Pro Asn Gly Ile His Phe Arg Glu
290           295           300
Gln Ala Asp Gly Thr Tyr Ala Val Ala Pro Arg Ile Phe Thr Ser Ser
305           310           315           320
Ile Val Lys Asp Ser Phe Leu Leu Gly Pro Lys Phe Met His Leu Leu
325           330           335
Gly Gly Gly Glu Leu Pro Leu Glu Phe Ser Ile Gly Glu Asp Leu Phe
340           345           350
Asn Ser Phe Lys Met Pro Thr Ser Trp Asn Leu Asp Glu Lys Thr Pro
355           360           365
Phe Glu Gln Phe Arg Val Ala Thr Ala Thr Gln Asn Thr Gln His Leu
370           375           380
Asp Ala Val Phe Gln Arg Met Lys Thr Glu Phe Pro Val Phe Glu Lys
385           390           395           400
Ser Glu Val Val Glu Arg Trp Gly Ala Val Val Ser Pro Thr Phe Asp

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405	410	415
Glu Leu Pro Ile Ile Ser Glu Val Lys Glu Tyr Pro Gly Leu Val Ile		
420	425	430
Asn Thr Ala Thr Val Trp Gly Met Thr Glu Gly Pro Ala Ala Gly Glu		
435	440	445
Val Thr Ala Asp Ile Val Met Gly Lys Lys Pro Val Ile Asp Pro Thr		
450	455	460
Pro Phe Ser Leu Asp Arg Phe Lys Lys		
465	470	
<210> SEQ ID NO 7		
<211> LENGTH: 471		
<212> TYPE: PRT		
<213> ORGANISM: Proteus vulgaris		
<400> SEQUENCE: 7		
Met Ala Ile Ser Arg Arg Lys Phe Ile Ile Gly Gly Thr Val Val Ala		
1	5	10
Val Ala Ala Gly Ala Gly Ile Leu Thr Pro Met Leu Thr Arg Glu Gly		
20	25	30
Arg Phe Val Pro Gly Thr Pro Arg His Gly Phe Val Glu Gly Thr Glu		
35	40	45
Gly Ala Leu Pro Lys Gln Ala Asp Val Val Val Val Gly Ala Gly Ile		
50	55	60
Leu Gly Ile Met Thr Ala Ile Asn Leu Val Glu Arg Gly Leu Ser Val		
65	70	75
Val Ile Val Glu Lys Gly Asn Ile Ala Gly Glu Gln Ser Ser Arg Phe		
85	90	95
Tyr Gly Gln Ala Ile Ser Tyr Lys Met Pro Asp Glu Thr Phe Leu Leu		
100	105	110
His His Leu Gly Lys His Arg Trp Arg Glu Met Asn Ala Lys Val Gly		
115	120	125
Ile Asp Thr Thr Tyr Arg Thr Gln Gly Arg Val Glu Val Pro Leu Asp		
130	135	140
Glu Glu Asp Leu Val Asn Val Arg Lys Trp Ile Asp Glu Arg Ser Lys		
145	150	155
Asn Val Gly Ser Asp Ile Pro Phe Lys Thr Arg Ile Ile Glu Gly Ala		
165	170	175
Glu Leu Asn Gln Arg Leu Arg Gly Ala Thr Thr Asp Trp Lys Ile Ala		
180	185	190
Gly Phe Glu Glu Asp Ser Gly Ser Phe Asp Pro Glu Val Ala Thr Phe		
195	200	205
Val Met Ala Glu Tyr Ala Lys Lys Met Gly Val Arg Ile Tyr Thr Gln		
210	215	220
Cys Ala Ala Arg Gly Leu Glu Thr Gln Ala Gly Val Ile Ser Asp Val		
225	230	235
Val Thr Glu Lys Gly Ala Ile Lys Thr Ser Gln Val Val Val Ala Gly		
245	250	255
Gly Val Trp Ser Arg Leu Phe Met Gln Asn Leu Asn Val Asp Val Pro		
260	265	270
Thr Leu Pro Ala Tyr Gln Ser Gln Gln Leu Ile Ser Gly Ser Pro Thr		
275	280	285
Ala Pro Gly Gly Asn Val Ala Leu Pro Gly Gly Ile Phe Phe Arg Glu		
290	295	300

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Gln Ala Asp Gly Thr Tyr Ala Thr Ser Pro Arg Val Ile Val Ala Pro
 305 310 315 320
 Val Val Lys Glu Ser Phe Thr Tyr Gly Tyr Lys Tyr Leu Pro Leu Leu
 325 330 335
 Ala Leu Pro Asp Phe Pro Val His Ile Ser Leu Asn Glu Gln Leu Ile
 340 345 350
 Asn Ser Phe Met Gln Ser Thr His Trp Asn Leu Asp Glu Val Ser Pro
 355 360 365
 Phe Glu Gln Phe Arg Asn Met Thr Ala Leu Pro Asp Leu Pro Glu Leu
 370 375 380
 Asn Ala Ser Leu Glu Lys Leu Lys Ala Glu Phe Pro Ala Phe Lys Glu
 385 390 395 400
 Ser Lys Leu Ile Asp Gln Trp Ser Gly Ala Met Ala Ile Ala Pro Asp
 405 410 415
 Glu Asn Pro Ile Ile Ser Glu Val Lys Glu Tyr Pro Gly Leu Val Ile
 420 425 430
 Asn Thr Ala Thr Gly Trp Gly Met Thr Glu Ser Pro Val Ser Ala Glu
 435 440 445
 Leu Thr Ala Asp Leu Leu Leu Gly Lys Lys Pro Val Leu Asp Pro Lys
 450 455 460
 Pro Phe Ser Leu Tyr Arg Phe
 465 470

<210> SEQ ID NO 8
 <211> LENGTH: 452
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 8

Met Ser Thr Ala Val Leu Glu Asn Pro Gly Leu Gly Arg Lys Leu Ser
 1 5 10 15
 Asp Phe Gly Gln Glu Thr Ser Tyr Ile Glu Asp Asn Cys Asn Gln Asn
 20 25 30
 Gly Ala Ile Ser Leu Ile Phe Ser Leu Lys Glu Glu Val Gly Ala Leu
 35 40 45
 Ala Lys Val Leu Arg Leu Phe Glu Glu Asn Asp Val Asn Leu Thr His
 50 55 60
 Ile Glu Ser Arg Pro Ser Arg Leu Lys Lys Asp Glu Tyr Glu Phe Phe
 65 70 75 80
 Thr His Leu Asp Lys Arg Ser Leu Pro Ala Leu Thr Asn Ile Ile Lys
 85 90 95
 Ile Leu Arg His Asp Ile Gly Ala Thr Val His Glu Leu Ser Arg Asp
 100 105 110
 Lys Lys Lys Asp Thr Val Pro Trp Phe Pro Arg Thr Ile Gln Glu Leu
 115 120 125
 Asp Arg Phe Ala Asn Gln Ile Leu Ser Tyr Gly Ala Glu Leu Asp Ala
 130 135 140
 Asp His Pro Gly Phe Lys Asp Pro Val Tyr Arg Ala Arg Arg Lys Gln
 145 150 155 160
 Phe Ala Asp Ile Ala Tyr Asn Tyr Arg His Gly Gln Pro Ile Pro Arg
 165 170 175
 Val Glu Tyr Met Glu Glu Gly Lys Lys Thr Trp Gly Thr Val Phe Lys
 180 185 190
 Thr Leu Lys Ser Leu Tyr Lys Thr His Ala Cys Tyr Glu Tyr Asn His
 195 200 205

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Ile Phe Pro Leu Leu Glu Lys Tyr Cys Gly Phe His Glu Asp Asn Ile
 210 215 220
 Pro Gln Leu Glu Asp Val Ser Gln Phe Leu Gln Thr Cys Thr Gly Phe
 225 230 235 240
 Arg Leu Arg Pro Val Ala Gly Leu Leu Ser Ser Arg Asp Phe Leu Gly
 245 250 255
 Gly Leu Ala Phe Arg Val Phe His Cys Thr Gln Tyr Ile Arg His Gly
 260 265 270
 Ser Lys Pro Met Tyr Thr Pro Glu Pro Asp Ile Cys His Glu Leu Leu
 275 280 285
 Gly His Val Pro Leu Phe Ser Asp Arg Ser Phe Ala Gln Phe Ser Gln
 290 295 300
 Glu Ile Gly Leu Ala Ser Leu Gly Ala Pro Asp Glu Tyr Ile Glu Lys
 305 310 315 320
 Leu Ala Thr Ile Tyr Trp Phe Thr Val Glu Phe Gly Leu Cys Lys Gln
 325 330 335
 Gly Asp Ser Ile Lys Ala Tyr Gly Ala Gly Leu Leu Ser Ser Phe Gly
 340 345 350
 Glu Leu Gln Tyr Cys Leu Ser Glu Lys Pro Lys Leu Leu Pro Leu Glu
 355 360 365
 Leu Glu Lys Thr Ala Ile Gln Asn Tyr Thr Val Thr Glu Phe Gln Pro
 370 375 380
 Leu Tyr Tyr Val Ala Glu Ser Phe Asn Asp Ala Lys Glu Lys Val Arg
 385 390 395 400
 Asn Phe Ala Ala Thr Ile Pro Arg Pro Phe Ser Val Arg Tyr Asp Pro
 405 410 415
 Tyr Thr Gln Arg Ile Glu Val Leu Asp Asn Thr Gln Gln Leu Lys Ile
 420 425 430
 Leu Ala Asp Ser Ile Asn Ser Glu Ile Gly Ile Leu Cys Ser Ala Leu
 435 440 445
 Gln Lys Ile Lys
 450

<210> SEQ ID NO 9
 <211> LENGTH: 117
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 9

atccccatca ctcttgatgg agatcaattc cccaagctgc tagagcggtta ccttgccctt 60
 aaacattagc aatgtcgatt tatcagaggg cgcacaggct cccacaggag aaaaccg 117

<210> SEQ ID NO 10
 <211> LENGTH: 108
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polynucleotide

<400> SEQUENCE: 10

ctcttgatcg ttatcaattc ccacgctggt tcagagcggtt accttgccct taaacattag 60
 caatgtcgat ttatcagagg gccacaggc tccacagga gaaaaccg 108

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<210> SEQ ID NO 11
<211> LENGTH: 290
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 11

gtcagcataa caccctgacc tctcattaat tggtcatgcc gggcggcact atcgctgtcc      60
ggccttttcc tctcttactc tgctacgtac atctatttct ataaatccgt tcaatttgtc      120
tgttttttgc acaaacatga aatatcagac aattccgtga cttaagaaaa ttatacaaaa      180
tcagcaatat accccttaag gagtatataa aggtgaattt gatttacatc aataagcggg      240
gttgctgaat cgtaagga ggcggtaata gaaaagaaat cgaggcaaaa      290


<210> SEQ ID NO 12
<211> LENGTH: 433
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 12

cggcccgatc gttgaacata gcggtccgca ggcggcactg cttacagcaa acggtctgta      60
cgctgtcgtc tttgtgatgt gcttctgtt aggtttcgtc agccgtcacc gtcagcataa      120
caccctgacc tctcattaat tgctcatgcc ggacggcact atcgctgtcc ggccttttcc      180
tctcttcccc cgtacgtgc atctatttct ataaaccgcg tcattttgtc tattttttgc      240
acaaacatga aatatcagac aattccgtga cttaagaaaa ttatacaaaa tcagcaatat      300
accattaag gagtatataa aggtgaattt gatttacatc aataagcggg gttgctgaat      360
cgtaagga ggcggtaata gaaaagaaat cgaggcaaaa atgtttgttt aactttaaga      420
aggagatata cat      433


<210> SEQ ID NO 13
<211> LENGTH: 290
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 13

gtcagcataa caccctgacc tctcattaat tgctcatgcc ggacggcact atcgctgtcc      60
ggccttttcc tctcttcccc cgtacgtgc atctatttct ataaaccgcg tcattttgtc      120
tattttttgc acaaacatga aatatcagac aattccgtga cttaagaaaa ttatacaaaa      180
tcagcaatat acccattaag gagtatataa aggtgaattt gatttacatc aataagcggg      240
gttgctgaat cgtaagga ggcggtaata gaaaagaaat cgaggcaaaa      290


<210> SEQ ID NO 14
<211> LENGTH: 173
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 14

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atttcctctc atcccatccg ggggtgagagt cttttccccc gacttatggc tcatgcatgc    60
atcaaaaaag atgtgagcct gatcaaaaac aaaaaatatt tcaactcgaca ggagtattta    120
tattgcgccc gttacgtggg ctctgactgt aaatcagaaa ggagaaaaca cct          173

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<210> SEQ ID NO 15
<211> LENGTH: 305
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

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<400> SEQUENCE: 15

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gtcagcataa caccctgacc tctcattaat tgttcatgcc gggcggcact atcgtcgtcc    60
ggccttttcc tctcttactc tgctacgtac atctatttct ataaatccgt tcaatttgtc    120
tgttttttgc acaaacatga aatatcagac aattccgtga cttaagaaaa ttatacaaaa    180
tcagcaatat accccttaag gagtatataa aggtgaattt gatttacatc aataagcggg    240
gttgctgaat cgtaaggat cctcttagaa ataattttgt ttaactttaa gaaggagata    300
tacat                                             305

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<210> SEQ ID NO 16
<211> LENGTH: 180
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

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<400> SEQUENCE: 16

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catttcctct catcccatcc ggggtgagag tcttttcccc cgacttatgg ctcatgcatg    60
catcaaaaaa gatgtgagct tgatcaaaaa caaaaaatat ttcactcgac aggagtattt    120
atattgcgcc cggatccctc tagaaataat ttgtttaac ttaagaagg agatatacat    180

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<210> SEQ ID NO 17
<211> LENGTH: 199
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

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<400> SEQUENCE: 17

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agttgttctt attggtgggtg ttgctttatg gttgcatcgt agtaaatggt tgtaacaaaa    60
gcaatttttc cggctgtctg tatacaaaaa cgccgtaaag ttgagcgaa gtcaataaac    120
tctctacca ttcagggcaa tatctctctt ggatccctct agaaataatt ttgtttaact    180
ttaagaagga gatatacat                             199

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<210> SEQ ID NO 18
<211> LENGTH: 207
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

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<400> SEQUENCE: 18

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agttgttctt attggtgggtg ttgctttatg gttgcatcgt agtaaatggt tgtaacaaaa    60

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gcaatttttc cggtgtctcg tatacaaaaa cgccgcaaag tttgagcgaa gtcaataaac	120
tctctaccca ttcagggcaa tatctctctt ggatccaaag tgaactctag aaataatttt	180
gtttaacttt aagaaggaga tatacat	207

<210> SEQ ID NO 19
 <211> LENGTH: 390
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 19
 tcgtctttgt gatgtgcttc ctgttaggtt tcgtcagccg tcaccgtcag cataacaccc 60
 tgacctctca ttaattgtct atgccggacg gcactatcgt cgcccgccct tttctctct 120
 tcccccgcta cgtgcactta tttctataaa cccgctcatt ttgtctattt ttgacacaaa 180
 catgaaatat cagacaattc cgtgacttaa gaaaatttat acaaatcagc aatataacca 240
 ttaaggagta tataaagggt aatttgattt acatcaataa gcgggggttg tgaatcgta 300
 aggtagaaat gtgatctagt tcacatttgc ggtaatagaa aagaaatcga ggcaaaaatg 360
 ttgttttaac ttaagaagg agatatacat 390

<210> SEQ ID NO 20
 <211> LENGTH: 200
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 20
 agttgttctt attggtgggt ttgctttatg gttgcatcgt agtaaatggt tgtaacaaaa 60
 gcaatttttc cggtgtctcg tatacaaaaa cgccgcaaag tttgagcgaa gtcaataaac 120
 tctctaccca ttcagggcaa tatctctcaa atgtgatcta gttcacattt ttgttttaac 180
 ttaagaagg agatatacat 200

<210> SEQ ID NO 21
 <211> LENGTH: 1485
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 21
 ctcttgatcg ttatcaattc ccacgtgtt tcagagcggt accttgccct taaacattag 60
 caatgtcgat ttatcagagg gccgacaggc tcccacagga gaaaaccgat gaaaaacgcg 120
 tcaaccgtat cggaagatac tgcgtcgaat caagagccga cgcttcacg cggttaccat 180
 aaccgtcata ttcaactgat tgcgttgggt ggcgcaattg gtactggtct gtttcttggc 240
 attggcccg cgattcagat ggcgggtccg gctgtattgc tgggctacgg cgtcgccggg 300
 atcatcgctt tctgtattat gcgccagctt ggcgaaatgg tgggtgagga gccggtatcc 360
 ggttcatttg cccactttgc ctataaatac tggggaccgt ttgcgggctt cctctctggc 420
 tggaaactact gggtaatggt cgtgctgggt ggaatggcag agctgaccgc tgcgggcatc 480
 tatatgcagt actggttccc ggaatgtcca acgtggattt gggtgcgcg cttctttatt 540

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atcatcaacg cegttaacct ggtgaacgtg cgettatatg gcgaaccga gttctgggtt	600
gcgttgatta aagtgcggc aatcatcggt atgatcggt ttggcctgtg gctgctgtt	660
tctggtcacg gcggcgagaa agccagtatc gacaacctct ggcgctacgg tggttcttc	720
gccaccggct ggaatgggct gattttgtcg ctggcggtaa ttatgttctc ctteggcggt	780
ctggagctga ttgggattac tgcgctgaa gcgcgcgac cggaaaaaag cattccaaaa	840
gcggtaaatc aggtggtgta tcgcctctg ctgttttaca tcggttcact ggtggtttta	900
ctggcgctct atccgtgggt ggaagtgaat tccaacagta gcccgtttgt gatgattttc	960
cataatctcg acagcaacgt ggtagcttct gcgctgaact tcgtcattct ggtagcatcg	1020
ctgtcagtgt ataacagcgg ggtttactct aacagccgca tgctgtttgg cctttctgtg	1080
cagggtaatg gcgcgaagt tttgactcgc gtcagccgct gcggtgtgcc gattaactcg	1140
ctgatgcttt ccggagcgat cacttcgctg gtggtgttaa tcaactatct gctgcgcaa	1200
aaagcgtttg gtctgtgat ggcgctgggt gtagcaacgc tgctgttgaa ctggattatg	1260
atctgtctgg cgcactctcg tttctgtgca gcgatgcgac gtcaggggcg tgaacacag	1320
tttaaggcgc tgctctatcc gttcggcaac tatctctgca ttgccttctc cgcatgatt	1380
ttgtctgta tgtgcacgat ggatgatatg cgttctgac cgatcctgct gccggtgtgg	1440
attgtattcc tgtttatggc atttaaacg ctgcgtcga aataa	1485

<210> SEQ ID NO 22

<211> LENGTH: 4428

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 22

ctcttgatcg ttatcaattc ccacgtggt tcagagcgtt acctgcoct taaacattag	60
caatgtcgat ttatcagagg gccgacaggc tcccacagga gaaaaccgat gaaaacacta	120
tcacaggccc aatctaaac ttcttcacag caattcagct ttaccgggaa ctgctctgcg	180
aatgtaatta tcggcaatca aaagctgacc attaatgatg tagctcgcgt tgcccggat	240
ggcacttttg tgtcactgac gaacaatacc gacattctgc aaggtattca agctagctgc	300
gattatatca ataacgcgt tgaatctggc gagccaatct acggggtaac aagcggtttt	360
ggtgggatgg cgaacgttgc cattagccgt gaacaggcga gcgaacttca gaccaacctc	420
gtttggttcc taaagacagg agctggtaat aagttacctc tggctgacgt aagagccgcg	480
atgctgcttc gcgctaatag tcacatgcgc ggccgacgtg gtatccgtct tgagcttata	540
aagaggatgg aaatcttctc caacgcgggt gtcacaccat atgtttatga gtttggtagt	600
atcggagcca gtggtgatct tgttccccctg agttatatta cgggttcatt gattggttta	660
gaccgcctct ttaaagtgga ttttaacggg aaagaaatgg acgccccgac cgctttacga	720
cagcttaatc tgagccact tactttgctc cetaaagaag gtcttgccat gatgaatggc	780
acctctgtga tgactggaat tgccgcgaat tgtgtgtatg acacgcagat cctaaccggcc	840
attgccatgg gtgttcacgc gttggacatt caagccctga atggtacaaa ccagtcgttt	900
catccgttta tccataattc aaaaccccat ccgggacagc tttgggctgc tgatcagatg	960
atctcactcc tggccaatag tcaactggtt cgggacgagc tcgacggcaa acatgattat	1020
cgcgatcatg agtcaccca ggaccggtat tcacttcgtt gtctcccaca atacctgggg	1080

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cctatcggtg atggtatatc tcaaattgcy aagcaaattg aaattgagat caatagcgta	1140
accgacaacc cgcttatcga tgttgataat caggcctctt atcacgggtg caattttctg	1200
ggccagtatg ttggtatggg gatggatcac ctgcgggtact atattgggct tctggctaaa	1260
catcttgatg tgcagattgc cttattagct tcaccagaat tttcaaatgg actgccgcca	1320
tcattgctcg gtaacagaga aaggaaagta aatatgggccc ttaagggcct tcagatatgt	1380
ggtaactcaa tcatgccctt cctgacctt tatgggaact caattgctga tegttttccg	1440
acacatgctg aacagttaa ccaaacatt aactcacagg gctatacatc cgcgacgtta	1500
gcgcgtcggt ccgtggatat ctccagaat tatgttgcta tcgctctgat gttcggcgta	1560
caggccgttg atttgcgcac ttataaaaa accggctact acgatgctcg ggcttgctg	1620
tcgctgcca ccgagcggct ttatagcgcc gtacgtcatg ttgtgggtca gaaaccgacg	1680
tcggaccgcc cctatatattg gaatgataat gaacaagggc tggatgaaca catcgcccg	1740
atatctgcgc atattgcgc cggaggtgtc atcgctcagg cggtaacaaga catacttct	1800
tgctgcatt aagcttggcg taatcatggt catagctgtt tctgtgtga aattgttatc	1860
cgctcacaat tccacacaac atacgagccg gaagcataaa gtgtaaagcc tggggtgcct	1920
aatgagttag ctaactcaca ttaattgctg tgcgctcact gcccgcttcc cagtcgggaa	1980
acctgtcgtg ccagctgcat taatgaatcg gccaacgcgc ggggagaggg ggtttgcgta	2040
ttgggcgctc ttccgcttcc tcgctcactg actcgctcgc ctccggtcgtt cggctgcggc	2100
gagcgggtatc agctcactca aaggcggtaa tacggttatc cacagaatca ggggataacg	2160
caggaaagaa catgtgagca aaaggccagc aaaaggccag gaaccgtaaa aaggccgcgt	2220
tgctggcggtt ttccataggt ctccgcccc ctgacgagca tcacaaaaat cgacgctcaa	2280
gtcagaggtg gcgaaaccgc acaggactat aaagatacca ggcgtttccc cctggaagct	2340
ccctcgtgcg ctctcctgtt ccgaccctgc cgttaccgg atacctgtcc gcctttctcc	2400
cttcgggaag cgtggcgctt tctcatagct cacgctgtag gtatctcagt tcggtgtagg	2460
tcgttcgctc caagctgggc tgtgtgcacg aacccccgt tcagcccgac cgtgcgcct	2520
tatccggtaa ctatcgtctt gagtccaacc cggtaagaca cgacttatcg ccaactggcag	2580
cagccactgg taacaggatt agcagagcga ggtatgtagg cgggtctaca gagttcttga	2640
agtgtgggcc taactacggc tacactagaa gaacagtatt tggatatctg cctctgctga	2700
agccagttac cttcgaaaa agagttggta gctcttgatc cggcaaaaa accaccgctg	2760
gtagcggtag tttttttgtt tgcaagcagc agattacgcg cagaaaaaa ggatctcaag	2820
aagatccttt gatcttttct acgggggtctg acgctcagtg gaacgaaaa tcacgttaa	2880
ggattttgtt catgagatta tcaaaaagga tcttcacct gatcctttta aattaaaaat	2940
gaagttttta atcaatctaa agtatatatg agtaaaactg gtctgacagt taccaatgct	3000
taatcagtga ggcacctatc tcagcgatct gtctatctcg ttcacctata gttgcctgac	3060
tccccgctgt gtagataact acgatacggg agggcttacc atctggcccc agtgctgcaa	3120
tgataaccgc agacccacgc tcaccggctc cagatttatc agcaataaac cagccagccg	3180
gaagggccga gcgcagaagt ggtcctgcaa ctttatccgc ctccatccag tctattaatt	3240
gttgccggga agctagagta agtagttcgc cagttaatag tttgcgcaac gttgttgcca	3300
ttgctacagg catcgtgggtg tcacgctcgt cgtttggtat ggcttcattc agtcgggtt	3360
cccaacgacg aaggcgagtt acatgatccc ccatgttggtg caaaaaagcg gttagctcct	3420
tcggtcctcc gatcgtgtgc agaagtaagt tggccgcagt gttatcactc atggttatgg	3480

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cagcactgca taattctctt actgtcatgc catccgtaag atgcttttct gtgactggtg 3540
agtactcaac caagtcattc tgagaatagt gtatgcggcg accgagttgc tcttgcccgg 3600
cgtcaatacg ggataatacc gcgccacata gcagaacttt aaaagtgtc atcattggaa 3660
aacgttcttc ggggcgaaaa ctctcaagga tcttaccgct gttgagatcc agttcgatgt 3720
aaccactcgc tgcaccaaac tgatcttcag catcttttac tttcaccagc gtttctgggt 3780
gagcaaaaaa aggaaggcaa aatgccgcaa aaaagggaat aaggcgaca cggaaatgtt 3840
gaatactcat actcttcctt tttcaatatt attgaagcat ttatcagggt tattgtctca 3900
tgagcggata catatttgaa tgtatttaga aaaataaaca aataggggtt ccgcgcacat 3960
ttccccgaaa agtgccacct gacgtctaag aaaccattat tatcatgaca ttaacctata 4020
aaaaataggc tatcacgagg ccttttcgtc tcgcgcgttt cggatgatgc ggtgaaaacc 4080
tctgacacat gcagctcccg gagacgggtc cagcttgtct gtaagcggat gccgggagca 4140
gacaagcccc tcaggcgcg tcagcgggtg ttggcgggtg tcggggctgg cttaactatg 4200
cggcatcaga gcagattgta ctgagagtgc accatattgc gtgtgaaata ccgcacagat 4260
gcgtaaggag aaaataccgc atcaggcgcc attcgccatt caggctgcgc aactgttggg 4320
aaggcgatc ggtgcgggcc tcttcgctat tacgccagct ggcgaaaggg ggatgtgctg 4380
caaggcgatt aagttgggta acgccagggt tttccagtc acgacgtt 4428

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<210> SEQ ID NO 23

<211> LENGTH: 4323

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 23

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 24

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<210> SEQ ID NO 25

<211> LENGTH: 4987

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<400> SEQUENCE: 26

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<211> LENGTH: 5857

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

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<210> SEQ ID NO 28

<211> LENGTH: 6602

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 28

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

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actgtttttc	tgtaggccgt	gtacctaaat	gtacttttgc	tccatcgcca	tgacttagta	6180
aagcacatct	aaaactttta	gcgttattac	gtaaaaaato	ttgccagctt	tccccttcta	6240
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cccgccttatt ttttacatgc caatacaatg taggctgctc tacacctagc ttctgggcga 6360
gtttacgggt tgttaaacct tcgattccga cctcattaag cagctctaag gcgctgttaa 6420
tcactttact tttatctaag ctgagacatca ttaattccta atttttgttg acactctatc 6480
attgatagag ttatttt 6497

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<210> SEQ ID NO 30

<211> LENGTH: 2137

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 30

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ttgtcgtaat aatggcggca tactatcagt agtaggtgtt tccctttctt ctttagcgac 180
ttgatgctct tgatcttcca atacgcaacc taaagtaaaa tgccccacag cgctgagtgc 240
atataatgca ttctctagtg aaaaaccttg ttggcataaa aaggctaatt gattttcgag 300
agtttcatac tgttttctcg taggcctgt acctaaatgt acttttgctc catcgcgatg 360
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cccttctaaa gggcaaaagt gagtatgggt cctatctaac atctcaatgg ctaaggcgctc 480
gagcaaaagg cgcttatttt ttacatgcc aataaatgta ggctgctcta cacctagctt 540
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gctgttaate actttacttt tatctaact agacatcatt aattcctaatt tttgttgac 660
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gtataacagc ggggtttact ctaacagccg catgctgttt ggcctttctg tgcagggtaa 1740
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gatgtgcacg atggatgata tgcgcttgtc agcgatcctg ctgccggtgt ggattgtatt	2100
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<210> SEQ ID NO 31

<211> LENGTH: 3383

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 31

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tcgtccggcc ttttctcttc ttactctgct acgtacatct atttctataa atccgttcaa	120
tttgtctgtt ttttgcacaa acatgaaata tcagacaatt ccgtgactta agaaaattta	180
tacaaatcag caatataccc cttaaggagt atataaagg gaatttgatt tacatcaata	240
agcggggttg ctgaatcggt aaggtaggcg gtaatagaaa agaaatcgag gcaaaatga	300
gcaaagtcag actcgcaatt atggatcttc tggccgtcgt attacaacgt cgtgactggg	360
aaaacccttg cgttacccaa cttaatcgcc ttgcggcaca tcccccttc gccagctggc	420
gtaatagcga agaggccgc accgatcgcc ctccccaa gttgcgcagc ctgaatggcg	480
aatggcgctt tgcttggttt ccggcaccag aagcggtgcc ggaaagctgg ctggagtgcg	540
atcttctcga cgccgatact gtcgtcgtcc cctcaaaactg gcagatgcac ggttacgatg	600
cgctatctca caccaacgtg acctatccca ttacgggtcaa tccgcggttt gttcccgcg	660
agaatccgac aggttggttac tcgctcacat ttaattattga tgaaagctgg ctacaggaag	720
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cggtttcttt gtggcagggt gaaacgcagg tcgccagcgg caccgcgcct ttcggcggtg	1140
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tgtggtacac gctgtgcgac cgctacggcc tgtatgtggt ggatgaagcc aatattgaaa	1560
cccacggcat ggtgccaatg aatcgtctga ccgatgatcc gcgctggcta cccgcgatga	1620
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actcctggag ccgctcagta tcggcggaat tccagctgag gcgcggtcgc taccattacc 3360
agttgggtctg gtgtcaaaaa taa 3383

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<210> SEQ ID NO 32

<211> LENGTH: 3258

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 32

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gtatttatat tgcgccggtt acgtgggctt cgactgtaaa tcagaaagga gaaaacacct 180
atgacgacct acgatcggga tcctctggcc gtcgtattac aacgtcgtga ctgggaaaac 240
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cctgacgcgc	atactgtcgt	cgtccctcca	aactggcaga	tgacgggta	cgatgcgcct	480
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gccacgcgt ggcagcatca ggggaaaacc ttatttatca gccggaaaac ctaccggatt 2760
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gaagaaggca catggctgaa tategacggg ttccatatgg ggattggtgg cgacgactcc 3180
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<210> SEQ ID NO 33

<211> LENGTH: 3386

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 33

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gagatataca tatgactatg attacggatt ctctggccgt cgtattacaa cgtcgtgact 360
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aaaccacg catggtgcc atgaatcgtc tgaccgatga tccgcgctgg ctaccgcga	1620
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cgctgacgga agcaaacac caacagcagt atttccagt ccgtttatcc gggcgaacca	2220
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<210> SEQ ID NO 34

<211> LENGTH: 3261

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 34

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tgcattcatc	aaaaaagatg	tgagcttgat	caaaaaacaa	aaatatttca	ctcgacagga	120
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cagttgattg	aactgcctga	actgcgcgag	ccggagagcg	ccggacaact	ctggctaacg	2280
gtacgcgtag	tgacaacaaa	cgcgaccgca	tggtcagaag	ccggacacat	cagcgccctg	2340
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<210> SEQ ID NO 35

<211> LENGTH: 3279

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 35

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ataaactctc taccattca gggcaatctc tctcttgat ccctotagaa ataattttgt 180
ttaactttaa gaaggagata tacatatgct atgattacgg attctctggc cgtcgtatta 240
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catcgccatc tgctgcacgc ggaagaaggc acatggctga atatcgacgg ttcccatatg	3180
gggattggtg gcgacgactc ctggagcccc tcagtatcgg cggaattcca gctgagcgcc	3240
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<210> SEQ ID NO 36

<211> LENGTH: 1910

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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polynucleotide

<400> SEQUENCE: 36

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tacaaatcag caatataccc cttaaggagt atataaagg gaatttgatt tacatcaata	240
agcgggggtg ctgaatcggt aaggatccct ctagaataa ttttgtttaa ctttaagaag	300
gagatataca tatgaaagct aaagatgttc agccaacat tattattaat aaaaatggcc	360
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tcagttctcc tttgatcact gatcgtgctg tggatgaaga tataatccgc attgcggatg	1860
caattattaa tgatcaactt cctctgccag aaatcatgct ggaagaataa	1910

<210> SEQ ID NO 37

<211> LENGTH: 1785

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 37

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atacatatga aagctaaaga tgttcagcca accattatta ttaataaaaa tggccttatc    240
tctttggaag atatctatga cattgcgata aaacaaaaaa aagtagaagt atcaacggag    300
atcactgaac ttttgacgca tggctgtgaa aaattagagg aaaaattaaa ttcaggagag    360
gttatatatg gaatcaatac aggatttga ggaatgccca atttagttgt gccatttgag    420
aaaatcgcag agcatcagca aaatctgtta acttttcttt ctgctggtac tggggactat    480
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tctcctttga tcaatgatcg tgcgttgat gaagatataa tccgcattgc ggatgcaatt    1740
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<210> SEQ ID NO 38
<211> LENGTH: 1804
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polynucleotide

<400> SEQUENCE: 38

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ataaactctc taccaattca gggcaatctc tctcttgat ccctctagaa ataattttgt    180
ttaactttaa gaaggagata tacatatgaa agctaaagat gttcagccaa ccattattat    240

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taataaaaaat ggccttatct ctttgaaga tatctatgac attgcgataa aacaaaaaaa	300
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aaaattaaat tcaggagagg ttatatatgg aatcaataga ggatttggag ggaatgcca	420
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tgctggtact ggggactata tgcacaaacc ttgtattaaa gcgcacaaat ttactatgtt	540
actttctgtt tgcaagggtt ggtctgcaac cagaccaatt gtcgtcaag caattgttga	600
tcataataat catgacattg ttctctgtgt tcctcgctat ggctcagtgg gtgcaagcgg	660
tgatttaatt cctttatctt atattgcag agcattatgt ggatcggca aagtttatta	720
tatgggcgca gaaattgacg ctgctgaagc aattaaacgt gcagggttga caccattatc	780
gttaaaagcc aaagaaggtc ttgctctgat taacggcacc cgggtaatgt caggaaatcag	840
tgcaatcacc gtcattaac tggaacaaact atttaaacgc tcaatttctg cgattgccct	900
tgctgttgaa gcattacttg catctcatga acattatgat gcccgattc aacaagtaaa	960
aaatcatcct ggtcaaacg cggtggcaag tgcattgctt aatttattgg caggttcaac	1020
gcaggttaat ctattatctg ggggttaaga acaagccaat aaagcttgct gtcacaaaga	1080
aattacccaa ctaaatgata cttacagga agtttattca attcgtctgt caccacaagt	1140
attaggtata gtccagaat ctttagctac cgtcggaaa atattggaac ggaagttat	1200
ctcagctaata gataatccat tgatagatcc agaaaatggc gatgttctac acggtggaaa	1260
ttttatggg caatatgtcg ccgaacaat ggatgcatta aaactggata ttgctttaat	1320
tgcaaatcat cttcacgcca ttgtggctct tatgatggat aaccgtttct ctctggatt	1380
acctaattca ctgagtcga caccggcat gtatcaaggt tttaaggcg tccaactttc	1440
tcaaacctgt tttagttgtg caattcgcca tgattgtgct gcacagga ttcataccct	1500
cgcacagaa caatacaatc aagatattgt cagtttaggt ctgcagccg ctcaagatgt	1560
tttagagatg gagcagaat tacgcaatat tgtttcaatg acaattctgg tagtttgta	1620
ggccattcat cttcgggca atattagtga aattgcgcct gaaactgcta aattttacca	1680
tgcagtacgc gaaatcagtt ctcctttgat cactgatcgt gcgttgatg aagatataat	1740
ccgcatctgc gatcaatta ttaatgatca acttcctctg ccagaaatca tgctggaaga	1800
ataa	1804

<210> SEQ ID NO 39

<211> LENGTH: 2170

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 39

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gaataagaag gctggctctg caccttggtg atcaataat tcgatagctt gtcgtaataa	120
tggcggcata ctatcagtag taggtgttct cttttcttct ttagcgactt gatgctcttg	180
atcttccaat acgcaaccta aagtaaaatg cccacagcg ctgagtgcatt ataattgcatt	240
ctctagttaa aaacctgtt ggcataaaaa ggctaattga ttttcagagag ttccatactg	300
ttttctgtga ggccgtgtac ctaaatgtac ttttctcca tcgcgatgac ttagtaaaagc	360
acatctaaaa ctttttagctt tattacgtaa aaaatcttgc cagctttccc cttctaaagg	420

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gcaaaagtga gtatggtgcc tatctaacat ctcaatggct aaggcgctga gcaaagcccg	480
cttatTTTTT acatgccaat acaatgtagg ctgctctaca cctagcttct gggcgagttt	540
acgggttggt aaacottcga ttcgacctc attaaagcagc tctaatagcg tgtaatacac	600
tttactTTTA tctaacttag acatcattaa ttectaattt ttgttgacac tctatcattg	660
atagagttaa tttaccactc cctatcagtg atagagaaaa gtgaactcta gaaataattt	720
tgTTtaactt taagaaggag atatacatat gaacatttca aggagaaagc tacttttagg	780
tgTTggtgct gcgggcgttt tagcaggtgg tgcggcttta gttccaatgg ttcgccgtga	840
cggcaaatTT gtggaagcta aatcaagagc atcatttggt gaaggtagcg aaggggctct	900
tcctaagaa gcagatgtag tgattattgg tgcgggtatt caagggatca tgaccgctat	960
taaccttgct gaacgtggta tgagtgtcac tatcttagaa aagggtcaga ttgccggtga	1020
gcaatcaggc cgtgcataca gccaaattat tagttaccaa acatcgccag aaatcttccc	1080
attacaccat tatgggaaaa tattatggcg tggcatgaat gagaaaattg gtgcggatac	1140
cagttatcgt actcaaggtc gtgtagaagc gctggcagat gaaaaagcat tagataaagc	1200
tcaagcgtgg atcaaaacag ctaagaagc ggcaggTTTT gatacaccat taaataactcg	1260
catcattaaa ggtgaagagc tatcaaatcg cttagtcggt gctcaaacgc catggactgt	1320
tgctgcattt gaagaagatt caggctctgt tgatcctgaa acaggcacac ctgcactcgc	1380
tcgTtatgcc aaacaaatcg gtgtgaaaat ttataccaac tgtgcagtaa gaggtattga	1440
aaactgcgggt ggtaaaatct ctgatgtggt gagtgagaaa ggggcgatta aaacgtctca	1500
agttgtactc gctgggggta tctggtcgcg tttatttatg ggcaatatgg gtattgatat	1560
cccaacgctc aatgtatatc tatcacaaca acgtgtctca ggggttcctg gtgcaccacg	1620
tggtaatgtg catTTaccta atgggtattca tttccgcgaa caagcggatg gtacttatgc	1680
cgttgacca cgtatcttta cgagttcaat agtcaaagat agcttcctgc tagggcctaa	1740
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atttaattca tttaaaatgc cgacctcttg gaatttagat gaaaaaacac cattcgaaca	1860
attccgagtt gccacggcaa cacaaaatac gcaacactta gatgctgttt tccaaagaat	1920
gaaaacagaa ttcccagtat ttgaaaaatc agaagttgtt gaacgttggg gtgccgttgt	1980
gagtccaaca tttgatgaat tacctatcat ttctgaggtc aaagaatacc caggcttagt	2040
gattaacacg gcaacagtgt ggggtatgac agaaggcccg gcagcgggtg aagtgaaccg	2100
tgatattgtc atgggcaaga aaacctgttat tgatccaacg ccgttttagt ttggtcgttt	2160
taagaagtaa	2170

<210> SEQ ID NO 40

<211> LENGTH: 2624

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 40

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cgagaaatag agttgatcgt caaaaccgac attgcgaccg acggtggcga taggcatccg	120
ggtggtgctc aaaagcagct tcgcctgact gatcgctgg tccctgcgcc agcttaatac	180
gctaaccct aactgctggc ggaacaaatg cgacagacgc gacggcgaca ggacagacatg	240

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ctgtgcgacg	ctggcgatat	caaaattact	gtctgccagg	tgatecgtga	tgtactgaca	300
agcctcgcg	acccgattat	ccatcggtgg	atggagcgac	togttaatcg	cttccatg	360
ccgcagtaac	aattgctcaa	gcagatttat	cggcagcaat	tccgaatagc	gcccttcccc	420
ttgtccggca	ttaatgattt	gccccaaacag	gtcgtgaaa	tgcggctggg	gcgcttcac	480
cgggcgaaa	aaaccggtat	tggcaaatat	cgacggccag	ttaagccatt	catgccagta	540
ggcgcgcgga	cgaagtaaa	cccactggtg	ataccattcg	tgagcctccg	gatgacgacc	600
gtagtgatga	atctctccag	gcgggaacag	caaaatatca	cccggtcggc	agacaaattc	660
tcgctccctga	tttttcacca	ccccctgacc	gcgaatgggtg	agattgagaa	tataaccttt	720
cattcccagc	ggtcggtcga	taaaaaatc	gagataaccg	ttggcctcaa	tcggcggttaa	780
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agccatactt	ttcatactcc	cgccattcag	agaagaaacc	aattgtccat	attgcatcag	900
acattgccgt	cactgcgtct	tttactggct	cttctcgcta	acccaaccgg	taaccccgct	960
tattaaaagc	attctgtaac	aaagcgggac	caaagccatg	acaaaaacgc	gtaacaaaag	1020
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tgccatagca	tttttatcca	taagattagc	ggatccagcc	tgacgctttt	tttcgcaact	1140
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atatgaacat	ttcaaggaga	aagctacttt	taggtgttgg	tgctgcgggc	gttttagcag	1260
gtggtgcggc	tttagttcca	atgggtcgcc	gtgacggcaa	atttgtggaa	gctaaatcaa	1320
gagcatcatt	tggtgaaggt	acgcaagggg	ctcttcctaa	agaagcagat	gtagtgatta	1380
ttggtgcggg	tattcaaggg	atcatgaccg	ctattaacct	tgctgaacgt	ggtatgagtg	1440
tcactatctt	agaaaagggt	cagattgccg	gtgagcaatc	aggccgtgca	tacagccaaa	1500
ttattagtta	ccaaacatcg	ccagaaatct	tcccattaca	ccattatggg	aaaatattat	1560
ggcgtggcat	gaatgagaaa	attggtgcgg	ataccagtta	tcgtactcaa	ggtcgtgtag	1620
aagcgctggc	agatgaaaaa	gcattagata	aagctcaagc	gtggatcaaa	acagctaaa	1680
aagcggcagg	ttttgataca	ccattaaata	ctcgcatcat	taaagggtgaa	gagctatcaa	1740
atcgcttagt	cgggtgctcaa	acgccatgga	ctgttgctgc	atttgaagaa	gattcaggct	1800
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tggtgagtga	gaaagggcg	attaaaacgt	ctcaagttgt	actcgctggg	ggtatctggt	1980
cgcgtttatt	tatgggcaat	atgggtattg	atatcccaac	gctcaatgta	tatctatcac	2040
aacaacgtgt	ctcagggggt	cctggtgcac	cacgtggtaa	tgtgcattta	cctaattggta	2100
ttcatttccg	cgaacaagcg	gatggctactt	atgccgttgc	accacgtatc	tttacgagtt	2160
caatagtcaa	agatagcttc	ctgctagggc	ctaaatttat	gcacttatta	ggtggcggag	2220
agttaccgtt	ggaattctct	attggtgaag	atctatttaa	ttcattttaa	atgccgacct	2280
cttggaattt	agatgaaaaa	acaccattcg	aacaattccg	agttgccacg	gcaacacaaa	2340
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aatcagaagt	tgttgaacgt	tggggtgcgg	ttgtgagtc	aacatttgat	gaattaccta	2460
tcatttctga	ggcacaagaa	taccaggct	tagtgattaa	cacggcaaca	gtgtggggta	2520
tgacagaagg	cccggcagcg	ggtgaagtga	ccgctgatat	tgtcatgggc	aagaaacctg	2580
ttattgatcc	aacgcggtt	agtttggtgc	gttttaagaa	gtaa		2624

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<210> SEQ ID NO 41
<211> LENGTH: 3206
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 41

ggtaccagtt gttcttattg gtggtgttgc tttatggttg catcgtagta aatggttgta      60
acaaaagcaa tttttccggc tgtctgtata caaaaacgcc gcaaagtttg agcgaagtca      120
ataaactctc taccattca gggcaatata tctcttgat ccaaagtga ctctagaaat      180
aattttgttt aactttaaga aggagatata catatgaaag ctaaagatgt tcagccaacc      240
attattatta ataaaaatgg ccttatctct ttggaagata tctatgacat tgcgataaaa      300
caaaaaaag tagaatatc aacggagatc actgaacttt tgacgcatgg tcgtgaaaaa      360
ttagaggaaa aattaaattc aggagaggtt atatatggaa tcaatacagg atttgagggg      420
aatgccaatt tagtgtgcc atttgagaaa atcgagagc atcgcaaaa tctgttaact      480
tttctttctg ctggtactgg ggactatatg tccaaacctt gtattaaagc gtcacaattt      540
actatgttac tttctgtttg caaaggttgg tctgcaacca gaccaattgt cgctcaagca      600
attgttgatc atattaatca tgacattgtt cctctgggtc ctgctatgg ctgagtggt      660
gcaagcggtg atttaattcc tttatcttat attgcacgag cattatgttg tatcgcaaaa      720
gtttattata tgggcgcaga aattgacgct gctgaagcaa ttaaacgtgc aggggtgaca      780
ccattatcgt taaaagccaa agaaggcttt gctctgatta acggcacccg ggtaatgtca      840
ggaatcagtg caatcacgt cattaaactg gaaaaactat taaagcctc aatttctgcg      900
attgcccttg ctgttgaagc attacttga tctcatgaac attatgatgc ccgattcaa      960
caagtaaaaa atcatcctgg tcaaaacgcg gtggcaagtg cattgcgtaa tttattggca      1020
ggttcaacgc aggttaatct attatctggg gttaaagaac aagccaataa agcttgctgt      1080
catcaagaaa ttaccaact aaatgatacc ttacaggaag tttattcaat tcgctgtgca      1140
ccacaagtat taggtatagt gccagaatct ttagctaccg ctcggaatat attggaacgg      1200
gaagtatatc cagctaata gaatccattg atagatccag aaaatggcga tgttctacac      1260
ggtggaaatt ttatggggca atatgtcgcc cgaacaatgg atgcattaaa actggatatt      1320
gctttaattg ccaatcatct tcacgccatt gtggctctta tgatggataa ccgtttctct      1380
cgtggattac ctaattcact gagtcgcaca cccggcatgt atcaaggttt taaaggcgtc      1440
caactttctc aaacgcgttt agttgctgca attcgccatg attgtgctgc atcaggtatt      1500
cataccctcg ccacagaaca atacaatcaa gatattgtca gtttaggtct gcatgccgct      1560
caagatgttt tagagatgga gcagaaatta cgcaatattg tttcaatgac aattctggta      1620
gtttgtcagg ccattcatct tcgcggcaat attagtgaat ttgcgcctga aactgctaaa      1680
ttttaccatg cagtacgcga aatcagttct cctttgatca ctgacgtgc gttggatgaa      1740
gatataatcc gcattgcgga tgcaattatt aatgatcaac ttcctctgcc agaaatcatg      1800
ctggaagaat aaaagaagga gatatacata tgaaaaacgc gtcaaccgta tcggaagata      1860
ctgcgtcgaa tcaagagcgc acgcttcac gcggattaca taaccgtcat attcaactga      1920
ttgcgttggg tggcgcaatt ggtactggtc tgtttcttgg cattggcccg gcgattcaga      1980
tggcgggtcc ggctgtattg ctgggctacg gcgtcgccgg gatcatcgct ttcctgatta      2040

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tgccagcgt	tggcgaaatg	gtgggtgagg	agccggatc	cggttcattt	gccactttg	2100
cctataaata	ctggggaccg	tttgcgggct	tcctctctgg	ctggaactac	tgggtaatgt	2160
tcgtgctggt	gggaatggca	gagctgaccg	ctgcgggcat	ctatatgcag	tactggttcc	2220
cggtggttcc	aacgtggatt	tgggtgccc	ccttctttat	tatcatcaac	gccgttaacc	2280
tggagaacgt	gcgcttatat	ggcgaaaccg	agtctctggt	tgcgttgatt	aaagtgcctg	2340
caatcatcgg	tatgatcggc	tttggcctgt	ggctgctggt	ttctggtcac	ggcggcgaga	2400
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tggagtgaa	atccaacagt	agcccgtttg	tgatgatttt	ccataatctc	gacagcaacg	2700
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gggtttactc	taacagccgc	atgctgtttg	gcctttctgt	gcagggtaat	gcgccgaagt	2820
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gttttcgtgc	agcgatgcga	cgtcaggggc	gtgaaacaca	gtttaaggcg	ctgctctatc	3060
cgttcggcaa	ctatctctgc	attgccttcc	tcggcatgat	tttctgctg	atgtgcacga	3120
tggatgatat	gcgcttgta	gcgatcctgc	tgcgggtgtg	gattgtattc	ctgtttatgg	3180
catttaaaac	gctgcgtcgg	aaataa				3206

<210> SEQ ID NO 42

<211> LENGTH: 3741

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 42

ttaggtacgg	gctgccatt	tgattttaac	gcgttcac	ccatcaaacg	gacgaccacg	60
ctggcctttt	gcaacccaaa	tttcatcgat	gcaggatca	ataattgcat	tacgcatggt	120
cggggttgca	cgcagccaca	gttcttcata	atcgctgcta	tcaacaatcc	agctaacatc	180
aaactgctgc	cttgcgctgc	tttcgctaac	tgcactcttg	gctgctgca	gggtgctcag	240
tgctttctga	tatgcagggg	caaaaaactg	ttctgccgga	ccatcataaa	caccattctg	300
acgatcacgc	agcaggcgac	ccagattttt	ttcggttca	cgaactgcgg	cttttgcata	360
cttttcatct	tcgcttgctt	gcggatgggt	cagtgtgcc	cagcgatctg	caactgcaat	420
aacaaacgga	tcacccggtt	cgcttgctgc	taattttgct	gccaacgaa	atgcaacata	480
ttcttcaacg	cttttaactg	caacataggt	cggtgccgga	caaccacctt	tcacactgct	540
acgccaacaa	cgataacat	taccgctata	gctacagcta	ccaccacaac	ccggacaacg	600
catacgaccg	ctcagcagat	gtttgcgacg	ggtatcatga	tcgctacat	ccagcggaac	660
accaacacca	tcttcacctt	taacggctgc	ttttgoggt	tcttggtctt	catcggtcac	720
cagcggagga	ccatgcataa	cgctaacacg	tttaccttca	ccgttataaa	aggtcagacg	780
acgtgtttta	ccatcctgac	gacctgtggt	ctgccaaacc	gcatatgccg	gattctgaat	840

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catatcacgc	acggtaactg	caatccacgg	accacgggtc	gggctcggaa	tttcacgggt	900
attcattgca	tgtgcgggtgc	ctgcatagct	cagacgatcg	gtaaccggca	gggtaaaaac	960
cagacgggct	gcttctgctt	tggtcagacc	atcaggacca	cccgcattct	catcatctgc	1020
tgccagttta	cgttcacat	attcatcacc	ctcttcacat	ctaacggtaa	ccagaacaac	1080
acgcagacca	tacgggtgcac	gggcattaac	ccattcacca	ttttcacgct	gatgtgcttt	1140
ggtatcacga	acacgttcgc	tcagtttttc	tgcttcttcg	cgtgcttctt	ctgcacgacg	1200
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accgctttcc	agggttttca	gtgctttgct	ctgatcatca	cgtgctttac	ctttacgaaa	1440
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cggattatat	cttcatccaa	cgcacgatca	gtgatcaaa	gagaactgat	ttcgctact	2160
gcattggtaa	atttagcagt	ttcaggcgca	atttcactaa	tattgccgcg	aagatgaatg	2220
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ggtaatccac	gagagaaacg	gttatccatc	ataagagcca	caatggcggtg	aagatgattg	2520
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aaatttccac	cgtgtagaac	atcgccattt	tctggatcta	tcaatggatt	atcattagct	2640
gagataactt	cccgttccaa	tattttccga	gcggtagcta	aagattcttg	cactatacct	2700
aatacttggt	gtgcacagcg	aattgaataa	acttcctgta	aggtatcatt	tagttgggta	2760
atttcttgat	gacgacaagc	tttattggct	tgttctttta	cccagataa	tagattaacc	2820
tgcttgtaac	ctgccataa	attacgcaat	gcacttgcca	ccgcgttttg	accaggatga	2880
ttttttactt	gttgaatccg	ggcatcataa	tgttcattgag	atgcaagtaa	tgcttcaaca	2940
gcaagggcaa	tcgcagaaat	tgaggcttta	aatagttttt	ccagtttaat	gacggtgatt	3000
gcactgattc	ctgacattac	ccgggtgccc	ttaatcagag	caagaccttc	tttggttttt	3060
aacgataatg	gtgtcaaccc	tgacggttta	attgcttcag	cagcgtcaat	ttctgcgccc	3120
atataataaa	ctttgcgat	accacataat	gctcgtgcaa	tataagataa	aggaattaaa	3180

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tcaccgcttg caccactga gccatagcga ggaaccagag gaacaatgtc atgattaata	3240
tgatcaacaa ttgcttgagc gacaattggt ctgggtgcag accaaccttt gcaaacagaa	3300
agtaacatag taaattgtga cgctttaata caaggtttg acatatagtc cccagtacca	3360
gcagaaagaa aagttaacag attttgcgtga tgcctgcga tttctcaaa tggcacaact	3420
aaattggcat tccctccaaa tctgtattg attccatata taacctctcc tgaatttaat	3480
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actttttttt gttttatgcg aatgtcatag atatcttcca aagagataag gccattttta	3600
ttaataataa tgggtggtcg aacatcttta gctttcatat gtatatctcc ttcttaaagt	3660
taaaaaaat tatttctaga gcagatcagg gtgcgcaagt tgtcaacgct cccaggagag	3720
ttatcgactt gcgtattagg g	3741

<210> SEQ ID NO 43

<211> LENGTH: 4794

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 43

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cggggttgca cgcagccaca gttcttcata atcgtgcta tcaacaatcc agctaacatc	180
aactgctgag cttgcgtgc tttcgttaac tgcctcttg gctgctgca ggggtgctcag	240
tgcttcttga tatgcagggg caaaaaactg ttctgccgga ccatcataaa caccattctg	300
acgatcacgc agcaggcgac ccagattttt ttccgcttca cgaactgcgg cttttgcata	360
cttttcatct tcgcttgct gcggatgggt cagtgtgcc cagcgatctg caactgcaat	420
aacaaacgga tcacccggtt cgcttgcgtc taattttgct gcccaacgaa atgcaacata	480
ttcttcaacg cttttacgtg caacataggt cggtgccgga caaccacctt tcacactgct	540
acgccaacaa cgataacat taccgctata gctacagcta ccaccacaac cgggacaacg	600
catacgaccg ctacgagat gtttgcgac ggtatcatga tcgctacat ccagcggaac	660
accaacacca tcttccactt taacggctgc ttttgcggct tctgttctt catcggtcac	720
cagcggagga ccatgcataa cgtaaacag tttaccttca ccgttataaa aggtcagacg	780
acgtgttta ccatcctgac gacctgtggt ctgccaaacc gcatatgccg gattctgaat	840
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cagacgggct gcttctgctt tggtcagacc atcaggacca cccgcattct catcatctgc	1020
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tgacccggca ccaccacgat ccaacgac cagacgataa caccacagtg caccaacttc	1380
accgctttcc agggctttca gtgctttgct ctgatcatca cgtgctttac ctttacgaaa	1440

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acggcttgcg	ctaccaactt	ctttccaaac	atgacgaacc	tgcataccca	gcagtgtctgc	1500
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tgcaccattt	gcacggcttt	taactgtctt	gcttttacga	caaacaggt	caatcagacc	1620
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acaactggta	cgggatattc	atatggacca	tggcagctag	ccctgcaggg	tgcactcaga	1920
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caactgggtg	ttacgttgge	gaccgctagg	actttcttgt	tgattttcca	tgcggtgttt	3720
tgcgcaatgt	taatcgcttt	gtacacctca	ggcatgtaaa	cgtcttcgta	gcgcacagtc	3780
gctttcttac	tgtgagtacg	caccagcgcc	agaggacgac	gaccgttagc	ccaatagcca	3840

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ccaccagtaa tgccagtcca cggcttagga ggaactacgc aaggttggaa catcggagag 3900
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atagtctcag agtcttgacc tactacgcca gcattttggc ggtgtaagct aaccattccg 4020
gttgactcaa tgagcatctc gatgcagcgt actcctacat gaatagagtc ttccttatgc 4080
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agagtgttga acgggatagc agccagttcg atgtcagaga agtcgttctt agcgatgtta 4680
atcgtgttca tatgtatc tcttctttaa agttaacaaa aattatttct agagcagatc 4740
aggggtgcga agttgtcaac gctcccagga gagttatcga cttgcgtatt aggg 4794

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<210> SEQ ID NO 44

<211> LENGTH: 1672

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 44

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taatacgact cactataggg agaagtgaa ctctagaaat aattttgttt aactttaaga 60
aggagatata catatgaaag ctaaagatgt tcagccaacc attattatta ataaaaatgg 120
ccttatctct ttggaagata tctatgacat tgcgataaaa caaaaaaag tagaaatc 180
aacggagatc actgaacttt tgacgcatgg tcgtgaaaaa ttagaggaaa aattaaattc 240
aggagaggtt atatatggaa tcaatacagg atttggaggg aatgccaat tagttgtgcc 300
at ttgagaaa atcgcagagc atcagcaaaa tctgttaact tttctttctg ctggtactgg 360
ggactatatg tccaaacctt gtattaaagc gtcacaattt actatgttac tttctgtttg 420
caaagggttg tctgcaacca gaccaattgt cgctcaagca attgttgatc atattaatca 480
tgacattgtt cctctggttc ctcgctatgg ctcaagtggg gcaagcggtg atttaattcc 540
tttatcttat attgcacgag cattatgtgg tatcggcaaa gtttattata tgggcgcaga 600
aattgacgct gctgaagcaa ttaaacgtgc aggggttgaca ccattatcgt taaaagccaa 660
agaaggctct gctctgatta acggcacccg ggtaatgtca ggaatcagtg caatcacctg 720
cattaaactg gaaaaactat taaagcctc aatttctgcg attgcccttg ctgttgaagc 780
attacttgca tctcatgaac attatgatgc ccgattcaa caagtaaaaa atcatcctgg 840
tcaaaacgcg gtggcaagtg cattgcgtaa tttattggca gggtcaacgc aggttaatct 900
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aaatgatacc ttacaggaag tttattcaat tcgctgtgca ccacaagtat taggtatagt 1020
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taatccattg atagatccag aaaatggcga tgttctacac ggtggaaatt ttatggggca	1140
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tcacgccatt gtggctctta tgatggataa ccgtttctct cgtggattac ctaattcact	1260
gagtcgcaca cccggcatgt atcaagggtt taaaggcgtc caactttctc aaaccgcttt	1320
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atacaatcaa gatattgtca gtttaggtct gcatgccgct caagatgttt tagagatgga	1440
gcagaaatta cgcaatattg tttcaatgac aattctggta gtttgcagg ccattcatct	1500
tcgcggcaat attagtgaat ttgcgcctga aactgctaaa ttttaccatg cagtacgcga	1560
aatcagttct cctttgatca ctgatcgtgc gttggatgaa gatataatcc gcattgcgga	1620
tgcaattatt aatgatcaac ttctctgccc agaaatcatg ctggaagaat aa	1672

<210> SEQ ID NO 45

<211> LENGTH: 2861

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 45

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cgagaaatag agttgatcgt caaaaccgac attgcgaccg acggtggcga taggcacccg	120
ggtggtgctc aaaagcagct tcgcctgact gatgcgctgg tcctcgcgcc agcttaatac	180
gctaatacct aactgctggc ggaacaaatg cgacagacgc gacggcgaca ggacacatg	240
ctgtgcgacg ctggcgatat caaaattact gtctgccagg tgatcgctga tgtactgaca	300
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ccgcagtaac aattgctcaa gcagatttat ccgcagcaat tcggaatagc gcccttcccc	420
ttgtccggca ttaatgatth gcccaaacag gtcgctgaaa tgcggctggg gcgcttcac	480
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ggcgcgcgga cgaaagtaaa cccactgggt ataccattcg tgagcctccg gatgacgacc	600
gtagtgatga atctctccag gcgggaacag caaaatatca cccggtcggc agacaaatc	660
tcgtccctga tttttcacca cccctgacc gcgaatgggt agattgagaa tataaccttt	720
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tgtctataat cacggcagaa aagtccacat tgattatttg cacggcgta cactttgcta	1080
tgccatagca tttttatcca taagattagc ggatccagcc tgacgctttt tttcgcaact	1140
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gtgatgatca gagcaaagca ctgaaagccc tggaaagcgg tgaagttggt gcaactgtggt 1500
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cggaagatgg tatgcctcgt cgtctgctgt ttggttggga tgaagatacc ggtcgtccgg 1620
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cgaccgggtg tccgtggatt gcagttaccg tgcgtgatat gattcagaat ccggcatatg 2040
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<210> SEQ ID NO 46
<211> LENGTH: 60
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      primer

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<400> SEQUENCE: 46

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tagaactgat gcaaaaagtg ctcgacgaag gcacacagat gtgtaggctg gagctgcttc 60

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<210> SEQ ID NO 47
<211> LENGTH: 60
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      primer

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<400> SEQUENCE: 47

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gtttcgtaat tagatagcca ccggcgcttt aatgcccgga catatgaata tcctccttag 60

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<210> SEQ ID NO 48
<211> LENGTH: 52
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        primer

<400> SEQUENCE: 48

caacacgttt cctgaggaac catgaaacag tatttagaac tgatgcaaaa ag          52

<210> SEQ ID NO 49
<211> LENGTH: 46
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        primer

<400> SEQUENCE: 49

cgcacactgg cgtcggctct ggcaggatgt ttcgtaatta gatagc          46

<210> SEQ ID NO 50
<211> LENGTH: 36
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        primer

<400> SEQUENCE: 50

atatcgtcgc agcccacagc aacacgtttc ctgagg          36

<210> SEQ ID NO 51
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        primer

<400> SEQUENCE: 51

aagaatttaa cggaggggcaa aaaaaaccga cgcacactgg cgtcggc          47

<210> SEQ ID NO 52
<211> LENGTH: 52
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

<400> SEQUENCE: 52

Leu Asn Pro Leu Ile Asn Glu Ile Ser Lys Ile Ile Ser Ala Ala Gly
1      5      10      15

Asn Phe Asp Val Lys Glu Glu Arg Ala Ala Ser Leu Leu Gln Leu
20     25     30

Ser Gly Asn Ala Ser Asp Phe Ser Tyr Gly Arg Asn Ser Ile Thr Leu
35     40     45

Thr Ala Ser Ala
50

<210> SEQ ID NO 53
<211> LENGTH: 584
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polypeptide

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<400> SEQUENCE: 53

Met Lys Ala Lys Asp Val Gln Pro Thr Ile Ile Ile Asn Lys Asn Gly
 1 5 10 15
 Leu Ile Ser Leu Glu Asp Ile Tyr Asp Ile Ala Ile Lys Gln Lys Lys
 20 25 30
 Val Glu Ile Ser Thr Glu Ile Thr Glu Leu Leu Thr His Gly Arg Glu
 35 40 45
 Lys Leu Glu Glu Lys Leu Asn Ser Gly Glu Val Ile Tyr Gly Ile Asn
 50 55 60
 Thr Gly Phe Gly Gly Asn Ala Asn Leu Val Val Pro Phe Glu Lys Ile
 65 70 75 80
 Ala Glu His Gln Gln Asn Leu Leu Thr Phe Leu Ser Ala Gly Thr Gly
 85 90 95
 Asp Tyr Met Ser Lys Pro Cys Ile Lys Ala Ser Gln Phe Thr Met Leu
 100 105 110
 Leu Ser Val Cys Lys Gly Trp Ser Ala Thr Arg Pro Ile Val Ala Gln
 115 120 125
 Ala Ile Val Asp His Ile Asn His Asp Ile Val Pro Leu Val Pro Arg
 130 135 140
 Tyr Gly Ser Val Gly Ala Ser Gly Asp Leu Ile Pro Leu Ser Tyr Ile
 145 150 155 160
 Ala Arg Ala Leu Cys Gly Ile Gly Lys Val Tyr Tyr Met Gly Ala Glu
 165 170 175
 Ile Asp Ala Ala Glu Ala Ile Lys Arg Ala Gly Leu Thr Pro Leu Ser
 180 185 190
 Leu Lys Ala Lys Glu Gly Leu Ala Leu Ile Asn Gly Thr Arg Val Met
 195 200 205
 Ser Gly Ile Ser Ala Ile Thr Val Ile Lys Leu Glu Lys Leu Phe Lys
 210 215 220
 Ala Ser Ile Ser Ala Ile Ala Leu Ala Val Glu Ala Leu Leu Ala Ser
 225 230 235 240
 His Glu His Tyr Asp Ala Arg Ile Gln Gln Val Lys Asn His Pro Gly
 245 250 255
 Gln Asn Ala Val Ala Ser Ala Leu Arg Asn Leu Leu Ala Gly Ser Thr
 260 265 270
 Gln Val Asn Leu Leu Ser Gly Val Lys Glu Gln Ala Asn Lys Ala Cys
 275 280 285
 Arg His Gln Glu Ile Thr Gln Leu Asn Asp Thr Leu Gln Glu Val Tyr
 290 295 300
 Ser Ile Arg Cys Ala Pro Gln Val Leu Gly Ile Val Pro Glu Ser Leu
 305 310 315 320
 Ala Thr Ala Arg Lys Ile Leu Glu Arg Glu Val Ile Ser Ala Asn Asp
 325 330 335
 Asn Pro Leu Ile Asp Pro Glu Asn Gly Asp Val Leu His Gly Gly Asn
 340 345 350
 Phe Met Gly Gln Tyr Val Ala Arg Thr Met Asp Ala Leu Lys Leu Asp
 355 360 365
 Ile Ala Leu Ile Ala Asn His Leu His Ala Ile Val Ala Leu Met Met
 370 375 380
 Asp Asn Arg Phe Ser Arg Gly Leu Pro Asn Ser Leu Ser Pro Thr Pro
 385 390 395 400
 Gly Met Tyr Gln Gly Phe Lys Gly Val Gln Leu Ser Gln Thr Ala Leu

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405	410	415
Val Ala Ala Ile Arg His Asp Cys Ala Ala Ser Gly Ile His Thr Leu		
420	425	430
Ala Thr Glu Gln Tyr Asn Gln Asp Ile Val Ser Leu Gly Leu His Ala		
435	440	445
Ala Gln Asp Val Leu Glu Met Glu Gln Lys Leu Arg Asn Ile Val Ser		
450	455	460
Met Thr Ile Leu Val Val Cys Gln Ala Ile His Leu Arg Gly Asn Ile		
465	470	475
Ser Glu Ile Ala Pro Glu Thr Ala Lys Phe Tyr His Ala Val Arg Glu		
485	490	495
Ile Ser Ser Pro Leu Ile Thr Asp Arg Ala Leu Asp Glu Asp Ile Ile		
500	505	510
Arg Ile Ala Asp Ala Ile Ile Asn Asp Gln Leu Pro Leu Pro Glu Ile		
515	520	525
Met Leu Glu Glu Leu Asn Pro Leu Ile Asn Glu Ile Ser Lys Ile Ile		
530	535	540
Ser Ala Ala Gly Asn Phe Asp Val Lys Glu Glu Arg Ala Ala Ala Ser		
545	550	555
Leu Leu Gln Leu Ser Gly Asn Ala Ser Asp Phe Ser Tyr Gly Arg Asn		
565	570	575
Ser Ile Thr Leu Thr Ala Ser Ala		
580		

<210> SEQ ID NO 54

<211> LENGTH: 525

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 54

Met Asn Ile Ser Arg Arg Lys Leu Leu Leu Gly Val Gly Ala Ala Gly			
1	5	10	15
Val Leu Ala Gly Gly Ala Ala Leu Val Pro Met Val Arg Arg Asp Gly			
20	25	30	
Lys Phe Val Glu Ala Lys Ser Arg Ala Ser Phe Val Glu Gly Thr Gln			
35	40	45	
Gly Ala Leu Pro Lys Glu Ala Asp Val Val Ile Ile Gly Ala Gly Ile			
50	55	60	
Gln Gly Ile Met Thr Ala Ile Asn Leu Ala Glu Arg Gly Met Ser Val			
65	70	75	80
Thr Ile Leu Glu Lys Gly Gln Ile Ala Gly Glu Gln Ser Gly Arg Ala			
85	90	95	
Tyr Ser Gln Ile Ile Ser Tyr Gln Thr Ser Pro Glu Ile Phe Pro Leu			
100	105	110	
His His Tyr Gly Lys Ile Leu Trp Arg Gly Met Asn Glu Lys Ile Gly			
115	120	125	
Ala Asp Thr Ser Tyr Arg Thr Gln Gly Arg Val Glu Ala Leu Ala Asp			
130	135	140	
Glu Lys Ala Leu Asp Lys Ala Gln Ala Trp Ile Lys Thr Ala Lys Glu			
145	150	155	160
Ala Ala Gly Phe Asp Thr Pro Leu Asn Thr Arg Ile Ile Lys Gly Glu			
165	170	175	

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Glu Leu Ser Asn Arg Leu Val Gly Ala Gln Thr Pro Trp Thr Val Ala
 180 185 190
 Ala Phe Glu Glu Asp Ser Gly Ser Val Asp Pro Glu Thr Gly Thr Pro
 195 200 205
 Ala Leu Ala Arg Tyr Ala Lys Gln Ile Gly Val Lys Ile Tyr Thr Asn
 210 215 220
 Cys Ala Val Arg Gly Ile Glu Thr Ala Gly Gly Lys Ile Ser Asp Val
 225 230 235 240
 Val Ser Glu Lys Gly Ala Ile Lys Thr Ser Gln Val Val Leu Ala Gly
 245 250 255
 Gly Ile Trp Ser Arg Leu Phe Met Gly Asn Met Gly Ile Asp Ile Pro
 260 265 270
 Thr Leu Asn Val Tyr Leu Ser Gln Gln Arg Val Ser Gly Val Pro Gly
 275 280 285
 Ala Pro Arg Gly Asn Val His Leu Pro Asn Gly Ile His Phe Arg Glu
 290 295 300
 Gln Ala Asp Gly Thr Tyr Ala Val Ala Pro Arg Ile Phe Thr Ser Ser
 305 310 315 320
 Ile Val Lys Asp Ser Phe Leu Leu Gly Pro Lys Phe Met His Leu Leu
 325 330 335
 Gly Gly Gly Glu Leu Pro Leu Glu Phe Ser Ile Gly Glu Asp Leu Phe
 340 345 350
 Asn Ser Phe Lys Met Pro Thr Ser Trp Asn Leu Asp Glu Lys Thr Pro
 355 360 365
 Phe Glu Gln Phe Arg Val Ala Thr Ala Thr Gln Asn Thr Gln His Leu
 370 375 380
 Asp Ala Val Phe Gln Arg Met Lys Thr Glu Phe Pro Val Phe Glu Lys
 385 390 395 400
 Ser Glu Val Val Glu Arg Trp Gly Ala Val Val Ser Pro Thr Phe Asp
 405 410 415
 Glu Leu Pro Ile Ile Ser Glu Val Lys Glu Tyr Pro Gly Leu Val Ile
 420 425 430
 Asn Thr Ala Thr Val Trp Gly Met Thr Glu Gly Pro Ala Ala Gly Glu
 435 440 445
 Val Thr Ala Asp Ile Val Met Gly Lys Lys Pro Val Ile Asp Pro Thr
 450 455 460
 Pro Phe Ser Leu Asp Arg Phe Lys Lys Leu Asn Pro Leu Ile Asn Glu
 465 470 475 480
 Ile Ser Lys Ile Ile Ser Ala Ala Gly Asn Phe Asp Val Lys Glu Glu
 485 490 495
 Arg Ala Ala Ala Ser Leu Leu Gln Leu Ser Gly Asn Ala Ser Asp Phe
 500 505 510
 Ser Tyr Gly Arg Asn Ser Ile Thr Leu Thr Ala Ser Ala
 515 520 525

<210> SEQ ID NO 55

<211> LENGTH: 159

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 55

Cys Thr Thr Ala Ala Thr Cys Cys Ala Thr Thr Ala Ala Thr Thr Ala
 1 5 10 15

-continued

Ala Thr Gly Ala Ala Ala Thr Cys Ala Gly Cys Ala Ala Ala Thr
 20 25 30
 Cys Ala Thr Thr Thr Cys Ala Gly Cys Thr Gly Cys Ala Gly Gly Thr
 35 40 45
 Ala Ala Thr Thr Thr Thr Gly Ala Thr Gly Thr Thr Ala Ala Ala Gly
 50 55 60
 Ala Gly Gly Ala Ala Ala Gly Ala Gly Cys Thr Gly Cys Ala Gly Cys
 65 70 75 80
 Thr Thr Cys Thr Thr Thr Ala Thr Thr Gly Cys Ala Gly Thr Thr Gly
 85 90 95
 Thr Cys Cys Gly Gly Thr Ala Ala Thr Gly Cys Cys Ala Gly Thr Gly
 100 105 110
 Ala Thr Thr Thr Thr Thr Cys Ala Thr Ala Thr Gly Gly Ala Cys Gly
 115 120 125
 Gly Ala Ala Cys Thr Cys Ala Ala Thr Ala Ala Cys Thr Thr Thr Gly
 130 135 140
 Ala Cys Ala Gly Cys Ala Thr Cys Ala Gly Cys Ala Thr Ala Ala
 145 150 155

<210> SEQ ID NO 56

<211> LENGTH: 1777

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 56

```

acttttcata ctcccgccat tcagagaaga aaccaattgt ccatattgca tcagacattg      60
ccgtcactgc gtcttttact ggctcttctc gctaacccaa ccggtaaccc cgcttattaa      120
aagcattctg taacaaagcg ggaccaaagc catgacaaaa acgcgtaaca aaagtgtcta      180
taatcacggc agaaaaatga acatttcaag gagaaagcta cttttagggtg ttggtgctgc      240
ggcggtttta gcagggtggtg cggctttagt tccaatggtt cgccgtgacg gcaaatttgt      300
ggaagctaaa tcaagagcat catttggtga aggtacgcaa ggggctcttc ctaaagaagc      360
agatgtagtg attattgggtg ccggtattca agggatcatg accgctatta accttgetga      420
acgtgggatg agtgtcacta tcttagaaaa gggtcagatt gccggtgagc aatcaggccg      480
tgcatacagc caaattatta gttaccaaac atcgccagaa atcttcccat tacaccatta      540
tgggaaaata ttatggcgtg gcatgaatga gaaaattggt gcggatacca gttatcgtac      600
tcaaggctcg gtagaagcgc tggcagatga aaaagcatta gataaagctc aagcgtggat      660
caaaacagct aaagaagcgg cagggtttga tacaccatta aatactcgca tcattaaagg      720
tgaagagcta tcaaatcgct tagtcggtgc tcaaacgcca tggactgttg ctgcatttga      780
agaagattca ggctctgttg atcctgaaac aggcacacct gcactcgctc gttatgccaa      840
acaaatcggt gtgaaaatth ataccaactg tgcagtaaga ggtattgaaa ctgcgggtgg      900
taaaatctct gatgtggtga gtgagaaagg ggcgattaaa acgtctcaag ttgtactcgc      960
tgggggtatc tggtcgctgt tattttatggg caatatgggt attgatattc caacgctcaa     1020
tgtatatcta tcacaacaac gtgtctcagg ggttcctggt gcaccacgtg gtaatgtgca     1080
tttacctaag ggtattcatt tccgcgaaca agcggatggt acctatgccg ttgcaccacg     1140
tatctttacg agttcaatag tcaaagatag cttcctgcta gggcctaaat ttatgcactt     1200

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attaggtggc ggagagttac cgttgaatt ctctattggt gaagatctat ttaattcatt	1260
taaaatgccg acctcttggga atttagatga aaaaacacca ttcgaacaat tccgagttgc	1320
cacggcaaca caaaatacgc aacacttaga tgctgttttc caaagaatga aaacagaatt	1380
cccagtattt gaaaaatcag aagttgttga acgttggggg gccgttgtga gtccaacatt	1440
tgatgaatta cctatcattt ctgagggtcaa agaataccca ggcttagtga ttaacacggc	1500
aacagtgtgg ggtatgacag aaggcccgcc agcgggtgaa gtgaccgctg atattgtcat	1560
gggcaagaaa cctgttattg atccaacgcc gtttagtttg gatcgtttta agaagtaact	1620
taatccatta attaataaaa tcagcaaaat catttcagct gcaggtaatt ttgatgttaa	1680
agaggaaaaga gctgcagctt ctttattgca gttgtccggt aatgccagtg atttttcata	1740
tggaacggaac tcaataactt tgacagcatc agcataa	1777

<210> SEQ ID NO 57

<211> LENGTH: 1963

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 57

ggtaccagtt gttcttattg gtgggtgtgc tttatggttg catcgtagta aatggttgta	60
acaaaagcaa tttttccggc tgtctgtata caaaaacgcc gtaaagtttg agcgaagtca	120
ataaactctc taccatttca gggcaatatc tctcttgat ccctctagaa ataattttgt	180
ttaactttta gaaggagata tacatatgaa agctaaagat gttcagccaa ccattattat	240
taataaaaaat ggccttatct ctttgaaga tatctatgac attgcgataa aacaaaaaaa	300
agtagaaata tcaacggaga tcaactgaact tttgacgcat ggctcgtgaa aattagagga	360
aaaattaaat tcaggagagg ttatatatgg aatcaatata ggatttgagg ggaatgccaa	420
tttagttgtg ccatttgaga aaatcgaga gcatcagcaa aatctgttaa cttttctttc	480
tgctgggtact ggggactata tgtccaaacc ttgtattaaa gcgtcacaat ttactatgtt	540
actttctgtt tgcaagggtt ggtctgcaac cagaccaatt gtgctcaag caattgttga	600
tcatattaat catgacattg ttctctggt tctctgctat ggctcagtg gtgcaagcgg	660
tgatttaatt cctttatctt atattgcacg agcattatgt ggtatcggca aagtttatta	720
tatgggcgca gaaattgacg ctgctgaagc aattaaacgt gcagggttga caccattatc	780
gttaaaagcc aaagaaggtc ttgctctgat taacggcacc cgggtaatgt caggaaatcag	840
tgcaatcacc gtcattaaac tggaaaaact atttaaagcc tcaatttctg cgattgccct	900
tgctgttgaa gcattacttg catctcatga acattatgat gcccggttc aacaagtaaa	960
aaatcatcct ggtcaaaacg cggtggaag tgcattgcgt aatttatttg caggttcaac	1020
gcaggttaat ctattatctg ggggttaaaga acaagccaat aaagcttgtc gtcataaaga	1080
aattacccaa ctaaatgata ccttacagga agtttattca attcgtgtg caccacaagt	1140
attaggtata gtgccagaat ctttagctac cgctcggaat atattggaac gggaagtatt	1200
ctcagctaata gataatccat tgatagatcc agaaaaatggc gatgttctac acggtggaaa	1260
ttttatgggg caatatgtcg cccgaacaat ggatgcatta aaactggata ttgctttaat	1320
tgccaatcat cttcacgcca ttgtggctct tatgatggat aaccgtttct ctctgggatt	1380
acctaattca ctgagtcga caccggcat gtatcaaggt tttaaaggcg tccaactttc	1440

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tcaaaccgct ttagttgctg caattcgcca tgattgtgct gcatcaggta ttcataccct 1500
cgccacagaa caatacaatc aagatattgt cagtttaggt ctgcatgccg ctcaagatgt 1560
ttagagatg gagcagaaat tacgcaatat tgtttcaatg acaattctgg tagtttgtca 1620
ggccattcat cttcgcgcca atattagtga aattgcgcct gaaactgcta aattttacca 1680
tgcagtacgc gaaatcagtt ctcccttgat cactgatcgt gcgttgatg aagatataat 1740
ccgcattgcg gatgcaatta ttaatgatca acttcctctg ccagaaatca tgctggaaga 1800
ataacttaat ccattaatta atgaaatcag caaaatcatt tcagctgcag gtaattttga 1860
tgtaaagag gaaagagctg cagcttcttt attgcagttg tccggtaatg ccagtgattt 1920
ttcatatgga cggaactcaa taactttgac agcatcagca taa 1963

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<210> SEQ ID NO 58
<211> LENGTH: 707
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                             polypeptide

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<400> SEQUENCE: 58

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Met Asp Ser Cys His Lys Ile Asp Tyr Gly Leu Tyr Ala Leu Glu Ile
 1           5           10           15
Leu Ala Gln Tyr His Asn Val Ser Val Asn Pro Glu Glu Ile Lys His
          20           25           30
Arg Phe Asp Thr Asp Gly Thr Gly Leu Gly Leu Thr Ser Trp Leu Leu
          35           40           45
Ala Ala Lys Ser Leu Glu Leu Lys Val Lys Gln Val Lys Lys Thr Ile
          50           55           60
Asp Arg Leu Asn Phe Ile Ser Leu Pro Ala Leu Val Trp Arg Glu Asp
          65           70           75           80
Gly Arg His Phe Ile Leu Thr Lys Val Ser Lys Glu Ala Asn Arg Tyr
          85           90           95
Leu Ile Phe Asp Leu Glu Gln Arg Asn Pro Arg Val Leu Glu Gln Ser
          100          105          110
Glu Phe Glu Ala Leu Tyr Gln Gly His Ile Ile Leu Ile Ala Ser Arg
          115          120          125
Ser Ser Val Thr Gly Lys Leu Ala Lys Phe Asp Phe Thr Trp Phe Ile
          130          135          140
Pro Ala Ile Ile Lys Tyr Arg Lys Ile Phe Ile Glu Thr Leu Val Val
          145          150          155          160
Ser Val Phe Leu Gln Leu Phe Ala Leu Ile Thr Pro Leu Phe Phe Gln
          165          170          175
Val Val Met Asp Lys Val Leu Val His Arg Gly Phe Ser Thr Leu Asn
          180          185          190
Val Ile Thr Val Ala Leu Ser Val Val Val Val Phe Glu Ile Ile Leu
          195          200          205
Ser Gly Leu Arg Thr Tyr Ile Phe Ala His Ser Thr Ser Arg Ile Asp
          210          215          220
Val Glu Leu Gly Ala Lys Leu Phe Arg His Leu Leu Ala Leu Pro Ile
          225          230          235          240
Ser Tyr Phe Glu Ser Arg Arg Val Gly Asp Thr Val Ala Arg Val Arg
          245          250          255
Glu Leu Asp Gln Ile Arg Asn Phe Leu Thr Gly Gln Ala Leu Thr Ser
          260          265          270

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Val	Leu	Asp	Leu	Leu	Phe	Ser	Phe	Ile	Phe	Phe	Ala	Val	Met	Trp	Tyr
275						280						285			
Tyr	Ser	Pro	Lys	Leu	Thr	Leu	Val	Ile	Leu	Phe	Ser	Leu	Pro	Cys	Tyr
290						295				300					
Ala	Ala	Trp	Ser	Val	Phe	Ile	Ser	Pro	Ile	Leu	Arg	Arg	Arg	Leu	Asp
305				310						315				320	
Asp	Lys	Phe	Ser	Arg	Asn	Ala	Asp	Asn	Gln	Ser	Phe	Leu	Val	Glu	Ser
				325				330						335	
Val	Thr	Ala	Ile	Asn	Thr	Ile	Lys	Ala	Met	Ala	Val	Ser	Pro	Gln	Met
		340				345						350			
Thr	Asn	Ile	Trp	Asp	Lys	Gln	Leu	Ala	Gly	Tyr	Val	Ala	Ala	Gly	Phe
355						360						365			
Lys	Val	Thr	Val	Leu	Ala	Thr	Ile	Gly	Gln	Gln	Gly	Ile	Gln	Leu	Ile
370						375				380					
Gln	Lys	Thr	Val	Met	Ile	Ile	Asn	Leu	Trp	Leu	Gly	Ala	His	Leu	Val
385				390						395				400	
Ile	Ser	Gly	Asp	Leu	Ser	Ile	Gly	Gln	Leu	Ile	Ala	Phe	Asn	Met	Leu
				405				410						415	
Ala	Gly	Gln	Ile	Val	Ala	Pro	Val	Ile	Arg	Leu	Ala	Gln	Ile	Trp	Gln
		420				425						430			
Asp	Phe	Gln	Gln	Val	Gly	Ile	Ser	Val	Thr	Arg	Leu	Gly	Asp	Val	Leu
435						440						445			
Asn	Ser	Pro	Thr	Glu	Ser	Tyr	His	Gly	Lys	Leu	Ala	Leu	Pro	Glu	Ile
450						455				460					
Asn	Gly	Asn	Ile	Thr	Phe	Arg	Asn	Ile	Arg	Phe	Arg	Tyr	Lys	Pro	Asp
465				470						475				480	
Ser	Pro	Val	Ile	Leu	Asp	Asn	Ile	Asn	Leu	Ser	Ile	Lys	Gln	Gly	Glu
		485						490						495	
Val	Ile	Gly	Ile	Val	Gly	Arg	Ser	Gly	Ser	Gly	Lys	Ser	Thr	Leu	Thr
		500				505						510			
Lys	Leu	Ile	Gln	Arg	Phe	Tyr	Ile	Pro	Glu	Asn	Gly	Gln	Val	Leu	Ile
515						520				525					
Asp	Gly	His	Asp	Leu	Ala	Leu	Ala	Asp	Pro	Asn	Trp	Leu	Arg	Arg	Gln
530				535						540					
Val	Gly	Val	Val	Leu	Gln	Asp	Asn	Val	Leu	Leu	Asn	Arg	Ser	Ile	Ile
545				550				555						560	
Asp	Asn	Ile	Ser	Leu	Ala	Asn	Pro	Gly	Met	Ser	Val	Glu	Lys	Val	Ile
		565						570						575	
Tyr	Ala	Ala	Lys	Leu	Ala	Gly	Ala	His	Asp	Phe	Ile	Ser	Glu	Leu	Arg
		580				585						590			
Glu	Gly	Tyr	Asn	Thr	Ile	Val	Gly	Glu	Gln	Gly	Ala	Gly	Leu	Ser	Gly
595						600						605			
Gly	Gln	Arg	Gln	Arg	Ile	Ala	Ile	Ala	Arg	Ala	Leu	Val	Asn	Asn	Pro
610						615				620					
Lys	Ile	Leu	Ile	Phe	Asp	Glu	Ala	Thr	Ser	Ala	Leu	Asp	Tyr	Glu	Ser
625				630						635				640	

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Glu Leu Leu Ser Glu Pro Glu Ser Leu Tyr Ser Tyr Leu Tyr Gln Leu
690 695 700

Gln Ser Asp
705

<210> SEQ ID NO 59
 <211> LENGTH: 478
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 59

Met Lys Thr Trp Leu Met Gly Phe Ser Glu Phe Leu Leu Arg Tyr Lys
1 5 10 15
 Leu Val Trp Ser Glu Thr Trp Lys Ile Arg Lys Gln Leu Asp Thr Pro
20 25 30
 Val Arg Glu Lys Asp Glu Asn Glu Phe Leu Pro Ala His Leu Glu Leu
35 40 45
 Ile Glu Thr Pro Val Ser Arg Arg Pro Arg Leu Val Ala Tyr Phe Ile
50 55 60
 Met Gly Phe Leu Val Ile Ala Val Ile Leu Ser Val Leu Gly Gln Val
65 70 75 80
 Glu Ile Val Ala Thr Ala Asn Gly Lys Leu Thr Leu Ser Gly Arg Ser
85 90 95
 Lys Glu Ile Lys Pro Ile Glu Asn Ser Ile Val Lys Glu Ile Ile Val
100 105 110
 Lys Glu Gly Glu Ser Val Arg Lys Gly Asp Val Leu Leu Lys Leu Thr
115 120 125
 Ala Leu Gly Ala Glu Ala Asp Thr Leu Lys Thr Gln Ser Ser Leu Leu
130 135 140
 Gln Thr Arg Leu Glu Gln Thr Arg Tyr Gln Ile Leu Ser Arg Ser Ile
145 150 155 160
 Glu Leu Asn Lys Leu Pro Glu Leu Lys Leu Pro Asp Glu Pro Tyr Phe
165 170 175
 Gln Asn Val Ser Glu Glu Glu Val Leu Arg Leu Thr Ser Leu Ile Lys
180 185 190
 Glu Gln Phe Ser Thr Trp Gln Asn Gln Lys Tyr Gln Lys Glu Leu Asn
195 200 205
 Leu Asp Lys Lys Arg Ala Glu Arg Leu Thr Ile Leu Ala Arg Ile Asn
210 215 220
 Arg Tyr Glu Asn Leu Ser Arg Val Glu Lys Ser Arg Leu Asp Asp Phe
225 230 235 240
 Arg Ser Leu Leu His Lys Gln Ala Ile Ala Lys His Ala Val Leu Glu
245 250 255
 Gln Glu Asn Lys Tyr Val Glu Ala Ala Asn Glu Leu Arg Val Tyr Lys
260 265 270
 Ser Gln Leu Glu Gln Ile Glu Ser Glu Ile Leu Ser Ala Lys Glu Glu
275 280 285
 Tyr Gln Leu Val Thr Gln Leu Phe Lys Asn Glu Ile Leu Asp Lys Leu
290 295 300
 Arg Gln Thr Thr Asp Asn Ile Glu Leu Leu Thr Leu Glu Leu Glu Lys
305 310 315 320
 Asn Glu Glu Arg Gln Gln Ala Ser Val Ile Arg Ala Pro Val Ser Gly
325 330 335

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Lys Val Gln Gln Leu Lys Val His Thr Glu Gly Gly Val Val Thr Thr
 340 345 350
 Ala Glu Thr Leu Met Val Ile Val Pro Glu Asp Asp Thr Leu Glu Val
 355 360 365
 Thr Ala Leu Val Gln Asn Lys Asp Ile Gly Phe Ile Asn Val Gly Gln
 370 375 380
 Asn Ala Ile Ile Lys Val Glu Ala Phe Pro Tyr Thr Arg Tyr Gly Tyr
 385 390 395 400
 Leu Val Gly Lys Val Lys Asn Ile Asn Leu Asp Ala Ile Glu Asp Gln
 405 410 415
 Lys Leu Gly Leu Val Phe Asn Val Ile Val Ser Val Glu Glu Asn Asp
 420 425 430
 Leu Ser Thr Gly Asn Lys His Ile Pro Leu Ser Ser Gly Met Ala Val
 435 440 445
 Thr Ala Glu Ile Lys Thr Gly Met Arg Ser Val Ile Ser Tyr Leu Leu
 450 455 460
 Ser Pro Leu Glu Glu Ser Val Thr Glu Ser Leu His Glu Arg
 465 470 475

<210> SEQ ID NO 60
 <211> LENGTH: 458
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 60

Met Lys Asn Ala Ser Thr Val Ser Glu Asp Thr Ala Ser Asn Gln Glu
 1 5 10 15
 Pro Thr Leu His Arg Gly Leu His Asn Arg His Ile Gln Leu Ile Ala
 20 25 30
 Leu Gly Gly Ala Ile Gly Thr Gly Leu Phe Leu Gly Ile Gly Pro Ala
 35 40 45
 Ile Gln Met Ala Gly Pro Ala Val Leu Leu Gly Tyr Gly Val Ala Gly
 50 55 60
 Ile Ile Ala Phe Leu Ile Met Arg Gln Leu Gly Glu Met Val Val Glu
 65 70 75 80
 Glu Pro Val Ser Gly Ser Phe Ala His Phe Ala Tyr Lys Tyr Trp Gly
 85 90 95
 Pro Phe Ala Gly Phe Leu Ser Gly Trp Asn Tyr Trp Val Met Phe Val
 100 105 110
 Leu Val Gly Met Ala Glu Leu Thr Ala Ala Gly Ile Tyr Met Gln Tyr
 115 120 125
 Trp Phe Pro Asp Val Pro Thr Trp Ile Trp Ala Ala Phe Phe Ile
 130 135 140
 Ile Ile Asn Ala Val Asn Leu Val Asn Val Arg Leu Tyr Gly Glu Thr
 145 150 155 160
 Glu Phe Trp Phe Ala Leu Ile Lys Val Leu Ala Ile Ile Gly Met Ile
 165 170 175
 Gly Phe Gly Leu Trp Leu Leu Phe Ser Gly His Gly Gly Glu Lys Ala
 180 185 190
 Ser Ile Asp Asn Leu Trp Arg Tyr Gly Gly Phe Phe Ala Thr Gly Trp
 195 200 205
 Asn Gly Leu Ile Leu Ser Leu Ala Val Ile Met Phe Ser Phe Gly Gly

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210	215	220
Leu Glu Leu Ile Gly Ile Thr Ala Ala Glu Ala Arg Asp Pro Glu Lys		
225	230	235 240
Ser Ile Pro Lys Ala Val Asn Gln Val Val Tyr Arg Ile Leu Leu Phe		
	245	250 255
Tyr Ile Gly Ser Leu Val Val Leu Leu Ala Leu Tyr Pro Trp Val Glu		
	260	265 270
Val Lys Ser Asn Ser Ser Pro Phe Val Met Ile Phe His Asn Leu Asp		
	275	280 285
Ser Asn Val Val Ala Ser Ala Leu Asn Phe Val Ile Leu Val Ala Ser		
	290	295 300
Leu Ser Val Tyr Asn Ser Gly Val Tyr Ser Asn Ser Arg Met Leu Phe		
305	310	315 320
Gly Leu Ser Val Gln Gly Asn Ala Pro Lys Phe Leu Thr Arg Val Ser		
	325	330 335
Arg Arg Gly Val Pro Ile Asn Ser Leu Met Leu Ser Gly Ala Ile Thr		
	340	345 350
Ser Leu Val Val Leu Ile Asn Tyr Leu Leu Pro Gln Lys Ala Phe Gly		
	355	360 365
Leu Leu Met Ala Leu Val Val Ala Thr Leu Leu Leu Asn Trp Ile Met		
	370	375 380
Ile Cys Leu Ala His Leu Arg Phe Arg Ala Ala Met Arg Arg Gln Gly		
385	390	395 400
Arg Glu Thr Gln Phe Lys Ala Leu Leu Tyr Pro Phe Gly Asn Tyr Leu		
	405	410 415
Cys Ile Ala Phe Leu Gly Met Ile Leu Leu Leu Met Cys Thr Met Asp		
	420	425 430
Asp Met Arg Leu Ser Ala Ile Leu Leu Pro Val Trp Ile Val Phe Leu		
	435	440 445
Phe Met Ala Phe Lys Thr Leu Arg Arg Lys		
450	455	

<210> SEQ ID NO 61

<211> LENGTH: 456

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 61

Met Glu Gly Gln Gln His Gly Glu Gln Leu Lys Arg Gly Leu Lys Asn		
1	5	10 15
Arg His Ile Gln Leu Ile Ala Leu Gly Gly Ala Ile Gly Thr Gly Leu		
	20	25 30
Phe Leu Gly Ser Ala Ser Val Ile Gln Ser Ala Gly Pro Gly Ile Ile		
	35	40 45
Leu Gly Tyr Ala Ile Ala Gly Phe Ile Ala Phe Leu Ile Met Arg Gln		
	50	55 60
Leu Gly Glu Met Val Val Glu Glu Pro Val Ala Gly Ser Phe Ser His		
	65	70 75 80
Phe Ala Tyr Lys Tyr Trp Gly Ser Phe Ala Gly Phe Ala Ser Gly Trp		
	85	90 95
Asn Tyr Trp Val Leu Tyr Val Leu Val Ala Met Ala Glu Leu Thr Ala		
	100	105 110

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Val	Gly	Lys	Tyr	Ile	Gln	Phe	Trp	Tyr	Pro	Glu	Ile	Pro	Thr	Trp	Val
	115						120					125			
Ser	Ala	Ala	Val	Phe	Phe	Val	Val	Ile	Asn	Ala	Ile	Asn	Leu	Thr	Asn
	130					135					140				
Val	Lys	Val	Phe	Gly	Glu	Met	Glu	Phe	Trp	Phe	Ala	Ile	Ile	Lys	Val
	145				150					155					160
Ile	Ala	Val	Val	Ala	Met	Ile	Ile	Phe	Gly	Ala	Trp	Leu	Leu	Phe	Ser
				165					170					175	
Gly	Asn	Gly	Gly	Pro	Gln	Ala	Ser	Val	Ser	Asn	Leu	Trp	Asp	Gln	Gly
			180					185					190		
Gly	Phe	Leu	Pro	His	Gly	Phe	Thr	Gly	Leu	Val	Met	Met	Met	Ala	Ile
		195					200					205			
Ile	Met	Phe	Ser	Phe	Gly	Gly	Leu	Glu	Leu	Val	Gly	Ile	Thr	Ala	Ala
	210					215					220				
Glu	Ala	Asp	Asn	Pro	Glu	Gln	Ser	Ile	Pro	Lys	Ala	Thr	Asn	Gln	Val
	225				230					235					240
Ile	Tyr	Arg	Ile	Leu	Ile	Phe	Tyr	Ile	Gly	Ser	Leu	Ala	Val	Leu	Leu
			245						250					255	
Ser	Leu	Met	Pro	Trp	Thr	Arg	Val	Thr	Ala	Asp	Thr	Ser	Pro	Phe	Val
			260					265					270		
Leu	Ile	Phe	His	Glu	Leu	Gly	Asp	Thr	Phe	Val	Ala	Asn	Ala	Leu	Asn
		275					280					285			
Ile	Val	Val	Leu	Thr	Ala	Ala	Leu	Ser	Val	Tyr	Asn	Ser	Cys	Val	Tyr
	290					295					300				
Cys	Asn	Ser	Arg	Met	Leu	Phe	Gly	Leu	Ala	Gln	Gln	Gly	Asn	Ala	Pro
	305				310					315					320
Lys	Ala	Leu	Ala	Ser	Val	Asp	Lys	Arg	Gly	Val	Pro	Val	Asn	Thr	Ile
			325						330					335	
Leu	Val	Ser	Ala	Leu	Val	Thr	Ala	Leu	Cys	Val	Leu	Ile	Asn	Tyr	Leu
			340					345					350		
Ala	Pro	Glu	Ser	Ala	Phe	Gly	Leu	Leu	Met	Ala	Leu	Val	Val	Ser	Ala
		355					360					365			
Leu	Val	Ile	Asn	Trp	Ala	Met	Ile	Ser	Leu	Ala	His	Met	Lys	Phe	Arg
	370					375					380				
Arg	Ala	Lys	Gln	Glu	Gln	Gly	Val	Val	Thr	Arg	Phe	Pro	Ala	Leu	Leu
	385				390					395					400
Tyr	Pro	Leu	Gly	Asn	Trp	Val	Cys	Leu	Leu	Phe	Met	Ala	Ala	Val	Leu
				405					410					415	
Val	Ile	Met	Leu	Met	Thr	Pro	Gly	Met	Ala	Ile	Ser	Val	Tyr	Leu	Ile
			420					425					430		
Pro	Val	Trp	Leu	Ile	Val	Leu	Gly	Ile	Gly	Tyr	Leu	Phe	Lys	Glu	Lys
		435					440					445			
Thr	Ala	Lys	Ala	Val	Lys	Ala	His								
	450					455									

<210> SEQ ID NO 62

<211> LENGTH: 1582

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 62

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60

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acaaaagcaa tttttccggc tgtctgtata caaaaacgcc gtaaagtttg agcgaagtca	120
ataaactctc taccattca gggcaatata tctcttgat ccctctagaa ataattttgt	180
ttaactttaa gaaggagata tacatatgaa aaacgcgtca accgtatcgg aagatactgc	240
gtcgaatcaa gagccgacgc ttcacgcggg attacataac cgtcatattc aactgattgc	300
gttgggtggc gcaattggta ctggctctgt tcttgccatt ggcccggcga ttcagatggc	360
gggtccggct gtattgtcgg gctacggcgt cgcggggatc atcgctttcc tgattatgcg	420
ccagcttggc gaaatggtgg ttgaggagcc ggtatccggt tcatttgccc actttgecta	480
taaatactgg ggaccgtttg cgggcttcct ctctggctgg aactactggg taatgttcgt	540
gctggtggga atggcagagc tgaccgctgc gggcatctat atgcagtact ggttcccgga	600
tgttccaacg tggatttggg ctgccgctt ctttattatc atcaacgccg ttaacctggt	660
gaacgtgcgc ttatatggcg aaaccgagtt ctggtttcg ttgattaaag tgctggcaat	720
catcggtatg atcggtttg gctgtggct gctgttttct ggtcacggcg gcgagaaagc	780
cagtatcgac aacctctggc gctacggcgg tttcttcgcc accggctgga atgggctgat	840
tttgtcgtg cgggtaatta tgttctcctt cggcggctct gagctgattg ggattactgc	900
cgtgaagcg cgcgatccgg aaaaaagcat tccaaaagcg gtaaatcagg tgggtgatcg	960
catcctgctg ttttacatcg gttcactggt ggttttactg gcgctctatc cgtgggtgga	1020
agtgaatcc aacagtagcc cgtttgtgat gattttccat aatctcgaca gcaacgtggt	1080
agcttctcgc ctgaacttcg tcattctggt agcatcgctg tcagtgtata acagcggggt	1140
ttactctaac agccgcagtc tgtttggcct ttctgtgcag ggtaatgcgc cgaagttttt	1200
gactcgcgtc agccgtcgcg gtgtgccgat taaactcgtg atgctttccg gagcgatcac	1260
ttcgtgggtg gtgttaatca actatctgct gccgcaaaaa gcgtttggtc tgetgatggc	1320
gctggtggta gcaacgctgc tgttgaactg gattatgata tgtctggcgc atctgcgttt	1380
tcgtgcagcg atgcgacgtc aggggcgtga aacacagttt aaggcgcgtc tctatccggt	1440
cggcaactat ctctgcattg ccttctcgg catgattttg ctgctgatgt gcacgatgga	1500
tgatatgcgc ttgtcagcga tctgtctgcc ggtgtggatt gtattcctgt ttatggcatt	1560
taaaacgctg cgtcggaaat aa	1582

<210> SEQ ID NO 63

<211> LENGTH: 1573

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 63

ggtaccagtt gttcttattg gtgggtgtgc tttatgggtg catcgtagta aatgggtgta	60
acaaaagcaa tttttccggc tgtctgtata caaaaacgcc gtaaagtttg agcgaagtca	120
ataaactctc taccattca gggcaatata tctcttgat ccctctagaa ataattttgt	180
ttaactttaa gaaggagata tacatatgga ggggcagcag catggggagc aactgaagcg	240
cgggttaaaa aatcgtcaca ttcaattaat cgcgctgggc ggagcaattg gtacgggatt	300
gttctcgggt tcagcgagcg tcattccaatc ggcaggtcca gggatcatct tgggatatgc	360
gatcgcagcg tttatcgett ttcttattat gcgccaatta ggtgagatgg tggtcgagga	420
gcctgtagct ggctccttct cacatttcgc gtacaagtat tggggatcct ttgcgggatt	480

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tgcttctggt	tggaactatt	gggttcttta	tgctctggtg	gccatggcgg	agctgaccgc	540
ggttggaata	tatatccagt	tctggtaccc	cgagatcccg	acgtgggtct	cagccgcggg	600
attctttggt	gttatcaatg	caatcaattt	aaccaacgta	aaagtatttg	gtgaaatgga	660
gttctggttc	gcgattatca	aagtaattgc	cgtagtgtct	atgattatct	ttggggcatg	720
gttgcttttc	tcaggaaatg	gcggaccaca	agcgtcgggt	tcaaacctgt	gggatcaagg	780
gggattcctg	cgcacgggat	ttacgggctt	ggtgatgatg	atggctatca	ttatgttttc	840
tttcggtggt	cttgaattag	tgggtattac	cgcagcagag	gcagataatc	ccgaacaaag	900
catcccaaaa	gctactaacc	aagttattta	cgtatccctg	atctttttata	ttgggtctct	960
ggcagtcctg	ctttccttaa	tgccctggac	acgtgtaacg	gccgatacat	ccccctttgt	1020
acttatcttt	cacgaactgg	gagacacgtt	cgtcgccaat	gcattaaaca	ttgttgtgct	1080
gacagctgcc	ttatctgtgt	ataatagctg	cgtttattgc	aattcacgta	tgttattcgg	1140
gcttgctcag	cagggtaacg	cgccaaaggc	gttgacctca	gtagataagc	gcggagtgcc	1200
tgtaaataca	atcttggtca	gcgcattagt	cacggctctt	tgctttctga	ttactatctt	1260
ggctcctgaa	agcgcattcg	gattacttat	ggccctggtt	gtttccgccc	tggttatcaa	1320
ttgggcaatg	attagtttgg	cacatatgaa	gttcgcgcgt	gctaacaagc	aacaagggtg	1380
cgtaactcgt	ttccctgect	tattgtatcc	gctggggaat	tgggtatgcc	ttctttttat	1440
ggccgcagta	ctggaatta	tgttgatgac	gcccgccatg	gctattagtg	tataccttat	1500
tccggtatgg	ttaatcgtct	tgggtatcgg	ctacttattt	aaagaaaaaa	cagcaaaaagc	1560
cgtaaaggct	cat					1573

<210> SEQ ID NO 64

<211> LENGTH: 967

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 64

caaatatcac	ataatcttaa	catatcaata	aacacagtaa	agtttcatgt	gaaaaacatc	60
aaacataaaa	tacaagctcg	gaatacgaat	cacgctatac	acattgctaa	caggaaatgag	120
attatctaaa	tgaggattga	tatattaatt	ggacatacta	gtttttttca	tcaaaccagt	180
agagataact	tccttcacta	tctcaatgag	gaagaaataa	aacgctatga	tcagtttcat	240
tttgtgagtg	ataaagaact	ctatatctta	agccgtatcc	tgctcaaaac	agcactaaaa	300
agatatcaac	ctgatgtctc	attacaatca	tggcaattta	gtacgtgcaa	atatggcaaa	360
ccattttatg	tttttcttca	gttggcaaaa	aagatttttt	ttaacctttc	ccatactata	420
gatacagtag	cgttgcttat	tagttctcac	tgcgagcttg	gtgtcgatat	tgaacaaata	480
agagatttag	acaactctta	tctgaatata	agtcagcatt	tttttactcc	acaggaagct	540
actaacatag	tttcaacttc	tcgttatgaa	ggtcaattac	ttttttggaa	aatgtggacg	600
ctcaaagaag	cttacatcaa	atatcgaggt	aaaggccctat	ctttaggact	ggattgtatt	660
gaatttcatt	taacaaataa	aaaactaact	tcaaaaataa	gaggttcacc	tgtttatctc	720
tctcaatgga	aaatatgtaa	ctcattttct	gcattagcct	ctccactcat	caccctataa	780
ataactattg	agctattttc	tatgcagtcc	caactttatc	accacgacta	tcagtaattt	840
cattcgtcaa	atgggcagaa	ttgaatcgcc	acggataatc	tagacacttc	tgagccgtcg	900

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ataatattga ttttcatatt cgcgcggtgg tgtaagtatc cgcataatc gtgccattca 960
catttag 967
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<210> SEQ ID NO 65
<211> LENGTH: 424
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide
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<400> SEQUENCE: 65
ggatgggggg aaacatggat aagttcaaag aaaaaaaccc gttatctctg cgtgaaagac 60
aagtattgcg catgctggca caaggtgatg agtactctca aatatcacat aatcttaaca 120
tatcaataaa cacagtaaag tttcatgtga aaaacatcaa acataaaata caagctcgga 180
atacgaatca cgctatacac attgctaaca ggaatgagat tatctaaatg aggattgatg 240
tgtaggctgg agctgcttcg aagttcctat actttctaga gaataggaac ttcggaatag 300
gaacttcgga ataggaacta aggaggatat tcatatgtcg tcaaatgggc agaattgaat 360
cgccacggat aatctagaca cttctgagcc gtcgataata ttgattttca tattccgctg 420
gtgg 424
```

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<210> SEQ ID NO 66
<211> LENGTH: 14
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      consensus sequence
<220> FEATURE:
<221> NAME/KEY: modified_base
<222> LOCATION: (6)..(9)
<223> OTHER INFORMATION: a, c, t, g, unknown or other
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<400> SEQUENCE: 66
ttgatnnnna tcaa 14
```

The invention claimed is:

1. A nucleic acid sequence comprising one or more gene(s) encoding a phenylalanine ammonia lyase (PAL) and one or more gene(s) encoding a phenylalanine transporter, wherein the PAL is from *Anabaena variabilis* (PAL1) or *Photorhabdus luminescens* (PAL3), wherein the one or more genes encoding the PAL is operably linked to an inducible promoter that is not in nature associated with the one or more genes encoding the PAL, wherein the inducible promoter is induced by low oxygen or anaerobic conditions to enable expression of the PAL in an *Escherichia* probiotic bacterium; wherein the one or more genes encoding the phenylalanine transporter are from *Acinetobacter calcoaceticus*, *Salmonella enterica*, or *Escherichia coli*; and wherein the one or more genes encoding the phenylalanine transporter is operably linked to an inducible promoter that is not in nature associated with the one or more genes encoding the phenylalanine transporter, wherein the inducible promoter is induced by low oxygen or anaerobic conditions to enable expression of the phenylalanine transporter in the bacterium.
2. The nucleic acid sequence of claim 1, wherein the promoter operably linked to the one or more gene(s) encoding the PAL and the promoter operably linked to the one or more gene(s) encoding the phenylalanine transporter are separate copies of the same promoter sequence.
3. The nucleic acid sequence of claim 1, wherein the promoter operably linked to the one or more gene(s) encoding the PAL and the promoter operably linked to the one or more gene(s) encoding a phenylalanine transporter are different promoters.
4. The nucleic acid sequence of claim 1, wherein the promoter operably linked to the one or more gene(s) encoding the PAL and the promoter operably linked to the one or more gene(s) encoding the phenylalanine transporter are induced by environmental conditions found in a mammalian gut.
5. The nucleic acid sequence of claim 4, wherein the promoter is induced by low-oxygen or anaerobic conditions.
6. The nucleic acid sequence of claim 5, wherein the promoter is selected from the group consisting of an FNR-responsive promoter, an ANR-responsive promoter, and a DNR-responsive promoter.
7. The nucleic acid sequence of claim 1, wherein the phenylalanine transporter is PheP.
8. The nucleic acid sequence of claim 1, wherein the one or more gene(s) encoding the PAL and the one or more gene(s) encoding the phenylalanine transporter are located on a chromosome in the bacterium.

345**346**

9. The nucleic acid sequence of claim 1, wherein the one or more gene(s) encoding the PAL and the one or more gene(s) encoding the phenylalanine transporter are located on a plasmid in the bacterium.

* * * * *